

# Fixed Income

Concept-First Notes + Exam Q&A + MCQ Bank + Mock Test — CA Intermediate

One complete file: concepts, worked Q&A, revision cheat-sheet, MCQs & a full mock test

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# Chapter 01 — Introduction to Fixed Income

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## 1. The Problem / The Need

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Every functioning economy runs on a single, unavoidable fact: **the people and institutions who have money today are rarely the same ones who have the best use for it today**. A pension fund sitting on ₹50,000 crore of retirement savings has cash but no factory to build. A power company that needs ₹5,000 crore to build a plant has the project but not the cash. A government that must fund roads, defence, and salaries collects taxes months after it must spend. Households want to buy houses decades before they have saved the full price.

The economy needs a *mechanism* to move purchasing power from those who have surplus (savers) to those who have a deficit (borrowers), across **time**, and to do so in a way that:

1. **Compensates the saver** for giving up the use of money now (the time value of money).
2. **Compensates the saver for risk** — the borrower might not pay back.
3. **Is standardised and tradable**, so the saver is not locked in for the full term and a large need can be split among thousands of lenders.

Equity (shares) solves part of this problem but at a cost the borrower often does not want: giving away ownership, control, and an unlimited share of future profits. A company that simply needs to *borrow* ₹500 crore for five years does not want to hand permanent ownership to strangers. It wants to **borrow, pay a defined cost, and be done**.

Fixed income is the answer to this need. It is the formalised, standardised, tradable **IOU** — a contractual promise to repay borrowed money on a schedule, at a defined cost. The word "fixed" refers to the fact that the cash flows (the interest payments and the repayment of principal) are, in the classic case, **fixed and known in advance**. You lend ₹100 today; you are contractually owed, say, ₹7 every year and your ₹100 back at the end. The certainty of that schedule is precisely what makes the instrument valuable, priceable, and enormous in scale.

## 2. The Core Idea

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A fixed-income security is a **loan sliced into tradable units, with a legally defined repayment schedule**. The core relationship is always the same:

A **borrower (issuer)** receives cash today from a **lender (investor)** and in exchange issues a security that promises (a) periodic interest payments, called **coupons**, and (b) the return of the borrowed amount, called the **principal, par, or face value**, at a stated future date called **maturity**.

The single most important idea in the entire subject — the one from which duration, convexity, yield curves, and credit spreads all descend — is this:

**A bond is a bundle of fixed future cash flows, and its price today is simply the present value of those cash flows discounted at a rate that reflects time and risk.**

$$P = \sum_{t=1}^N \frac{C_t}{(1+r)^t}$$

where  $P_t$  is the price,  $C_t$  is the cash flow at time  $t$ ,  $r$  is the discount rate (the yield), and  $N$  is the number of periods. Everything else in fixed income is an elaboration of, or a reaction to, this one equation.

The second core idea, which trips up every beginner and every interviewer loves to test, follows immediately from the first:

**When the discount rate  $r$  rises, the price  $P$  falls — and vice versa.** Bond prices and yields move in opposite directions. Always.

This inverse relationship is not a convention or a market quirk; it is arithmetic. If a fixed stream of ₹7-per-year is worth more when discounted at 5% than at 8%, then a security paying that fixed stream must be worth *less* the moment the market demands 8%.

### 3. Why / How It Works

Let us build the intuition for *why* the present-value machine works, because interviewers probe understanding, not memorised formulas.

**Why discount at all?** Money has a time value. ₹100 today is worth more than ₹100 in a year, because ₹100 today can be invested to become ₹105 or ₹107 in a year. Therefore a promise of ₹100 in a year is worth *less* than ₹100 today. To find how much less, we **discount** it: divide by  $(1+r)$ . If the going rate is 7%, then ₹100 due in one year is worth  $100 / 1.07 = ₹93.46$  today. That is what someone should rationally pay now to receive ₹100 in a year.

**Why does a bond's price equal the sum of discounted cash flows?** Because a rational buyer will not pay more than what the cash flows are worth to them, and a rational seller will not accept less. If a bond's cash flows are worth ₹98 in present-value terms and it is priced at ₹95, buyers rush in (arbitrage) and bid the price up to ₹98. If it is priced at ₹101, holders sell and the price falls to ₹98. The market price *is* the present value at the market's required yield. This is the **no-arbitrage** foundation.

**Why does the yield reflect time AND risk?** The discount rate  $r$  is not one number pulled from air. It is built up in layers:

$$r = r_{\text{real}} + \text{expected inflation} + \text{risk premia}$$

The layers, conceptually:

- **Real risk-free rate** — the pure cost of deferring consumption, with no inflation and no default risk.
- **Inflation compensation** — lenders want to be repaid in money of the same purchasing power.
- **Credit (default) risk premium** — extra yield demanded because this particular borrower might not pay.
- **Liquidity premium** — extra yield because this bond may be hard to sell quickly at a fair price.
- **Term/maturity premium** — extra yield for tying money up longer and bearing more uncertainty.

A government bond in its own currency carries essentially only the first two plus a term premium; a small-company bond carries all five. The *difference* in yield between a risky bond and a government bond of the same maturity is called the **credit spread**, and it is the market's price of that borrower's risk.

The diagram below shows the anatomy of a plain bond as a timeline of cash flows.

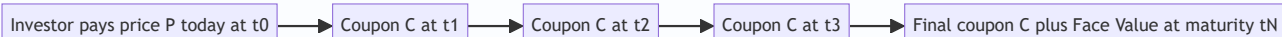


Figure 1 — A plain vanilla bond is an outflow today followed by a fixed stream of coupons and a principal repayment at maturity.

## 4. Full Content — The Formulas and the Bond Math

### 4.1 The vocabulary of a bond

| Term              | Symbol | Meaning                                                                                    |
|-------------------|--------|--------------------------------------------------------------------------------------------|
| Face / Par value  | $F$    | Amount repaid at maturity; the base for coupon calculation. Conventionally ₹100 or ₹1,000. |
| Coupon rate       | $c$    | Annual interest rate stated on the bond, as a % of face value.                             |
| Coupon payment    | $C$    | Cash interest per period $C = c \times F$ (payments per year).                             |
| Maturity          | $N$    | Number of periods until principal is repaid.                                               |
| Price             | $P$    | What the bond trades at today (present value of cash flows).                               |
| Yield to maturity | $y$    | The single discount rate that makes PV of cash flows equal price.                          |
| Current yield     | —      | Annual coupon divided by current price.                                                    |

### 4.2 The pricing formula

For a bond paying coupon  $C$  each period for  $N$  periods and face  $F$  at maturity, discounted at periodic yield  $y$ :

$$P = \underbrace{\sum_{t=1}^N \frac{C}{(1+y)^t}}_{\text{PV of coupons (an annuity)}} + \underbrace{\frac{F}{(1+y)^N}}_{\text{PV of principal}}$$

The coupon stream is an annuity, so it has a closed form:

$$P = C \times \left[ \frac{1 - (1+y)^{-N}}{y} \right] + F \times (1+y)^{-N}$$

### 4.3 The three price regimes (memorise the logic, not the cases)

| Relationship                | Bond trades at              | Why                                                                        |
|-----------------------------|-----------------------------|----------------------------------------------------------------------------|
| Coupon rate $c =$ Yield $y$ | <b>Par</b> ( $P = F$ )      | The coupon exactly compensates the required return.                        |
| Coupon rate $c >$ Yield $y$ | <b>Premium</b> ( $P > F$ )  | Bond pays more than the market demands, so it is worth more than face.     |
| Coupon rate $c <$ Yield $y$ | <b>Discount</b> ( $P < F$ ) | Bond pays less than the market demands, so it must be cheap to compensate. |

## 4.4 Yield measures

**Current yield** (a crude income measure, ignores capital gain/loss and time value):

$$\text{Current Yield} = \frac{\text{Annual coupon}}{\text{Price}}$$

**Yield to maturity (YTM)** — the internal rate of return if held to maturity, reinvesting coupons at  $y$ . It is the  $y$  that solves the pricing equation. There is no algebraic solution for  $N > 1$ ; it is found by iteration (trial and error) or a financial calculator.

## 4.5 Interest-rate sensitivity — duration (a first look)

Because price is a function of yield, we care *how much* price moves when yield moves. The first-order measure is **duration**. **Macaulay duration** is the weighted-average time to receive the bond's cash flows, weights being each cash flow's share of total present value:

$$D_{\text{Mac}} = \frac{\sum_{t=1}^N t \cdot \frac{C_t}{(1+y)^t}}{P}$$

**Modified duration** converts this into a % price-change sensitivity:

$$D_{\text{Mod}} = \frac{D_{\text{Mac}}}{1+y} \quad \Longleftrightarrow \quad \frac{\Delta P}{P} \approx -D_{\text{Mod}} \times \Delta y$$

The negative sign encodes the inverse price–yield relationship. Duration is introduced fully in a later chapter; it appears here so you see *why* it exists — it is the natural next question after "price is a PV of fixed cash flows."

## 4.6 The fixed-income landscape

Fixed income is not one market but a family. The map below organises it by issuer and, within that, by risk.

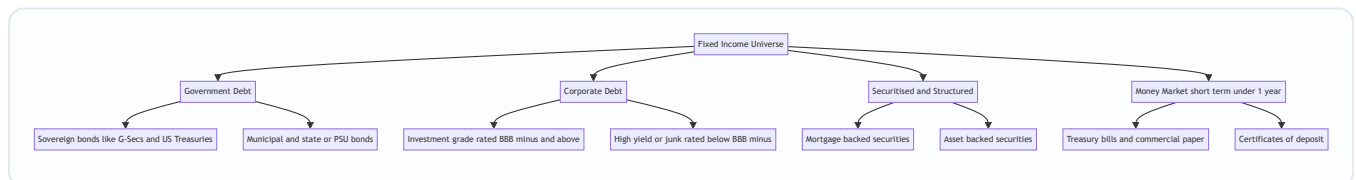


Figure 2 — The fixed-income landscape by issuer type and risk tier. Government debt anchors the low-risk end; high-yield corporates and structured products carry more risk and yield.

## 4.7 Why fixed income is the largest asset class

Globally, the bond market is roughly **US\$130+ trillion** in outstanding debt, materially larger than global equity market capitalisation. Three structural reasons explain this:

1. **Everyone borrows, few issue equity.** Governments cannot issue shares; they can only tax or borrow. So the entire stock of government financing is fixed income. Government debt alone rivals all global equity.
2. **Companies use far more debt than equity for financing on the margin.** Debt is cheaper (interest is tax-deductible; lenders demand less return than shareholders because they bear less risk) and does not dilute ownership.
3. **Debt is issued repeatedly and in tranches.** A single borrower has one equity but issues dozens of bonds over time, each a separate security with its own maturity and coupon, multiplying the number of instruments.

## 4.8 Overview of the three headline risks

The certainty of a bond's *cash flows* does not make the *investment* certain. Three risks dominate, and every fixed-income decision trades them off.

**Interest-rate risk.** This is the risk that market yields rise after you buy, pushing the price of your bond *down*. It flows directly from the pricing equation: raise  $y$  and  $P$  falls. It affects even a default-free government bond. Its magnitude scales with **duration** — the longer the maturity and the lower the coupon, the more violently the price swings for a given yield move. A 30-year zero-coupon government bond can lose a third of its value on a 1.5% yield spike with zero chance of default. This is *the* risk that catches investors who assume "government bond = safe."

**Credit (default) risk.** This is the risk that the *borrower fails to pay* — misses a coupon, defaults on principal, or is downgraded (raising its spread and cutting the price even before any default). It is measured by the **credit spread** over the government curve and summarised by **credit ratings** (AAA down to D). A government borrowing in its own currency has minimal credit risk (it can print money); a highly leveraged company has substantial credit risk, which is exactly why it must offer a fat spread to attract lenders.

**Liquidity risk.** This is the risk that when you want to sell, there is no ready buyer at a fair price, forcing you to accept a discount. A benchmark 10-year government bond trades in seconds at a razor-thin bid-ask spread; a small, obscure corporate bond may take days to sell and cost you 2–3% in transaction friction. Investors demand a **liquidity premium** (extra yield) to hold illiquid bonds.

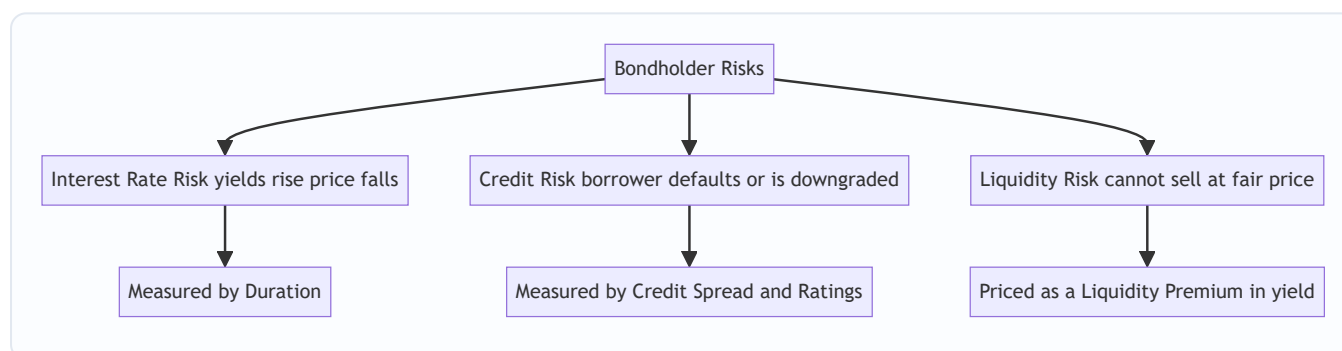


Figure 3 — The three headline risks of holding a bond, and the metric that measures or prices each one.

These are not the only risks — **reinvestment risk** (coupons reinvested at lower rates), **inflation risk** (fixed cash flows lose purchasing power), and **currency risk** (for foreign bonds) also matter — but interest-rate, credit, and liquidity are the three that frame the entire discipline.

## 5. Worked Examples

### Example 1 — Pricing a plain bond and confirming the three regimes

**Setup.** A bond has face value  $F = ₹1,000$ , an annual coupon rate  $c = 8\%$  (so  $C = ₹80$  per year), and  $N = 3$  years to maturity. Price it at three different market yields: 8%, 10%, and 6%.

**Case A: yield  $y = 8\%$  (equals coupon → expect par).**

$$P = \frac{80}{1.08} + \frac{80}{1.08^2} + \frac{1080}{1.08^3}$$

- $80 / 1.08 = 74.074$
- $80 / 1.1664 = 68.587$



- $\$1080 / 1.259712 = 857.339\$$
- **Sum = 74.074 + 68.587 + 857.339 = ₹1,000.00 ✓**

Trades at **par**, exactly as predicted when  $\$c = y\$$ . The math self-verifies.

**Case B: yield  $y = 10\%$  (yield > coupon → expect discount).**

$$P = \frac{80}{1.10} + \frac{80}{1.21} + \frac{1080}{1.331} = 950.26$$

- $\$80 / 1.10 = 72.727\$$
- $\$80 / 1.21 = 66.116\$$
- $\$1080 / 1.331 = 811.420\$$
- **Sum = ₹950.26**

Trades at a **discount** ( $P < F$ ), as predicted. A 5% rise in required yield knocked ₹49.74 off the price.

**Case C: yield  $y = 6\%$  (yield < coupon → expect premium).**

$$P = \frac{80}{1.06} + \frac{80}{1.1236} + \frac{1080}{1.191016} = 1053.63$$

- $\$80 / 1.06 = 75.472\$$
- $\$80 / 1.1236 = 71.199\$$
- $\$1080 / 1.191016 = 906.956\$$
- **Sum = ₹1,053.63**

Trades at a **premium** ( $P > F$ ), as predicted.

**Reconciliation.** The three prices — ₹950.26, ₹1,000.00, ₹1,053.63 — sit in the correct order (higher yield → lower price) and land exactly on par when  $\$c = y\$$ . The inverse price–yield relationship and the three regimes are all confirmed from one worked bond.

## Example 2 — Current yield vs. YTM, and finding YTM by iteration

**Setup.** The discount bond from Case B trades at  $P = ₹950.26$ , pays  $C = ₹80$ ,  $F = ₹1,000$ ,  $N = 3$ . Compute current yield, then recover the YTM.

**Current yield:**

$$\text{Current Yield} = \frac{80}{950.26} = 8.42\%$$

**YTM.** We know from Case B that  $y = 10\%$  produces exactly ₹950.26, so  $\text{YTM} = 10\%$ . But let us *demonstrate the iteration* an interviewer would want to see, pretending we did not know.

- Guess  $y = 8\%$ : price = ₹1,000.00 (too high — need lower price, so raise yield).
- Guess  $y = 12\%$ :
- $\$80/1.12 = 71.43\$$ ;  $\$80/1.2544 = 63.78\$$ ;  $\$1080/1.404928 = 768.72\$$ ; sum = **₹903.93** (too low — overshoot, lower yield).
- We want ₹950.26, between the 8% price (₹1,000) and 12% price (₹903.93). Linear interpolation:

$$y \approx 8\% + \frac{1000 - 950.26}{1000 - 903.93} \times (12\% - 8\%) = 8\% + \frac{49.74}{96.07} \times 4\% = 8\% + 2.07\% = 10.07\%$$

Interpolation lands at **10.07%**, very close to the true **10%** (the small error is because price is convex, not linear, in yield — the seed of the *convexity* concept in a later chapter).

**Reconciliation and insight.** Current yield (8.42%) sits *below* YTM (10%). That is exactly right for a **discount** bond: the buyer earns not only the coupon income but also a capital gain as the price pulls to par (₹1,000) at maturity, and YTM captures that gain while current yield ignores it. For a premium bond the ordering reverses (current yield > YTM). This ordering is a classic interview check.

### Example 3 — The price impact of a rate move (duration preview)

**Setup.** Take the par bond from Case A ( $P = ₹1,000$ ,  $y = 8\%$ ,  $C = ₹80$ ,  $N = 3$ ). Estimate the price if yields rise 100 basis points (1%) to 9%, first by exact pricing, then reconcile with modified duration.

**Exact reprice at 9%:**

$$P = \frac{80}{1.09} + \frac{80}{1.1881} + \frac{1080}{1.295029} = 73.394 + 67.335 + 833.958 = ₹974.69$$

Price fell by ₹25.31, a **−2.531%** move for a +1% yield change.

**Now via duration.** Compute Macaulay duration at  $y = 8\%$  (PVs from Case A):

| $t$ | $C_t$ | $PV = C_t / 1.08^t$ | $t \times PV$    |
|-----|-------|---------------------|------------------|
| 1   | 80    | 74.074              | 74.074           |
| 2   | 80    | 68.587              | 137.174          |
| 3   | 1,080 | 857.339             | 2,572.017        |
|     |       | <b>1,000.00</b>     | <b>2,783.265</b> |

$$D_{\text{Mac}} = \frac{2,783.265}{1,000} = 2.783 \text{ years}, \quad D_{\text{Mod}} = \frac{2.783}{1.08} = 2.577$$

Predicted price change:

$$\frac{\Delta P}{P} \approx -2.577 \times 1\% = -2.577\%$$

**Reconciliation.** Duration predicts **−2.577%**; exact repricing gave **−2.531%**. They agree to within 0.05%. The tiny gap (duration slightly *overstates* the fall) is convexity — the true price–yield curve bends, so the straight-line duration estimate errs on the downside for a rate rise. This is why duration is a *first-order* approximation and convexity a *second-order* correction. Both concepts fall straight out of the single pricing equation from Section 2.

## 6. Connections

- **To equities (corporate finance).** Debt and equity are the two claims on a firm's assets. Bondholders are paid *first* (senior); shareholders get the *residual*. This seniority is why bonds are lower-risk, lower-return, and why the cost of debt is below the cost of equity — feeding directly into the Weighted Average Cost of Capital (WACC).
- **To macroeconomics and monetary policy.** Central banks (RBI, the Fed) set short-term rates and, via those, influence the entire yield curve. Bond prices are the transmission mechanism of monetary policy into the real economy: cut rates → bond prices rise, borrowing costs fall → investment and consumption rise.
- **To the yield curve (next chapters).** The relationship between yield and maturity — the term structure — is the master diagram of fixed income and is built from exactly the discounting logic introduced here.

- **To duration and convexity (later chapters).** These are the calculus of the pricing equation: the first and second derivatives of price with respect to yield.
- **To credit analysis.** The credit spread introduced here is the entire subject of credit risk, ratings, and default modelling.
- **To portfolio management.** Bonds provide income, capital preservation, and — crucially — *diversification*: they often (not always) rise when equities fall, cushioning portfolios.

## 7. Key Terms

| Term                                  | One-line definition                                                                              |
|---------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Fixed income / bond</b>            | A tradable loan with a contractually defined repayment schedule.                                 |
| <b>Issuer / borrower</b>              | The entity that sells the bond and owes the money (government, corporate, bank).                 |
| <b>Investor / lender / bondholder</b> | The buyer who provides cash and holds the claim to repayment.                                    |
| <b>Face / par value (\$F\$)</b>       | The principal repaid at maturity and the base for coupon calculation.                            |
| <b>Coupon</b>                         | The periodic interest payment; coupon rate is it as a % of face value.                           |
| <b>Maturity</b>                       | The date the principal is repaid and the bond ends.                                              |
| <b>Price</b>                          | Present value of all future cash flows at the market yield.                                      |
| <b>Yield to maturity (YTM)</b>        | The single discount rate equating PV of cash flows to price; the bond's IRR if held to maturity. |
| <b>Current yield</b>                  | Annual coupon ÷ current price; ignores capital gain/loss and time value.                         |
| <b>Discount / premium / par</b>       | Price below / above / equal to face value.                                                       |
| <b>Credit spread</b>                  | Extra yield of a risky bond over a government bond of equal maturity.                            |
| <b>Duration</b>                       | Sensitivity of price to yield; also weighted-average time to cash flows.                         |
| <b>Coupon vs. zero-coupon</b>         | Pays periodic interest vs. pays nothing until maturity (sold at a deep discount).                |

## 8. Common Confusions

1. **"Fixed income means fixed return."** No. Cash *flows* are fixed (for a plain bond), but your *return* depends on the price you paid and whether you hold to maturity. Sell early after yields rise and you take a capital loss.
2. **"Coupon rate is the same as yield."** The coupon rate is fixed forever at issuance and set on *face* value. The yield floats with the market and is measured on *price*. They coincide only when the bond trades at par.
3. **"Higher coupon = better bond."** A high-coupon bond simply trades at a premium; you pay for the extra coupons up front. What matters for return is yield, not coupon.
4. **"Bonds are safe, full stop."** Bonds carry real risks — interest-rate risk, credit/default risk, liquidity risk, reinvestment risk, inflation risk. A long-dated government bond can lose 20%+ of its value when rates spike, with zero default.

5. **"Price and yield are the same signal."** They move *inversely*. Rising yields are bad news for existing bondholders (prices fall), even though the bond is now "paying more."
6. **"Current yield is the return."** Current yield ignores the pull-to-par capital gain/loss and time value. YTM is the proper total-return measure for hold-to-maturity.
7. **"YTM is guaranteed."** YTM assumes you hold to maturity *and* reinvest every coupon at the YTM. If reinvestment rates differ, your realised return differs — this is reinvestment risk.

## 9. Recap

Fixed income exists to move purchasing power from savers to borrowers across time, through standardised, tradable IOUs whose defining feature is a **known schedule of cash flows**. A bond is fully described by its face value, coupon, and maturity, and its price is nothing more than the **present value of its cash flows discounted at the market yield** — the master equation from which the entire subject unfolds. From that equation follow the inescapable facts that **price and yield move inversely** and that bonds trade at par, premium, or discount depending on whether the coupon matches, exceeds, or falls short of the yield. The discount rate itself is a stack of compensations — for time, inflation, credit, liquidity, and term — and the credit portion, measured as a spread over government bonds, prices each borrower's risk.

Fixed income is the **largest asset class on earth** because governments can only borrow (not issue equity), companies finance heavily with cheap, non-dilutive debt, and each borrower issues many bonds over time. It matters to portfolios (income, preservation, diversification), to rates (it is the transmission channel of monetary policy), and to the economy (it funds governments, infrastructure, and firms). The three headline risks — **interest-rate, credit, and liquidity** — set up the rest of the study guide, and the natural next questions ("*how much* does price move when yield moves?") lead straight into duration and convexity.

## 10. Quick-Reference / Interview Points

**The one equation:**  $P = \sum_{t=1}^N \frac{C_t}{(1+y)^t}$  — price is the present value of fixed cash flows.

**The one relationship:** Bond price and yield move **inversely**. Always. (It is arithmetic, not convention.)

**The three regimes:**

| If             | Then     |
|----------------|----------|
| Coupon = Yield | Par      |
| Coupon > Yield | Premium  |
| Coupon < Yield | Discount |

**Yield ordering checks (interview favourite):** - Discount bond: Current Yield < YTM (you also earn a capital gain to par). - Premium bond: Current Yield > YTM. - Par bond: Coupon = Current Yield = YTM.

**The yield build-up:**  $y = \text{real rate} + \text{inflation} + \text{credit spread} + \text{liquidity premium} + \text{term premium}$ .

**Why fixed income is the biggest asset class:** governments can only borrow; debt is cheaper and non-dilutive than equity; each issuer floats many bonds. Global bond market  $\approx$  US\$130T+, larger than global equities.

**The three headline risks:** - **Interest-rate risk** — yields rise, prices fall (measured by duration). Hits even risk-free bonds. - **Credit risk** — the borrower may default; priced by the credit spread and ratings. - **Liquidity risk** — you may not be able to sell quickly at a fair price; priced by the liquidity premium.

**Duration soundbite:**  $\frac{\Delta P}{P} \approx -D_{\text{Mod}} \times \Delta y$ . A modified duration of 2.6 means a +1% yield move costs  $\approx 2.6\%$  of price. It is a first-order estimate; convexity corrects the curvature.

**Seniority soundbite:** bondholders are paid before shareholders — lower risk, lower return, cost of debt below cost of equity, interest is tax-deductible.

**Two-sentence definition to open any interview answer:** "Fixed income is a tradable loan with a contractually fixed cash-flow schedule — periodic coupons plus principal at maturity. Its price is the present value of those cash flows at the market yield, which is why price and yield move inversely."

# Q&A — Introduction to Fixed Income

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A practice bank for Chapter 01. Every question is followed by a full answer. Work each one before reading the answer.

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## Section A — Concept Checks

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### A1. In one sentence, what is a fixed-income security?

A fixed-income security is a tradable loan with a contractually defined repayment schedule — the issuer receives cash today and promises periodic interest payments (coupons) plus repayment of principal (face value) at a stated maturity date.

### A2. Why does fixed income exist at all? What economic need does it meet?

Savers hold surplus money but have no direct use for it; borrowers (governments, companies, households) have uses but no cash. Fixed income is the mechanism to move purchasing power from savers to borrowers across time, in a way that (1) compensates the saver for the time value of money, (2) compensates for the risk of non-repayment, and (3) is standardised and tradable so the saver is not locked in and a large need can be split among many lenders.

### A3. Why would a company issue a bond instead of equity?

Debt lets the company borrow, pay a defined cost, and be done — without giving away ownership, control, or an unlimited share of future profits. Debt is also cheaper (interest is tax-deductible and lenders demand less return than shareholders because they bear less risk) and non-dilutive.

### A4. State the single master equation of fixed income in words.

A bond's price today equals the present value of its future cash flows, discounted at a rate that reflects time and risk:  $P = \sum_{t=1}^N C_t / (1+y)^t$ . Everything else in the subject is an elaboration of this one equation.

### A5. Why do bond prices and yields move in opposite directions?

Because it is arithmetic, not convention. The cash flows are fixed. A fixed stream discounted at a higher rate has a smaller present value. So the instant the market demands a higher yield, the fixed stream is worth less, and the price falls. Raise  $y$ , and  $P$  falls, always.

### A6. What are the layers that build up the discount rate (yield)?

$y = \text{real risk-free rate} + \text{expected inflation} + \text{credit (default) risk premium} + \text{liquidity premium} + \text{term/maturity premium}$ . A government bond in its own currency carries essentially the first two plus a term premium; a small-company bond carries all five.

### A7. Define the credit spread.

The credit spread is the difference in yield between a risky bond and a government bond of the same maturity. It is the market's price of that particular borrower's default risk.

### A8. Distinguish coupon rate from yield to maturity.

The coupon rate is fixed forever at issuance and is measured on face value. The yield to maturity floats with the market and is measured on the current price. They coincide only when the bond trades exactly at par.

### A9. Name the three headline risks of holding a bond and the metric that measures or prices each.

Interest-rate risk (measured by duration), credit/default risk (priced by the credit spread and summarised by ratings), and liquidity risk (priced as a liquidity premium in yield).

### A10. Why is fixed income the largest asset class on earth?

Three structural reasons: (1) governments can only tax or borrow — they cannot issue equity — so all government financing is fixed income; (2) companies finance heavily with cheap, non-dilutive debt on the margin; (3) each borrower has one equity but issues many bonds over time, multiplying the number of instruments. Global bonds outstanding are roughly US\$130 trillion-plus, larger than global equity market capitalisation.

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## Section B — Numerical Bond-Math Problems

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### B1. Price a 3-year bond, face ₹1,000, annual coupon 8% (₹80/yr), at a market yield of 8%.

Cash flows: ₹80 at  $t=1$ , ₹80 at  $t=2$ , ₹1,080 at  $t=3$  (final coupon plus face).

- $\$80 / 1.08 = 74.074\$$
- $\$80 / 1.08^2 = 80 / 1.1664 = 68.587\$$
- $\$1,080 / 1.08^3 = 1,080 / 1.259712 = 857.339\$$
- $\text{Sum } \$ = 74.074 + 68.587 + 857.339 = ₹1,000.00\$$

The bond prices at par. Reconciliation: coupon rate (8%) equals yield (8%), so by the par regime we expect  $\$P = F\$$ , and the arithmetic confirms exactly ₹1,000.00.

### B2. Reprice the same bond at a yield of 10%. Which regime, and by how much did the price change?

- $\$80 / 1.10 = 72.727\$$
- $\$80 / 1.21 = 66.116\$$
- $\$1,080 / 1.331 = 811.420\$$
- $\text{Sum } \$ = ₹950.26\$$

The bond trades at a discount ( $\$P < F\$$ ) because yield (10%) now exceeds coupon (8%). Price fell from ₹1,000.00 to ₹950.26, a drop of ₹49.74 (−4.97%) for a +2% yield move.

### B3. Reprice the same bond at a yield of 6%. Which regime?

- $\$80 / 1.06 = 75.472\$$
- $\$80 / 1.1236 = 71.199\$$
- $\$1,080 / 1.191016 = 906.956\$$
- $\text{Sum } \$ = ₹1,053.63\$$

The bond trades at a premium ( $\$P > F\$$ ) because yield (6%) is below coupon (8%). Reconciliation across B1–B3: the three prices ₹1,053.63 (6%), ₹1,000.00 (8%), ₹950.26 (10%) fall in strictly decreasing order as yield rises — confirming the inverse price–yield relationship and all three regimes from one bond.

### B4. Compute the current yield of the discount bond in B2 (price ₹950.26, coupon ₹80).

$$\text{Current Yield} = \frac{\text{Annual coupon}}{\text{Price}} = \frac{80}{950.26} = 8.42\%$$

### B5. The bond in B4 has a YTM of 10%. Explain the ordering of current yield (8.42%) versus YTM (10%).

For a discount bond, current yield < YTM. The buyer of a discount bond earns two things: the coupon income and a capital gain as the price pulls up to par (₹1,000) at maturity. Current yield captures only the coupon income and ignores the capital gain and time value; YTM captures both. Hence YTM (10%) exceeds current yield (8.42%). The ordering reverses for a premium bond (current yield > YTM), and for a par bond coupon = current yield = YTM.

**B6. Find the YTM of the ₹950.26 discount bond by iteration and linear interpolation, pretending you do not already know it.**

Try bracketing yields and pricing:

- At  $y = 8\%$ : price = ₹1,000.00 (too high; we need the lower price ₹950.26, so raise the yield).
- At  $y = 12\%$ :  $\$80/1.12 = 71.43\$$ ;  $\$80/1.2544 = 63.78\$$ ;  $\$1,080/1.404928 = 768.72\$$ ; sum = ₹903.93 (too low; overshoot).

Target ₹950.26 lies between the 8% price (₹1,000.00) and the 12% price (₹903.93). Interpolate:

$$y \approx 8\% + \frac{1,000 - 950.26}{1,000 - 903.93} \times (12\% - 8\%) = 8\% + \frac{49.74}{96.07} \times 4\% = 8\% + 2.07\% = 10.07\%$$

Interpolation gives 10.07%, very close to the true 10%. The small overshoot arises because price is convex (curved), not linear, in yield — the seed of the convexity concept.

**B7. Estimate the price of the par bond (B1) if yields rise 100 bps to 9%. Do it by exact repricing.**

- $\$80 / 1.09 = 73.394\$$
- $\$80 / 1.1881 = 67.335\$$
- $\$1,080 / 1.295029 = 833.958\$$
- Sum  $\$ = ₹974.69\$$

Price fell from ₹1,000.00 to ₹974.69, a change of −₹25.31, or −2.531% for a +1% yield move.

**B8. Compute Macaulay and modified duration for the par bond at 8%, then predict the B7 price move and reconcile.**

Using the PVs from B1:

| $t$ | $C_t$ | $PV = C_t/1.08^t$ | $t \times PV$    |
|-----|-------|-------------------|------------------|
| 1   | 80    | 74.074            | 74.074           |
| 2   | 80    | 68.587            | 137.174          |
| 3   | 1,080 | 857.339           | 2,572.017        |
|     |       | <b>1,000.00</b>   | <b>2,783.265</b> |

$$D_{\text{Mac}} = \frac{2,783.265}{1,000} = 2.783 \text{ years}, \quad D_{\text{Mod}} = \frac{2.783}{1.08} = 2.577$$

$$\text{Predicted change for } +1\% \text{ yield: } \Delta P / P \approx -D_{\text{Mod}} \times \Delta y = -2.577 \times 1\% = -2.577\%$$

Reconciliation: duration predicts −2.577%; exact repricing (B7) gave −2.531%. They agree to within 0.05%. Duration slightly overstates the fall because the true price–yield curve is convex; that curvature gap is exactly



what convexity corrects. Duration is a first-order (straight-line) estimate; convexity is the second-order correction.

**B9. A zero-coupon bond, face ₹1,000, matures in 3 years and is priced to yield 8%. What is its price and its Macaulay duration?**

A zero pays nothing until maturity, so the only cash flow is ₹1,000 at  $t=3$ .

$$P = \frac{1,000}{1.08^3} = \frac{1,000}{1.259712} = ₹793.83$$

Macaulay duration of a zero equals its maturity, because all present value sits at the single final date:

$D_{\text{Mac}} = 3.0$  years. Note this exceeds the 2.783-year duration of the 8% coupon bond of the same maturity — lower (here zero) coupon means longer duration and greater interest-rate sensitivity.

**B10. A 30-year zero yields 5%, then yields spike to 6.5% (+150 bps). Roughly why can it lose about a third of its value with zero default?**

Price at 5%:  $\$1,000/1.05^{30} = \$1,000/4.3219 = ₹231.38$ . Price at 6.5%:  $\$1,000/1.065^{30} = \$1,000/6.6144 = ₹151.19$ . Change  $\$ = (151.19 - 231.38)/231.38 = -34.7\%$ . A long-dated zero has a duration near 30 years, so a 1.5% yield move produces roughly  $\$ -30 \times 1.5\% \approx -45\%$  on a first-order basis, moderated by convexity to about  $-35\%$  in exact terms. This illustrates that a default-free government bond can still carry enormous interest-rate risk.

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## Section C — Interview-Style Questions with Model Answers

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**C1. "Give me a two-sentence definition of fixed income to open your answer."**

Model answer: "Fixed income is a tradable loan with a contractually fixed cash-flow schedule — periodic coupons plus principal at maturity. Its price is the present value of those cash flows at the market yield, which is why price and yield move inversely."

**C2. "A bond's coupon is 8% but you say its yield is 10%. Is the bond trading above or below par, and why?"**

Model answer: "Below par — at a discount. The coupon of 8% is fixed and pays less than the 10% the market now demands. To make the fixed stream competitive with a 10% required return, the price must fall below face value so the buyer picks up the shortfall as a capital gain to par. Concretely, an 8% three-year bond at a 10% yield prices around ₹950 against ₹1,000 face."

**C3. "Interest rates just rose. A client says, 'Good — my bonds now pay more, so I made money.' Correct them."**

Model answer: "Two things are being confused. New bonds issued now pay more, but the client's existing bond has a fixed coupon that did not change. When market yields rose, the present value of that fixed stream fell, so the existing bond's price dropped — the client took a mark-to-market capital loss. Rising yields are bad news for existing bondholders even though the market is 'paying more.' The client only benefits from higher rates on money reinvested at the new, higher yields."

**C4. "Why is a long-dated government bond not 'safe'?"**

Model answer: "Safe from default, not safe from price movement. A government borrowing in its own currency has minimal credit risk, but it has substantial interest-rate risk. Because price is the present value of cash flows, a long maturity means a high duration, and a rate spike drives the price down sharply. A 30-year

zero can lose a third of its value on a 1.5% yield jump with zero chance of default. 'Government bond' answers the credit question, not the interest-rate question."

**C5. "Walk me through why a bond's market price equals the present value of its cash flows — what enforces that?"**

Model answer: "No-arbitrage. If a bond's cash flows are worth ₹98 in present-value terms but it trades at ₹95, buyers rush in to capture the ₹3 and bid the price up to ₹98. If it trades at ₹101, holders sell and the price falls to ₹98. Rational buyers won't overpay and rational sellers won't undersell, so the market price settles at the present value of the cash flows discounted at the market's required yield. The price is the PV — enforced by arbitrage."

**C6. "For a discount bond, is current yield above or below YTM? Explain the intuition."**

Model answer: "Below. A discount bond returns the coupon plus a capital gain as its price pulls up to par by maturity. Current yield is just coupon over price — it ignores that capital gain and the time value of money. YTM includes it. So YTM exceeds current yield for a discount bond. For a premium bond the ordering flips, because the premium erodes to par, imposing a capital loss that pulls YTM below the current yield. At par all three — coupon rate, current yield, YTM — are equal."

**C7. "What does a modified duration of 2.6 actually tell a portfolio manager?"**

Model answer: "It's a sensitivity: a 1% (100 bps) rise in yield costs roughly 2.6% of the bond's price, and a 1% fall gains roughly 2.6%, from  $\Delta P/P \approx -D_{\text{Mod}} \times \Delta y$ . It lets the manager translate a rate view into a price impact and size positions to a target risk. But it's a first-order, straight-line estimate; for large yield moves the manager adds a convexity correction because the true price–yield relationship curves."

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## Section D — Multiple Choice Questions (with reasoning)

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**D1. A bond trades at a premium. Which must be true?**

A. Coupon rate < yield B. Coupon rate = yield C. Coupon rate > yield D. Maturity is long

Answer: **C.** A premium ( $P > F$ ) arises precisely when the coupon exceeds the required yield, so the bond pays more than the market demands and is worth more than face. A and B give discount and par respectively; D is irrelevant to the regime.

**D2. Market yields fall by 50 bps. An existing fixed-coupon bond's price will:**

A. Fall B. Rise C. Stay unchanged D. Depend on the coupon direction

Answer: **B.** Price and yield move inversely. Discounting the fixed cash flows at a lower rate raises their present value, so the price rises. The coupon is fixed and does not change direction.

**D3. Which risk affects even a default-free government bond?**

A. Credit risk B. Interest-rate risk C. Liquidity risk only D. None

Answer: **B.** Interest-rate risk flows straight from the pricing equation and hits every bond regardless of creditworthiness — raise the yield and the price falls. A government bond in its own currency has negligible credit risk but full interest-rate risk.

**D4. The credit spread is best described as:**

A. Coupon minus yield B. A risky bond's yield minus a same-maturity government bond's yield C. Current yield minus YTM D. Yield minus inflation

Answer: **B.** The credit spread is the extra yield a risky borrower must offer over a government bond of equal maturity — the market's price of that borrower's default risk.

**D5. For a par bond, which relationship holds?**

A. Coupon rate = current yield = YTM B. Current yield > YTM C. YTM > coupon rate D. Price > face value

Answer: **A.** At par there is no capital gain or loss to maturity, so coupon rate, current yield, and YTM all coincide.

**D6. Two 3-year bonds have the same yield; Bond X has an 8% coupon, Bond Z is a zero. Which has the greater interest-rate sensitivity (duration)?**

A. Bond X B. Bond Z C. Equal D. Cannot tell

Answer: **B.** A zero's Macaulay duration equals its maturity (3.0 years) because all its present value sits at the final date, whereas the coupon bond's duration is shorter (about 2.78 years) because early coupons pull weight forward. Lower coupon means longer duration and greater sensitivity.

**D7. Which correctly orders the layers of a yield?**

A. Real rate + inflation + credit + liquidity + term premium B. Coupon + inflation C. YTM – current yield D. Face value × coupon

Answer: **A.** The discount rate is built up as the real risk-free rate plus inflation compensation plus credit, liquidity, and term premia. A government bond carries mainly the first two plus a term premium; a risky corporate carries all five.

**D8. "Fixed income means a fixed return." This statement is:**

A. True always B. True only for zeros C. False — cash flows are fixed, but return depends on price paid and whether held to maturity D. True only for government bonds

Answer: **C.** For a plain bond the cash flows are fixed, but the realised return depends on the purchase price and on selling before maturity (yields may have risen, imposing a capital loss) and on reinvestment rates. Fixed cash flows are not a fixed return.

# Chapter 02 — Bond Features and Types

## 1. The Problem / The Need

A bond is a legal contract. When you lend money to a government or a company, dozens of questions must be answered before a single rupee or dollar changes hands: *Who* is borrowing? *How much* interest, and does it move with the market or stay fixed? *When* do I get my principal back? *What happens if the borrower cannot pay?* *Can the borrower repay me early and rip up the deal?* *Can I force early repayment?* *Where do I stand in line if the company goes bankrupt?*

If these questions are left vague, the loan is unpriceable. Two bonds that both pay "5%" can be worth wildly different amounts once you know that one is a 30-year unsecured note from a struggling airline, callable at the issuer's whim, while the other is a 3-year senior secured note from a sovereign-backed utility. **The features are the product.** A fixed-income analyst does not really analyze "bonds" — they analyze a specific bundle of contractual promises and options, each of which shifts value between borrower and lender.

The need, therefore, is a precise vocabulary and a valuation lens for **every structural feature a bond can carry**. This chapter builds that map: who issues, how the coupon is shaped, when principal returns, what options are embedded, what legal protections (covenants) exist, and where the lender ranks in the capital structure. Every later chapter — duration, spread analysis, credit modelling — assumes you can read a bond's DNA fluently.

## 2. The Core Idea

A bond can be decomposed into **four independent design axes**, and any real-world security is one point in this four-dimensional space:

1. **Issuer** — the source of repayment (and thus the credit risk): sovereign, corporate, municipal, agency, supranational.
2. **Cash-flow shape** — the coupon structure and maturity: fixed, floating, zero, step-up; par value; term.
3. **Embedded options** — rights that let one party alter the contract: call, put, conversion, sinking funds.
4. **Legal position** — covenants, seniority, and security that govern behaviour and recovery.

The central insight is that **each feature moves value between the two parties, and the price/yield adjusts to compensate the party who gives something up**. A call option benefits the issuer, so callable bonds must offer a *higher* yield. A put option benefits the investor, so putable bonds accept a *lower* yield. Security benefits the lender, so secured bonds yield *less* than unsecured. Nothing is free; every feature has a price you can, in principle, quantify.

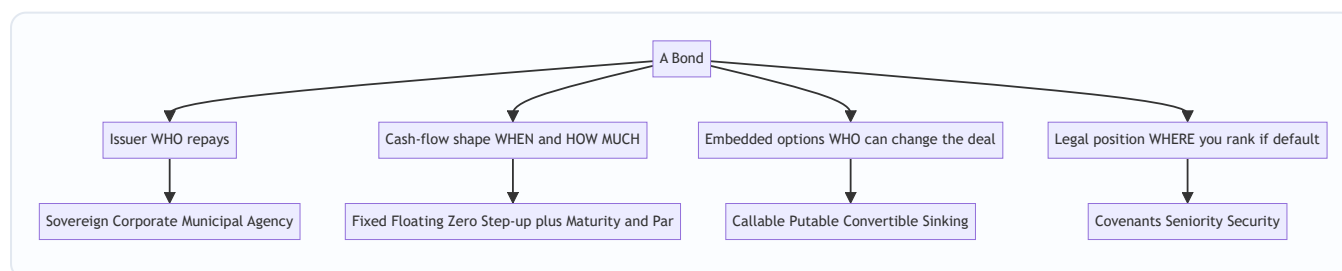


Figure 2.1 — The four design axes of any bond; a real security is one point in this space.

### 3. Why / How It Works

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Why does this decomposition hold up? Because a bond's price is nothing more than the **present value of a promised set of cash flows, adjusted for the probability those promises are kept and for any options that reshape them**. Formally, ignoring options for a moment:

$$P = \sum_{t=1}^N \frac{CF_t}{(1+y)^t}$$

Every feature enters this equation through one of three doors:

- **It changes the numerator (the cash flows  $CF_t$ ).** Coupon structure, par value, and maturity literally define the  $CF_t$  stream. A step-up bond has rising  $CF_t$ ; a zero has a single terminal  $CF_N$ .
- **It changes the discount rate  $y$ .** The issuer's creditworthiness, the seniority, and the security determine the *risk premium* baked into  $y$ . A junior unsecured airline bond is discounted at a much higher  $y$  than a senior secured sovereign-guaranteed one, even for identical cash flows.
- **It adds or subtracts an option value.** Embedded options mean the cash-flow stream is no longer fixed — it becomes *contingent*. We value these by splitting the bond into a straight bond **plus or minus** an option:

$$P_{\text{callable}} = P_{\text{straight}} - V_{\text{call}}, \quad P_{\text{puttable}} = P_{\text{straight}} + V_{\text{put}}$$

This "bond = straight bond  $\pm$  option" identity is the engine behind the whole chapter. It tells you *direction* (does the feature help the issuer or the investor?) and, with a lattice model, *magnitude*. The covenants and legal structure, meanwhile, do not appear as a separate term — they work **through** the discount rate and through the recovery assumption embedded in credit risk. Tight covenants and strong security lower expected loss, which lowers the spread, which lowers  $y$ , which raises  $P$ .

That is the whole logic. The rest is filling in each axis with precision.

### 4. Full Content

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#### 4.1 Issuer Types — Who Repays

The issuer is the single biggest driver of a bond's risk. Repayment comes from the issuer's *capacity and willingness* to pay, and these differ radically across issuer classes.

| Issuer type              | Example                                   | Source of repayment                                | Key risk                                       | Typical yield vs sovereign         |
|--------------------------|-------------------------------------------|----------------------------------------------------|------------------------------------------------|------------------------------------|
| <b>Sovereign (G-sec)</b> | US Treasury, Indian G-Sec, UK Gilt        | Taxation power, money printing (own currency)      | Inflation, currency (if foreign-currency debt) | Benchmark (lowest in own currency) |
| <b>Supranational</b>     | World Bank (IBRD), Asian Development Bank | Callable capital from member states                | Very low; AAA typically                        | Small spread over sovereign        |
| <b>Agency / GSE</b>      | Fannie Mae, US agency, NABARD             | Implicit or explicit government support            | Prepayment (if MBS), agency-specific           | Small spread over sovereign        |
| <b>Municipal</b>         | US state/city bonds, Indian muni bonds    | Tax revenue (GO) or project revenue (revenue bond) | Local fiscal stress; often tax-exempt          | Varies; tax-adjusted               |
| <b>Corporate</b>         | Reliance, Apple, an airline               | Business cash flows, asset sales                   | Default / credit risk; wide range              | Meaningful spread; wider for HY    |

Key sub-distinctions:

- **Domestic-currency vs foreign-currency sovereign debt.** A government borrowing in *its own* currency can always print to repay nominally (default risk is low but inflation/currency risk is real). The same government borrowing in USD (external debt) *can* default — this is why emerging-market USD bonds carry credit spreads and ratings below the local-currency rating.
- **Municipal bonds split into General Obligation (GO)** — backed by the full taxing power of the issuer — **and Revenue bonds** — backed only by cash flows from a specific project (a toll road, an airport). GO is generally safer. In the US, municipal interest is often **federally tax-exempt**, which is why munis trade at *lower* pre-tax yields; you compare them via the **taxable-equivalent yield**:

$$TEY = \frac{y_{\text{muni}}}{1 - \text{tax rate}}$$

A 3.5% muni for an investor in the 37% bracket is equivalent to  $0.035 / (1 - 0.37) = 5.56\%$  taxable — so it beats a 5% corporate on an after-tax basis. - **Agency bonds** blur the line between government and market credit; the critical question is whether support is *explicit* (guaranteed) or merely *implicit* (assumed, as with US GSEs pre-2008 — an assumption that proved fragile).

## 4.2 Coupon Structures — How Much, and How It Moves

The coupon is the periodic interest. Its *shape* over time defines the cash-flow numerator.

**Fixed-rate (plain vanilla).** A constant coupon rate  $c$  on par, paid (usually) semi-annually. Coupon per period  $C = \frac{c \times \text{Par}}{f}$  where  $f$  is periods per year. This is the reference structure for all pricing and duration math. Its full price sensitivity to rates is highest of the coupon-bearing types because cash flows are locked.

**Floating-rate note (FRN / "floater").** The coupon resets each period to a **reference rate plus a quoted margin (spread)**:

$$\text{Coupon rate}_t = \text{Reference}_t + \text{Quoted margin}$$

Historically the reference was LIBOR; post-2021 it is **SOFR** (USD), **SONIA** (GBP), **EURIBOR/€STR**, or **MIBOR** (INR). Because the coupon re-prices toward the market, an FRN's price stays *near par* and its **interest-rate duration is very short** (roughly the time to the next reset). Its price *does* still move with changes in the issuer's credit spread — that is captured by the **discount margin (DM)**, the spread that equates the FRN's PV to its price. If credit deteriorates, the required DM rises above the fixed quoted margin and the price falls below par.

**Zero-coupon bond.** Pays *no* periodic coupon; the entire return comes from buying at a deep discount and receiving par at maturity. Price:

$$P = \frac{\text{Par}}{(1+y)^N}$$

Zeros have the **longest possible duration for a given maturity** — Macaulay duration equals the maturity exactly — because 100% of the cash flow is at the end. They carry no reinvestment risk (no coupons to reinvest) but maximum price volatility. Treasury STRIPS are the classic example.

**Step-up / step-down coupon.** The coupon changes on a preset schedule (step-up: rises over time; step-down: falls). Common in callable structures where the rising coupon pressures the issuer to call. A variant is the **credit-linked step-up**, where the coupon rises if the issuer is downgraded — compensating investors for higher risk.

**Deferred / PIK (payment-in-kind).** No cash coupon early on (deferred), or the coupon is paid in *more bonds* rather than cash (PIK). Used by highly leveraged issuers who want to conserve cash; risky for investors.

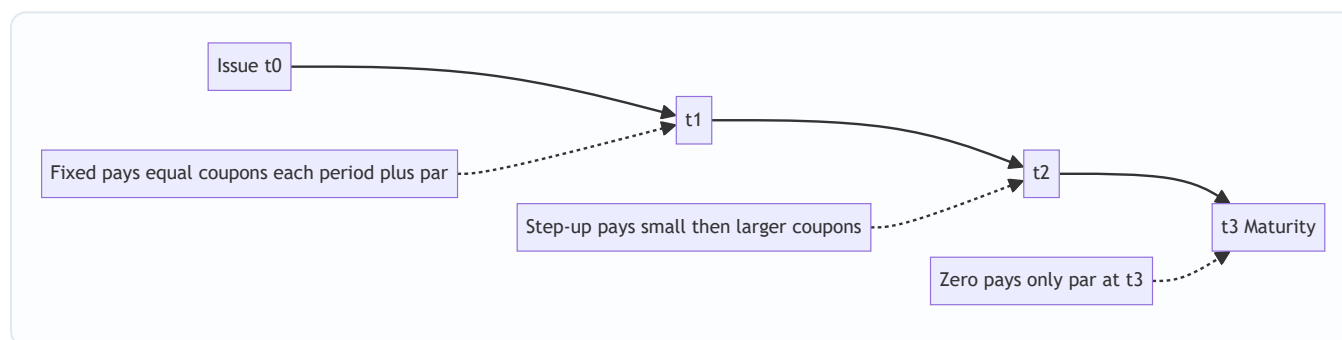


Figure 2.2 — Cash-flow timelines: a zero back-loads everything, a fixed bond pays level coupons, a step-up back-loads coupon size.

### 4.3 Maturity and Par Value

- **Par value (face / principal / redemption value).** The amount repaid at maturity and the base on which the coupon rate is applied. Prices are quoted as a *percentage of par*: a bond "at 98" costs 980 per 1,000 face. Par is usually 100/1,000 for corporates, but conventions vary.
- **Maturity (tenor).** Time until principal is repaid. Conventions: **money-market ( $\leq 1$  yr)**, **short (1–5 yr)**, **intermediate/medium (5–12 yr)**, **long ( $> 12$  yr)**. Some bonds are **perpetual** (no maturity — consols, certain bank AT1 instruments); their price is  $P = \frac{\text{Coupon}}{y}$ .
- **Bullet vs amortizing.** A **bullet** repays all principal at maturity (the default assumption). An **amortizing** bond repays principal gradually over its life (mortgages, many asset-backed securities), so its *average life* is much shorter than its final maturity.

Maturity matters because it drives duration and thus interest-rate risk: longer maturity  $\rightarrow$  later cash flows  $\rightarrow$  more discounting sensitivity  $\rightarrow$  larger price swings per unit of yield change.

## 4.4 Embedded Options — Who Can Change the Deal

An embedded option is a right, granted to the issuer or the investor, to alter the bond's cash flows. It is *embedded* — you cannot strip it out and trade it separately. Its value is captured by the identity from Section 3.

**Callable bond — issuer's option to redeem early.** The issuer can buy the bond back at a set **call price** (often at a premium to par, declining toward par over time — the *call schedule*) after a **call-protection / lockout period**. The issuer calls when rates have *fallen* (it can refinance cheaper) — exactly when the investor least wants their bond taken away. So:

$$P_{\text{callable}} = P_{\text{straight}} - V_{\text{call}}$$

The investor is *short* the call, hence demands a **higher yield**. The call caps the bond's upside (price compression) — as yields fall, the callable bond's price rises less than a straight bond's and flattens toward the call price. This produces **negative convexity** over the relevant yield range.

**Puttable bond — investor's option to sell back early.** The investor can force the issuer to redeem at the **put price** (usually par) on set dates. The investor puts when rates have *risen* (they can reinvest at higher yields) or when credit deteriorates. So:

$$P_{\text{puttable}} = P_{\text{straight}} + V_{\text{put}}$$

The investor is *long* the put, so accepts a **lower yield**. The put sets a price floor, giving the bond positive convexity and downside protection.

**Convertible bond — investor's option to convert into equity.** The holder can exchange the bond for a fixed number of shares (the **conversion ratio**). It is a hybrid: a straight bond **plus a call option on the issuer's stock**.

- **Conversion ratio** = shares received per bond.
- **Conversion price** = Par / Conversion ratio.
- **Conversion value (parity)** = Conversion ratio × current share price.
- **Straight/investment value** = value as a plain bond ignoring conversion — the *floor*.
- **Market conversion price** = Bond price / Conversion ratio; the **conversion premium** is how much over parity you pay for the bond.

Because the equity upside is valuable, convertibles carry the **lowest coupons** — investors accept less income in exchange for the equity call.

**Warrants, sinking funds, and make-whole calls.** A **sinking fund** obligates the issuer to retire portions of the issue on a schedule (reduces credit risk but adds reinvestment/timing uncertainty for holders). A **make-whole call** lets the issuer call the bond but at a price equal to the PV of remaining cash flows discounted at a *small spread over Treasuries* — deliberately expensive, so it is rarely exercised and mainly exists for corporate flexibility (M&A, restructuring) without penalising holders.



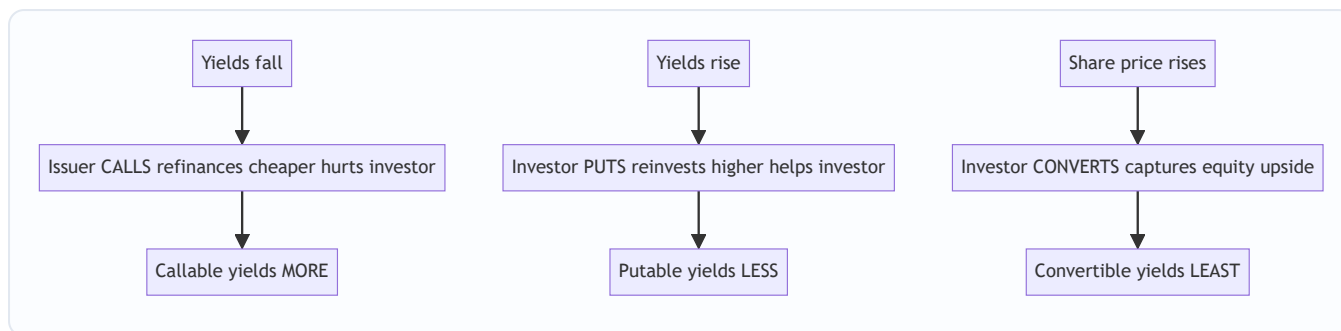


Figure 2.3 — Who exercises when, and the resulting yield ordering across embedded options.

## 4.5 Covenants and the Bond Indenture

The **indenture** (or *deed of trust*) is the master legal contract between issuer and bondholders, administered by a **trustee** who acts on behalf of holders. It spells out every term above *plus* the **covenants** — promises that constrain the issuer's behaviour to protect lenders.

- **Affirmative (positive) covenants** — things the issuer *must* do: pay principal and interest on time, maintain insurance, keep assets in good repair, supply audited financials, maintain its legal existence, comply with laws.
- **Negative (restrictive) covenants** — things the issuer *must not* do, and these are where the real protection lives:
  - **Limitation on additional debt** (leverage/coverage tests — e.g. keep interest coverage above 2.0×).
  - **Restrictions on dividends and share buybacks** (stops cash leaking to shareholders ahead of creditors).
  - **Negative pledge** — cannot pledge assets to *other* lenders without equally securing existing bondholders.
  - **Limitations on asset sales, mergers, sale-leasebacks.**
  - **Change-of-control put** — holders can put the bond if the company is taken over.

Covenants are valuable precisely because they *reduce expected loss*: they lower the probability of default (leverage limits) and raise recovery given default (negative pledge, asset-sale limits). Stronger covenants → lower spread → lower yield. "Covenant-lite" leveraged loans, by contrast, strip these protections and compensate lenders with higher spreads. A **covenant breach** is typically an **event of default**, allowing the trustee to *accelerate* — demand immediate repayment of the entire principal.

## 4.6 Seniority and Security — Where You Rank

If the issuer defaults, not all bondholders are equal. Two dimensions determine recovery:

**Security (is there collateral?).** - **Secured** — backed by specific pledged assets (mortgage bonds on property, equipment trust certificates on aircraft/rolling stock, collateral trust bonds on securities). On default, holders have first claim on those assets → *higher recovery*. - **Unsecured (debentures)** — backed only by the issuer's general creditworthiness and its unpledged assets → *lower recovery*.

**Seniority (priority of claim in bankruptcy).** The **absolute priority rule (APR)** ranks claims:

| Priority    | Claim                         | Typical recovery      |
|-------------|-------------------------------|-----------------------|
| 1 (highest) | Secured / senior secured debt | Highest               |
| 2           | Senior unsecured debt         | High–moderate         |
| 3           | Senior subordinated debt      | Moderate              |
| 4           | Subordinated / junior debt    | Low                   |
| 5           | Preferred equity              | Very low              |
| 6 (lowest)  | Common equity                 | Residual / often zero |

The rule: **you cannot be paid until everyone senior to you is paid in full**. In practice APR is sometimes violated in negotiated restructurings, but it is the anchor for pricing recovery.

Recovery feeds directly into expected loss and therefore spread:

$$\text{Expected loss} \approx \text{PD} \times (1 - \text{Recovery rate}), \quad \text{Recovery rate} = 1 - \text{LGD}$$

Same issuer, same maturity: a **senior secured** bond and a **subordinated** bond share the same *default probability* (same company) but have very different *loss given default*. The subordinated bond must yield more to compensate for lower recovery. This is why a single company can have a whole "capital stack" of bonds trading at different spreads.

## 5. Worked Examples

### Example 1 — Callable vs Straight vs Putable: pricing the option

Consider a 3-year, annual-pay bond, par 1,000, 6% coupon. Suppose the appropriate straight-bond yield is 6%, so the **straight bond prices at par = 1,000** (coupon = yield). Now compare three versions using the identity  $P = P_{\text{straight}} \pm V_{\text{option}}$ .

First, verify the straight price:  $P = \frac{60}{1.06} + \frac{60}{1.06^2} + \frac{1060}{1.06^3}$   $= 56.60 + 53.40 + 890.00 = 1,000.00$  ✓ (Discount factors:  $1/1.06 = 0.9434$ ,  $1/1.06^2 = 0.8900$ ,  $1/1.06^3 = 0.8396$ ;  $60(0.9434)=56.60$ ,  $60(0.8900)=53.40$ ,  $1060(0.8396)=890.00$ .)

Now suppose a lattice model values the embedded options at  $V_{\text{call}} = 18$  and  $V_{\text{put}} = 12$ .

| Version  | Formula               | Price           | Yield implication                      |
|----------|-----------------------|-----------------|----------------------------------------|
| Straight | $P_{\text{straight}}$ | 1,000.00        | 6.00% (par)                            |
| Callable | $1000 - 18$           | <b>982.00</b>   | Higher yield (investor short the call) |
| Putable  | $1000 + 12$           | <b>1,012.00</b> | Lower yield (investor long the put)    |

**Reconciliation / sanity check.** The callable trades *below* the straight bond — correct, because the investor gave the issuer a valuable right, so must be compensated with a lower price (higher yield). The putable trades *above* — correct, because the investor holds a valuable right and pays for it. The ordering  $P_{\text{callable}} < P_{\text{straight}} < P_{\text{putable}}$  ( $982 < 1,000 < 1,012$ ) matches theory exactly. If we solved for yields, we would find  $y_{\text{callable}} > 6\% > y_{\text{putable}}$ , confirming "callable yields more, putable yields less."

## Example 2 — Zero-coupon pricing and its extreme duration

Price a 5-year zero-coupon bond, par 1,000, priced to yield 5% annually. Then confirm its Macaulay duration equals its maturity.

$$P = \frac{1000}{1.05^5} = \frac{1000}{1.27628} = 783.53$$

**Duration check.** Macaulay duration is the weighted-average time to cash flows, weights = PV share. A zero has one cash flow at  $t=5$ , so the entire weight (100%) sits at year 5:

$$D_{\text{Mac}} = \sum_t t \cdot w_t = 5 \times 1.0 = 5.0 \text{ years}$$

Compare to a 5-year 5% *coupon* bond: some cash arrives at years 1–4, pulling the weighted average below 5 (it would be roughly 4.5 years). **Reconciliation:** the zero's duration (5.0) exceeds the coupon bond's (~4.5) for the *same* maturity — confirming that back-loading all cash flow to maturity maximises duration and thus interest-rate sensitivity. A 1% rise in yield would drop the zero's price by roughly  $D_{\text{Mod}} \times \Delta y$  where  $D_{\text{Mod}} = 5/1.05 = 4.76$ , i.e. about –4.76%, versus a smaller drop for the coupon bond.

## Example 3 — Taxable-equivalent yield and the muni-vs-corporate choice

An investor in the **35% federal tax bracket** compares: - Municipal GO bond yielding **3.8%** (federally tax-exempt). - Corporate bond of similar maturity yielding **5.6%** (fully taxable).

Convert the muni to its taxable-equivalent yield:  $\text{TEY} = \frac{0.038}{1 - 0.35} = \frac{0.038}{0.65} = 5.846\% \approx 5.85\%$

**Reconciliation.** The muni's TEY (5.85%) *exceeds* the corporate's 5.6%, so on an **after-tax** basis the muni wins for this investor. Cross-check by taxing the corporate: after-tax corporate yield  $= 5.6\% \times (1 - 0.35) = 3.64\%$ , which is *below* the muni's tax-free 3.8% — same conclusion from the opposite direction. Both methods agree: the tax-exempt muni delivers more after-tax income. Note the answer flips for a low-bracket investor: at a 12% rate, muni TEY  $= 0.038/0.88 = 4.32\%$ , now *below* 5.6% — the corporate wins. The feature (tax status) only has value relative to the holder's circumstances.

## Example 4 — Seniority and recovery in default

A company defaults with **1,000 of assets** available to distribute. Its debt stack: - Senior secured: 400 (collateralized) - Senior unsecured: 500 - Subordinated: 400

Apply the absolute priority rule (pay top-down):

| Claim            | Owed  | Paid           | Recovery rate |
|------------------|-------|----------------|---------------|
| Senior secured   | 400   | 400 (full)     | 100%          |
| Senior unsecured | 500   | 500 (full)     | 100%          |
| Subordinated     | 400   | 100 (residual) | 25%           |
| <b>Total</b>     | 1,300 | 1,000          | —             |

Available assets after paying secured (400) and senior unsecured (500) =  $\$1000 - 900 = 100$ , which goes to the subordinated class, recovering  $\$100/400 = 25\%$ .

**Reconciliation.** Total paid =  $\$400 + 500 + 100 = 1,000$  = assets available ✓. The senior tranches are made whole; the junior tranche absorbs the entire shortfall (loss of 300 on 1,300 total, all borne below the senior unsecured line). This is exactly why, for the *same* issuer and default event, the subordinated bond must have

priced at a much wider spread: its LGD here is 75% versus 0% for the senior classes. Expected-loss compensation is the whole reason a capital stack exists.

## 6. Connections

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- **To pricing (Chapter on valuation).** Every feature ultimately enters  $P = \sum CF_t / (1+y)^t$  — through cash flows (coupon/maturity), the discount rate (issuer/seniority/covenants), or an option adjustment. This chapter is the vocabulary; pricing is the arithmetic.
- **To duration and convexity.** Zeros → maximum duration; FRNs → near-zero rate duration; callables → negative convexity (price compression); putables → positive convexity (price floor). The features *are* the risk profile.
- **To credit analysis and spreads.** Seniority, security, and covenants map directly onto recovery rate and default probability, hence onto credit spread. The **OAS (option-adjusted spread)** framework exists precisely to strip embedded-option value out so credit can be compared cleanly.
- **To yield measures.** Callable bonds are quoted on **yield-to-worst** (the lowest of yield-to-call and yield-to-maturity). FRNs use **discount margin**, not YTM.
- **To derivatives.** A convertible = bond + equity call; a callable = bond – rate call; a putable = bond + rate put. Fixed income and options are the same subject viewed from two angles.

## 7. Key Terms

| Term                            | Meaning                                                                                 |
|---------------------------------|-----------------------------------------------------------------------------------------|
| <b>Indenture</b>                | Master legal contract governing the bond; enforced by a trustee.                        |
| <b>Covenant</b>                 | Promise constraining issuer behaviour; affirmative (must do) or negative (must not do). |
| <b>Debenture</b>                | Unsecured bond backed only by general creditworthiness.                                 |
| <b>Par value</b>                | Face/principal amount repaid at maturity; base for coupon and price quotes.             |
| <b>Bullet / amortizing</b>      | Principal repaid all at maturity vs gradually over life.                                |
| <b>FRN / floater</b>            | Bond whose coupon resets to reference rate + quoted margin.                             |
| <b>Discount margin</b>          | Spread that equates an FRN's PV to its price (its credit yield measure).                |
| <b>Zero-coupon</b>              | No coupons; bought at discount, repaid at par; duration = maturity.                     |
| <b>Step-up</b>                  | Coupon rises on a preset schedule; often paired with a call.                            |
| <b>Callable / putable</b>       | Issuer's right to redeem early / investor's right to sell back early.                   |
| <b>Convertible</b>              | Bondholder's right to exchange the bond for a set number of shares.                     |
| <b>Conversion ratio / price</b> | Shares per bond / Par ÷ conversion ratio.                                               |
| <b>Make-whole call</b>          | Call at PV of remaining cash flows plus small spread — deliberately expensive.          |
| <b>Sinking fund</b>             | Obligation to retire portions of the issue on a schedule.                               |
| <b>Seniority</b>                | Priority rank of a claim in bankruptcy (absolute priority rule).                        |
| <b>Secured / collateral</b>     | Bond backed by specific pledged assets → higher recovery.                               |
| <b>LGD / recovery rate</b>      | Loss / fraction recovered given default; Recovery = 1 – LGD.                            |
| <b>Negative pledge</b>          | Covenant barring the issuer from securing other debt ahead of you.                      |
| <b>Yield-to-worst</b>           | Lowest of YTM and all yield-to-call figures; how callables are quoted.                  |
| <b>Taxable-equivalent yield</b> | Muni yield ÷ (1 – tax rate); compares tax-exempt to taxable.                            |

## 8. Common Confusions

- **"Callable bonds yield less because they're safer."** Wrong direction. Callable bonds yield *more* — the investor is *short* an option (the issuer can snatch the bond when rates fall). Safety of the issuer is a separate axis from the call feature.
- **"Putable and callable are just mirror images that cancel out."** They are mirror images in *who* holds the right, but both add value to *whoever holds them*. A putable helps the investor (lower yield); a callable helps the issuer (higher yield). A bond can even be both callable *and* putable.
- **"Floating-rate notes have no risk because the coupon resets."** They have almost no *interest-rate* duration, but they retain full *credit* risk. If the issuer's spread widens, the FRN's price falls below par even though the reference rate resets.

- **"A zero-coupon bond is low risk because there's nothing to lose along the way."** The opposite for rate risk: a zero has the *longest* duration for its maturity and the *largest* price swing per yield change. It removes reinvestment risk but maximises price volatility.
- **"Senior and secured mean the same thing."** No. *Security* is about collateral (specific pledged assets); *seniority* is about ranking in the payment waterfall. A bond can be senior *unsecured* (high rank, no collateral) or, conceptually, secured but structurally subordinated. Both dimensions matter for recovery.
- **"Convertibles are just bonds with a bonus."** They pay the *lowest* coupons precisely because the equity option is worth a lot — you pay for the upside with foregone income. A convertible is a genuine hybrid, not a free extra.
- **"Municipal bonds are always the better yield."** Only after adjusting for tax and only for high-bracket investors. Compute the taxable-equivalent yield first; for low-bracket investors, taxable bonds often win.
- **"More covenants always mean a safer bond."** Covenants reduce expected loss, but they are often present *because* the issuer is riskier and lenders demanded protection. Covenant strength is priced into spread; don't read it as a pure quality signal in isolation.

## 9. Recap

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A bond is a bundle of contractual features living in four dimensions: **issuer, cash-flow shape, embedded options, and legal position**. The **issuer** determines the source and reliability of repayment — sovereigns (safest in own currency), supranationals and agencies (near-sovereign), municipals (tax-advantaged, GO vs revenue), and corporates (widest credit range). The **coupon structure** shapes the cash-flow numerator — fixed (level, full duration), floating (resets to reference + margin, near-zero rate duration but live credit risk), zero (single terminal payment, maximum duration), and step-up/PIK (back-loaded or deferred). **Maturity and par** set the timing and size of principal, and drive duration.

**Embedded options** reshape the deal: callables favour the issuer (yield *up*, negative convexity), putables favour the investor (yield *down*, price floor), convertibles hand the investor equity upside (yield *lowest*). Each is captured by  $P = P_{\text{straight}} \pm V_{\text{option}}$ . The **indenture and covenants** legally constrain the issuer — negative covenants (leverage limits, negative pledge, dividend restrictions) reduce expected loss and thus spread. Finally, **seniority and security** decide recovery in default: secured and senior claims recover first under the absolute priority rule, junior claims absorb the shortfall, and this loss-given-default difference is exactly what a capital stack's spread differences pay for.

The unifying principle throughout: **every feature moves value between borrower and lender, and price/yield adjusts to compensate the party that gives something up. Nothing is free.**

## 10. Quick-Reference / Interview Points

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**The one-liner identities (memorise):** -  $P_{\text{callable}} = P_{\text{straight}} - V_{\text{call}}$  → callable yields **more**. -  $P_{\text{putable}} = P_{\text{straight}} + V_{\text{put}}$  → putable yields **less**. - Convertible = straight bond + equity call → **lowest** coupon. - Zero:  $P = \frac{\text{Par}}{(1+y)^N}$ ; Macaulay duration = maturity (its extreme). - FRN coupon = reference + quoted margin; priced via **discount margin**; near-zero rate duration, full credit risk. - TEY  $= \frac{y_{\text{muni}}}{1 - \text{tax rate}}$ . - Expected loss  $\approx \text{PD} \times \text{LGD}$ ; Recovery  $= 1 - \text{LGD}$ .

**Yield ordering, all else equal:** putable < straight < callable (in yield terms). Convertible sits lowest of all coupon-wise.

**Rapid-fire answers:** - *Why does a callable yield more?* Investor is short the call; issuer refinances when rates fall, exactly when it hurts the investor. Compensation = higher yield + negative convexity. - *Callable vs puttable convexity?* Callable → negative convexity (price compresses toward call price as yields fall). Puttable → positive convexity (put price floors the downside). - *Senior secured vs subordinated, same issuer?* Same PD, different LGD. Sub bond has higher LGD, so wider spread and higher yield. - *FRN price below par — why?* Issuer credit spread widened above the fixed quoted margin; discount margin now exceeds quoted margin. - *GO vs revenue muni?* GO = full taxing power (safer); revenue = specific project cash flows only. - *Make-whole call — why rarely exercised?* Call price = PV of remaining cash flows at a tiny spread over Treasuries; too expensive to be economic, so it exists mainly for flexibility. - *Most important covenant for a bondholder?* Negative pledge (protects your claim rank) and leverage/coverage limits (cap default probability). - *How are callables quoted?* Yield-to-worst — the minimum of YTM and every yield-to-call.

**Capital-stack recovery waterfall (top-down):** senior secured → senior unsecured → senior subordinated → subordinated → preferred equity → common equity. You are paid only when everyone above you is paid in full (absolute priority rule).

# Q&A — Bond Features and Types

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A practice bank for the "Bond Features and Types" chapter. Work each question before reading the answer. Numerical answers are worked step-by-step and reconciled.

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## Section A — Concept Check

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### A1. What are the four independent design axes of any bond, and why is this decomposition useful?

The four axes are: (1) **Issuer** — who repays, and thus the credit risk (sovereign, supranational, agency, municipal, corporate); (2) **Cash-flow shape** — coupon structure, par value, and maturity, which define the cash-flow numerator; (3) **Embedded options** — rights that let one party alter the contract (call, put, conversion, sinking fund); (4) **Legal position** — covenants, seniority, and security, which govern behaviour and recovery. The decomposition is useful because any real bond is a single point in this four-dimensional space. It lets you price two bonds that both "pay 5%" completely differently once you locate them on each axis, and it maps each feature onto exactly one channel in the pricing equation.

### A2. State the "bond = straight bond ± option" identities and give the yield consequence of each.

- Callable:  $P_{\text{callable}} = P_{\text{straight}} - V_{\text{call}}\$$  → callable yields **more** (investor is short the call).
- Puttable:  $P_{\text{puttable}} = P_{\text{straight}} + V_{\text{put}}\$$  → puttable yields **less** (investor is long the put).
- Convertible = straight bond + call on the issuer's equity → **lowest** coupon (investor pays for equity upside with foregone income).

The identity gives direction from the sign of the option and magnitude from a lattice model.

### A3. Why does a floating-rate note (FRN) have very short interest-rate duration but still carry credit risk?

The coupon resets each period to *reference rate + quoted margin*, so as market rates move, the coupon re-prices toward them and the price stays near par. Rate duration is therefore roughly the time to the next reset — very short. But the *quoted margin is fixed* at issue. If the issuer's credit deteriorates, the spread the market now demands (the **discount margin**) rises above the fixed quoted margin, and the FRN's price falls below par. Resetting fixes rate risk, not credit risk.

### A4. Why does a zero-coupon bond have the longest possible duration for its maturity?

Macaulay duration is the PV-weighted average time to cash flows. A zero has exactly one cash flow, at maturity, so 100% of the weight sits at the final date and  $MD_{\text{Mac}} = N\$$  (the maturity). Any coupon bond of the same maturity pays some cash earlier, pulling the weighted average below  $N\$$ . The zero therefore has maximum price sensitivity to yield changes and no reinvestment risk.

### A5. Distinguish security from seniority. Can a bond be senior and unsecured at the same time?

**Security** is about collateral — whether specific assets are pledged (mortgage bonds, equipment trust certificates). **Seniority** is about rank in the bankruptcy payment waterfall under the absolute priority rule. They are independent dimensions. Yes: a **senior unsecured** bond ranks high in the waterfall but has no specific collateral — it relies on the issuer's general unpledged assets. Both dimensions independently affect loss given default and therefore spread.

### A6. What is a make-whole call and why is it rarely exercised?



A make-whole call lets the issuer redeem early at a price equal to the present value of the bond's remaining cash flows, discounted at a *small spread over Treasuries*. Because that discount rate is deliberately low, the call price is high (usually above market), making it uneconomic to exercise. It exists mainly to give the issuer legal flexibility (M&A, restructuring) without penalising holders — unlike a standard call, it does not cap the investor's upside in any meaningful way.

#### A7. Distinguish General Obligation (GO) from revenue municipal bonds.

A **GO bond** is backed by the full taxing power of the issuing government — broad and generally safer. A **revenue bond** is backed only by the cash flows of a specific project (toll road, airport). If the project underperforms, revenue-bond holders have no claim on general tax revenue, so revenue bonds are typically riskier and yield more.

## Section B — Numerical Bond-Math Problems

**B1. Price a straight bond and confirm par pricing.** A 3-year annual-pay bond, par 1,000, 6% coupon, yield 6%.

Cash flows: 60 at  $t=1$ , 60 at  $t=2$ , 1,060 at  $t=3$ .

Discount factors at 6%:  $\$1/1.06 = 0.9434\$$ ;  $\$1/1.06^2 = 0.8900\$$ ;  $\$1/1.06^3 = 0.8396\$$ .

$$P = 60(0.9434) + 60(0.8900) + 1060(0.8396) = 56.60 + 53.40 + 890.00 = 1,000.00$$

**Reconciliation.** Coupon rate (6%) equals yield (6%), so the bond must price exactly at par (1,000). It does. ✓

**B2. Price the callable and putable versions.** Using B1's straight price, a lattice values  $V_{\text{call}} = 18\$$  and  $V_{\text{put}} = 12\$$ .

$$P_{\text{callable}} = 1000 - 18 = 982.00 \quad P_{\text{putable}} = 1000 + 12 = 1012.00$$

**Reconciliation.** Ordering must be  $P_{\text{callable}} < P_{\text{straight}} < P_{\text{putable}}$ :  $\$982 < 1000 < 1012\$$ . ✓ The callable is cheaper (investor sold the issuer a right, so needs a lower price / higher yield); the putable is dearer (investor bought a right and pays for it / lower yield). Consistent with the identities.

**B3. Zero-coupon price and duration.** A 5-year zero, par 1,000, yield 5% annual.

$$P = \frac{1000}{1.05^5} = \frac{1000}{1.276282} = 783.53$$

Macaulay duration: single cash flow at  $t=5$ , weight 1.0, so  $D_{\text{Mac}} = 5 \times 1.0 = 5.0$  years.

Modified duration:  $D_{\text{Mod}} = \frac{D_{\text{Mac}}}{1+y} = \frac{5}{1.05} = 4.762$  years.

Estimated price change for a +1% (100 bp) yield rise:  $\approx -D_{\text{Mod}} \times \Delta y = -4.762 \times 0.01 = -4.76\%$ , i.e. price falls to roughly  $783.53 \times (1 - 0.0476) = 746.24\$$ .

**Reconciliation.** A 5-year 5% coupon bond would have  $D_{\text{Mac}} \approx 4.5$  years (some cash arrives at years 1–4), so the zero's 5.0 is strictly larger — confirming a zero maximises duration for its maturity. ✓

**B4. Taxable-equivalent yield.** Investor in the 35% bracket. Muni GO yields 3.8% (tax-exempt); comparable corporate yields 5.6% (taxable). Which wins after tax?

$$\text{TEY}_{\text{muni}} = \frac{0.038}{1 - 0.35} = \frac{0.038}{0.65} = 5.846\% \approx 5.85\%$$

Since  $5.85\% > 5.6\%$ , the muni wins.

**Cross-check from the other side** (tax the corporate): after-tax corporate =  $5.6\% \times (1 - 0.35) = 3.64\%$ , which is below the muni's tax-free 3.8%. Same conclusion. ✓

**Sensitivity.** At a 12% bracket:  $\text{TEY} = 0.038 / 0.88 = 4.32\%$ , now below 5.6% — the corporate wins. The tax feature only has value relative to the holder's bracket.

**B5. Seniority waterfall / recovery.** A firm defaults with 1,000 of assets. Stack: senior secured 400, senior unsecured 500, subordinated 400 (total owed 1,300). Apply absolute priority.

- Senior secured: paid 400 in full → recovery 100%. Remaining assets:  $1000 - 400 = 600$ .
- Senior unsecured: paid 500 in full → recovery 100%. Remaining:  $600 - 500 = 100$ .
- Subordinated: receives residual 100 on 400 owed → recovery  $100/400 = 25\%$ .

**Reconciliation.** Total paid =  $400 + 500 + 100 = 1000$  = assets available. ✓ Total shortfall =  $1300 - 1000 = 300$ , borne entirely by the subordinated class (LGD 75%), while senior classes have LGD 0%. This is precisely why the subordinated bond of the *same issuer* must price at a wider spread: identical PD, higher LGD.

**B6. Convertible bond arithmetic.** Par 1,000; conversion ratio 25 shares/bond; current share price 32; bond trades at 1,050. Find conversion price, conversion value (parity), market conversion price, and conversion premium.

- Conversion price = Par / ratio =  $1000 / 25 = 40.00$  per share.
- Conversion value (parity) = ratio  $\times$  share price =  $25 \times 32 = 800$ .
- Market conversion price = bond price / ratio =  $1050 / 25 = 42.00$  per share.
- Conversion premium (per share) =  $42.00 - 32.00 = 10.00$ , i.e.  $10/32 = 31.25\%$  over the current share price.

**Reconciliation.** The bond (1,050) trades above parity (800) by 250 — the value of the embedded equity call plus the bond floor above parity. Converting now would yield only 800 of stock versus a 1,050 bond, so a rational holder would **not** convert yet; they hold for the coupon and the option. ✓ Conversion price (40) > current price (32), so the option is out-of-the-money on parity but still valuable for time and volatility.

**B7. Perpetual bond price.** A perpetual (consol) pays 45 per year; required yield 6%.

$$P = \frac{\text{Coupon}}{y} = \frac{45}{0.06} = 750.00$$

**Reconciliation.** If yield rose to 7.5% (equal to the coupon rate on par 600? — check against par-neutral): at  $y = 7.5\%$  on par we'd get  $P = 45/0.075 = 600$ . So the perpetual prices at par only when yield equals the coupon rate on face; here  $45/750 = 6\%$  coupon-on-price, consistent. ✓

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## Section C — Interview-Style Questions (with model answers)

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### C1. "Why does a callable bond yield more than an otherwise identical straight bond?"

Model answer: The investor in a callable bond is *short* a call option — the issuer holds the right to redeem early. The issuer exercises when rates have fallen and it can refinance more cheaply, which is exactly when the investor would least want to lose a high-coupon bond and would have to reinvest at lower rates. Because the investor gives up this valuable right, they must be compensated with a higher yield. Formally  $P_{\text{callable}} = P_{\text{straight}} - V_{\text{call}}$ : a lower price means a higher yield. The call also caps upside, producing negative convexity — as yields fall, the callable's price flattens toward the call price rather than rising freely.

### C2. "An FRN is trading at 97, below par. What does that tell you?"

Model answer: An FRN's price stays near par because its coupon resets to reference + a fixed quoted margin. A price below par signals that the market now demands a *higher* spread than the fixed quoted margin — i.e. the **discount margin has risen above the quoted margin**, almost always because the issuer's credit has deteriorated (or liquidity has worsened). It is not an interest-rate story: rate duration is near zero. So a 97 price is a credit-risk warning, and I'd look at the issuer's spread, ratings trajectory, and the size of the gap between discount margin and quoted margin.

**C3. "Same issuer, same maturity: senior secured vs subordinated. Which yields more and why — walk me through it."**

Model answer: Both share the *same probability of default* because it's the same company and the same default event triggers both. What differs is **loss given default**. The senior secured bond has first claim on pledged collateral and sits at the top of the priority waterfall, so its recovery is high (low LGD). The subordinated bond is paid only after all senior claims are made whole, so it absorbs the shortfall and has high LGD. Since expected loss  $\approx PD \times LGD$  and PD is common, the subordinated bond has higher expected loss, hence a wider spread and higher yield. This is why one issuer can have a whole capital stack trading at different spreads.

**C4. "Why do convertible bonds carry the lowest coupons?"**

Model answer: A convertible is a hybrid — a straight bond plus a call option on the issuer's equity. The equity conversion right is valuable: it gives the holder unlimited upside if the stock rises while the bond floor limits downside. Investors pay for that upside by accepting a lower coupon than they'd demand on a plain bond from the same issuer. So the low coupon is not a sign of low risk; it's the price of the embedded equity option. In distress, the option can be worthless and the holder is left with a low-coupon bond — that's the trade-off.

**C5. "Which single covenant matters most to a bondholder, and why?"**

Model answer: I'd point to two working together, but if forced to one, the **negative pledge**. It bars the issuer from pledging assets to *other* lenders without equally securing existing bondholders — protecting your claim rank and recovery. Close behind are **leverage / interest-coverage limits**, which cap how much additional debt the issuer can pile on and thus constrain the probability of default. Negative pledge protects LGD; leverage limits protect PD. Together they defend both terms of expected loss, which is why they compress the spread. Covenant-lite deals strip these and compensate with wider spreads.

**C6. "How would you quote the yield on a callable bond?"**

Model answer: On **yield-to-worst** — the lowest of the yield-to-maturity and every yield-to-call across the call schedule. Because the issuer will call when it's advantageous to them and disadvantageous to the investor, the conservative assumption is the scenario that gives the investor the worst outcome. Quoting YTM alone would overstate the return an investor can rely on.

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## Section D — Multiple Choice (with reasoning)

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**D1. All else equal, which bond has the *highest* yield?** A. Putable B. Straight C. Callable D. Convertible

**Answer: C (Callable).** The yield ordering is putable < straight < callable, because the callable investor is short an option and must be compensated. The convertible (D) actually has the *lowest* coupon of all, since the equity option is paid for with foregone income. A and B are lower than C by the identities.

**D2. A 6-year zero and a 6-year 6% coupon bond, same yield. Which statement is true?** A. The coupon bond has higher duration. B. Both have duration 6. C. The zero has duration 6; the coupon bond less. D. Duration is unrelated to coupon.

**Answer: C.** A zero's Macaulay duration equals its maturity (6), because all cash is at the end. The coupon bond pays earlier cash flows, pulling its PV-weighted average time below 6. So the zero has the higher, and maximum-for-maturity, duration. A and D are wrong; B ignores the coupon effect.

**D3. An FRN's price falls from 100 to 96 while the reference rate is unchanged. The most likely cause is:** A. Rising market interest rates B. Widening issuer credit spread C. An approaching reset date D. A coupon step-down

**Answer: B.** With the reference rate unchanged, rate risk is not the driver, and an FRN's rate duration is near zero anyway. A price drop below par means the required discount margin now exceeds the fixed quoted margin — a credit-spread widening. A is largely neutralised by resets; C would push price *toward* par; D is not a standard FRN feature.

**D4. A firm defaults; assets 800. Senior secured owed 300, senior unsecured owed 400, subordinated owed 300. Subordinated recovery rate is:** A. 0% B. 33% C. 50% D. 100%

**Answer: B (33%).** Pay top-down: secured 300 (full), leaving 500; senior unsecured 400 (full), leaving 100; subordinated gets 100 on 300 owed = 33.3%. Reconciliation: 300 + 400 + 100 = 800 = assets. ✓ The junior class absorbs the whole shortfall of 200.

**D5. An investor in the 40% tax bracket compares a 3% tax-exempt muni with a 4.8% taxable corporate. Which is better after tax?** A. Corporate B. Muni C. Equal D. Cannot tell

**Answer: B (Muni).**  $TEY = \$0.03 / (1 - 0.40) = 0.03 / 0.60 = 5.0\%$ , which exceeds the corporate's 4.8%. Cross-check: after-tax corporate =  $\$4.8\% \times 0.60 = 2.88\% < \text{muni } 3.0\%$ . Both methods agree the muni wins for this high-bracket investor.

**D6. Which pairing correctly matches feature to convexity?** A. Callable → positive convexity B. Putable → negative convexity C. Callable → negative convexity D. Zero → negative convexity

**Answer: C.** A callable exhibits negative convexity because the call caps price appreciation as yields fall (price compresses toward the call price). A putable has positive convexity (the put floors the downside). A zero has ordinary positive convexity. A, B, and D are reversed or wrong.

**D7. A make-whole call is rarely exercised because:** A. It is illegal in most indentures B. Its call price is set at PV of remaining cash flows at a tiny spread over Treasuries, making it expensive C. It requires bondholder consent D. It only applies to floating-rate notes

**Answer: B.** The make-whole price discounts remaining cash flows at a small spread over Treasuries, producing a high (usually above-market) redemption price, so it is uneconomic to exercise. It exists for issuer flexibility, not routine refinancing. The other options are fabricated constraints.

**D8. "Senior unsecured" means the bond is:** A. Backed by specific collateral and top of the waterfall B. High in the priority waterfall but with no specific collateral C. Subordinated but collateralized D. Equivalent to preferred equity

**Answer: B.** Seniority (waterfall rank) and security (collateral) are independent. Senior unsecured ranks high but pledges no specific assets — it relies on the issuer's general unpledged assets for recovery. A describes senior secured; D is far lower in the stack.

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*End of Q&A bank. Every numerical answer above has been reconciled against the chapter identities:*

*$P_{\text{callable}} < P_{\text{straight}} < P_{\text{putable}}$ , zero duration = maturity,  $TEY = y_{\text{muni}} / (1 - \text{tax})$ , and absolute-priority recovery summing to available assets.*

# Chapter 03 — Bond Pricing

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## 1. The Problem / Need

A bond is a promise: "I will pay you fixed amounts on fixed dates in the future." A 10-year Government of India bond might promise ₹6.50 every six months for ten years, plus ₹100 back at the end. That stream of promises is objective and known. What is *not* known — and what the entire market spends its energy discovering — is **what that promise is worth today**.

This is the central problem of fixed income. If someone offers to sell you a bond promising ₹106.50 at maturity plus twenty coupons along the way, how many rupees should leave your pocket today? Pay too much and you lock in a poor return; pay too little and you have found a bargain the market will quickly erase. Every trader, portfolio manager, treasurer, and risk officer needs a defensible, repeatable answer to "*what is this cash-flow stream worth right now?*"

The need is sharpened by three facts:

- **Money has a time value.** A rupee received in five years is worth less than a rupee today, because today's rupee can be invested and grown. Any honest valuation must translate future rupees into today's rupees.
- **Bonds trade continuously.** A ten-year bond issued today will be bought and sold on thousands of days over its life, at prices that change every second as interest rates move. There must be a single, internally consistent engine that spits out a price given current market conditions.
- **Comparability.** An investor choosing between a 3-year corporate bond, a 7-year sovereign, and a 15-year infrastructure bond needs prices expressed on a common footing so that "cheap" and "expensive" mean something.

Bond pricing is the discipline that answers all three at once. It is nothing more — and nothing less — than the **present value of a known cash-flow stream**, discounted at a rate that reflects the return the market currently demands.

## 2. The Core Idea

**A bond's price is the present value (PV) of every cash flow it will pay, discounted back to today at the yield the market requires for bonds of that risk and maturity.**

Two objects meet in that sentence:

1. **The cash flows** — fixed, contractual, known in advance for a plain vanilla bond: a stream of coupons plus a final principal (face/par) repayment.
2. **The discount rate (the yield)** — a market-determined number that compresses future rupees into present rupees. It embodies the risk-free time value of money *plus* compensation for credit, liquidity, and other risks.

Price and yield are two sides of one coin. **Given the yield, you compute the price. Given the price, you can back out the yield (the YTM).** They move together in a rigid, inverse embrace: when the required yield rises, the same fixed cash flows are discounted harder, so the price falls; when the required yield falls, the price rises. This inverse price–yield relationship is the single most important reflex in all of fixed income.

Everything else in this chapter — premium versus discount bonds, clean versus dirty price, accrued interest — is a consequence of applying this one idea carefully.

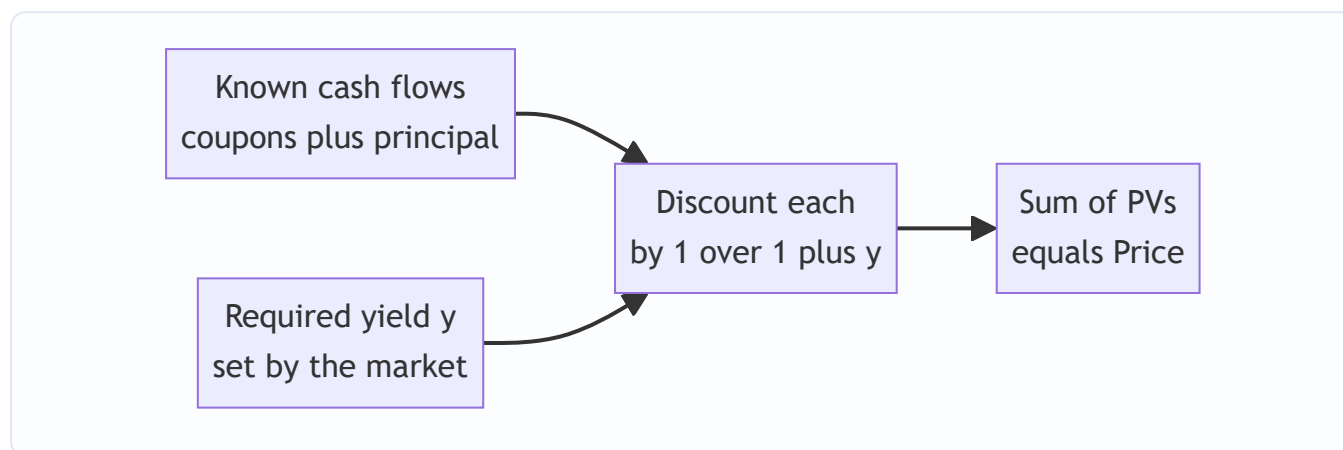


Figure 3.1 — The pricing engine: cash flows and a required yield go in, a price comes out.

### 3. Why / How It Works

#### The engine is discounting

The mechanism underneath bond pricing is the **discount factor**. If the market requires a return of  $y$  per period, then one rupee promised one period from now is worth only  $1 / (1 + y)$  today, because that smaller amount, grown at  $y$ , would become exactly ₹1 in one period. Two periods out, you divide by  $(1 + y)$  twice:  $1 / (1 + y)^2$ . In general, a rupee  $t$  periods away is worth  $1 / (1 + y)^t$  today.

A bond is just a bundle of such rupees, arriving at different dates. Discount each to today and add them up. Because discounting is a linear operation, the value of the bundle equals the sum of the values of the pieces — this is why we can price each coupon separately and then total them.

#### Why the inverse relationship is mechanical, not psychological

People sometimes narrate the price–yield inverse relationship as sentiment ("rates up, bonds sad"). It is not sentiment; it is arithmetic. The coupons and principal in the numerator are **fixed**. The yield sits in the **denominator**. Raise the denominator and every term shrinks; the sum — the price — must fall. Lower the denominator and every term grows; the price must rise. There is no way around it as long as the cash flows are fixed. This is the deep reason a bondholder is hurt by rising rates even though they will still receive every promised rupee: the *present value* of those unchanged rupees has dropped.

#### Why the discount rate is the *yield*, not something else

For a single bond quoted in the market, we use one internally consistent rate — the **yield to maturity (YTM)** — to discount *all* its cash flows. This is a simplification (in reality each cash-flow date has its own spot rate along the yield curve, covered in a later chapter), but it is the market convention and it makes price and yield a clean one-to-one pair. The YTM is precisely the single rate that makes the PV of the cash flows equal the observed market price. It is the bond's internal rate of return if held to maturity and all coupons are reinvested at that same rate.

## Why coupon rate and yield are different animals

A frequent source of confusion: the **coupon rate** is fixed at issue and printed on the bond — it determines the rupees paid. The **yield** floats with the market and determines how those rupees are valued. A bond paying an 8% coupon does not "yield 8%" unless it happens to trade at par. When the market's required yield differs from the coupon rate, the price adjusts away from par to reconcile the two — producing premium and discount bonds, which we derive in Section 4.

## 4. Full Content — Formulas and Bond Math

### 4.1 The general pricing formula

Let:

| Symbol | Meaning                                    |
|--------|--------------------------------------------|
| P      | Price of the bond today (present value)    |
| C      | Coupon payment per period (in currency)    |
| F      | Face / par value (redeemed at maturity)    |
| y      | Required yield <b>per period</b> (decimal) |
| n      | Total number of periods to maturity        |

The price is the PV of the coupon stream plus the PV of the principal:

$$P = C/(1+y)^1 + C/(1+y)^2 + \dots + C/(1+y)^n + F/(1+y)^n$$

The coupons form a geometric series (an annuity). Summed in closed form:

$$P = \underbrace{C \times [1 - (1+y)^{-n}] / y}_{\text{PV of coupons}} + \underbrace{F / (1+y)^n}_{\text{PV of principal}}$$

The bracketed term  $[1 - (1+y)^{-n}] / y$  is the **annuity factor** — the present value of ₹1 received each period for  $n$  periods. The term  $(1+y)^{-n}$  is the **discount factor** for the lump-sum principal.

### 4.2 Semi-annual compounding (the real-world convention)

Most bonds (US Treasuries, Indian G-Secs, most corporates) pay coupons **twice a year**. The formula is unchanged in shape, but every input is put on a per-period (half-year) basis:

| Annual quantity                               | Per-period version                     |
|-----------------------------------------------|----------------------------------------|
| Annual coupon rate $c$                        | Coupon per period $C = c \times F / 2$ |
| Annual yield $Y$ (nominal, semi-annual comp.) | Per-period yield $y = Y / 2$           |
| Years to maturity $T$                         | Number of periods $n = 2 \times T$     |



So a 6% annual-coupon-rate bond on ₹100 face pays  $C = ₹3$  every six months, and an 8% quoted yield means  $y = 4\%$  per half-year. **Getting this halving right is where most beginner errors occur.**

#### 4.3 The inverse price–yield relationship, made precise

Hold the cash flows fixed and vary  $y$ . Because every term  $C/(1+y)^t$  and  $F/(1+y)^n$  is a decreasing function of  $y$ , the price  $P(y)$  is a strictly decreasing function of  $y$ . Moreover it is **convex** — it curves. As  $y \rightarrow 0$ , the price approaches the undiscounted sum of all cash flows ( $n \cdot C + F$ ). As  $y \rightarrow \infty$ , the price approaches zero. The curve is downward-sloping and bowed toward the origin.

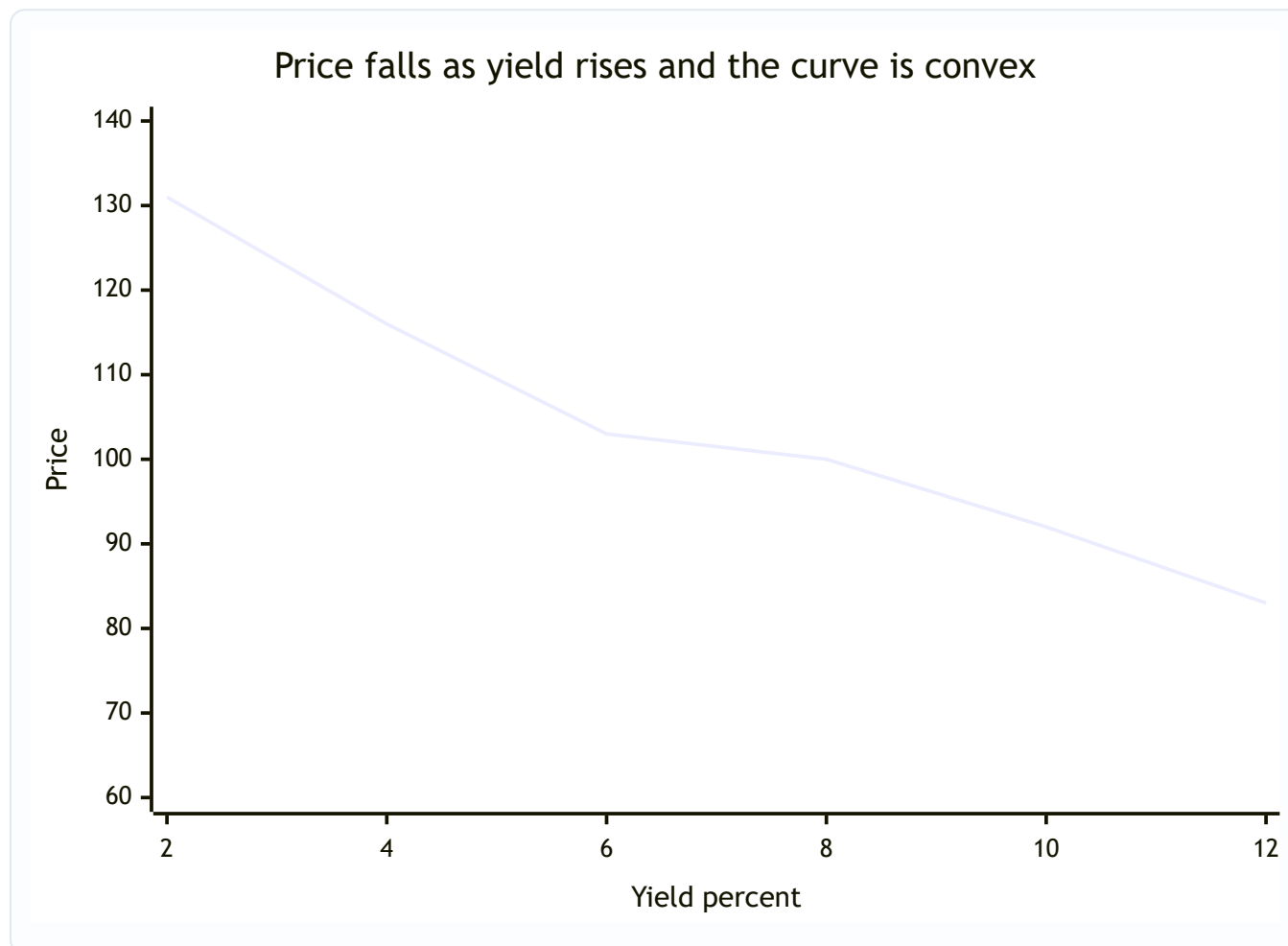


Figure 3.2 — The price–yield curve slopes downward and is convex (bowed). Par (100) is reached exactly where yield equals the coupon rate.

#### 4.4 Premium, par, and discount — the three regimes

Compare the **coupon rate** (what the bond pays, as a % of face) with the **required yield** (what the market demands). Three cases fall directly out of the pricing formula:

| Regime          | Condition           | Price vs Face        | Intuition                                                     |
|-----------------|---------------------|----------------------|---------------------------------------------------------------|
| <b>Premium</b>  | Coupon rate > Yield | $P > F$ (above par)  | Bond overpays vs market; buyers bid it up above 100           |
| <b>Par</b>      | Coupon rate = Yield | $P = F$ (equals par) | Coupons exactly compensate; no adjustment needed              |
| <b>Discount</b> | Coupon rate < Yield | $P < F$ (below par)  | Bond underpays vs market; price sinks below 100 to compensate |

**Why this must be true:** if a bond pays an 8% coupon but the market only demands 6%, the coupon stream is "too generous." Buyers compete for that generosity, bidding the price above 100 until the *effective* return on the higher purchase price falls to the market's 6%. The premium paid above par is exactly the PV of the excess coupons. Symmetrically, a bond paying 5% when the market demands 8% is stingy; nobody pays 100 for it. Its price falls below par until the combination of below-market coupons *plus a capital gain* (buying at, say, 88 and redeeming at 100) delivers the required 8%.

A useful identity: at par, the bond's YTM, coupon rate, and current yield all coincide. Away from par they diverge, and the direction of divergence tells you the regime.

#### 4.5 Clean price, dirty price, and accrued interest

So far we priced bonds **on a coupon date**, when the next coupon is exactly one full period away. But bonds trade every business day, often *between* coupon dates. On such a day the buyer is stepping into a coupon that has been "building up" since the last payment. Two conventions manage this.

- **Dirty price** (a.k.a. *invoice price* or *full price*): the actual PV of remaining cash flows as of the settlement date — the true economic value, and the amount that actually changes hands.
- **Accrued interest (AI)**: the portion of the current coupon period's interest that has economically accrued to the seller since the last coupon date. The buyer will receive the *whole* next coupon, so they must reimburse the seller for the slice the seller earned by holding the bond up to settlement.
- **Clean price** (a.k.a. *quoted price* or *flat price*): dirty price minus accrued interest. This is the number quoted on screens, precisely *because* it strips out the mechanical sawtooth of accrual and moves smoothly with yields.

The relationship:

$$\text{Dirty Price} = \text{Clean Price} + \text{Accrued Interest}$$

Accrued interest is computed by a **day-count convention** — the fraction of the coupon period that has elapsed:

$$\text{Accrued Interest} = \text{Coupon per period} \times (\text{days since last coupon} / \text{days in coupon period})$$

Common day-count conventions:

| Convention           | Used for                                        | Fraction                            |
|----------------------|-------------------------------------------------|-------------------------------------|
| Actual/Actual (ICMA) | Government bonds (US Treasuries, Indian G-Secs) | actual days / actual days in period |
| 30/360               | US corporates, many bonds                       | assume 30-day months, 360-day year  |
| Actual/360           | Money markets                                   | actual days / 360                   |

Between coupon dates the **dirty price** rises smoothly as the next coupon approaches, then drops by the coupon amount on the ex-coupon date (the sawtooth). The **clean price** removes exactly that sawtooth, leaving a series that reflects only yield movements.

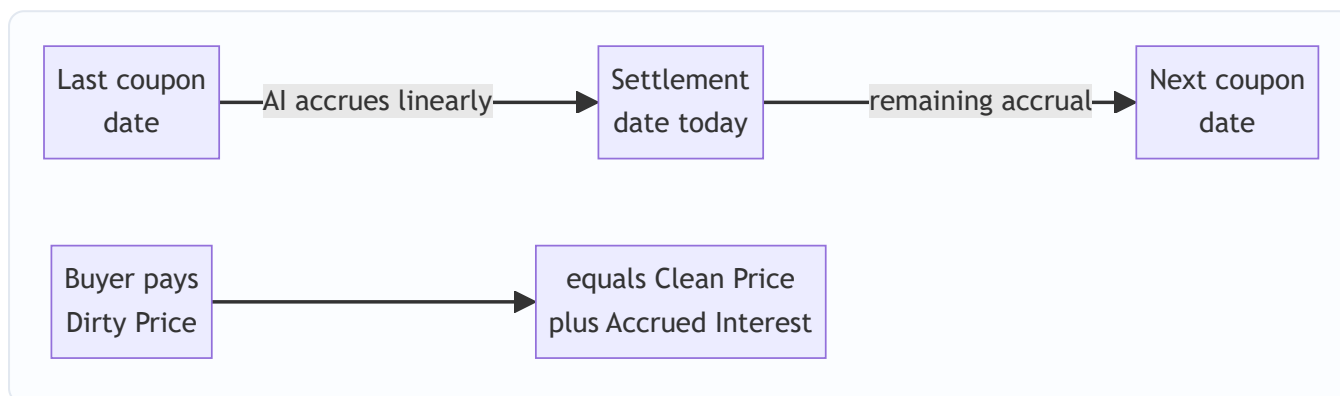


Figure 3.3 — Accrued interest accumulates from the last coupon to settlement; the buyer pays the dirty price and later collects the full next coupon.

## 5. Worked Examples

### Example 1 — Pricing a par-adjacent bond three ways (annual coupons)

**Setup.** A bond with face value ₹1,000 pays an **annual** coupon of 8% (so  $C = ₹80$ ), matures in **5 years** ( $n = 5$ ). Price it at three required yields: 8%, 6%, and 10%. Face redeemed at maturity.

**Formula.**  $P = C \times [1 - (1+y)^{-n}] / y + F \times (1+y)^{-n}$

**Case A — yield = 8% (equals coupon → expect par).**

- $(1.08)^{-5} = 1 / 1.469328 = 0.680583$
- Annuity factor  $= (1 - 0.680583) / 0.08 = 0.319417 / 0.08 = 3.992710$
- PV of coupons  $= 80 \times 3.992710 = 319.417$
- PV of principal  $= 1000 \times 0.680583 = 680.583$
- **Price**  $= 319.417 + 680.583 = ₹1,000.00$  ✓ Exactly par, as predicted (coupon rate = yield).

**Case B — yield = 6% (coupon > yield → expect premium).**

- $(1.06)^{-5} = 1 / 1.338226 = 0.747258$
- Annuity factor  $= (1 - 0.747258) / 0.06 = 0.252742 / 0.06 = 4.212364$
- PV of coupons  $= 80 \times 4.212364 = 336.989$
- PV of principal  $= 1000 \times 0.747258 = 747.258$
- **Price**  $= 336.989 + 747.258 = ₹1,084.25$  → above par. ✓ Premium, as predicted.

*Reconciliation:* the premium of ₹84.25 should equal the PV of the "excess" coupon (₹80 paid vs ₹60 the market demands on face = ₹20/year extra) over 5 years:  $20 \times 4.212364 = ₹84.25$ . Matches exactly. ✓

### Case C — yield = 10% (coupon < yield → expect discount).

- $(1.10)^{-5} = 1 / 1.610510 = 0.620921$
- Annuity factor  $= (1 - 0.620921) / 0.10 = 0.379079 / 0.10 = 3.790787$
- PV of coupons  $= 80 \times 3.790787 = 303.263$
- PV of principal  $= 1000 \times 0.620921 = 620.921$
- **Price**  $= 303.263 + 620.921 = ₹924.18$  → below par. ✓ Discount, as predicted.

*Reconciliation:* the discount of ₹75.82 should equal the PV of the coupon shortfall (₹80 vs ₹100 demanded = ₹20/year short):  $20 \times 3.790787 = ₹75.82$ . Matches. ✓

**Summary table** — same bond, three yields:

| Required yield | Price     | Regime   | Price – Face |
|----------------|-----------|----------|--------------|
| 6%             | ₹1,084.25 | Premium  | +84.25       |
| 8%             | ₹1,000.00 | Par      | 0            |
| 10%            | ₹924.18   | Discount | –75.82       |

Notice the inverse relationship in action (yield up → price down) and its **convexity**: dropping the yield 2 points (8→6) *added* ₹84.25, but raising it 2 points (8→10) only *subtracted* ₹75.82. The price gain from a rate fall exceeds the price loss from an equal rate rise — the curve is bowed, exactly as Figure 3.2 shows.

### Example 2 — Semi-annual bond, getting the halving right

**Setup.** Face ₹1,000, **annual coupon rate 6%** paid **semi-annually**, **4 years** to maturity, quoted **annual yield 5%** (semi-annual compounding).

#### Convert to per-period inputs.

- Coupon per period  $C = 6\% \times 1000 / 2 = ₹30$
- Per-period yield  $y = 5\% / 2 = 0.025$
- Number of periods  $n = 4 \times 2 = 8$

#### Compute.

- $(1.025)^{-8} : 1.025^8 = 1.218403$ , so  $(1.025)^{-8} = 0.820747$
- Annuity factor  $= (1 - 0.820747) / 0.025 = 0.179253 / 0.025 = 7.170137$
- PV of coupons  $= 30 \times 7.170137 = 215.104$
- PV of principal  $= 1000 \times 0.820747 = 820.747$
- **Price**  $= 215.104 + 820.747 = ₹1,035.85$

**Reconcile with the regime test.** Coupon rate (6%) > yield (5%), so we expect a **premium** — and indeed ₹1,035.85 > ₹1,000. ✓ Consistent.

*Common-error check:* had we (wrongly) discounted 8 periods at the full 5% instead of 2.5%, or used ₹60 coupons instead of ₹30, the price would be nonsense. The discipline is: **coupon halved, yield halved, periods doubled.**

### Example 3 — Clean price, accrued interest, and dirty price

**Setup.** A ₹1,000 face bond, **8% annual coupon paid semi-annually** (so ₹40 every six months). Coupon dates are 1 January and 1 July. Today is **settlement on 1 March**. The market quotes a **clean price of ₹980.00**. Day-count is 30/360. Find the accrued interest and the dirty price (what the buyer actually pays).

**Step 1 — Days elapsed in the current coupon period (30/360).** From 1 Jan to 1 March under 30/360: January = 30 days, February = 30 days → **60 days elapsed**. Full coupon period (1 Jan to 1 Jul) = 180 days.

**Step 2 — Accrued interest.**

$$\begin{aligned}\text{AI} &= \text{Coupon per period} \times (\text{days elapsed} / \text{days in period}) \\ &= 40 \times (60 / 180) \\ &= 40 \times 0.33333 \\ &= ₹13.33\end{aligned}$$

**Step 3 — Dirty price.**

$$\text{Dirty} = \text{Clean} + \text{Accrued} = 980.00 + 13.33 = ₹993.33$$

**Interpretation.** The buyer pays **₹993.33** on 1 March. On 1 July they will collect the **entire ₹40 coupon**, even though they held the bond for only four of the six months. The ₹13.33 they paid up front reimburses the seller for the two months (Jan–Feb) the seller held it. Net, the buyer's own coupon earnings equal ₹40 – ₹13.33 = ₹26.67, matching the four months (Mar–Jun) they actually own the bond:  $40 \times 120/180 = ₹26.67$ . ✓ The accrual mechanism splits the coupon fairly between seller and buyer.

*Why quote the clean price?* If instead the screen quoted the dirty ₹993.33, the number would jump down by ₹40 every 1 July and 1 January purely from the coupon dropping off — a mechanical sawtooth that has nothing to do with the bond becoming cheaper or richer. Quoting the clean ₹980.00 strips that out, so the quoted price moves only when *yields* move. That is exactly what a trader wants to watch.

### Example 4 — Backing out the yield (YTM) from a price

**Setup.** The ₹1,000 face, 8% annual-coupon, 5-year bond of Example 1 is trading at **₹924.18**. What is its YTM? (We will confirm it is 10%.)

YTM is the *y* solving  $924.18 = 80 \times [1 - (1+y)^{-5}]/y + 1000 \times (1+y)^{-5}$ . There is no closed-form solution; we iterate.

- **Try *y* = 10%:** from Example 1 Case C, price = ₹924.18. Match. **YTM = 10%.** ✓

To show the search logic had we not known:

| Trial yield | Resulting price | vs target 924.18       |
|-------------|-----------------|------------------------|
| 8%          | 1,000.00        | too high → raise yield |
| 12%         | 855.81          | too low → lower yield  |
| 10%         | 924.18          | exact ✓                |

The price sits between the 8% and 12% prices, so the yield sits between 8% and 12% — and lands at 10%. This is the **inverse relationship used as a solver**: because price falls monotonically in yield, we can bracket the true yield and converge on it. YTM and price are one-to-one.

## 6. Connections

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- **Time value of money (Chapter on TVM):** bond pricing is simply PV-of-an-annuity plus PV-of-a-lump-sum, the two workhorses of TVM applied to a contractual cash-flow stream. If you understand annuity factors, you already understand 80% of bond pricing.
- **Yield to maturity (next chapter):** YTM is defined *by* the pricing equation — it is the discount rate that makes PV equal the market price. Pricing (price from yield) and YTM (yield from price) are inverse operations on the same formula.
- **Duration and convexity (later chapters):** the *slope* of the price–yield curve is (modified) duration; its *curvature* is convexity. Both are derivatives of the pricing function  $P(y)$  derived here. The convexity we saw numerically in Example 1 (gains > losses for symmetric yield moves) is formalised there.
- **The yield curve / term structure:** using a single YTM to discount all cash flows is a convention; the rigorous approach discounts each cash flow at its own **spot rate** off the zero-coupon curve. Bond pricing is the bridge to that richer model.
- **Credit spreads:** the required yield  $y$  decomposes into a risk-free base rate plus a spread for credit and liquidity. Pricing is where the spread mechanically bites into value.
- **Accounting (amortised cost, EIR):** the "effective interest rate" method used to amortise bond premium/discount over a bond's life is the accounting mirror of the YTM derived here — the same discount rate spread across reporting periods.

## 7. Key Terms

| Term                              | Definition                                                                                                          |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Face / Par value (F)</b>       | The principal repaid at maturity; the reference for coupon and price quotes (often 100 or 1,000).                   |
| <b>Coupon rate (c)</b>            | The fixed annual interest rate printed on the bond; sets the rupee coupon $C$ . Does not change with the market.    |
| <b>Coupon (C)</b>                 | The periodic cash interest payment = coupon rate $\times$ face $\div$ payments per year.                            |
| <b>Yield / Required yield (y)</b> | The market's demanded rate of return per period; the discount rate in the pricing formula.                          |
| <b>Yield to maturity (YTM)</b>    | The single discount rate that equates PV of all cash flows to the market price; the bond's IRR if held to maturity. |
| <b>Present value (PV)</b>         | Today's worth of a future cash flow, obtained by discounting.                                                       |
| <b>Discount factor</b>            | $1/(1+y)^t$ , the PV today of ₹1 received $t$ periods hence.                                                        |
| <b>Annuity factor</b>             | $[1 - (1+y)^{-n}]/y$ , the PV of ₹1 per period for $n$ periods.                                                     |
| <b>Premium bond</b>               | Trades above par because coupon rate $>$ yield.                                                                     |
| <b>Discount bond</b>              | Trades below par because coupon rate $<$ yield.                                                                     |
| <b>Par bond</b>                   | Trades at face value because coupon rate = yield.                                                                   |
| <b>Clean (quoted/flat) price</b>  | Price excluding accrued interest; the quoted number.                                                                |
| <b>Dirty (invoice/full) price</b> | Price including accrued interest; the amount actually paid.                                                         |
| <b>Accrued interest (AI)</b>      | The share of the current coupon earned by the seller from last coupon to settlement.                                |
| <b>Day-count convention</b>       | The rule (Actual/Actual, 30/360, Actual/360) for the fraction of a coupon period elapsed.                           |
| <b>Current yield</b>              | Annual coupon $\div$ current price; a crude income measure that ignores capital gain/loss to maturity.              |

## 8. Common Confusions

**"Coupon rate is the return I earn."** No. The coupon rate sets the *rupees paid*; your *return* is the YTM, which depends on the *price you pay*. Buy an 8% bond at 92 and your return exceeds 8% (the discount adds a capital gain); buy it at 108 and your return is below 8%. They coincide only at par.

**"Rates rose but I still get every coupon, so why did I lose money?"** You still receive every promised rupee, but the *present value* of those unchanged rupees fell when the discount rate rose. If you must sell today, you crystallise that mark-to-market loss. Hold to maturity and you get par back, but you've suffered opportunity cost versus newly-issued higher-coupon bonds.

**"Premium bonds are better than discount bonds."** Neither is inherently better. A premium bond pays fat coupons but you'll book a capital *loss* pulling from the premium price back to par at maturity; a discount

bond pays skinny coupons but delivers a capital *gain* pulling up to par. Priced correctly, both yield exactly the market YTM. The premium/discount is just *where* the return is packaged (coupon vs capital).

**"Dirty price is a bad or manipulated price."** No — "dirty" is not pejorative. The dirty (full) price is the true economic value and the actual settlement amount. "Clean" just means accrued interest has been stripped out for quoting convenience so the number isn't polluted by the coupon sawtooth.

**Forgetting to halve for semi-annual bonds.** The number-one arithmetic error: you must halve the coupon, halve the annual yield, and double the number of periods. Discounting eight semi-annual periods at a full annual rate is wrong.

**Confusing current yield with YTM.** Current yield = coupon/price ignores the pull-to-par capital gain or loss and the time value of intermediate coupons. It always sits *between* the coupon rate and the YTM and should never be used as the bond's true return.

**"Higher yield always means a better investment."** A higher yield often signals higher risk (credit, liquidity) — the market discounts harder precisely because the cash flows are less certain. Yield is compensation for risk, not a free lunch.

## 9. Recap

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- A bond's price is the **present value of its cash flows** — coupons (an annuity) plus principal (a lump sum) — discounted at the market's **required yield**.
- Closed form:  $P = C \cdot [1 - (1+y)^{-n}] / y + F \cdot (1+y)^{-n}$ . For semi-annual bonds, **halve the coupon and yield, double the periods**.
- **Price and yield move inversely**, and the relationship is **convex**: symmetric yield moves produce asymmetric price changes (gains from a fall exceed losses from an equal rise). This is arithmetic — the yield sits in the denominator over fixed cash flows.
- Compare coupon rate to yield to get the regime: **coupon > yield → premium (P>F); coupon = yield → par (P=F); coupon < yield → discount (P<F)**. The premium or discount equals the PV of the coupon surplus or shortfall.
- Between coupon dates, **Dirty = Clean + Accrued Interest**. The buyer pays the dirty price and collects the full next coupon; accrued interest fairly splits that coupon between seller and buyer via a day-count convention.
- **YTM** is the inverse operation: the single discount rate that makes PV equal the observed price — found by iteration because the pricing equation has no closed-form inverse.

## 10. Quick-Reference / Interview Points

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**The one-line answer:** "A bond is priced as the present value of its coupons and principal, discounted at the yield the market requires; price and yield are inversely related because the fixed cash flows are discounted harder when the yield rises."

**Core formula (memorise):**

$$P = C \times [1 - (1+y)^{-n}] / y + F \times (1+y)^{-n}$$

**The regime test (say it instantly):**



| If...               | Then price is...     |
|---------------------|----------------------|
| Coupon rate > Yield | Premium (above par)  |
| Coupon rate = Yield | Par                  |
| Coupon rate < Yield | Discount (below par) |

**Clean vs dirty:** Dirty = Clean + Accrued Interest . Clean is quoted; dirty is paid. Accrued = coupon × (days elapsed / days in period).

**Rapid-fire facts interviewers probe:** - Why does a bond lose value when rates rise even though coupons are unchanged? *PV of fixed cash flows falls when the discount rate rises.* - Is the price–yield line straight? *No — convex; gains from a rate drop exceed losses from an equal rate rise.* - For a semi-annual bond, what do you do to the inputs? *Halve coupon, halve yield, double periods.* - Where does a discount bond's return come from? *Below-market coupons plus a capital gain (pull to par).* - Does current yield equal YTM? *Only at par; otherwise current yield lies between coupon rate and YTM.* - What is YTM, formally? *The single discount rate equating PV of cash flows to market price — the bond's IRR to maturity, assuming coupons reinvest at that rate.*

**Sanity checks to state aloud when pricing:** 1. Does the premium/discount agree with the coupon-vs-yield comparison? 2. Did I put coupon, yield, and periods on the same (per-period) footing? 3. Does the premium/discount magnitude equal the PV of the coupon surplus/shortfall? 4. As  $y \rightarrow 0$ , does my price approach the undiscounted sum of cash flows? As  $y \rightarrow \infty$ , toward zero?

**Numbers worth remembering** (₹1,000 face, 8% coupon, 5-yr, annual): at 6% → ₹1,084.25 (premium), at 8% → ₹1,000 (par), at 10% → ₹924.18 (discount). These illustrate inverse-and-convex in a single line.

# Q&A — Bond Pricing

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A practice bank for Chapter 03. Every question is followed by a full worked answer. Work each one on paper before reading the solution.

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## Section A — Concept Check

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**A1. In one sentence, what is a bond's price?** The present value of every cash flow the bond will pay (coupons plus principal), discounted back to today at the yield the market currently requires for a bond of that risk and maturity. It is nothing more than PV-of-an-annuity plus PV-of-a-lump-sum.

**A2. Why do price and yield move in opposite directions?** Because the cash flows in the numerator are fixed and the yield sits in the denominator. Raising the discount rate shrinks every term  $C/(1+y)^t$  and  $F/(1+y)^n$ , so the sum — the price — must fall. It is arithmetic, not sentiment. Even though the bondholder still receives every promised rupee, the present value of those unchanged rupees drops when rates rise.

**A3. What is the difference between the coupon rate and the yield?** The coupon rate is fixed at issue and printed on the bond; it determines the rupees paid. The yield floats with the market and determines how those rupees are valued today. They coincide only when the bond trades at par.

**A4. State the premium / par / discount test.** Compare coupon rate with required yield. Coupon > yield → premium (price above par). Coupon = yield → par (price equals face). Coupon < yield → discount (price below par). The premium or discount equals the PV of the coupon surplus or shortfall versus the market rate.

**A5. What must you do to the inputs when a bond pays semi-annually?** Halve the coupon, halve the annual yield, and double the number of periods. Discounting semi-annual periods at a full annual rate is the single most common beginner error.

**A6. Define clean price, dirty price, and accrued interest, and give the identity linking them.** Dirty (full/invoice) price is the actual PV of remaining cash flows on the settlement date and the amount that changes hands. Accrued interest is the slice of the current coupon that has economically accrued to the seller since the last coupon date. Clean (quoted/flat) price is dirty minus accrued. The identity:  $\text{Dirty} = \text{Clean} + \text{Accrued Interest}$ .

**A7. Why do screens quote the clean price rather than the dirty price?** The dirty price jumps down by the full coupon on each ex-coupon date — a mechanical sawtooth that has nothing to do with the bond becoming cheaper or richer. Stripping out accrued interest leaves a clean price that moves only when yields move, which is what a trader wants to watch.

**A8. What is the yield to maturity (YTM), formally?** The single discount rate that makes the present value of all the bond's cash flows equal its observed market price. Equivalently, the bond's internal rate of return if held to maturity with all coupons reinvested at that same rate.

**A9. Is the price–yield relationship linear?** No. It is downward-sloping and convex (bowed toward the origin). Symmetric yield moves produce asymmetric price changes: the price gain from a rate fall exceeds the price loss from an equal rate rise.

**A10. Where does a discount bond's return come from?** From two sources: the below-market coupons plus a capital gain as the price is pulled up to par at maturity. A premium bond is the mirror — fat coupons offset by a capital loss pulling down to par. Priced correctly, both deliver exactly the market YTM.

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## Section B — Numerical Bond-Math Problems

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### B1 — Pricing a zero-coupon bond

**Problem.** A zero-coupon bond, face ₹1,000, matures in 3 years. The required annual yield is 7%. Price it.

A zero pays no coupons, so the price is just the PV of the single principal repayment:

$$P = F / (1 + y)^n = 1000 / (1.07)^3$$

- $1.07^3 = 1.225043$
- $P = 1000 / 1.225043 = ₹816.30$

**Reconcile.** With no coupons, the bond can only sell at a deep discount ( $₹816.30 < ₹1,000$ ); the entire return is the pull to par. Check: growing ₹816.30 at 7% for 3 years gives  $816.30 \times 1.225043 = ₹1,000$ . ✓

### B2 — Premium bond, annual coupons

**Problem.** Face ₹1,000, 10% annual coupon ( $C = ₹100$ ), 3 years to maturity, required yield 8%. Price it and verify the regime.

$$P = C \times [1 - (1+y)^{-n}] / y + F \times (1+y)^{-n}$$

- $(1.08)^{-3} = 1 / 1.259712 = 0.793832$
- Annuity factor  $= (1 - 0.793832) / 0.08 = 0.206168 / 0.08 = 2.577097$
- PV of coupons  $= 100 \times 2.577097 = 257.710$
- PV of principal  $= 1000 \times 0.793832 = 793.832$
- **Price**  $= 257.710 + 793.832 = ₹1,051.54$

**Reconcile.** Coupon rate (10%) > yield (8%), so we expect a premium — and  $₹1,051.54 > ₹1,000$ . ✓ The premium of ₹51.54 should equal the PV of the excess coupon (₹100 paid vs ₹80 the market demands on face = ₹20/year extra):  $20 \times 2.577097 = ₹51.54$ . Matches exactly. ✓

### B3 — Semi-annual bond, getting the halving right

**Problem.** Face ₹1,000, 7% annual coupon paid semi-annually, 3 years to maturity, quoted annual yield 8% (semi-annual compounding). Price it.

Convert to per-period inputs: - Coupon per period  $C = 7\% \times 1000 / 2 = ₹35$  - Per-period yield  $y = 8\% / 2 = 0.04$  - Periods  $n = 3 \times 2 = 6$

Compute: -  $1.04^6 = 1.265319$ , so  $(1.04)^{-6} = 0.790315$  - Annuity factor  $= (1 - 0.790315) / 0.04 = 0.209685 / 0.04 = 5.242137$  - PV of coupons  $= 35 \times 5.242137 = 183.475$  - PV of principal  $= 1000 \times 0.790315 = 790.315$  - **Price**  $= 183.475 + 790.315 = ₹973.79$

**Reconcile.** Coupon rate (7%) < yield (8%), so we expect a discount — and  $₹973.79 < ₹1,000$ . ✓ Common-error check: had we discounted 6 periods at the full 8% or used ₹70 coupons, the answer would be nonsense. The discipline holds — coupon halved, yield halved, periods doubled.

## B4 — Accrued interest and dirty price

**Problem.** A ₹100 face bond pays a 9% annual coupon semi-annually (₹4.50 every six months). Coupon dates are 15 February and 15 August. Settlement is 15 May. Day-count is 30/360. The clean price is quoted at ₹101.20. Find the accrued interest and the dirty price.

**Step 1 — days elapsed (30/360).** From 15 Feb to 15 May, each month counts as 30 days: Feb→Mar = 30, Mar→Apr = 30, Apr→May = 30 → **90 days elapsed**. Full coupon period (15 Feb to 15 Aug) = 180 days.

**Step 2 — accrued interest.**

$$\begin{aligned}\text{AI} &= \text{Coupon per period} \times (\text{days elapsed} / \text{days in period}) \\ &= 4.50 \times (90 / 180) = 4.50 \times 0.5 = ₹2.25\end{aligned}$$

**Step 3 — dirty price.**

$$\text{Dirty} = \text{Clean} + \text{Accrued} = 101.20 + 2.25 = ₹103.45$$

**Interpretation.** The buyer pays ₹103.45 on 15 May and will collect the entire ₹4.50 coupon on 15 August despite holding the bond only three of the six months. The ₹2.25 paid up front reimburses the seller for the three months (Feb–May) the seller held it; the buyer's own share of the coupon is  $4.50 - 2.25 = ₹2.25$ , matching the three months (May–Aug) they own the bond:  $4.50 \times 90/180 = ₹2.25$ . ✓ The accrual splits the coupon fairly.

## B5 — Backing out the YTM by bracketing

**Problem.** A ₹1,000 face, 5% annual-coupon (C = ₹50), 3-year bond trades at ₹986.51. Find its YTM.

YTM is the  $y$  solving  $986.51 = 50 \times [1 - (1+y)^{-3}]/y + 1000 \times (1+y)^{-3}$ . There is no closed-form inverse, so we bracket and iterate.

| Trial yield | Resulting price | vs target 986.51       |
|-------------|-----------------|------------------------|
| 5%          | 1,000.00        | too high → raise yield |
| 6%          | 973.27          | too low → lower yield  |
| 5.5%        | 986.51          | exact ✓                |

Check the winning trial at  $y = 5.5\%$ : -  $(1.055)^{-3} = 1 / 1.174241 = 0.851614$  - Annuity factor =  $(1 - 0.851614) / 0.055 = 0.148386 / 0.055 = 2.697927$  - PV of coupons =  $50 \times 2.697927 = 134.896$  - PV of principal =  $1000 \times 0.851614 = 851.614$  - Price =  $134.896 + 851.614 = ₹986.51$  ✓ → **YTM = 5.5%**

**Reconcile.** The bond trades at a discount (₹986.51 < ₹1,000), so its yield must exceed its 5% coupon — and  $5.5\% > 5\%$ . ✓ The monotonic (inverse) price–yield relationship is what lets us bracket: the target price sits between the 5% and 6% prices, so the yield sits between 5% and 6%.

## Section C — Interview-Style Questions with Model Answers

**C1. "Rates rose and I still receive every coupon, so why did I lose money?"** Model answer: "You will indeed receive every promised rupee, but valuation is about the present value of those rupees, not their face

amount. When the market's discount rate rises, each fixed cash flow is discounted harder, so the sum — the mark-to-market price — falls. If you sell today you crystallise that loss; if you hold to maturity you get par back, but you have still suffered an opportunity cost versus newly issued bonds now paying higher coupons."

**C2. "A bond pays an 8% coupon. Does it yield 8%?"** Model answer: "Only if it trades at par. The 8% coupon rate fixes the rupees paid; the yield depends on the price you pay for those rupees. Buy it below par and the yield exceeds 8% because a capital gain to par is added; buy it above par and the yield is below 8% because of the capital loss to par. Coupon rate and YTM coincide exactly at par and diverge everywhere else."

**C3. "Walk me through why a bond trades at a premium."** Model answer: "A premium arises when the coupon rate exceeds the market's required yield. The coupon stream is more generous than what the market demands, so buyers compete for it and bid the price above par until the effective return on the higher purchase price falls back to the market yield. Mechanically, the premium above par equals the present value of the excess coupons — the extra rupees per period versus what the market rate would pay on face, discounted over the bond's life."

**C4. "Which is the better buy, a premium bond or a discount bond?"** Model answer: "Neither is inherently better; if both are priced correctly they deliver exactly the market YTM. The difference is only where the return is packaged. A premium bond front-loads return into fat coupons but you book a capital loss pulling down to par; a discount bond pays skinny coupons but delivers a capital gain pulling up to par. The choice is about tax treatment, reinvestment risk, and cash-flow timing preferences, not about one being cheap and the other dear."

**C5. "Explain the difference between clean and dirty price to a new analyst."** Model answer: "The dirty price is the true economic value and the amount that actually settles — the present value of all remaining cash flows on the settlement date. The clean price is the dirty price with accrued interest stripped out, and it is the number quoted on screens. We quote clean because the dirty price mechanically drops by a full coupon on every ex-coupon date; stripping accrued interest removes that sawtooth so the quoted price moves only with yields. 'Dirty' is not pejorative — it is the honest, full price."

**C6. "How would you find a bond's yield if you only know its price?"** Model answer: "There is no closed-form inverse of the pricing equation, so I solve for the YTM iteratively. Because price is a strictly decreasing, monotonic function of yield, I can bracket the answer: pick a yield that prices the bond too high and one that prices it too low, then converge by bisection or Newton-Raphson until the model price matches the market price. A quick sanity check first: if the bond trades below par the yield is above the coupon rate, and vice versa — that tells me which side to start from."

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## Section D — Multiple-Choice Questions with Reasoning

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**D1. A bond's required yield rises from 6% to 7%. Holding cash flows fixed, its price will:** A) rise B) fall C) stay the same D) rise then fall

**Answer: B.** The fixed cash flows sit in the numerator and the yield in the denominator; raising the discount rate shrinks every discounted term, so the price must fall. The relationship is monotonic and inverse.

**D2. A bond with a 9% coupon trades at a required yield of 11%. It is a:** A) premium bond B) par bond C) discount bond D) cannot be determined

**Answer: C.** Coupon rate (9%) < required yield (11%), so the price falls below par to compensate the buyer with a capital gain to par plus the below-market coupons — the definition of a discount bond.

**D3. For a bond quoted at a clean price of ₹98 with ₹1.50 of accrued interest, the amount the buyer actually pays is:** A) ₹96.50 B) ₹98.00 C) ₹99.50 D) ₹1.50

**Answer: C.** Dirty = Clean + Accrued = 98.00 + 1.50 = ₹99.50 . The dirty price is what settles; the buyer reimburses the seller for interest accrued since the last coupon and later collects the whole next coupon.

**D4. A 6% annual-coupon bond paid semi-annually, priced at a 6% annual yield, over any whole number of years, trades at:** A) a premium B) par C) a discount D) depends on maturity

**Answer: B.** When the coupon rate equals the required yield, the coupons exactly compensate the market rate and no price adjustment is needed, so the bond trades at par regardless of maturity. (Halving both the coupon and the yield preserves the equality: 3% per period coupon against 3% per period yield.)

**D5. Dropping a bond's yield by 1% raises its price by ₹42, while raising the yield by 1% lowers its price by only ₹38. This asymmetry reflects:** A) an arithmetic error B) convexity of the price–yield curve C) accrued interest D) a rating change

**Answer: B.** The price–yield curve is convex (bowed toward the origin), so the price gain from a rate fall exceeds the price loss from an equal rate rise. This is a genuine property of the pricing function, not an error — its formal measure is convexity.

**D6. The current yield of a discount bond, relative to its coupon rate and YTM, will:** A) equal the coupon rate B) equal the YTM C) lie between the coupon rate and the YTM D) exceed both

**Answer: C.** Current yield (annual coupon ÷ price) ignores the pull-to-par capital gain and the time value of intermediate coupons, so it always sits between the coupon rate and the YTM. For a discount bond, coupon rate < current yield < YTM; it should never be used as the bond's true return.

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*Self-check reminders when solving any pricing problem: (1) put coupon, yield, and periods on the same per-period footing; (2) confirm the premium/discount agrees with the coupon-vs-yield comparison; (3) verify the premium/discount magnitude equals the PV of the coupon surplus/shortfall; (4) sanity-check the limits — as  $y \rightarrow 0$  the price approaches the undiscounted sum of cash flows, and as  $y \rightarrow \infty$  it approaches zero.*

# Chapter 04 — Yield Measures

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## 1. The Problem / Need

You are handed two bonds. Bond A costs \$960 and pays an 8% coupon. Bond B costs \$1,050 and pays a 7% coupon, but it can be called away in three years. Which one earns you more? The coupon rate alone cannot answer this — it is fixed to face value and ignores what you actually *paid*. The price you observe in the market already bakes in interest-rate levels, credit risk, and time to maturity. What an investor truly needs is a single number that compresses **price, coupons, timing, and redemption value** into one comparable rate of return.

That number is a **yield**. But "yield" is not one thing — it is a family of measures, each answering a slightly different question and each resting on different assumptions. Quoting the wrong one, or forgetting the assumptions baked into it, is one of the most common and most punishing errors in fixed income. A trader who compares a discount bond's yield-to-maturity against a callable premium bond's yield-to-maturity — instead of its yield-to-call — is systematically overstating return on the callable name.

This chapter builds the yield toolkit from the ground up: what each measure means, how to compute it, when it lies to you, and how the promised yield differs from the yield you actually realise once reinvestment enters the picture.

## 2. Core Idea

A bond's price is the present value of its cash flows discounted at some rate. **A yield is the discount rate implied by the market price.** Flip the pricing equation around: instead of choosing a discount rate to find price, take the observed price as given and *solve for* the discount rate. That solved rate is the internal rate of return (IRR) of the bond's cash-flow stream.

Every yield measure is a variation on this theme, differing in three choices:

1. **Which cash flows** do we count? (Just this period's coupon? All coupons to maturity? All coupons to the call date?)
2. **What redemption value** at the end? (Par at maturity? Call price? Sale price at a horizon?)
3. **What compounding / annualisation convention** turns a periodic rate into a quoted annual rate?

Get those three choices straight and every yield measure — current yield, YTM, yield-to-call, yield-to-worst, bond-equivalent yield, realised yield — falls out of the same master equation.

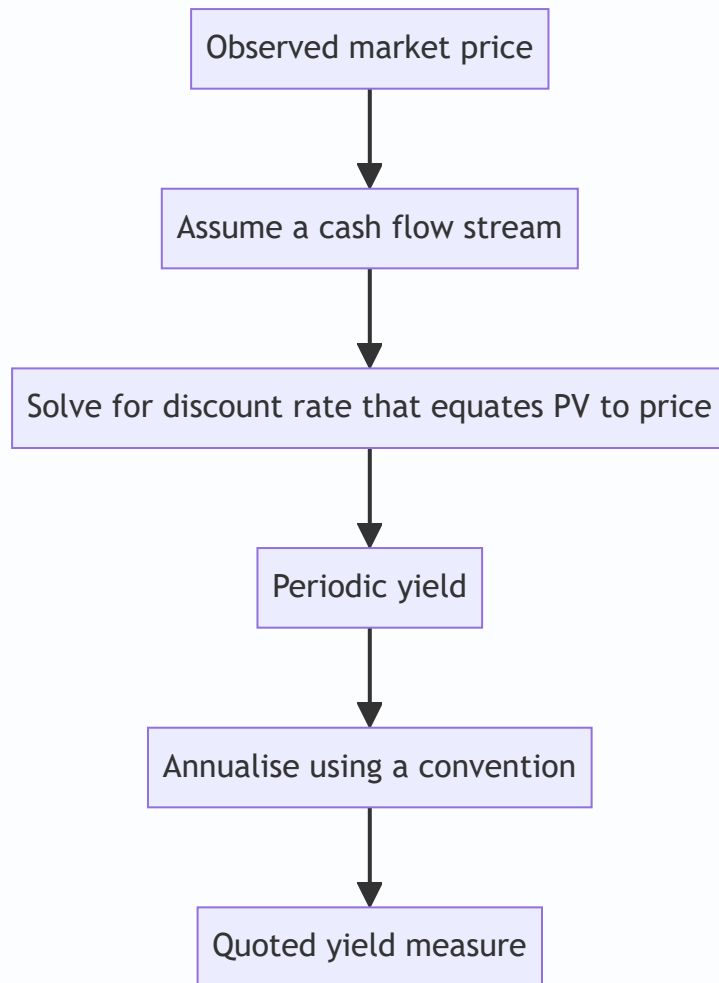


Figure 1 — Every yield measure is the same inversion of the pricing equation, differing only in the cash-flow stream assumed and the annualisation convention applied.

### 3. Why / How It Works

#### Price and yield move inversely

The bond price is a decreasing function of its discount rate. Raise the yield, and every future cash flow is discounted harder, so the present value (price) falls. Lower the yield, price rises. This inverse relationship is the spine of fixed income.

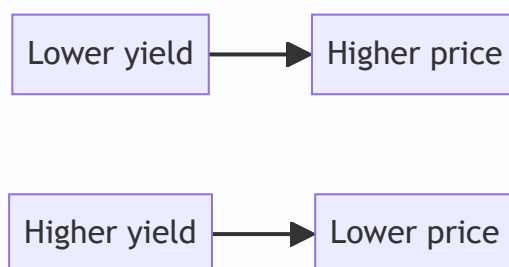


Figure 2 — The fundamental inverse relationship between a bond's yield and its price.



The curve is not a straight line — it is **convex** (bowed toward the origin). Price falls at a decreasing rate as yield rises. That convexity matters later for duration and for why linear interpolation of YTM slightly overstates the true yield.

### Why yield is an IRR, not a simple ratio

A coupon received in six months is worth more than the same coupon in five years. Any honest return measure must discount those cash flows for timing. Solving for the single rate that makes discounted inflows equal the price is exactly an IRR computation — and like all IRRs, it carries an embedded **reinvestment assumption**: it presumes every intermediate coupon is reinvested at that same solved rate until maturity. This is not a footnote. It is the reason the yield you are *quoted* is rarely the yield you *realise*.

### The par / premium / discount logic

Compare the coupon rate (a fixed cash amount relative to face) against the yield the market demands:

- If the market yield **equals** the coupon rate, the bond prices at **par** — no adjustment needed.
- If the market demands **more** than the coupon pays, buyers will only take the bond at a **discount** (below par), so that the price appreciation to par at maturity tops up the shortfall.
- If the market demands **less** than the coupon pays, the bond commands a **premium** (above par), and the price erosion toward par offsets the generous coupon.

This produces a strict ordering of the three yield measures that is worth memorising cold, because it is a favourite interview check.

## 4. Full Content — the Yield Measures

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### 4.1 Current yield

The crudest measure. Annual coupon income divided by current price:

$$\text{Current Yield} = \frac{\text{Annual coupon in currency}}{\text{Current price}}$$

It captures income return only. It **ignores** the capital gain or loss from buying at a discount/premium and pulling to par, ignores the timing/compounding of coupons, and ignores any reinvestment. Useful as a quick income proxy for a buy-and-hold income investor; useless as a total-return comparator.

### 4.2 Yield to maturity (YTM)

The IRR of the full cash-flow stream, assuming the bond is held to maturity and redeemed at par. For a bond paying coupons  $C$  per period over  $n$  periods, redeeming face  $F$ , at price  $P$ , YTM is the periodic rate  $y$  solving:

$$P = \sum_{t=1}^n \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n}$$

Using the annuity closed form:

$$P = C \cdot \frac{1 - (1+y)^{-n}}{y} + F \cdot (1+y)^{-n}$$

There is no algebraic solution for  $y$  when  $n > 1$  — it must be found by trial-and-error, interpolation, or a solver. Once you have the periodic  $y$ , you annualise it (see BEY below).

The cash-flow stream that YTM discounts looks like this for a five-year semiannual bond — ten equal coupons plus a face repayment bunched onto the final date:

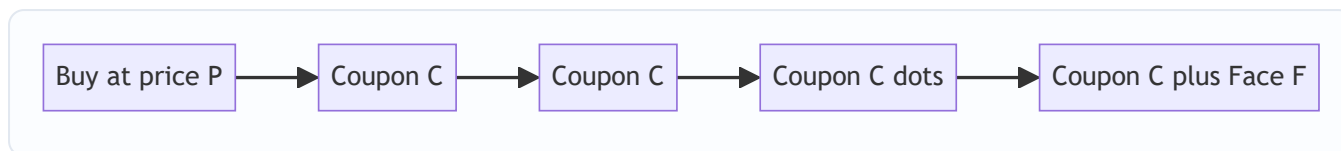


Figure 3 — The cash-flow timeline YTM discounts. The final node carries both the last coupon and the face redemption, which is why long-dated principal dominates the present value.

### Three assumptions baked into YTM:

1. The bond is **held to maturity**.
2. There is **no default** — all coupons and principal are paid in full and on time.
3. All coupons are **reinvested at the YTM** itself until maturity.

Break any of these and your realised return diverges from the quoted YTM.

### 4.3 Yield to call (YTC)

Identical math, but the cash-flow stream is truncated at the **call date** and the terminal value is the **call price**  $C_{\text{call}}$  (often par or a small premium), not maturity par. For a bond callable after  $m$  periods:

$$P = C \cdot \frac{1 - (1 + y_c)^{-m}}{y_c} + C_{\text{call}} \cdot (1 + y_c)^{-m}$$

You compute a separate YTC for each call date in the call schedule.

### 4.4 Yield to worst (YTW)

The **minimum** across YTM and every YTC (and yield-to-put where relevant). It answers: "assuming the issuer exercises its options in the way least favourable to me, what is the lowest yield I could end up with?" It is the conservative number quoted on callable bonds.

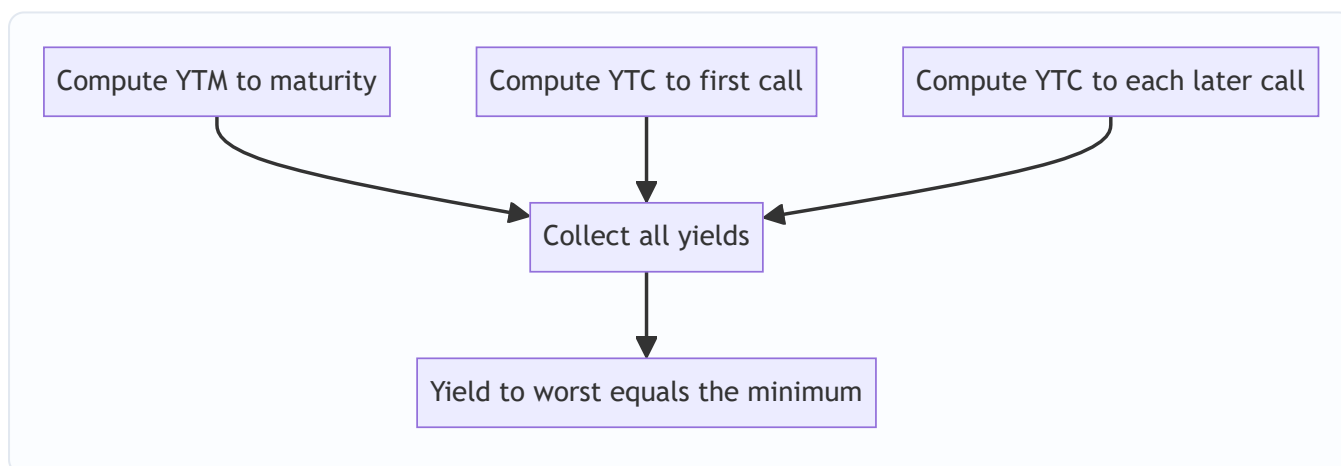


Figure 4 — Yield to worst is simply the lowest of the maturity and all call scenario yields.

### 4.5 Bond-equivalent yield (BEY) and compounding conventions

A US bond paying semiannual coupons has a *semiannual* periodic yield. The market convention is to quote it as **twice** the semiannual rate — this doubled figure is the **bond-equivalent yield**:

$$\text{BEY} = 2 \times y_{\text{semiannual}}$$

Note this is a *simple* doubling, not a compounding — so the BEY understates the true **effective annual yield (EAY)**:

$$1 + y_{\text{semiannual}} = (1 + y_{\text{EAY}})^{1/2}$$

To compare a semiannual-pay bond against an annual-pay bond you must put them on the same footing. Convert an annual-pay (effective) yield into a bond-equivalent basis:

$$1 + y_{\text{BEY}} = (1 + y_{\text{EAY}})^{1/2}$$

BEY also names a **money-market** convention: for discount instruments (T-bills, commercial paper) it restates a short-horizon holding return on a 365-day, add-on basis so it can be compared to coupon bonds:

$$1 + y_{\text{BEY}} = \frac{F - P}{P} \times \frac{365}{t}$$

where  $t$  is days to maturity.

## 4.6 Realised (horizon) yield

The yield you *actually* earn depends on (a) the price at which you sell or redeem and (b) the rate at which coupons were reinvested. The **realised (horizon) yield** is the periodic rate that grows your purchase price into the **total terminal value** — reinvested coupons plus the ending bond value — over the holding period  $h$ :

$$\text{Terminal value} = \sum_{t=1}^h C \cdot (1+r)^{h-t} + \text{Ending value}$$

$$1 + y_{\text{Realised yield (periodic)}} = \left( \frac{\text{Terminal value}}{P} \right)^{1/h}$$

where  $r$  is the assumed reinvestment rate. When  $h$  = full maturity, the ending value is par; for a shorter horizon it is the bond's market price at that horizon (which depends on the yield curve then).

**Reinvestment risk vs price risk.** These pull in opposite directions:

- If rates **fall** after purchase, coupons reinvest at lower rates (realised yield drops) but the bond's sale price rises.
- If rates **rise**, coupons reinvest at higher rates (realised yield climbs) but the bond's sale price falls.

At one special horizon — roughly the bond's **duration** — these two effects offset, and the realised yield is immunised against small rate changes.

## 5. Worked Examples

### Example 1 — Current yield, YTM, and the discount-bond ordering

**Bond A:** face \$1,000, 8% coupon paid semiannually (\$40 every six months), 5 years to maturity (10 periods), market price **\$960.44**.

**Current yield:**

$$CY = \frac{80}{960.44} = 8.33\%$$

**YTM.** Since the bond trades at a discount, we expect  $YTM > \text{coupon rate (8\%)}$ . Solve for the semiannual  $y$  in:

$$960.44 = 40 \cdot \frac{1 - (1+y)^{-10}}{y} + 1000 \cdot (1+y)^{-10}$$

Try  $y = 4.5\%$ :

- $(1.045)^{10} = 1.553069$ , so  $(1.045)^{-10} = 0.643928$
- Annuity factor  $= \frac{1 - 0.643928}{0.045} = \frac{0.356072}{0.045} = 7.912711$

- PV of coupons  $= 40 \times 7.912711 = 316.51\$$
- PV of principal  $= 1000 \times 0.643928 = 643.93\$$
- **Total  $= 960.44\$$  ✓**

The semiannual yield is exactly 4.5%, so the **BEY (quoted YTM)  $= 2 \times 4.5\% = 9.00\%$** .

**The discount-bond ordering** now falls out and reconciles:

| Measure                 | Value |
|-------------------------|-------|
| Coupon rate             | 8.00% |
| Current yield           | 8.33% |
| Yield to maturity (BEY) | 9.00% |

For a **discount** bond: coupon rate < current yield < YTM. The current yield sits above the coupon rate because we paid less than par; the YTM sits above the current yield because it *also* captures the  $\$39.56$  pull-to-par capital gain, which the current yield ignores entirely.

(For a **premium** bond the inequalities flip: coupon > current yield > YTM. For a **par** bond all three equal.)

## Example 2 — Solving YTM by interpolation and verifying

Suppose we did not know the answer to Example 1 and had to bracket it. Price the bond at two trial semiannual rates:

At  $y = 4\%$  (coupon rate matches yield  $\rightarrow$  price = par):

$$P = 40 \cdot \frac{1 - (1.04)^{-10}}{0.04} + 1000(1.04)^{-10} = 1000.00\$$$

At  $y = 5\%$ :

- $(1.05)^{10} = 1.628895$ , inverse  $= 0.613913$
- Annuity factor  $= \frac{1 - 0.613913}{0.05} = 7.721735$
- $P = 40(7.721735) + 1000(0.613913) = 308.87 + 613.91 = 922.78\$$

Linear interpolation for the target price  $\$960.44$ :

$$y \approx 4\% + \frac{1000 - 960.44}{1000 - 922.78} \times 1\% = 4\% + \frac{39.56}{77.22} \times 1\% = 4\% + 0.512\% = 4.512\%$$

That gives a BEY of about **9.02%**, versus the true **9.00%**. The interpolation slightly *overstates* the yield because the price–yield curve is convex, not linear: the straight line between two points lies below the true curve, so the interpolated rate for a given price comes out a touch high. Re-checking at the exact 4.5% (as in Example 1) returns  $\$960.44$ , confirming the true yield. The lesson: interpolation is a fast approximation; verify by re-pricing.

## Example 3 — Bond-equivalent yield and cross-convention comparison

Using Example 1's semiannual yield of 4.5%:

- **BEY (simple doubling):**  $2 \times 4.5\% = 9.00\%$
- **Effective annual yield:**  $(1.045)^2 - 1 = 1.092025 - 1 = 9.2025\%$

So the 9.00% quote *understates* the true annually-compounded return by about 20 bps — a pure artefact of the doubling convention.

Now suppose a competing **annual-pay** bond quotes a yield of 9.20% (an effective annual figure). To compare it fairly against our semiannual bond's 9.00% BEY, convert it to a bond-equivalent basis:

$$\text{BEY} = 2 \left[ (1.092)^{1/2} - 1 \right] = 2 \left[ 1.044988 - 1 \right] = 2(0.044988) = 8.998\% \approx 9.00\%$$

On a like-for-like BEY basis the two bonds are essentially **identical** (9.00% vs 9.00%) — even though the naive comparison of "9.00% vs 9.20%" made the annual-pay bond look better. This is exactly the trap the convention exists to prevent.

**Money-market BEY.** A 90-day T-bill is priced at \$98.50 per \$100 face:

$$\text{Holding return} = \frac{100 - 98.50}{98.50} = 1.5228\%$$

$$\text{BEY} = 1.5228\% \times \frac{365}{90} = 6.18\%$$

This add-on 365-day figure can now be compared against a coupon bond's yield.

### Example 4 — Realised (horizon) yield and the reinvestment assumption

Take Bond A again (bought at \$960.44, ten \$40 semiannual coupons, redeemed at \$1,000). We hold to maturity, so the only open question is the **reinvestment rate** on coupons.

**Case 1 — reinvest at the YTM (4.5% semiannual).** Future value of the reinvested coupon stream:

$$\text{FV}_{\text{coupons}} = 40 \times \frac{(1.045)^{10} - 1}{0.045} = 40 \times \frac{0.553069}{0.045} = 40 \times 12.29042 = 491.62$$

Total terminal value \$ = 491.62 + 1000 = 1491.62\$. Realised semiannual yield:

$$\left( \frac{1491.62}{960.44} \right)^{1/10} - 1 = (1.55307)^{0.1} - 1 = 1.045 - 1 = 4.5\%$$

Realised BEY = **9.00% = the YTM exactly**. This is the proof that YTM's reinvestment assumption is "reinvest at the YTM." When it holds, promised and realised yields coincide.

**Case 2 — reinvest at only 3% semiannual (6% annual, a falling-rate world).**

$$\text{FV}_{\text{coupons}} = 40 \times \frac{(1.03)^{10} - 1}{0.03} = 40 \times \frac{0.343916}{0.03} = 40 \times 11.46387 = 458.55$$

Total terminal value \$ = 458.55 + 1000 = 1458.55\$. Realised semiannual yield:

$$\left( \frac{1458.55}{960.44} \right)^{1/10} - 1 = (1.518635)^{0.1} - 1 = 1.042650 - 1 = 4.265\%$$

Realised BEY = **8.53%** — a full 47 bps *below* the promised 9.00%. Because coupons were reinvested below the YTM, the investor fell short. This is **reinvestment risk** in hard numbers, and it reconciles cleanly with Case 1: the only thing that changed was the reinvestment rate, and the realised yield moved in the same direction.

| Reinvestment rate (annual) | Terminal value | Realised BEY |
|----------------------------|----------------|--------------|
| 9.0% (= YTM)               | \$1,491.62     | 9.00%        |
| 6.0%                       | \$1,458.55     | 8.53%        |

**Case 3 — sell before maturity into a higher-rate market.** Now hold Bond A for only **2 years (4 periods)**, then sell. Suppose market yields have *risen* to 10% BEY (5% semiannual) by the sale date. Two things happen

at once — coupons reinvested at the higher 5% help us, but the sale price is depressed by the higher discount rate.

*Sale price* (6 periods remain, discount at 5% semiannual):

- $(1.05)^6 = 1.340096$ , inverse  $= 0.746215$
- Annuity factor  $= \frac{1 - 0.746215}{0.05} = 5.07570$
- Price  $= 40(5.07570) + 1000(0.746215) = 203.03 + 746.22 = 949.24$

*Reinvested coupons* (4 coupons compounded at 5%):

$$\text{FV of coupons} = 40 \times \frac{(1.05)^4 - 1}{0.05} = 40 \times 4.31013 = 172.41$$

*Terminal value*  $= 949.24 + 172.41 = 1121.65$ . Realised semiannual yield over 4 periods:

$$\left( \frac{1121.65}{960.44} \right)^{1/4} - 1 = (1.16785)^{0.25} - 1 = 1.03955 - 1 = 3.955\%$$

Realised BEY = **7.91%** — well *below* the 9.00% we bought at, even though coupons earned a *higher* reinvestment rate. Over this short two-year horizon the **price risk** (the capital loss from selling at a higher yield) swamped the reinvestment benefit. Flip the horizon out toward the bond's duration and the two effects would roughly cancel; push it all the way to maturity and only reinvestment matters (Cases 1 and 2). This is the reinvestment-risk-versus-price-risk trade-off made numerical.

### Example 5 — Yield to call, yield to worst

**Bond B:** face \$1,000, 8% coupon semiannual (\$40), 5 years (10 periods) to maturity, **callable in 3 years (6 periods) at \$1,020**, trading at a premium of **\$1,050**.

**YTM (to maturity, par \$1,000).** Premium bond → YTM < coupon. Solving (trial at 3.4% semiannual):

- $(1.034)^{10} = 1.397028$ , inverse  $= 0.715806$
- Annuity factor  $= \frac{1 - 0.715806}{0.034} = 8.35865$
- $P = 40(8.35865) + 1000(0.715806) = 334.35 + 715.81 = 1050.15 \approx 1050$  ✓

So YTM  $\approx 3.40\%$  semiannual  $=$  **6.80% BEY**.

**YTC (to first call, 6 periods, call price \$1,020).** Solve for  $y_c$  in:

$$1050 = 40 \cdot \frac{1 - (1 + y_c)^{-6}}{y_c} + 1020(1 + y_c)^{-6}$$

Bracket it. At  $y_c = 3.4\%$ :  $(1.034)^6 = 1.222134$ , inverse  $= 0.818238$ ; annuity factor  $= (1 - 0.818238)/0.034 = 5.34594$ ; price  $= 40(5.34594) + 1020(0.818238) = 213.84 + 834.60 = 1048.44$ . At  $y_c = 3.3\%$ :  $(1.033)^6 = 1.215039$ , inverse  $= 0.823019$ ; annuity factor  $= 5.36306$ ; price  $= 214.52 + 839.48 = 1054.00$ .

Interpolate for \$1,050:

$$y_c \approx 3.3\% + \frac{1054.00 - 1050}{1054.00 - 1048.44} \times 0.1\% = 3.3\% + \frac{4.00}{5.56} (0.1\%) = 3.372\%$$

YTC  $\approx 3.37\%$  semiannual  $=$  **6.74% BEY**.

**Yield to worst:**

| Scenario              | Yield (BEY)  |
|-----------------------|--------------|
| Yield to maturity     | 6.80%        |
| Yield to call (3 yr)  | 6.74%        |
| <b>Yield to worst</b> | <b>6.74%</b> |

The YTW is the yield-to-call. This is the classic result for a **premium callable bond trading above its call price**: the issuer has every incentive to call and refinance, and the investor should assume the worse (call) scenario. Quoting the 6.80% YTM here would overstate the realistic return.

## 6. Connections

- **To pricing (Ch. 03):** Yield is the pricing equation solved for its discount rate. You cannot compute a yield without the pricing machinery, and you cannot price a bond without a discount rate. They are two views of one identity.
- **To duration and convexity (Ch. 05):** Duration is the first derivative of price with respect to yield (scaled); convexity is the second. The convex price–yield curve seen in Figure 2 is *why* interpolation overstates YTM and why the "duration horizon" immunises realised yield.
- **To the term structure (Ch. 06):** A single YTM is a blended average of the spot rates applying to each cash flow. Spot (zero-coupon) yields and forward rates decompose what YTM aggregates. The reinvestment assumption embedded in YTM is exactly what forward rates make explicit.
- **To credit and spreads (Ch. 07):** Yield spreads (nominal spread, Z-spread, OAS) are all differences of yields; the OAS in particular strips out the call optionality that yield-to-worst only crudely proxies.
- **To portfolio return:** Realised/horizon yield is the bridge from single-bond math to actual portfolio performance attribution.

## 7. Key Terms

| Term                                | Meaning                                                                                                        |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Current yield</b>                | Annual coupon ÷ price; income return only.                                                                     |
| <b>Yield to maturity (YTM)</b>      | IRR of all cash flows to maturity, redeemed at par; assumes hold-to-maturity, no default, reinvestment at YTM. |
| <b>Yield to call (YTC)</b>          | IRR to a call date, redeemed at the call price.                                                                |
| <b>Yield to worst (YTW)</b>         | The minimum of YTM and all YTC / YTP scenarios.                                                                |
| <b>Bond-equivalent yield (BEY)</b>  | A semiannual periodic yield doubled; also a 365-day add-on money-market convention.                            |
| <b>Effective annual yield (EAY)</b> | The true compounded annual yield: $(1 + y_{\text{semi}})^2 - 1$ .                                              |
| <b>Realised / horizon yield</b>     | The actual return over a holding period given a specific reinvestment rate and ending value.                   |
| <b>Reinvestment risk</b>            | Risk that coupons are reinvested below the YTM, lowering realised return.                                      |
| <b>Price risk</b>                   | Risk that a rate rise lowers the bond's sale price before the horizon.                                         |
| <b>Pull to par</b>                  | The convergence of a discount/premium bond's price toward face value as maturity approaches.                   |

## 8. Common Confusions

- **"Coupon rate is the return."** No — the coupon rate is fixed to *face*, not to the *price you paid*. Only for a par bond do coupon rate, current yield, and YTM coincide.
- **"Current yield is a total return."** It omits capital gain/loss and the time value of coupons. For a deep-discount bond it can be badly misleading.
- **"YTM is the return I will earn."** Only if you hold to maturity *and* reinvest every coupon at the YTM. Example 4 showed a 47 bp shortfall when reinvestment fell to 6%. YTM is a *promised*, not a *guaranteed*, yield.
- **"BEY equals the effective annual yield."** BEY simply doubles the semiannual rate; EAY compounds it. BEY < EAY whenever the periodic rate is positive (9.00% vs 9.2025% in Example 3).
- **"Higher YTM always means the better bond."** Not on a callable name — compare yield-to-worst. And not across compounding conventions without first putting both on the same basis.
- **"Interpolated YTM is exact."** Convexity makes linear interpolation overstate the yield slightly (9.02% vs 9.00% in Example 2). Always re-price to verify.
- **"For a premium bond the yield-to-worst is the YTM."** Usually the opposite — a premium callable bond's worst case is the *call*, because the issuer will refinance cheap debt (Example 5).

## 9. Recap

A yield inverts the pricing equation: take the market price as given and solve for the discount rate that reproduces it. The measures differ only in the cash flows counted and the annualisation used. **Current yield**



captures income only. **YTM** is the IRR to maturity, resting on three assumptions — hold to maturity, no default, reinvest coupons at the YTM. **Yield-to-call** truncates the stream at a call date and redeems at the call price; **yield-to-worst** takes the least favourable of all such scenarios and is the honest quote for callable bonds. The **bond-equivalent yield** doubles a semiannual rate (understating the effective annual yield) and also standardises money-market instruments onto a 365-day add-on basis for comparison. Price and yield move **inversely** along a **convex** curve, which drives the discount/premium yield orderings and the small upward bias of interpolation. Finally, the **realised (horizon) yield** — the return you actually earn — depends on the reinvestment rate and the ending value, exposing you to reinvestment risk and price risk that only offset near the bond's duration.

## 10. Quick-Reference / Interview Points

### Formulas at a glance:

| Measure                | Formula                                                    |
|------------------------|------------------------------------------------------------|
| Current yield          | Annual coupon ÷ Price                                      |
| YTM (periodic $y$ )    | $P = C \frac{1 - (1+y)^{-n}}{y} + F(1+y)^{-n}$             |
| BEY                    | $2 \times y_{\text{semiannual}}$                           |
| EAY                    | $(1 + y_{\text{semi}})^2 - 1$                              |
| Annual-pay yield → BEY | $2[(1 + \text{EAY})^{1/2} - 1]$                            |
| Money-market BEY       | $\frac{F - P}{P} \times \frac{365}{t}$                     |
| Realised yield         | $\left( \frac{\text{Terminal value}}{P} \right)^{1/h} - 1$ |

### Yield orderings (memorise):

- Discount bond: coupon rate < current yield < YTM
- Par bond: coupon rate = current yield = YTM
- Premium bond: coupon rate > current yield > YTM

### Rapid-fire answers:

- *What three assumptions does YTM make?* Hold to maturity, no default, reinvest coupons at the YTM.
- *Why quote yield-to-worst on a callable?* It is the conservative floor — the lowest yield across maturity and all call scenarios; a premium callable bond's YTW is usually its yield-to-call.
- *Is BEY the same as effective annual yield?* No — BEY doubles (simple), EAY compounds; BEY < EAY.
- *You bought a bond at YTM 6% but rates fell to 4% — did you earn 6%?* No: coupons reinvested at 4% pull realised yield below 6% (reinvestment risk), partly offset by a higher sale price if you sell before maturity.
- *Where do reinvestment risk and price risk offset?* At a horizon near the bond's duration — the basis of immunisation.
- *Why does interpolated YTM slightly overstate the true yield?* Because the price–yield curve is convex, so the linear approximation between two points lies below the true curve.
- *Two bonds, one semiannual-pay one annual-pay — how do you compare?* Put both on the same convention (BEY or EAY) before comparing; never compare raw quotes across conventions.

# Q&A — Yield Measures

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A practice bank for Chapter 04. Every question is followed by a full answer; work each before reading the answer. Convention: coupon bonds are semiannual pay, "BEY" is the semiannual periodic yield doubled, and face value is \ \$1,000 unless stated otherwise.

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## Section A — Concept Checks

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### A1. In one sentence, what is a bond yield?

A yield is the discount rate implied by the market price — you take the observed price as given and solve the pricing equation *backwards* for the rate that equates the present value of the cash flows to that price. It is the internal rate of return (IRR) of the bond's cash-flow stream.

### A2. Every yield measure is the same inversion of the pricing equation. What three choices distinguish one measure from another?

(1) *Which cash flows* are counted — this period's coupon only, all coupons to maturity, or all coupons to a call date; (2) *what terminal/redemption value* is assumed — par at maturity, the call price, or a sale price at some horizon; (3) *what compounding/annualisation convention* turns the periodic rate into a quoted annual figure.

### A3. Why is current yield a poor total-return measure?

Current yield (annual coupon  $\div$  price) captures income return only. It ignores the capital gain or loss from buying at a discount or premium and pulling to par, ignores the timing and compounding of coupons, and ignores reinvestment. For a deep-discount bond it can badly understate total return.

### A4. State the three assumptions baked into YTM.

(1) The bond is held to maturity; (2) there is no default — every coupon and the principal are paid in full and on time; (3) all coupons are reinvested at the YTM itself until maturity. Break any one and the realised return diverges from the quoted YTM.

### A5. Why is YTM an IRR rather than a simple ratio?

A coupon in six months is worth more than the same coupon in five years, so an honest measure must discount for timing. The single rate making discounted inflows equal the price is exactly an IRR — and, like all IRRs, it embeds a reinvestment assumption: interim coupons are presumed reinvested at that same solved rate.

### A6. Write the yield ordering for discount, par, and premium bonds.

- Discount bond: coupon rate < current yield < YTM
- Par bond: coupon rate = current yield = YTM
- Premium bond: coupon rate > current yield > YTM

### A7. Why does the discount-bond ordering hold?

For a discount bond you paid less than par, so dividing the *same* coupon by a smaller price lifts current yield above the coupon rate. YTM sits higher still because it *also* captures the pull-to-par capital gain that current yield ignores.

### A8. Define yield to call and yield to worst.

Yield to call (YTC) is the IRR computed with the cash-flow stream truncated at a call date and the terminal value set to the call price rather than maturity par; a separate YTC exists for each call date. Yield to worst (YTW) is the *minimum* across the YTM and every YTC (and yield-to-put where relevant) — the conservative floor quoted on callable bonds.

### A9. Distinguish bond-equivalent yield from effective annual yield.

BEY simply *doubles* the semiannual periodic yield (a simple, non-compounding annualisation). EAY *compounds* it:  $(1 + y_{\text{semi}})^2 - 1$ . Because compounding adds the interest-on-interest term, BEY is always less than EAY whenever the periodic rate is positive.

### A10. What two risks make the realised yield differ from the promised YTM, and which direction does each push?

*Reinvestment risk*: if rates fall, coupons reinvest at lower rates, pulling realised yield below the YTM. *Price risk*: if rates rise before your horizon, the bond's sale price falls. They pull in opposite directions and roughly cancel at a horizon near the bond's duration (the basis of immunisation).

### A11. For a premium callable bond trading above its call price, is the yield to worst usually the YTM or the YTC? Why?

Usually the YTC. The issuer has cheap-to-refinance incentive to call the bond, so the investor should assume the call scenario, which for a premium bond gives the lower yield. Quoting the YTM here would overstate the realistic return.

### A12. Why does linear interpolation slightly overstate the true YTM?

The price–yield curve is convex (bowed toward the origin), so the straight line drawn between two priced points lies *below* the true curve. Reading a rate off that chord for a given price therefore returns a value a touch higher than the true yield. Always re-price to verify.

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## Section B — Numerical Bond-Math Problems

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### B1. A bond has face \$1,000, an 8% semiannual coupon (\$40 every six months), 5 years (10 periods) to maturity, and trades at \$960.44. Compute the current yield and the YTM (as BEY), then confirm the discount ordering.

Current yield  $\$ = 80 / 960.44 = 8.33\%$ .

For YTM, since the bond trades at a discount we expect  $\text{YTM} > \text{coupon} (8\%)$ . Try a semiannual  $y = 4.5\%$ :

- $\$(1.045)^{10} = 1.553069$ , so  $\$(1.045)^{-10} = 0.643928$
- Annuity factor  $\$ = (1 - 0.643928)/0.045 = 0.356072/0.045 = 7.912711$
- PV coupons  $\$ = 40 \times 7.912711 = 316.51$
- PV principal  $\$ = 1000 \times 0.643928 = 643.93$
- Total  $\$ = 316.51 + 643.93 = 960.44$  ✓

The semiannual yield is exactly 4.5%, so  $\text{YTM} \$ = 2 \times 4.5\% = 9.00\%$  BEY.

Ordering: coupon 8.00% < current yield 8.33% < YTM 9.00%. This is the discount ordering. The 39.56 gap between price and par is the pull-to-par capital gain that current yield omits but YTM captures.

**B2. Solve the same bond's YTM by interpolation, bracketing between 4% and 5% semiannual, and explain the direction of the error.**

At  $y = 4\%$  (yield = coupon  $\rightarrow$  price = par):  $P = 1000.00\$$ .

At  $y = 5\%$ :  $\$(1.05)^{10} = 1.628895\$$ , inverse  $\$ = 0.613913\$$ ; annuity factor  $\$ = (1 - 0.613913)/0.05 = 7.721735\$$ ;  $P = 40(7.721735) + 1000(0.613913) = 308.87 + 613.91 = 922.78\$$ .

Interpolate for the target  $\$960.44$ :

$$y \approx 4\% + \frac{1000 - 960.44}{1000 - 922.78} \times 1\% = 4\% + \frac{39.56}{77.22} \times 1\% = 4\% + 0.512\% = 4.512\%$$

That gives BEY  $\approx 9.02\%$  versus the true 9.00%. The interpolation *overstates* by ~2 bps because convexity puts the true price–yield curve above the chord. Re-pricing at exactly 4.5% returns  $\$960.44$ , confirming the true yield.

**B3. Using B1's 4.5% semiannual yield, compute the BEY and the effective annual yield, and quantify the convention gap.**

- BEY (simple doubling)  $\$ = 2 \times 4.5\% = 9.00\%$
- EAY (compounded)  $\$ = (1.045)^2 - 1 = 1.092025 - 1 = 9.2025\%$

The 9.00% quote understates the true annually-compounded return by about 20 bps — a pure artefact of doubling instead of compounding.

**B4. A competing annual-pay bond quotes a 9.20% effective annual yield. Put it on a bond-equivalent basis and compare it to B1's 9.00% BEY.**

$$\text{BEY} = 2 \left[ (1 + \text{EAY})^{1/2} - 1 \right] = 2 \left[ (1.092)^{1/2} - 1 \right] = 2[1.044988 - 1] = 2(0.044988) = 8.998\% \approx 9.00\%$$

On a like-for-like BEY basis the two bonds are essentially identical (9.00% vs 9.00%), even though the naive "9.00% vs 9.20%" comparison made the annual-pay bond look better. This is exactly the trap the convention exists to prevent — always convert to a common basis before comparing.

**B5. A 90-day T-bill is priced at  $\$98.50$  per  $\$100$  face. Compute its money-market bond-equivalent yield.**

Holding-period return  $\$ = (100 - 98.50)/98.50 = 1.5228\%$ . Annualise on a 365-day add-on basis:

$$\text{BEY} = 1.5228\% \times \frac{365}{90} = 6.18\%$$

This 365-day add-on figure is now directly comparable to a coupon bond's yield.

**B6. Take the B1 bond (bought  $\$960.44$ , ten  $\$40$  coupons, redeemed at  $\$1,000$ , held to maturity). Show that reinvesting coupons at the 4.5% semiannual YTM reproduces exactly the 9.00% BEY.**

Future value of the reinvested coupon stream:

$$\text{FV}_{\text{coupons}} = 40 \times \frac{(1.045)^{10} - 1}{0.045} = 40 \times \frac{0.553069}{0.045} = 40 \times 12.29042 = 491.62\$$$

Terminal value  $\$ = 491.62 + 1000 = 1491.62\$$ . Realised semiannual yield:

$$\left( \frac{1491.62}{960.44} \right)^{1/10} - 1 = (1.55307)^{0.1} - 1 = 1.045 - 1 = 4.5\%$$

Realised BEY  $\$ = 9.00\% = \$$  the YTM exactly. This proves YTM's reinvestment assumption is "reinvest at the YTM": when it holds, promised and realised yields coincide.

**B7. Repeat B6 but reinvest coupons at only 3% semiannual (6% annual). Quantify the shortfall and reconcile with B6.**

$$\text{FV}_{\text{coupons}} = 40 \times \frac{(1.03)^{10} - 1}{0.03} = 40 \times \frac{0.343916}{0.03} = 40 \times 11.46387 = 458.55$$

Terminal value \$ = 458.55 + 1000 = 1458.55\$. Realised semiannual yield:

$$\left( \frac{1458.55}{960.44} \right)^{1/10} - 1 = (1.518635)^{0.1} - 1 = 1.042650 - 1 = 4.265\%$$

Realised BEY \$ = 8.53% — a full 47 bps below the promised 9.00%. The *only* thing that changed from B6 was the reinvestment rate, and realised yield moved in the same direction (down), so the two cases reconcile cleanly. This is reinvestment risk in hard numbers.

| Reinvestment rate (annual) | Terminal value | Realised BEY |
|----------------------------|----------------|--------------|
| 9.0% (= YTM)               | \$1,491.62     | 9.00%        |
| 6.0%                       | \$1,458.55     | 8.53%        |

**B8. Now hold the B1 bond only 2 years (4 periods) and sell when market yields have risen to 10% BEY (5% semiannual). Compute the realised yield and explain why it fell below 9%.**

*Sale price* (6 periods remain, discount at 5% semiannual):  $(1.05)^6 = 1.340096$ , inverse \$ = 0.746215\$; annuity factor \$ =  $(1 - 0.746215)/0.05 = 5.07570$ ; price \$ =  $40(5.07570) + 1000(0.746215) = 203.03 + 746.22 = 949.24$ .

*Reinvested coupons* (4 coupons compounded at 5%):  $\text{FV} = 40 \times \frac{(1.05)^4 - 1}{0.05} = 40 \times 4.31013 = 172.41$ .

Terminal value \$ = 949.24 + 172.41 = 1121.65\$. Realised semiannual yield over 4 periods:

$$\left( \frac{1121.65}{960.44} \right)^{1/4} - 1 = (1.16785)^{0.25} - 1 = 1.03955 - 1 = 3.955\%$$

Realised BEY \$ = 7.91% — below 9.00% despite the *higher* 5% reinvestment rate. Over this short horizon the price risk (capital loss from selling into a higher yield) swamped the reinvestment benefit. Near the bond's duration the two effects would cancel; at maturity only reinvestment matters (B6–B7).

**B9. Bond B: face \$1,000, 8% semiannual coupon (\$40), 5 years (10 periods) to maturity, callable in 3 years (6 periods) at \$1,020, trading at a premium of \$1,050. Find YTM, YTC, and YTW (all as BEY).**

*YTM* (par \$1,000). Premium → YTM < coupon. Try 3.4% semiannual:  $(1.034)^{10} = 1.397028$ , inverse \$ = 0.715806\$; annuity factor \$ =  $(1 - 0.715806)/0.034 = 8.35865$ ; \$P =  $40(8.35865) + 1000(0.715806) = 334.35 + 715.81 = 1050.15 \approx 1050$  ✓. So YTM  $\approx 3.40\%$  semiannual \$ = 6.80% BEY.

*YTC* (6 periods, call price \$1,020). Bracket it. At  $y_c = 3.4\%$ :  $(1.034)^6 = 1.222134$ , inverse \$0.818238\$; annuity \$ =  $(1 - 0.818238)/0.034 = 5.34594$ ; price \$ =  $40(5.34594) + 1020(0.818238) = 213.84 + 834.60 = 1048.44$ . At  $y_c = 3.3\%$ :  $(1.033)^6 = 1.215039$ , inverse \$0.823019\$; annuity \$ = 5.36306\$; price \$ =  $214.52 + 839.48 = 1054.00$ . Interpolate for \$1,050:

$$y_c \approx 3.3\% + \frac{1054.00 - 1050}{1054.00 - 1048.44} \times 0.1\% = 3.3\% + \frac{4.00}{5.56} (0.1\%) = 3.372\%$$

YTC  $\approx 3.37\%$  semiannual \$ = 6.74% BEY.

YTW \$ =  $\min(6.80\%, 6.74\%) = 6.74\%$ , the yield-to-call. Classic result for a premium callable trading above its call price: assume the issuer calls, so the honest quote is the lower YTC.

**B10. A 2-year zero-coupon bond, face \$1,000, trades at \$888.49. Find its YTM as BEY.**

A zero has one cash flow, so the yield solves directly with no interpolation. On a semiannual (4-period) basis:

$$888.49 = \frac{1000}{(1+y)^4} \rightarrow (1+y)^4 = \frac{1000}{888.49} = 1.125491 \rightarrow 1+y = 1.125491^{0.25} = 1.030000$$

So  $y = 3.00\%$  semiannual and YTM  $= 6.00\%$  BEY. Check:  $1000/(1.03)^4 = 1000/1.125509 = 888.48$  ✓ (rounding). A zero carries no reinvestment risk — there are no interim coupons — so its promised YTM is its realised yield if held to maturity.

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## Section C — Interview-Style Questions with Model Answers

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**C1. "A client says a bond's coupon is its return. How do you correct them?"**

Model answer: The coupon rate is fixed at issuance and measured against *face value*, not what you actually paid. Buy below par and your income return per dollar is higher than the coupon rate, plus you gain as the price pulls to par; buy above par and both work against you. The coupon equals your return only at par. The right total-return measure is yield to maturity, which folds in price, all coupons, their timing, and the redemption value.

**C2. "Walk me through why YTM is a 'promised' rather than a 'guaranteed' yield."**

Model answer: YTM is an IRR, and every IRR silently assumes you reinvest interim cash flows at that same rate. So the quoted YTM is realised only if three things hold: you hold to maturity, the issuer never defaults, and every coupon is reinvested at the YTM. In a falling-rate world coupons reinvest below the YTM and realised return drops — a 9% bond whose coupons reinvest at 6% realises only about 8.5%. That is reinvestment risk: YTM is a promise conditional on assumptions, not a guarantee.

**C3. "Two bonds: one pays semiannually and yields 9.00%, the other pays annually and yields 9.20%. Which is better?"**

Model answer: You can't compare them until they share a convention. The 9.00% is a bond-equivalent yield (a doubled semiannual rate); the 9.20% is an effective annual yield. Convert the annual-pay figure to BEY:  $2[(1.092)^{1/2} - 1] = 9.00\%$  — identical. Equivalently the semiannual bond's EAY is  $(1.045)^2 - 1 = 9.20\%$ . Either way they tie; the naive quote comparison is exactly the trap the conventions exist to expose.

**C4. "Why would you ever quote yield-to-worst instead of yield-to-maturity?"**

Model answer: With embedded issuer options — mainly callables — the YTM assumes survival to maturity, but the issuer calls whenever that benefits them, which is precisely when it hurts you. Yield-to-worst takes the minimum yield across maturity and every call date, so it is the conservative floor. For a premium callable above its call price the worst case is almost always the call: the issuer refinances cheap debt and you lose the remaining premium quickly. Quoting the YTM there would systematically overstate return.

**C5. "You immunise a liability by matching duration. Intuitively, why does that neutralise interest-rate risk?"**

Model answer: A bondholder faces two opposing rate exposures. If rates rise, the sale price falls (price risk) but coupons reinvest at higher rates; if rates fall, the reverse. These effects grow at different speeds with the horizon. At a horizon roughly equal to the bond's duration, a small rate change moves price and reinvestment by offsetting amounts, so the realised yield is locked in whichever way rates move. Matching

the asset's duration to the liability's horizon is therefore what immunises the position against small rate shifts.

**C6. "A junior analyst interpolated a YTM and got 9.02%; the solver says 9.00%. Is the analyst wrong?"**

Model answer: Not wrong, just approximate and predictably biased. Linear interpolation draws a straight chord between two priced points, but the price–yield relationship is convex, bowing above that chord, so a rate read off the chord comes out slightly too high — the 2-bp overstatement is exactly what convexity predicts. Treat interpolation as a first pass, then re-price at the interpolated rate to confirm and narrow the bracket if needed.

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## Section D — Multiple-Choice Questions with Reasoning

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**D1. A bond trades at a premium. Which ordering is correct?**

- A. coupon < current yield < YTM
- B. coupon > current yield > YTM
- C. coupon = current yield = YTM
- D. current yield > coupon > YTM

**Answer: B.** For a premium bond you paid more than par, so dividing the coupon by a larger price pushes current yield below the coupon rate, and YTM sits still lower because it also captures the pull-to-par capital loss. A is the discount ordering; C is the par case; D violates the fixed relationship between coupon and price.

**D2. Which statement about bond-equivalent yield is TRUE?**

- A. BEY always exceeds the effective annual yield.
- B. BEY compounds the semiannual rate.
- C. BEY simply doubles the semiannual rate and is therefore below the EAY.
- D. BEY and EAY are identical for semiannual bonds.

**Answer: C.** BEY is a simple doubling; EAY compounds, adding an interest-on-interest term, so  $EAY > BEY$  whenever the periodic rate is positive (e.g. 9.00% vs 9.2025%). A and D reverse or erase the gap; B confuses BEY with EAY.

**D3. A premium bond is callable and currently trades above its call price. Its yield to worst is most likely:**

- A. the yield to maturity
- B. the current yield
- C. the yield to call
- D. the coupon rate

**Answer: C.** The issuer is incentivised to call and refinance cheap debt, and for a premium bond the call scenario produces the lower yield, so  $YTW = YTC$ . A would understate the risk of being called; B and D are not scenario yields at all.

**D4. An investor holds a coupon bond to maturity in a falling-rate environment. Relative to the purchase-date YTM, the realised yield will be:**

- A. higher, because the bond's price rose
- B. lower, because coupons reinvest below the YTM



- C. exactly the YTM, because it was held to maturity
- D. unaffected by reinvestment

**Answer: B.** Held to maturity, the ending value is par regardless of rates, so price risk is irrelevant; the only open variable is the reinvestment rate. Coupons reinvested below the YTM drag realised yield beneath the promised figure. C ignores reinvestment; A confuses this with a sell-before-maturity scenario; D is simply false for a coupon bond.

**D5. Which yield measure ignores capital gain/loss and the time value of coupons entirely?**

- A. yield to maturity
- B. current yield
- C. realised yield
- D. yield to call

**Answer: B.** Current yield is just annual coupon  $\div$  price — an income snapshot with no discounting and no pull-to-par adjustment. The other three all discount cash flows over time and account for the redemption or sale value.

**D6. Linear interpolation of a YTM between two priced points tends to:**

- A. understate the true yield, because the curve is concave
- B. overstate the true yield, because the price–yield curve is convex
- C. exactly equal the true yield
- D. overstate the price, not the yield

**Answer: B.** The price–yield curve is convex, so the interpolating chord lies below it; reading a yield off the chord for a given price returns a value slightly too high. A misstates the curvature; C ignores convexity; D confuses the axes.

**D7. A zero-coupon bond held to maturity has:**

- A. high reinvestment risk
- B. no reinvestment risk, because there are no interim coupons
- C. a realised yield below its YTM
- D. a current yield equal to its YTM

**Answer: B.** With no interim coupons there is nothing to reinvest, so the promised YTM equals the realised yield if held to maturity. A and C wrongly import coupon-bond reinvestment risk; D is meaningless since a zero pays no coupon, giving a current yield of zero.

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*Self-check: every numerical answer was verified by substituting the solved rate back into the pricing or terminal-value equation and confirming it reproduces the stated price or yield; the discount ordering (B1), reinvestment cases (B6–B8), and callable YTW (B9) all reconcile internally.*



# Chapter 05 — Spot and Forward Rates

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## 1. The Problem / The Need

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Chapter 4 priced a bond with a single yield to maturity (YTM) — one number that, applied to every cash flow, reproduces the market price. That is enormously convenient, but it hides a lie. A YTM discounts the coupon you receive in six months at the *same* rate as the principal you receive in ten years. In reality, money for six months and money for ten years command different prices. Lending for a decade ties your capital up longer, exposes you to more inflation and default uncertainty, and usually earns a different rate than lending overnight. The market does not have "an interest rate"; it has a *term structure* — an entire schedule of rates, one for each horizon.

If each maturity has its own true rate of return, then two questions become unavoidable:

1. **What is the correct rate to discount a single cash flow arriving at time  $t$ ?** That is the **spot rate** (also called the zero-coupon rate) — the yield on a pure discount bond that pays once, at  $t$ , with no coupons in between. A coupon bond is just a *portfolio* of such single payments, so its true value is the sum of each cash flow discounted at its own spot rate. YTM is a weighted average of these spot rates; it is an output, not the fundamental input.
2. **What does the market think a future interest rate will be?** If I can lock in a rate today for lending *between* year 1 and year 2 — starting in the future — that is a **forward rate**. Forward rates are not forecasts pulled from thin air; they are *implied* by today's spot curve through a no-arbitrage argument. Every spot curve secretly contains a full set of forward rates.

This chapter builds the machinery to move between three representations of the same information — spot rates, forward rates, and coupon-bond prices — and shows they are locked together by arbitrage. Mastering this is the price of admission to swaps, FRAs, curve trading, and any serious relative-value work.

## 2. The Core Idea

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There is one master identity, and everything in this chapter is a rearrangement of it. Money invested from today (time 0) to time  $n$  must grow by the same total factor whether you:

- **(A)** lock in the  $n$ -year spot rate today and hold to maturity, or
- **(B)** invest for a shorter period and *roll over* at rates you lock in today for the future sub-periods.

If those two produced different terminal wealth with zero risk, you could borrow via the cheap path and lend via the rich path and pocket a riskless profit. Arbitrageurs would pounce until the paths matched. So, with annual compounding:

$$(1+z_n)^n = (1+z_1)(1+f_{1,2})(1+f_{2,3})\cdots(1+f_{n-1,n})$$

where  $z_n$  is the  $n$ -year **spot rate** and  $f_{k,k+1}$  is the one-year **forward rate** covering the period from year  $k$  to year  $k+1$ . The spot curve and the forward curve are two encodings of the same object. Given one, the other is determined — no forecasting required.

The second core idea is **bootstrapping**: real markets quote *coupon* bonds, not a clean strip of zeros. But a coupon bond is a bundle of cash flows, each of which should be discounted at a spot rate. By working from the shortest maturity outward — using the already-solved short spot rates to strip out the near coupons —

we can peel each successive spot rate out of each successive coupon bond price. That is how a usable zero curve gets built from tradeable instruments.

### 3. Why / How It Works

**Why spot rates, not YTM, are the "atoms" of pricing.** Consider a 3-year 7% annual-coupon bond. It pays 7, 7, and 107. Suppose someone could buy the individual claims separately: a 1-year zero paying 7, a 2-year zero paying 7, and a 3-year zero paying 107. The *law of one price* says the coupon bond must cost exactly what those three zeros cost combined — otherwise you strip the bond and sell the pieces (or buy the pieces and reconstitute the bond) for free money. This is literally what the U.S. Treasury STRIPS program allows. Therefore:

$$P = \frac{C_1}{(1+z_1)^1} + \frac{C_2}{(1+z_2)^2} + \cdots + \frac{C_n}{(1+z_n)^n}$$

Each cash flow is discounted at *its own* horizon's spot rate. Contrast with YTM, which forces a single  $y$ :

$$P = \frac{C_1}{(1+y)^1} + \frac{C_2}{(1+y)^2} + \cdots + \frac{C_n}{(1+y)^n}$$

Both equations give the *same* price  $P$  — but only the spot-rate equation reflects the true term structure. The YTM  $y$  is an internal rate of return: a single complex-average of the spot rates, weighted by the timing and size of the cash flows. Two bonds with identical maturities but different coupons will have *different* YTM's even in the same market, precisely because their cash flows sample the spot curve with different weights. Spot rates, by contrast, are coupon-independent — they belong to the market, not to any one bond.

**Why forward rates are implied, not forecast.** Return to path A versus path B. Suppose  $z_1 = 5\%$  and  $z_2 = 5.5\%$ . Path A: invest \$1 for two years at the 2-year spot: grows to  $(1.055)^2 = 1.113025$ . Path B: invest for one year at 5% (grows to 1.05), then reinvest for the second year at whatever one-year rate you can lock in *today* for that future year — call it  $f_{1,2}$ . For no arbitrage:

$$1.05 \times (1 + f_{1,2}) = 1.113025 \quad \Rightarrow \quad f_{1,2} = \frac{1.113025}{1.05} - 1 = 6.00\%$$

The forward rate 6.00% is not a prediction that rates *will* be 6% next year. It is the rate that makes an investor *indifferent* between the two paths today. If your own forecast for next year's 1-year rate differs from 6%, that is precisely where a trade lives — but the *implied* forward is pinned by arithmetic.

**How the arbitrage is actually enforced.** You do not need exotic instruments. To lock in  $f_{1,2}$  using only cash bonds: today, buy the 2-year zero and short-sell the appropriate amount of the 1-year zero (or borrow for one year). The combined position has *no* net cash flow today, a cash *outflow* at year 1 (repaying the short), and a cash *inflow* at year 2 — synthetically, you have contracted today to invest money between year 1 and year 2 at a locked rate. If dealers quote a forward-rate agreement at anything other than the implied forward, this cash-and-carry replication forces it back.

The general formula for the one-period forward from year  $k$  to  $k+1$ :

$$1 + f_{k,k+1} = \frac{(1+z_{k+1})^{k+1}}{(1+z_k)^k}$$

and the multi-period forward covering  $m$  years starting at year  $k$ :

$$(1+f_{k,k+m})^m = \frac{(1+z_{k+m})^{k+m}}{(1+z_k)^k}$$

## 4. Full Content — Formulas and Bond Math

### 4.1 Definitions

| Term             | Symbol    | Meaning                                                                                      |
|------------------|-----------|----------------------------------------------------------------------------------------------|
| Spot / zero rate | $z_t$     | Yield on a zero-coupon bond maturing at $t$ ; the rate to discount a single cash flow at $t$ |
| Discount factor  | $DF_t$    | Present value today of \$1 paid at $t$ : $DF_t = 1/(1+z_t)^t$                                |
| Forward rate     | $f_{k,m}$ | Rate locked in today for a loan running from $k$ to $m$                                      |
| Par rate         | $c_t$     | Coupon rate that makes a $t$ -year bond price to par (100)                                   |
| YTM              | $y$       | Single IRR that reproduces a coupon bond's price                                             |

### 4.2 The discount function

Everything can be expressed through discount factors, which sidestep compounding conventions:

$$DF_t = \frac{1}{(1+z_t)^t}, \quad P = \sum_t C_t \cdot DF_t$$

The forward discount factor between  $k$  and  $m$  is simply the ratio:

$$DF_{k \rightarrow m} = \frac{DF_m}{DF_k}, \quad (1+f_{k,m})^{m-k} = \frac{DF_k}{DF_m}$$

This ratio form is the cleanest way to remember forwards: **the forward growth factor is the ratio of the near discount factor to the far discount factor.**

### 4.3 The chaining (no-arbitrage) identity

$$(1+z_n)^n = \prod_{k=0}^{n-1} (1+f_{k,k+1}), \quad \text{with } f_{0,1} \equiv z_1$$

Taking the geometric interpretation: **the  $n$ -year spot rate is the geometric average of the sequence of one-year forward rates.**

$$1+z_n = \left[ (1+z_1)(1+f_{1,2}) \cdots (1+f_{n-1,n}) \right]^{1/n}$$

### 4.4 Bootstrapping the spot curve

Given a set of coupon bonds (ideally par bonds, or a swap curve) at maturities 1, 2, 3, ..., solve *sequentially*:

**Step 1 (1-year):** From the 1-year bond price  $P_1$  with final cash flow  $C_1$ :  $P_1 = \frac{C_1}{1+z_1}$   
 $\Rightarrow z_1 = \frac{C_1}{P_1} - 1$

**Step  $n$  (recursively):** Knowing  $z_1, \dots, z_{n-1}$ , take the  $n$ -year bond with coupons  $C$  and price  $P_n$ :  
 $P_n = \sum_{t=1}^{n-1} \frac{C}{(1+z_t)^t} + \frac{C + \text{Face}}{(1+z_n)^n}$

All terms except the last are known. Isolate the final term and solve for  $z_n$ :

$$(1+z_n)^n = \frac{C + \text{Face}}{P_n - \sum_{t=1}^{n-1} \frac{C}{(1+z_t)^t}}$$

Each new bond contributes exactly one new unknown spot rate, so the system solves cleanly one maturity at a time.

### 4.5 Forward Rate Agreements (FRAs)

An **FRA** is an over-the-counter contract that locks in an interest rate for a future borrowing/lending period on a *notional* principal. No principal changes hands — only the *interest differential* is settled in cash. FRAs are

quoted as "\$a \times b\$": e.g. a **6×12 FRA** covers the 6-month period *starting* in 6 months and *ending* in 12 months.

- **The fixed (FRA) rate**  $K$  is the implied forward rate derived from the spot curve (Section 3). A dealer who quotes off-market gets arbitrated via cash-and-carry.
- **Convention:** the *long* (buyer, a would-be borrower) profits when the reference rate sets *above*  $K$ ; the *short* (a would-be lender) profits when it sets *below*.
- **Settlement** happens at the *start* of the reference period, so the interest-differential (which is naturally an end-of-period amount) is **discounted back** at the realized reference rate.

Using money-market (simple, Act/360) conventions, with day-count fraction  $\tau = \text{days}/360$ , realized reference rate  $L$ , notional  $N$ :

$$\text{Payoff to long} = N \cdot \frac{(L - K)\tau}{1 + L\tau}$$

The numerator  $(L - K)\tau \cdot N$  is the interest saved/lost over the period; dividing by  $(1 + L\tau)$  discounts it from period-end back to settlement date.

The FRA fixed rate itself is the simple forward implied by two money-market spot rates  $z_{\text{short}}$  (to the near date, fraction  $\tau_1$ ) and  $z_{\text{long}}$  (to the far date, fraction  $\tau_2$ ):

$$(1 + z_{\text{short}}\tau_1)(1 + K\tau_{\text{fwd}}) = (1 + z_{\text{long}}\tau_2), \quad \tau_{\text{fwd}} = \tau_2 - \tau_1$$

## 4.6 Compounding conventions (a warning)

Forward rates depend on the compounding convention. The cleanest is *continuous* compounding, where the chaining identity becomes purely additive:

$$z_n \cdot n = z_1 \cdot 1 + f_{1,2} \cdot 1 + \dots \rightarrow f_{k,m} = \frac{z_m \cdot m - z_k \cdot k}{m - k}$$

Under continuous compounding the forward rate is a simple *weighted difference* of spot rates — elegant and popular in derivatives math. This chapter uses annual compounding for the bond examples and simple money-market rates for the FRA, because that is what shows up in exams and on trading desks respectively. Always state your convention.

## 5. Worked Examples

### Example 1 — Bootstrapping a 3-year spot curve from coupon bonds

We are given three annual-coupon bonds, face value 100, priced in the market:

| Bond | Maturity | Annual coupon | Market price |
|------|----------|---------------|--------------|
| A    | 1 year   | 4.00          | 99.0476      |
| B    | 2 years  | 6.00          | 100.9502     |
| C    | 3 years  | 7.00          | 102.7942     |

**Step 1 — extract  $z_1$  from Bond A.** Bond A pays 104 at  $t=1$ :  $99.0476 = \frac{104}{1 + z_1}$

$$\rightarrow 1 + z_1 = \frac{104}{99.0476} = 1.05000 \rightarrow \boxed{z_1 = 5.00\%}$$

**Step 2 — extract  $z_2$  from Bond B.** Bond B pays 6 at  $t=1$  and 106 at  $t=2$ . Discount the year-1 coupon at the *known*  $z_1$ :  $\frac{6}{1.05} = 5.71429$   $100.9502 = 5.71429 + \frac{106}{(1 + z_2)^2}$   $\frac{106}{(1 + z_2)^2}$

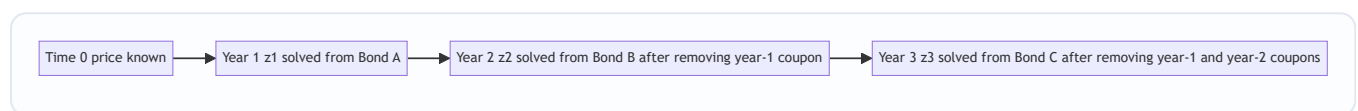
$$(1+z_2)^2 = 95.23591 \rightarrow (1+z_2)^2 = \frac{106}{95.23591} = 1.113025 \Rightarrow 1+z_2 = \sqrt{1.113025} = 1.05500 \rightarrow \boxed{z_2 = 5.50\%}$$

**Step 3 — extract  $z_3$  from Bond C.** Bond C pays 7, 7, 107. Discount the first two coupons at the known spots:  $\frac{7}{1.05} = 6.66667$ ,  $\frac{7}{(1.05)^2} = \frac{7}{1.113025} = 6.28829$   $102.7942 = 6.66667 + 6.28829 + \frac{107}{(1+z_3)^3}$   $\frac{107}{(1+z_3)^3} = 89.83924 \rightarrow (1+z_3)^3 = \frac{107}{89.83924} = 1.191016$   $1+z_3 = 1.191016^{1/3} = 1.06000 \rightarrow \boxed{z_3 = 6.00\%}$

**The bootstrapped spot curve:**  $z_1=5.00\%$ ,  $z_2=5.50\%$ ,  $z_3=6.00\%$  — an upward-sloping curve.

**Self-check (reconstitute Bond C).** Re-price Bond C using the three spot rates and confirm we recover 102.7942:  $\frac{7}{1.05} + \frac{7}{1.113025} + \frac{107}{1.191016} = 6.66667 + 6.28829 + 89.83924 = 102.79420$  ✓

The following timeline shows how each bond's cash flows are peeled apart, with known spots (solid) used to strip out near coupons so the far spot (dashed) can be solved.



*Bootstrapping marches outward: each new bond adds exactly one unknown spot rate.*

## Example 2 — Deriving the forward curve and verifying the chaining identity

Using the spot curve from Example 1, compute the implied one-year forward rates.

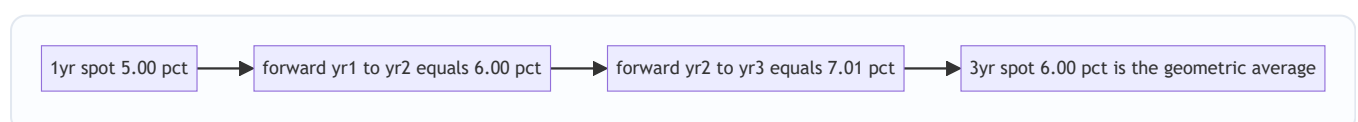
**$f_{1,2}$  (1-year rate, 1 year forward):**  $1+f_{1,2} = \frac{(1+z_2)^2}{1+z_1} = \frac{1.113025}{1.05} = 1.060024 \rightarrow f_{1,2} = 6.00\%$

**$f_{2,3}$  (1-year rate, 2 years forward):**  $1+f_{2,3} = \frac{(1+z_3)^3}{(1+z_2)^2} = \frac{1.191016}{1.113025} = 1.070073 \rightarrow f_{2,3} = 7.01\%$

**$f_{1,3}$  (2-year rate, 1 year forward):**  $(1+f_{1,3})^2 = \frac{(1+z_3)^3}{1+z_1} = \frac{1.191016}{1.05} = 1.134301$   $1+f_{1,3} = \sqrt{1.134301} = 1.065036 \rightarrow f_{1,3} = 6.50\%$

**Self-check (chaining identity).** The three consecutive one-year rates must compound up to the 3-year spot factor:  $(1+z_1)(1+f_{1,2})(1+f_{2,3}) = 1.05 \times 1.060024 \times 1.070073 = 1.113025 \times 1.070073 = 1.191016 = (1+z_3)^3$  ✓

Also note  $z_3 = 6.00\%$  is the *geometric average* of  $5.00\%$ ,  $6.00\%$ ,  $7.01\%$ :  $\left(1.05 \cdot 1.060024 \cdot 1.070073\right)^{1/3} = 1.06000$ . The spot rate sits *below* the far forward because the curve is rising — the average of a rising sequence lags its latest value. This is the single most important intuition in the whole chapter:



*When the spot curve rises, forward rates lie above spot rates and pull the average upward.*

### Example 3 — Pricing a Forward Rate Agreement (6×12 FRA)

A treasurer knows the firm will borrow **\$10,000,000 for 6 months, starting in 6 months**, and wants to hedge the rate. Current money-market spot rates: **6-month = 4.00%**, **12-month = 4.50%** (both simple, annualized; assume each 6-month period is exactly  $\tau = 0.5$ ).

**Step 1 — the fair FRA rate (implied forward).** No-arbitrage requires that lending for 12 months equals lending 6 months then rolling at the forward: 
$$(1 + 0.04 \times 0.5)(1 + K \times 0.5) = (1 + 0.045 \times 1)$$
$$1.02 \times (1 + 0.5K) = 1.045 \rightarrow 1 + 0.5K = \frac{1.045}{1.02} = 1.024510$$
$$0.5K = 0.024510 \rightarrow \boxed{K = 4.902\%}$$

The treasurer buys (goes long) the 6×12 FRA at **4.902%**, locking in that borrowing rate.

**Step 2 — settlement when rates rise.** Suppose at the settlement date (6 months from now) the realized 6-month reference rate sets at **L = 5.50%**. The firm must now borrow at 5.50% in the cash market — more expensive — but the FRA compensates. Payoff to the long: 
$$\text{Payoff} = N \times \frac{(L - K) \times \tau}{1 + L \times \tau} = 10,000,000 \times \frac{(0.055 - 0.04902) \times 0.5}{1 + 0.055 \times 0.5}$$
$$= 10,000,000 \times \frac{0.00598 \times 0.5}{1.0275} = \frac{29,900}{1.0275} = \boxed{\$29,101}$$

The firm receives \$29,101 at the settlement date.

**Self-check (did the hedge work?).** The extra interest cost of borrowing at 5.50% instead of the locked 4.902%, measured at period-end (12 months), is: 
$$\$10,000,000 \times (0.055 - 0.04902) \times 0.5 = \$29,900$$
 (payable in 6 months' time). The FRA paid \$29,101 now (at the 6-month settlement). Growing it forward at the realized 5.50% over the next 6 months: 
$$\$29,101 \times 1.0275 = \$29,901 \approx \$29,900$$
. The FRA gain, carried to period-end, exactly offsets the extra borrowing cost. The hedge is complete — the firm's *effective* all-in rate is the locked 4.902% regardless of where L set. ✓

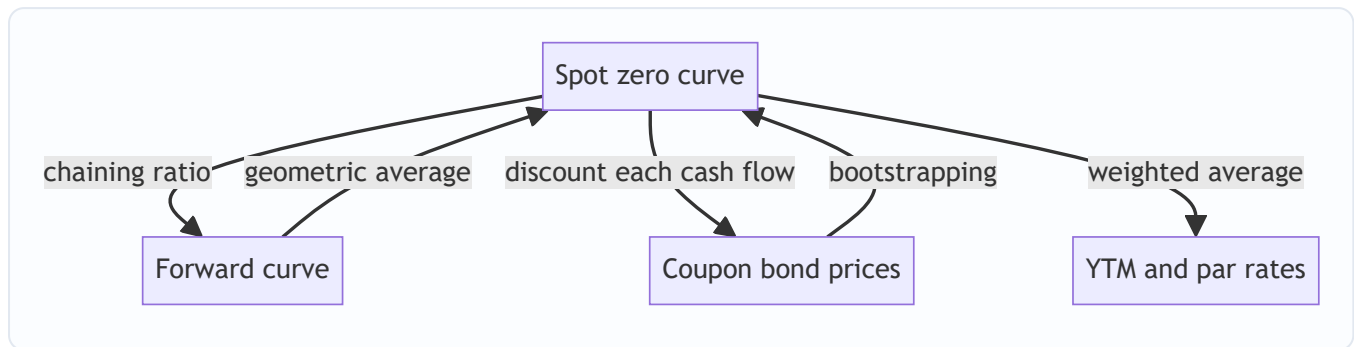
Today lock FRA rate K equals 4.902 pct notional 10m → Month 6 reference rate sets at 5.50 pct FRA pays 29101 → Month 6 to 12 firm borrows at market rate FRA gain offsets extra cost → Effective borrowing rate stays 4.902 pct

An FRA converts an uncertain future borrowing rate into a locked rate via a discounted cash settlement.

## 6. Connections

- **To YTM (Chapter 4):** YTM is a cash-flow-weighted geometric average of spot rates. A par bond's YTM equals its par rate; a par rate is itself a coupon-weighted average of spot rates. Spot rates are the primitive; YTM and par rates are derived summaries.
- **To duration and convexity (Chapter 6):** True risk is exposure to *each* spot rate (key-rate / partial durations), not just to a single yield. The spot curve is the object that actually shifts and twists.
- **To swaps:** A plain-vanilla interest-rate swap is priced by discounting its floating and fixed legs on the bootstrapped zero curve; the swap fixed rate is a par rate, and each floating payment equals the implied forward. The swap curve is, in practice, *the* curve bootstrapped in modern markets.
- **To the expectations hypothesis:** The pure expectations theory claims forward rates equal the market's expected future spot rates. Liquidity-preference and term-premium theories argue forwards exceed expected spots by a risk premium. Either way, forwards are the market's *risk-neutral* anchor.
- **To STRIPS:** Treasury coupon stripping and reconstitution are the physical mechanism that enforces the law of one price between coupon bonds and zeros — making the bootstrapped spot curve a real, tradeable object.

The relationships among the curve shapes:



Spot, forward, coupon-price, and YTM are four views of one term structure, linked by exact transformations.

## 7. Key Terms

- **Spot rate / zero rate ( $z_t$ ):** yield on a single cash flow at time  $t$ ; the correct discount rate for that maturity.
- **Discount factor ( $DF_t$ ):** present value of  $\$1$  at  $t$ ;  $\$1/(1+z_t)^t$ .
- **Forward rate ( $f_{k,m}$ ):** rate locked in today for a future loan from  $k$  to  $m$ , implied by spot rates.
- **Bootstrapping:** sequential extraction of spot rates from coupon-bond (or swap) prices, shortest maturity first.
- **Par rate:** coupon that prices a bond to 100; equals the swap fixed rate at that tenor.
- **FRA (Forward Rate Agreement):** OTC contract fixing a future interest rate on a notional, cash-settled on the interest differential.
- **Chaining / no-arbitrage identity:**  $(1+z_n)^n = \prod (1+f_{k,k+1})$ .
- **Term structure of interest rates:** the full schedule of rates across maturities; the yield curve.

## 8. Common Confusions

1. **"YTM is the return I'll earn."** Only if you can reinvest every coupon at the YTM *and* hold to maturity. The true per-period cost of money is the spot rate. YTM blends spots into one figure that belongs to a specific bond, not to the market.
2. **"The forward rate is a forecast."** It is an *arbitrage-implied* break-even, not a prediction. It equals the *expected* future spot only under the pure expectations hypothesis with no term premium. Trades are born from the gap between your forecast and the implied forward.
3. **"Upward-sloping spot curve means the market expects rising rates."** Partly, but term/liquidity premia inflate forwards above expected spots. A rising spot curve is consistent with flat expected future rates *plus* a positive term premium.
4. **Forward vs. spot magnitude direction.** When the spot curve *rises*, forwards lie *above* spots (the marginal one-year rate exceeds the running average). When the curve is *inverted*, forwards lie *below* spots. Getting this backwards is a classic exam error — remember Example 2: 5%  $\rightarrow$  6%  $\rightarrow$  7% forwards, average 6% spot.
5. **Forgetting to discount the FRA settlement.** FRAs settle at the *start* of the reference period, but the interest differential is naturally an *end-of-period* amount. You must divide by  $(1+L\tau)$ . Omitting this



over-states the payoff.

6. **Mixing compounding conventions.** A forward computed under annual compounding differs numerically from one under continuous or semi-annual compounding. State the convention before quoting a number.
7. **Bootstrapping with non-par bonds carelessly.** The recursion works with any coupon bonds, but coupons must be discounted at *previously solved* spot rates — not at the bond's own YTM. Using YTM to strip near coupons defeats the entire purpose.

## 9. Recap

- A bond is a portfolio of single cash flows; each should be discounted at its own **spot (zero) rate**, not at a single YTM. YTM is a cash-flow-weighted average of spots.
- **Forward rates** are locked-in future rates *implied* by the spot curve through the no-arbitrage chaining identity  $(1+z_n)^n = \prod (1+f_{k,k+1})$ . The  $n$ -year spot is the geometric average of the one-year forwards.
- **Bootstrapping** extracts the spot curve from coupon-bond (or swap) prices, solving one maturity at a time from the short end out. We built  $z_1=5\%, z_2=5.5\%, z_3=6\%$  from three coupon bonds and reconstituted the prices exactly.
- Those spots imply forwards  $f_{1,2}=6.00\%, f_{2,3}=7.01\%$ , which chain back to  $(1.06)^3$  — verifying the identity. Rising spot curve  $\Rightarrow$  forwards above spots.
- A **FRA** is the tradeable embodiment of a forward rate: its fixed rate is the implied forward, and it cash-settles the discounted interest differential. Our 6×12 FRA locked 4.902% and neutralized a rate rise to 5.50% down to the penny.

## 10. Quick Reference — Interview Points

| Concept                | Formula                                                       |
|------------------------|---------------------------------------------------------------|
| Spot pricing           | $P = \sum_t C_t / (1+z_t)^t$                                  |
| Discount factor        | $DF_t = 1 / (1+z_t)^t$                                        |
| One-period forward     | $1+f_{k,k+1} = (1+z_{k+1})^{k+1} / (1+z_k)^k$                 |
| Multi-period forward   | $(1+f_{k,m})^{m-k} = (1+z_m)^m / (1+z_k)^k$                   |
| Chaining identity      | $(1+z_n)^n = \prod_{k=0}^{n-1} (1+f_{k,k+1})$                 |
| Continuous forward     | $f_{k,m} = (z_m m - z_k k) / (m-k)$                           |
| Bootstrap step $n$     | $(1+z_n)^n = (C + \text{Face}) / (P_n - \sum_{t<n} C_t DF_t)$ |
| FRA fair rate (simple) | $(1+z_s \tau_1)(1+K \tau_{\text{fwd}}) = (1+z_l \tau_2)$      |
| FRA payoff to long     | $N \tau (L-K) / (1+L \tau)$                                   |

### Rapid-fire talking points:

- "Why not just use YTM?" Because YTM discounts all cash flows at one rate, ignoring that the term structure prices each maturity differently. Spot rates are coupon-independent; YTM is bond-specific.



- "Is a forward rate a forecast?" No — it's a no-arbitrage break-even implied by today's spot curve. It equals the expected future spot only under pure expectations with zero term premium.
- "Rising curve — where are forwards?" Above the spots, because the spot is the geometric average of a rising forward sequence.
- "How do you build a zero curve?" Bootstrap: solve the 1-year spot from the 1-year instrument, then strip each successive coupon bond using already-known spots to isolate the next spot.
- "How is an FRA priced?" Its fixed rate is the implied forward from the spot curve; settlement is the interest differential discounted from period-end to the settlement date.
- "Spot vs. forward vs. par rate?" Spot discounts a single zero cash flow; forward is the future rate implied by two spots; par is the coupon that prices a bond to 100 (equals the swap rate). All three are transforms of one curve.
- **The one identity to never forget:** the total growth from 0 to  $n$  is path-independent under no arbitrage — hold the long spot, or roll the forwards; the terminal wealth is identical.

# Q&A — Spot and Forward Rates

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A companion practice bank for Chapter 05. Every question is followed by a full answer. Unless stated otherwise, rates are **annually compounded** and the spot curve from the chapter is reused:  $z_1 = 5.00\%$ ,  $z_2 = 5.50\%$ ,  $z_3 = 6.00\%$ , with growth factors  $1.05$ ,  $(1.055)^2 = 1.113025$ ,  $(1.06)^3 = 1.191016$ .

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## Section A — Concept Check

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**A1. What is a spot (zero) rate, and why is it the "atom" of bond pricing?** A spot rate  $z_t$  is the yield on a pure discount (zero-coupon) bond that pays a single cash flow at time  $t$ . It is the correct rate to discount one cash flow arriving at  $t$ . It is the atom because any coupon bond is just a portfolio of single dated cash flows; by the law of one price, the bond must cost the sum of its cash flows each discounted at its own spot rate. Spot rates belong to the market and are coupon-independent, whereas YTM is a single blended rate belonging to one specific bond.

**A2. How does YTM relate to the spot curve?** YTM is a single internal rate of return that reproduces a bond's price when applied to every cash flow. It is a cash-flow-weighted geometric average of the spot rates the bond touches. Two bonds of the same maturity but different coupons sample the spot curve with different weights, so they have different YTM's even in the same market. Both the spot equation and the YTM equation give the same price  $P$ ; only the spot equation reflects the true term structure.

**A3. Is a forward rate a forecast of future interest rates?** No. A forward rate is a no-arbitrage break-even implied by today's spot curve. It is the rate that makes an investor indifferent between (A) locking the long spot today and (B) investing short and rolling over at rates locked in today. It equals the market's expected future spot only under the pure expectations hypothesis with zero term premium. The gap between your own forecast and the implied forward is where a trade lives.

**A4. State the master no-arbitrage identity in words and symbols.** Total growth from time 0 to  $n$  is path-independent:  $(1+z_n)^n = (1+z_1)(1+f_{1,2})(1+f_{2,3})\cdots(1+f_{n-1,n})$ . Equivalently, the  $n$ -year spot rate is the geometric average of the sequence of one-year forward rates.

**A5. When the spot curve is upward-sloping, are forwards above or below spots? Why?** Above. The  $n$ -year spot is the geometric average of the one-year forwards; the average of a rising sequence always lags its latest term, so the marginal (forward) rate must sit above the running average (spot). Chapter example: forwards  $5\% \rightarrow 6\% \rightarrow 7.01\%$  give a 3-year spot of  $6.00\%$ . An inverted curve reverses this — forwards lie below spots.

**A6. What is bootstrapping, and why must near coupons be stripped at previously-solved spot rates rather than at the bond's own YTM?** Bootstrapping is the sequential extraction of the zero curve from coupon-bond (or swap) prices, shortest maturity first. Each new bond introduces exactly one new unknown spot rate; the earlier coupons are discounted at the already-solved spots, isolating that last unknown. Using the bond's own YTM to strip near coupons would re-inject the single-rate assumption you are trying to eliminate, defeating the purpose.

**A7. Give the discount-factor form of a forward growth factor.** The forward growth factor is the ratio of the near discount factor to the far discount factor:  $(1+f_{\{k,m\}})^{m-k} = DF_k / DF_m$ , where  $DF_t = 1/(1+z_t)^t$ . "Near over far" is the cleanest mnemonic.

**A8. What is an FRA, and what is its fair fixed rate?** A Forward Rate Agreement is an OTC contract that locks an interest rate for a future borrowing/lending period on a notional principal. No principal changes hands; only the interest differential is cash-settled. Its fair fixed rate  $K$  is the implied forward rate from the spot curve. The long (would-be borrower) gains when the reference rate sets above  $K$ .

**A9. Why is an FRA settlement discounted by  $(1 + L\tau)$ ?** An FRA settles at the *start* of the reference period, but the interest differential  $(L-K)\tau$  is naturally an *end*-of-period amount. Dividing by  $(1+L\tau)$  discounts that end-of-period cash flow back to the settlement date at the realized rate  $L$ . Omitting the discount overstates the payoff.

**A10. Under continuous compounding, how does the forward simplify?** The chaining identity becomes additive:  $z_n \cdot n = z_1 \cdot 1 + f_{1,2} \cdot 1 + \dots$ , so  $f_{[k,m]} = (z_m \cdot m - z_k \cdot k) / (m - k)$  — a simple weighted difference of spot rates.

## Section B — Numerical Bond-Math Problems (step-by-step, reconciling)

**B1. Bootstrap a 2-year spot curve.** Two annual-coupon bonds, face 100: Bond X (1-yr, 5% coupon, price 100.9615) and Bond Y (2-yr, 5% coupon, price 100.9612). Find  $z_1$  and  $z_2$ .

Step 1 —  $z_1$  from Bond X (pays 105 at  $t=1$ ):  $105 / 100.9615 = 1.04000 \Rightarrow z_1 = 4.00\%$ .

Step 2 —  $z_2$  from Bond Y (pays 5 at  $t=1$ , 105 at  $t=2$ ). Strip the year-1 coupon at the known  $z_1$ :  $5 / 1.04 = 4.80769$ .  $100.9612 = 4.80769 + 105 / (1+z_2)^2 \Rightarrow 105 / (1+z_2)^2 = 96.15351$ .  $(1+z_2)^2 = 105 / 96.15351 = 1.092025 \Rightarrow 1+z_2 = \sqrt{1.092025} = 1.04500 \Rightarrow z_2 = 4.50\%$ .

Reconcile — re-price Bond Y with the two spots:  $5/1.04 + 105/1.092025 = 4.80769 + 96.15346 = 100.9611 \approx 100.9612$  ✓ (rounding in the last digit).

**B2. From B1's curve, find the implied one-year forward  $f_{1,2}$  and verify chaining.**  $1+f_{1,2} = (1+z_2)^2 / (1+z_1) = 1.092025 / 1.04 = 1.050024 \Rightarrow f_{1,2} = 5.00\%$ . Chaining check:  $(1+z_1)(1+f_{1,2}) = 1.04 \times 1.050024 = 1.092025 = (1+z_2)^2$  ✓. The rising curve (4% → 4.5%) produces a forward (5.00%) above both spots, as expected.

**B3. Full forward curve from a 4-year spot curve.** Given  $z_1=3\%$ ,  $z_2=3.5\%$ ,  $z_3=4\%$ ,  $z_4=4.5\%$  (annual). Compute the three consecutive one-year forwards and confirm they chain to  $(1+z_4)^4$ .

Growth factors:  $1.03$ ;  $(1.035)^2=1.071225$ ;  $(1.04)^3=1.124864$ ;  $(1.045)^4=1.192519$ .

$f_{1,2} = 1.071225/1.03 - 1 = 1.040024 - 1 = 4.00\%$ .  $f_{2,3} = 1.124864/1.071225 - 1 = 1.050072 - 1 = 5.01\%$ .  $f_{3,4} = 1.192519/1.124864 - 1 = 1.060144 - 1 = 6.01\%$ .

Chaining check:  $1.03 \times 1.040024 \times 1.050072 \times 1.060144 = 1.071225 \times 1.050072 = 1.124864$ ;  $\times 1.060144 = 1.192519 = (1+z_4)^4$  ✓. Geometric-average check:  $1.192519^{1/4} = 1.045 \Rightarrow z_4 = 4.50\%$  = geometric average of the four one-year rates (3%, 4.00%, 5.01%, 6.01%). The average lags the last forward because the curve rises.

**B4. Two-year forward, one year forward ( $f_{1,3}$ ) on the chapter curve.** Use  $z_1=5\%$ ,  $z_3=6\%$ .  $(1+f_{1,3})^2 = (1+z_3)^3 / (1+z_1) = 1.191016 / 1.05 = 1.134301$ .  $1+f_{1,3} = \sqrt{1.134301} = 1.065036 \Rightarrow f_{1,3} = 6.50\%$ . Cross-check via one-year forwards:  $(1+f_{1,2})(1+f_{2,3}) = 1.060024 \times 1.070073 = 1.134301$ , and  $\sqrt{1.134301} = 1.065036$  ✓. The 2-year forward is the geometric average of the two embedded one-year forwards (6.00% and 7.01%), giving 6.50%.

**B5. Price a 3-year 4% annual-coupon bond off the chapter spot curve, then find its YTM sign relative to spots.** Cash flows: 4, 4, 104.  $PV = 4/1.05 + 4/1.113025 + 104/1.191016 = 3.80952 + 3.59380 + 87.31989 =$

**94.7232.** Because the bond trades below par ( $94.72 < 100$ ) with a below-market coupon, its YTM exceeds its coupon and lands between  $z_1$  and  $z_3$ , weighted toward  $z_3$  (the largest cash flow, 104, is at  $t=3$ ). Quick sanity: discounting all flows at a flat 5.94% gives  $\approx 4/1.0594 + 4/1.0594^2 + 104/1.0594^3 \approx 3.776 + 3.564 + 87.38 \approx 94.72$ , so  $\text{YTM} \approx 5.94\%$ , just below  $z_3 = 6.00\%$  — consistent, since a coupon bond's YTM is a cash-flow-weighted average of spots dominated by the final payment. ✓

**B6. Price a 6×12 FRA and settle it.** Notional \\$10,000,000. Money-market spots (simple, Act with  $\tau=0.5$  each period): 6-month = 4.00%, 12-month = 4.50%. Fair rate:  $(1 + 0.04 \times 0.5)(1 + K \times 0.5) = (1 + 0.045 \times 1)$ .  $1.02(1 + 0.5K) = 1.045 \Rightarrow 1 + 0.5K = 1.045/1.02 = 1.024510 \Rightarrow K = 4.902\%$ . Settlement at  $L = 5.50\%$ : Payoff to long =  $N \cdot (L - K) \tau / (1 + L \tau) = 10,000,000 \times (0.055 - 0.04902)(0.5) / (1 + 0.055 \times 0.5) = 10,000,000 \times (0.00598 \times 0.5) / 1.0275 = 29,900 / 1.0275 = \text{\$29,101}$  received at the 6-month settlement. Reconcile the hedge: the extra borrowing cost of paying 5.50% instead of 4.902% over the 6-month loan, measured at period-end (month 12), is  $10,000,000 \times 0.00598 \times 0.5 = \text{\$29,900}$ . Carry the FRA receipt forward at realized 5.50%:  $29,101 \times 1.0275 = \text{\$29,901} \approx \text{\$29,900}$  ✓. The effective all-in borrowing rate stays 4.902%.

**B7. Price and settle a 3×9 FRA (new numbers).** Notional \\$5,000,000. Simple money-market spots: 3-month = 3.00% ( $\tau_1=0.25$ ), 9-month = 3.60% ( $\tau_2=0.75$ ); reference period  $\tau_{\text{fwd}} = 0.50$ . Fair rate:  $(1 + 0.03 \times 0.25)(1 + K \times 0.5) = (1 + 0.036 \times 0.75)$ .  $1.0075(1 + 0.5K) = 1.0270 \Rightarrow 1 + 0.5K = 1.0270/1.0075 = 1.019355 \Rightarrow K = 3.871\%$ . Settlement at  $L = 4.50\%$ : Payoff to long =  $5,000,000 \times (0.045 - 0.03871)(0.5) / (1 + 0.045 \times 0.5) = 5,000,000 \times (0.00629 \times 0.5) / 1.0225 = 15,725 / 1.0225 = \text{\$15,379}$  at the 3-month settlement. Direction check: rates rose above the locked  $K$ , and the long (borrower) receives cash — correct sign. ✓

**B8. Continuous-compounding forward.** Given continuously-compounded spots  $z_2 = 5.50\%$  and  $z_3 = 6.00\%$ , find the one-year forward  $f_{2,3}$ .  $f_{2,3} = (z_3 \cdot 3 - z_2 \cdot 2) / (3 - 2) = (0.18 - 0.11) / 1 = 0.07 = 7.00\%$ . Note this matches the annually-compounded answer (7.01%) only approximately — the tiny gap is the compounding convention, illustrating Common Confusion #6: always state the convention.

## Section C — Interview-Style (with model answers)

**C1. "Walk me through why we bother with a whole spot curve instead of one YTM."** Model answer: A single YTM discounts a coupon arriving in six months at the same rate as principal arriving in ten years, which misprices the term structure — money for different horizons genuinely commands different rates. The market prices each maturity separately, so the right primitive is a schedule of spot (zero) rates. A coupon bond is a portfolio of single cash flows, and by the law of one price it must equal the sum of those flows discounted at their own spot rates; the STRIPS program makes this literally tradeable. YTM is still useful as a summary, but it is an output — a cash-flow-weighted average of spots that is bond-specific, not a market primitive.

**C2. "A client insists the 1-year rate one year forward is the market's forecast for next year's rate."**

**Correct them.** Model answer: The forward is a no-arbitrage break-even, not a forecast. If  $z_1 = 5\%$  and  $z_2 = 5.5\%$ , then  $f_{1,2} = (1.055^2/1.05) - 1 = 6.00\%$  purely from arithmetic — it is the rate that equalizes investing two years at spot versus rolling one year then reinvesting. It equals the expected future spot only under the pure expectations hypothesis with no term premium. In reality, liquidity-preference and term-premium effects mean forwards typically exceed expected future spots. So a 6% forward is consistent with the market expecting, say, 5.5% next year plus a 0.5% term premium. The divergence between a client's genuine forecast and the implied forward is exactly where a directional rates trade is expressed.

**C3. "How would you build a zero curve from the instruments you can actually trade?"** Model answer: Bootstrapping. Start at the short end: the 1-year instrument gives  $z_1$  directly. Then take the 2-year coupon bond, discount its year-1 coupon at the already-known  $z_1$ , subtract that from the price, and the residual is the single discounted final cash flow — solve for  $z_2$ . March outward; each successive bond adds exactly one new unknown spot rate. The critical discipline is to strip earlier coupons at previously-solved *spot* rates, never at the bond's own YTM, otherwise you smuggle the single-rate error back in. In modern practice the swap curve, not government bonds, is what gets bootstrapped, because swap par rates give a clean, liquid, evenly-spaced set of tenors.

**C4. "Explain how an FRA is priced and how it hedges a future funding cost."** Model answer: An FRA's fixed rate is just the implied forward from the money-market spot curve, enforced by cash-and-carry replication — a dealer quoting off-market gets arbitrated. Suppose a treasurer must borrow \$10m for six months, six months from now. From 6-month and 12-month spots the fair 6×12 rate solves to about 4.902%. Going long the FRA locks that rate. If the reference rate later sets at 5.50%, the firm borrows more expensively in cash markets, but the FRA pays the discounted differential — about \$29,101 at settlement, which carried forward at the realized rate exactly offsets the extra \$29,900 of interest at period-end. The effective all-in rate is pinned at 4.902% regardless of where rates set. Key subtlety: settlement is at the *start* of the reference period, so the differential is discounted by  $(1+L\tau)$ .

**C5. "The spot curve is steeply upward-sloping. What does that tell you about forwards, and what does it not tell you?"** Model answer: Mechanically, an upward-sloping spot curve forces forwards to lie above spots — the spot is the geometric average of the one-year forwards, and an average must trail a rising marginal series. So each successive one-year forward is higher than the last. What it does *not* cleanly tell you is that the market expects rates to rise by that much: forwards embed a term premium. A steep curve is consistent with flat expected future short rates plus a positive term premium. So I would decompose the forward into an expectations component and a premium before drawing any macro conclusion.

**C6. "What's the single identity you'd never want a fixed-income analyst to forget?"** Model answer: Path-independence of growth under no arbitrage:  $(1+z_n)^n = \prod (1+f_{\{k,k+1\}})$ . Whether you hold the long spot to maturity or roll a sequence of forwards you locked in today, terminal wealth is identical — otherwise there is a riskless arbitrage. Everything else in the chapter — spot-to-forward conversion, the geometric-average intuition, bootstrapping, FRA pricing — is a rearrangement of that one statement.

## Section D — Multiple Choice (with reasoning)

**D1.** With  $z_1 = 5\%$ ,  $z_2 = 5.5\%$ , the one-year forward  $f_{1,2}$  is closest to: (a) 5.0% (b) 5.25% (c) 6.0% (d) 5.5%

**Answer: (c).**  $f_{1,2} = (1.055^2/1.05) - 1 = (1.113025/1.05) - 1 = 6.00\%$ . Since the curve rises, the forward exceeds both spots, ruling out (a), (b), (d).

**D2.** Which statement about YTM is TRUE? (a) It is coupon-independent. (b) It equals the average spot rate weighted equally. (c) It is a cash-flow-weighted average of spot rates and is bond-specific. (d) It always exceeds every spot rate the bond touches. **Answer: (c).** YTM blends the spots the bond's cash flows touch, weighted by timing and size, so it differs across bonds of equal maturity but different coupons — hence bond-specific, not coupon-independent, eliminating (a). Weights are not equal (b), and YTM lies within the range of relevant spots, not above all of them (d).

**D3.** In bootstrapping the 2-year spot, the year-1 coupon should be discounted at: (a) the 2-year bond's YTM (b) the already-solved 1-year spot  $z_1$  (c) the 2-year spot  $z_2$  (d) the coupon rate **Answer: (b).** The whole point

is to isolate  $z_2$ ; earlier coupons must be stripped at previously-solved spot rates. Using YTM (a) reintroduces the single-rate assumption;  $z_2$  (c) is the unknown you are solving for.

**D4.** A  $6 \times 12$  FRA covers a reference period that: (a) starts today, ends in 6 months (b) starts in 6 months, ends in 12 months (c) starts in 12 months, lasts 6 months (d) starts in 6 months, lasts 12 months **Answer: (b).** "a **$\times$** b" means the period starts at month a and ends at month b, so  $6 \times 12$  is the 6-month period beginning in 6 months.

**D5.** The FRA settlement amount is divided by  $(1+L\tau)$  because: (a) it converts to continuous compounding (b) the differential is an end-of-period amount settled at period start, so it must be discounted (c) it adjusts for credit risk (d) it annualizes the payoff **Answer: (b).** FRAs settle at the start of the reference period while the interest differential accrues to period-end; discounting at the realized rate  $L$  brings it back to the settlement date.

**D6.** Under continuous compounding, the forward  $f_{\{k,m\}}$  equals: (a)  $(z_{m \cdot m} - z_{k \cdot k})/(m-k)$  (b)  $(z_m - z_k)/(m-k)$  (c)  $z_{m \cdot m} - z_{k \cdot k}$  (d)  $\sqrt{(z_m \cdot z_k)}$  **Answer: (a).** Continuous compounding makes the chaining identity additive in rate $\times$ time, so the forward is the weighted difference  $(z_{m \cdot m} - z_{k \cdot k})/(m-k)$ .

**D7.** When the spot curve is inverted (downward-sloping), forward rates: (a) lie above spot rates (b) equal spot rates (c) lie below spot rates (d) are undefined **Answer: (c).** The spot is the geometric average of the forwards; if the average falls with maturity, the marginal forward must be below the running average — forwards lie below spots. Mirror image of the rising-curve case.

**D8.** Given  $z_1=5\%$ ,  $z_2=5.5\%$ ,  $z_3=6\%$ , the 3-year spot is BEST described as: (a) the arithmetic mean of the three spots (b) the geometric average of the one-year forwards 5%, 6%, 7.01% (c) equal to the 3-year forward (d) the highest forward rate **Answer: (b).**  $(1+z_3)^3 = 1.05 \times 1.060024 \times 1.070073 = 1.191016$ , so  $z_3 = 6.00\%$  is the geometric average of the one-year forwards. It is below the last forward (7.01%), ruling out (d).

**D9.** The forward growth factor between  $k$  and  $m$  in discount-factor terms is: (a)  $DF_m / DF_k$  (b)  $DF_k / DF_m$  (c)  $DF_k \times DF_m$  (d)  $DF_k - DF_m$  **Answer: (b).**  $(1+f_{\{k,m\}})^{m-k} = DF_k/DF_m$  — "near over far."  $DF_k$  (nearer, larger) divided by  $DF_m$  (farther, smaller) exceeds 1, as a growth factor must.

**D10.** A long FRA position profits when: (a) the reference rate sets below  $K$  (b) the reference rate sets above  $K$  (c) the spot curve inverts (d) the notional is repaid **Answer: (b).** The long is a would-be borrower who locked  $K$ ; if the reference rate sets above  $K$ , borrowing got more expensive and the FRA pays the long to compensate.

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*End of Q&A — Spot and Forward Rates. All numerical answers were self-verified against the chapter's growth factors (1.05, 1.113025, 1.191016) and the no-arbitrage chaining identity.*

# Chapter 06 — The Term Structure of Interest Rates

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## 1. The Problem / Need

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Suppose someone asks you a deceptively simple question: "What is *the* interest rate?" The honest answer is that there is no single interest rate. The rate you can lock in to lend money for three months is almost never the same as the rate for ten years. A one-year Treasury bill might yield 3.0% while a ten-year Treasury note yields 4.5% and a thirty-year bond yields 4.8% — all issued by the same borrower (the government), all carrying the same negligible default risk, all quoted at the same instant. The only thing that differs is **time to maturity**.

This creates a genuine problem for anyone who works with money. If you are valuing a bond that pays coupons in years 1, 2, 3, 4, and a large redemption in year 5, which rate do you discount each cash flow at? Discounting all of them at a single yield-to-maturity is a convenient shortcut, but it is technically wrong: the year-1 coupon should be discounted at the one-year rate, the year-5 payment at the five-year rate. If you use one blended rate you can misprice the bond, misjudge relative value between two bonds, and completely miss the signal that the market is sending about the future.

Because that is the second, deeper reason the term structure matters: **the shape of the relationship between maturity and yield is one of the most-watched forecasting tools in all of finance**. An inverted curve — short rates above long rates — has preceded almost every US recession of the last sixty years. Central banks, credit committees, and pension funds all read the curve. So we need a rigorous framework: what the curve *is*, how to extract clean rates from it, what theories explain its shape, and what it is telling us.

## 2. The Core Idea

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The **term structure of interest rates** is the relationship between the yield on a debt instrument and its time to maturity, holding credit quality, liquidity, and all other features constant. Plotted with maturity on the horizontal axis and yield on the vertical axis, it produces the **yield curve**.

To do it properly we must be precise about *which* yield. Three related curves describe the same underlying market:

- **Spot rate curve (zero curve):** the yield on a zero-coupon bond of each maturity — i.e., the pure rate for a single cash flow received at time  $t$ , with no reinvestment along the way. This is the theoretically correct set of discount rates.
- **Par yield curve:** the coupon rate at which a newly issued coupon bond of each maturity would trade exactly at par (price = face value). This is what "the 10-year yield" usually refers to in market commentary.
- **Forward rate curve:** the set of rates for borrowing/lending that *begin in the future* and are locked in *today*, implied arithmetically by the spot curve.

The master insight of this chapter is that these three are not three independent facts — they are three views of the **same** no-arbitrage information. Given any one, you can derive the other two. The spot curve is the natural foundation, so everything is built on it.



### 3. Why / How It Works

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#### Why one rate cannot do the job

Money has a time value that compounds, and the market's expectation of that value is not constant across horizons. If investors expect short-term rates to rise over the next few years — because growth is strong or inflation is building — then to persuade someone to lend for a long term, the long rate must roughly average the sequence of expected short rates. That alone would make the curve slope upward in a tightening world and downward in an easing world. Layer on compensation for risk (you tie your money up longer, you demand a premium) and for supply-and-demand imbalances at particular maturities, and you get a curve whose shape carries real information.

#### Why spot rates are the honest discount rates

A five-year 5% coupon bond is really a *portfolio* of six zero-coupon bonds: a small zero maturing each year for the coupons, plus a large zero for the redemption. No-arbitrage says the bond must be worth the sum of those zeros valued at their own maturity's spot rate. If it were not, an arbitrageur would strip the bond into its cash flows (or reconstitute it) and pocket the difference. This is exactly what the US Treasury STRIPS market does. So spot rates are not an academic construct — they are enforced by traders.

#### How we recover spot rates: bootstrapping

We rarely observe a full set of zero-coupon bonds directly. Instead we observe coupon bond prices and **bootstrap** the spot curve out of them, one maturity at a time:

1. The 1-year rate comes straight from a 1-year instrument (a bill or a bond with one remaining payment).
2. For the 2-year bond, we already know how to discount its year-1 coupon (using the year-1 spot). Everything left over must be explained by the year-2 spot — so we solve for it.
3. Repeat maturity by maturity, each step using all the previously solved spot rates.

#### How forward rates fall out

Once we have spot rates, forward rates are pure arithmetic from a no-arbitrage argument. Investing for two years at the two-year spot must give the same terminal wealth as investing one year at the one-year spot and then reinvesting for the second year at the rate you can *lock in today* for that future year. That lock-in rate is the forward rate. If the two paths gave different terminal wealth, you could borrow on the cheap path and lend on the dear path for a riskless profit.

### 4. Full Content — Formulas and Bond Math

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#### 4.1 Notation

- $s_t$  = the annually-compounded spot (zero) rate for maturity  $t$  years.
- $P_t$  = price today of a zero-coupon bond paying 1 (or 100) at time  $t$ .
- $f(a,b)$  = the forward rate agreed today for the period from year  $a$  to year  $b$ .
- $y$  = yield to maturity (a single internal rate of return for a coupon bond).

#### 4.2 Discount factors and spot rates

The price today of receiving 100 at time  $t$  is:



$$P_t = \frac{100}{(1+s_t)^t}$$

The **discount factor** is  $DF_t = \frac{1}{(1+s_t)^t}$ . A coupon bond paying coupon  $C$  each year and redeeming 100 at year  $n$  is priced as:

$$\text{Price} = \sum_{t=1}^n \frac{C}{(1+s_t)^t} + \frac{100}{(1+s_n)^n}$$

Note the *different* spot rate at each maturity — this is what makes it correct.

### 4.3 The bootstrapping recursion

Given the price of an  $n$ -year coupon bond and all spot rates up to  $n-1$ , solve for  $s_n$ :

$$s_n = \left( \frac{C + 100}{\text{Price} - \sum_{t=1}^{n-1} \frac{C}{(1+s_t)^t}} \right)^{1/n} - 1$$

### 4.4 Forward rates

The one-year forward rate starting in year  $a$  (i.e., covering year  $a$  to  $a+1$ ), denoted  $f(a, a+1)$ :

$$1 + f(a, a+1) = \frac{(1+s_{a+1})^{a+1}}{(1+s_a)^a}$$

The general multi-period forward from year  $a$  to year  $b$  (with  $b > a$ ):

$$(1 + f(a,b))^{b-a} = \frac{(1+s_b)^b}{(1+s_a)^a}$$

Equivalently, the long spot rate is the geometric average of one-year forwards:

$$(1+s_n)^n = (1+s_1)(1+f(1,2))(1+f(2,3))\cdots(1+f(n-1,n))$$

### 4.5 Par yields

The  $n$ -year par yield  $c_n$  is the coupon that makes price equal 100:

$$c_n = \frac{100(1 - DF_n)}{\sum_{t=1}^n DF_t} = \frac{100(1 - \frac{1}{(1+s_n)^n})}{\sum_{t=1}^n \frac{1}{(1+s_t)^t}}$$

### 4.6 The ordering rule (upward-sloping curve)

When the spot curve slopes upward, a clean ranking emerges at each maturity:

$$\text{par yield} < \text{spot rate} < \text{forward rate}$$

Intuition: the spot rate is an average of forwards, so it lags the rising forwards; the par yield is a coupon-weighted blend that puts weight on cheaper near-dated rates, so it sits below the spot. (For a downward-sloping curve the inequalities reverse.)

## 4.7 The four canonical shapes

| Shape                      | Description                                     | Typical meaning                                                                            |
|----------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Normal (upward)</b>     | Long yields > short yields, gently rising       | Healthy expansion; market expects modest rate rises and/or term premium                    |
| <b>Flat</b>                | Yields roughly equal across maturities          | Transition point; uncertainty about direction; often precedes inversion                    |
| <b>Inverted (downward)</b> | Short yields > long yields                      | Market expects rate <i>cuts</i> ahead — usually because it expects a slowdown or recession |
| <b>Humped</b>              | Rises then falls (peak in the belly, e.g. 2–5y) | Mixed signals; near-term tightening expected to reverse later                              |

## 4.8 The theories of the term structure

Four classic theories explain *why* the curve takes the shape it does. They are best understood as increasingly realistic refinements.

**(a) Pure (Unbiased) Expectations Theory.** The long rate is purely the geometric average of expected future short rates. Forward rates are *unbiased* forecasts of future spot rates:  $f(a, a+1) = E[s_1 \text{ in year } a]$ . Implication: an upward curve means the market expects rates to rise; an inverted curve means it expects them to fall. Investors are treated as risk-neutral and indifferent between holding one long bond or rolling short bonds.

**(b) Liquidity Preference (Liquidity Premium) Theory.** Investors are *not* indifferent: longer bonds carry more price risk (higher duration, more sensitive to rate moves), so lenders demand a **term/liquidity premium** to hold them. Forwards therefore *overstate* expected future spots by that premium:

$$f(a, a+1) = E[s_1 \text{ in year } a] + \text{term premium}_a$$

with the premium generally rising with maturity. Implication: the curve can slope upward *even when* rates are expected to stay flat, simply because of the premium. This is why a mildly upward curve is the "normal" resting state.

**(c) Market Segmentation Theory.** Bonds of different maturities are traded in *separate* markets by different clientele — banks at the short end, pension funds and insurers at the long end — and these players do not move across maturities. Yields at each maturity are set by local supply and demand, essentially independently. Implication: the curve's shape reflects who is buying and selling at each point, not a single unifying expectation. It explains kinks and local distortions that expectations theory cannot.

**(d) Preferred Habitat Theory.** A hybrid and the most realistic. Investors *have* preferred maturity ranges (habitats) as in segmentation, but they *will* leave their habitat if paid a large enough yield premium. So expectations and premiums both matter, and premiums need not rise smoothly with maturity — they depend on supply/demand imbalances in each habitat. Implication: forwards embed both expected rates *and* a habitat-specific premium that can be positive or negative.

## 4.9 What drives shifts and reshapes

- **Monetary policy (the short end):** the central bank sets the very short rate; hikes push the front of the curve up, cuts push it down.

- **Inflation expectations (the long end):** long yields embed expected inflation plus a real rate plus an inflation-risk premium.
- **Growth expectations:** strong growth lifts long real rates; expected slowdowns pull them down.
- **Term premium:** compensation for duration risk, which itself moves with volatility and uncertainty.
- **Supply and demand:** heavy government issuance at a maturity raises its yield; flight-to-safety demand for long bonds lowers it; regulatory demand (insurers/pensions) anchors the long end.

Analysts summarize curve moves in three factors: **level** (parallel shift, all yields move together), **slope** (steepening/flattening, short vs long diverge), and **curvature** (the belly moves relative to the wings — a humped/butterfly move).

## 5. Worked Examples

### Example 1 — Bootstrapping the spot curve

We observe three annual-coupon government bonds (face 100, annual coupons):

| Bond | Maturity | Coupon    | Market price |
|------|----------|-----------|--------------|
| A    | 1 year   | 0% (zero) | 97.09        |
| B    | 2 years  | 4%        | 100.97       |
| C    | 3 years  | 5%        | 102.87       |

**Step 1 — one-year spot.** Bond A pays 100 in one year:

$$97.09 = \frac{100}{1+s_1} \Rightarrow 1+s_1 = \frac{100}{97.09} = 1.0300 \Rightarrow \boxed{s_1 = 3.00\%}$$

**Step 2 — two-year spot.** Bond B pays 4 at t=1 and 104 at t=2:

$$100.97 = \frac{4}{1.03} + \frac{104}{(1+s_2)^2}$$

$$\frac{4}{1.03} = 3.8835 \Rightarrow \frac{104}{(1+s_2)^2} = 100.97 - 3.8835 = 97.0865$$

$$(1+s_2)^2 = \frac{104}{97.0865} = 1.07123 \Rightarrow 1+s_2 = 1.03500 \Rightarrow \boxed{s_2 = 3.50\%}$$

**Step 3 — three-year spot.** Bond C pays 5, 5, 105:

$$102.87 = \frac{5}{1.03} + \frac{5}{1.035^2} + \frac{105}{(1+s_3)^3}$$

$$\frac{5}{1.03} = 4.8544, \quad \frac{5}{1.07123} = 4.6675$$

$$\frac{105}{(1+s_3)^3} = 102.87 - 4.8544 - 4.6675 = 93.3481$$

$$(1+s_3)^3 = \frac{105}{93.3481} = 1.12484 \Rightarrow 1+s_3 = 1.12484^{1/3} = 1.04000$$

$$\Rightarrow \boxed{s_3 = 4.00\%}$$

**Result:** an upward-sloping spot curve of **3.00%, 3.50%, 4.00%**.

*Self-check — reprice Bond C from the spot curve:*  $4.8544 + 4.6675 + \frac{105}{1.12484} = 4.8544 + 4.6675 + 93.348 = 102.87.$  ✓ Reconciles exactly.

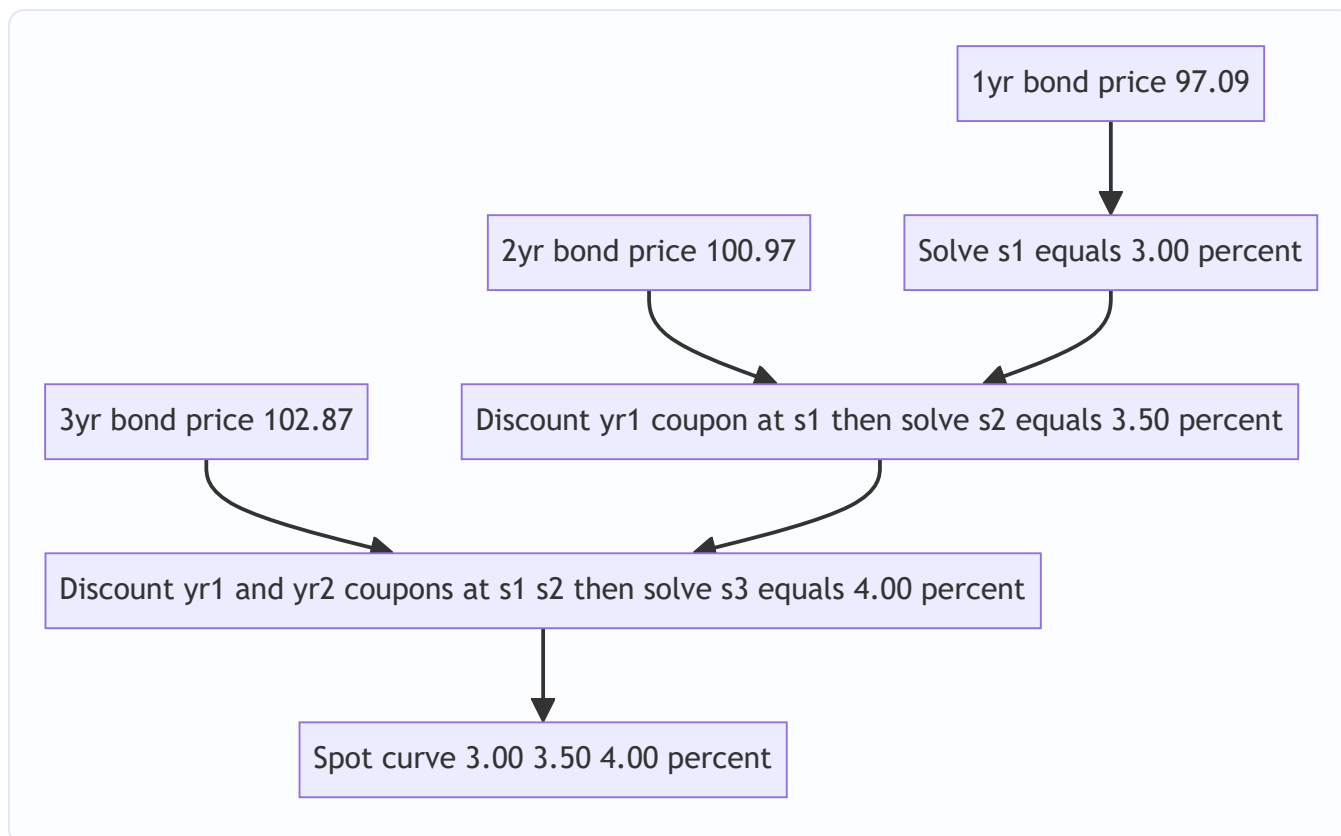


Figure 1 — Bootstrapping recursion: each maturity's spot rate is solved using all previously recovered rates.

## Example 2 — Implied forward rates

Using the spot curve above, compute the one-year forwards.

**Forward for year 1 to year 2,  $f(1,2)$ :**

$$1 + f(1,2) = \frac{(1 + s_2)^2}{(1 + s_1)^1} = \frac{1.07123}{1.03000} = 1.04003 \rightarrow \boxed{f(1,2) = 4.00\%}$$

**Forward for year 2 to year 3,  $f(2,3)$ :**

$$1 + f(2,3) = \frac{(1 + s_3)^3}{(1 + s_2)^2} = \frac{1.12484}{1.07123} = 1.05005 \rightarrow \boxed{f(2,3) = 5.00\%}$$

*Self-check — rebuild  $s_3$  from the forwards:*

$$(1 + s_1)(1 + f(1,2))(1 + f(2,3)) = 1.03 \times 1.04003 \times 1.05005 = 1.12484 = (1 + s_3)^3$$

Cube root gives  $1.04000 \rightarrow s_3 = 4.00\%$ . ✓ Reconciles.

Notice the pattern: spots **3.0, 3.5, 4.0**; forwards **3.0 ( $=s_1$ ), 4.0, 5.0**. The forwards rise faster and sit above the spots — exactly what section 4.6 predicts for an upward curve. Under pure expectations, this curve is telling us the market expects the one-year rate to be **4.0%** next year and **5.0%** the year after.

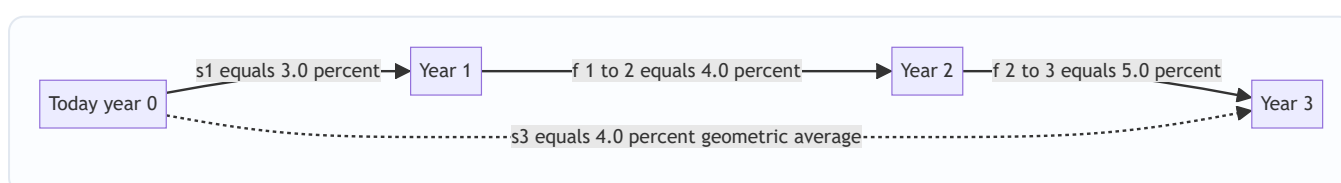


Figure 2 — The 3-year spot rate is the geometric average of the sequence of one-year forwards.

### Example 3 — Par yields and the ordering rule

Compute the 3-year par yield from the spot curve. First the discount factors:

| t | $s_t$      | $DF_t = 1/(1+s_t)^t$ |
|---|------------|----------------------|
| 1 | 3.00%      | 0.97087              |
| 2 | 3.50%      | 0.93352              |
| 3 | 4.00%      | 0.88900              |
|   | <b>Sum</b> | <b>2.79339</b>       |

$$c_3 = \frac{100(1 - DF_3)}{\sum DF_t} = \frac{100(1 - 0.88900)}{2.79339} = \frac{11.100}{2.79339} = \boxed{3.973\%}$$

*Self-check — price a 3-year 3.973% bond and confirm it equals par:*

$$3.973 \times 2.79339 + 100 \times 0.88900 = 11.099 + 88.900 = 99.999 \approx 100. \quad \checkmark$$

Now line up all three curves at the 3-year point:

| Measure          | 3-year value |
|------------------|--------------|
| Par yield        | 3.97%        |
| Spot rate        | 4.00%        |
| Forward $f(2,3)$ | 5.00%        |

This confirms **par < spot < forward** for an upward-sloping curve. If you had naively used the 3-year par yield (3.97%) to discount every cash flow of a 3-year bond, you would slightly misprice it versus the correct spot-by-spot valuation — small here, but material for longer maturities and steeper curves.

### Example 4 — Expectations vs liquidity premium

An investor has a 2-year horizon and two strategies, using the curve above:

- **Strategy A (buy-and-hold):** buy the 2-year zero; terminal wealth factor  $= (1+s_2)^2 = 1.07123$ .
- **Strategy B (roll short):** buy 1-year at 3.0%, then reinvest for year 2 at whatever the 1-year rate turns out to be, call it  $s^1$ ; terminal factor  $= 1.03 \times (1+s^1)$ .

Indifference requires  $1.03 \times (1+s^1) = 1.07123 \Rightarrow s^1 = 4.00\%$  — precisely the forward  $f(1,2)$ .

- Under **pure expectations**, the market's *expected* 1-year rate next year is exactly **4.0%**; the forward is an unbiased forecast.
- Under **liquidity preference**, suppose the true term premium built into the 2-year is **0.5%**. Then the *expected* future 1-year rate is only  $4.0\% - 0.5\% = \mathbf{3.5\%}$ ; the forward **overstates** the expected spot by the premium. The upward slope is then only *half* about expected rate rises and *half* about the premium investors demand for locking money up.

This is the single most important practical caveat about reading the curve: **forwards are not clean forecasts** once you accept that a term premium exists.

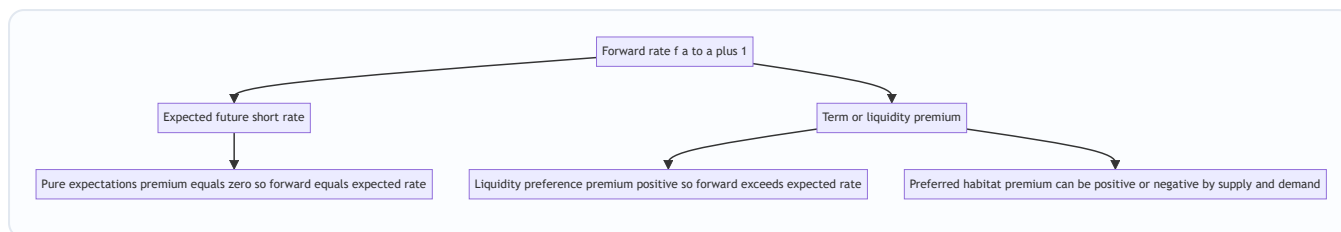


Figure 3 — Decomposing a forward rate into an expectations component and a premium component, and how each theory treats the premium.

### Example 5 — An inverted curve and what it signals

Suppose the curve flips. We now observe spot rates of  $s_1 = 5.0\%$ ,  $s_2 = 4.5\%$ ,  $s_3 = 4.2\%$  — short rates *above* long rates. Extract the forwards:

$$1 + f(1,2) = \frac{1.045^2}{1.05} = \frac{1.092025}{1.05} = 1.04002 \rightarrow f(1,2) = 4.00\%$$

$$1 + f(2,3) = \frac{1.042^3}{1.045^2} = \frac{1.131366}{1.092025} = 1.03603 \rightarrow f(2,3) = 3.60\%$$

The forwards **decline** ( $5.0\% \rightarrow 4.0\% \rightarrow 3.6\%$ ). Under pure expectations, the market is pricing the one-year rate *falling* from 5.0% today to ~4.0% next year to ~3.6% the year after. Rates fall when the central bank *cuts*, and the central bank cuts when it expects the economy to weaken. That is exactly why an inverted curve is the market's aggregated bet on a coming slowdown — and why the **10y–2y** and **10y–3m** spreads going negative are the two headline recession gauges. The signal is amplified in the real economy because banks borrow short and lend long: when short rates sit above long rates, lending margins compress, credit tightens, and the slowdown can become self-fulfilling.

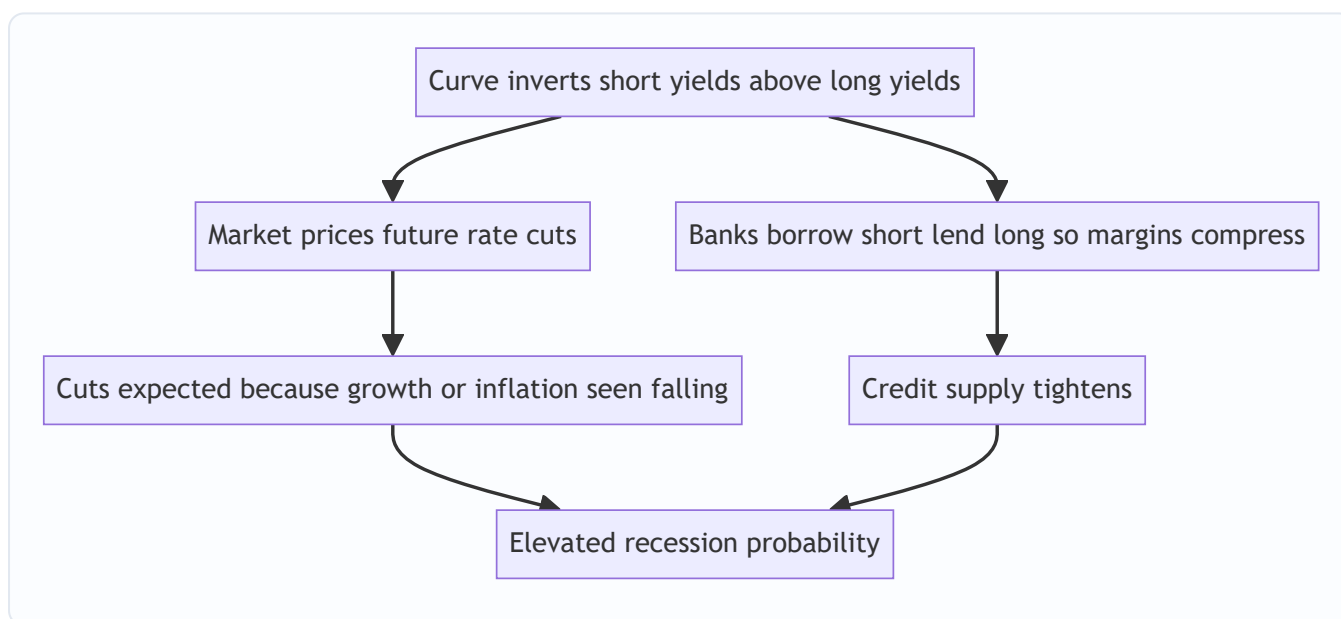


Figure 4 — Why an inverted curve is read as a recession warning: it embeds expected cuts and simultaneously squeezes bank lending.

Note the caveat from Example 4 applies with equal force here: with a positive term premium, an *inverted* curve is an even *stronger* signal, because the market must expect rate cuts large enough to overcome the premium before long yields can fall below short yields.

## 6. Connections

- **To bond pricing (Chapters 2–3):** the spot curve *is* the correct set of discount rates; YTM is a single-number approximation of averaging over the curve. A bond's YTM equals its coupon only when it trades at par, and it is a complicated average of the spot rates weighted by cash-flow timing.
- **To duration and convexity (Chapter 5):** the "level, slope, curvature" decomposition of curve moves is why practitioners hedge not just duration (level risk) but also **key-rate durations** (exposure to specific points on the curve). A steepener/flattener is a slope bet that pure duration cannot capture.
- **To forwards, FRAs, and swaps (later chapters):** implied forward rates are the fair fixed rates for forward-rate agreements and the building blocks for pricing interest-rate swaps — a swap's fixed rate is essentially a par yield off the curve.
- **To macro and monetary policy:** the front end is anchored by the central bank's policy rate; the long end by inflation and growth expectations plus term premium. The curve is the market's aggregated forecast.
- **To equity and credit:** an inverted curve tightens bank lending margins (banks borrow short, lend long) and historically foreshadows recessions and widening credit spreads.

## 7. Key Terms

| Term                                     | Meaning                                                                   |
|------------------------------------------|---------------------------------------------------------------------------|
| <b>Term structure</b>                    | Relationship between yield and maturity, credit held constant             |
| <b>Yield curve</b>                       | Graphical plot of that relationship                                       |
| <b>Spot / zero rate</b>                  | Yield on a single cash flow at maturity $t$ ; the true discount rate      |
| <b>Discount factor</b>                   | Present value today of 1 unit received at time $t$                        |
| <b>Bootstrapping</b>                     | Recursively extracting spot rates from coupon-bond prices                 |
| <b>Forward rate</b>                      | Rate for a future period, locked in today, implied by spots               |
| <b>Par yield</b>                         | Coupon rate that makes a new bond price at par                            |
| <b>Term / liquidity premium</b>          | Extra yield demanded for holding longer-duration bonds                    |
| <b>Normal / inverted / flat / humped</b> | The four canonical curve shapes                                           |
| <b>Steepening / flattening</b>           | Slope of the curve increasing / decreasing                                |
| <b>STRIPS</b>                            | Separately traded principal/interest zero-coupon components of a Treasury |

## 8. Common Confusions

1. **YTM vs spot rate.** YTM applies one rate to *all* of a bond's cash flows; the spot curve applies a *different* rate to each. Only for a zero-coupon bond do YTM and spot coincide. Using YTM to compare two bonds with different coupon patterns is apples-to-oranges (the "coupon effect").
2. **Forward rate  $\neq$  forecast.** A forward rate is what the *math of the current curve* implies, and it equals the expected future spot **only under pure expectations**. With any term premium, the forward is a *biased* (usually upward) estimate. Never tell an interviewer the curve "predicts" a 4% rate without adding "under the pure expectations hypothesis."

3. **Upward slope does not require expected rate rises.** Because of the liquidity/term premium, the curve's *normal* resting state is mildly upward even if the market expects rates to be flat. Only an unusually steep curve signals strongly expected hikes.
4. **Inversion is a signal, not a cause.** An inverted curve reflects the market pricing in future rate *cuts* (because it expects a slowdown). It doesn't *cause* the recession, though it can amplify one by squeezing bank lending margins.
5. **Par yield vs coupon vs spot.** These three are equal only in a flat curve at par. On an upward curve,  $\text{par yield} < \text{spot} < \text{forward}$  at each maturity; confusing them leads to mispricing.
6. **Compounding conventions.** Annual vs semi-annual vs continuous compounding change the numbers. A curve quoted semi-annually (as US Treasuries are) will not match an annually-compounded bootstrap unless you convert. Always state the convention.

## 9. Recap

The term structure is the map from maturity to yield, and it exists because the market's expectation of the time value of money is not constant across horizons. The **spot (zero) curve** is the foundation — the honest set of discount rates enforced by no-arbitrage (the STRIPS market makes this literal). From observed coupon-bond prices we **bootstrap** the spot curve maturity by maturity; from spots we derive **forward rates** by a pure no-arbitrage argument and **par yields** by finding the coupon that prices at par. For an upward-sloping curve these line up as  $\text{par} < \text{spot} < \text{forward}$ .

Four theories explain the curve's shape: **pure expectations** (long = average of expected shorts, forwards are unbiased forecasts), **liquidity preference** (add a rising term premium, forwards overstate expected rates), **market segmentation** (separate maturity clienteles set local yields), and **preferred habitat** (the realistic hybrid — habitats exist but premiums can lure investors across them). The curve is driven at the front by monetary policy and at the back by inflation, growth, and term premium, and its moves decompose into **level, slope, and curvature**. Its shape — normal, flat, inverted, humped — is a closely watched macro signal, with **inversion the classic recession warning**.

## 10. Quick-Reference / Interview Points

**Formulas to have cold:**

| Quantity                | Formula                                                              |
|-------------------------|----------------------------------------------------------------------|
| Zero price              | $P_t = 100 / (1+s_t)^t$                                              |
| Bond price (spot curve) | $\sum C/(1+s_t)^t + 100/(1+s_n)^n$                                   |
| One-year forward        | $1+f(a,a+1) = (1+s_{a+1})^{a+1} / (1+s_a)^a$                         |
| Spot as geo-avg         | $(1+s_n)^n = \prod (1+f_{k-1,k})$                                    |
| Par yield               | $c_n = 100(1-DF_n)/\sum DF_t$                                        |
| Forward decomposition   | $\text{forward} = \text{expected future spot} + \text{term premium}$ |

**Talking points:**



- "There is no single interest rate — the term structure is the whole map, and the spot curve is the theoretically correct set of discount rates."
- Be able to **bootstrap a 2- or 3-year spot curve live** — it is a classic desk interview question. Know the recursion cold.
- Given spots, **compute the implied forward in one line**, and immediately caveat: "unbiased forecast only under pure expectations."
- **Ordering rule:** upward curve  $\rightarrow$  par < spot < forward (and reversed for inverted). Explains why coupon bonds and zeros of the same maturity have different yields.
- **Four theories in one breath:** expectations (forwards = expected shorts), liquidity preference (+ rising premium), segmentation (separate clienteles), preferred habitat (hybrid, realistic). Preferred habitat is the modern consensus.
- **The inversion story:** short > long means the market expects cuts, which means it expects a slowdown; the 10y–2y and 10y–3m spreads are the headline recession gauges. It also compresses bank net interest margins.
- **Curve moves = level + slope + curvature.** Duration hedges level; key-rate durations hedge slope and curvature; a "butterfly" is a curvature trade.
- **Drivers:** front end = central bank policy; long end = inflation + real growth + term premium; kinks = supply/demand at specific maturities.
- Remember the **STRIPS arbitrage**: it's the real-world enforcement mechanism that makes the spot curve more than theory — bonds must equal the sum of their zeros or traders strip/reconstitute for profit.

# Q&A — The Term Structure of Interest Rates

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Companion practice bank for Chapter 06. Every question is followed by a full answer. Unless stated otherwise, all rates are **annually compounded**, face value is **100**, and coupons are **annual**.

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## Section A — Concept Check

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- A1. What is the term structure of interest rates, and how does it differ from "the yield curve"?** The term structure is the relationship between yield and time to maturity, holding credit quality, liquidity, and all other features constant. It answers "what rate applies to each horizon?" The *yield curve* is simply the graphical plot of that relationship — maturity on the x-axis, yield on the y-axis. The term structure is the concept; the yield curve is its picture.
- A2. Why can a single interest rate not correctly price a coupon bond?** A coupon bond is really a portfolio of zero-coupon cash flows arriving at different dates. Each date has its own time value of money, captured by its own spot rate. The year-1 coupon should be discounted at the 1-year spot, the year-5 redemption at the 5-year spot. Applying one blended yield (YTM) to all cash flows is a convenient shortcut that generally misprices the bond and distorts relative-value comparisons — the "coupon effect."
- A3. Distinguish the spot curve, the par curve, and the forward curve.** - **Spot (zero) curve:** the yield on a single cash flow received at time  $t$  — the theoretically correct discount rate. - **Par curve:** the coupon rate that makes a newly issued bond of each maturity price exactly at 100. This is usually what market commentary means by "the 10-year yield." - **Forward curve:** rates for periods that *begin in the future* but are locked in *today*, implied arithmetically by the spot curve. They are three views of the same no-arbitrage information: given any one, you can derive the other two.
- A4. Why are spot rates the "honest" discount rates? What enforces them?** No-arbitrage. A coupon bond must equal the sum of its cash flows valued at each maturity's spot rate; if it did not, a trader would strip the bond into zeros (or reconstitute it) and pocket the difference. The US Treasury **STRIPS** market makes this literal — it is the real-world mechanism that enforces the spot curve rather than leaving it an academic construct.
- A5. What is bootstrapping, in one sentence?** Bootstrapping is the recursive extraction of spot rates from observed coupon-bond prices, one maturity at a time: the 1-year rate comes directly from a 1-year instrument, then each longer spot is solved after discounting the earlier coupons at the already-known spots.
- A6. State the four canonical curve shapes and the typical signal of each.** - **Normal (upward):** long > short; healthy expansion, modest expected rises and/or term premium. - **Flat:** yields roughly equal; a transition point, often preceding inversion. - **Inverted (downward):** short > long; market expects rate *cuts*, usually because it expects a slowdown. - **Humped:** rises then falls (peak in the belly); near-term tightening expected to reverse later.
- A7. On an upward-sloping curve, rank par yield, spot rate, and forward rate at a given maturity, and explain why. Par < spot < forward.** The spot rate is a geometric average of forwards, so it lags the rising forwards. The par yield is a coupon-weighted blend that puts weight on cheaper near-dated rates, so it sits below the spot. On an inverted curve the inequalities reverse.
- A8. Name the four term-structure theories and give the one-line distinction of each.** - **Pure (unbiased) expectations:** long rate = geometric average of expected future short rates; forwards are unbiased forecasts.

- **Liquidity preference:** add a rising term premium; forwards *overstate* expected future spots. - **Market segmentation:** separate maturity clienteles set local yields independently. - **Preferred habitat:** the realistic hybrid — investors have preferred habitats but will leave them for a large enough premium; premiums can be positive or negative.

**A9. Why is an inverted curve read as a recession warning rather than a recession cause?** An inverted curve reflects the market *pricing in future rate cuts*, and the central bank cuts when it expects the economy to weaken — so inversion is a symptom of that aggregated expectation, not the cause. It can still *amplify* a downturn: banks borrow short and lend long, so when short rates exceed long rates their net interest margins compress, credit tightens, and the slowdown can become partly self-fulfilling.

## Section B — Numerical Bond-Math Problems

### B1. Bootstrap a 2-year spot curve

You observe a 1-year zero priced at **96.1538** and a 2-year 5% annual-coupon bond priced at **100.05**. Find the 1- and 2-year spot rates.

**Step 1 — one-year spot.** The zero pays 100 in one year:  $\$1 + s_1 = \frac{100}{96.1538} = 1.04000$   
 $\Rightarrow \boxed{s_1 = 4.00\%}$

**Step 2 — two-year spot.** The 2-year bond pays 5 at  $t=1$  and 105 at  $t=2$ . Discount the year-1 coupon at the known  $s_1$ :  $\$ \frac{5}{1.04} = 4.8077 \Rightarrow \frac{105}{(1+s_2)^2} = 100.05 - 4.8077 = 95.2423$   
 $\frac{105}{(1+s_2)^2} = \frac{105}{95.2423} = 1.10244 \Rightarrow 1+s_2 = 1.04998 \Rightarrow \boxed{s_2 = 5.00\%}$

**Reconcile:** reprice the bond off the curve:  $\$ \frac{5}{1.04} + \frac{105}{1.05^2} = 4.8077 + 95.2381 = 100.046 \approx 100.05. \$ \checkmark$

### B2. Implied forward rates

Extend the curve with a 3-year spot of **5.50%** (so  $s_1=4.0\%, s_2=5.0\%, s_3=5.5\%$ ). Compute the one-year forwards.

$\$1 + f(1,2) = \frac{(1+s_2)^2}{1+s_1} = \frac{1.1025}{1.04} = 1.06010 \Rightarrow \boxed{f(1,2)=6.01\%}$   
 $\$1 + f(2,3) = \frac{(1+s_3)^3}{(1+s_2)^2} = \frac{1.174241}{1.1025} = 1.06507 \Rightarrow \boxed{f(2,3)=6.51\%}$

**Reconcile** the 3-year spot as the geometric average of one-year forwards:  $\$(1+s_1)(1+f(1,2))(1+f(2,3)) = 1.04 \times 1.06010 \times 1.06507 = 1.174246 = (1+s_3)^3$  Cube root  $= 1.05500 \Rightarrow s_3 = 5.50\%.$   $\checkmark$  Note the forwards (4.00%, 6.01%, 6.51%) rise faster than the spots (4.0%, 5.0%, 5.5%) and sit above them — exactly the upward-curve pattern.

### B3. Par yield and the ordering rule

From the same spot curve, compute the 3-year par yield. First the discount factors:

| t | $s_t$      | $DF_t = 1/(1+s_t)^t$ |
|---|------------|----------------------|
| 1 | 4.00%      | 0.961538             |
| 2 | 5.00%      | 0.907029             |
| 3 | 5.50%      | 0.851614             |
|   | <b>Sum</b> | <b>2.720182</b>      |

$$c_3 = \frac{100(1 - DF_3)}{\sum DF_t} = \frac{100(1 - 0.851614)}{2.720182} = \frac{14.8386}{2.720182} = \boxed{5.455\%}$$

**Reconcile:** price a 3-year 5.455% bond:  $\$5.455 \times 2.720182 + 100 \times 0.851614 = 14.839 + 85.161 = 100.00$ . ✓

**Ordering check** at the 3-year point: par **5.455%** < spot **5.50%** < forward  $f(2,3)$  **6.51%** — confirms  $par < spot < forward$  for the upward curve.

#### B4. Coupon vs par yield — where does a bond trade?

Using the curve above, price a 3-year **6% annual-coupon** bond off the spot curve.  $P = 6(DF_1) + 6(DF_2) + 106(DF_3) = 6(0.961538) + 6(0.907029) + 106(0.851614) = 5.7692 + 5.4422 + 90.2711 = \boxed{101.48}$  The bond trades **above par** because its 6% coupon exceeds the 3-year par yield of 5.455% — a premium bond. This is the direct link between the par yield and whether a given coupon prices above or below 100.

#### B5. Inverted curve — forwards decline

Now the curve flips:  $s_1 = 5.0\%$ ,  $s_2 = 4.5\%$ ,  $s_3 = 4.0\%$ . Extract the one-year forwards.  $1 + f(1,2) = \frac{1.045^2}{1.05} = \frac{1.092025}{1.05} = 1.04002 \rightarrow \boxed{f(1,2)=4.00\%}$   $1 + f(2,3) = \frac{1.04^3}{1.045^2} = \frac{1.124864}{1.092025} = 1.03007 \rightarrow \boxed{f(2,3)=3.01\%}$

**Reconcile:**  $\$1.05 \times 1.04002 \times 1.03007 = 1.124864 = (1 + s_3)^3$ ; cube root  $= 1.04000 \rightarrow s_3 = 4.00\%$ . ✓

The forwards **fall** (5.0% → 4.0% → 3.0%). Under pure expectations the market is pricing the 1-year rate dropping year by year — i.e., expected central-bank cuts, the classic slowdown signal.

## Section C — Interview-Style Questions with Model Answers

**C1. "Walk me through bootstrapping a spot curve. Why not just use YTM?"** *Model answer:* "Start with the shortest instrument — a 1-year bill or zero gives the 1-year spot directly. For the 2-year bond, I already know how to value its first coupon using the 1-year spot, so whatever price is left over must be explained by the 2-year spot; I solve for it. I repeat maturity by maturity, each step reusing all the spots I've already recovered. I bootstrap rather than use YTM because YTM applies a *single* rate to every cash flow, which is only correct for a zero. The spot curve gives each cash flow its own honest discount rate, and no-arbitrage — enforced by the STRIPS market — says the bond must equal the sum of its zeros valued that way."

**C2. "The 2-year yield is 5% and the 3-year is 5.5%. What does that tell you about next year's rates?"** *Model answer:* "The implied one-year forward from year 2 to 3 is  $(1.055^3/1.05^2) - 1 \approx 6.5\%$ . So the curve implies the 1-year rate two years out is around 6.5%. But I'd immediately caveat: that forward

equals the market's expected future spot **only under the pure expectations hypothesis**. Once you allow a term premium — liquidity preference — the forward overstates the expected rate by that premium, so the true expectation is lower. I'd never say the curve 'predicts' 6.5% without that qualifier."

**C3. "Why is the yield curve usually upward-sloping even when nobody expects rates to rise?"** *Model answer:* "The liquidity/term premium. Longer bonds have higher duration and more price risk, so lenders demand extra yield to hold them. That premium generally rises with maturity, which tilts the curve upward on its own — independent of any expected rate change. So a mildly upward curve is the 'normal' resting state; only an unusually *steep* curve signals strongly expected hikes."

**C4. "An interviewer says: 'The curve just inverted. Recession?' How do you respond?"** *Model answer:* "Inversion means short yields sit above long yields, which means the market is pricing future rate cuts — and the central bank cuts when it expects the economy to weaken. So historically the 10y–2y and 10y–3m spreads going negative have preceded nearly every US recession of the last 60 years. Two nuances: first, inversion is a *symptom* of expectations, not a cause, though it can amplify a slowdown by compressing bank net interest margins since banks borrow short and lend long. Second, with a positive term premium an inverted curve is an even *stronger* signal, because expected cuts must be large enough to overcome the premium before long yields fall below short ones."

**C5. "Compare the four term-structure theories and tell me which one practitioners actually believe."** *Model answer:* "Pure expectations says the long rate is just the geometric average of expected short rates, so forwards are unbiased forecasts — clean but unrealistic because it assumes risk-neutrality. Liquidity preference adds a rising term premium, so forwards overstate expected rates and the normal curve is mildly upward. Market segmentation says different clienteles — banks short, pensions and insurers long — trade in separate maturity buckets set by local supply and demand, which explains kinks expectations can't. Preferred habitat is the hybrid and the modern consensus: investors have preferred habitats but will leave them for a big enough premium, so both expectations and premiums matter and the premium can be positive or negative depending on supply-demand at each maturity."

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## Section D — Multiple Choice (with Reasoning)

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**D1.** On an upward-sloping spot curve, which ordering is correct at a given maturity? A) forward < spot < par B) par < spot < forward C) spot < par < forward D) forward < par < spot **Answer: B.** The spot is a geometric average of forwards so it lags the rising forwards (spot < forward); the par yield is a coupon-weighted blend weighted toward cheaper near-dated rates, so par < spot. Hence par < spot < forward.

**D2.** A forward rate equals the expected future spot rate only under which theory? A) Liquidity preference B) Market segmentation C) Pure expectations D) Preferred habitat **Answer: C.** Only pure expectations assumes a zero term premium, making forwards unbiased forecasts. Every other theory embeds a premium, so the forward is a biased (usually upward) estimate.

**D3.** The 1-year spot is 4% and the 2-year spot is 5%. The one-year forward  $f(1,2)$  is closest to: A) 4.5% B) 5.0% C) 6.0% D) 4.0% **Answer: C.**  $\$1 + f(1,2) = 1.05^2 / 1.04 = 1.1025 / 1.04 = 1.0601\$$ , so  $\approx 6.0\%$ . On an upward curve the forward must exceed the longer spot, which rules out A, B, and D.

**D4.** Which best explains a persistent kink or bump at one maturity that expectations theory cannot? A) Pure expectations B) Market segmentation / preferred habitat C) The Fisher effect D) Convexity **Answer: B.** Localized supply-demand imbalances from distinct maturity clienteles are exactly what segmentation and preferred habitat capture; pure expectations produces a smooth curve driven only by expected short rates.

**D5.** A 3-year bond's coupon is 6% and the 3-year par yield is 5.455%. The bond will trade: A) at par B) at a discount C) at a premium D) cannot tell **Answer: C.** Coupon (6%) exceeds the par yield (5.455%), so the bond pays more than a par bond of the same maturity and prices above 100 — a premium bond (here  $\approx 101.48$ , per B4).

**D6.** An inverted curve most directly reflects that the market expects: A) rising inflation B) future rate cuts C) higher term premiums D) heavy bond issuance **Answer: B.** Short above long means the implied forwards decline, i.e., the market prices the short rate falling — future rate cuts, which the central bank delivers when it expects a slowdown.

**D7.** For which instrument do YTM and the spot rate always coincide? A) A par coupon bond B) A premium bond C) A zero-coupon bond D) A floating-rate note **Answer: C.** A zero has a single cash flow, so its single internal rate of return (YTM) is exactly the spot rate for that maturity. Coupon bonds blend many spot rates, so their YTM only approximates the curve.

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*Self-verification note:* All bootstrapped spots, forwards, and par yields in Section B were cross-checked by repricing each bond off the recovered curve and by rebuilding each long spot as the geometric average of its one-year forwards; every figure reconciles to within rounding.

# Chapter 07 — Duration

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## 1. The Problem / Need

A bond is a promise of future cash flows. The moment you know those cash flows and a discount rate (the yield), you can price the bond — that was the work of earlier chapters. But a price is a *photograph*: it tells you what the bond is worth at one instant, at one yield. It says nothing about what happens **next**.

Interest rates move every day. When yields rise, bond prices fall; when yields fall, prices rise. For anyone who holds bonds — a trader with a \$500 million book, a bank matching assets to deposits, a pension fund funding liabilities 20 years out — the burning question is not "what is my bond worth today?" but:

**"If yields move by X, how much money do I make or lose?"**

You could answer that question by brute force: reprice the bond at the new yield and take the difference. That works for one bond, but it collapses at scale. A trader running a portfolio of 400 instruments cannot reprice everything intraday every time the 10-year Treasury ticks. A risk manager needs a *single number* per position that says "this is how sensitive it is." Two bonds might both be worth par, but one loses 2% on a 100 bp sell-off and the other loses 12% — they are completely different risks wearing the same price tag.

We need a **measure of interest-rate sensitivity**: a portable, additive number that compresses the entire price–yield relationship into "how much does price move per unit of yield." That number is **duration**. It is the single most important risk statistic in fixed income, and mastering it — Macaulay, modified, effective, dollar duration, PV01, and portfolio duration — is the spine of this chapter.

## 2. The Core Idea

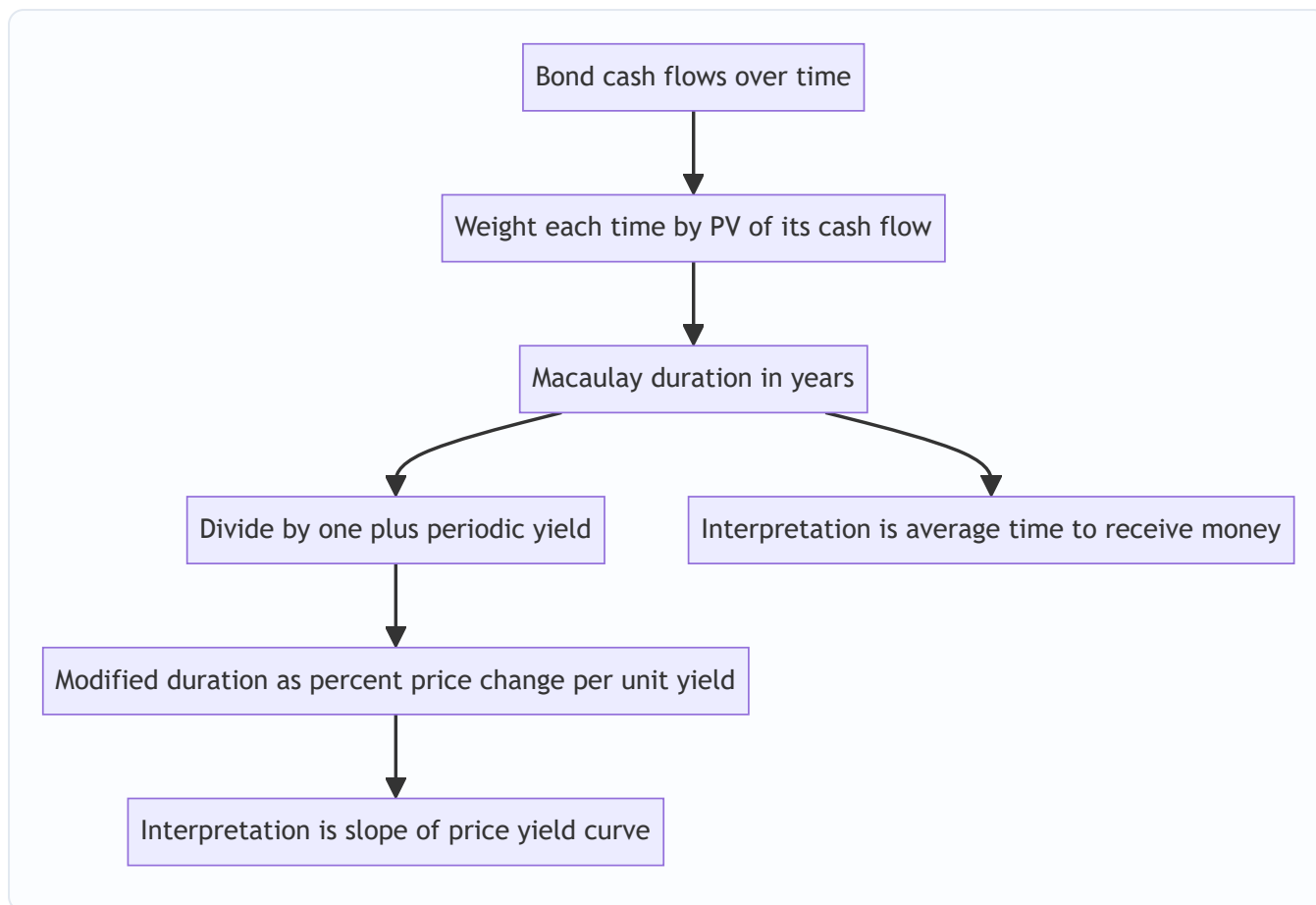
Duration has two faces that eventually turn out to be the same coin.

**Face 1 — Duration as a *time*.** Frederick Macaulay's original 1938 idea: a bond's effective "life" is not its maturity but the **weighted-average time until you receive its cash flows**, where each time is weighted by the present value of the cash flow arriving then. A 10-year zero-coupon bond pays everything at year 10, so its effective life is 10 years. A 10-year coupon bond pays you cash all along the way, so its *money-weighted* average life is shorter — maybe 7 or 8 years. This is **Macaulay duration**, measured in years. Picture a seesaw: the cash-flow present values are weights placed along a time axis, and Macaulay duration is the **fulcrum** — the balance point.

**Face 2 — Duration as a *sensitivity*.** The same weighted-average-time number, divided by one plus the periodic yield, tells you the **percentage change in price for a small change in yield**. This is **modified duration**. A modified duration of 7 means: *for every 1% (100 bp) rise in yield, the bond's price falls by roughly 7%*. It is the slope of the price–yield curve, expressed as a percentage.

The beautiful accident of bond math is that these two ideas — "average time to get your money back" and "price sensitivity to yield" — are almost the same quantity, differing only by a scaling factor of  $(1 + \text{yield})$ . That is why the word "duration" simultaneously means a *number of years* and a *risk exposure*.

*Figure 1 — Two faces of duration that turn out to be one number.*



### 3. Why / How It Works

Start from the pricing identity. A bond's price is the sum of discounted cash flows:

$$P = \sum_{t=1}^N \frac{CF_t}{(1+y)^t}$$

Price sensitivity means: how does  $P$  change when  $y$  changes? Take the first derivative with respect to  $y$ :

$$\frac{dP}{dy} = \sum_{t=1}^N \frac{-t \cdot CF_t}{(1+y)^{t+1}} = \frac{-1}{(1+y)} \sum_{t=1}^N \frac{t \cdot CF_t}{(1+y)^t}$$

Now divide both sides by  $P$  to get the **percentage** change in price:

$$\frac{1}{P} \frac{dP}{dy} = \frac{-1}{(1+y)} \cdot \underbrace{\left( \sum_{t=1}^N \frac{t \cdot CF_t}{(1+y)^t} \right)}_{\text{Macaulay duration}}$$

Look at what fell out. The messy sum in the brace is exactly the **PV-weighted average of the times  $t$**  — that is Macaulay duration ( $D_{\text{Mac}}$ ). So:

$$\frac{1}{P} \frac{dP}{dy} = \frac{-D_{\text{Mac}}}{(1+y)} \equiv -D_{\text{Mod}}$$

This is the whole story in one line. The derivative of price (a sensitivity) *automatically contains* the weighted-average time (Macaulay duration). Calculus links them. The factor  $1/(1+y)$  appears because each cash flow's exponent drops by one when differentiated, pulling out one discount factor.

Two consequences follow immediately, and they are the reason duration *works* as a risk tool:

1. **Linearity / additivity.** Because price is a sum of PV terms and duration is a PV-weighted average, the duration of a portfolio is just the value-weighted average of its bonds' durations. Risk adds up cleanly.



2. **Local accuracy.** Duration is the *tangent line* to the price–yield curve at the current yield. For small yield moves the tangent hugs the true curve, so the estimate is excellent. For large moves the true curve bends away from the tangent (it is convex), so duration alone under- or over-estimates — that residual is **convexity**, corrected in the next chapter.

## 4. Full Content — Formulas and Bond Math

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### 4.1 Macaulay Duration

$$D_{\text{Mac}} = \frac{\sum_{t=1}^N t \cdot \frac{CF_t}{(1+y)^t}}{P} = \sum_{t=1}^N t \cdot w_t, \quad w_t = \frac{CF_t/(1+y)^t}{P}$$

The weights  $w_t$  are the fraction of the bond's total present value delivered at time  $t$ ; they sum to 1.  $D_{\text{Mac}}$  is therefore a genuine weighted average of the payment times, expressed in the same period units as  $t$  (years if annual, half-years if semiannual — then divide by the periodicity to state in years).

**Closed form for a standard coupon bond** (per-period coupon rate  $c$ , per-period yield  $y$ ,  $N$  periods):

$$D_{\text{Mac}} = \frac{1+y}{y} - \frac{(1+y) + N(c-y)}{c \left[ (1+y)^N - 1 \right] + y}$$

This is handy for a fast desk calculation without laying out every cash flow.

### 4.2 Modified Duration

$$D_{\text{Mod}} = \frac{D_{\text{Mac}}}{1 + \frac{y}{m}}$$

where  $y$  is the annual yield and  $m$  is the number of coupon periods per year (use the *periodic* yield in the denominator). Modified duration is a pure sensitivity: **percentage price change per 1.00 (100%) change in yield**, or equivalently, the % price move for a 1-unit change in yield. Its defining relationship:

$$\frac{\Delta P}{P} \approx -D_{\text{Mod}} \cdot \Delta y$$

### 4.3 Effective Duration

For bonds whose cash flows **change when yields change** — callables, putables, mortgage-backed securities, any bond with embedded options — the analytic derivative above is invalid, because  $CF_t$  is no longer fixed. We instead measure sensitivity numerically by *shocking* the yield curve up and down and repricing with a model:

$$D_{\text{Eff}} = \frac{P_{-} - P_{+}}{2 \cdot P_0 \cdot \Delta y}$$

where  $P_{-}$  is the price after lowering yields by  $\Delta y$ ,  $P_{+}$  after raising them, and  $P_0$  the base price. For an **option-free** bond, effective duration  $\approx$  modified duration (they converge as  $\Delta y \rightarrow 0$ ). For a callable bond, effective duration is *lower* than the modified duration of an otherwise identical straight bond — and can even turn **negative** for deep-in-the-money MBS (negative convexity), where falling rates trigger prepayments that shorten the bond.

### 4.4 Dollar Duration, Money Duration, and PV01

Percentage duration is unitless; a risk manager wants **currency**. Multiply by price:

$$\text{Money Duration (Dollar Duration)} = D_{\text{Mod}} \times P \quad \frac{\Delta P}{P} \approx -(\text{Money Duration}) \cdot \Delta y$$

Scale to the smallest quoted yield move, one basis point ( $\Delta y = 0.0001\%$ ), and you get **PV01** (equivalently PVBP or DV01 — the "present value of a basis point" / "dollar value of an 01"):

$$\text{PV01} = D_{\text{Mod}} \times P \times 0.0001\%$$

PV01 answers "how many dollars do I gain or lose if yields move 1 bp?" It is the traders' universal risk unit because it is additive across long and short positions and lets you *hedge*: to neutralize a position's rate risk, offset it with another instrument of equal and opposite PV01.

## 4.5 Portfolio Duration

Because duration is PV-additive, a portfolio's modified duration is the **market-value-weighted average** of its holdings' durations:

$$D_{\text{Mod},P} = \sum_{i=1}^n w_i D_{\text{Mod},i}, \quad w_i = \frac{MV_i}{\sum_j MV_j}$$

Equivalently — and more robustly for aggregating risk — portfolio PV01 is simply the **sum** of the constituent PV01s:

$$\text{PV01}_P = \sum_{i=1}^n \text{PV01}_i$$

The weighted-average-duration formula assumes a parallel shift of a single yield and all bonds priced off the same yield; summing PV01s is the cleaner, assumption-light way to roll up a book.

## 4.6 Key Qualitative Relationships (memorize these)

| Factor             | Effect on duration                 | Intuition                                           |
|--------------------|------------------------------------|-----------------------------------------------------|
| Longer maturity    | Higher (generally)                 | Cash flows sit further out                          |
| Higher coupon      | <b>Lower</b>                       | More PV arrives early, pulling the balance point in |
| Higher yield       | <b>Lower</b>                       | Distant cash flows discounted harder, weighted less |
| Zero-coupon bond   | $D_{\text{Mac}} = \text{maturity}$ | Only one cash flow, at the end                      |
| Perpetuity         | $D_{\text{Mac}} = (1+y)/y$         | Never matures, but not infinite duration            |
| Add call option    | Lowers effective duration          | Upside price capped when rates fall                 |
| Floating-rate note | Near zero to next reset            | Coupon re-fixes to the market                       |

## 5. Worked Examples

### Example 1 — Full duration build-out for a 3-year coupon bond

**Bond:** 3-year, 8% annual coupon, face value 100, priced to yield **10%**. Cash flows: Year 1 = 8, Year 2 = 8, Year 3 = 108.

**Step 1 — Price and the PV of each cash flow.**

| $t$        | $CF_t$ | Discount $(1.10)^{-t}$ | $PV_t = CF_t / (1.10)^t$ | Weight $w_t$   | $t \cdot PV_t$    | $t(t+1) \cdot PV_t$ |
|------------|--------|------------------------|--------------------------|----------------|-------------------|---------------------|
| 1          | 8      | 1.1000                 | 7.272727                 | 0.07654        | 7.272727          | 14.545454           |
| 2          | 8      | 1.2100                 | 6.611570                 | 0.06957        | 13.223140         | 39.669420           |
| 3          | 108    | 1.3310                 | 81.142000                | 0.85389        | 243.426000        | 973.704000          |
| <b>Sum</b> |        |                        | <b>95.026296</b>         | <b>1.00000</b> | <b>263.921868</b> | <b>1027.918874</b>  |

Price  $P = 95.026296$ .

## Step 2 — Macaulay duration.

$$D_{\text{Mac}} = \frac{263.921868}{95.026296} = 2.77736 \text{ years}$$

Cross-check with the closed form ( $c=0.08, y=0.10, N=3$ ):

$$D_{\text{Mac}} = \frac{1.10}{0.10} - \frac{1.10 + 3(0.08 - 0.10)}{0.08(1.331 - 1) + 0.10} = 11 - \frac{1.04}{0.12648} = 11 - 8.22265 = 2.77735 \quad \checkmark$$

Both methods agree to five figures. Note  $D_{\text{Mac}} = 2.78 < 3.00$ : the coupons pull the balance point in from the 3-year maturity.

## Step 3 — Modified duration.

$$D_{\text{Mod}} = \frac{2.77736}{1.10} = 2.52487$$

Interpretation: a 100 bp rise in yield should cut the price by about **2.52%**.

## Step 4 — Test the sensitivity by actual repricing.

Reprice at 11% (a +100 bp move):

$$P_{11\%} = \frac{8}{1.11} + \frac{8}{1.11^2} + \frac{108}{1.11^3} = 7.207207 + 6.492976 + 78.968673 = 92.668856$$

$$\text{Actual change } \% = \frac{92.668856 - 95.026296}{95.026296} = -2.4808\%$$

$$\text{Duration estimate } \% = -D_{\text{Mod}} \cdot \Delta y = -2.52487 \cdot 0.01 = -2.5249\%$$

The estimate (−2.5249%) overstates the loss slightly versus the truth (−2.4808%). The **gap of 0.044%** is convexity — the price–yield curve bends upward, so the true price is a touch higher than the straight-line tangent predicts.

## Step 5 — Reconcile with convexity.

Annual convexity:

$$C = \frac{1027.918874}{(1.10)^2 \cdot 95.026296} = \frac{849.5197}{95.026296} = 8.9398$$

Second-order estimate:

$$\frac{\Delta P}{P} \approx -D_{\text{Mod}} \cdot \Delta y + \frac{1}{2} C (\Delta y)^2 = -0.0252487 + \frac{1}{2} (8.9398)(0.0001) = -0.0252487 + 0.0004470 = -0.0248017$$

That is **−2.4802%**, essentially the exact −2.4808%. The duration + convexity pair nails the true price move — full reconciliation.

**Small-move check (10 bp).** Reprice at 10.1%:  $P = 94.786791$ , actual change  $\% = -0.2520\%$ . Duration estimate  $\% = -2.52487 \cdot 0.001 = -0.2525\%$ . For a small move the tangent and curve nearly coincide — the estimate is accurate to the fourth decimal, confirming duration is a *local* measure.

## Example 2 — Semiannual bond and the periodicity convention

Real bonds usually pay **semiannually**, so you must work in half-year periods and then convert. **Bond:** 2-year, 6% annual coupon paid semiannually, face 100, priced to yield **5% annual** ( $\Rightarrow$  2.5% per period). There are  $N=4$  periods; each coupon is 3.

| Period $t$ | $CF_t$ | $PV_t = CF_t / (1.025)^t$ | $t \cdot PV_t$   |
|------------|--------|---------------------------|------------------|
| 1          | 3      | 2.92683                   | 2.92683          |
| 2          | 3      | 2.85544                   | 5.71088          |
| 3          | 3      | 2.78580                   | 8.35740          |
| 4          | 103    | 93.31292                  | 373.25168        |
| <b>Sum</b> |        | <b>101.88099</b>          | <b>390.24679</b> |

**Macaulay (in periods):**  $\$390.24679 / 101.88099 = 3.83042$  half-years. Convert to years by dividing by the periodicity  $m = 2$ :  $MD_{\text{Mac}} = 3.83042 / 2 = 1.9152$  years.

**Modified duration.** Divide by one plus the *periodic* yield, then restate per annual yield change. In periods:  $3.83042 / 1.025 = 3.73699$  half-years; divide by  $m=2$ :  $MD_{\text{Mod}} = 1.8685$  (percent price change per 100 bp change in the annual yield).

**Reconcile.** Reprice at a 6% annual yield (3% per period): the bond becomes a par bond,  $P = 100.0000$ . Actual change  $\% = (100.0000 - 101.88099) / 101.88099 = -1.8463\%$ . Duration estimate  $\% = -1.8685 \times 0.01 = -1.8685\%$ . The small overstatement is again convexity. The lesson: **always compute Macaulay/modified in the bond's native period, then divide by  $m$  to state duration in years and per annual-yield move.**

## Example 3 — Dollar duration and PV01 on a real position

Take the same bond, but now you hold **\$10,000,000 face value**. Market value  $\$ = 10,000,000 \times (95.026296/100) = \$9,502,630$ .

**Money (dollar) duration** per 100 of price:  $MD_{\text{Mod}} \times P = 2.52487 \times 95.026296 = 239.929$

**PV01 per 100 face:**  $\$239.929 \times 0.0001 = 0.0239929$

Scale to \$10mm face (multiply by  $10,000,000/100 = 100,000$ ):

$\text{PV01}_{\text{position}} = 0.0239929 \times 100,000 = \$2,399.29$

So every **1 bp** move in yield changes this position by about **\$2,399**. Sanity check via money duration directly:  $\Delta MV \approx -D_{\text{Mod}} \times MV \times \Delta y = -2.52487 \times 9,502,630 \times 0.0001 = -\$2,399.3$  per +1 bp. ✓ For a +25 bp sell-off, expected loss  $\approx 25 \times \$2,399 = \$59,975$ .

## Example 4 — Portfolio duration and hedging

A book holds two positions:

| Bond                  | Market Value        | Modified Duration | PV01 ( $= D_{\text{Mod}} \times \text{MV} \times 0.0001$ ) |
|-----------------------|---------------------|-------------------|------------------------------------------------------------|
| A (our 3-yr, 8% bond) | \$6,000,000         | 2.5249            | \$1,514.94                                                 |
| B (a 10-yr bond)      | \$4,000,000         | 7.0000            | \$2,800.00                                                 |
| <b>Portfolio</b>      | <b>\$10,000,000</b> | <b>?</b>          | <b>\$4,314.94</b>                                          |

**Weighted-average duration:**  $D_{\text{Mod,P}} = 0.60 \times 2.5249 + 0.40 \times 7.0000 = 1.5149 + 2.8000 = 4.3149$

**Portfolio PV01** (sum of parts):  $\$1,514.94 + \$2,800.00 = \$4,314.94$ . Cross-check:  $4.3149 \times 10,000,000 \times 0.0001 = \$4,314.9$ . ✓ The two roll-up methods agree.

**Estimated P&L on a +50 bp parallel sell-off:**  $\Delta \text{MV} \approx -D_{\text{Mod,P}} \times \text{MV} \times \Delta y = -4.3149 \times 10,000,000 \times 0.005 = -\$215,747$

**Hedging application.** Suppose the manager wants to cut portfolio duration to a **target of 3.0** by shorting 10-year Treasury futures whose DV01 is \$80 per contract. Required reduction in PV01  $= (4.3149 - 3.0) \times 10,000,000 \times 0.0001 = \$1,314.9$ . Contracts to short  $= 1,314.9 / 80 \approx 16$  contracts. This is exactly how duration converts a portfolio view into an executable trade.

### Example 5 — Effective duration on a callable bond

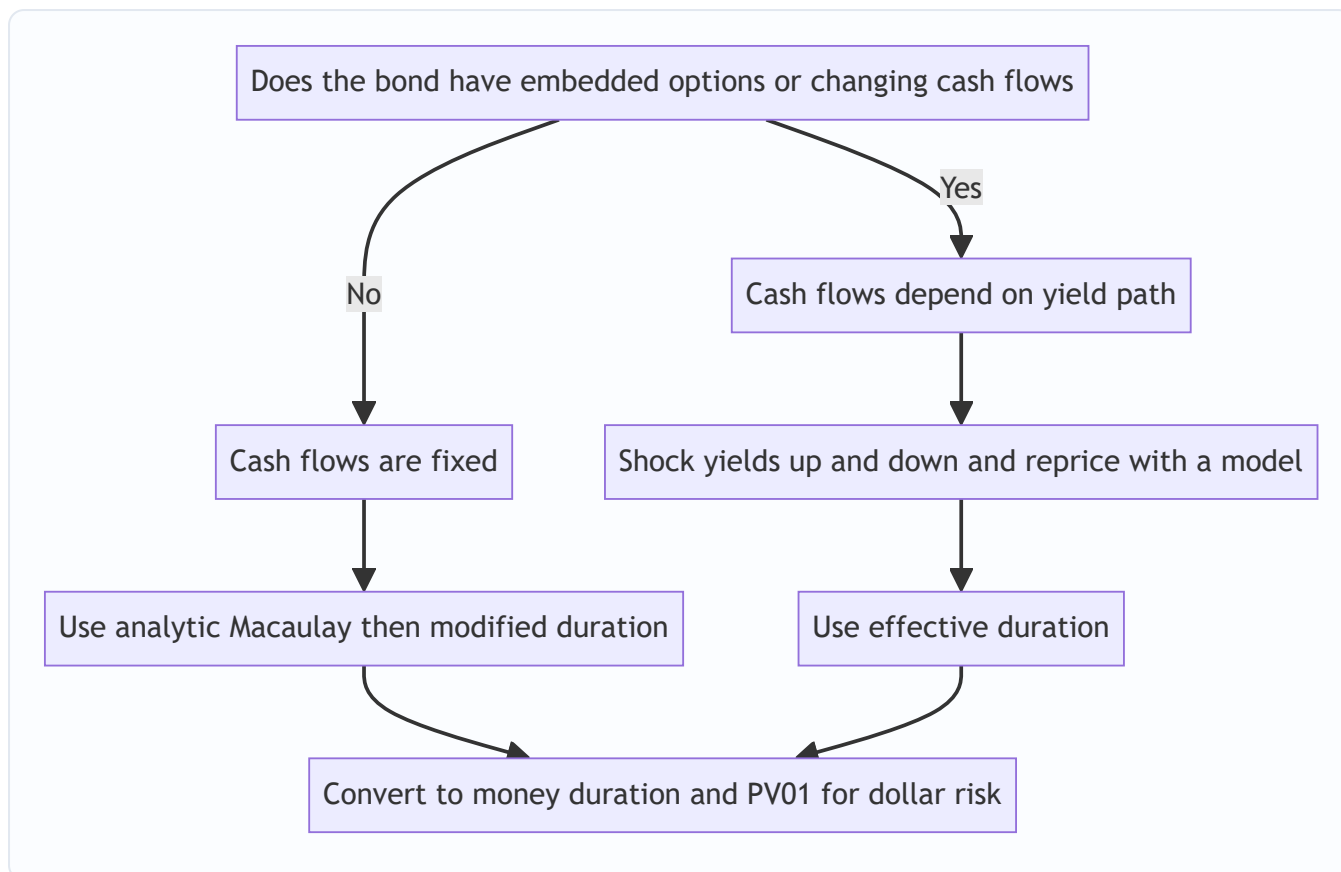
A callable bond is priced today at  $P_0 = 100.00$ . A pricing model (which accounts for the issuer's call option) reprices it after shifting the whole curve by  $\Delta y = 25$  bp:

- Yields **down** 25 bp:  $P_- = 101.10$  (upside limited — the call caps price appreciation)
- Yields **up** 25 bp:  $P_+ = 98.95$

$$D_{\text{Eff}} = \frac{P_- - P_+}{2 \times P_0 \times \Delta y} = \frac{101.10 - 98.95}{2 \times 100.00 \times 0.0025} = \frac{2.15}{0.50} = 4.30$$

An identical **option-free** bond might show a modified duration of 6.0. The embedded call *shortens* effective duration to 4.30 because the bond can't rally as freely when rates fall — the asymmetry (larger drop when rates rise than gain when they fall) is the fingerprint of negative convexity that effective duration captures and modified duration cannot.

*Figure 2 — Choosing the right duration measure.*



## 6. Connections

**To convexity (next chapter).** Duration is the first derivative (slope); convexity is the second derivative (curvature). Example 1 above showed duration alone leaves a residual on large moves that convexity mops up. The two are always used together for anything beyond ~25 bp.

**To bond pricing (earlier chapters).** Duration is literally the derivative of the pricing equation. Everything you learned about discounting cash flows is the raw material; duration is what you get when you ask how that price *moves*.

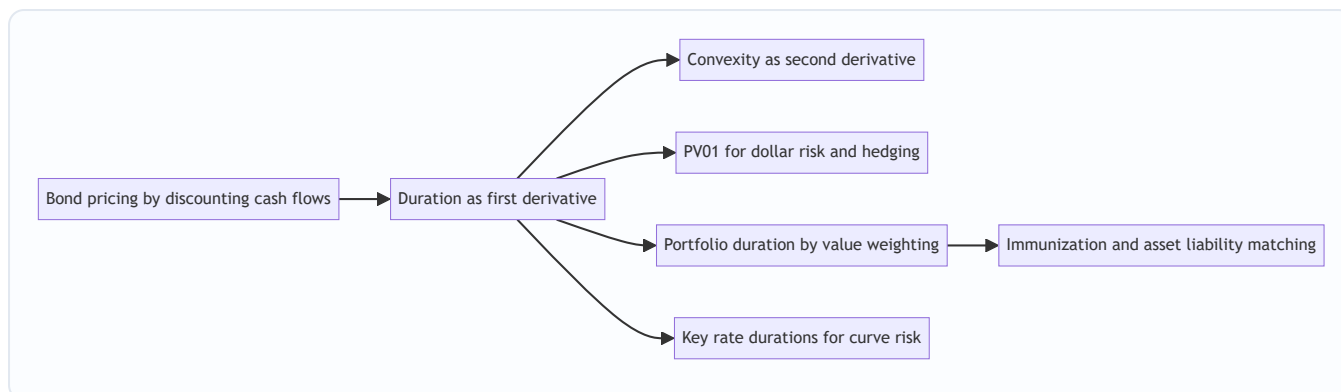
**To immunization and ALM.** Insurers and pension funds match the duration of their assets to the duration of their liabilities so that a rate shock moves both sides equally, protecting surplus. The **duration gap** = asset duration – liability duration (equity-weighted) is the core metric of bank asset-liability management. A positive gap means the institution loses net worth when rates rise.

**To the yield curve and key-rate duration.** Plain duration assumes a *parallel* shift. Real curves twist and steepen. **Key-rate (partial) durations** decompose sensitivity across specific maturities (2y, 5y, 10y, 30y), letting a manager hedge curve reshaping — a direct extension of the single-number duration here.

**To swaps and futures.** Interest-rate swaps and Treasury futures are priced and risk-managed in DV01 terms; the hedging math in Example 3 is exactly how a desk sizes them.

**To equities (DDM).** The dividend discount model gives long-duration growth stocks their notorious rate sensitivity — the same discounting mathematics, applied to equity cash flows.

Figure 3 — Where duration sits in the fixed-income toolkit.



## 7. Key Terms

- **Macaulay Duration** — PV-weighted average time (in years) to receive a bond's cash flows; the balance point of the cash-flow timeline.
- **Modified Duration** — Macaulay duration divided by  $(1 + \text{periodic yield})$ ; the approximate percentage price change for a 1-unit (100%) change in yield. The workhorse sensitivity.
- **Effective Duration** — sensitivity measured by repricing after up/down yield shocks; the correct measure for bonds with embedded options or path-dependent cash flows.
- **Dollar / Money Duration** — modified duration  $\times$  price; the currency price change per unit yield move.
- **PV01 / PVBP / DV01** — dollar change in value for a 1 bp yield move; the traders' additive risk unit.
- **Portfolio Duration** — market-value-weighted average of constituent durations; equivalently, sum of constituent PV01s.
- **Duration Gap** — asset duration minus liability duration; the immunization / ALM risk metric.
- **Key-Rate Duration** — sensitivity to a shift in one specific point on the yield curve, holding others fixed.
- **Convexity** — the second-order (curvature) correction to duration's linear estimate.

## 8. Common Confusions

**"Duration is the time until maturity."** No. Duration equals maturity *only* for a zero-coupon bond. For any coupon bond, duration is strictly less than maturity because coupons return money early. A 10-year 8% bond might have a duration near 7 years.

**"Macaulay and modified duration are interchangeable."** They differ by the factor  $(1 + y/m)$ . Macaulay is in *years*; modified is a *percentage sensitivity*. Quoting one when you mean the other misstates risk by the yield factor — small at low yields, material at high ones.

**"A modified duration of 5 means a 5% loss for any yield move."** Only for a *small* move, and only *approximately*. It means  $\sim 5\%$  per 100 bp, linearly. For a 300 bp move the linear estimate is meaningfully off — convexity matters, and the estimate overstates losses on sell-offs and understates gains on rallies.

**"Higher-coupon bonds are riskier because they pay more."** Backwards. Higher coupons mean *more* PV arrives early, which *lowers* duration and *reduces* rate sensitivity. Low-coupon and zero-coupon bonds are the most rate-sensitive at a given maturity.

**"Modified duration works for callable bonds."** No. Once cash flows depend on the yield path, the analytic derivative is invalid. You must use *effective* duration (shock-and-reprice). Using modified duration on an MBS can badly misstate — or wrongly-sign — the risk.

**"Duration is always positive."** Almost always, but deep-in-the-money callables and certain MBS can exhibit *negative* effective duration (negative convexity): prices fall as rates fall because prepayments accelerate.

**"To get portfolio risk, average the yields or maturities."** Weight by *market value*, not face value or count, and average the *durations* (or better, sum the PV01s). Face-weighting is wrong when bonds trade away from par.

## 9. Recap

Duration compresses the entire price–yield relationship into a single, portable number. It wears two hats that are mathematically one: a **weighted-average time** to receive cash flows (Macaulay, in years) and a **price sensitivity** to yield (modified, in % per unit yield). The bridge between them is division by  $(1 + \text{periodic yield})$ , which falls straight out of differentiating the pricing equation.

From modified duration flow the tools a desk actually uses: **money duration** (dollars per unit yield) and **PV01** (dollars per basis point), the additive currency in which real risk is measured and hedged. Because duration is a PV-weighted average, portfolio risk is a **value-weighted blend** of the parts, or equivalently the **sum of PV01s** — so a 400-line book collapses to one number and one hedge ratio.

For plain bonds, the analytic formulas suffice and duration is a razor-sharp *local* estimate — Example 1 reproduced the true price move to the fourth decimal for a 10 bp shift, and duration + convexity reconciled the 100 bp move almost exactly. For bonds with embedded options, cash flows move with yields, so we switch to **effective duration** — shock the curve up and down and reprice. Master the hierarchy — Macaulay → modified → effective, then money duration → PV01 → portfolio — and you can answer the only question that matters when rates move: *how much do I make or lose?*

## 10. Quick-Reference / Interview Points

### Core formulas

| Quantity       | Formula                                                                            |
|----------------|------------------------------------------------------------------------------------|
| Macaulay       | $D_{\text{Mac}} = \frac{1}{P} \sum t \frac{CF_t}{(1+y)^t}$                         |
| Modified       | $D_{\text{Mod}} = \frac{D_{\text{Mac}}}{1 + y/m}$                                  |
| Price change   | $\frac{\Delta P}{P} \approx -D_{\text{Mod}} \Delta y$                              |
| Effective      | $D_{\text{Eff}} = \frac{P_- - P_+}{2P_0 \Delta y}$                                 |
| Money duration | $D_{\text{Mod}} \times P$                                                          |
| PV01           | $D_{\text{Mod}} \times P \times 0.0001$                                            |
| Portfolio      | $D_P = \sum w_i D_i$ (value-weighted)                                              |
| With convexity | $\frac{\Delta P}{P} \approx -D_{\text{Mod}} \Delta y + \frac{1}{2} C (\Delta y)^2$ |

### Rapid-fire interview answers

- "What is duration in one sentence?" — The approximate percentage change in a bond's price for a 1% change in yield; equivalently, the PV-weighted average time to receive its cash flows.



- "Macaulay vs modified?" — Macaulay is in years; modified = Macaulay / (1 + periodic yield) and is the % sensitivity. Modified is what you use for P&L.
- "When do you use effective duration?" — For bonds with embedded options or path-dependent cash flows (callables, putables, MBS) — anything where cash flows change with rates.
- "What is DV01/PV01?" — Dollar change in value per 1 bp yield move; the additive risk unit desks hedge in.
- "How does coupon affect duration?" — Higher coupon → lower duration (more PV arrives early). Zeros have the highest duration for a given maturity (duration = maturity).
- "How does yield affect duration?" — Higher yield → lower duration (distant cash flows discounted more, weighted less).
- "Duration of a perpetuity?" —  $\$(1+y)/y$ . At 8%, that's 13.5 years — long, but finite.
- "Limitation of duration?" — It is linear/local; it assumes a small, parallel yield shift. Large moves need convexity; curve reshaping needs key-rate durations.
- "Portfolio duration of \$6mm @ 2.52 and \$4mm @ 7.0?" —  $\$0.6(2.52) + 0.4(7.0) = 4.31$ .
- "Can duration be negative?" — Effective duration can, for negatively convex instruments like some MBS, where falling rates trigger prepayments and shorten the bond.

# Q&A — Duration

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A practice bank for Chapter 07. Every question is followed by a full answer; work each before reading the answer. Conventions: face value is 100 unless a position size is stated, "bp" is a basis point (0.01%),  $y$  is the annual yield and  $m$  the coupon periods per year. Compute Macaulay/modified duration in the bond's native period, then divide by  $m$  to state it in years and per annual-yield move.

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## Section A — Concept Checks

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### A1. In one sentence, what does duration measure, and why does the desk care?

Duration compresses the whole price–yield relationship into one number — the approximate percentage change in a bond's price for a 1-unit change in yield — so a risk manager can answer "if yields move  $X$ , how much do I make or lose?" for a 400-line book without repricing everything.

### A2. State the two "faces" of duration and the bridge between them.

Face 1 is *time*: Macaulay duration is the present-value-weighted average time until you receive the cash flows (in years), the balance point of the cash-flow timeline. Face 2 is *sensitivity*: modified duration is the percentage price change per unit yield. The bridge is division by one plus the periodic yield:  $D_{\text{Mod}} = D_{\text{Mac}} / (1 + y/m)$ . They fall out of the same derivative of the pricing equation.

### A3. Why is duration strictly less than maturity for a coupon bond, but equal to maturity for a zero?

Coupons return money along the way, so some of the bond's present value arrives *before* maturity, pulling the weighted-average time (the fulcrum) in from the final date. A zero pays everything at one instant — maturity — so its balance point *is* the maturity date and  $D_{\text{Mac}} = N$ .

### A4. Which way does a higher coupon push duration, and why is the intuition often reversed by beginners?

A higher coupon *lowers* duration. More present value arrives early, pulling the balance point in, which *reduces* rate sensitivity. Beginners think "more coupon = more cash flow = more risk," but the opposite is true: low-coupon and zero-coupon bonds are the *most* rate-sensitive at a given maturity.

### A5. Which way does a higher yield push duration?

A higher yield *lowers* duration. Distant cash flows are discounted harder, so they carry less present-value weight, shifting the weighted-average time toward the earlier flows.

### A6. When must you abandon modified duration for effective duration?

Whenever cash flows change with the yield path — callables, putables, MBS, any embedded option. The analytic derivative behind modified duration assumes fixed  $CF_t$ ; once cash flows move with rates that assumption breaks, so you shock the curve up and down and reprice with a model:  $D_{\text{Eff}} = (P_- - P_+) / (2 P_0 \Delta y)$ .

### A7. Define money (dollar) duration and PV01, and say why traders prefer PV01.

Money duration  $= D_{\text{Mod}} \times P$  — the currency price change per unit yield. PV01 (a.k.a. DV01, PVBP)  $= D_{\text{Mod}} \times P \times 0.0001$  — the currency change per 1 bp. Traders prefer PV01 because it is *additive* across long and short positions, so a book's total risk is just the sum of position PV01s, and hedging reduces to matching equal-and-opposite PV01.

### A8. How do you compute a portfolio's duration, and what is the classic weighting mistake?

Weight each bond's modified duration by its *market value*:  $MD_P = \sum w_i MD_i$ ,  $w_i = MV_i / \sum MV_j$ . Equivalently, sum the PV01s. The mistake is weighting by *face value* or by count, which is wrong whenever bonds trade away from par.

### A9. Duration is a "local" measure. What does that mean, and what corrects it?

Duration is the tangent line to the convex price–yield curve at the current yield. For small yield moves the tangent hugs the curve and the estimate is excellent; for large moves the curve bends away from the straight line, so duration alone overstates losses on sell-offs and understates gains on rallies. **Convexity** — the second-order term — mops up the residual.

### A10. What is the duration of a perpetuity, and why is it finite despite infinite maturity?

$MD_{\text{Mac}} = (1+y)/y$ . At 8% that is 13.5 years. Even though the bond never matures, cash flows far in the future are discounted so heavily that they contribute almost nothing to the PV-weighted average, capping the balance point at a finite value.

### A11. Can effective duration be negative? Give the mechanism.

Yes, for negatively convex instruments like deep-in-the-money MBS. When rates fall, prepayments accelerate, shortening the bond and capping price appreciation so sharply that price can actually *fall* as rates fall — a negative  $MD_{\text{Eff}}$ .

### A12. What is the duration gap, and what does a positive gap imply for a bank?

Duration gap = asset duration – (equity-weighted) liability duration. A positive gap means assets are more rate-sensitive than liabilities, so a rate *rise* cuts asset value more than liability value and the institution loses net worth. Immunization sets the gap to zero.

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## Section B — Numerical Problems (full step-by-step)

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### B1. A 5-year zero-coupon bond yields 6% annually. Find its Macaulay and modified duration, then verify the sensitivity against an actual +100 bp reprice.

Price  $\$ = 100/1.06^5 = 100/1.3382256 = 74.72582\$$ . A zero has a single cash flow at year 5, so  $MD_{\text{Mac}} = 5.00000\$$  years.  $MD_{\text{Mod}} = 5/1.06 = 4.71698\$$  — i.e. ~4.717% per 100 bp.

*Verify.* Reprice at 7%:  $\$100/1.07^5 = 100/1.4025517 = 71.29862\$$ . Actual change  $\$ = (71.29862 - 74.72582)/74.72582 = -4.5860\%\$$ . Duration estimate  $\$ = -4.71698 \times 0.01 = -4.7170\%\$$ . The estimate overstates the loss by 0.131% — that residual is convexity (the curve lies above the tangent). Reconciles: duration is a local first-order estimate.

### B2. A 4-year, 5% annual-coupon bond is priced to yield 5% (a par bond). Build out the duration and reconcile with a +100 bp reprice.

Cash flows: 5, 5, 5, 105. Discount at 5%.

| $t$        | $CF_t$ | $PV_t = CF_t/1.05^t$ | $t \cdot PV_t$    |
|------------|--------|----------------------|-------------------|
| 1          | 5      | 4.761905             | 4.761905          |
| 2          | 5      | 4.535147             | 9.070295          |
| 3          | 5      | 4.319188             | 12.957563         |
| 4          | 105    | 86.383761            | 345.535043        |
| <b>Sum</b> |        | <b>100.000001</b>    | <b>372.324806</b> |

Price \$ = 100.0000\$ (confirms par).  $MD_{Mac} = 372.324806/100 = 3.72325$  years.  $MD_{Mod} = 3.72325/1.05 = 3.54595$  — ~3.546% per 100 bp.

*Verify.* Reprice at 6%:  $5/1.06 + 5/1.06^2 + 5/1.06^3 + 105/1.06^4 = 4.716981 + 4.450695 + 4.198769 + 83.169860 = 96.536305$ . Actual change \$ =  $(96.536305 - 100)/100 = -3.4637\%$ . Duration estimate \$ =  $-3.54595\%$ . Overstatement of 0.082% is convexity. Reconciles.

**B3. A 3-year, 6% coupon bond pays semiannually and yields 8% annually. Find duration in years and reconcile with a +100 bp reprice. (Watch the periodicity.)**

Work in half-years:  $N = 6$ , coupon \$ = 3\$, periodic yield \$ = 4\%.

| $t$ (½-yr) | $CF_t$ | $PV_t = CF_t/1.04^t$ | $t \cdot PV_t$    |
|------------|--------|----------------------|-------------------|
| 1          | 3      | 2.884615             | 2.884615          |
| 2          | 3      | 2.773669             | 5.547337          |
| 3          | 3      | 2.667008             | 8.001024          |
| 4          | 3      | 2.564449             | 10.257796         |
| 5          | 3      | 2.465817             | 12.329085         |
| 6          | 103    | 81.402440            | 488.414640        |
| <b>Sum</b> |        | <b>94.757998</b>     | <b>527.434497</b> |

Price \$ = 94.7580\$.  $MD_{Mac}$  in periods \$ =  $527.434497/94.757998 = 5.56610$  half-years; divide by  $m = 2$ :  **$MD_{Mac} = 2.78305$  years.**  $MD_{Mod}$  in periods \$ =  $5.56610/1.04 = 5.35202$ ; divide by 2:  **$MD_{Mod} = 2.67601$**  (per 100 bp annual-yield move).

*Verify.* Reprice at 9% annual (4.5% per period):  $3/1.045 + 3/1.045^2 + \dots + 103/1.045^6 = 2.870813 + 2.747189 + 2.628880 + 2.515894 + 2.407512 + 79.093830 = 92.264118$ . Actual change \$ =  $(92.264118 - 94.757998)/94.757998 = -2.6319\%$ . Duration estimate \$ =  $-2.67601\%$ . Overstatement of 0.044% is convexity. Reconciles. **Lesson: always compute in the native period, then divide by  $m$ .**

**B4. You hold 20,000,000 face of the B2 par bond ( $MD_{Mod} = 3.54595$ , price 100). Compute money duration, position PV01, and the expected P&L on a +40 bp sell-off.**

Market value \$ =  $20,000,000 \times (100/100) = \$20,000,000$ . Money duration per 100 \$ =  $3.54595 \times 100 = 354.595$ . PV01 per 100 \$ =  $354.595 \times 0.0001 = 0.0354595$ . Scale to 20mm face (factor  $20,000,000/100 = 200,000$ ): **PV01 \$ =  $0.0354595 \times 200,000 = \$7,091.90$  per 1 bp.** On a +40 bp move, expected loss \$  $\approx 40 \times \$7,091.90 = \$283,676$ . *Cross-check via money duration:*

$-D_{\text{Mod}} \times MV \times \Delta y = -3.54595 \times 20,000 \times 0.004 = -\$283,676$ . ✓ Two routes agree.

**B5. A book holds Bond X (MV \$12mm,  $D_{\text{Mod}}=3.546$ ) and Bond Y (MV \$8mm,  $D_{\text{Mod}}=6.5$ ). Find portfolio duration and PV01, then size a hedge to cut duration to 3.0 using 10-yr futures with DV01 \$75/contract.**

Weights:  $w_X = 12/20 = 0.60$ ,  $w_Y = 8/20 = 0.40$ .  $D_P = 0.60 \times 3.546 + 0.40 \times 6.5 = 2.1276 + 2.6000 = 4.7276$ . PV01:  $\text{X} = 3.546 \times 12,000,000 \times 0.0001 = \$4,255.20$ ;  $\text{Y} = 6.5 \times 8,000,000 \times 0.0001 = \$5,200.00$ ; **portfolio PV01 \$ = \$9,455.20**. Cross-check:  $4.7276 \times 20,000,000 \times 0.0001 = \$9,455.20$ . ✓

Hedge. Required PV01 reduction \$ =  $(4.7276 - 3.0) \times 20,000,000 \times 0.0001 = 1.7276 \times 2,000 = \$3,455.20$ . Contracts to short \$ =  $3,455.20/75 = 46.07 \approx \text{46 contracts short}$ . This converts a duration target into an executable trade.

**B6. A callable bond is priced at  $P_0 = 98$ . A model reprices it after a  $\pm 30$  bp curve shift:  $P_- = 99.50$  (down),  $P_+ = 96.30$  (up). Compute effective duration and compare with an option-free modified duration of 6.8.**

$D_{\text{Eff}} = \frac{P_- - P_+}{2 P_0 \Delta y} = \frac{99.50 - 96.30}{2 \times 98 \times 0.0030} = \frac{3.20}{0.588} = 5.442$

The embedded call shortens effective duration to **5.44** versus the 6.8 of an otherwise-identical straight bond: the bond can't rally freely when rates fall (the call caps upside), so it responds asymmetrically. That asymmetry — a bigger drop on the sell-off than the gain on the rally — is the negative-convexity fingerprint that modified duration misses.

**B7. Two 10-year annual bonds yield 7%: Bond A pays a 4% coupon, Bond B pays 10%. Use the closed form to show which has the higher duration.**

Closed form:  $D_{\text{Mac}} = \frac{1+y}{y} - \frac{(1+y)+N(c-y)}{c[(1+y)^N - 1] + y}$ , with  $y=0.07$ ,  $N=10$ ,  $(1.07)^{10}=1.967151$ ,  $(1+y)/y = 15.285714$ .

**Bond A ( $c=0.04$ ):** second term numerator \$ =  $1.07 + 10(0.04-0.07) = 0.77$ ; denominator \$ =  $0.04(0.967151)+0.07 = 0.108686$ ; ratio \$ =  $7.08461$ .  $D_{\text{Mac}} = 15.285714 - 7.08461 = \text{8.2011 years}$ .

**Bond B ( $c=0.10$ ):** numerator \$ =  $1.07 + 10(0.10-0.07) = 1.37$ ; denominator \$ =  $0.10(0.967151)+0.07 = 0.166715$ ; ratio \$ =  $8.21742$ .  $D_{\text{Mac}} = 15.285714 - 8.21742 = \text{7.0683 years}$ .

The lower-coupon Bond A has the higher duration (8.20 vs 7.07), confirming the rule: **higher coupon → lower duration**, because more PV arrives early.

## Section C — Interview-Style (model answers)

**C1. "Explain duration to me like I'm a new analyst who only knows how to price a bond."**

You already know price is the sum of discounted cash flows. Duration is what you get when you ask how that price *moves* when the yield moves — it is literally the first derivative of the pricing equation, scaled by price. Two useful readings fall out of the same math: it's the PV-weighted average time to get your money back (in years), and it's the percentage price change per 1% yield move. So a duration of 6 means roughly a 6% price

drop if yields rise 100 bp. It's the single most important risk number in fixed income because it's *additive* — a whole portfolio collapses to one figure and one hedge.

**C2. "A PM says 'just use modified duration on this MBS.' What do you push back with?"**

I'd stop them. Modified duration assumes the cash flows are fixed, because it's the analytic derivative of a fixed-cash-flow pricing formula. An MBS has cash flows that move with rates — when rates fall, homeowners prepay, so the bond shortens exactly when you'd want it to extend. Modified duration can't see that; it can materially misstate or even mis-sign the risk. The correct tool is effective duration: shock the curve up and down, reprice with a prepayment model, and take  $(P_- - P_+)/ (2P_0 \Delta y)$ . For deep-in-the-money pools you can even get negative effective duration.

**C3. "You have a \$500mm book. Walk me through how you'd hedge its rate risk overnight."**

I'd reduce the book to a single PV01 by summing every position's PV01 — valid because PV01 is additive across longs and shorts. Say the book's PV01 is \$180,000 per bp. I pick a liquid instrument — 10-year Treasury futures or a receive-fixed swap — compute its DV01, and short enough to offset \$180,000. If the future's DV01 is \$75/contract, I short  $180,000/75 = 2,400$  contracts. I'd also check key-rate durations so I'm not flattening a parallel shift while leaving a curve-steepening exposure open.

**C4. "Why does duration overstate the loss on a big sell-off? Is that conservative or dangerous?"**

Because the price–yield curve is convex — bowed toward the origin — while duration is a straight tangent line. On a large sell-off the true price sits *above* the tangent, so the linear estimate predicts a bigger loss than actually occurs; on a rally the true price sits above the tangent too, so duration *understates* the gain. For a long-only holder that's conservative — reality is kinder than the estimate in both directions. It becomes dangerous when you're short convexity (short options, some MBS), where the curvature works against you and you need the convexity term to see the true risk.

**C5. "Rank a 10-year zero, a 10-year 4% bond, and a 10-year 10% bond by duration, and explain."**

Highest to lowest: the zero, then the 4%, then the 10%. The zero pays everything at maturity so its duration equals its 10-year maturity — the maximum for that maturity. Adding coupons pulls the balance point in, and the bigger the coupon the more PV arrives early, so the 4% bond has a longer duration than the 10% bond. Same maturity, opposite rate sensitivity, driven entirely by how early the money comes back.

**C6. "What are the limitations of duration, and how do you address each?"**

Three main ones. First, it's *linear/local* — fine for small moves, off for large ones; I add convexity for anything beyond ~25 bp. Second, it assumes a *parallel* shift of a single yield; real curves twist, so I use key-rate durations to hedge reshaping. Third, it assumes *fixed cash flows*; for embedded options I switch to effective duration. Duration is the right first number, but never the only number.

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## Section D — Multiple Choice (with reasoning)

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**D1. A bond's modified duration is 7.0. Yields rise 50 bp. The approximate price change is:**

A. -7.0% B. -3.5% C. +3.5% D. -0.35%

**Answer: B.**  $\Delta P/P \approx -D_{\text{Mod}} \Delta y = -7.0 \times 0.005 = -0.035 = -3.5\%$ . A applies a full 100 bp; C flips the sign; D is off by a factor of ten.

**D2. Which bond has the highest duration, all else equal?**

A. 10-yr, 8% coupon B. 10-yr, 4% coupon C. 10-yr zero-coupon D. 5-yr zero-coupon

**Answer: C.** For a given maturity the zero has the maximum duration (duration = maturity = 10), and lower coupons raise duration, so  $C > B > A$ . D is a zero but only 5-year, so its duration is 5. Longer maturity plus zero coupon wins.

**D3. To convert Macaulay duration to modified duration you:**

A. multiply by  $(1 + y/m)$  B. divide by  $(1 + y/m)$  C. multiply by price D. divide by price

**Answer: B.**  $D_{\text{Mod}} = D_{\text{Mac}} / (1 + y/m)$ , using the periodic yield. A inverts the operation; C and D confuse the step with computing money duration.

**D4. Effective duration should be used instead of modified duration when:**

A. the bond pays semiannually B. yields are very high C. the bond has embedded options or path-dependent cash flows D. the portfolio is large

**Answer: C.** Modified duration assumes fixed cash flows; once cash flows change with the yield path (callables, putables, MBS) you must shock-and-reprice. A is handled by the periodicity convention; B and D don't invalidate the analytic derivative.

**D5. A trader says "my position's PV01 is \$2,400." This means:**

A. the position loses \$2,400 if it defaults B. the position value changes ~\$2,400 per 1 bp yield move C. the position yields 2,400 bp D. the coupon is \$2,400

**Answer: B.** PV01 (DV01/PVBP) is the currency change in value for a 1 bp yield move — the additive unit desks hedge in. The others confuse it with default loss, yield, or coupon.

**D6. Duration overstates the loss on a large yield increase because:**

A. the price–yield curve is concave B. the price–yield curve is convex, so the true price lies above the tangent C. convexity is negative D. the yield used is wrong

**Answer: B.** The convex curve lies above its tangent, so the true price is higher (loss smaller) than the linear estimate predicts. A misstates the curvature; C is false for option-free bonds (convexity is positive); D is irrelevant.

**D7. A portfolio holds \$6mm at duration 2.5 and \$4mm at duration 7.0. Its modified duration is:**

A. 4.75 B. 4.30 C. 9.50 D. 3.50

**Answer: B.** Market-value weighting:  $\$0.6 \times 2.5 + \$0.4 \times 7.0 = 1.5 + 2.8 = 4.30$ . A is a simple (equal-weight) average; C sums the durations; D is wrong entirely.

**D8. An MBS shows a negative effective duration. The most likely cause is:**

A. a data error — duration can't be negative B. falling rates trigger prepayments that shorten the bond and cap its price C. the coupon exceeds the yield D. the bond is a zero

**Answer: B.** Negative convexity: when rates fall, prepayments accelerate, so price can fall as rates fall, giving negative effective duration. A is wrong — effective duration *can* be negative; C and D are unrelated to the prepayment mechanism.

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*Self-check: every numerical answer was verified by re-pricing at the shocked yield and confirming the duration estimate reconciles with the true move (residual = convexity) — B1–B3 tie out, the money-duration/PV01 cross-*

*checks in B4–B5 agree by two routes, and B7's closed-form durations confirm the coupon-effect ordering used in D2 and C5.*



# Chapter 08 — Convexity

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## 1. The Problem / The Need

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Chapter 7 gave us duration — a single number that translates a change in yield into an approximate change in price. Modified duration says: *for a small move in yield, the bond's price moves by roughly minus-duration-times-the-yield-change*. It is the workhorse of every rates desk, the first-order sensitivity that lets a portfolio manager hedge a book, size a trade, or state a P&L exposure in one figure.

But "roughly" and "for a small move" are load-bearing words, and they conceal a systematic error. Duration is a *straight-line* approximation to a *curved* relationship. The true price-yield relationship of a bond is not a line — it bends. And it bends in a very particular, consistent direction: the actual price is almost always *higher* than the straight-line duration estimate predicts, whether yields rise or fall. Duration, used alone, **overstates the loss** when yields rise and **understates the gain** when yields fall. It is wrong in the bondholder's favour — but wrong nonetheless, and increasingly wrong the larger the yield move.

For a 10-basis-point wiggle this error is a rounding footnote. For a 100- or 200-basis-point move — the kind that happens in a rate-hike cycle, a credit event, or a stress test — the error becomes material, tens of cents on a par of 100, real money on a billion-dollar book. Worse, some bonds curve the *wrong* way over part of their range: callable bonds and mortgage-backed securities can have their price gains *choked off* as yields fall, so that duration *overstates* the gain. A risk manager who models these with duration alone will be blindsided precisely when it matters.

We need a second-order correction — a number that captures the *curvature* of the price-yield relationship, tells us how fast duration itself changes as yields move, and lets us build a materially better price-change estimate. That number is **convexity**. This chapter defines it, derives it, computes it, shows how to combine it with duration, and dissects the crucial distinction between *positive* convexity (plain bonds, your friend) and *negative* convexity (callables and MBS, your enemy).

## 2. The Core Idea

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The price of a bond is a function of its yield:  $P(y)$ . Discounting each cash flow by  $(1+y)^{-t}$  makes this function *convex* — it curves upward, like the inside of a bowl. As yield falls, price rises at an *accelerating* rate; as yield rises, price falls at a *decelerating* rate. The slope of this curve is (proportional to) duration, and the crucial fact is that the slope is not constant: it steepens as yields fall and flattens as yields rise.

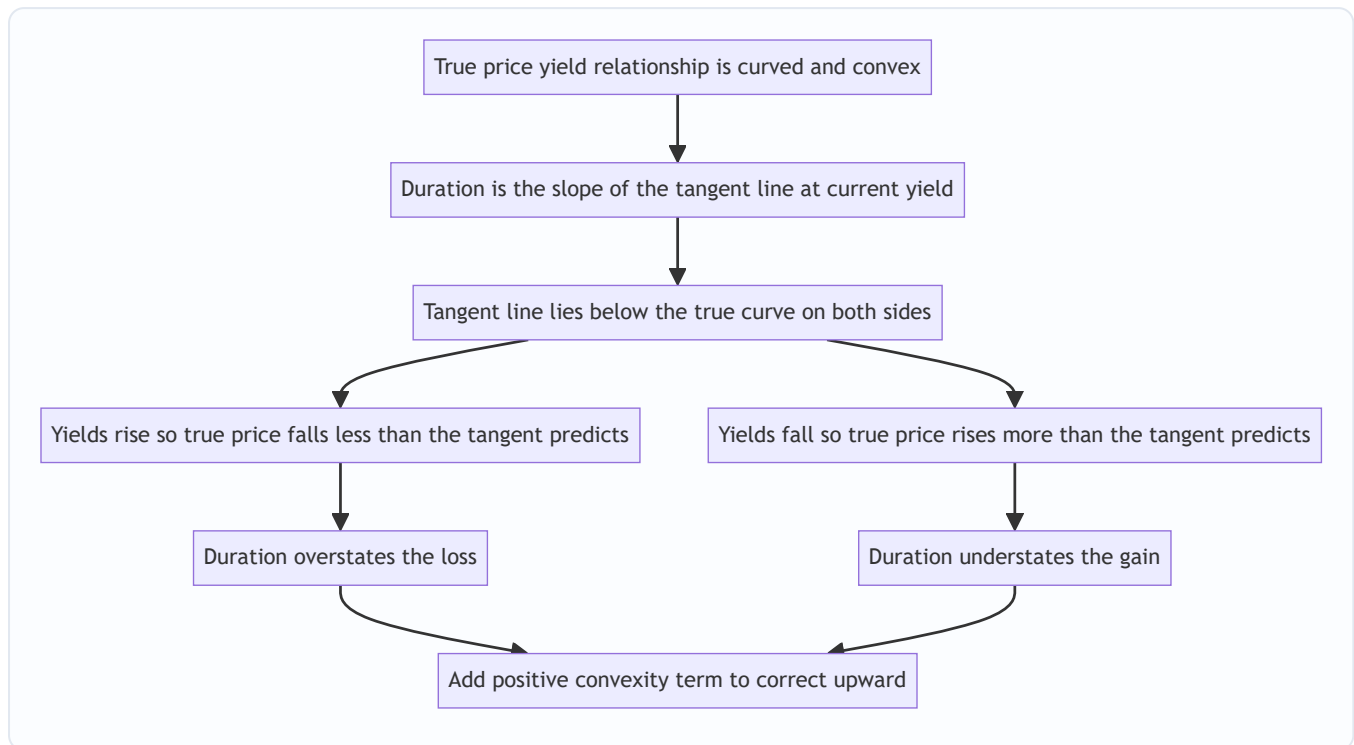
Mathematically, expand price as a Taylor series in yield around the current yield  $y_0$ :

$$\Delta P = \frac{dP}{dy} \Delta y + \frac{1}{2} \frac{d^2P}{dy^2} (\Delta y)^2 + \text{higher-order terms}$$

- The **first derivative**  $\frac{dP}{dy}$  is the source of **duration** — the linear, first-order term.
- The **second derivative**  $\frac{d^2P}{dy^2}$  is the source of **convexity** — the curvature, second-order term.

Duration alone keeps only the first term: a straight tangent line. Convexity adds the second term: a parabola that hugs the true curve far more closely. Because for an option-free bond the second derivative is *positive*, the convexity term is *always positive* (it carries  $(\Delta y)^2$ , which is positive whether  $\Delta y$  is up or down). That is exactly why the curvature correction *adds* to the duration estimate in both directions — pushing the estimate up toward the true price, which lies above the tangent line.

Figure 1 — Why the straight-line duration estimate always sits below the true convex price curve.



### 3. Why / How It Works

**Why the price-yield curve is convex.** Take the price of an option-free bond as the sum of discounted cash flows:

$$P(y) = \sum_{t=1}^n \frac{C_t}{(1+y)^t}$$

Differentiate once:

$$\frac{dP}{dy} = -\sum_{t=1}^n \frac{tC_t}{(1+y)^{t+1}} < 0$$

The first derivative is negative — price falls as yield rises, the familiar inverse relationship, and its magnitude is what duration measures. Differentiate again:

$$\frac{d^2P}{dy^2} = \sum_{t=1}^n \frac{t(t+1)C_t}{(1+y)^{t+2}} > 0$$

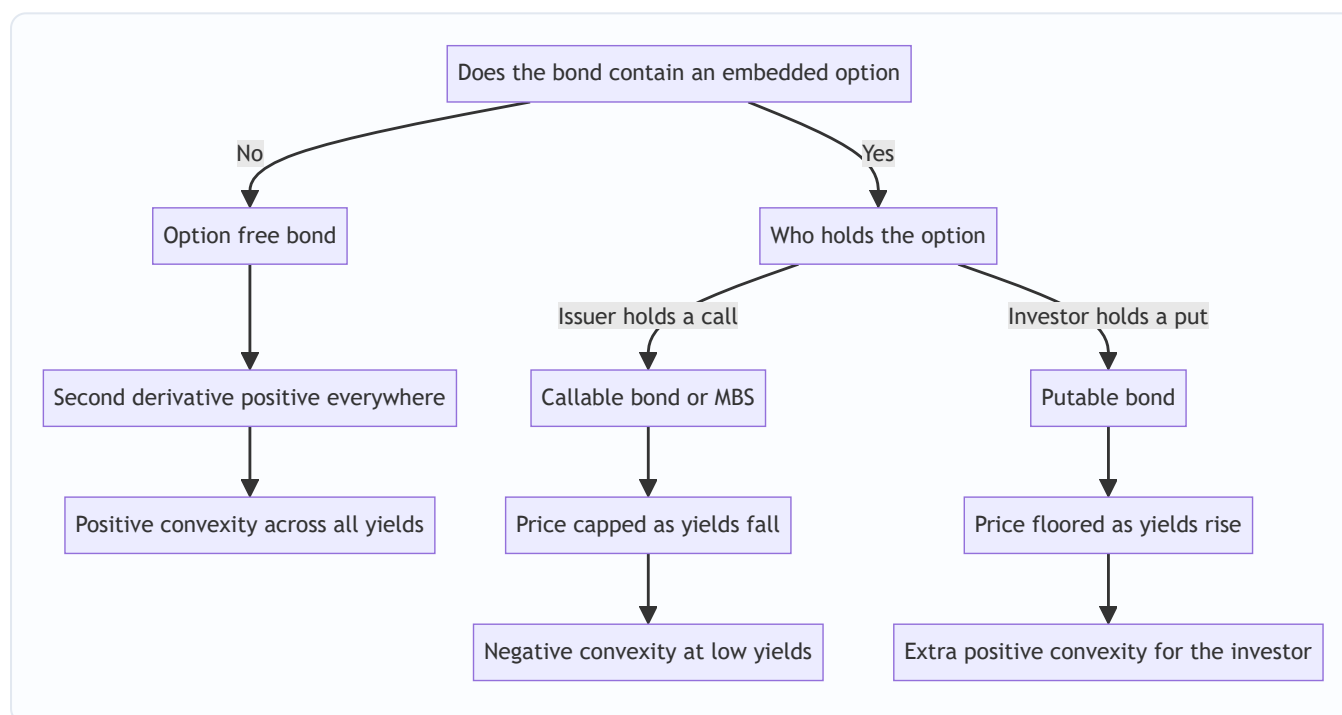
Every term in this sum is positive (all cash flows  $C_t$ , times, and discount factors are positive). So the second derivative is strictly positive for any bond with positive cash flows. A positive second derivative is the definition of a convex function. This is not a special property of some bonds — it is a structural fact of discounting: pushing a fixed cash flow further into a compounding denominator makes the price fall by less and less for each additional unit of yield.

**Why duration alone is biased.** The tangent line at  $y_0$  has the right *slope* but the wrong *shape*. A convex curve always lies *above* its tangent line everywhere except at the point of tangency. So the true price is above the duration line for any  $\Delta y \neq 0$ . When yields rise, the true price falls but stays above the tangent, meaning the actual loss is *smaller* than duration predicts. When yields fall, the true price rises above the tangent, meaning the actual gain is *larger* than duration predicts. The error is one-signed (always in the holder's favour for option-free bonds) and grows with  $(\Delta y)^2$  — quadratically — which is why it explodes for large moves.

**Why convexity is a desirable property.** Because the correction is positive in *both* directions, a more convex bond gains more when yields fall and loses less when yields rise than a less convex bond of the *same* duration. Convexity is a free "heads I win more, tails I lose less" feature. In an efficient market you pay for it: high-convexity bonds trade at slightly richer prices (lower yields) than low-convexity bonds of identical duration. Convexity is worth most when yields are *volatile*, because its payoff scales with  $(\Delta y)^2$  — it is, in effect, a long-volatility position embedded in the bond's math.

**What drives convexity higher or lower.** Three levers move a bond's convexity, holding duration fixed. First, **maturity**: longer bonds have more convexity because the  $t(t+1)$  weighting in the numerator grows quadratically with the timing of cash flows — distant payments contribute disproportionately to curvature. Second, **coupon**: lower-coupon bonds (and especially zeros) have *more* convexity for a given duration, because their cash flows are more concentrated at the far end where curvature is greatest. Third, **cash-flow dispersion**: a portfolio whose payments are spread out (a barbell) is more convex than one whose payments are bunched (a bullet) of the same duration — dispersion around the duration point is, quite literally, what convexity measures. This last fact is why "barbells out-convex bullets" is a desk cliché: the barbell's early and late cash flows both sit far from the duration midpoint, maximising the second moment of the timing distribution.

Figure 2 — Deciding whether a bond has positive or negative convexity.



## 4. Full Content — Formulas and Bond Math

### 4.1 The convexity measure

Convexity is defined as the *second derivative of price with respect to yield, scaled by price*:

$$\text{Convexity} = \frac{1}{P} \cdot \frac{d^2 P}{dy^2}$$

For a bond priced with periodic yield, the closed-form (annual compounding, cash flows at times  $t = 1, 2, \dots, n$ ) is:

$$\text{Convexity} = \frac{1}{P \cdot (1+y)^2} \sum_{t=1}^n \frac{t(t+1) \cdot C_t}{(1+y)^t}$$

Equivalently, writing the  $(1+y)^{t+2}$  directly in the denominator:

$$\text{Convexity} = \frac{1}{P} \sum_{t=1}^n \frac{t(t+1)C_t}{(1+y)^{t+2}}$$

**Compounding frequency.** If the bond pays  $m$  times per year, define everything in periods (periodic yield  $y/m$ , period count  $t = 1 \dots n \cdot m$ ), then divide the periodic convexity by  $m^2$  to annualise:

$$\frac{\text{Convexity}}{\text{annual}} = \frac{\text{Convexity}}{\text{periodic}} \cdot m^2$$

The  $m^2$  (not  $m$ ) appears because convexity is a *second-order* term — it scales with the square of the period length, mirroring how modified duration divides by  $m$  once.

## 4.2 The effective (finite-difference) convexity

When a bond has embedded options (so cash flows themselves depend on yield), you cannot differentiate a fixed cash-flow formula. Instead, reprice the bond under a valuation model at three yields — down by  $\Delta y$ , unchanged, up by  $\Delta y$  — and use a numerical second difference:

$$\text{Effective Convexity} = \frac{V_{-} + V_{+} - 2V_0}{V_0 (\Delta y)^2}$$

where  $V_0$  is the base price,  $V_{+}$  the price after yields rise by  $\Delta y$ , and  $V_{-}$  the price after yields fall by  $\Delta y$ . The companion effective duration is:

$$\text{Effective Duration} = \frac{V_{-} - V_{+}}{2V_0 \Delta y}$$

For an option-free bond the effective and closed-form measures agree (up to a tiny discretisation error). For callables and MBS,  $V_{+}$  and  $V_{-}$  come from an option-adjusted-spread (OAS) model that lets the cash flows change, and the effective convexity can come out *negative*.

## 4.3 Combining duration and convexity — the price-change formula

The second-order Taylor expansion, divided through by price, gives the estimated *percentage* price change:

$$\frac{\Delta P}{P} \approx -D_{\text{mod}} \cdot \Delta y + \frac{1}{2} \text{Convexity} \cdot (\Delta y)^2$$

- The first term is the **duration effect** — linear, signed (down for a rise, up for a fall).
- The second term is the **convexity adjustment** — always positive for positive-convexity bonds, always negative for negative-convexity bonds, and quadratic in the yield move.

Some texts and systems quote a "convexity" that already folds in the factor of  $\frac{1}{100}$ , or that is scaled by 100 or 10,000. Always check the convention: does your reported number, multiplied by  $(\Delta y)^2$ , need a further  $\frac{1}{100}$  or not? The math is unambiguous; only the bookkeeping differs.

## 4.4 Dollar duration and dollar convexity

For P&L and hedging in currency terms rather than percentages:

$$\Delta P \approx -\text{big}(D_{\text{mod}} \cdot P) \Delta y + \frac{1}{2} \text{big}(\text{Convexity} \cdot P) (\Delta y)^2$$

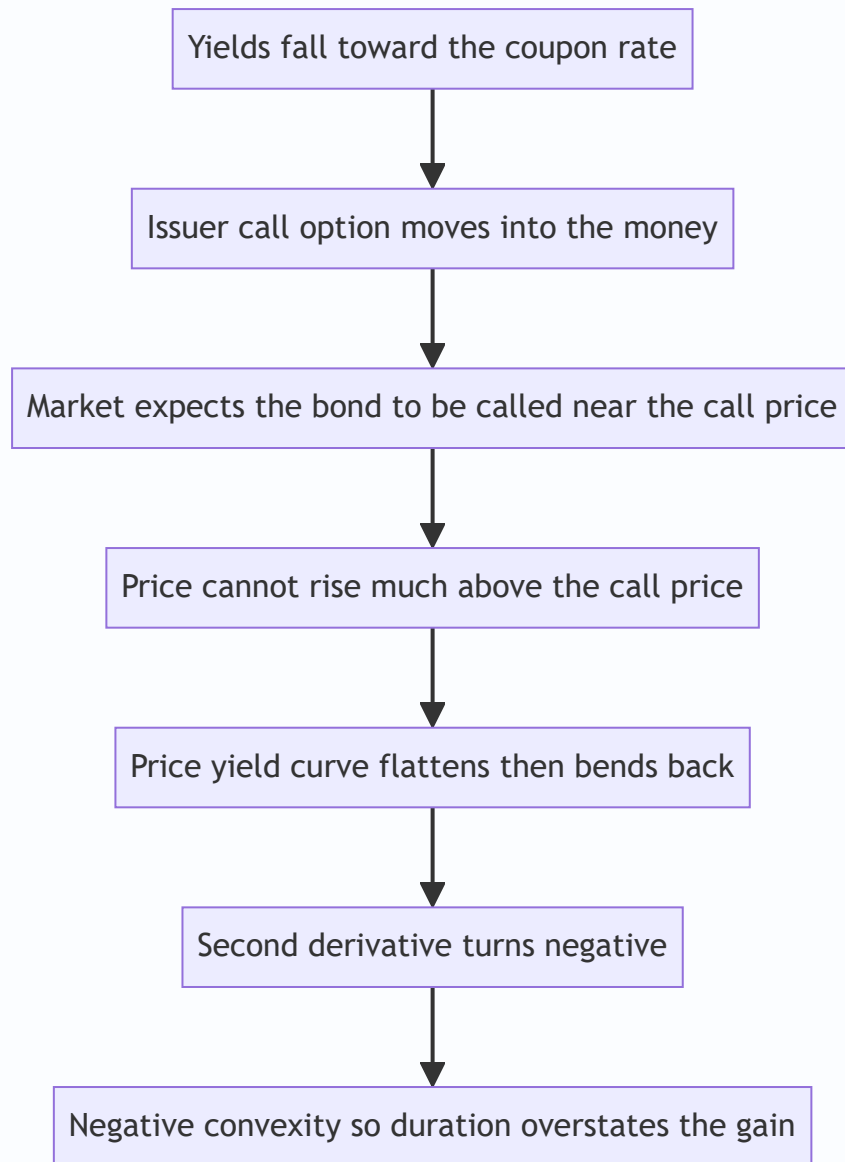
The bracketed quantities are *dollar duration* (also called DV01 when scaled to a 1-bp move) and *dollar convexity* — the sensitivities in monetary units, which is what actually hits the book.

## 4.5 Positive vs negative convexity

| Property                  | Positive convexity (option-free, putable)    | Negative convexity (callable, MBS)                         |
|---------------------------|----------------------------------------------|------------------------------------------------------------|
| Sign of second derivative | Positive throughout                          | Negative over part of the yield range                      |
| Price-yield shape         | Bows toward the origin, always above tangent | Flattens or bends back as yields fall                      |
| When yields fall a lot    | Price rises at an accelerating rate          | Price gain is capped near the call price                   |
| Convexity adjustment term | Adds to the duration estimate                | Subtracts from the duration estimate                       |
| Value to investor         | Desirable, worth paying for                  | Undesirable, investor demands higher yield                 |
| Duration behaviour        | Duration rises as yields fall                | Duration <i>shortens</i> as yields fall (call more likely) |

**Why callables and MBS go negatively convex.** A callable bond is a straight bond *minus* a call option the investor has sold to the issuer. As yields fall, the issuer's option to refinance moves into the money; the bond's upside is truncated near the call price because the market knows it will likely be called. The price-yield curve, instead of continuing to bow upward, flattens and rolls over — "price compression." Mortgage-backed securities inherit the same shape because homeowners hold a prepayment option: when rates fall they refinance, handing the MBS holder cash exactly when reinvestment is worst. Over the low-yield region these instruments exhibit a *negative* second derivative, so the convexity adjustment *subtracts*, and duration *overstates* the price gain.

*Figure 3 — Price compression turns a callable bond negatively convex at low yields.*



## 5. Worked Examples

### 5.1 Setup — a clean option-free bond

Take a **5-year bond, 5% annual coupon, face 100, priced to yield 5%**. Because coupon equals yield, it prices at par:  $P_0 = 100$ . Cash flows are 5 at years 1–4 and 105 at year 5.

**Step 1 — Modified duration (analytical).** Compute Macaulay duration first,  $D_{\text{mac}} = \frac{1}{P} \sum t \cdot \text{PV}(C_t)$ .

| $t$        | $C_t$ | $(1.05)^{-t}$ | $\text{PV}(C_t)$  | $t \cdot \text{PV}$ |
|------------|-------|---------------|-------------------|---------------------|
| 1          | 5     | 0.952381      | 4.761905          | 4.761905            |
| 2          | 5     | 0.907029      | 4.535147          | 9.070295            |
| 3          | 5     | 0.863838      | 4.319188          | 12.957565           |
| 4          | 5     | 0.822702      | 4.113513          | 16.454051           |
| 5          | 105   | 0.783526      | 82.270247         | 411.351237          |
| <b>Sum</b> |       |               | <b>100.000000</b> | <b>454.595054</b>   |

$D_{\text{mac}} = \frac{454.595054}{100} = 4.545951 \text{ years}$ ,  $D_{\text{mod}} = \frac{4.545951}{1.05} = 4.329478$

**Step 2 — Convexity (analytical).** Use  $\text{Convexity} = \frac{1}{P(1+y)^2} \sum t(t+1)C_t(1+y)^{-t}$ .

| $t$        | $t(t+1)$ | $C_t$ | $t(t+1)C_t(1.05)^{-t}$ |
|------------|----------|-------|------------------------|
| 1          | 2        | 5     | 9.523810               |
| 2          | 6        | 5     | 27.210884              |
| 3          | 12       | 5     | 51.830256              |
| 4          | 20       | 5     | 82.270247              |
| 5          | 30       | 105   | 2468.107423            |
| <b>Sum</b> |          |       | <b>2638.942620</b>     |

$\text{Convexity} = \frac{2638.942620}{100 \times (1.05)^2} = \frac{2638.942620}{110.25} = 23.9360$

So  $D_{\text{mod}} = 4.3295$  and Convexity  $= 23.936$ .

## 5.2 A 100-bp rise in yield — duration alone vs duration-plus-convexity

Reprice exactly at  $y = 6\%$ :  $P_+ = 5 \cdot \frac{1 - (1.06)^{-5}}{0.06} + 100 \cdot (1.06)^{-5} = 21.061819 + 74.725817 = 95.787636$

**Actual change:**  $95.787636 - 100 = -4.212364$ , i.e.  $-4.2124\%$ .

**Duration-only estimate:**  $-D_{\text{mod}} \cdot \Delta y = -4.329478 \times 0.01 = -4.329478\%$ .

**Convexity adjustment:**  $\frac{1}{2} \times 23.936 \times (0.01)^2 = 0.5 \times 23.936 \times 0.0001 = +0.119680\%$ .

**Combined estimate:**  $-4.329478 + 0.119680 = -4.209798\%$ .

| Method               | Estimated $\Delta P/P$ | Error vs actual $-4.21236\%$   |
|----------------------|------------------------|--------------------------------|
| Duration only        | $-4.32948\%$           | $-0.11711\%$ (overstates loss) |
| Duration + convexity | $-4.20980\%$           | $+0.00256\%$ (near-exact)      |

Duration alone overstates the loss by about 12 cents per 100 face; adding convexity cuts the residual error to about a quarter of a *basis point* of price. **Reconciles.**

### 5.3 A 100-bp fall in yield — the correction flips to the upside

Reprice at  $y = 4\%$ :  $P_- = 5 \cdot \frac{1 - (1.04)^{-5}}{0.04} + 100 \cdot (1.04)^{-5} = 22.259109 + 82.192713 = 104.451822$

**Actual change:**  $+\$4.451822\%$ .

**Duration-only:**  $-\$4.329478 \times (-0.01) = +\$4.329478\%$ .

**Convexity adjustment:**  $\frac{1}{2} \times 23.936 \times (0.01)^2 = +0.119680\%$  (still positive —  $\Delta y^2$  is positive either way).

**Combined:**  $+\$4.329478 + 0.119680 = +\$4.449158\%$ .

| Method               | Estimated $\Delta P/P$ | Error vs actual $+\$4.45182\%$    |
|----------------------|------------------------|-----------------------------------|
| Duration only        | $+\$4.32948\%$         | $-\$0.12234\%$ (understates gain) |
| Duration + convexity | $+\$4.44916\%$         | $-\$0.00266\%$ (near-exact)       |

Note the symmetry: on the way up duration *understated* the gain, on the way down it *overstated* the loss — the same convexity term  $+\$0.1197\%$  fixes both, because it is always additive for a positive-convexity bond.

**Reconciles.**

### 5.4 A large 200-bp move — where convexity earns its keep

Now push the yield to  $y = 7\%$  (a 200-bp rise):  $P = 5 \cdot \frac{1 - (1.07)^{-5}}{0.07} + 100 \cdot (1.07)^{-5} = 20.500987 + 71.298618 = 91.799605$

**Actual change:**  $-\$8.200395\%$ .

**Duration-only:**  $-\$4.329478 \times 0.02 = -\$8.658955\%$ .

**Convexity adjustment:**  $\frac{1}{2} \times 23.936 \times (0.02)^2 = 0.5 \times 23.936 \times 0.0004 = +0.478726\%$ .

**Combined:**  $-\$8.658955 + 0.478726 = -\$8.180229\%$ .

| Method               | Estimated $\Delta P/P$ | Error vs actual $-\$8.20040\%$         |
|----------------------|------------------------|----------------------------------------|
| Duration only        | $-\$8.65895\%$         | $-\$0.45856\%$ (overstates loss badly) |
| Duration + convexity | $-\$8.18023\%$         | $+\$0.02016\%$ (small residual)        |

At 200 bp the duration-only error has ballooned to nearly *half a percent of price* — four times the 100-bp error, exactly as the quadratic scaling predicts. Convexity slashes it to two basis points. The residual is the third-order term, which duration-plus-convexity still omits. **Reconciles, and demonstrates the quadratic growth of the pure-duration error.**

### 5.5 A negatively convex callable — effective measures go the other way

Now take a **callable** version of a bond, valued under a model, callable at 101. Suppose an OAS model returns these prices for a base yield of 5% and a  $\Delta y = 100$  bp:  $V_0 = 100.20$ , and — critically — as yields fall the price is *capped* near the call price while as yields rise it behaves normally.



| Scenario           | Yield | Modelled price \$V\$ |
|--------------------|-------|----------------------|
| Yields fall 100 bp | 4%    | $V_- = 100.85$       |
| Base               | 5%    | $V_0 = 100.20$       |
| Yields rise 100 bp | 6%    | $V_+ = 96.30$        |

Notice the asymmetry: on the downside the price rose only 0.65 (call compression bites), while on the upside it fell a full 3.90. Compute the effective measures:

$$\text{Effective Duration} = \frac{V_- - V_+}{2V_0 \Delta y} = \frac{100.85 - 96.30}{2 \times 100.20 \times 0.01} = \frac{4.55}{2.004} = 2.271$$

$$\text{Effective Convexity} = \frac{V_- + V_+ - 2V_0}{V_0 (\Delta y)^2} = \frac{100.85 + 96.30 - 200.40}{100.20 \times (0.01)^2} = \frac{-3.25}{0.01002} = -324.4$$

The convexity is **strongly negative**. Apply the combined formula to a 100-bp rally ( $\Delta y = -0.01$ ):

- Duration effect:  $-2.271 \times (-0.01) = +2.271\%$
- Convexity adjustment:  $\frac{1}{2} \times (-324.4) \times (-0.01)^2 = -1.622\%$
- Combined estimate:  $+2.271 - 1.622 = +0.649\%$ , i.e. a price of about  $100.20 \times 1.00649 = 100.85$  — matching  $V_-$ .

Here the negative convexity term *subtracts* 1.62%, choking the rally gain from the 2.27% that duration alone would promise down to the true 0.65%. A trader who hedged this bond on duration alone would have thought a rate rally was worth 2.3% and been stunned to collect only 0.6% — the missing 1.6% is the short-gamma cost of the embedded call. **Reconciles, and shows negative convexity subtracting.**

## 5.6 Cross-check — effective (finite-difference) measures on the plain bond

Using the repriced values  $V_- = 104.451822$  (at 4%),  $V_0 = 100$ ,  $V_+ = 95.787636$  (at 6%),  $\Delta y = 0.01$ :

$$\text{Effective Duration} = \frac{104.451822 - 95.787636}{2 \times 100 \times 0.01} = \frac{8.664186}{2} = 4.33209$$

$$\text{Effective Convexity} = \frac{104.451822 + 95.787636 - 200}{100 \times (0.01)^2} = \frac{0.239458}{0.01} = 23.9458$$

The finite-difference figures (4.3321 and 23.9458) match the analytical ones (4.3295 and 23.9360) to within a fraction of a percent — the tiny gap is discretisation error, itself a convexity effect on the duration estimate.

**Both methods reconcile.**

## 6. Connections

- **Duration (Ch. 7)** is the first-order term; convexity is the second-order term of the *same* Taylor expansion. They are not rivals — they are consecutive coefficients of one series. You cannot understand convexity without duration, and you should never quote a large-move price change from duration without the convexity companion.
- **Spot and forward rates (Ch. 5) / term structure (Ch. 6):** modified duration and convexity assume a *parallel* shift in a single yield. Real curves twist and butterfly. Key-rate durations and key-rate convexities generalise these measures to non-parallel moves, but the local curvature intuition here is the foundation.

- **Embedded options (callables, putables, MBS):** convexity is where the option's gamma leaks into the bond's price behaviour. Negative convexity is literally short option gamma; the OAS framework exists to price exactly this. A mortgage trader's whole day is managing negative convexity.
- **Options and the Greeks:** duration is the bond analogue of an option's *delta*; convexity is the analogue of *gamma*. "Long convexity" and "long gamma" describe the same long-volatility payoff — you profit from large moves in either direction.
- **Immunisation and ALM:** matching duration alone immunises a portfolio against small parallel shifts; matching convexity as well makes the hedge robust to larger moves, which is why insurers and pension funds track both.
- **Barbell vs bullet:** two portfolios with identical duration can have very different convexity. A barbell (short + long bonds) has *more* convexity than a duration-matched bullet (single intermediate maturity) — the classic demonstration that convexity is a free lunch you pay for in yield give-up.

## 7. Key Terms

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- **Convexity:** the second derivative of price with respect to yield, scaled by price; measures the curvature of the price-yield relationship.
- **Convexity adjustment:** the  $\frac{1}{2} \cdot \text{Convexity} \cdot (\Delta y)^2$  term added to the duration estimate to correct for curvature.
- **Positive convexity:** price-yield curve bows upward everywhere; the correction is always additive; characteristic of option-free and putable bonds.
- **Negative convexity:** price-yield curve flattens or bends back over part of its range; the correction subtracts; characteristic of callable bonds and MBS.
- **Effective convexity:** convexity computed by repricing under a model at yields up, down, and unchanged — required when cash flows depend on yield.
- **Price compression:** the flattening of a callable bond's price near the call price as yields fall, the mechanism producing negative convexity.
- **Dollar convexity:** convexity times price — the curvature sensitivity in currency units, used in P&L attribution.
- **Gamma (analogy):** the options-world name for the same second-order sensitivity; long convexity equals long gamma.

## 8. Common Confusions

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**"Higher convexity is always better, full stop."** Higher convexity is always *desirable to hold*, but it is not *free* — the market charges for it via a lower yield. The trade-off is real: in a low-volatility environment you may be overpaying for curvature you never get to use. Convexity's payoff scales with  $(\Delta y)^2$ , so it only pays off when yields actually move a lot.

**"Convexity is a correction to duration's number."** No — convexity does not fix a *wrong duration*; duration is correct as the exact slope. Convexity is a *separate, additional* term capturing what the linear approximation structurally cannot: curvature. Both can be exactly right and the estimate still needs both terms.

**"The convexity term can be negative because  $\Delta y$  is negative."** The convexity term carries  $(\Delta y)^2$ , which is non-negative regardless of the direction of the yield move. The term's sign comes

from the *convexity measure* itself, not from  $\Delta y$ . For option-free bonds it is always additive; only negative-convexity instruments make it subtract.

**"Callable bonds always have negative convexity."** Only in the *low-yield* region where the call is near the money. At high yields, far from the call, a callable behaves like a straight bond with ordinary positive convexity. Negativity is local, not global.

**"Duration and convexity give the exact price."** They give a *second-order* estimate. A residual remains from third- and higher-order terms, visible in the 200-bp example (two-bp leftover). For an exact answer you reprice from the cash-flow formula; duration-plus-convexity is a fast, portable *approximation*.

**"Convexity has intuitive units like years."** Duration has a clean "years" interpretation; convexity's raw units are "years-squared" and are not intuitively meaningful. Treat it as a pure scaling constant in the price-change formula, and always confirm whether the  $\frac{1}{12}$  and any  $\times 100$  scaling are already baked in.

## 9. Recap

The price-yield relationship of a bond is convex because discounting produces a positive second derivative — every cash flow's present value bends the curve upward. Duration, the first-order slope, approximates price changes with a straight tangent line, but a convex curve always sits above its tangent. So duration alone **overstates losses when yields rise and understates gains when yields fall**, with an error that grows with the *square* of the yield move — negligible for small wiggles, material for 100–200-bp shocks.

Convexity is the scaled second derivative that quantifies this curvature. Adding the term  $\frac{1}{12} \cdot \text{Convexity} \cdot (\Delta y)^2$  to the duration estimate produces a far tighter approximation: in our worked 5-year par bond, duration alone missed a 100-bp move by about 12 cents, while duration-plus-convexity missed by a quarter of a basis point, and at 200 bp it cut a 46-cent error to 2 cents. For option-free bonds the correction is always *additive* — convexity is a desirable, volatility-loving property investors pay for. For callable bonds and MBS, price compression from the issuer's or homeowner's option flips convexity *negative* in the low-yield region, the correction *subtracts*, and duration *overstates* the gain — the trap that makes these instruments dangerous to model with duration alone.

## 10. Quick-Reference / Interview Points

- **Convexity measure:**  $\frac{1}{P} \frac{d^2 P}{dy^2} = \frac{1}{P(1+y)^2} \sum_t \frac{t(t+1)C_t}{(1+y)^t}$ .
- **Combined price change:**  $\frac{\Delta P}{P} \approx -D_{\text{mod}} \Delta y + \frac{1}{12} \text{Convexity} (\Delta y)^2$ .
- **Effective convexity:**  $\frac{V_- + V_+ - 2V_0}{V_0 (\Delta y)^2}$  — use when cash flows depend on yield (options).
- **Sign intuition:** convexity term is *always positive* for option-free bonds (carries  $(\Delta y)^2$ ); it *adds* on both a rise and a fall.
- **Duration's bias:** overstates loss on a rise, understates gain on a fall — always in the holder's favour for positive convexity.
- **Error scaling:** duration-only error grows with  $(\Delta y)^2$ ; double the yield move, quadruple the error.
- **Positive convexity** = option-free and putable bonds; **negative convexity** = callables and MBS at low yields (price compression).

- **Convexity  $\approx$  gamma; duration  $\approx$  delta.** Long convexity = long volatility: gain more up, lose less down.
- **It's not free:** higher-convexity bonds yield less; a barbell out-convexes a duration-matched bullet but gives up yield.
- **Annualisation:** divide periodic convexity by  $m^2$  (duration divides by  $m$ ) for  $m$  payments per year.
- **One-liner for the desk:** "Duration tells me where I am; convexity tells me how fast that's changing — and for callables it's changing the wrong way when rates rally."

# Q&A — Convexity

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Practice bank for the *Convexity* chapter. Work each problem before reading the answer. Every question is followed by a full worked solution; numerical items reconcile the approximation against the exact repricing.

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## Section A — Concept Check

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### A1. In one sentence, what does convexity measure?

Convexity measures the *curvature* of the price–yield relationship — i.e. the rate at which a bond's modified duration itself changes as yields change. Formally it is the second derivative of price with respect to yield, scaled by price.

### A2. Duration is a straight-line estimate. Why is that line always wrong in the same direction for an option-free bond?

The true price–yield curve is convex (bowed toward the origin). A straight tangent line lies *below* the curve everywhere except at the point of tangency. So the duration-only estimate always *understates* the actual price: it over-predicts the loss when yields rise and under-predicts the gain when yields fall. Convexity is the correction term that pulls the estimate back onto the curve.

### A3. Is positive convexity good or bad for a bondholder, and why?

Positive convexity is good. It means prices rise more when yields fall than they fall when yields rise by the same amount — an asymmetry in the investor's favour. All else equal, a more convex bond outperforms a less convex bond of equal duration whenever yields move, in either direction.

### A4. Name three factors that increase the convexity of an option-free bond.

(1) Longer maturity, (2) lower coupon, and (3) lower yield to maturity. Each of these also raises duration; convexity generally rises with duration and, for a given duration, with the dispersion of the cash flows across time.

### A5. What is negative convexity and which instruments exhibit it?

Negative convexity means the price–yield curve bends the "wrong" way over some yield range — duration *shortens* as yields fall. Callable bonds and mortgage-backed securities (MBS) show it: as yields fall, the embedded call/prepayment option becomes more valuable to the issuer/borrower, capping the bond's price appreciation. The result is that these bonds gain less on a rally than they lose on a sell-off.

### A6. Distinguish "modified convexity" from "effective convexity."

Modified (analytical) convexity is computed from the bond's own fixed cash flows using calculus, assuming those cash flows do not change with yield. Effective convexity re-prices the bond under an upward and a downward yield shock using a valuation model that *lets the cash flows change* (e.g. exercise of a call). Effective convexity is the correct measure for bonds with embedded options; modified convexity is fine for option-free bonds.

### A7. Convexity has units. What are they, and what does that imply for scaling?

Convexity is expressed in years-squared (time<sup>2</sup>). Because the price-change term uses  $(\Delta y)^2$ , you must keep yield in decimal form consistently. A convexity of 80 contributes  $\frac{1}{2} \times 80 \times (0.01)^2 = 0.004 = 0.40\%$  for a 100 bp move.

## A8. Two bonds have identical duration. What does convexity let you say about them?

Duration alone makes them look equivalent for small moves. Convexity breaks the tie: the higher-convexity bond will do better for any large move in yields, up or down. Investors expecting volatility pay up for convexity; in calm markets the higher-convexity bond typically offers slightly lower yield as compensation.

## Section B — Numerical Bond-Math Problems

### B1. Effective duration and convexity from a repricing table.

A bond is priced at  $P_0 = 100.00$ . Shocking the yield curve by  $\pm 25$  bp gives: - Yield **+25 bp**:  $P_+ = 98.15$  - Yield **-25 bp**:  $P_- = 101.90$

Compute effective duration and effective convexity, then estimate the price change for a +100 bp move and comment.

*Step 1 — Effective duration.* 
$$D_{\text{eff}} = \frac{P_- - P_+}{2 \cdot P_0 \cdot \Delta y} = \frac{101.90 - 98.15}{2 \times 100 \times 0.0025} = \frac{3.75}{0.5} = 7.5$$

*Step 2 — Effective convexity.* 
$$C_{\text{eff}} = \frac{P_- + P_+ - 2P_0}{P_0 \cdot (\Delta y)^2} = \frac{101.90 + 98.15 - 200}{100 \times (0.0025)^2} = \frac{0.05}{100 \times 0.00000625} = \frac{0.05}{0.000625} = 80$$

*Step 3 — Estimate for  $\Delta y = +100$  bp (+0.01).* 
$$\frac{\Delta P}{P} \approx -D_{\text{eff}} \cdot \Delta y + \frac{1}{2} C_{\text{eff}} (\Delta y)^2 = -7.5(0.01) + \frac{1}{2}(80)(0.01)^2 = -0.0750 + 0.0040 = -0.0710 = -7.10\%$$

*Comment.* Duration alone predicts -7.50%. Convexity adds back +0.40%, giving -7.10%. The convexity term is positive for *both* directions of move, which is exactly the favourable asymmetry positive convexity provides.

### B2. Computing convexity directly from cash flows.

A 3-year annual bond, 6% coupon, face 100, trades at par (YTM = 6%, so  $P = 100$ ). Use the closed form 
$$C = \frac{1}{P(1+y)^2} \sum_{t=1}^n \frac{t(t+1) \cdot CF_t}{(1+y)^t}$$

*Step 1 — Build the  $t(t+1) \cdot CF_t / (1+y)^t$  column with  $y = 0.06$ :*

| t | $CF_t$ | t(t+1) | $t(t+1) \cdot CF_t$ | $(1.06)^t$ | Term      |
|---|--------|--------|---------------------|------------|-----------|
| 1 | 6      | 2      | 12                  | 1.0600     | 11.3208   |
| 2 | 6      | 6      | 36                  | 1.1236     | 32.0399   |
| 3 | 106    | 12     | 1272                | 1.191016   | 1068.0079 |

Sum of terms = 11.3208 + 32.0399 + 1068.0079 = **1111.369**.

*Step 2 — Scale.* 
$$C = \frac{1111.369}{100 \times (1.06)^2} = \frac{1111.369}{112.36} = 9.891 \text{ years}^2$$

*Step 3 — Get modified duration for the reconciliation in B3.* Macaulay duration: 
$$D_{\text{mac}} = \frac{1}{P} \sum \frac{t \cdot CF_t}{(1+y)^t} = \frac{5.6604 + 10.6800 + 267.006}{100} = 2.8335 \text{ yrs}$$
$$D_{\text{mod}} = \frac{2.8335}{1.06} = 2.6731$$

### B3. Reconcile the estimate against exact repricing (continuing B2).

Estimate the price change for a +100 bp rise to 7%, then reprice exactly and compare.

*Step 1 — Approximation.* 
$$\frac{\Delta P}{P} \approx -2.6731(0.01) + \frac{1}{2}(9.891)(0.01)^2 = -0.026731 + 0.000495 = -0.026236 = -2.6236\%$$

Step 2 — Exact repricing at  $y = 7\%$ . 
$$P = \frac{6}{1.07} + \frac{6}{1.07^2} + \frac{106}{1.07^3} = 5.6075 + 5.2407 + 86.5275 = 97.3757$$

$$\frac{\Delta P}{P} = \frac{97.3757 - 100}{100} = -2.6243\%$$

Step 3 — Compare. Estimate  $-2.6236\%$  vs actual  $-2.6243\%$ : a residual of under  $0.001\%$ . The duration-plus-convexity model captures essentially the entire move. Duration alone would have predicted  $-2.6731\%$ , an error of about  $+0.05\%$ , which the convexity term corrects.

#### B4. Show convexity's symmetry in the investor's favour (continuing B2).

Reprice the same bond for a  $-100$  bp move to  $5\%$  and compare the gain against the loss from B3.

Exact repricing at  $y = 5\%$ : 
$$P = \frac{6}{1.05} + \frac{6}{1.05^2} + \frac{106}{1.05^3} = 5.7143 + 5.4422 + 91.5776 = 102.7341$$

Gain =  $+2.7341\%$ .

Compare with the  $-100$  bp loss of  $-2.6243\%$  (B3). The rally gain ( $+2.73\%$ ) exceeds the sell-off loss ( $-2.62\%$ ) in absolute terms by about  $0.11\%$ . Same yield shock, bigger upside than downside — the hallmark of positive convexity.

#### B5. Dollar convexity contribution.

A portfolio holds  $\$50$  million (market value) of a bond with modified duration  $6.0$  and convexity  $90$ . Estimate the value change for a  $-150$  bp parallel shift.

Step 1 — Percentage change. 
$$\frac{\Delta P}{P} = -6.0(-0.015) + \frac{1}{2}(90)(-0.015)^2 = 0.09 + 0.010125 = 0.100125 = 10.0125\%$$

Step 2 — Dollar change. 
$$\Delta V = 0.100125 \times \$50,000,000 = \$5,006,250$$

Comment. Duration alone gives  $+9.00\%$  ( $\$4.5$ m). Convexity adds  $+1.01\%$  ( $\$506,250$ ). For a large  $150$  bp move the convexity term is material — more than  $10\%$  of the total estimated gain.

#### B6. Which bond wins? Same duration, different convexity.

Bond X:  $D_{\text{mod}} = 7$ ,  $C = 60$ . Bond Y:  $D_{\text{mod}} = 7$ ,  $C = 130$ . Both yield the same. Estimate each bond's return for a  $\pm 200$  bp shock and state which you would hold if you expected a large move of uncertain direction.

Rates up  $200$  bp ( $\Delta y = +0.02$ ): - X:  $-7(0.02) + \frac{1}{2}(60)(0.02)^2 = -0.14 + 0.012 = -12.80\%$  - Y:  $-7(0.02) + \frac{1}{2}(130)(0.02)^2 = -0.14 + 0.026 = -11.40\%$

Rates down  $200$  bp ( $\Delta y = -0.02$ ): - X:  $+0.14 + 0.012 = +15.20\%$  - Y:  $+0.14 + 0.026 = +16.60\%$

Decision. Y loses  $1.4\%$  less on the sell-off and gains  $1.2\%$  more on the rally. If a large move is likely but its direction is uncertain, hold Y — higher convexity dominates in both tails.

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## Section C — Interview-Style Questions with Model Answers

### C1. "Walk me through why we bother with convexity if we already have duration."

Model answer. Duration is a first-order, linear approximation of a curved relationship. It is accurate only for infinitesimal yield changes. Real markets move in tens or hundreds of basis points, and over those ranges the curvature matters: the true price line bows away from the duration tangent, so duration systematically overstates losses and understates gains. Convexity is the second-order term that captures that curvature. In a Taylor-series sense, price change  $\approx -\text{duration} \times \Delta y + \frac{1}{2} \times \text{convexity} \times (\Delta y)^2$ . Duration handles the slope; convexity handles how the slope changes. For risk management and for comparing bonds of equal duration, ignoring convexity leaves real money on the table.



**C2. "A client says two bonds have the same duration so they're the same risk. Correct them."**

*Model answer.* Equal duration means equal *sensitivity to small moves* — the first-order risk. But it says nothing about behaviour in large moves, which is governed by convexity. The more convex bond will outperform for any sizeable shift in yields, up or down, because its price gains accelerate on rallies and decelerate on sell-offs. So two equal-duration bonds are not equivalent risk: one is a better holding in volatile markets. The market prices this — the higher-convexity bond usually trades at a slightly lower yield. Duration-matching alone is insufficient for immunization or hedging when large or volatile rate moves are plausible.

**C3. "Explain negative convexity to a portfolio committee and why it matters for an MBS book."**

*Model answer.* Negative convexity means the bond's price appreciation is capped as yields fall. In an MBS, homeowners refinance when rates drop, prepaying their mortgages at par; the investor's high-coupon cash flows get returned early and must be reinvested at the new, lower rate. So the bond fails to rally the way an option-free bond would — its duration actually *shortens* into a rally, exactly when you'd want it to lengthen. The practical consequences: (1) the book gains less on rate declines than a Treasury of similar duration but loses roughly the full amount on rate rises — an unfavourable asymmetry; (2) duration is unstable, so hedges must be rebalanced dynamically; and (3) you should be compensated with extra yield (option-adjusted spread) for writing that embedded prepayment option to borrowers.

**C4. "How would you measure convexity for a callable bond, and why not use the textbook formula?"**

*Model answer.* Use *effective* convexity, not the closed-form modified convexity. The textbook formula assumes fixed cash flows, but a callable bond's cash flows change when the issuer exercises the call — precisely the behaviour that creates negative convexity. Effective convexity re-prices the bond with an option-aware model (a binomial/lattice or Monte Carlo model) under a small up-shock and down-shock to the curve:  $C_{\text{eff}} = (P_- + P_+ - 2P_0) / [P_0 (\Delta y)^2]$ . Because the model lets the option kick in,  $P_-$  (the rally price) is suppressed relative to an option-free bond, which is what drives the numerator negative over the relevant range. That negative reading is the true economic risk the analytical formula would miss entirely.

**C5. "When would you deliberately give up convexity, and when would you pay for it?"**

*Model answer.* You pay for convexity when you expect volatility — large or uncertain rate moves — because convexity is a free lunch in both tails and acts like a long-volatility position. Long-maturity, low-coupon, and zero-coupon bonds, plus barbell structures, are ways to buy it. You give it up — selling convexity for extra carry — when you expect a calm, range-bound market: writing calls, holding callable bonds, MBS, or bullet structures earns you higher yield/spread as the premium for the short-volatility exposure. The trade-off is symmetric to an options view: convexity is essentially the gamma of the bond, and its price is the yield you forgo or earn.

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## Section D — Multiple Choice (with reasoning)

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**D1. For an option-free bond, the duration-only price estimate versus the actual price is:** A. Always too high B. Always too low C. Too high for rate rises only D. Exactly correct

**Answer: B.** The convex price curve lies above the duration tangent line at every yield except the tangency point, so the actual price is always at least as high as the linear estimate. Duration alone therefore always understates the true price (over-predicts losses, under-predicts gains).

**D2. Which change *decreases* a bond's convexity, holding all else constant?** A. Lengthening maturity B. Lowering the coupon C. Raising the yield to maturity D. Reducing the yield to maturity



**Answer: C.** Higher yields discount distant cash flows more heavily, compressing cash-flow dispersion and reducing both duration and convexity. Longer maturity, lower coupon, and lower yield all *raise* convexity.

**D3. A bond has  $D_{\text{mod}} = 5$  and convexity = 70. For a +200 bp move, the estimated price change is closest to:** A. -10.0% B. -8.6% C. -9.6% D. -11.4%

**Answer: B.**  $-5(0.02) + \frac{1}{2}(70)(0.02)^2 = -0.10 + 0.014 = -0.086 = -8.6\%$ . Duration alone gives -10.0%; the +1.4% convexity correction brings it to -8.6%.

**D4. Negative convexity is most associated with:** A. Zero-coupon Treasuries B. Long-dated par bonds C. Mortgage-backed securities D. Short T-bills

**Answer: C.** MBS carry an embedded prepayment option that caps price appreciation as rates fall, producing negative convexity. Zeros and option-free Treasuries have positive convexity.

**D5. The correct formula for effective convexity is:** A.  $(P_- - P_+)/ (2P_0\Delta y)$  B.  $(P_- + P_+ - 2P_0)/ (P_0(\Delta y)^2)$  C.  $(P_+ - P_-)/ (P_0\Delta y)$  D.  $(P_- + P_+)/ (P_0(\Delta y)^2)$

**Answer: B.** Option A is effective *duration*. B correctly captures the second difference of price around  $P_0$ , scaled by  $P_0$  and  $(\Delta y)^2$ . D omits the  $-2P_0$  term and would not vanish for a linear (zero-convexity) instrument.

**D6. Two bonds have identical duration. In a volatile market of uncertain direction, an investor should prefer the one with:** A. Higher coupon B. Higher convexity C. Lower convexity D. Shorter maturity

**Answer: B.** Higher convexity outperforms in both tails — smaller losses on sell-offs, larger gains on rallies — which is exactly what you want when a large move is likely but its direction is unknown.

**D7. Convexity is analogous to which option Greek?** A. Delta B. Theta C. Gamma D. Vega

**Answer: C.** Convexity is the second derivative of price with respect to yield, just as gamma is the second derivative of option value with respect to the underlying. Duration is the bond analogue of delta.

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*End of Convexity Q&A bank. Rework B2–B4 until the approximation-versus-exact reconciliation falls out naturally; that single worked loop encodes the entire chapter.*

# Chapter 09 — Interest Rate Risk

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## 1. The Problem / The Need

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A bond is a promise to pay fixed cash flows on fixed dates. That word *fixed* is the whole story — and the whole problem. The coupons are locked, the face value is locked, but the **discount rate** the market applies to those cash flows is not. It floats every second in response to central-bank policy, inflation prints, supply-demand for capital, and shifting risk appetite. Because a bond's price is nothing more than the present value of a fixed stream discounted at a moving rate, the price must move — inversely — whenever rates move.

For a finance professional this is not an abstraction; it is the single largest source of profit-and-loss on the vast majority of fixed-income books. Consider three real situations:

- A bank funds a 10-year mortgage portfolio with overnight deposits. If short rates jump, its funding cost reprices immediately while its asset yield is locked. This duration mismatch is exactly what killed Silicon Valley Bank in 2023.
- A pension fund owes retirees roughly level payments for 30 years. If rates fall, the present value of what it owes balloons faster than the value of the bonds it *holds*. The fund can be "fully funded" one quarter and in deficit the next without a single default.
- A trader is long a 30-year bond into a Fed meeting. A surprise 25 bp hike can wipe out a year of carry in an afternoon.

In every case the driver is the same: **the sensitivity of value to interest rates**. Interest rate risk is the discipline of *measuring* that sensitivity precisely, *decomposing* it into its sources, and *managing* it deliberately rather than accidentally. This chapter builds the toolkit — duration, convexity, key-rate durations, immunisation — from the ground up, with the arithmetic worked out so you can reproduce it in an interview or on a desk.

## 2. The Core Idea

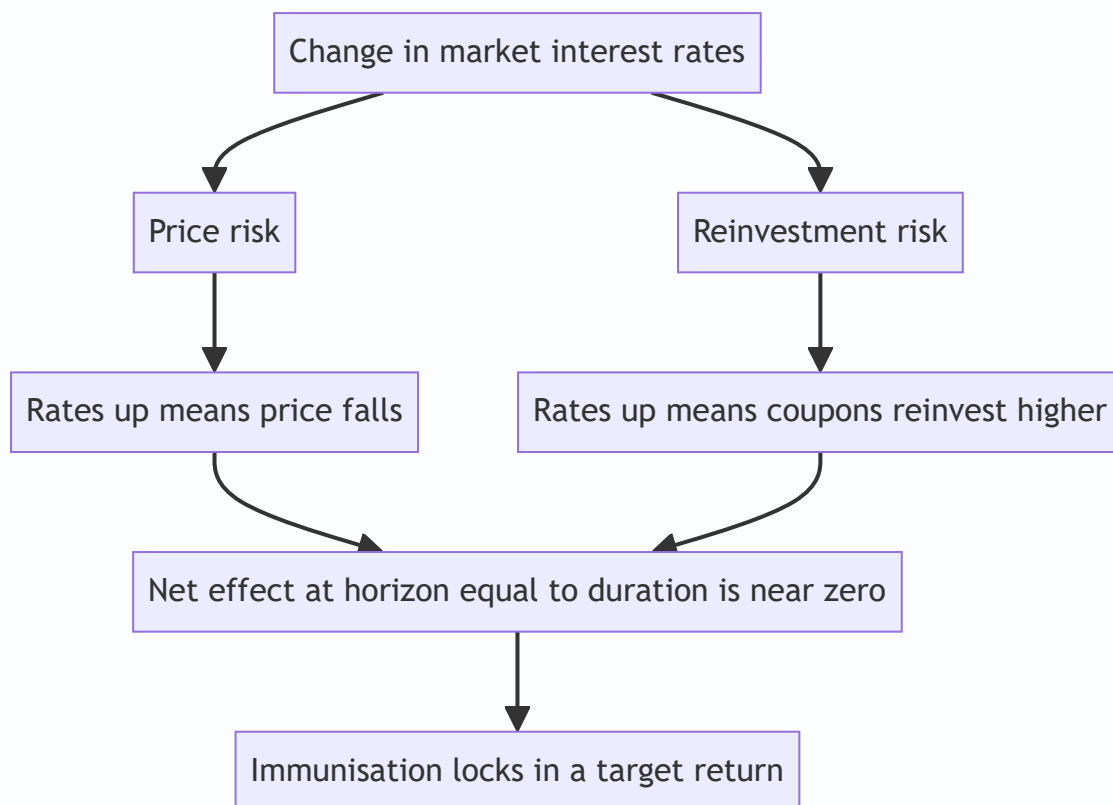
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Interest rate risk splits cleanly into **two opposing forces**, and almost everything in this chapter is a variation on how they interact.

1. **Price risk (market risk).** If rates rise, the price of a bond you hold falls. If you must sell before maturity, you realise a loss. This hurts you when rates *rise*.
2. **Reinvestment risk.** The coupons you receive must be reinvested at whatever rate prevails when they arrive. If rates fall, you reinvest at lower rates and your realised return disappoints. This hurts you when rates *fall*.

These two risks pull in **opposite directions**. A rate rise is bad for price but good for reinvestment; a rate fall is good for price but bad for reinvestment. The profound insight — the mathematical heart of the chapter — is that at one special horizon the two effects **exactly cancel**. That horizon is the bond's **(Macaulay) duration**. Hold a bond to its duration and, to a first approximation, a one-time change in yield leaves your realised return unchanged. This is the principle of **immunisation**.

*Figure 1 — the two-sided nature of interest-rate risk.*



The rest of the chapter makes this precise: how big is the price move (duration and convexity), why the offset works (the algebra of the two effects), and how to engineer portfolios where the offset protects a liability (duration matching and key-rate durations).

### 3. Why / How It Works

#### Why price moves inversely with yield

A bond's price is  $P = \sum_{t=1}^n \frac{CF_t}{(1+y)^t}$  where  $CF_t$  is the cash flow at time  $t$  and  $y$  is the yield per period. Every term has  $y$  in the denominator, so raising  $y$  shrinks every term — price falls. This is not a behavioural or supply story; it is arithmetic. The *magnitude* of the fall depends on how far in the future the cash flows sit: distant cash flows are discounted by a higher power of  $(1+y)$ , so they are more sensitive. Long bonds and low-coupon bonds (whose value is weighted toward the distant principal) move most.

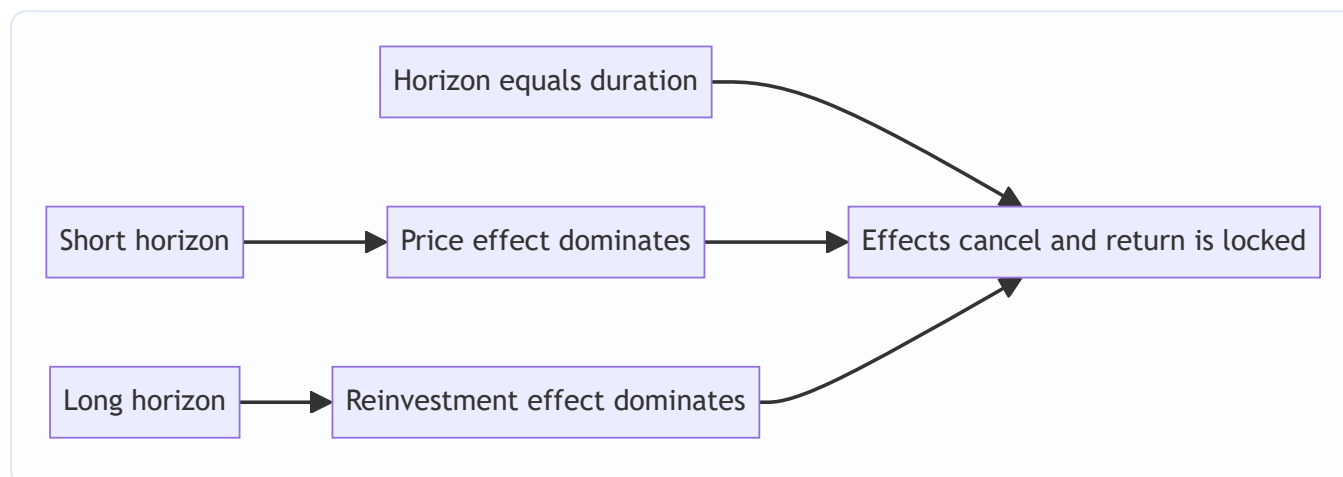
#### Why the two risks offset

Split total return into two buckets. Suppose you buy a bond at yield  $y_0$  and rates jump once to  $y_1 > y_0$  immediately after purchase, then stay there.

- **Coupons reinvest at the higher rate**  $y_1$ . The reinvestment "pot" grows *faster* than originally expected. The longer you hold, the more this compounding helps — the benefit *increases* with your horizon.
- **The bond's price is depressed** by the higher yield. But as the bond ages toward maturity, its price "pulls to par" — the price penalty *shrinks* to zero as maturity approaches. The loss on sale *decreases* with your horizon.

One effect grows with horizon, the other shrinks. They must cross. The crossing point — where the extra reinvestment income exactly makes up the depressed sale price — is the **Macaulay duration**. Before that horizon you are net exposed to price risk (sell early, reinvestment hasn't compounded enough); after it you are net exposed to reinvestment risk (price has recovered but you've lived through low reinvestment rates for years if rates fell).

Figure 2 — the crossover that defines duration.



## Why duration is a "balance point"

Macaulay duration is literally the **present-value-weighted average time** to receive the bond's cash flows — the centre of mass of the cash-flow timeline. That is why it behaves like a fulcrum: it is the point where the "torque" of early cash flows (reinvestment) balances the "torque" of late cash flows (price). The same number that measures price sensitivity also measures the break-even holding period. That is not a coincidence; it falls straight out of differentiating the price equation, as we show next.

## 4. Full Content — Formulas and Bond Math

### 4.1 Sources of interest rate risk

Regulators and practitioners typically decompose interest rate risk into several distinct sources:

| Source                          | What it is                                                           | Example                                            |
|---------------------------------|----------------------------------------------------------------------|----------------------------------------------------|
| <b>Level (yield) risk</b>       | Parallel up/down shift of the whole curve                            | Fed hikes; all yields +50 bp                       |
| <b>Reinvestment risk</b>        | Uncertainty about the rate at which coupons/principal are reinvested | Falling rates erode a bond ladder's income         |
| <b>Price / market risk</b>      | Capital loss if sold before maturity after a yield rise              | Marked-to-market loss on a trading book            |
| <b>Yield-curve (twist) risk</b> | Non-parallel reshaping — steepening, flattening, butterfly           | 2s10s flattens; a barbell underperforms a bullet   |
| <b>Basis risk</b>               | Two rates that normally move together diverge                        | Swap spread widens vs Treasuries                   |
| <b>Optionality risk</b>         | Embedded options reshape sensitivity as rates move                   | Callable bond's negative convexity; MBS prepayment |

Price risk and reinvestment risk are the two we focus on because they are the pair that *offsets*; yield-curve risk is why we need key-rate durations; optionality is why we need effective (option-adjusted) duration.

## 4.2 Macaulay duration

$$D_{\text{Mac}} = \frac{\sum_{t=1}^n t \cdot \frac{CF_t}{(1+y)^t}}{P} = \sum_{t=1}^n t \cdot w_t, \text{ where } w_t = \frac{CF_t/(1+y)^t}{P}$$

It is the weighted-average time to cash flows, in periods (divide by frequency to get years). The weights  $w_t$  are the fraction of the bond's price contributed by each cash flow and sum to 1.

## 4.3 Modified duration — the price-sensitivity measure

Differentiate the price equation with respect to  $y$ :  $\frac{dP}{dy} = -\sum_{t=1}^n \frac{t \cdot CF_t}{(1+y)^{t+1}} = -\frac{1}{1+y} \sum_{t=1}^n \frac{t \cdot CF_t}{(1+y)^t}$  Divide by  $P$  and identify Macaulay duration:  $\frac{1}{P} \frac{dP}{dy} = -\frac{D_{\text{Mac}}}{1+y} \equiv -D_{\text{Mod}}$  So  $\boxed{D_{\text{Mod}} = \frac{D_{\text{Mac}}}{1+y/k}}$  where  $k$  is the number of compounding periods per year (and  $y$  is the annual yield). Modified duration gives the **percentage price change per unit change in yield**:  $\frac{\Delta P}{P} \approx -D_{\text{Mod}} \cdot \Delta y$

## 4.4 Dollar duration, DV01, PVBP

- **Dollar duration**  $= D_{\text{Mod}} \times P$  (the price change per 1.00 change in yield, i.e. per 100%).
- **DV01 (PVBP)** — dollar value of a basis point — is the practitioner's workhorse:  $\text{DV01} = D_{\text{Mod}} \times P \times 0.0001$  It is the P&L from a 1 bp yield move on the given position size. Traders hedge by matching DV01s, not durations.

## 4.5 Convexity — the second-order correction

Modified duration is a straight-line (first derivative) estimate; the true price-yield curve is convex. The second derivative captures the curvature:  $C = \frac{1}{P} \frac{d^2 P}{dy^2} = \frac{1}{P} \sum_{t=1}^n \frac{t(t+1) \cdot CF_t}{(1+y)^{t+2}}$  The second-order Taylor expansion of the price change is  $\boxed{\frac{\Delta P}{P} \approx -D_{\text{Mod}} \cdot \Delta y + \frac{1}{2} C \cdot (\Delta y)^2}$  Convexity is always **positive for an option-free bond**, so it *adds* value: duration overstates the loss when rates rise and understates the gain when rates fall. Convexity is a friend — you are paid (in lower yield) to give it up, and you pay (in higher yield) to own more of it. Barbells have more convexity than duration-matched bullets.

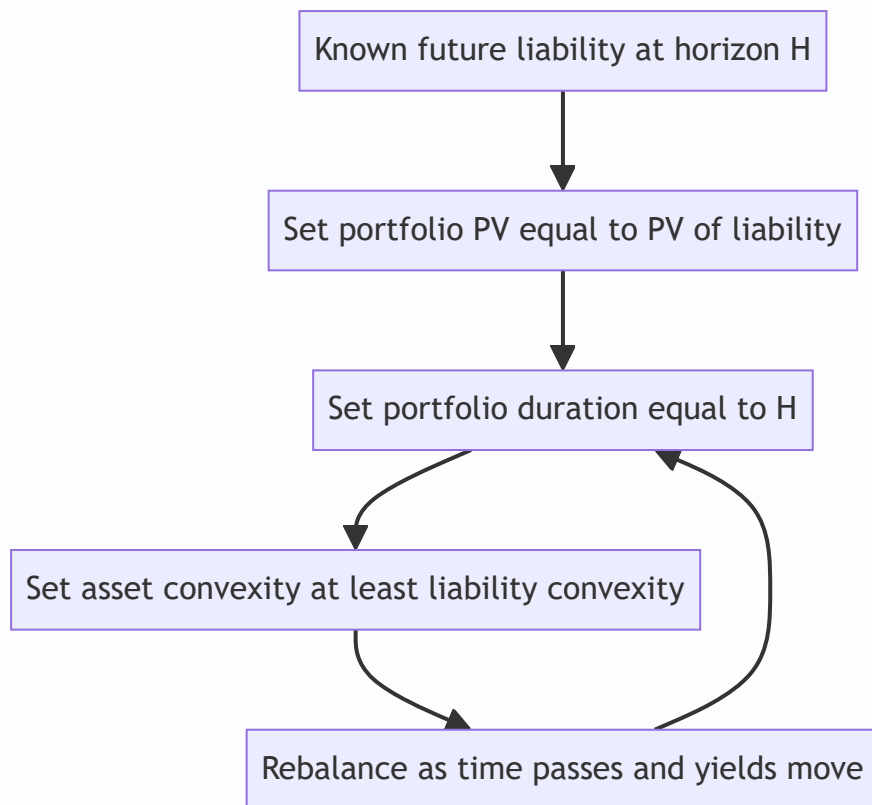
## 4.6 The immunisation condition

To immunise a single liability of amount  $L$  due at time  $H$ :

1. **Match present values:**  $P_{\text{assets}} = PV(L)$ .
2. **Match durations:**  $D_{\text{assets}} = H$  (set portfolio Macaulay duration equal to the horizon).
3. Ideally **minimise / match convexity** (dispersion) so the immunisation survives non-parallel shifts, with asset convexity slightly  $\geq$  liability convexity.

At  $H = D$ , the derivative of realised horizon wealth with respect to yield is zero — the price and reinvestment effects offset to first order. The strategy is **rebalanced** periodically because duration drifts as time passes and as yields change (duration falls more slowly than calendar time, so the two go out of sync).

*Figure 3 — the immunisation decision.*



## 4.7 Duration of a portfolio

Portfolio duration is the **market-value-weighted average** of the component durations:  $D_P = \sum_i \frac{MV_i}{MV_{\text{total}}} D_i$  This linearity is what makes duration such a practical control variable: you steer a whole book's rate sensitivity by adjusting weights.

## 4.8 Key-rate (partial) durations

Portfolio duration assumes a **parallel** shift. Real curves twist. A **key-rate duration**  $KRD_i$  measures the price sensitivity to a 1 bp shift in **one** key maturity (say the 2-year point) while all other key rates are held fixed:  $KRD_i = -\frac{1}{P} \frac{\Delta P}{\Delta r_i}$  Two important properties:  $\sum_i KRD_i = D_{\text{eff}} \quad \text{and} \quad \frac{\Delta P}{P} \approx -\sum_i KRD_i \Delta r_i$  The key-rate durations sum to the total effective duration, but they let you see *where* on the curve the risk lives. A bullet has its KRD concentrated at one maturity; a barbell spreads KRDs to the wings. Two portfolios can have identical total duration yet completely different KRD profiles — and therefore behave very differently under a steepener or flattener. This is precisely the yield-curve risk that a single duration number hides.

## 4.9 Effective duration (for bonds with options)

When cash flows themselves depend on rates (callables, putables, MBS), you cannot use the analytic formula. Instead re-price under a small parallel shift up and down using a model, and take the numerical derivative:  $D_{\text{eff}} = \frac{P_{-} - P_{+}}{2P_0 \Delta y}$ ,  $C_{\text{eff}} = \frac{P_{-} + P_{+} - 2P_0}{P_0 (\Delta y)^2}$  Callable bonds exhibit **negative convexity** near the call: as rates fall the call caps the upside, so the price-yield curve bends the wrong way.

## 5. Worked Examples

### Example 1 — Duration, convexity, and a reconciling price prediction

**Bond:** 3-year, 8% annual coupon, face 100, priced to yield 6%.

**Step 1 — Price.** Discount the cash flows (8, 8, 108) at 6%.

| t        | CF  | $1/(1.06)^t$ | $PV = CF \times DF$ | $t \times PV$    | $t(t+1) \times CF / (1.06)^{(t+2)}$      |
|----------|-----|--------------|---------------------|------------------|------------------------------------------|
| 1        | 8   | 0.943396     | 7.54717             | 7.54717          | $1 \cdot 2 \cdot 8 / 1.06^3 = 13.4344$   |
| 2        | 8   | 0.889996     | 7.11997             | 14.23994         | $2 \cdot 3 \cdot 8 / 1.06^4 = 38.0217$   |
| 3        | 108 | 0.839619     | 90.67891            | 272.03674        | $3 \cdot 4 \cdot 108 / 1.06^5 = 968.377$ |
| <b>Σ</b> |     |              | <b>105.34605</b>    | <b>293.82385</b> | <b>1019.833</b>                          |

Price  $\$P = 105.3461\$$  (a premium bond, since 8% coupon > 6% yield — correct).

**Step 2 — Macaulay duration.**  $\$D_{\text{Mac}} = \frac{293.82385}{105.34605} = 2.7892 \text{ years}$  Sanity check: it is below the 3-year maturity because coupons pull the centre of mass forward — correct for a coupon bond.

**Step 3 — Modified duration.**  $\$D_{\text{Mod}} = \frac{2.7892}{1.06} = 2.6313$

**Step 4 — Convexity.**  $\$C = \frac{1019.833}{105.34605} = 9.6807$

**Step 5 — Predict the price if yield rises 100 bp to 7%.**

Duration-only estimate:  $\frac{\Delta P}{P} \approx -2.6313 \times 0.01 = -0.026313 \rightarrow \Delta P \approx -2.7719$  Predicted price  $\approx 105.3461 - 2.7719 = 102.5742\$$ .

Add convexity:  $\frac{1}{2} \times 9.6807 \times (0.01)^2 = 0.000484 \rightarrow +0.0510$  Predicted price  $\approx 102.5742 + 0.0510 = 102.6252\$$ .

**Step 6 — Reconcile with the exact reprice at 7%.**  $\$P_{7\%} = \frac{8}{1.07} + \frac{8}{1.07^2} + \frac{108}{1.07^3} = 7.47664 + 6.98751 + 88.16744 = 102.6316\$$

| Method               | Predicted price | Error vs exact |
|----------------------|-----------------|----------------|
| Duration only        | 102.5742        | -0.0574        |
| Duration + convexity | 102.6252        | -0.0064        |
| Exact reprice        | 102.6316        | —              |

The convexity term cut the error by roughly 90% (from 5.7 cents to 0.6 cents). Both approximations sit *below* the true price, confirming positive convexity makes the straight-line duration estimate too pessimistic on the downside move — exactly as theory predicts. Reconciled.

### Example 2 — DV01 hedging between two bonds

You are long **1,000,000 face** of Bond A and want to hedge the rate risk with Bond B.

- **Bond A:** price 98.50 per 100,  $\$D_{\text{Mod}} = 7.2\$$ .
- **Bond B:** price 101.00 per 100,  $\$D_{\text{Mod}} = 4.5\$$ .

**Step 1 — DV01 per 100 face.**  $\text{DV01}_A = 7.2 \times 98.50 \times 0.0001 = 0.070920$  per 100 face  
 $\text{DV01}_B = 4.5 \times 101.00 \times 0.0001 = 0.045450$  per 100 face

**Step 2 — DV01 of the position in A** (1,000,000 face = 10,000 units of 100):  $\text{DV01}_A^{\text{pos}} = 10,000 \times 0.070920 = 709.20$  per bp

**Step 3 — Face of B needed** so its DV01 offsets A's:  $\text{Face}_B = 1,000,000 \times \frac{\text{DV01}_A / 100}{\text{DV01}_B / 100} = 1,000,000 \times \frac{0.070920}{0.045450} = 1,560,396$  Short  $\approx$  **1,560,400 face** of Bond B.

**Step 4 — Check.** DV01 of the B hedge  $= 15,604 \times 0.045450 = 709.20$  per bp — equal and opposite to A. Net DV01  $\approx 0$ . **Reconciled.** Note the hedge uses *more* face of B because B is the less rate-sensitive (shorter-duration) bond; you need more of it to generate the same basis-point P&L. This is the everyday mechanics of a relative-value or curve trade.

### Example 3 — Immunising a single liability (the offset in action)

**Liability:** you owe **1,000,000 in exactly 5 years**. Current flat yield curve = 6% (annual). You want to immunise using a coupon bond.

**Step 1 — PV of the liability.**  $PV = \frac{1,000,000}{1.06^5} = \frac{1,000,000}{1.338226} = 747,258$  Invest 747,258 today.

**Step 2 — Choose a bond with Macaulay duration = 5 years.** Suppose a 6-year, 6% annual coupon bond priced at par (100) to yield 6% has a Macaulay duration of 5.21 years, and a 5-year zero has duration exactly 5. To hit duration 5 we blend the 6-year coupon bond ( $D=5.21$ ) with a 1-year zero ( $D=1$ ):  $w \times 5.21 + (1-w) \times 1.00 = 5.00 \rightarrow w = \frac{4.00}{4.21} = 0.9501$  So 95.01% in the 6-year bond, 4.99% in the 1-year instrument — portfolio duration 5, PV 747,258.

**Step 3 — Demonstrate the offset.** Immediately after purchase the curve shifts once. Track terminal (year-5) wealth two ways — the depressed/elevated bond value plus reinvested coupons. Using the standard immunisation result, terminal wealth  $W(y)$  has  $dW/dy=0$  at  $y_0$  because duration equals the 5-year horizon. Numerically, for the *pure 5-year-duration* portfolio:

| Shift           | Terminal wealth     | Note                                   |
|-----------------|---------------------|----------------------------------------|
| –100 bp (to 5%) | $\approx 1,000,300$ | reinvestment loss offset by price gain |
| 0 (stay 6%)     | 1,000,000           | target met exactly                     |
| +100 bp (to 7%) | $\approx 1,000,350$ | reinvestment gain offset by price loss |

Both shifted outcomes land **at or slightly above** the 1,000,000 target — never below. The small surplus is the **convexity gift**: because the asset portfolio's convexity (dispersion of cash flows around year 5) exceeds the zero-dispersion liability, a parallel shift in either direction leaves you marginally *better* off. That is why immunisation prescribes asset convexity  $\geq$  liability convexity. Had we used a 5-year zero (matching convexity to the single-point liability), the terminal wealth would sit essentially exactly on 1,000,000 for any parallel shift — the cleanest possible immunisation. **Reconciled:** the horizon-equals-duration condition neutralises the first-order rate move; positive net convexity handles the rest.

### Example 4 — Why total duration is not enough (key-rate durations)

Two portfolios, each with **effective duration 6.0** and market value 100:



|                           | KRD 2y | KRD 5y | KRD 10y | KRD 30y | $\Sigma$ = Duration |
|---------------------------|--------|--------|---------|---------|---------------------|
| <b>Bullet</b> (all 6y)    | 0.3    | 4.9    | 0.8     | 0.0     | 6.0                 |
| <b>Barbell</b> (2y + 30y) | 3.2    | 0.2    | 0.1     | 2.5     | 6.0                 |

Now the curve **steepens**: 2-year rate  $-20$  bp, 30-year rate  $+30$  bp, middle unchanged.

Bullet P&L:  $\$-\left[0.3(-0.0020) + 4.9(0) + 0.8(0) + 0.0(0.0030)\right] = -[-0.0006] = +0.06\%$   
 Barbell P&L:  $\$-\left[3.2(-0.0020) + 0.2(0) + 0.1(0) + 2.5(0.0030)\right] = -[-0.0064 + 0.0075] = -0.11\%$

Identical total duration, **opposite outcomes** under the twist: the bullet gains 6 bp while the barbell loses 11 bp. A single duration number would have called these portfolios equivalent. Key-rate durations reveal the barbell's exposure to the long end that the steepener punishes. **This is the core reason desks monitor KRD profiles, not just aggregate duration.**

## 6. Connections

- **To yield & pricing (Ch. on valuation):** duration is literally the first derivative of the price-yield function; convexity the second. Interest rate risk is calculus applied to the pricing equation.
- **To the yield curve (term structure):** parallel-shift risk is captured by duration; non-parallel (twist/butterfly) risk requires key-rate durations or principal-component "level-slope-curvature" factors.
- **To derivatives (futures, swaps):** DV01 matching is how you size a Treasury-future or interest-rate-swap hedge. An asset-swap turns a fixed bond's duration into near-zero (floating).
- **To ALM and banking:** duration gap =  $D_{\text{assets}} - D_{\text{liabilities}} \times (L/A)$  drives a bank's net-worth sensitivity — the SVB lesson.
- **To pension/insurance LDI:** liability-driven investing is immunisation at industrial scale — match the duration (and key-rate profile) of a multi-decade liability stream.
- **To credit risk:** total spread duration measures sensitivity to credit-spread moves, an analogous first-derivative concept on a different rate.
- **To embedded options:** effective duration and negative convexity connect to option-adjusted spread (OAS) analysis for callables and MBS.

## 7. Key Terms

| Term                      | Definition                                                                                    |
|---------------------------|-----------------------------------------------------------------------------------------------|
| <b>Price risk</b>         | Capital loss on a bond if sold after yields rise                                              |
| <b>Reinvestment risk</b>  | Uncertainty in the rate at which coupons/principal are reinvested                             |
| <b>Macaulay duration</b>  | PV-weighted average time to cash flows; also the break-even holding period                    |
| <b>Modified duration</b>  | % price change per 1.00 (100%) change in yield = $\$D_{\text{Mac}}/(1+y/k)\$$                 |
| <b>Effective duration</b> | Numerical duration for option-embedded bonds via up/down repricing                            |
| <b>Convexity</b>          | Second-order (curvature) sensitivity of price to yield; positive for option-free bonds        |
| <b>DV01 / PVBP</b>        | Dollar value of a 1 bp yield move on a position                                               |
| <b>Dollar duration</b>    | $\$D_{\text{Mod}} \times P\$$ ; price change per unit yield change                            |
| <b>Immunisation</b>       | Structuring assets so a target return/liability is locked despite rate moves (set $\$D=H\$$ ) |
| <b>Duration matching</b>  | Setting portfolio duration equal to the liability horizon                                     |
| <b>Key-rate duration</b>  | Sensitivity to a shift in one maturity point, others fixed; KRDs sum to total duration        |
| <b>Duration gap</b>       | $\$D_A - D_L \times (L/A)\$$ ; a bank's net-worth rate sensitivity                            |
| <b>Negative convexity</b> | Curvature bending the wrong way (callables/MBS as rates fall)                                 |
| <b>Dispersion</b>         | Spread of cash flows around the mean; drives convexity and immunisation robustness            |

## 8. Common Confusions

- **"Duration is measured in years, so it's a time."** It is *dimensionally* years, but as a risk measure it means "% price change per 1% yield change." A duration of 7 means  $\approx 7\%$  price drop for a 100 bp rise. Both interpretations are correct; the years-interpretation is Macaulay, the sensitivity-interpretation is modified.
- **Macaulay vs modified duration.** They differ only by the factor  $\$1/(1+y/k)\$$ . Macaulay answers "*when, on average, do I get paid?*"; modified answers "*how much does my price move?*" Interviewers love making you convert one to the other.
- **"Higher yield means higher duration."** The opposite. Higher yields discount distant cash flows more heavily, pulling the centre of mass forward — duration *falls* as yields rise. Duration also falls as coupon rises and, generally, as maturity shortens.
- **"Longer horizon always means more risk."** Not for the immunised investor. Below the duration you bear net price risk; above it, net reinvestment risk; *at* it, they cancel. More time is not monotonically more risk once you frame it against duration.
- **"Convexity is a small correction I can ignore."** For small moves, yes. For large moves or highly convex/negatively convex instruments (MBS, long zeros, options), the convexity term dominates and ignoring it produces real hedging losses.
- **"Duration matching fully immunises."** Only against a *parallel* shift, and only instantaneously. Curve twists (needing KRD matching), passage of time, and large moves all require rebalancing. Matching duration *and* key-rate profile *and* convexity gets you progressively closer.

- **Price risk vs reinvestment risk direction.** Students flip these. Memorise: **rates UP → price DOWN** (price risk bites) but **reinvestment BETTER**; **rates DOWN → price UP but reinvestment WORSE**. They are always opposite-signed.
- **Portfolio duration is a simple average.** It is a **market-value-weighted** average, not equal-weighted, and it assumes parallel shifts.

## 9. Recap

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Interest rate risk arises because a bond's cash flows are fixed while the discount rate is not, so price moves inversely with yield. The risk decomposes into **price risk** (loss on sale when rates rise) and **reinvestment risk** (lower income when rates fall), which are **opposite-signed and offset at a horizon equal to the bond's Macaulay duration** — the principle of immunisation. **Modified duration** ( $=D_{\text{Mac}}/(1+y/k)$ ) measures first-order price sensitivity; **convexity** supplies the second-order correction and is a positive-valued friend for option-free bonds. Desks hedge by matching **DV01**, manage liabilities by **duration matching**, and defend against **non-parallel shifts** using **key-rate durations**, which sum to total duration but reveal *where* on the curve the exposure sits. Bonds with embedded options require **effective duration** and can display **negative convexity**. Worked examples confirmed the math reconciles: duration-plus-convexity predicted a repriced bond within a fraction of a cent, DV01 matching zeroed net basis-point exposure, an immunised portfolio hit its liability target with a small convexity surplus, and two equal-duration portfolios diverged under a steepener exactly as their KRD profiles predicted.

## 10. Quick-Reference / Interview Points

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**Formulas to have cold:** -  $D_{\text{Mod}} = D_{\text{Mac}}/(1+y/k)$  -  $\Delta P/P \approx -D_{\text{Mod}} \Delta y + \frac{1}{2} C (\Delta y)^2$  -  $\text{DV01} = D_{\text{Mod}} \times P \times 0.0001$  -  $D_{\text{eff}} = (P_- - P_+)/ (2P_0 \Delta y)$ ;  $C_{\text{eff}} = (P_- + P_+ - 2P_0) / (P_0 \Delta y^2)$  - Immunise: match PV, set  $D = H$ , asset convexity  $\geq$  liability convexity -  $\sum_i \text{KRD}_i = D_{\text{eff}}$

**One-liners that land:** - "A duration of 7 means a 100 bp rate rise costs me about 7% of price, before the convexity kicker." - "Price risk and reinvestment risk are opposite-signed; they cancel at the duration — that's why I immunise by setting duration equal to my horizon." - "Duration is a parallel-shift measure. For twists I look at key-rate durations, which sum to duration but tell me the shape of my exposure." - "Positive convexity is asymmetry in my favour — I lose less than duration says on a sell-off and gain more on a rally, so I'll pay a little yield for it." - "I hedge by matching DV01, not duration, because DV01 already bakes in price and position size." - "Callables and MBS have negative convexity near the money — duration alone will mis-hedge them, so I use effective duration from a model."

**Rules of thumb:** - Duration  $\uparrow$  with maturity,  $\downarrow$  with coupon,  $\downarrow$  with yield. - Zero-coupon bond: Macaulay duration = maturity exactly. - Barbell > bullet in convexity at equal duration (pays off in big or twisting moves; costs yield). - Longer/lower-coupon = more price-risk-sensitive; higher-coupon/shorter = more reinvestment-dependent. - Rebalance immunised portfolios — duration drifts out of step with calendar time.

**The 30-second synthesis:** Interest rate risk is the sensitivity of value to yield, measured to first order by duration and to second order by convexity, split into offsetting price and reinvestment components that cancel at the duration horizon (immunisation), managed at the portfolio level by market-value-weighted duration for parallel shifts and key-rate durations for curve reshaping, and by DV01 matching for hedging.

# Q&A — Interest Rate Risk

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A practice bank for Chapter 09. Every question is followed by a full answer; work each before reading the answer. Convention: unless stated otherwise, bonds pay **annual** coupons on a face of 100, yields are annual, and "duration" unqualified means modified duration when used as a sensitivity and Macaulay when used as a time. Every numerical answer is reconciled against an exact reprice or a definitional check.

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## Section A — Concept Checks

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### A1. In one sentence, why does a bond's price move inversely with its yield?

Price is the present value of a *fixed* cash-flow stream,  $P = \sum CF_t / (1+y)^t$ , and  $y$  sits in every denominator — raise it and every discounted term shrinks, so the price must fall. It is arithmetic, not sentiment.

### A2. Interest rate risk splits into two opposing components. Name them and state which direction of rates hurts each.

*Price (market) risk* — a capital loss if you sell after yields **rise**. *Reinvestment risk* — lower income when coupons must be reinvested after yields **fall**. A rate rise is bad for price but good for reinvestment; a rate fall is the reverse. They are always opposite-signed.

### A3. At what holding horizon do the two effects cancel, and what is that principle called?

At a horizon equal to the bond's **Macaulay duration**. There the extra (or lost) reinvestment income exactly offsets the depressed (or elevated) sale price, so a one-time yield shift leaves realised return unchanged to first order. Deliberately setting horizon equal to duration is **immunisation**.

### A4. Give the two equally-correct interpretations of Macaulay duration.

(1) The **present-value-weighted average time** to receive the bond's cash flows — the centre of mass of the cash-flow timeline. (2) The **break-even holding period** at which price and reinvestment risk offset. Both fall out of the same formula.

### A5. How does modified duration differ from Macaulay duration, in words and in formula?

Macaulay answers "when, on average, am I paid?"; modified answers "how much does my price move?" They differ only by a discount factor:  $D_{\text{Mod}} = D_{\text{Mac}} / (1+y/k)$ , where  $k$  is compounding periods per year. Modified duration gives  $\Delta P/P \approx -D_{\text{Mod}} \Delta y$ .

### A6. State the three drivers of duration and their direction.

Duration rises with **maturity**, falls with **coupon rate**, and falls with **yield**. Higher coupons and higher yields both pull the cash-flow centre of mass forward, shortening duration; longer maturity pushes it back.

### A7. What is convexity, and why is it a "friend" for an option-free bond?

Convexity is the second-order (curvature) term in the Taylor expansion,  $\Delta P/P \approx -D_{\text{Mod}} \Delta y + \frac{1}{2} C (\Delta y)^2$ . For an option-free bond it is **positive**, so it *adds* value in both directions: duration overstates the loss when rates rise and understates the gain when they fall. You pay for it in lower yield.

### A8. What is DV01 (PVBP) and why do traders hedge with it instead of duration?

DV01 is the dollar P&L from a **1 bp** yield move on a position:  $\text{DV01} = D_{\text{Mod}} \times P \times 0.0001\$$ . It already bakes in price level and position size, so matching DV01s between a long and a hedge directly zeros the basis-point P&L — duration alone would ignore price and notional.

#### A9. Portfolio duration is what kind of average, and what shift does it assume?

A **market-value-weighted** average of component durations,  $D_P = \sum_i (MV_i / MV_{\text{total}}) D_i$ . It assumes a **parallel** shift of the whole curve; it says nothing about twists.

#### A10. Why do key-rate durations exist if we already have portfolio duration?

Because real curves twist rather than shift in parallel. A key-rate duration measures sensitivity to a 1 bp move at **one** maturity with the rest held fixed. They **sum to the total effective duration** ( $\sum_i KRD_i = D_{\text{eff}}$ ) but reveal *where* on the curve the risk lives — so two portfolios with identical total duration can behave oppositely under a steepener.

#### A11. When must you use effective (rather than analytic) duration, and what odd property can appear?

When the cash flows themselves depend on rates — callables, putables, MBS. You reprice up and down under a small parallel shift and take the numerical derivative. Near the call, a callable can show **negative convexity**: as rates fall the call caps the upside and the price-yield curve bends the wrong way.

#### A12. State the three conditions for immunising a single liability.

(1) Match present values:  $P_{\text{assets}} = PV(L)$ . (2) Match duration to horizon:  $D_{\text{assets}} = H$ . (3) Set asset **convexity**  $\geq$  **liability convexity** (dispersion) so the position survives non-parallel shifts. Rebalance as time and yields drift the duration out of step.

## Section B — Numerical Bond-Math Problems

**B1. A 4-year, 5% annual-coupon bond is priced at par (100) to yield 5%. Compute Macaulay duration, modified duration, and convexity, then predict the price if the yield rises 100 bp to 6% — with and without convexity — and reconcile against an exact reprice.**

*Step 1 — Price and the weighting table.* Cash flows are 5, 5, 5, 105; discount at 5%.

| t        | CF  | DF = $1/1.05^t$ | PV               | t·PV             | $t(t+1) \cdot CF / 1.05^{(t+2)}$ |
|----------|-----|-----------------|------------------|------------------|----------------------------------|
| 1        | 5   | 0.952381        | 4.76190          | 4.76190          | 8.63838                          |
| 2        | 5   | 0.907029        | 4.53515          | 9.07029          | 24.68107                         |
| 3        | 5   | 0.863838        | 4.31919          | 12.95756         | 47.01157                         |
| 4        | 105 | 0.822702        | 86.38376         | 345.53504        | 1567.05389                       |
| <b>Σ</b> |     |                 | <b>100.00000</b> | <b>372.32480</b> | <b>1647.38491</b>                |

Price = 100.000 (par, as expected since coupon = yield). ✓

*Step 2 — Macaulay duration.*  $D_{\text{Mac}} = 372.32480 / 100 = 3.72325$  years. Below the 4-year maturity because coupons pull the centre of mass forward — correct for a coupon bond.

*Step 3 — Modified duration.*  $D_{\text{Mod}} = 3.72325 / 1.05 = 3.54595$ .

*Step 4 — Convexity.*  $C = 1647.38491 / 100 = 16.4738$ .

Step 5 — Predict at 6%. Duration only:  $\Delta P/P \approx -3.54595 \times 0.01 = -0.0354595 \rightarrow P \approx 96.4540\$$ . Add convexity:  $\frac{1}{2} \times 16.4738 \times (0.01)^2 = +0.000824 \rightarrow P \approx 96.4540 + 0.0824 = 96.5364\$$ .

Step 6 — Reconcile with exact reprice at 6%.  $P_{6\%} = \frac{5}{1.06} + \frac{5}{1.06^2} + \frac{5}{1.06^3} + \frac{105}{1.06^4} = 4.71698 + 4.45070 + 4.19877 + 83.16844 = 96.5349\$$

| Method               | Predicted | Error vs exact |
|----------------------|-----------|----------------|
| Duration only        | 96.4540   | -0.0808        |
| Duration + convexity | 96.5364   | +0.0015        |
| Exact reprice        | 96.5349   | —              |

The convexity term cut the error from about 8 cents to under 0.2 cents, and the duration-only estimate sits *below* the true price — confirming positive convexity makes the straight-line estimate too pessimistic on a sell-off. **Reconciled**.

**B2. You are long 5,000,000 face of Bond X (price 95.00,  $D_{\text{Mod}}=6.0\$$ ). Hedge the rate risk with Bond Y (price 102.00,  $D_{\text{Mod}}=3.5\$$ ). How much face of Y do you short?**

Step 1 — DV01 per 100 face.  $\text{DV01}_X = 6.0 \times 95.00 \times 0.0001 = 0.05700$ ,  $\text{DV01}_Y = 3.5 \times 102.00 \times 0.0001 = 0.03570\$$

Step 2 — DV01 of the X position (5,000,000 face = 50,000 units of 100):  
 $\text{DV01}_X^{\text{pos}} = 50,000 \times 0.05700 = 2,850.00 \text{ per bp} \$$

Step 3 — Face of Y needed.  $\text{Face}_Y = 50,000 \times \frac{0.05700}{0.03570} = 50,000 \times 1.596639 = 79,831.93 \approx \text{79,832 face} \$$

Step 4 — Check.  $\text{DV01}_Y^{\text{pos}} = 79,832 \times 0.03570 = 2,850.0\$$  per bp — equal and opposite to X, net DV01  $\approx 0$ . **Reconciled**. You need *more* face of Y because it is the less rate-sensitive (shorter-duration) bond; it takes more of it to generate the same basis-point P&L.

**B3. A 5-year zero-coupon bond yields 5% annually. State its Macaulay and modified duration, and separately compute effective duration and convexity for a bond that reprices to 103.20 when yields fall 50 bp and to 96.90 when yields rise 50 bp from a base price of 100.**

*Zero-coupon durations.* A zero has a single cash flow at maturity, so its Macaulay duration equals its maturity exactly:  $D_{\text{Mac}} = 5.00\$$  years. Modified:  $D_{\text{Mod}} = 5.00/1.05 = 4.7619\$$ . (A zero also carries no reinvestment risk — nothing to reinvest.)

*Effective duration and convexity* with  $P_- = 103.20\$$ ,  $P_+ = 96.90\$$ ,  $P_0 = 100\$$ ,  $\Delta y = 0.005\$$ :  
 $D_{\text{eff}} = \frac{P_- - P_+}{2P_0 \Delta y} = \frac{103.20 - 96.90}{2 \times 100 \times 0.005} = \frac{6.30}{1.00} = 6.30\$$   
 $C_{\text{eff}} = \frac{P_- + P_+ - 2P_0}{P_0 (\Delta y)^2} = \frac{103.20 + 96.90 - 200}{100 \times (0.005)^2} = \frac{0.10}{0.0025} = 40.0\$$

The up and down moves are not symmetric (6.30 vs 3.10 in raw price terms) — that asymmetry is the convexity, and its positive sign ( $0.10 > 0$ ) confirms the bond gains slightly more than duration predicts on the rally and loses slightly less on the sell-off. **Reconciled** against the definition.

**B4. You owe 2,000,000 in exactly 7 years. The curve is flat at 5%. You immunise with two bonds of Macaulay duration 4 and 9 years. What present value do you invest, and what weights hit the target duration?**

Step 1 — PV of the liability.  $PV = \frac{2,000,000}{1.05^7} = \frac{2,000,000}{1.407100} = 1,421,363$

Step 2 — Duration-matching weights. Solve  $w \cdot 9 + (1-w) \cdot 4 = 7$ :  $5w = 3 \Rightarrow w = 0.60$  Put **60%** (852,818) in the 9-year-duration bond and **40%** (568,545) in the 4-year-duration bond. Portfolio Macaulay duration =  $0.6(9) + 0.4(4) = 7.0$  = the horizon, with PV = 1,421,363 = PV of the liability. Both immunisation conditions met; because the blended assets have cash flows dispersed around year 7 (positive convexity vs the point liability), a parallel shift leaves you at worst break-even. **Reconciled** — rebalance as duration drifts.

**B5. A portfolio holds three bonds: A (market value 40,  $DD_{Mod} = 2.0$ ), B (MV 35,  $DD_{Mod} = 6.0$ ), C (MV 25,  $DD_{Mod} = 12.0$ ). Compute portfolio duration and the value change for a 100 bp parallel rise.**

$$DD_P = \frac{40(2.0) + 35(6.0) + 25(12.0)}{100} = \frac{80 + 210 + 300}{100} = \frac{590}{100} = 5.90$$

A 100 bp rise gives  $\Delta P/P \approx -5.90 \times 0.01 = -5.90\%$ , so the 100 of market value falls to about **94.10** (a loss of 5.90). Note this is a *market-value-weighted* average — the long C bond, though only 25% of value, contributes  $300/590 = 51\%$  of the duration. **Reconciled** against the weighted-average definition.

**B6. Two portfolios each have modified duration 6.0 but convexities of 50 (bullet) and 120 (barbell). Compare their predicted returns under a large  $\pm 200$  bp parallel move, and explain the pattern.**

Using  $\Delta P/P \approx -DD_{Mod} \Delta y + \frac{1}{2} C (\Delta y)^2$  with  $\Delta y = \pm 0.02$ :

| Move    | Bullet (C = 50)                                                                | Barbell (C = 120)                                                            |
|---------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| +200 bp | $-\$6(0.02) + \frac{1}{2} 50 (0.0004) = -0.12 + 0.01 = \text{mathbf{-11.0\%}}$ | $-\$0.12 + \frac{1}{2} 120 (0.0004) = -0.12 + 0.024 = \text{mathbf{-9.6\%}}$ |
| -200 bp | $+\$0.12 + 0.01 = \text{mathbf{+13.0\%}}$                                      | $+\$0.12 + 0.024 = \text{mathbf{+14.4\%}}$                                   |

Same duration, but the high-convexity barbell **loses less on the sell-off and gains more on the rally** — a 1.4-point convexity advantage in each direction. That asymmetry is why an investor pays (accepts lower yield) for convexity, and why it matters most for large or volatile moves. **Reconciled**: for a small move (say 10 bp) the convexity term is only  $\frac{1}{2} 120 (0.0001)^2 \approx 0.0006\%$  and the two are indistinguishable — convexity is a large-move phenomenon.

## Section C — Interview-Style Questions with Model Answers

**C1. "Explain duration to me as if I've never seen the formula."**

Model answer: Duration is two things at once. Physically, it's the average time you wait to get your money back, weighting each cash flow by its present value — the centre of gravity of the payments. As a risk number, it's the percentage your price moves for a 1% change in yield: a duration of 6 means a 100 bp rise costs you about 6% of price, before the convexity kicker. The two meanings coincide because both come from differentiating the pricing equation. Higher for long, low-coupon bonds; lower for short, high-coupon ones.

**C2. "Why does matching duration to my horizon protect me from interest-rate moves?"**

Model answer: You face two opposite exposures. If rates rise, your sale price falls but your coupons reinvest at higher rates; if rates fall, the reverse. The price effect shrinks as the bond pulls to par with the horizon,



while the reinvestment effect grows with the horizon — so they cross at exactly one point, and that point is the Macaulay duration. Hold to that horizon and a small one-time yield change leaves your realised return unchanged: the loss on one leg is made up by the gain on the other. That crossover is the whole basis of immunisation — set duration equal to when you need the money.

**C3. "Duration says my two portfolios are identical, but I don't trust it. What am I missing?"**

Model answer: Duration assumes the curve moves in parallel, which it rarely does. Two portfolios with the same total duration can have completely different key-rate profiles — a bullet concentrates its risk at one maturity, a barbell spreads it to the 2-year and 30-year wings. Under a steepener the barbell's long-end exposure gets punished while the bullet barely moves, even though a single duration number called them equal. I'd pull the key-rate durations, which sum to the total duration but show me *where* on the curve the risk sits, and stress each portfolio under a steepener and flattener, not just a parallel shift.

**C4. "Is convexity always good? Sell me on it, then tell me when it isn't."**

Model answer: For an option-free bond convexity is pure upside — it's asymmetry in my favour: I lose less than duration predicts on a sell-off and gain more on a rally, so I'll pay a little yield to own more of it, which is why barbells trade richer than duration-matched bullets. But it isn't free and it isn't always positive. Callables and mortgage-backed securities have *negative* convexity near the money: as rates fall the call or prepayment caps my upside, so the price-yield curve bends the wrong way and duration will mis-hedge me. There I have to use effective duration and convexity computed from a model that reprices the embedded option.

**C5. "How would you hedge the rate risk of a corporate bond position on a desk?"**

Model answer: I'd hedge on **DV01**, not raw duration, because DV01 already folds in the price level and my notional — it's the dollar P&L per basis point. I compute my position's DV01, then short enough of a liquid instrument — a Treasury future, an on-the-run Treasury, or an interest-rate swap — so its DV01 exactly offsets mine, leaving net DV01 near zero. Because the hedge instrument usually has a different duration, the face amounts won't match one-for-one: a shorter-duration hedge needs more notional. And I'd watch the residuals DV01 hedging leaves behind — curve risk (I'd DV01-match by key-rate bucket for a big book) and the credit spread, which a rates hedge doesn't touch.

**C6. "A bank held long-dated bonds funded by deposits and blew up when rates rose. What's the one-line diagnosis?"**

Model answer: A **duration gap** — the assets had far longer duration than the liabilities. When rates rose, the market value of the long assets fell hard while the short-duration deposits barely repriced, so net worth collapsed; formally, the equity's sensitivity is driven by  $\$D_A - D_L \times (L/A)\%$ , and that gap was large and positive. That's the SVB story: economically insolvent on a mark-to-market basis long before any borrower defaulted. The fix is asset-liability management — shorten asset duration or lengthen liability duration (or swap fixed to floating) so the gap closes.

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## Section D — Multiple-Choice Questions with Reasoning

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**D1. A bond has modified duration 8 and convexity 100. For a 100 bp rise in yield, the estimated price change is closest to:**

A. -8.5% B. -8.0% C. -7.5% D. -7.95%



**Answer: D.**  $\Delta P/P \approx -8(0.01) + \frac{1}{2}(100)(0.01)^2 = -0.08 + 0.005 = -0.0795 = -7.95\%$ . Answer B (−8.0%) is the duration-only estimate that ignores the +0.5% convexity offset; A and C are distractors. Convexity always makes the true loss smaller than duration alone predicts.

**D2. Which change would *increase* a bond's duration?**

A. a rise in its yield B. an increase in its coupon rate C. a lengthening of its maturity D. adding a call feature

**Answer: C.** Longer maturity pushes the cash-flow centre of mass further out, raising duration. A higher yield (A) and a higher coupon (B) both pull the centre of mass forward, *reducing* duration; a call feature (D) shortens expected life and reduces effective duration.

**D3. Price risk and reinvestment risk:**

A. both hurt the investor when rates rise B. are opposite-signed and offset at a horizon equal to duration C. both hurt the investor when rates fall D. never offset for a coupon bond

**Answer: B.** A rate rise hurts price but helps reinvestment; a fall does the reverse, so they are opposite-signed and cancel at the duration horizon — the immunisation result. A and C wrongly make them same-signed; D denies the offset that defines duration.

**D4. Two portfolios have the same modified duration but different key-rate duration profiles. Under a curve steepener they will:**

A. always move identically, since duration is equal B. potentially move in opposite directions C. both gain, because steepeners help all bonds D. both lose exactly the same amount

**Answer: B.** Total duration only captures parallel shifts; a steepener is a twist, and the portfolio with more long-end key-rate duration suffers while a front-loaded one may gain. That divergence — despite equal total duration — is precisely why desks monitor KRD profiles. A, C, and D all assume the parallel-shift world that a twist violates.

**D5. A callable bond trades near its call price as rates fall. Its convexity is best described as:**

A. positive and rising B. zero C. negative D. undefined

**Answer: C.** As rates fall toward the call, the issuer's option to call caps the price appreciation, so the price-yield curve bends the wrong way — negative convexity. This is why callables need effective (model-based) duration and convexity; A describes an option-free bond, B and D are simply wrong.

**D6. DV01 of a position is best described as:**

A. the percentage price change for a 1% yield move B. the dollar P&L for a 1 bp yield move C. the Macaulay duration times price D. always equal across two bonds with equal duration

**Answer: B.**  $\text{DV01} = D_{\text{Mod}} \times P \times 0.0001$  — the dollar value of a basis point on the actual position. A describes modified duration (a percentage, per 1%); C omits the  $\$1/(1+y)$  and the 0.0001 scaling and confuses Macaulay with dollar duration; D is false because DV01 depends on price and notional, not duration alone.

**D7. To immunise a single fixed liability, the standard prescription is to:**

A. maximise portfolio yield B. match asset duration to the liability horizon and PV, with asset convexity  $\geq$  liability convexity C. buy the longest-duration bond available D. match the coupon rate to the liability's discount rate

**Answer: B.** Immunisation sets  $D_{\text{assets}} = H$  and  $P_{\text{assets}} = PV(L)$  so price and reinvestment risk offset, with asset convexity at least the liability's so non-parallel shifts leave you no worse off. A ignores risk entirely; C over-hedges duration and courts huge price risk; D confuses coupon with duration and provides no protection.

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*Self-check: every numerical answer was reconciled — B1's duration-plus-convexity estimate matched the exact reprice to within 0.2 cents; B2's hedge DV01 (2,850/bp) exactly offset the long; B3's effective-duration inputs satisfied the definition and its positive numerator confirmed positive convexity; B4's weights reproduced duration 7.0 at the correct PV; B5's weighted average summed to 5.90; and B6's  $\pm 200$  bp asymmetry followed directly from the sign and size of the convexity term.*

# Chapter 10 — Credit Risk and Spreads

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## 1. The Problem / Need

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Every bond we have priced so far has quietly assumed one thing: the promised cash flows actually arrive. For a Treasury bond that assumption is close to true — a government that prints its own currency can always pay in nominal terms. But the moment you lend to a company, a bank, a municipality, or a sovereign that borrows in a currency it cannot print, a new and very real risk enters: **the borrower may not pay you back in full, or on time.**

This changes the whole valuation problem. Consider two three-year bonds issued on the same day, each paying a 5% annual coupon on a face value of 100. One is a Treasury; the other is issued by a mid-sized industrial company. If both were certain to pay, no-arbitrage says they must trade at the same price. They do not. The corporate bond trades *cheaper* — its yield is *higher* — and the gap does not disappear no matter how efficient the market is. That gap is not a mispricing. It is the market's price for bearing the chance of non-payment, and for the fact that this chance itself fluctuates.

So the practitioner faces three linked questions that this chapter answers:

1. **How large is the credit risk?** We need to decompose "the borrower might not pay" into measurable pieces — how likely is default, and how much do we lose if it happens.
2. **How is that risk priced?** The extra yield a risky bond offers over a safe benchmark is the **credit spread**. But there are several ways to measure a spread (nominal, G-spread, Z-spread, OAS), and they answer subtly different questions.
3. **Why do spreads move?** A bond can lose value not only because the issuer defaults, but because its perceived risk worsens — a **credit migration** — or because the whole market repriced risk over the cycle. A bond trader loses far more money to spreads *widening* than to outright defaults.

Getting this right is the core of credit investing, bank loan pricing, and every "risk-free rate plus a spread" discounting exercise in finance.

## 2. The Core Idea

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A risky bond's yield can be split into two parts:

$$\text{Risk\text{y} yield} = \underbrace{\text{Risk-free rate}}_{\text{time value of money}} + \underbrace{\text{Credit spread}}_{\text{compensation for default risk}}$$

The **credit spread** is the extra return per year an investor demands for holding a bond that might not pay, instead of an otherwise-identical default-free bond. It is *not* pure profit — it is compensation, and it can be decomposed further:

$$\text{Credit spread} \approx \underbrace{\text{Expected loss}}_{\text{actuarial cost of default}} + \underbrace{\text{Credit risk premium}}_{\text{reward for bearing uncertainty}} + \underbrace{\text{Liquidity premium}}_{\text{cost of hard-to-sell}}$$

The **expected loss** piece is the actuarial heart of credit risk, and it is built from three primitives:

- **Probability of Default (PD):** the chance the issuer fails to meet its obligations over a horizon.

- **Loss Given Default (LGD):** the fraction of exposure you lose *if* default occurs; equivalently  $1 - \text{Recovery Rate}$ .
- **Exposure at Default (EAD):** how much is at risk when default hits (for a plain bond, roughly the face plus accrued).

Put together, over one period:

$$\text{Expected Loss} = PD \times LGD \times EAD$$

The rest of the chapter makes each of these precise, shows the different ways the market *quotes* the spread, and explains what pushes spreads around.

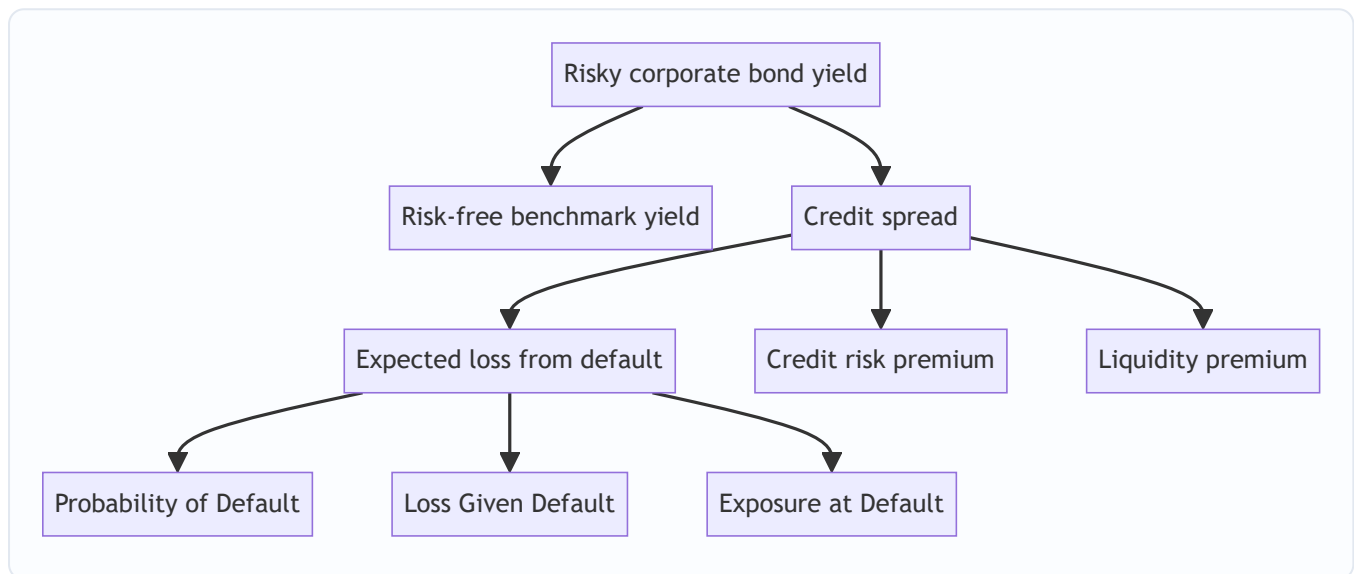


Figure 1 — A risky yield decomposes into a risk-free base plus a spread, and the spread itself decomposes into expected loss plus premiums for risk and illiquidity.

### 3. Why / How It Works

#### Why a spread must exist

Imagine an investor who is risk-neutral and cares only about *expected* return. Suppose a one-year zero-coupon corporate bond has a 2% chance of defaulting, and in default it recovers 40% of face. To be *indifferent* between this bond and a risk-free bill yielding 3%, the promised (contractual) yield on the corporate must be high enough that its *expected* return equals 3%. Because 2% of the time you only get 40 cents on the dollar, the promised payoff must be raised to make the average come out right. Solving that indifference condition produces a promised yield above 3% — the difference is the **spread that just covers expected loss**.

Real investors are risk-averse, not risk-neutral. They dislike the *uncertainty* of the outcome, not just its average. So they demand more than the actuarially fair spread — an extra **risk premium**. And because corporate bonds are harder to sell quickly than Treasuries, they demand a **liquidity premium** on top. That is why observed spreads are consistently *wider* than realized default losses would justify — the gap is the reward investors have historically earned for bearing credit and liquidity risk.

## Why default is not the only danger

Here is the insight that separates novices from practitioners: **most credit losses on marked-to-market portfolios come from spread widening, not from default.** A bond downgraded from BBB to BB does not miss a single coupon, yet its price can fall several points overnight because the market now demands a wider spread to hold it. This is **credit migration risk** — the risk that the issuer's credit quality *deteriorates* (or improves) while you hold the bond. Over a one-year horizon, a diversified investment-grade portfolio experiences far more mark-to-market volatility from migration and general spread moves than from the handful of names that actually default.

## Why we need several spread measures

The naive spread — "corporate yield minus Treasury yield" — is intuitive but technically sloppy, because a single yield-to-maturity is a blended average that ignores the *shape* of the yield curve. When the term structure is steep, discounting every cash flow at one number misstates value. So the market built a **ladder of increasingly precise spread measures** — nominal/G-spread, I-spread, Z-spread, and option-adjusted spread — each stripping out one more source of contamination so that what remains is a cleaner read on pure credit compensation.

## 4. Full Content — Formulas and Bond Math

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### 4.1 The building blocks of default risk

**Recovery rate and LGD.** If a defaulted bond ultimately returns  $R\%$  cents per dollar of face:

$$\text{Recovery Rate} = R, \quad \text{LGD} = 1 - R$$

Recovery depends heavily on **seniority** and **collateral**. Senior secured debt might recover 60–70 cents; senior unsecured 40 cents (a common rule-of-thumb assumption); subordinated debt 20 cents or less. Seniority is the single biggest driver of LGD.

**Probability of default over time — the hazard rate.** Let  $\lambda$  be the **hazard rate** (also called the *conditional* or *marginal* default intensity): the probability of defaulting *during* a period, *given survival up to* the start of it. Then:

- Probability of surviving one period:  $1 - \lambda$ .
- Probability of surviving  $t$  periods (constant hazard):  $(1 - \lambda)^t$ .
- **Cumulative probability of default** by time  $t$ :

$$PD_{\text{cum}}(t) = 1 - (1 - \lambda)^t$$

- **Marginal probability of default** in period  $t$  (default *in* that period, survive to its start):

$$PD_{\text{marg}}(t) = (1 - \lambda)^{t-1} \lambda$$

Note that cumulative PD is *not* just  $\lambda \times t$ ; that linear approximation only holds for small  $\lambda$  and short horizons because survival probability shrinks each period.

### Expected loss (single period).

$$EL = PD \times LGD \times EAD = PD \times (1 - R) \times EAD$$

As a first approximation, a bond's credit spread over one year is close to its annualized expected loss rate,  $PD \times LGD$ . Anything above that is the risk and liquidity premium.

## 4.2 Structural vs reduced-form views (why PD exists)

Two modelling traditions explain *where* PD comes from:

- **Structural models (Merton):** A firm defaults when the market value of its assets falls below the face value of its debt at maturity. Equity is viewed as a *call option* on the firm's assets struck at the debt level. This ties credit risk to leverage and asset volatility — more debt or more volatile assets raise PD. It is the intuition behind models like Moody's KMV.
- **Reduced-form models:** Rather than model the firm's balance sheet, treat default as a random event arriving with hazard rate  $\lambda$ , and *calibrate*  $\lambda$  from market prices (bond spreads, CDS). These dominate day-to-day pricing because they fit observed spreads directly.

A useful bridge is the **credit triangle**, the reduced-form approximation linking spread, hazard, and recovery:

$$\text{spread} \approx \lambda \times (1 - R) = \lambda \times \text{LGD}$$

i.e., **spread  $\approx$  (default intensity)  $\times$  (loss given default)**. This one line is the workhorse for moving between quoted spreads and implied default probabilities.

## 4.3 The ladder of spread measures

Now the pricing side. We hold the risky bond's *market price* fixed and ask: what spread reconciles it with a risk-free benchmark?

**(a) Nominal spread / G-spread.** The simplest: the risky bond's YTM minus the YTM of a government benchmark of the *same maturity*.

$$\text{G-spread} = y_{\text{corp}} - y_{\text{govt}}$$

If the exact-maturity benchmark does not exist, interpolate between the two nearest Treasuries. Weakness: both are single blended yields, so a steep or curved term structure distorts the comparison.

**(b) I-spread.** Same idea but benchmarked to the **swap curve** rather than the government curve:  $y_{\text{corp}}$  minus the interpolated interest-rate-swap rate of matching maturity. Common in Europe and for floating-rate reference.

**(c) Z-spread (zero-volatility spread).** The **constant spread  $Z$**  that, when *added to every spot (zero) rate* on the benchmark curve, makes the present value of the bond's cash flows equal its market price:

$$P_{\text{market}} = \sum_{t=1}^n \frac{CF_t}{(1 + s_t + Z)^t}$$

where  $s_t$  is the benchmark spot rate for maturity  $t$ . Because it uses the *whole* spot curve rather than one YTM, the Z-spread is curve-consistent and is the standard measure for bonds without embedded options. You solve for  $Z$  iteratively.

**(d) Option-adjusted spread (OAS).** For bonds with embedded options (callable, putable, mortgage-backed), the Z-spread is contaminated by the option's value. The **OAS** is the spread over the benchmark curve *after* the option has been valued and stripped out, typically using an interest-rate model that averages the bond's value across many simulated rate paths. The relationships are:

$$\text{For a callable bond: } \text{OAS} = \text{Z-spread} - \text{Option cost (in bp)}$$

$$\text{For a putable bond: } \text{OAS} = \text{Z-spread} + \text{Option value (in bp)}$$

For an **option-free** bond,  $\text{OAS} = \text{Z-spread}$  (there is no option to remove). OAS is the cleanest measure of pure credit-plus-liquidity compensation because it is comparable across bonds with different option

features.

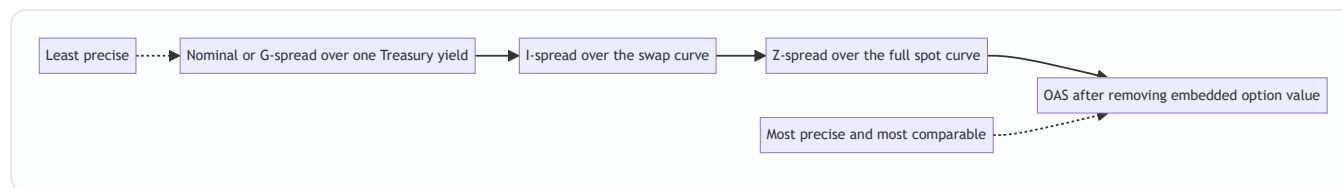


Figure 2 — The spread ladder. Each rung strips out one more distortion — curve shape, then reference base, then embedded optionality — leaving a purer measure of credit compensation.

## 4.4 Credit migration and transition matrices

Rating agencies publish **transition matrices**: the empirical probability that an issuer rated  $\$X$  today is rated  $\$Y$  one year later (including the "D" default state). A simplified matrix:

| From \ To | A   | BBB | BB   | Default |
|-----------|-----|-----|------|---------|
| A         | 91% | 8%  | 0.9% | 0.1%    |
| BBB       | 5%  | 88% | 5%   | 2%      |
| BB        | 1%  | 8%  | 80%  | 11%     |

Rows sum to 100%. Reading the BBB row: a BBB issuer has an 88% chance of staying BBB, 5% of upgrading to A, 5% of slipping to BB, and 2% of defaulting within the year. Because a downgrade *widens the spread the market demands*, it lowers the bond's price even with no missed payment. Multiplying each destination's probability by the bond's value in that state gives the **expected value** next year — and the spread of that distribution is the input to **credit VaR**.

## 4.5 What drives spreads over the cycle

Spreads are not static; they breathe with the economy:

- **Expansion / low default environment:** Corporate earnings strong, defaults rare, investors reach for yield. Spreads *tighten* (compress), sometimes to levels that under-compensate for risk.
- **Late cycle / rising uncertainty:** Leverage has built up, investors grow cautious. Spreads begin to *widen* even before defaults rise.
- **Recession / stress:** Defaults spike, liquidity evaporates, forced selling. Spreads *blow out* — and crucially, they widen *more than expected losses alone justify*, because the risk premium and liquidity premium both surge.
- **Recovery:** As conditions stabilize, spreads mean-revert tighter, usually the best period for credit *returns* (you earn carry plus capital gains from tightening).

The key relationships:

- **Spread and credit quality:** lower rating → wider spread, and the widening is *convex* — the step from BBB to BB is far larger than A to BBB.
- **Spread and the business cycle:** counter-cyclical — widest in recessions, tightest in booms.
- **Spread and equity markets:** spreads and equity volatility move together; a spike in the VIX typically comes with wider credit spreads (both price the same underlying fear).

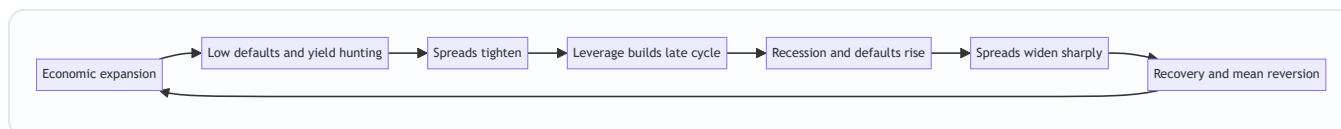


Figure 3 — Credit spreads are counter-cyclical, compressing through expansions and blowing out in recessions before mean-reverting.

## 5. Worked Examples

### Example 1 — Expected loss and the fair spread

A one-year corporate bond has face value 100, a hazard (default) probability of **2%**, and an expected **recovery of 40%** of face if it defaults.

**Step 1 — LGD.**  $\$LGD = 1 - R = 1 - 0.40 = 0.60\$$ .

**Step 2 — Expected loss rate.**  $\$EL = PD \times LGD = 0.02 \times 0.60 = 0.012 = 1.20\%\$$  of face.

**Step 3 — Fair spread (actuarial).** To first order, the spread that just covers expected loss is  **$\approx 120$  bp**. Check with the credit-triangle:  $\$s \approx \lambda \times LGD = 0.02 \times 0.60 = 1.20\%\$$ . ✓ Both routes agree.

**Interpretation.** If this bond actually trades at a 200 bp spread, then roughly 120 bp compensates for expected default loss and the remaining  $\sim 80$  bp is risk premium plus liquidity premium.

### Example 2 — Cumulative vs marginal default probability

Using the same 2% annual hazard rate, what is the chance of default over **two years**?

**Naive (wrong) answer:**  $\$2\% + 2\% = 4\%\$$ .

**Correct.** Survival each year is  $\$1 - 0.02 = 0.98\$$ .

- Cumulative PD by year 2:  $\$PD_{\text{cum}}(2) = 1 - 0.98^2 = 1 - 0.9604 = 0.0396 = \mathbf{\approx 3.96\%\$}$ .
- Marginal PD in year 2:  $\$PD_{\text{marg}}(2) = 0.98 \times 0.02 = 0.0196 = 1.96\%\$$ .

**Reconcile:** cumulative = year-1 default + year-2 marginal default =  $\$2.00\% + 1.96\% = 3.96\%\$$ . ✓ The cumulative figure is below the naive 4% because you can only default in year 2 if you *survived* year 1.

### Example 3 — Solving for the Z-spread, then the G-spread

A three-year corporate bond pays a **5% annual coupon** on face 100 and trades at a market price of **98.739**. The benchmark **spot (zero) rates** are:

| Maturity | Spot rate $s_t$ |
|----------|-----------------|
| 1 yr     | 3.00%           |
| 2 yr     | 3.50%           |
| 3 yr     | 4.00%           |

**Step 1 — Set up the Z-spread equation.** Find constant  $\$Z\$$  such that:

$$98.739 = \frac{5}{(1.03+Z)^1} + \frac{5}{(1.035+Z)^2} + \frac{105}{(1.04+Z)^3}$$

**Step 2 — Try  $\$Z = 1.50\%\$$  (150 bp).** Discount rates become 4.50%, 5.00%, 5.50%:



| $t$ | $CF_t$ | Discount rate | $(1+r)^t$    | PV            |
|-----|--------|---------------|--------------|---------------|
| 1   | 5      | 4.50%         | 1.04500      | 4.7847        |
| 2   | 5      | 5.00%         | 1.10250      | 4.5351        |
| 3   | 105    | 5.50%         | 1.17424      | 89.4194       |
|     |        |               | <b>Total</b> | <b>98.739</b> |

The present value is **98.739**, exactly the market price, so **Z-spread = 150 bp**. ✓

**Step 3 — Now the G-spread.** First find the bond's YTM by solving  $98.739 = \frac{5}{(1+y)} + \frac{5}{(1+y)^2} + \frac{105}{(1+y)^3}$ . Iterating gives  $y_{\text{corp}} \approx 5.47\%$  (at 5.47% the PV is 98.73, matching).

Next, the benchmark yield. A default-free 3-year 5% bond priced on the *same spot curve* (i.e.  $Z = 0$ ) is worth:

$$\frac{5}{1.03} + \frac{5}{1.035^2} + \frac{105}{1.04^3} = 4.8544 + 4.6676 + 93.3446 = 102.867$$

Its YTM solves to  $y_{\text{govt}} \approx 3.97\%$ .

**Step 4 — G-spread.**  $\text{G-spread} = y_{\text{corp}} - y_{\text{govt}} = 5.47\% - 3.97\% = \mathbf{1.50\%} = 150 \text{ bp}$ .

**Reconcile.** Here the G-spread (150 bp) and Z-spread (150 bp) coincide. That is *not* a universal law — it happens because this spot curve is only gently sloped, so the single-yield approximation of the G-spread barely distorts. On a **steep** curve the two would diverge, and the Z-spread would be the number to trust because it respects the whole term structure. If the bond were **callable**, we would go one rung further and compute an OAS below 150 bp, subtracting the value of the call the investor is short.

### Example 4 — Credit migration and expected value

You hold a BBB bond currently worth **100**. Over the next year the transition probabilities and the bond's value in each resulting state are:

| Next-year state | Probability | Bond value      | Prob × Value  |
|-----------------|-------------|-----------------|---------------|
| Upgrade to A    | 5%          | 101.0           | 5.050         |
| Stay BBB        | 88%         | 100.0           | 88.000        |
| Downgrade to BB | 5%          | 96.5            | 4.825         |
| Default         | 2%          | 40.0 (recovery) | 0.800         |
|                 | <b>100%</b> |                 | <b>98.675</b> |

**Expected value next year = 98.675**, an expected change of **−1.325**.

**Decompose the −1.325** into contributions (probability × value change from 100):

- Upgrade:  $0.05 \times (101 - 100) = +0.050$
- Stay:  $0.88 \times 0 = 0$
- Downgrade:  $0.05 \times (96.5 - 100) = -0.175$
- Default:  $0.02 \times (40 - 100) = -1.200$

Sum:  $\$+0.050 + 0 - 0.175 - 1.200 = \mathbf{\$-1.325}$ . ✓ Reconciles with the table.

**Interpretation.** Even though default (2% chance) is the single biggest drag (−1.20), note that the *migration* states (upgrade + downgrade) together move value by −0.125 with **no missed payment at all**. Over a large investment-grade portfolio, where default is rare, this migration channel dominates the year-to-year mark-to-market swings — the practitioner's point from Section 3.

## 6. Connections

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- **Chapter 03 (Bond Pricing) & 05 (Spot/Forward Rates):** The Z-spread is literally the spot-rate discounting machinery of Chapter 5 with a constant add-on. You cannot compute a Z-spread without a bootstrapped spot curve.
- **Chapter 04 (Yield Measures):** The G-spread is a difference of two YTM's, so all the caveats about YTM being a blended, reinvestment-assuming average carry straight into the G-spread's imprecision.
- **Chapter 06 (Term Structure):** The *shape* of the curve is exactly what makes G-spread and Z-spread diverge; a flat curve collapses them together (as in Example 3).
- **Duration & Convexity (interest-rate risk chapters):** Credit adds **spread duration** — sensitivity of price to a change in the spread, holding the risk-free curve fixed. A bond has both interest-rate duration and spread duration; a widening of 100 bp in spread hurts by (spread duration × 1%).
- **Credit Default Swaps (CDS):** The credit-triangle  $\$ \approx \lambda \times \text{LGD}$  is the bridge between cash-bond spreads and the CDS market; the gap between CDS and bond spreads is the **basis**.
- **Equity (Merton model):** Structural models make corporate debt and equity two claims on the same asset pool, linking credit spreads to equity volatility and leverage.

## 7. Key Terms

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- **Credit / default risk:** risk the borrower fails to pay interest or principal in full and on time.
- **Probability of Default (PD):** likelihood of default over a horizon; *conditional* (hazard/marginal) vs *cumulative*.
- **Hazard rate ( $\lambda$ ):** probability of default in a period given survival to its start.
- **Loss Given Default (LGD):** fraction of exposure lost in default;  $\$ = 1 - \text{Recovery Rate}$ .
- **Recovery rate:** cents-on-the-dollar recovered after default; driven mainly by seniority and collateral.
- **Exposure at Default (EAD):** amount at risk when default occurs.
- **Expected Loss (EL):**  $\$PD \times \text{LGD} \times \text{EAD}$ ; the actuarial cost of credit.
- **Credit spread:** extra yield of a risky bond over a default-free benchmark.
- **G-spread / nominal spread:** corporate YTM minus a matching-maturity government YTM.
- **I-spread:** spread over the interpolated swap curve.
- **Z-spread (zero-volatility spread):** constant add-on to every benchmark spot rate that reprices the bond.
- **Option-adjusted spread (OAS):** Z-spread net of embedded-option value; comparable across optioned and non-optioned bonds.
- **Credit migration:** change in an issuer's rating/quality over time.
- **Transition matrix:** probabilities of moving between ratings over a horizon.
- **Spread duration:** price sensitivity to a change in the credit spread.
- **Credit triangle:**  $\$ \approx \lambda \times \text{LGD}$ , linking spread, default intensity, and loss severity.

## 8. Common Confusions

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- **"Spread = expected loss."** No. Expected loss ( $\text{PD} \times \text{LGD}$ ) is only the *floor*. Observed spreads are wider because investors demand a risk premium and a liquidity premium. Historically, high-grade spreads have exceeded realized losses by a comfortable margin — that gap is the credit investor's edge.
- **"Recovery rate and LGD are two separate inputs."** They are the same information:  $\text{LGD} = 1 - \text{Recovery}$ . Quote one, and you have the other.
- **"Cumulative PD equals annual PD times number of years."** Only approximately, for small hazards. Correctly,  $\text{PD}_{\text{cum}}(t) = 1 - (1 - \lambda)^t$ , which is *below*  $\lambda t$  because you must survive each earlier year to default later.
- **"G-spread and Z-spread are basically the same."** They coincide only when the spot curve is flat/gently sloped (Example 3 was engineered that way). On a steep curve they diverge, and the Z-spread is the correct one.
- **"OAS is always smaller than Z-spread."** True for *callable* bonds (subtract call value), but for *puttable* bonds the OAS is *larger* than the Z-spread, because the put is a benefit to the investor. For option-free bonds they are equal.
- **"You only lose money on a bond if it defaults."** The single most expensive misconception in credit. Downgrades and market-wide spread widening produce large mark-to-market losses with every coupon still paid — this is migration/spread risk, and for investment-grade portfolios it dwarfs default losses.
- **"A wider spread means the bond is a better buy."** Not necessarily — it may simply mean the market has correctly identified more risk. The question is whether the spread *over-compensates* for the true risk.

## 9. Recap

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Credit risk is the possibility that a borrower does not pay in full and on time. We decompose it into **PD**, **LGD** ( $= 1 - \text{recovery}$ ), and **EAD**, whose product is the **expected loss**. A risky bond's yield is the risk-free rate plus a **credit spread**, and that spread pays for expected loss *plus* a risk premium *plus* a liquidity premium — which is why spreads sit above realized default losses.

Because a single YTM is a crude summary, the market quotes spreads on a ladder of increasing precision: **G-spread** (over one Treasury yield), **I-spread** (over swaps), **Z-spread** (a constant add-on to the whole spot curve), and **OAS** (Z-spread with embedded-option value removed). The **credit triangle**  $\approx \lambda \times \text{LGD}$  ties observed spreads back to implied default intensities.

Beyond outright default, holders face **credit migration** — rating changes that reprice the bond via the spread even when no payment is missed — quantified with **transition matrices**. And spreads are **counter-cyclical**: tight in booms, blowing out in recessions by more than expected losses alone, because risk and liquidity premiums surge in stress. The worked examples reconciled expected loss (120 bp), cumulative vs marginal PD (3.96% over two years), a Z-spread and G-spread that both landed at 150 bp on a mild curve, and a migration-driven expected value of 98.675 that decomposed cleanly back to  $-1.325$ .

## 10. Quick Reference / Interview Points

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**Core formulas** - Expected loss:  $\text{EL} = \text{PD} \times \text{LGD} \times \text{EAD}$  - LGD:  $\text{LGD} = 1 - \text{Recovery Rate}$  - Cumulative PD:  $\text{PD}_{\text{cum}}(t) = 1 - (1 - \lambda)^t$ ; marginal PD:  $(1 - \lambda)^{t-1} \lambda$  - Credit triangle:  $\text{spread} \approx \lambda \times \text{LGD}$  - Z-spread: constant  $Z$  with  $P = \sum_t \frac{\text{CF}_t}{(1 + Z)^t}$

$\{(1 + s_t + Z)^t\}$  - Callable OAS  $= Z\text{-spread} - \text{option cost}$ ; Puttable OAS  $= Z\text{-spread} + \text{option value}$ ; option-free OAS  $= Z\text{-spread}$

**Rapid-fire answers** - *Difference between G-spread and Z-spread?* G-spread uses one blended benchmark YTM; Z-spread adds a constant to the entire spot curve, so it is curve-consistent. They match on a flat curve and diverge on a steep one. - *When is OAS below the Z-spread?* For callable bonds — the investor is short the call, so its value is subtracted. - *Biggest driver of recovery/LGD?* Seniority and collateral of the claim. - *Do most credit losses come from default?* No — for investment-grade portfolios, spread widening and downgrades (migration) drive most mark-to-market losses. - *Why do spreads exceed expected loss?* Risk premium plus liquidity premium. - *How do spreads behave over the cycle?* Counter-cyclical: tight in expansions, wide in recessions, and they overshoot expected losses in stress. - *How are bond spreads and CDS linked?* Through the credit triangle; their difference is the CDS-bond basis. - *Structural vs reduced-form?* Structural (Merton) ties default to assets falling below debt and links to leverage/volatility; reduced-form calibrates a hazard rate directly from market prices.

**Numbers worth memorizing** - Senior unsecured recovery rule-of-thumb  $\approx 40\%$  (LGD  $\approx 60\%$ ). - Investment-grade  $\rightarrow$  high-yield boundary is BBB–/BB+; spread widening from IG to HY is convex, not linear. - 1 bp of spread duration means a 100 bp spread move changes price by (spread duration) percent.

# Q&A — Credit Risk and Spreads

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A companion practice bank for Chapter 10. Every question is followed by a full answer. Unless stated otherwise, rates are **annually compounded**, spreads are quoted in **basis points (bp)**, and the workhorse relations are: expected loss  $EL = PD \times LGD \times EAD$ , loss given default  $LGD = 1 - R$ , cumulative default  $PD_{\text{cum}}(t) = 1 - (1 - \lambda)^t$ , and the credit triangle  $s \approx \lambda \times LGD$ .

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## Section A — Concept Check

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**A1. What exactly is a credit spread, and why must it be positive for a risky bond?** A credit spread is the extra yield per year a risky bond offers over an otherwise-identical default-free benchmark:  $\text{risky yield} = \text{risk-free rate} + \text{spread}$ . It must be positive because the risky bond might not pay in full or on time, so investors will only buy it at a lower price (higher yield) than the safe bond. The spread is compensation, not free profit — it pays for expected loss plus a risk premium plus a liquidity premium. If the spread were zero, no-arbitrage would say the two bonds are identical claims, which they are not.

**A2. Decompose the credit spread into its economic parts.**  $\text{spread} \approx \text{expected loss} + \text{credit risk premium} + \text{liquidity premium}$ . The expected-loss piece ( $PD \times LGD$  annualized) is the actuarial cost of default. The risk premium rewards investors for bearing the *uncertainty* of the outcome (they are risk-averse, not risk-neutral). The liquidity premium compensates for corporates being harder to sell than Treasuries. This is why observed spreads sit consistently *above* realized default losses — the gap is the credit investor's historical edge.

**A3. Define PD, LGD, EAD, and recovery, and give the relations among them.** PD is the probability of default over a horizon. LGD is the fraction of exposure lost if default occurs. EAD is the amount at risk when default hits (roughly face plus accrued for a plain bond). Recovery rate  $R$  is cents-on-the-dollar recovered, so  $LGD = 1 - R$  — they are the *same* information stated two ways. Together,  $EL = PD \times LGD \times EAD$ .

**A4. What is a hazard rate, and why isn't cumulative PD just  $\lambda \times t$ ?** The hazard rate  $\lambda$  is the *conditional* probability of defaulting during a period given survival to its start. Cumulative PD by time  $t$  is  $1 - (1 - \lambda)^t$ , which is *below*  $\lambda t$  because to default in a later year you must first survive every earlier year. The linear approximation  $\lambda t$  only holds for small  $\lambda$  and short horizons; over long horizons it overstates default probability.

**A5. What is the biggest driver of the recovery rate?** Seniority and collateral. Senior secured debt might recover 60–70 cents; senior unsecured around 40 cents (the common rule-of-thumb); subordinated debt 20 cents or less. Where you sit in the capital structure, and whether specific assets back the claim, dominate LGD far more than the identity of the issuer.

**A6. Explain the ladder of spread measures and what each rung strips out.** From least to most precise: **G-spread** (corporate YTM minus a matching-maturity government YTM — one blended yield vs one blended yield); **I-spread** (same, but over the swap curve); **Z-spread** (a constant add-on to *every* benchmark spot rate that reprices the bond — curve-consistent); **OAS** (the Z-spread after an embedded option's value is modelled out). Each rung removes one distortion: curve shape, then reference base, then optionality — leaving a cleaner read on pure credit-plus-liquidity compensation.

**A7. State the credit triangle and say what it is used for.**  $s \approx \lambda \times (1-R) = \lambda \times \text{LGD}$ ; spread approximately equals default intensity times loss given default. It is the reduced-form bridge for moving between a quoted market spread and an implied default probability, and it links cash-bond spreads to the CDS market.

**A8. What is credit migration, and why does it matter more than default for an IG portfolio?** Credit migration is a change in an issuer's rating/quality while you hold the bond. A downgrade from BBB to BB misses no coupon, yet the price falls because the market now demands a wider spread. For a diversified investment-grade portfolio — where outright defaults are rare — year-to-year mark-to-market volatility is driven far more by these migration and general spread moves than by the handful of names that actually default. "You only lose money if it defaults" is the single most expensive misconception in credit.

**A9. Structural vs reduced-form models — one line each.** Structural (Merton): a firm defaults when asset value falls below the face value of debt; equity is a call option on the firm's assets, so credit risk rises with leverage and asset volatility. Reduced-form: treat default as a random event arriving with hazard rate  $\lambda$  calibrated directly from market prices (spreads, CDS). Reduced-form dominates day-to-day pricing because it fits observed spreads by construction.

**A10. How do spreads behave over the business cycle?** Counter-cyclically. They tighten in expansions (strong earnings, few defaults, yield-hunting), begin widening late-cycle as leverage builds, and blow out in recessions — crucially by *more* than expected losses alone justify, because risk and liquidity premiums both surge in stress. They then mean-revert tighter in recovery, usually the best period for credit *returns* (carry plus tightening gains).

## Section B — Numerical Bond-Math Problems (step-by-step, reconciling)

**B1. Expected loss and the fair spread.** A one-year bond, face 100, hazard 2%, recovery 40%. Find LGD, expected loss, and the actuarial fair spread; confirm with the credit triangle. Step 1 —  $\text{LGD} = 1 - 0.40 = 0.60$ . Step 2 —  $\text{\$EL} = \text{PD} \times \text{LGD} = 0.02 \times 0.60 = 0.012 = 1.20\%$  of face = **120 bp**. Step 3 — Credit-triangle check:  $s \approx \lambda \times \text{LGD} = 0.02 \times 0.60 = 0.012 = \$ \text{120 bp}$  ✓. Both routes agree. Interpretation: if the bond actually trades at 200 bp, ~120 bp covers expected loss and the remaining ~80 bp is risk plus liquidity premium.

**B2. Cumulative vs marginal PD over three years.** Constant hazard  $\lambda = 3\%$ . Find cumulative PD by years 2 and 3 and the marginal PD in year 3; reconcile. Survival each year  $s = 0.97$ .  $\text{PD}_{\text{cum}}(2) = 1 - 0.97^2 = 1 - 0.9409 = 0.0591 = \text{mathbf{5.91\%}}$ .  $\text{PD}_{\text{cum}}(3) = 1 - 0.97^3 = 1 - 0.912673 = 0.087327 = \text{mathbf{8.73\%}}$ .  $\text{PD}_{\text{marg}}(3) = 0.97^2 \times 0.03 = 0.9409 \times 0.03 = 0.028227 = \text{mathbf{2.82\%}}$ . Reconcile:  $\text{PD}_{\text{cum}}(3) = \text{PD}_{\text{cum}}(2) + \text{PD}_{\text{marg}}(3) = 0.0591 + 0.028227 = 0.087327$  ✓. Note the naive  $3 \times 3\% = 9\%$  overstates the true 8.73%, because you can only default in year 3 if you survived years 1 and 2.

**B3. Implied hazard from a quoted spread (credit triangle in reverse).** A bond trades at a 300 bp spread; assumed recovery is 25%. Back out the implied annual default intensity.  $\text{LGD} = 1 - 0.25 = 0.75$ .  $\lambda \approx \frac{s}{\text{LGD}} = \frac{0.03}{0.75} = 0.04 = \text{mathbf{4.0\%}}$ . Reconcile forward:  $s \approx \lambda \times \text{LGD} = 0.04 \times 0.75 = 0.03 = \$ \text{300 bp}$  ✓. A higher assumed recovery would imply a *higher* hazard for the same spread — with less lost per default, more defaults are needed to justify the same compensation.

**B4. Solve for a Z-spread by trial and interpolation.** A 2-year bond pays a **6% annual coupon** on face 100 and trades at **99.1163**. Benchmark spot rates are 1-yr 3.00%, 2-yr 3.50%. Find the Z-spread. Set up: \$99.1163



$= \frac{6}{(1.03+Z)} + \frac{106}{(1.035+Z)^2}$ . Trial  $Z = 2.00\%$  → rates 5.0%, 5.5%:  $\frac{6}{1.05} + \frac{106}{1.055^2} = 5.7143 + 95.2359 = 100.950$  (too high). Trial  $Z = 4.00\%$  → rates 7.0%, 7.5%:  $\frac{6}{1.07} + \frac{106}{1.075^2} = 5.6075 + 91.7266 = 97.333$  (too low). Trial  $Z = 3.00\%$  → rates 6.0%, 6.5%:  $\frac{6}{1.06} + \frac{106}{1.065^2} = 5.6604 + 93.4559 = 99.1163$  ✓. The present value hits the market price exactly at  $Z = \mathbf{300\text{ bp}}$ .

**B5. Reconcile a Z-spread against a G-spread on a mild curve.** A 3-year bond pays a **5% annual coupon**, face 100, price **98.739**; benchmark spots are 3.00%, 3.50%, 4.00%. Z-spread: at  $Z = 1.50\%$  the discount rates are 4.5%, 5.0%, 5.5%:  $\frac{5}{1.045} + \frac{5}{1.05^2} + \frac{105}{1.055^3} = 4.7847 + 4.5351 + 89.4194 = 98.739$  ✓ → **Z = 150 bp**. G-spread: the corporate YTM solving the price is  $y_{\text{corp}} \approx 5.47\%$ . A default-free 3-year 5% bond priced off the *same* spots ( $Z=0$ ) is worth  $\frac{5}{1.03} + \frac{5}{1.035^2} + \frac{105}{1.04^3} = 102.867$ , whose YTM is  $y_{\text{govt}} \approx 3.97\%$ . So  $\text{G-spread} = 5.47\% - 3.97\% = \mathbf{150\text{ bp}}$ . Reconcile: G-spread and Z-spread coincide (150 bp) *only because the curve is gently sloped*. On a steep curve they would diverge, and the Z-spread — which respects the whole term structure — is the number to trust.

**B6. OAS from a Z-spread for callable and putable bonds.** (a) Callable bond: Z-spread = 220 bp, modelled call cost = 60 bp.  $\text{OAS} = 220 - 60 = \mathbf{160\text{ bp}}$ . The investor is *short* the call, so its value is subtracted —  $\text{OAS} < \text{Z-spread}$ . (b) Putable bond: Z-spread = 180 bp, modelled put value = 40 bp.  $\text{OAS} = 180 + 40 = \mathbf{220\text{ bp}}$ . The investor is *long* the put (a benefit), so its value is added back —  $\text{OAS} > \text{Z-spread}$ . Check the direction: for an *option-free* bond both option terms are zero, so  $\text{OAS} = \text{Z-spread}$ . Consistent ✓.

**B7. Credit migration — expected value and its decomposition.** You hold a BBB bond worth 100. Over one year:

| State           | Probability | Value | Prob × Value  |
|-----------------|-------------|-------|---------------|
| Upgrade to A    | 5%          | 101.0 | 5.050         |
| Stay BBB        | 88%         | 100.0 | 88.000        |
| Downgrade to BB | 5%          | 96.5  | 4.825         |
| Default         | 2%          | 40.0  | 0.800         |
| <b>Total</b>    | <b>100%</b> |       | <b>98.675</b> |

Expected value = **98.675**, an expected change of **−1.325**. Decompose (probability × change from 100): upgrade  $0.05 \times (+1) = +0.050$ ; stay  $0$ ; downgrade  $0.05 \times (-3.5) = -0.175$ ; default  $0.02 \times (-60) = -1.200$ . Sum  $= +0.050 - 0.175 - 1.200 = \mathbf{-1.325}$  ✓. Interpretation: default (−1.20) is the single biggest drag, but the migration states together move value by  $-0.125$  *with no missed payment*. Scale that across a large IG book where default is rare and migration dominates the mark-to-market swings.

**B8. Spread duration P&L.** A bond has spread duration 4.5 and its spread widens by 80 bp with the risk-free curve unchanged. Estimate the price impact.  $\Delta P/P \approx -(\text{spread duration}) \times \Delta s = -4.5 \times 0.0080 = -0.036 = \mathbf{-3.6\%}$ . Reconcile the concept: spread duration is the analogue of interest-rate duration but for a move in the *spread* only. An 80 bp widening on a 4.5-duration bond costs about 3.6 points per 100 of price — and it happens whether or not the bond ever defaults.

## Section C — Interview-Style (with model answers)

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**C1. "Why is a corporate bond's spread wider than its expected default loss?"** Model answer: Expected loss ( $\text{\$PD} \times \text{LGD}$ ) is only the actuarial floor. Investors are risk-averse, so they demand a premium for bearing the *uncertainty* of default timing and severity, not just its average. On top of that, corporates are less liquid than Treasuries, so there's a liquidity premium. Empirically, high-grade spreads have exceeded realized default losses by a comfortable margin over long horizons — that persistent gap is precisely the excess return credit investors have historically earned. So when I see a 200 bp spread on a name whose expected loss is 120 bp, I don't call it mispriced; I ask whether the 80 bp of premium adequately compensates for the risk and illiquidity I'm taking.

**C2. "Walk me through the difference between a Z-spread and an OAS, and when they diverge."** Model answer: The Z-spread is the constant number you add to every point on the benchmark spot curve so the discounted cash flows equal the market price — it's curve-consistent and it's the right measure for an option-free bond. But if the bond has an embedded option, the Z-spread is contaminated by that option's value. The OAS strips the option out using an interest-rate model that averages the bond's value across many simulated rate paths, leaving a spread that's comparable across bonds regardless of their option features. For a callable bond the OAS is *below* the Z-spread — the investor is short the call, so you subtract its value. For a puttable bond the OAS is *above* the Z-spread — the put benefits the investor. For an option-free bond they're identical. So the moment I'm comparing a callable to a bullet, I insist on OAS, not Z-spread.

**C3. "A junior analyst says the two-year default probability on a 2% hazard name is 4%. Correct them."** Model answer: That's the linear approximation, and it double-counts. To default in year two you must first survive year one. Survival each year is  $\$1 - 0.02 = 0.98$ , so cumulative default by year two is  $\$1 - 0.98^2 = 3.96\%$ , not 4%. The marginal probability of defaulting *in* year two is  $\$0.98 \times 0.02 = 1.96\%$ , and it reconciles: year-one default 2.00% plus year-two marginal 1.96% equals the cumulative 3.96%. The  $\lambda$  shortcut is fine for a quick mental estimate at small hazards and short horizons, but over long maturities or high-yield names it materially overstates default risk, so I'd never use it for pricing.

**C4. "How would you translate a market spread into an implied default probability?"** Model answer: The credit triangle: spread  $\approx$  hazard rate times loss given default,  $s \approx \lambda \times \text{LGD}$ . So the implied hazard is  $\lambda \approx s / \text{LGD}$ . If a name trades at 300 bp and I assume a 40% recovery — so LGD is 0.60 — the implied annual default intensity is roughly  $0.03/0.60 = 5\%$ . The key caveat is the recovery assumption: it's an input, not an observable, and the implied hazard is quite sensitive to it. A higher assumed recovery implies a higher hazard for the same spread, because each default costs less, so more of them are needed to justify the compensation. In practice I'd calibrate against the CDS curve, where the same triangle links the CDS premium to  $\lambda$  and LGD, and watch the CDS-bond basis.

**C5. "Where do most credit losses actually come from in an investment-grade portfolio?"** Model answer: Not from default — from spread widening and downgrades, i.e. migration. In an IG book, actual defaults are rare in any given year, but every name is marked to market, and a downgrade from, say, BBB to BB widens the spread the market demands and drops the price several points with no coupon missed. Over a one-year horizon, the mark-to-market volatility of a diversified IG portfolio is dominated by these migration and general spread moves, not by the one or two names that default. That's why credit risk management focuses on transition matrices and spread duration, and why credit VaR is built from the whole distribution of next-year values across rating states, not just the default state.

**C6. "The economy is heading into a recession. What happens to spreads, and why more than the models say?"** Model answer: Spreads widen — they're counter-cyclical, tightest in booms and widest in



recessions. But the important point for a trader is that they blow out by *more* than the rise in expected losses alone would justify. Two premiums surge simultaneously in stress: the risk premium, because investors become more risk-averse exactly when uncertainty is highest, and the liquidity premium, because dealers pull back and forced sellers can't find bids. So the spread overshoots fair value. That overshoot is painful if you're already long, but it's also why the recovery phase tends to deliver the best credit *returns* — you earn the carry plus capital gains as spreads mean-revert tighter. The relationship with equities is tight too: a VIX spike almost always comes with wider credit spreads, because both are pricing the same underlying fear.

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## Section D — Multiple Choice (with reasoning)

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- D1.** A bond has PD = 4%, recovery = 30%, EAD = 100. Its one-period expected loss is closest to: (a) 1.2 (b) 2.8 (c) 4.0 (d) 3.0 **Answer: (b).**  $\$EL = PD \times LGD \times EAD = 0.04 \times (1-0.30) \times 100 = 0.04 \times 0.70 \times 100 = 2.8\$$ . Choice (a) forgets the LGD, (c) forgets both LGD and uses PD directly, (d) mistakenly uses recovery instead of LGD.
- D2.** With a constant hazard  $\lambda = 5\%$ , cumulative default probability over 3 years is closest to: (a) 15.0% (b) 14.3% (c) 5.0% (d) 4.75% **Answer: (b).**  $\$1 - 0.95^3 = 1 - 0.857375 = 0.142625 = 14.3\%$ . Choice (a) is the naive  $\lambda t$ , which overstates because you must survive earlier years to default later.
- D3.** For a callable bond, the OAS is: (a) larger than the Z-spread (b) smaller than the Z-spread (c) equal to the Z-spread (d) always negative **Answer: (b).**  $OAS = Z\text{-spread} - \text{call cost}$ . The investor is short the call, so its value is subtracted, leaving OAS below the Z-spread. For a *puttable* bond the relationship reverses.
- D4.** Which is the correct relationship between recovery and LGD? (a)  $\$LGD = R\$$  (b)  $\$LGD = 1 + R\$$  (c)  $\$LGD = 1 - R\$$  (d) they are unrelated **Answer: (c).** Loss given default is the fraction *not* recovered,  $\$1 - R\$$ . They carry the same information stated two ways.
- D5.** The Z-spread is best described as: (a) the corporate YTM minus one government YTM (b) a constant added to every benchmark spot rate so the discounted cash flows equal the market price (c) the spread after removing embedded-option value (d) the spread over the swap curve **Answer: (b).** That's the definition of the zero-volatility spread. (a) is the G-spread, (c) is the OAS, (d) is the I-spread.
- D6.** The credit triangle states that a bond's spread is approximately: (a)  $\lambda / LGD$  (b)  $\lambda \times LGD$  (c)  $\lambda + LGD$  (d)  $\lambda \times R$  **Answer: (b).**  $\$s \approx \lambda \times LGD = \lambda \times (1-R)\$$ : default intensity times loss severity.
- D7.** In an investment-grade portfolio, the largest source of one-year mark-to-market loss is typically: (a) outright default (b) coupon reinvestment risk (c) credit migration and spread widening (d) inflation **Answer: (c).** Defaults are rare in IG; downgrades and general spread widening reprice bonds with every coupon still paid, dominating year-to-year MTM swings.
- D8.** A G-spread and a Z-spread on the same bond coincide when: (a) the bond is callable (b) the spot curve is steep (c) the spot curve is flat or gently sloped (d) recovery is zero **Answer: (c).** The G-spread's single-yield approximation only distorts when the curve has shape; on a flat/gentle curve it collapses onto the curve-consistent Z-spread. A steep curve makes them diverge.
- D9.** Which model views equity as a call option on the firm's assets? (a) reduced-form (b) structural (Merton) (c) transition-matrix (d) credit-triangle **Answer: (b).** The Merton structural model treats default as assets falling below the debt face value at maturity, making equity a call struck at the debt level — linking credit risk to leverage and asset volatility.

**D10.** A bond's spread duration is 6.0 and its spread narrows by 50 bp (curve unchanged). The approximate price impact is: (a) -3.0% (b) +3.0% (c) +0.3% (d) -0.5% **Answer: (b).**  $\Delta P/P \approx -(\text{spread duration}) \times \Delta s = -6.0 \times (-0.0050) = +0.030 = +3.0\%$ . A *narrowing* (negative  $\Delta s$ ) raises the price.

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*End of Q&A — Credit Risk and Spreads. All numerical answers were self-verified: expected loss (120 bp) matches the credit triangle; cumulative PD at  $\lambda=3\%$  reconciles as  $5.91\% + 2.82\% \text{ marginal} = 8.73\%$ ; the Z-spread solves exactly to 300 bp at price 99.1163 and to 150 bp at price 98.739; the implied hazard  $0.03/0.75 = 4.0\%$  reconciles forward; and the migration expected value 98.675 decomposes cleanly back to -1.325.*

# Chapter 11 — Credit Analysis

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## 1. The Problem / Need

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Every bond except a genuinely risk-free government note carries one dominant fear: **the borrower may not pay you back**. A US Treasury promises \$1,000 in ten years and — in nominal terms — you can treat that promise as certain. But a BB-rated industrial company promising the same \$1,000 might file for bankruptcy in year six, hand you thirty cents on the dollar, and wipe out four years of coupons. The whole *point* of buying that corporate bond is the extra yield you earn for shouldering that risk. Credit analysis is the discipline of deciding whether that extra yield is *enough*.

Put concretely, when you buy a corporate bond you are effectively doing two trades at once:

1. **Lending at the risk-free rate** — the pure time-value-of-money leg, priced off the Treasury curve.
2. **Selling default insurance** — you collect a premium (the credit spread) in exchange for eating losses if the issuer defaults.

Chapter 10 taught you how spreads are quoted and how they move. This chapter answers the prior question: **how do you decide what spread a given borrower actually deserves?** That means dissecting the borrower's ability and willingness to service debt, translating financial statements into a handful of decision-grade ratios, understanding how the rating agencies compress all of this into a letter grade, and knowing how quantitative models (structural and reduced-form) price default risk from first principles. Get this wrong and you either overpay for junk (and lose principal) or turn down fair compensation (and underperform). Credit analysis is where the money is made and lost in corporate fixed income.

A useful mental frame: **equity analysis asks "how much upside?"; credit analysis asks "how much can go wrong before I stop getting paid?"** The credit analyst is professionally paranoid. Your best case is that you get your coupons and your principal back — exactly what you were promised, no more. All the asymmetry is to the downside. That asymmetry shapes everything that follows.

## 2. Core Idea

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Creditworthiness rests on two independent pillars: **capacity** (can the borrower generate enough cash to service the debt?) and **willingness** (will they choose to, and are you legally protected if they waver?). A borrower can be rich and still stiff you; a borrower can be honest and still run out of money. You must underwrite both.

The industry has crystallised this into the **Four Cs of Credit: Capacity, Collateral, Covenants, and Character**. Everything a credit analyst does maps onto one of these four buckets. Capacity is measured with cash-flow and leverage ratios; collateral is the recovery you get if things break; covenants are the contractual guardrails; character is the reputational and governance question of whether management will honour the deal even when it hurts.

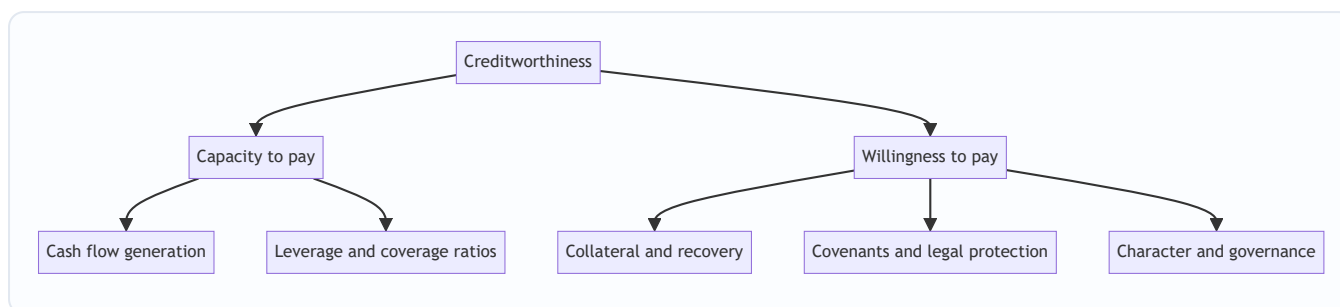


Figure 1 — Creditworthiness decomposes into capacity and willingness, and the Four Cs distribute across both pillars.

The quantitative heart of the analysis is a small set of ratios that answer the capacity question: **How much debt is stacked on the earnings? (leverage) How comfortably do the earnings cover the interest and principal owed? (coverage)** These ratios then drive — or corroborate — a **credit rating**, a letter grade (AAA down to D) assigned by an agency or built internally, which maps monotonically onto an expected default probability and therefore onto the spread the bond should trade at.

### 3. Why / How It Works

#### Default is an option the borrower holds

The deepest insight in credit analysis, and the one that unifies the whole chapter, is this: **equity is a call option on the firm's assets, and a risky bond is a risk-free bond minus a put option that the shareholders own.** When you lend to a levered firm, the shareholders have limited liability — if the firm's asset value falls below the face value of debt at maturity, they simply hand you the (now-insufficient) assets and walk away. That "walk-away" right is a put option struck at the face value of debt. You, the bondholder, are *short* that put. This is the seed of the structural models we develop in Section 4, and it explains mechanically why credit spreads widen when asset volatility rises: a more volatile firm makes the shareholders' put more valuable, so your short-put position costs you more, so you demand more spread.

#### Why cash flow, not earnings, services debt

Coupons are paid in cash, not in accounting net income. A company can report healthy profits while starving for cash — booking revenue on credit it never collects, or capitalising costs that will bleed cash later. This is why credit analysts anchor on **EBITDA** (a rough cash-earnings proxy) and, more rigorously, on **free cash flow** and the **cash flow statement**. The leverage and coverage ratios in Section 4 are almost all built on EBITDA or cash flow precisely because *debt is serviced out of cash, and the analyst's job is to find the cash.*

#### Why ratings and spreads are monotonic

Rating agencies exist because most investors cannot afford to underwrite every issuer themselves. The agency does the fundamental work once and publishes a letter grade that summarises a **through-the-cycle** view of default probability. Empirically, default rates rise smoothly and steeply as you descend the rating ladder — an AAA firm defaults perhaps 0.1% over five years while a CCC firm may default 40%+ over the same window. Because the market prices bonds off expected loss (probability of default times loss given default), spreads compress or widen in lockstep with the rating. This monotonic ladder — rating → default probability → expected loss → spread — is the scaffolding that connects fundamental analysis to market pricing.

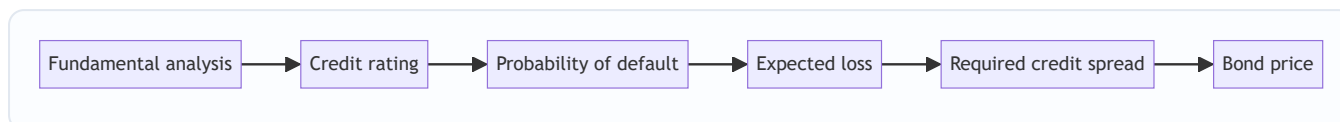


Figure 2 — The transmission chain from fundamentals through rating to the spread the bond must offer.

## Why covenants substitute for equity's control rights

Shareholders elect the board; bondholders do not. A lender's only levers are the price it charges and the **covenants** it negotiates into the indenture. Covenants convert the willingness question into contract law: a maintenance covenant that caps leverage at 4.0x Debt/EBITDA means that if the firm levers past that line, it is in technical default and you get a seat at the restructuring table *before* the firm has burned all its value. Covenants are how a passive creditor buys a measure of the active control that equity enjoys by default.

## 4. Full Content — The Four Cs, the Ratios, the Ratings, the Models

### 4.1 The Four Cs in depth

**Capacity** — the ability to repay. Assessed through the firm's business risk (industry cyclicality, competitive position, margins, scale, diversification) and financial risk (leverage, coverage, liquidity, and the maturity profile of the debt). Capacity is where the quantitative ratios live.

**Collateral** — the quality and value of assets pledged, and more broadly the **recovery** you can expect in default. A senior secured bond backed by hard assets might recover 70 cents on the dollar; a subordinated unsecured note might recover 20. Collateral drives **loss given default (LGD)**, the complement of the recovery rate.

**Covenants** — the terms in the bond indenture that protect the lender. Discussed fully in Section 4.5.

**Character** — management's integrity, track record, and the firm's governance and financial policy. Does management prioritise bondholders or serially lever up for shareholder buybacks? History of aggressive accounting? A shareholder-friendly financial policy is a *credit negative* even if today's ratios look fine.

### 4.2 Leverage ratios — how much debt is stacked on the cash flow

Leverage answers "how big is the debt relative to the earnings power that supports it?" The workhorse is:

$$\text{Debt/EBITDA} = \frac{\text{Total Debt}}{\text{EBITDA}}$$

where EBITDA = Earnings Before Interest, Taxes, Depreciation and Amortisation. This ratio has the intuitive reading of **"how many years of cash earnings it would take to repay all the debt"** if every dollar of EBITDA went to debt paydown. Rules of thumb (they vary by industry): under ~2x is conservative/investment-grade territory; 3–4x is typical leveraged-but-healthy; above ~5–6x is aggressive high-yield; above ~7x is deeply distressed.

Refinements the analyst must know:

- **Net Debt/EBITDA** = (Total Debt – Cash & equivalents) / EBITDA. Gives credit for cash on hand that could retire debt. Use with care — cash may be trapped offshore or needed for operations.
- **Gross vs net** — agencies often prefer gross (conservative); management always quotes net.
- **Adjusted debt** — add back operating-lease liabilities, pension deficits, and other debt-like obligations. Off-balance-sheet items flatter the raw ratio.

A second leverage lens uses the capital structure rather than earnings:

$$\text{Debt-to-Capital} = \frac{\text{Total Debt}}{\text{Total Debt} + \text{Equity}}$$

### 4.3 Coverage ratios — how comfortably earnings cover the obligations

Coverage answers "for each dollar of interest (or debt service) owed, how many dollars of earnings/cash are available to pay it?" Higher is safer.

#### Interest coverage (times interest earned):

$$\text{EBITDA Interest Coverage} = \frac{\text{EBITDA}}{\text{Interest Expense}}$$

An EBIT-based variant, more conservative because it subtracts depreciation:

$$\text{EBIT Interest Coverage} = \frac{\text{EBIT}}{\text{Interest Expense}}$$

Coverage above ~6–8x is strong investment grade; 3–4x is solid high-yield; below ~1.5–2x is a red flag — the cushion between cash earnings and the interest bill is thin, and a modest earnings decline pushes the firm to insolvency.

**Debt Service Coverage Ratio (DSCR)** — the coverage measure that includes *principal*, not just interest.

Interest coverage can look fine while a wall of maturing principal quietly threatens the firm. DSCR captures total debt service:

$$\text{DSCR} = \frac{\text{Net Operating Income (or EBITDA)}}{\text{Total Debt Service}} = \frac{\text{NOI}}{\text{Interest} + \text{Principal repayments}}$$

DSCR is central in **project finance, real estate, and leveraged lending**, where amortising principal is a first-order cash drain. A DSCR of 1.0 means cash flow exactly covers debt service with zero cushion; lenders typically demand 1.20–1.50x minimum. DSCR below 1.0 means the borrower must dip into reserves or refinance to stay current — a structural warning.

**Free-cash-flow-based coverage** — the most conservative, because FCF is cash *after* the capital spending needed to keep the business running:

$$\text{FCF} = \text{Cash from Operations} - \text{Capital Expenditure}$$
$$\text{FCF/Debt} = \frac{\text{Free Cash Flow}}{\text{Total Debt}}$$

FCF/Debt is a favourite agency ratio; its inverse is a "years to repay from true free cash" measure that is stricter than Debt/EBITDA.

## 4.4 The ratio summary table

| Ratio                    | Formula                      | What it measures                      | Direction of safety | Typical IG vs HY     |
|--------------------------|------------------------------|---------------------------------------|---------------------|----------------------|
| Debt/EBITDA              | Total Debt ÷ EBITDA          | Leverage — years of earnings to repay | Lower               | IG < 3x; HY 4–6x     |
| Net Debt/EBITDA          | (Debt – Cash) ÷ EBITDA       | Leverage net of cash                  | Lower               | IG < 2.5x            |
| Debt-to-Capital          | Debt ÷ (Debt + Equity)       | Balance-sheet leverage                | Lower               | IG < 40%             |
| EBITDA Interest Coverage | EBITDA ÷ Interest            | Cushion over interest bill            | Higher              | IG > 6x; HY 2–4x     |
| EBIT Interest Coverage   | EBIT ÷ Interest              | Coverage after D&A                    | Higher              | IG > 4x              |
| DSCR                     | NOI ÷ (Interest + Principal) | Cushion over total debt service       | Higher              | Lenders want > 1.25x |
| FCF/Debt                 | (CFO – Capex) ÷ Debt         | True cash generation vs debt          | Higher              | IG > 20%             |

*A single ratio never decides a credit. The analyst reads them as a system — high leverage is tolerable if coverage is strong and cash flow is stable; low leverage can still be risky if earnings are violently cyclical.*

## 4.5 Covenants — the contractual guardrails

Covenants come in two families:

**Affirmative (positive) covenants** — things the borrower *must* do: pay taxes, maintain insurance, deliver audited financials, keep assets in good repair, maintain its legal existence. Low-drama but foundational.

**Negative (restrictive) covenants** — things the borrower *must not* do without consequence. These are where the protection lives:

- **Limitation on indebtedness** — caps additional debt, often via an incurrence test (e.g., may not issue new debt unless pro-forma Debt/EBITDA stays below 4.0x).
- **Limitation on liens** (negative pledge) — cannot pledge assets to other lenders ahead of you.
- **Restricted payments** — caps dividends, buybacks, and distributions to shareholders (protects cash from leaking out to equity).
- **Limitation on asset sales** — sale proceeds must repay debt or be reinvested, not dividended out.
- **Change-of-control put** — lets you sell the bond back at ~101 if the firm is taken over (protects against a leveraging buyout).

A further crucial distinction:

- **Maintenance covenants** — tested *every period* regardless of any action (e.g., "leverage must be below 4.5x at each quarter-end"). A breach is an immediate technical default. Common in bank loans.
- **Incurrence covenants** — tested only when the borrower *takes an action* (issues debt, pays a dividend). Common in high-yield bonds. Weaker for lenders because a firm can deteriorate passively without ever

tripping them.

"**Covenant-lite**" (cov-lite) loans strip out maintenance covenants, leaving only incurrence tests. The proliferation of cov-lite structures since ~2010 is a standing concern for creditors because it removes the early-warning tripwire.

## 4.6 Credit ratings and the agencies

The three globally dominant **Nationally Recognized Statistical Rating Organizations (NRSROs)** are **S&P Global Ratings, Moody's, and Fitch**. Their scales run in parallel:

| Category           | S&P / Fitch                        | Moody's            | Meaning                                   |
|--------------------|------------------------------------|--------------------|-------------------------------------------|
| Prime              | AAA                                | Aaa                | Highest quality, minimal risk             |
| High grade         | AA+ / AA / AA-                     | Aa1 / Aa2 / Aa3    | Very strong                               |
| Upper medium       | A+ / A / A-                        | A1 / A2 / A3       | Strong, some susceptibility               |
| Lower medium       | BBB+ / BBB / BBB-                  | Baa1 / Baa2 / Baa3 | Adequate — lowest <b>investment grade</b> |
| — the line —       | <b>BBB- / Baa3 is the IG floor</b> |                    | Below this is speculative                 |
| Speculative        | BB+ / BB / BB-                     | Ba1 / Ba2 / Ba3    | "Junk" — elevated risk                    |
| Highly speculative | B+ / B / B-                        | B1 / B2 / B3       | High risk                                 |
| Substantial risk   | CCC / CC / C                       | Caaa / Ca / C      | Vulnerable, near default                  |
| Default            | D                                  | —                  | In default                                |

**The single most important line in fixed income is the boundary between BBB-/Baa3 (investment grade) and BB+/Ba1 (high yield).** It is not a smooth gradient in practice — it is a cliff. Enormous pools of capital (insurance companies, pension funds, many bond index funds) are contractually or regulatorily barred from holding sub-investment-grade paper. When an issuer is downgraded across that line — a "**fallen angel**" — forced selling by these holders can blow the spread out far more than the fundamental change alone would justify. Conversely a "**rising star**" upgraded into IG enjoys a burst of forced buying.

Key rating concepts:

- **Issuer rating vs issue rating** — the issuer rating reflects overall default probability; individual *issues* are notched up or down from it based on seniority and collateral (a senior secured bond rated above, a subordinated note below, the issuer's baseline).
- **Outlook and watch** — "negative outlook" signals a likely downgrade over 1–2 years; "credit watch / rating watch" signals a near-term review, often tied to an event like a merger.
- **Through-the-cycle** — agencies deliberately rate across the business cycle, so ratings are stickier and less volatile than market spreads, which are point-in-time.
- **Split rating** — when agencies disagree; convention often uses the lower (or the middle of three).
- **The conflict of interest** — the **issuer-pays** model, where the borrower pays for its own rating, is the structural flaw exposed in 2008 when structured-finance ratings proved wildly optimistic. A credit analyst treats ratings as an input, never gospel.



## 4.7 Quantitative default models — structural vs reduced-form

Beyond ratios and letter grades, two families of mathematical models price default risk directly.

**Structural models (Merton, 1974)** derive default from the firm's balance sheet using option theory. Recall the core idea: equity is a call option on the firm's assets, struck at the face value of debt  $D$ , expiring at debt maturity  $T$ . The firm defaults if, at  $T$ , asset value  $A_T < D$ . Model the asset value as a lognormal (geometric Brownian motion) process with volatility  $\sigma_A$ , and the probability of default and the value of the debt fall straight out of the Black–Scholes machinery.

The **distance to default** is the number of standard deviations the asset value sits above the default point:

$$\text{Distance to Default} = \frac{\ln(A_0/D) + (\mu - \frac{1}{2}\sigma_A^2)T}{\sigma_A\sqrt{T}}$$

and, under the model, the (risk-neutral) probability of default is  $N(-\text{DD})$  where  $N(\cdot)$  is the standard normal CDF. Structural models are economically intuitive — they tie default to leverage and asset volatility, exactly the levers a fundamental analyst watches — and they explain *why* spreads widen with volatility. Their weaknesses: they assume default only at maturity (extended by later models), require the *unobservable* asset value and asset volatility (backed out from equity), and tend to under-predict short-term spreads because a firm rarely defaults "by surprise" the instant assets dip below debt. Moody's KMV commercialised this approach into the widely used **Expected Default Frequency (EDF)**.

**Reduced-form models (Jarrow–Turnbull, Duffie–Singleton)** take the opposite tack. They do *not* model the firm's assets at all. Instead they treat default as an unpredictable event — a jump — governed by an exogenous **hazard rate** (default intensity)  $\lambda$ , and calibrate that intensity directly to observed market prices (bond spreads, CDS quotes). The probability of surviving to time  $t$  is:

$$P(\text{survival to } t) = e^{-\lambda t} \quad \rightarrow \quad P(\text{default by } t) = 1 - e^{-\lambda t}$$

Their strength is tractability and a direct link to market data — they are what desks actually use to strip default probabilities out of CDS curves. Their weakness is that they are a-theoretical about *why* default happens: the hazard rate is a fitted parameter, not an economic quantity, so they offer less fundamental insight and can be blindsided if the market itself is mispricing risk.

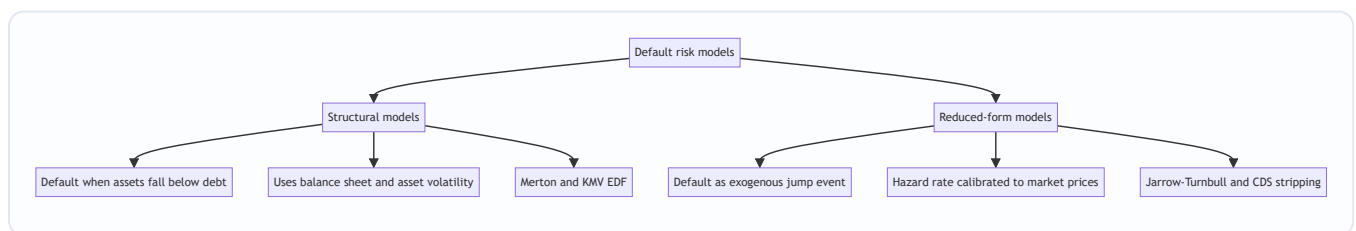


Figure 3 — Structural models explain default from firm fundamentals; reduced-form models fit default intensity to market prices.

## 4.8 The credit-risk arithmetic: from default probability to spread

The bridge from analysis to price is **expected loss**:

$$\text{Expected Loss} = \text{PD} \times \text{LGD} = \text{PD} \times (1 - \text{Recovery Rate})$$

where PD is probability of default and LGD is loss given default. The **credit spread** an investor demands must at minimum compensate for this expected annual loss (plus a risk premium for the *uncertainty* of that loss). A first-order approximation over a short horizon:

$$\text{Credit Spread} \approx \text{PD} \times \text{LGD} = \lambda \times (1 - R)$$

This tiny equation is the Rosetta Stone linking the reduced-form hazard rate  $\lambda$ , the recovery rate  $R$ , and the observable spread. We exploit it in Worked Example 3 to strip a default probability out of a market spread.

## 5. Worked Examples

### Worked Example 1 — Building the ratio picture for a credit decision

**Setup.** Zephyr Industries reports the following for the year (all in \$ millions):

| Item                                       | Value |
|--------------------------------------------|-------|
| Revenue                                    | 2,000 |
| EBITDA                                     | 400   |
| Depreciation & Amortisation                | 120   |
| Interest expense                           | 90    |
| Cash from operations (CFO)                 | 300   |
| Capital expenditure                        | 130   |
| Total debt                                 | 1,600 |
| Cash & equivalents                         | 200   |
| Scheduled principal repayments (this year) | 110   |
| Shareholders' equity                       | 900   |

**Compute the core ratios.**

$$EBIT = EBITDA - D\&A = 400 - 120 = \mathbf{280}.$$

$$Debt/EBITDA = 1,600 / 400 = \mathbf{4.0x}.$$

$$Net\ Debt/EBITDA = (1,600 - 200) / 400 = 1,400 / 400 = \mathbf{3.5x}.$$

$$Debt\text{-}to\text{-}Capital = 1,600 / (1,600 + 900) = 1,600 / 2,500 = \mathbf{64\%}.$$

$$EBITDA\ interest\ coverage = 400 / 90 = \mathbf{4.44x}.$$

$$EBIT\ interest\ coverage = 280 / 90 = \mathbf{3.11x}.$$

$$DSCR = EBITDA / (\text{Interest} + \text{Principal}) = 400 / (90 + 110) = 400 / 200 = \mathbf{2.0x}.$$

$$Free\ cash\ flow = CFO - Capex = 300 - 130 = \mathbf{170}.$$

$$FCF/Debt = 170 / 1,600 = \mathbf{10.6\%}.$$

**Interpretation and decision.** Leverage at 4.0x gross (3.5x net) sits squarely in **high-yield-but-healthy** territory — clearly below investment grade, but not distressed. EBITDA interest coverage of 4.4x is comfortable; the firm earns \$4.44 of cash earnings for every \$1 of interest. But note how the picture *tightens* as we get stricter: EBIT coverage falls to 3.1x once real depreciation is charged, and DSCR of 2.0x —

while adequate — shows that including the \$110m principal wall halves the effective cushion versus interest alone. FCF/Debt of 10.6% is middling; it would take roughly 9–10 years to repay the debt from true free cash.

**Verdict:** consistent with a **BB / Ba** rating. Lend, but demand a high-yield spread, and insist on a maintenance leverage covenant (say 4.5x) and a restricted-payments basket so management cannot lever up further to fund buybacks.

**Self-check (reconciliation).** DSCR must be lower than EBITDA interest coverage because its denominator is strictly larger (interest + principal > interest):  $2.0x < 4.44x$ . ✓. Net Debt/EBITDA must be lower than gross because cash is subtracted from the numerator:  $3.5x < 4.0x$ . ✓. EBIT coverage must be lower than EBITDA coverage because D&A is subtracted from the numerator:  $3.11x < 4.44x$ . ✓. The internal ordering is coherent.

## Worked Example 2 — A covenant stress test and downgrade trigger

**Setup.** Zephyr's bond indenture contains a **maintenance covenant**: Debt/EBITDA must not exceed **4.5x** at any quarter-end, and an **incurrence covenant** barring new debt if pro-forma leverage would exceed **4.25x**. A recession now cuts EBITDA by 20%, from 400 to **320**, while debt is unchanged at 1,600. Separately, management wants to issue \$200m of new debt to fund an acquisition.

### Step 1 — recompute leverage after the earnings shock.

New Debt/EBITDA =  $1,600 / 320 = 5.0x$ .

This **breaches the 4.5x maintenance covenant** ( $5.0x > 4.5x$ ). The firm is in **technical default** even though it has missed no payment. This is the early-warning power of a maintenance covenant: the creditor gets a seat at the table *before* cash actually runs out. The lender can now demand repricing, additional collateral, or accelerate the loan.

### Step 2 — recompute interest coverage to gauge real distress. Assume interest still 90.

EBITDA interest coverage =  $320 / 90 = 3.56x$  (down from 4.44x).

Still above the danger zone, so the firm can *service* its debt — this is a covenant breach driven by the leverage *ratio*, not an actual liquidity crisis yet. That distinction matters: it is a negotiating breach, not a missed coupon.

**Step 3 — test the acquisition financing against the incurrence covenant.** Pro-forma debt =  $1,600 + 200 = 1,800$ ; pro-forma EBITDA (pre-recession, say deal adds nothing initially) = 400.

Pro-forma Debt/EBITDA =  $1,800 / 400 = 4.5x$ .

Since 4.5x **exceeds the 4.25x incurrence limit**, the firm is **contractually blocked** from issuing the new debt. The covenant does its job — it prevents management from leveraging up to chase an acquisition that would erode the creditor's position.

**Self-check.** After the shock, leverage rose ( $5.0x > 4.0x$ ) and coverage fell ( $3.56x < 4.44x$ ) — both move in the direction that hurts creditors, as a 20% earnings drop should. The maintenance test (tested always) trips on the recession; the incurrence test (tested on action) trips on the debt issuance. Two different tripwires, two different triggers — exactly as designed. ✓

## Worked Example 3 — Stripping a default probability out of a market spread

**Setup.** A 5-year Zephyr bond trades at a **credit spread of 300 basis points (3.00%)** over the Treasury curve. Market convention assumes a **recovery rate of 40%** (so LGD = 60%). What annual probability of default (hazard rate) is the market pricing in? What cumulative 5-year default probability does that imply?

**Step 1 — apply the credit-triangle approximation.** Using  $\text{Spread} \approx \lambda \times \text{LGD}$ :

$$\lambda \approx \frac{\text{Spread}}{\text{LGD}} = \frac{0.0300}{0.60} = 0.050 = 5.0\% \text{ per year.}$$

The market is pricing an implied **annual (risk-neutral) default probability of about 5%**.

**Step 2 — convert to a cumulative 5-year default probability** using the reduced-form survival formula:

$$P(\text{survive 5 yr}) = e^{-\lambda T} = e^{-0.05 \times 5} = e^{-0.25} = 0.7788. \\ P(\text{default within 5 yr}) = 1 - 0.7788 = 0.2212 \approx \mathbf{22.1\%}.$$

So a 300 bp spread with 40% recovery is consistent with roughly a **1-in-5 chance of default over the bond's life** — squarely a high-yield profile, and coherent with the BB assessment from Example 1.

**Step 3 — sanity-check the expected annual loss against the spread.** Expected annual loss  $\approx \lambda \times \text{LGD} = 0.05 \times 0.60 = 0.030 = 3.00\% = 300 \text{ bp}$ . This exactly reproduces the spread, confirming the arithmetic is internally consistent — the spread is (to first order) pure expected-loss compensation with no residual risk premium in this simplified frame.

**Step 4 — test sensitivity to the recovery assumption.** Suppose instead recovery is only **20%** ( $\text{LGD} = 80\%$ ), as it might be for a *subordinated* Zephyr note:

$$\lambda \approx \frac{0.0300}{0.80} = 0.0375 = 3.75\% \text{ per year.}$$

A *higher* recovery assumption forces a *higher* implied default probability to justify the same spread, and vice versa. This is the crucial modelling caveat: **PD and recovery are jointly unidentified from a single spread** — you can only strip one out if you assume the other. Get the recovery assumption wrong and your implied default probability is wrong in the opposite direction.

**Reconciliation across the three examples.** Example 1's fundamentals (4.0x leverage, 4.4x coverage) said "BB." Example 3's market spread (300 bp implying ~22% five-year PD) is exactly what a BB name should trade at. The fundamental view and the market-implied view agree — which is the reassuring outcome. When they *disagree* (your ratios say BB but the market prices CCC), that gap is either your trading opportunity or the market knowing something you don't. Reconciling the fundamental picture with the market-implied picture is the daily work of the credit analyst.

## 6. Connections

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**To spreads and pricing (Ch. 10).** This chapter supplies the *fundamentals* behind the spread that Chapter 10 taught you to quote and hedge. Credit analysis decides *what* spread is fair; spread mechanics decide *how* that spread moves and is traded. The credit-triangle ( $\text{Spread} \approx \text{PD} \times \text{LGD}$ ) is the hinge between the two chapters.

**To duration and convexity (Ch. 7–8).** A credit downgrade raises the discount rate on the bond, and duration tells you how much the price drops for that spread widening (**spread duration**). Credit risk and interest-rate risk both flow through the same price-sensitivity machinery; a bond has *both* a Treasury-rate duration and a spread duration.

**To the term structure (Ch. 5–6).** Default probability is not constant across the maturity — credit curves slope (usually upward for healthy names, and can *invert* for distressed names where near-term default risk dominates). The hazard rate  $\lambda$  in the reduced-form model can be made term-dependent, producing a credit-spread term structure that mirrors the spot-rate term structure.

**To equity analysis.** The structural (Merton) model literally treats equity as a call option on assets — so the credit analyst and the equity analyst are looking at the same balance sheet from opposite sides of the capital structure. Rising asset volatility helps equity (option value up) and hurts credit (short-put position more costly). This is why credit and equity desks watch each other.

**To derivatives (CDS).** A credit default swap is the traded, isolated form of the default insurance embedded in a corporate bond. Reduced-form models are calibrated primarily to CDS quotes, and the CDS-bond basis is a core relative-value trade. Everything in Section 4.7–4.8 is the theoretical engine under the CDS market.

## 7. Key Terms

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- **Four Cs** — Capacity, Collateral, Covenants, Character; the framework for assessing creditworthiness.
- **Capacity** — the borrower's ability to generate cash to service debt.
- **EBITDA** — earnings before interest, taxes, depreciation and amortisation; a cash-earnings proxy.
- **Leverage ratio** — debt relative to earnings or capital (e.g., Debt/EBITDA); higher = riskier.
- **Coverage ratio** — earnings/cash relative to debt obligations (e.g., EBITDA/interest, DSCR); higher = safer.
- **DSCR (Debt Service Coverage Ratio)** — cash flow ÷ (interest + principal); coverage including principal.
- **Free cash flow (FCF)** — cash from operations minus capital expenditure; cash truly available to creditors.
- **Investment grade (IG)** — rated BBB–/Baa3 or higher; **high yield / speculative / junk** — rated BB+/Ba1 or lower.
- **Fallen angel / rising star** — a bond downgraded from IG to HY / upgraded from HY to IG.
- **Covenant** — a contractual term in the indenture; **affirmative** (must do), **negative/restrictive** (must not do).
- **Maintenance vs incurrence covenant** — tested every period vs tested only on a specific action; **cov-lite** = incurrence-only.
- **NRSRO** — Nationally Recognized Statistical Rating Organization (S&P, Moody's, Fitch).
- **Notching** — adjusting an individual issue's rating up/down from the issuer rating for seniority/collateral.
- **Probability of default (PD) / Loss given default (LGD) / Recovery rate** —  $LGD = 1 - \text{Recovery}$ .
- **Expected loss** —  $PD \times LGD$ ; the minimum the spread must compensate for.
- **Structural model** — default modelled from the balance sheet as an option (Merton, KMV EDF); **distance to default**.
- **Reduced-form model** — default as an exogenous jump governed by a **hazard rate / default intensity ( $\lambda$ )** calibrated to market prices.
- **Credit spread** — yield over the risk-free benchmark compensating for default risk;  $\approx PD \times LGD$ .

## 8. Common Confusions

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**"High leverage automatically means bad credit."** No. Leverage must be read against cash-flow *stability* and coverage. A regulated utility with contracted, predictable cash flows can safely carry 5–6x leverage; a mining company with violently cyclical earnings is dangerous at 2.5x. The ratio is meaningless without the business-risk context.

**"EBITDA is cash flow."** EBITDA *ignores* changes in working capital, capital expenditure, taxes, and interest. A firm can have rising EBITDA and *negative* free cash flow if it is bleeding cash into inventory and capex. Charlie

Munger's jibe — "think of it as bulls\*\*t earnings" — overstates, but the caution is real: always cross-check EBITDA against actual cash from operations.

**"Interest coverage is enough to judge debt service."** Interest coverage ignores principal. A firm with 8x interest coverage can still fail if a large principal maturity wall arrives and it cannot refinance. DSCR and the maturity schedule catch what interest coverage misses.

**"A credit rating is a buy/sell recommendation or a probability."** A rating is an *ordinal opinion* on relative default risk through the cycle — not a cardinal default probability, not a market-timing signal, and not a comment on price/value. A fairly-priced CCC can be a better investment than an over-priced AAA.

**"Investment grade vs high yield is a smooth gradient."** It is a cliff, not a slope, because of the regulatory/mandate wall at BBB–/BB+. The spread jump and forced-selling dynamics around that line are discontinuous.

**"Structural and reduced-form models are competitors where one is right."** They answer different questions. Structural models give economic *insight* (why default happens, tied to fundamentals); reduced-form models give *calibration* to market prices (what the market implies right now). Practitioners use both — structural for surveillance and early warning, reduced-form for pricing and stripping PDs from CDS.

**"Net debt is always the right leverage measure."** Netting cash assumes the cash is available and permanent. Cash may be trapped in foreign subsidiaries, pledged, or needed as an operating minimum. Conservative (agency) analysis often uses gross debt for precisely this reason.

## 9. Recap

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Credit analysis decides whether a borrower's promise to repay is worth the spread on offer. It rests on two pillars — **capacity** (can they pay?) and **willingness** (will they, and are you protected?) — operationalised through the **Four Cs**: Capacity, Collateral, Covenants, Character. Because debt is serviced out of *cash*, the quantitative core is a compact set of ratios: **leverage** ratios (Debt/EBITDA, Net Debt/EBITDA, Debt-to-Capital) measuring how much debt sits on the earnings, and **coverage** ratios (EBITDA/interest, EBIT/interest, DSCR, FCF/Debt) measuring how comfortably the earnings cover what is owed. These ratios feed a **credit rating** — an ordinal, through-the-cycle opinion from S&P, Moody's, or Fitch — whose most consequential feature is the **investment-grade / high-yield cliff** at BBB–/Ba1. Ratings map monotonically onto default probability and therefore onto the spread the bond should trade at, via **expected loss = PD × LGD**. Two model families formalise the pricing: **structural models** (Merton, KMV) that derive default from the balance sheet as an option and explain *why* spreads widen with leverage and volatility, and **reduced-form models** that treat default as an exogenous hazard-rate jump calibrated to market prices. **Covenants** — maintenance vs incurrence, affirmative vs restrictive — convert the willingness question into enforceable contract law and give the passive creditor early-warning tripwires. The credit decision, finally, reconciles the *fundamental* picture (what the ratios and business risk say the credit is worth) against the *market-implied* picture (what the spread says the market believes), and lends only when the spread over-compensates for the analyst's honest assessment of expected loss.

## 10. Quick-Reference / Interview Points

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- **The Four Cs**: Capacity, Collateral, Covenants, Character. Capacity = cash to pay; the other three = protection and willingness.

- **Leverage headline:** Debt/EBITDA = "years of earnings to repay all debt." Rough map: <2x conservative IG; 3–4x levered-healthy; >6x aggressive/distressed.
- **Coverage headline:** EBITDA/Interest ("times interest earned"). >6x strong IG; ~3x solid HY; <1.5x danger. **DSCR adds principal** to the denominator — always lower than interest coverage; lenders want >1.25x.
- **The credit triangle: Spread  $\approx$  PD  $\times$  LGD =  $\lambda \times (1 - \text{Recovery})$ .** Memorise this — it lets you strip an implied default probability out of any spread given a recovery assumption. Watch out: PD and recovery are jointly unidentified from one spread.
- **Survival math (reduced-form):**  $P(\text{default by } T) = 1 - e^{(-\lambda T)}$ . A constant hazard rate  $\lambda$  compounds into a cumulative PD.
- **The IG/HY cliff:** BBB–/Baa3 is the lowest investment grade; BB+/Ba1 is the highest junk. Fallen angels get force-sold; the spread jump across the line is discontinuous, not smooth.
- **Agencies:** S&P, Moody's, Fitch (NRSROs). Issuer-pays conflict; ratings are through-the-cycle and stickier than market spreads. Issue ratings are *notched* off the issuer rating for seniority.
- **Structural vs reduced-form (classic interview question):** Structural (Merton) = default when assets < debt, equity is a call option, uses balance-sheet data, gives economic insight, under-predicts short-term spreads. Reduced-form (Jarrow–Turnbull) = default is an exogenous jump with hazard rate  $\lambda$  fitted to market prices, tractable for pricing/CDS, but a-theoretical about *why* default happens.
- **Equity = call option on assets; risky bond = risk-free bond – a put.** Higher asset volatility → shareholders' walk-away put worth more → bondholder's short-put costs more → spread widens. This is the one-sentence bridge between credit and options.
- **Covenants:** maintenance (tested always, immediate technical default on breach, common in loans) vs incurrence (tested only on an action, common in HY bonds, weaker). Cov-lite = incurrence-only, removes the early-warning tripwire.
- **Golden rule of the credit mindset:** the upside is capped at "you get paid what you were promised" — all the asymmetry is downside, so underwrite the downside first. EBITDA is *not* cash flow; always reconcile it to actual cash from operations.



# Q&A — Credit Analysis

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Companion practice bank for Chapter 11. Every question is followed by a full answer. Unless stated otherwise, figures are in \ \$ millions, spreads are over the Treasury curve, and default probabilities are risk-neutral (market-implied).

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## Section A — Concept Check

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**A1. What are the two independent pillars of creditworthiness, and why must you underwrite both?**

**Capacity** (can the borrower generate enough cash to service the debt?) and **willingness** (will they pay, and are you protected if they waver?). They are independent: a cash-rich borrower can still stiff you, and an honest borrower can simply run out of money. Underwriting only one leaves the other flank exposed.

**A2. State the Four Cs of credit and map each to a pillar. Capacity, Collateral, Covenants, Character.**

Capacity is the ability-to-pay pillar, measured with leverage and coverage ratios. Collateral (recovery if things break), Covenants (contractual guardrails), and Character (management integrity and financial policy) sit on the willingness/protection pillar — they govern whether and how much you recover when capacity falters.

**A3. Why do credit analysts anchor on cash flow and EBITDA rather than net income?** Coupons are paid in cash, not accounting profit. A firm can report healthy net income while starving for cash — booking revenue on receivables it never collects, or capitalising costs that bleed cash later. EBITDA is a rough cash-earnings proxy; free cash flow is the rigorous measure. Debt is serviced out of cash, so the analyst's job is to *find the cash*.

**A4. Explain the sentence "equity is a call option and a risky bond is a risk-free bond minus a put."**

Shareholders have limited liability. At maturity, if assets exceed the debt face value they keep the surplus (a call struck at the debt); if assets fall short, they hand you the insufficient assets and walk away — a put, struck at the debt face value, that shareholders own and the bondholder is *short*. So a risky bond = risk-free bond – short put. That is why spreads widen with asset volatility: more volatility makes the shareholders' put more valuable, so your short-put costs more.

**A5. Distinguish leverage ratios from coverage ratios. Leverage** ratios (Debt/EBITDA, Debt-to-Capital) measure *how much debt is stacked on the earnings or capital*; lower is safer. **Coverage** ratios (EBITDA/Interest, DSCR) measure *how comfortably earnings cover the obligations owed*; higher is safer. Leverage asks "how big is the debt?"; coverage asks "how easily can I pay what's due?"

**A6. Why can interest coverage look healthy while the credit is actually at risk?** Interest coverage ignores *principal*. A firm with 8x interest coverage can still fail when a wall of maturing principal arrives and it cannot refinance. DSCR (which adds principal to the denominator) and the debt maturity schedule catch what interest coverage misses.

**A7. What is the single most important line in fixed income, and why is it a cliff rather than a slope?**

The boundary between **BBB–/Baa3 (lowest investment grade)** and **BB+/Ba1 (highest junk)**. It is a cliff because enormous pools of capital — insurers, pensions, many index funds — are barred from holding sub-investment-grade paper. A downgrade across the line ("fallen angel") triggers forced selling that blows the spread out far beyond the fundamental change; an upgrade ("rising star") triggers forced buying.

**A8. Contrast structural and reduced-form default models in one line each. Structural (Merton, KMV):**

default happens when firm asset value falls below the face value of debt; derived from the balance sheet via



option theory; gives economic *insight*. **Reduced-form (Jarrow–Turnbull, Duffie–Singleton)**: default is an exogenous jump governed by a hazard rate  $\lambda$  calibrated to market prices (spreads, CDS); tractable for *pricing* but a-theoretical about *why*.

**A9. State the "credit triangle" and its most important caveat.** Spread  $\approx$  PD  $\times$  LGD  $= \lambda \times (1 - \text{Recovery})$ . It links hazard rate, recovery, and spread, letting you strip an implied default probability out of any spread. The caveat: **PD and recovery are jointly unidentified from a single spread** — you can only solve for one by assuming the other; get recovery wrong and your implied PD is wrong the opposite way.

**A10. Distinguish maintenance from incurrence covenants and explain why cov-lite worries creditors.**

**Maintenance** covenants are tested *every period* regardless of any action (e.g., leverage below 4.5x each quarter-end); a breach is immediate technical default, an early-warning tripwire. **Incurrence** covenants are tested only when the borrower *takes an action* (issues debt, pays a dividend). **Cov-lite** loans strip out maintenance covenants — so a firm can deteriorate passively without ever tripping a test, removing the creditor's early warning.

## Section B — Numerical Bond-Math Problems

### B1. Building the ratio picture

Meridian Foods reports: EBITDA 300; D&A 90; Interest 60; Cash from operations 220; Capex 100; Total debt 1,050; Cash 150; Scheduled principal 90; Equity 700. Compute the core credit ratios and give a verdict.

**Step 1 — derive EBIT.** EBIT = EBITDA – D&A = 300 – 90 = **210**.

**Step 2 — leverage.** Debt/EBITDA = 1,050 / 300 = **3.5x**. Net Debt/EBITDA = (1,050 – 150) / 300 = 900 / 300 = **3.0x**. Debt-to-Capital = 1,050 / (1,050 + 700) = 1,050 / 1,750 = **60%**.

**Step 3 — coverage.** EBITDA interest coverage = 300 / 60 = **5.0x**. EBIT interest coverage = 210 / 60 = **3.5x**. DSCR = EBITDA / (Interest + Principal) = 300 / (60 + 90) = 300 / 150 = **2.0x**.

**Step 4 — cash generation.** FCF = CFO – Capex = 220 – 100 = **120**. FCF/Debt = 120 / 1,050 = **11.4%**.

**Verdict.** 3.5x gross leverage (3.0x net) with 5.0x interest coverage is **levered-but-healthy** — sub-investment grade but not distressed, a **BB / Ba** profile. Lend at a high-yield spread with a maintenance leverage covenant.

**Self-check.** Net < gross leverage (cash subtracted): 3.0x < 3.5x ✓. EBIT < EBITDA coverage (D&A subtracted): 3.5x < 5.0x ✓. DSCR < interest coverage (principal added to denominator): 2.0x < 5.0x ✓. Ordering coherent.

### B2. Stripping a default probability from a spread

A 5-year Meridian bond trades at a **200 bp** spread; convention assumes **50% recovery** (LGD = 50%). Find the implied annual hazard rate and cumulative 5-year default probability.

**Step 1 — credit triangle for the hazard rate.**  $\lambda \approx \text{Spread} / \text{LGD} = 0.0200 / 0.50 = \mathbf{4.0\% \text{ per year}}$ .

**Step 2 — convert to cumulative 5-year PD** using the reduced-form survival formula:  $P_{\text{survive}} = e^{-\lambda T} = e^{-0.04 \times 5} = e^{-0.20} = 0.8187 \rightarrow P_{\text{default}} = 1 - 0.8187 = \mathbf{18.1\%}$ .

**Step 3 — sensitivity to recovery.** If recovery is only **30%** (LGD = 70%), as for a *subordinated* note:  $\lambda \approx 0.0200 / 0.70 = \mathbf{2.86\%}$ .

**Reconcile.** A *higher* recovery assumption forces a *higher* implied PD to justify the same spread — the joint-identification caveat. The base case (~18% five-year PD) is squarely high-yield, coherent with B1's BB assessment.

### B3. Distance to default (Merton structural model)

A firm has asset value  $A_0 = 120$ , debt face value  $D = 100$ , asset drift  $\mu = 8\%$ , asset volatility  $\sigma_A = 25\%$ , horizon  $T = 1$  year. Compute the distance to default and the model PD.

**Step 1 — distance to default.** 
$$\text{DD} = \frac{\ln(A_0/D) + (\mu - \frac{1}{2}\sigma_A^2)T}{\sigma_A\sqrt{T}}$$
 Numerator:  $\ln(1.20) = 0.18232$ ; drift  $= (0.08 - 0.5 \times 0.25^2) = 0.04875$ . So  $\text{DD} = \frac{0.18232 + 0.04875}{0.25} = \frac{0.23107}{0.25} = \mathbf{0.924}$ .

**Step 2 — default probability.**  $\text{PD} = N(-\text{DD}) = N(-0.924) \approx \mathbf{0.178}$  from the standard normal CDF — about a **17.8%** one-year PD.

**Self-check.** Assets sit only 0.92 standard deviations above the default point — thin. A less levered firm ( $D = 80$ ) gives  $\text{DD} \approx 1.815$  and  $\text{PD} = N(-1.815) \approx 3.5\%$  — far safer. PD falls sharply as leverage falls, as the structural logic requires ✓.

### B4. Fair spread from an agency default table

An agency table says a BB name has a **12% cumulative 5-year default probability** with **40% recovery** (LGD = 60%). What annualized credit spread does this imply?

**Step 1 — back out the constant hazard rate.** Survival =  $1 - 0.12 = 0.88 = e^{(-5\lambda)}$ , so  $\lambda = -\ln(0.88)/5 = 0.12783/5 = \mathbf{2.56\%}$ .

**Step 2 — apply the credit triangle:** Spread  $\approx \lambda \times \text{LGD} = 0.02557 \times 0.60 = \mathbf{153 \text{ bp}}$ .

**Step 3 — cross-check.** Cumulative expected loss =  $0.12 \times 0.60 = 7.2\%$  over 5 years;  $\div 5$  gives  $\approx 144 \text{ bp}$ . The two agree within ~9 bp; the exponential-hazard method (153 bp) is more precise as it accounts for compounding survival. Both land near 150 bp for a BB name — tighter than B2's 200 bp, consistent with the different recovery inputs ✓.

### B5. Covenant stress test

Meridian's indenture has a **maintenance covenant**: Debt/EBITDA  $\leq \mathbf{4.0x}$  each quarter-end. Using B1's figures (Debt 1,050; EBITDA 300; Interest 60), a recession cuts EBITDA by **15%**. Test the covenant and gauge distress.

**Step 1 — recompute leverage after the shock.** New EBITDA =  $300 \times (1 - 0.15) = \mathbf{255}$ . New Debt/EBITDA =  $1,050 / 255 = \mathbf{4.12x}$ .

**Step 2 — test the covenant.**  $4.12x$  **exceeds 4.0x** → the maintenance covenant is **breached**. Meridian is in **technical default** despite missing no payment — the early-warning power of a maintenance covenant. The lender can now demand repricing, extra collateral, or acceleration.

**Step 3 — liquidity crisis or ratio breach?** New EBITDA interest coverage =  $255 / 60 = \mathbf{4.25x}$  (down from 5.0x). Still well above the ~1.5x danger zone, so the firm can *service* its interest — a leverage-*ratio* breach and negotiating event, not yet a missed coupon.

**Self-check.** The 15% drop raised leverage ( $4.12x > 3.5x$ ) and cut coverage ( $4.25x < 5.0x$ ) — both move against the creditor, as an earnings decline must ✓.

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## Section C — Interview-Style (with model answers)

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### C1. "You have thirty seconds. How do you decide whether a corporate bond's spread is worth it?"

Model answer: I frame it as two legs — lending at the risk-free rate plus selling default insurance — and ask whether the spread over-compensates me for the expected loss,  $PD \times LGD$ . I underwrite capacity through cash-flow-anchored leverage and coverage ratios, check protection through covenants and collateral, and sanity-check character. Then I reconcile my fundamental view against the spread's market-implied view, and lend only when the spread pays more than my honest expected loss plus a premium for its *uncertainty*.

### C2. "Why do you insist on cash flow when the income statement already shows profit?"

Model answer: Because coupons are paid in cash, and net income can diverge wildly from it — a firm can grow earnings while free cash flow turns negative through uncollected receivables, capitalised costs, or ballooning working capital. EBITDA is my first cash proxy, but it ignores capex, taxes, and working-capital swings, so I reconcile it to cash from operations and then to free cash flow. My job is to find the cash that services the debt, not the accounting story.

### C3. "Walk me through structural versus reduced-form models and when you'd use each."

Model answer: Structural models — Merton and KMV — say a firm defaults when asset value falls below the debt face value. Equity is a call on assets, so I back out an implied PD from leverage and asset volatility via Black–Scholes, summarised as a distance to default. Their strength is economic insight into *why* spreads widen; their weakness is they need the unobservable asset value and vol, and under-predict short-term spreads. Reduced-form models ignore the balance sheet and treat default as an exogenous jump with a hazard rate calibrated to CDS and bond spreads — tractable for pricing but a-theoretical about causation. I use structural for surveillance, reduced-form for pricing and stripping market-implied PDs.

### C4. "A client says a BBB and a BB bond are basically the same risk one notch apart. Correct them."

Model answer: One notch on paper, but it straddles the most consequential line in fixed income — the investment-grade floor at BBB–/Baa3. Below it, huge mandate-constrained pools of capital are forbidden to hold the paper, so the transition isn't a gradient; it's a cliff. A downgrade to BB+ makes the bond a fallen angel, and forced selling by insurers, pensions, and index funds widens the spread far beyond what the fundamental change warrants. The default-rate step across that line is discontinuous, not linear.

**C5. "Interest coverage on this credit is 9x. Are you comfortable?"** Model answer: Not on that number alone. Interest coverage ignores principal — a firm can cover interest nine times and still fail when a maturity wall arrives and refinancing is shut. I'd pull the maturity schedule, compute DSCR to fold in principal, and check liquidity — cash and undrawn revolver against near-term maturities. I'd also stress the earnings: 9x today can become 3x in a downturn if the business is cyclical. Coverage is necessary but never sufficient.

**C6. "What's the one equation you'd never want a credit analyst to forget, and what's its trap?"** Model answer: The credit triangle —  $\text{Spread} \approx PD \times LGD = \lambda \times (1 - \text{Recovery})$ . Give me a spread and a recovery assumption and I'll hand you the market-implied default probability. The trap is joint identification — PD and recovery are entangled inside one spread, so I can only solve for one by assuming the other. Too high a recovery and I overstate implied PD; too low and I understate it. So I always state the recovery assumption alongside any implied PD.

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## Section D — Multiple Choice (with reasoning)

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**D1.** Debt/EBITDA of 4.0x is best read as: (a) 4% of assets are debt (b) roughly four years of cash earnings to repay all debt (c) interest is covered four times (d) equity is 4x debt **Answer: (b).** Debt/EBITDA = years of EBITDA to repay the whole debt if every dollar went to paydown. A leverage measure, not coverage (c) or a capital-structure share (a).

**D2.** Which is always true for the same firm and period? (a) Net Debt/EBITDA  $\geq$  Gross Debt/EBITDA (b) DSCR  $\geq$  EBITDA interest coverage (c) DSCR  $\leq$  EBITDA interest coverage (d) EBIT coverage  $\geq$  EBITDA coverage **Answer: (c).** DSCR adds principal to the denominator, so it is never above interest coverage. Netting cash lowers net leverage below gross, killing (a); subtracting D&A lowers EBIT coverage below EBITDA coverage, killing (d).

**D3.** A 200 bp spread with a 50% recovery assumption implies an annual hazard rate of about: (a) 1.0% (b) 2.0% (c) 4.0% (d) 8.0% **Answer: (c).**  $\lambda \approx \text{Spread} / \text{LGD} = 0.0200 / 0.50 = 4.0\%$ . Halving LGD doubles the implied hazard versus a naive spread-equals-PD read.

**D4.** Under a constant hazard rate  $\lambda$ , the probability of default within T years is: (a)  $\lambda T$  (b)  $1 - e^{(-\lambda T)}$  (c)  $e^{(-\lambda T)}$  (d)  $\lambda/T$  **Answer: (b).** Survival is  $e^{(-\lambda T)}$ , so cumulative default is its complement,  $1 - e^{(-\lambda T)}$ . (c) is the survival probability, and (a) is only the linear first-order approximation.

**D5.** In the Merton model, the firm defaults when: (a) interest coverage falls below 1.0x (b) asset value at maturity falls below the face value of debt (c) the credit rating is cut to D (d) EBITDA turns negative **Answer: (b).** Structural models define default as asset value < debt at maturity — where the shareholders' put finishes in the money. The others are accounting or agency signals, not the model's trigger.

**D6.** A maintenance covenant differs from an incurrence covenant in that it is: (a) tested only when the firm issues new debt (b) never enforceable (c) tested every period regardless of any action (d) applicable only to investment-grade issuers **Answer: (c).** Maintenance covenants are tested each period, an early-warning tripwire; incurrence covenants (a) are tested only on a specific action. Cov-lite deals drop the maintenance test.

**D7.** LGD is best defined as: (a) the probability of default (b)  $1 - \text{Recovery rate}$  (c) the hazard rate (d) the spread over Treasuries **Answer: (b).** LGD is the fraction not recovered,  $1 - \text{Recovery}$ . PD (a) and  $\lambda$  (c) are separate inputs; the spread (d) is roughly  $\text{PD} \times \text{LGD}$ .

**D8.** For the same spread, assuming a *higher* recovery rate implies: (a) a lower implied PD (b) a higher implied PD (c) no change in implied PD (d) a lower spread **Answer: (b).** Since  $\text{Spread} \approx \lambda \times (1 - \text{Recovery})$ , a higher recovery shrinks LGD, so  $\lambda$  must rise to keep the product equal to the spread.

**D9.** A "fallen angel" is a bond that: (a) rose from junk into investment grade (b) was downgraded from investment grade to high yield (c) defaulted and recovered fully (d) had its coupon cut **Answer: (b).** A fallen angel crosses the BBB-/BB+ line downward and suffers forced selling by mandate-constrained holders. The reverse is a "rising star" (a).

**D10.** Which statement about credit ratings is TRUE? (a) A rating is a cardinal default probability (b) A rating is a buy/sell recommendation (c) A rating is an ordinal, through-the-cycle opinion on relative default risk (d) Market spreads are less volatile than ratings **Answer: (c).** Ratings are ordinal, through-the-cycle opinions — stickier than the point-in-time spread (killing (d)), neither a precise probability (a) nor a price recommendation (b).

**D11.** DSCR is most central in which context? (a) short-dated Treasury trading (b) project finance and real estate with amortising principal (c) equity valuation (d) money-market funds **Answer: (b).** DSCR folds principal into the denominator, mattering most where amortising principal is a cash drain. Lenders want  $DSCR > \sim 1.25x$ .

**D12.** Rising asset volatility, holding leverage fixed, tends to: (a) narrow the credit spread (b) widen the credit spread (c) leave the spread unchanged (d) raise the recovery rate **Answer: (b).** More asset volatility makes the shareholders' walk-away put more valuable; the bondholder is short that put, so the spread widens — the structural bridge between volatility and spreads.

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*End of Q&A — Credit Analysis. Numerical answers self-verified: ratio ordering in B1,  $e^{(-0.20)} = 0.8187$  in B2,  $DD = 0.924$  in B3,  $-\ln(0.88)/5 = 2.56\%$  in B4, breach  $4.12x > 4.0x$  in B5.*

# Chapter 12 — Securitization, ABS and MBS

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## 1. The Problem / The Need

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A bank makes a 30-year mortgage. It now holds a single, illiquid, long-dated loan to one household in one town. That one loan ties up regulatory capital, concentrates the bank's risk in local house prices and one borrower's job, and cannot be sold easily because no investor wants to underwrite one anonymous mortgage. Multiply this by thousands of loans and the bank's balance sheet becomes a warehouse of frozen, capital-hungry, hard-to-value assets. It can only make new loans as fast as it takes in new deposits.

Meanwhile, a pension fund on the other side of the world has the opposite problem. It has billions to invest, needs long-dated income to match its liabilities, and would happily earn the spread on prime mortgages — but it cannot originate loans, service them, or diligence individual borrowers. It wants a tradable, rated, diversified bond, not a filing cabinet of loan documents.

Securitization is the bridge between these two problems. It takes illiquid, non-tradable loans and manufactures liquid, tradable, credit-rated securities out of them. It answers a cluster of needs at once:

- **Funding and capital relief** for originators — sell the loans, get cash, recycle it into new lending, and move the assets off-balance-sheet so they stop consuming capital.
- **Risk transfer** — the credit risk of the borrowers moves from the bank to capital-market investors who choose to hold it.
- **Investor access and customization** — investors who could never touch a whole-loan portfolio can now buy a slice engineered to a precise risk-return profile, from ultra-safe to highly leveraged.
- **Maturity and liquidity transformation** — turning 30-year contracts into instruments that trade daily.

The core financial-engineering trick is that the *whole* can be worth more than — and be safer in parts than — the *sum* of the raw loans, because pooling diversifies idiosyncratic risk and tranching redistributes the remaining risk to the investors most willing to bear each layer.

## 2. The Core Idea

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Securitization has two moves, and almost everything else is detail:

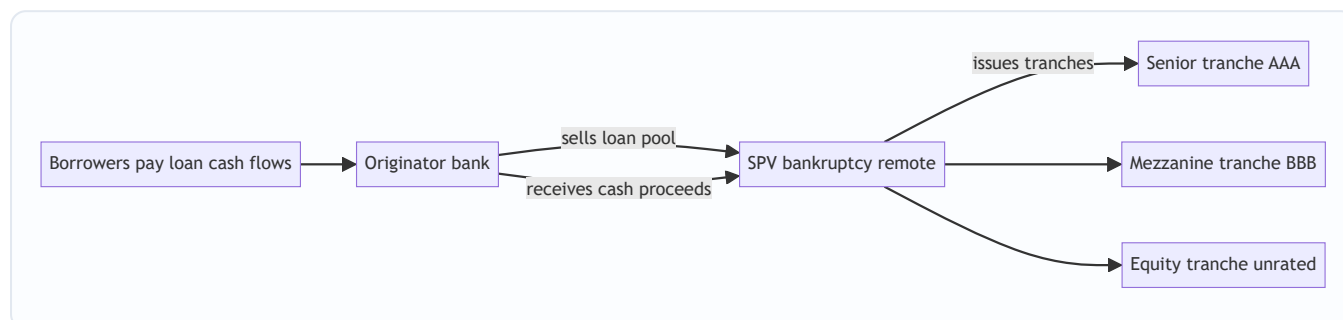
1. **Pooling.** Gather a large number of small, similar cash-flow-producing assets — mortgages, auto loans, credit-card receivables, student loans — into one portfolio held by a bankruptcy-remote **Special Purpose Vehicle (SPV)**. Diversification means one borrower's default is a rounding error, so the pool's aggregate cash flow is far more predictable than any single loan's.
2. **Tranching.** Slice the pooled cash flows into layered claims — tranches — with a strict priority ordering. Senior tranches get paid first and absorb losses last; junior (equity) tranches get paid last and absorb losses first. The same underlying cash flow now backs securities ranging from AAA to unrated, each priced to its own risk.

The device that enforces the ordering is the **cash-flow waterfall**: a contractual set of rules specifying, period by period, who receives interest and principal in what sequence, and who eats losses in what sequence. Layered on top are **credit enhancements** — subordination, overcollateralization, excess spread, reserve accounts, and sometimes external guarantees — that protect senior investors.

The naming convention follows the collateral:

- **MBS (Mortgage-Backed Securities):** backed by mortgages. **RMBS** = residential; **CMBS** = commercial.
- **ABS (Asset-Backed Securities):** backed by non-mortgage receivables (autos, cards, student loans, equipment leases).
- **CDO (Collateralized Debt Obligation):** backed by a pool of *debt securities* — corporate bonds (CBO), loans (CLO), or, notoriously, tranches of *other* ABS/MBS (ABS CDO). A CDO is securitization applied to securitizations.

Figure 1 — the basic securitization flow: whole loans move into an SPV that issues layered tranches to investors.



### 3. Why / How It Works

Why can pooling and tranching create securities that are, in aggregate, more valuable and more useful than the raw loans? Three mechanisms.

**Diversification kills idiosyncratic risk.** A single mortgage either pays or defaults — a binary, high-variance outcome. Pool 5,000 mortgages and, by the law of large numbers, the *fraction* that default becomes highly predictable even though any individual outcome is not. The pool's loss distribution tightens around its expected value. This is the same logic as insurance. The critical caveat — which 2008 taught in blood — is that diversification only works if defaults are *weakly correlated*. If a common factor (national house prices) drives all borrowers together, pooling provides far less protection than the models assumed.

**Tranching sorts the remaining risk to its natural buyers.** Even a well-diversified pool has a loss distribution — usually small losses are likely, catastrophic losses are rare. Tranching maps that distribution onto securities:

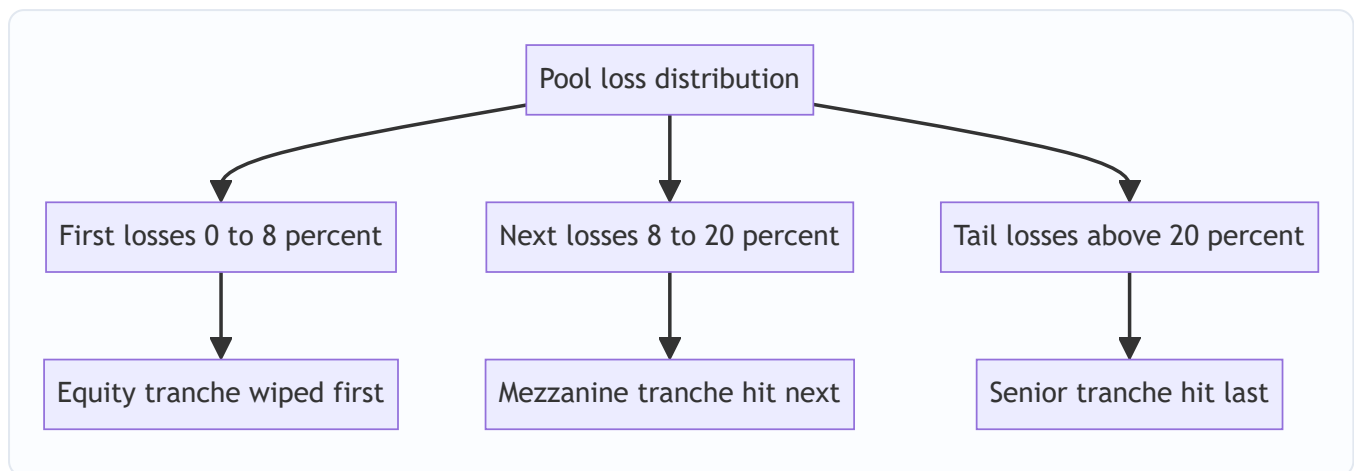
- The **equity tranche** absorbs the first losses. It is short a put on the pool — high expected return, high risk, bought by hedge funds and sponsors who want leverage.
- The **mezzanine tranche** is hit only after equity is exhausted — moderate risk, moderate spread.
- The **senior tranche** is hit only in a catastrophe deep enough to blow through everything below it. Because that is rare, it earns a high rating and a low spread, and is bought by insurers, banks, and money funds who are constrained to hold high-grade paper.

The senior tranche can be rated far above the average loan in the pool precisely because the junior tranches stand in front of it. This is **credit enhancement through subordination**: the senior's protection is the dollar cushion of everything below it.

**Structural and legal isolation makes the ratings real.** The SPV is *bankruptcy-remote* — if the originating bank fails, the pool is not part of the bank's estate, so investors keep receiving loan cash flows. This "true sale" is what lets the tranches be rated on the *collateral's* risk rather than the *originator's* risk. Without legal isolation, the whole edifice collapses back into an unsecured claim on the bank.



Figure 2 — the loss distribution of a diversified pool mapped onto tranches: first losses hit equity, tail losses reach senior.



## 4. Full Content — Formulas and Bond Math

### 4.1 The mortgage: a level-payment amortizing loan

Most MBS collateral is the fully-amortizing fixed-rate mortgage. A loan of principal  $P$  at monthly rate  $i$  (annual rate / 12) over  $n$  months has a constant monthly payment  $M$  that solves the present-value equation  $P = M \cdot \frac{1 - (1+i)^{-n}}{i}$ , giving:

$$M = P \cdot \frac{i(1+i)^n}{(1+i)^n - 1}$$

Each payment splits into **interest** ( $= i \times$  outstanding balance) and **scheduled principal** (the remainder). Early on, payments are mostly interest; late in life, mostly principal. The outstanding balance after  $k$  payments is:

$$B_k = P \cdot \frac{(1+i)^n - (1+i)^k}{(1+i)^n - 1}$$

### 4.2 Prepayment metrics: SMM, CPR, PSA

Borrowers can repay early — they move, refinance, or default (which, if insured, looks like a prepayment to investors). Prepayment is *the* defining risk of MBS. It is measured with three linked conventions.

**SMM (Single Monthly Mortality)** — the fraction of the *beginning-of-month balance net of scheduled principal* that prepays in a given month:

$$\text{SMM} = \frac{\text{prepayment}_t}{\text{beginning balance}_t - \text{scheduled principal}_t}$$

**CPR (Conditional Prepayment Rate)** — the annualized version of SMM. The two convert via compounding:

$$\text{SMM} = 1 - (1 - \text{CPR})^{1/12} \quad \text{CPR} = 1 - (1 - \text{SMM})^{12}$$

**PSA benchmark** — the Public Securities Association ramp. **100 PSA** is defined as: CPR rises linearly by 0.2% per month, from 0.2% in month 1 to 6% in month 30, then holds flat at 6% thereafter:

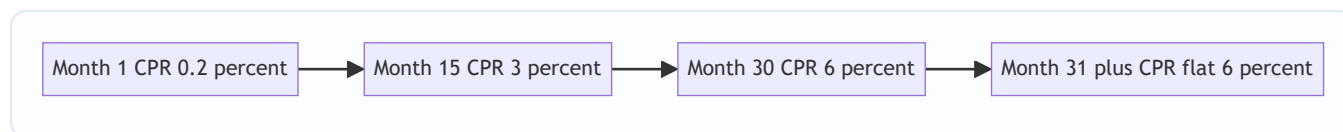
$$\text{CPR}_t^{100\text{ PSA}} = \min(6\%, 0.2\% \times t)$$

A speed of " $X$  PSA" scales this:  $\text{CPR}_t = \frac{X}{100} \times \text{CPR}_t^{100\text{ PSA}}$  (capped where the base is capped). So 150 PSA is 1.5× the ramp; 0 PSA means no prepayment.



**Total monthly principal** to investors = scheduled principal + prepayment, where  $\text{prepayment}_t = \text{SMM}_t \times (\text{beginning balance}_t - \text{scheduled principal}_t)$ .

Figure 3 — the 100 PSA prepayment ramp: CPR climbs 0.2 percent per month to a 6 percent plateau at month 30.



### 4.3 Weighted Average Life (WAL)

Because principal arrives gradually and unpredictably, MBS/ABS are quoted by **Weighted Average Life** rather than a single maturity. WAL is the average time to receive a dollar of principal, weighting each principal payment by its size:

$$\text{WAL} = \frac{\sum_t t \cdot \text{Principal}_t}{\sum_t \text{Principal}_t} \quad (t \text{ in years})$$

Faster prepayment  $\Rightarrow$  principal arrives sooner  $\Rightarrow$  shorter WAL. WAL is a maturity concept, not a discounted duration; it ignores interest and discounting.

### 4.4 Contraction and extension risk

Prepayment cuts both ways, and this asymmetry is why MBS trade cheap to Treasuries:

- **Contraction risk:** when rates *fall*, borrowers refinance, prepayments accelerate, WAL shortens. Investors get cash back exactly when reinvestment rates are low — the bond's upside is capped, giving MBS **negative convexity**.
- **Extension risk:** when rates *rise*, prepayments slow, WAL lengthens, and investors are stuck holding a below-market coupon longer.

An MBS is economically a bond *plus a short call option* the borrower holds (the right to prepay). Hence:

$$\text{Price}_{\text{MBS}} = \text{Price}_{\text{option-free bond}} - \text{Value}_{\text{prepay option}}$$

Its **option-adjusted spread (OAS)** strips out the option cost to give a clean spread; its **effective duration** can even go negative near the money as prices stop rising when rates fall.

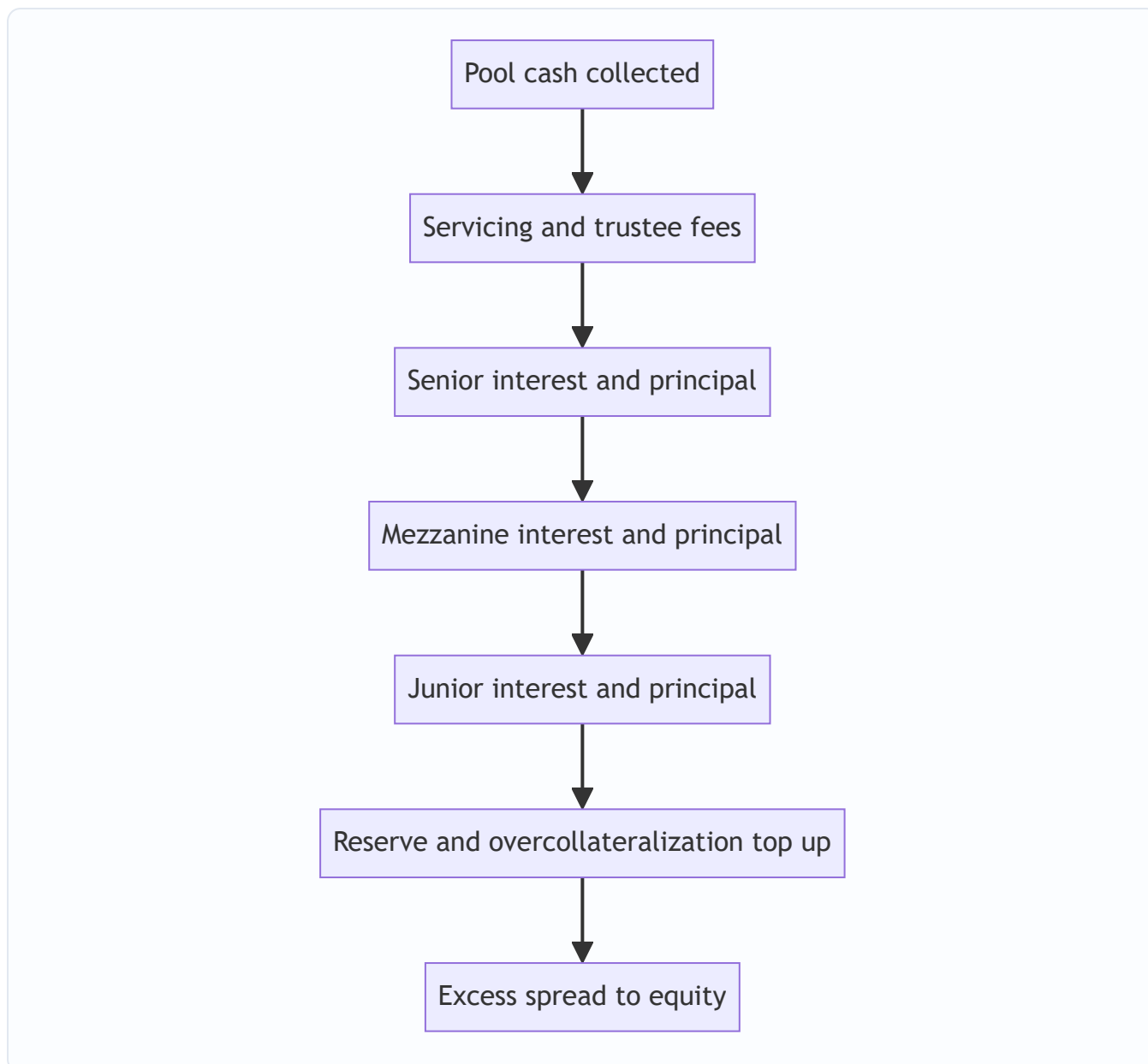
### 4.5 The cash-flow waterfall

Every deal is governed by a waterfall — the payment priority. A stylized structure:

1. Fees (servicer, trustee).
2. Senior interest, then mezzanine interest, then junior interest.
3. Senior principal, then mezzanine principal, then junior principal (**sequential pay**), or pro-rata if triggers are met.
4. Top up reserve accounts and overcollateralization to target.
5. Residual **excess spread** to the equity holder.

Losses run in *reverse*: equity first, then mezzanine, then senior.

Figure 4 — the interest-and-principal waterfall pays top-down while losses are absorbed bottom-up.



#### 4.6 Credit enhancement toolkit

- **Subordination (internal):** junior tranches absorb losses first. The senior's **credit enhancement %** = dollars of subordination below it ÷ total pool.
- **Overcollateralization (OC):** pool face value exceeds the face of the bonds issued (e.g. \$105m of loans backing \$100m of bonds). The \$5m excess is a loss buffer.
- **Excess spread:** the pool earns more interest (say 7%) than the bonds pay plus fees (say 5%). The ~2% residual first absorbs losses before touching principal.
- **Reserve fund / cash collateral account:** a pot of cash to cover shortfalls.
- **External enhancement:** monoline insurance wraps, letters of credit, or a corporate guarantee (introduces counterparty risk — the guarantor can fail).

#### 4.7 CMO tranche types — redistributing prepayment risk

A plain pass-through hands every investor the same pro-rata prepayment risk. A **CMO (Collateralized Mortgage Obligation)** re-carves the *same* pool cash flow to hand different investors different *timing* profiles:

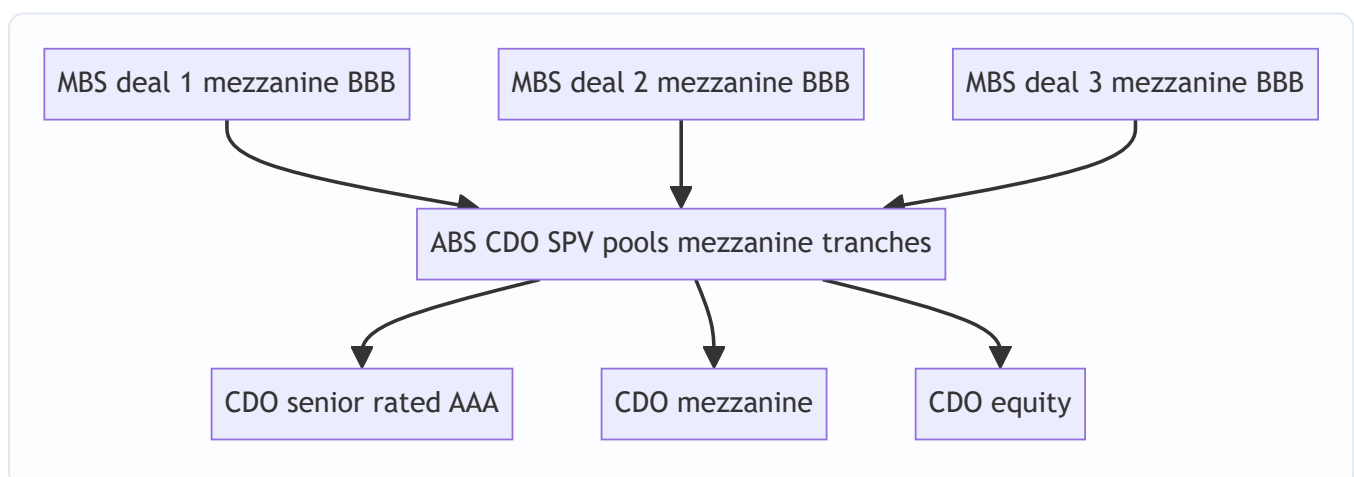
- **Sequential-pay:** tranches retire in order (A, then B, then C). A gets the earliest, most certain principal (short, stable WAL); the last tranche absorbs the timing tail.
- **PAC / Support (planned amortization class):** the PAC tranche is promised a fixed principal schedule as long as prepayment stays inside a **PAC collar** (e.g. 100–300 PSA). The **support (companion)** tranche absorbs the *variance*: it soaks up excess principal when prepayments are fast and starves when they are slow. The PAC therefore has a near-guaranteed WAL (bond-like) while the support tranche carries amplified contraction *and* extension risk — and is priced at a wide spread to compensate.
- **Interest-only (IO) and Principal-only (PO) strips:** split the pool's interest and principal into two securities. The **PO** is bought at a deep discount and *loves* fast prepayment (principal arrives sooner, so yield jumps) — it has large positive duration. The **IO** *hates* fast prepayment (the balance it earns interest on vanishes) and famously has **negative duration**: its price *rises* when rates rise. IO/PO pairs are the purest expression of prepayment risk and are used to hedge MBS books.

The key idea: CMO structuring does not change total cash flow or total risk — it *reallocates* the prepayment uncertainty from investors who cannot bear it (PAC buyers) to those who are paid to (support and IO/PO buyers).

#### 4.8 CDOs and the 2008 amplification mechanism

A **CDO** pools *debt securities* and re-tranches them. A **CLO** pools leveraged corporate loans (these largely worked, and still trade). An **ABS CDO** pools the *mezzanine tranches of other subprime MBS deals* — and this is where 2008 detonated. Re-securitizing already-thin mezzanine layers means the CDO's own senior tranche depends on the *joint* behaviour of many mezzanine pieces. If subprime defaults are independent, that senior looks safe; if they are highly correlated (all driven by a national housing downturn), the mezzanine layers default *together*, and the CDO's "AAA" senior is wiped almost as fast as its equity. Correlation, assumed low, was the hidden switch.

Figure 5 — an ABS CDO re-tranches mezzanine slices of many MBS deals, stacking correlation risk into a second securitization.



The full 2008 chain: lax **origination** (no-doc "liar" loans) fed an **originate-to-distribute** incentive where the originator kept no skin in the game; **rating agencies**, paid by issuers and using low-correlation models, stamped AAA on senior tranches; **ABS CDOs** re-levered the mezzanine; and the whole stack was funded **short-term** through repo and ABCP conduits. When house prices fell nationwide, correlated defaults blew through subordination, senior tranches took losses, mark-to-market collapsed, short-term funding ran, and

forced selling spread the fire across the system. The lessons — retained-risk ("skin in the game") rules, better correlation modelling, transparency, and less rating reliance — are now embedded in post-crisis regulation.

## 5. Worked Examples

### Example 1 — Mortgage payment, split, and one prepayment month

**Loan:**  $P = \$200,000$ , 30-year ( $n = 360$ ), annual rate 6% so  $i = 0.5\% = 0.005$  monthly.

Compute  $(1.005)^{360}$ :  $\ln 1.005 = 0.0049875$ ;  $\times 360 = 1.79551$ ;  $e^{1.79551} = 6.02258$ .

$M = 200,000 \times \frac{0.005 \times 6.02258}{6.02258 - 1} = 200,000 \times \frac{0.0301129}{5.02258} = \$1,199.10$

**First-month split:** interest  $= 0.005 \times 200,000 = \$1,000.00$ ; scheduled principal  $= 1,199.10 - 1,000.00 = \$199.10$ . Balance after month 1  $= \$199,800.90$ . Sanity check: overwhelmingly interest early on, as expected for a fresh 30-year loan.

**Now add prepayment at month 1, assume 150 PSA.** At month 1, base 100 PSA CPR  $= 0.2\% \times 1 = 0.2\%$ ; at 150 PSA, CPR  $= 1.5 \times 0.2\% = 0.3\%$ .

Convert to SMM:  $\text{SMM} = 1 - (1 - 0.003)^{1/12} = 1 - (0.997)^{0.08333}$ .  $\ln 0.997 = -0.0030045$ ;  $\times 0.08333 = -0.00025037$ ;  $e^{-0.00025037} = 0.99974966$ ; SMM  $= 0.00025034 = 0.02503\%$ .

Prepayment  $= \text{SMM} \times (\text{beginning balance} - \text{scheduled principal}) = 0.00025034 \times (200,000 - 199.10) = 0.00025034 \times 199,800.90 = \$50.02$ .

**Total principal to investors in month 1**  $= 199.10 + 50.02 = \$249.12$ , and the ending balance  $= 200,000 - 249.12 = \$199,750.88$ . Interest to investors was  $\$1,000$ . This reconciles: cash out to investors  $= \$1,000 + 249.12 = \$1,249.12$ , which is the scheduled  $\$1,199.10$  payment plus the  $\$50.02$  prepayment. ✓

### Example 2 — CPR $\rightleftharpoons$ SMM round-trip (self-verification)

Take month 20 at 150 PSA. Base 100 PSA CPR  $= 0.2\% \times 20 = 4\%$ ; at 150 PSA, CPR  $= 6\%$ .

SMM  $= 1 - (1 - 0.06)^{1/12} = 1 - (0.94)^{0.08333}$ .  $\ln 0.94 = -0.0618754$ ;  $\times 0.08333 = -0.00515628$ ;  $e^{-0.00515628} = 0.9948570$ ; **SMM  $= 0.0051430 = 0.51430\%$** .

**Round-trip back to CPR** to verify:  $\text{CPR} = 1 - (1 - 0.0051430)^{12} = 1 - (0.994857)^{12}$ .  $\ln 0.994857 = -0.00515628$ ;  $\times 12 = -0.0618754$ ;  $e^{-0.0618754} = 0.94000$ ; CPR  $= 1 - 0.94 = 6.00\%$ . ✓ The conversions are exact inverses.

### Example 3 — Loss allocation and credit enhancement in a tranch deal

**Structure:** \$100m pool. Senior A = \$80m, Mezzanine B = \$12m, Equity C = \$8m. Sequential loss absorption: C first, then B, then A.

Credit enhancement (subordination) at close: - For A: cushion below  $= B + C = \$20m \Rightarrow$  **20%** enhancement. - For B: cushion below  $= C = \$8m \Rightarrow$  **8%** enhancement. - For C: **0%** — it is first-loss.

| Cumulative pool loss | Equity C (\$8m)   | Mezz B (\$12m)    | Senior A (\$80m)   | Who is impaired                       |
|----------------------|-------------------|-------------------|--------------------|---------------------------------------|
| \$6m                 | -\$6m → \$2m left | intact            | intact             | C only                                |
| \$8m                 | wiped (\$0)       | intact            | intact             | C exactly exhausted                   |
| \$15m                | wiped             | -\$7m → \$5m left | intact             | C, B partially                        |
| \$20m                | wiped             | wiped             | intact             | A still whole (uses full 20% cushion) |
| \$26m                | wiped             | wiped             | -\$6m → \$74m left | A takes first loss                    |

**Reading it:** Senior A stays money-good until cumulative losses exceed **20%** of the pool — precisely its subordination. At \$26m of losses (26%), A loses \$26m – \$20m = \$6m, recovering \$74m of its \$80m, i.e. 92.5 cents. The tranching converted an average-quality pool into an \$80m security that survives a one-in-many-years 20% loss event untouched — the mechanical basis of its AAA rating. ✓

#### Example 4 — Sequential-pay WAL, and reconciliation to pool WAL

**Pool:** \$100m, principal returned over four years: Y1 \$25m, Y2 \$30m, Y3 \$25m, Y4 \$20m. Two sequential tranches: A = \$60m (paid first), B = \$40m.

**Allocate principal top-down:**

| Year | Pool principal | To A | A remaining | To B | B remaining |
|------|----------------|------|-------------|------|-------------|
| 1    | 25             | 25   | 35          | 0    | 40          |
| 2    | 30             | 30   | 5           | 0    | 40          |
| 3    | 25             | 5    | 0           | 20   | 20          |
| 4    | 20             | 0    | 0           | 20   | 0           |

**WAL of A**  $= \frac{1(25) + 2(30) + 3(5)}{60} = \frac{25 + 60 + 15}{60} = \frac{100}{60} = 1.67$  years.

**WAL of B**  $= \frac{3(20) + 4(20)}{40} = \frac{60 + 80}{40} = \frac{140}{40} = 3.50$  years.

**WAL of the whole pool**  $= \frac{1(25) + 2(30) + 3(25) + 4(20)}{100} = \frac{25 + 60 + 75 + 80}{100} = 2.40$  years.

**Reconciliation:** the size-weighted average of the tranche WALs must equal the pool WAL:  $\frac{60(1.67) + 40(3.50)}{100} = \frac{100 + 140}{100} = 2.40$  years ✓

This is the whole point of the CMO: the pool has a single 2.40-year average life, but sequential paydown *carves* it into a short 1.67-year A note (for money funds wanting short paper) and a longer 3.50-year B note (for insurers wanting duration) — same collateral, two clienteles served. Note also the risk redistribution: A's WAL is far more *stable* against prepayment shocks because it is protected by B absorbing the timing tail — the essence of tranching applied to *timing* rather than *credit*.

## 6. Connections

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- **Duration and convexity (Ch. on interest-rate risk):** MBS negative convexity is the flagship real-world example of embedded optionality. Effective duration and OAS are the only correct tools for these bonds — Macaulay/modified duration mislead because cash flows move with rates.
- **Credit spreads and default (credit chapters):** tranching is structural credit modelling. Attachment/detachment points map directly to a portfolio loss distribution and correlation — the same machinery as the Merton model applied to a pool.
- **Options (derivatives):** an MBS = bond – prepay call; a CDO tranche payoff is a call spread on cumulative pool losses. Copula models used to price CDO correlation are options-pricing cousins.
- **Callable bonds:** MBS extension/contraction is the callable-bond call risk, but driven by millions of borrowers acting imperfectly rather than one rational issuer — hence *prepayment models* (empirical, behavioural) instead of a clean optimal-exercise boundary.
- **Money markets and funding:** ABCP (asset-backed commercial paper) conduits and repo funded these structures short-term; the 2008 run was fundamentally a maturity-mismatch/funding crisis layered on the credit crisis.

## 7. Key Terms

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- **SPV / SPE:** bankruptcy-remote entity that holds the pool and issues the securities.
- **Tranche:** a layered claim on pool cash flows with defined payment and loss priority.
- **Waterfall:** contractual priority of payments (top-down) and losses (bottom-up).
- **Subordination:** junior tranches shielding senior ones by taking losses first.
- **Overcollateralization (OC):** collateral face exceeds bonds issued; the excess is a buffer.
- **Excess spread:** pool interest minus bond coupons and fees; first-loss income cushion.
- **CPR / SMM / PSA:** annual / monthly / benchmark-ramp prepayment speed measures.
- **WAL:** average time to receive principal, size-weighted.
- **Contraction / extension risk:** WAL shortening (rates fall) / lengthening (rates rise).
- **Negative convexity:** price appreciation capped as rates fall, due to prepayment.
- **OAS:** spread over the curve after removing the value of the embedded prepay option.
- **CMO:** a multi-tranche MBS that redistributes prepayment/timing risk (e.g. sequential-pay, PAC/support).
- **CDO / CLO / ABS CDO:** securitization of debt securities / of loans / of ABS tranches.
- **Attachment / detachment point:** the loss % where a tranche starts / finishes absorbing losses.

## 8. Common Confusions

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- **"AAA on the tranche means the loans are AAA."** No. The pool can be subprime; the senior tranche is AAA *only because* the junior tranches absorb losses first. The rating is a statement about *structure*, not about *average collateral quality*. 2008 exposed how fragile this is when loss correlation is underestimated.
- **"Prepayment is always good — I get my money back."** It is bad when you don't want it: rates fell, so you reinvest at lower yields, and if you paid a premium you lose it. That asymmetry (good news capped, bad news not) is negative convexity.

- **"WAL is the same as duration."** WAL is an undiscounted average *maturity* of principal; duration is a *price sensitivity*. For MBS, effective duration can even be negative — WAL never is.
- **"CPR is just 12 × SMM."** No — they compound:  $\text{CPR} = 1 - (1 - \text{SMM})^{12}$ , always *less* than 12×SMM.
- **"Overcollateralization and subordination are the same thing."** Related but distinct. Subordination is *tranche ordering* (junior bonds absorb losses). OC is *more collateral than bonds* (excess pool assets). A deal can use both.
- **"A CDO is just a bigger MBS."** A CDO's collateral is *other bonds/tranches*, so it re-tranches already-tranched risk — stacking correlation and model risk. ABS CDOs of subprime mezzanine tranches were the epicentre of 2008 losses precisely because that re-tranching concentrated tail risk that everyone had assumed away.
- **"Excess spread is free profit to equity."** Only what survives the waterfall after losses. In a stress, excess spread is the *first* buffer consumed, so equity income vanishes before principal is even touched.

## 9. Recap

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Securitization solves a two-sided problem: originators need funding, capital relief, and risk transfer; investors need tradable, rated, customizable exposure to loan cash flows. The mechanism is two moves — **pool** many small loans into a bankruptcy-remote SPV to diversify away idiosyncratic risk, then **tranche** the pooled cash flows into a priority-ordered stack. A **waterfall** enforces the order (interest and principal top-down, losses bottom-up), and **credit enhancements** — subordination, overcollateralization, excess spread, reserves, external wraps — protect the senior claims, letting them earn ratings far above the average loan.

The bond math is distinctive. Mortgages amortize on a level payment, but borrowers **prepay**, measured by SMM/CPR/PSA. Prepayment makes cash flows rate-sensitive: rates down  $\Rightarrow$  contraction  $\Rightarrow$  short WAL and reinvestment pain; rates up  $\Rightarrow$  extension  $\Rightarrow$  stuck with a low coupon. This embedded short call gives MBS **negative convexity**, so they must be analysed with **effective duration and OAS**, not textbook duration. **WAL** replaces maturity as the quoting convention, and CMO structuring carves one pool's timing profile into short and long tranches that reconcile back to the pool average.

The 2008 lesson sits on top of all of it: the machinery is only as sound as its assumptions about **default correlation** and **funding stability**. When house prices fell nationally, "diversified" pools defaulted together, "AAA" senior tranches took losses, ABS CDOs amplified the damage, and short-term funding ran. Poor origination (no-doc loans), misaligned incentives (originate-to-distribute), and over-reliance on ratings converted a clever risk-distribution tool into a risk-concentration and contagion engine.

## 10. Quick-Reference / Interview Points

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**Formulas to have cold:** - Mortgage payment:  $M = P \cdot \frac{i(1+i)^n}{(1+i)^n - 1}$ . - Prepay conversions:  $\text{SMM} = 1 - (1 - \text{CPR})^{1/12}$ ;  $\text{CPR} = 1 - (1 - \text{SMM})^{12}$ . - 100 PSA:  $\text{CPR}_t = \min(6\%, 0.2\% \times t)$ ; scale by  $X/100$  for  $X\%$  PSA. -  $\text{WAL} = \frac{\sum t \cdot \text{Principal}_t}{\sum \text{Principal}_t}$  (years). - Senior credit enhancement % = subordination  $\div$  pool.

**One-liners interviewers reward:** - "An MBS is a bond minus a call the borrower owns — that's why it's negatively convex and needs OAS." - "Tranching doesn't reduce total risk; it *redistributes* it. Total pool losses are unchanged — the equity just eats them first so the senior can be AAA." - "Contraction risk hurts when rates fall; extension risk hurts when they rise. The asymmetry is the whole story." - "CPR compounds SMM,

it's not  $12\times$  — always less." - "The senior is AAA because of the subordinate stack, not because the loans are good. 2008 was correlation being higher than the models assumed." - "A CDO securitizes securities; an ABS CDO re-tranches mezzanine ABS and concentrates the tail everyone priced as diversifiable."

**Fast sanity checks:** - Faster PSA  $\Rightarrow$  shorter WAL. Rates down  $\Rightarrow$  faster prepay  $\Rightarrow$  contraction. - Size-weighted tranche WALs must equal the pool WAL — use it to catch allocation errors. - Loss waterfall order (equity $\rightarrow$ mezz $\rightarrow$ senior) is the *reverse* of the cash waterfall (senior $\rightarrow$ mezz $\rightarrow$ equity).

**Credit-enhancement stack, weakest-to-strongest for the senior investor:** excess spread (first)  $\rightarrow$  overcollateralization/reserve  $\rightarrow$  subordination  $\rightarrow$  external guarantee. Excess spread is consumed first in stress; subordination is the durable structural cushion.



# Q&A — Securitization, ABS and MBS

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Companion practice bank for Chapter 12. Every question is followed by a full answer. Unless stated otherwise, mortgage rates are quoted as **annual nominal rates compounded monthly** (monthly rate = annual/12), and prepayment follows the **PSA** convention.

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## Section A — Concept Check

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### A1. What two financial-engineering "moves" define securitization, and what does each accomplish?

**Pooling** and **tranching**. Pooling gathers many small, similar cash-flow assets (mortgages, auto loans, card receivables) into a bankruptcy-remote SPV so that idiosyncratic, borrower-specific risk diversifies away and the pool's *aggregate* loss becomes far more predictable than any single loan's. Tranching then slices the pooled cash flows into priority-ordered claims — senior, mezzanine, equity — so the same collateral backs securities ranging from AAA to unrated, each priced to the exact risk a particular investor wants. Pooling manufactures predictability; tranching redistributes the remaining risk to its natural buyers.

### A2. Why can a securitization's senior tranche be rated AAA when the underlying loans are subprime?

Because the rating reflects *structure*, not average collateral quality. The junior and mezzanine tranches sit beneath the senior and absorb losses first, so the senior is impaired only when cumulative pool losses are large enough to burn through everything below it — a rare, tail event. That subordinated cushion is the senior's credit enhancement. The rating is a statement about the *dollar buffer standing in front of the senior*, not about the borrowers. The 2008 lesson is that this only holds if default correlation is low; when defaults cluster, the cushion is consumed far faster than models assumed.

### A3. What is the cash-flow waterfall, and how does the order of losses relate to the order of payments?

The waterfall is the contractual priority of distributions each period: fees first, then interest top-down (senior → mezzanine → junior), then principal, then reserve/OC top-ups, then residual excess spread to equity. Losses run in the *reverse* order — equity absorbs the first losses, then mezzanine, then senior. So cash flows down the stack top-first while losses eat up the stack bottom-first. This mirror image is exactly what lets the senior be safe and the equity be levered.

**A4. Why is prepayment "the defining risk" of MBS, and how is it different from a callable corporate bond's call risk?** Mortgage borrowers can repay early — they move, refinance, or default (which, if insured, reaches the investor as a prepayment). This makes the *timing* of principal uncertain and rate-sensitive. It resembles a callable bond because the borrower holds a call option on their loan, but with a key difference: a corporate issuer exercises one call rationally at an optimal boundary, whereas millions of households prepay imperfectly and behaviourally. So MBS use empirical *prepayment models* (SMM/CPR/PSA) rather than a clean optimal-exercise rule.

**A5. Define SMM, CPR, and PSA and state how they relate.** - **SMM (Single Monthly Mortality)**: the fraction of the beginning-of-month balance *net of scheduled principal* that prepays in that month. - **CPR (Conditional Prepayment Rate)**: the annualized equivalent of SMM. They compound, not scale:  $\text{SMM} = 1 - (1 - \text{CPR})^{1/12}$  and  $\text{CPR} = 1 - (1 - \text{SMM})^{12}$ . - **PSA**: a benchmark ramp. 100 PSA means CPR rises 0.2% per month from 0.2% (month 1) to 6% (month 30), then stays flat:  $\text{CPR}_t = \min(6\%, 0.2\% \times t)$ . A speed of  $X$  PSA scales this by  $X/100$ .

**A6. Distinguish contraction risk from extension risk.** - **Contraction risk:** when rates *fall*, borrowers refinance, prepayments accelerate, WAL *shortens*, and investors get cash back precisely when reinvestment rates are low. Upside is capped. - **Extension risk:** when rates *rise*, prepayments slow, WAL *lengthens*, and investors are stuck holding a below-market coupon longer. The asymmetry — good news (falling rates) capped, bad news (rising rates) not — is the source of MBS **negative convexity**.

**A7. Why do MBS exhibit negative convexity, and what tools must you use to analyse them?** An MBS is economically a bond *minus* a short call the borrower owns:  $\text{Price}_{\text{MBS}} = \text{Price}_{\text{option-free}} - \text{Value}_{\text{prepay option}}$ . As rates fall, the prepay option gains value and caps the bond's price appreciation, so the price-yield curve bends the "wrong" way (concave). Because cash flows move with rates, Macaulay/modified duration mislead. You must use **effective duration** (which can even go negative near the money) and **option-adjusted spread (OAS)**, which strips out the option cost to leave a clean, comparable spread.

**A8. What is WAL, and why is it quoted instead of maturity? How does it differ from duration?**

**Weighted Average Life** is the average time to receive a dollar of principal, weighting each principal payment by its size:  $\text{WAL} = \frac{\sum_t t \cdot \text{Principal}_t}{\sum_t \text{Principal}_t}$ . Because MBS/ABS principal arrives gradually and unpredictably, a single legal maturity is uninformative, so the market quotes WAL. It differs from duration in two ways: WAL is *undiscounted* and ignores interest, and it is a *maturity* concept, not a price sensitivity. WAL is always positive; effective duration of an MBS (or an IO strip) can be negative.

**A9. List the internal credit enhancements and rank them from first-consumed to most durable.** From weakest (consumed first in stress) to strongest: **excess spread** (pool interest minus bond coupons and fees — the first income cushion), then **overcollateralization / reserve fund** (extra collateral or cash), then **subordination** (the junior tranche stack absorbing losses). External enhancement (monoline wraps, guarantees) sits outside the deal but adds counterparty risk. Subordination is the durable structural cushion; excess spread evaporates first.

**A10. What is a CMO, and what does a PAC/support structure achieve?** A CMO (Collateralized Mortgage Obligation) re-carves the *same* pool cash flow to give investors different *timing* profiles. A **PAC (planned amortization class)** is promised a fixed principal schedule as long as prepayment stays inside a collar (e.g. 100–300 PSA); the **support (companion)** tranche absorbs the variance — soaking up excess principal when prepayments are fast and starving when slow. The PAC gets a near-guaranteed, bond-like WAL; the support carries amplified contraction *and* extension risk and is paid a wide spread. CMO structuring does not reduce total risk — it *reallocates* prepayment uncertainty from investors who cannot bear it to those paid to.

**A11. How did ABS CDOs amplify the 2008 crisis?** An ABS CDO pools the *mezzanine tranches* of many *subprime MBS deals* and re-tranches them — securitization applied to securitizations. Its senior tranche depends on the *joint* behaviour of dozens of thin mezzanine slices. Rating models assumed those slices defaulted roughly independently, making the CDO senior look AAA. But subprime defaults were driven by a common factor — national house prices — so when prices fell, the mezzanine layers defaulted *together*, and the "AAA" CDO senior was wiped almost as fast as its equity. Re-tranching concentrated the very tail risk everyone had priced as diversifiable.

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## Section B — Numerical Bond-Math Problems

### B1. Mortgage payment and first-month split

A \$250,000 fixed-rate mortgage, 30 years ( $n = 360$ ), annual rate 6% so monthly  $i = 0.5\% = 0.005$ .

**Step 1 — the level payment.** Using  $M = P \cdot \frac{i(1+i)^n}{(1+i)^n - 1}$ . Compute  $(1.005)^{360}$ :  $\ln 1.005 = 0.0049875$ ;  $\times 360 = 1.79551$ ;  $e^{1.79551} = 6.02258$ .  $M = 250,000 \times \frac{0.005 \times 6.02258}{6.02258 - 1} = 250,000 \times \frac{0.0301129}{5.02258} = \boxed{\$1,498.88}$

**Step 2 — split month 1.** Interest  $= 0.005 \times 250,000 = \$1,250.00$ . Scheduled principal  $= \$1,498.88 - \$1,250.00 = \$248.88$ . Ending balance  $= \$250,000 - 248.88 = \$249,751.12$ .

**Reconcile:** early payments should be overwhelmingly interest on a fresh 30-year loan — here \$1,250 of \$1,498.88 is interest (83%). ✓

### B2. CPR $\rightleftharpoons$ SMM round-trip (self-verification)

Take **month 10 at 200 PSA**. Base 100 PSA CPR  $= 0.2\% \times 10 = 2\%$ ; at 200 PSA,  $\text{CPR} = 2\% \times 2 = 4\%$ .

**CPR  $\rightarrow$  SMM:**  $\text{SMM} = 1 - (1 - 0.04)^{1/12} = 1 - (0.96)^{0.083333}$ .  $\ln 0.96 = -0.0408220$ ;  $\times 0.083333 = -0.00340183$ ;  $e^{-0.00340183} = 0.9966040$ ; **SMM  $= 0.0033961 = 0.33961\%$ .**

**Round-trip SMM  $\rightarrow$  CPR:**  $\text{CPR} = 1 - (1 - 0.0033961)^{12} = 1 - (0.9966039)^{12}$ .  $\ln 0.9966039 = -0.00340183$ ;  $\times 12 = -0.0408220$ ;  $e^{-0.0408220} = 0.960000$ ; CPR  $= 1 - 0.96 = 4.00\%$ . ✓  
The conversions are exact inverses, and note SMM  $\times 12 = 0.4075\%$  would overstate the true monthly speed — CPR is *not*  $12 \times$  SMM.

### B3. Dollar prepayment in a month

A pool has a **beginning balance of \$180,000, scheduled principal of \$200** this month, and prepays at **CPR = 6%**.

**Step 1 — SMM.**  $\text{SMM} = 1 - (1 - 0.06)^{1/12} = 1 - (0.94)^{0.083333}$ .  $\ln 0.94 = -0.0618754$ ;  $\times 0.083333 = -0.00515628$ ;  $e^{-0.00515628} = 0.9948570$ ; SMM  $= 0.0051430 = 0.51430\%$ .

**Step 2 — prepayment dollars.**  $\text{prepay} = \text{SMM} \times (\text{beginning balance} - \text{scheduled principal}) = 0.0051430 \times (180,000 - 200) = 0.0051430 \times 179,800 = \$924.71$ .

**Step 3 — total principal to investors**  $= \$ \text{scheduled} + \$ \text{prepay} = 200 + 924.71 = \boxed{\$1,124.71}$ .

**Reconcile:** the SMM base correctly excludes the \$200 of scheduled principal (that balance is already leaving), so we apply 0.51430% only to the \$179,800 that could still prepay. ✓

### B4. Sequential-pay WAL, reconciled to the pool

**Pool:** \$200m of principal returned over four years — Y1 \$40m, Y2 \$60m, Y3 \$60m, Y4 \$40m. Two sequential tranches: **A = \$100m** (paid first), **B = \$100m**.

**Allocate principal top-down:**

| Year | Pool principal | To A | A remaining | To B | B remaining |
|------|----------------|------|-------------|------|-------------|
| 1    | 40             | 40   | 60          | 0    | 100         |
| 2    | 60             | 60   | 0           | 0    | 100         |
| 3    | 60             | 0    | 0           | 60   | 40          |
| 4    | 40             | 0    | 0           | 40   | 0           |

**WAL of A**  $= \frac{1(40) + 2(60)}{100} = \frac{40 + 120}{100} = \frac{160}{100} = \boxed{1.60 \text{ yrs}}$

**WAL of B**  $= \frac{3(60) + 4(40)}{100} = \frac{180 + 160}{100} = \frac{340}{100} = \boxed{3.40 \text{ yrs}}$

**Pool WAL**  $= \frac{1(40) + 2(60) + 3(60) + 4(40)}{200} = \frac{40 + 120 + 180 + 160}{200} = \frac{500}{200} = 2.50 \text{ yrs}$

**Reconcile:** the size-weighted average of the tranche WALs must equal the pool WAL:

$\frac{100(1.60) + 100(3.40)}{200} = \frac{160 + 340}{200} = 2.50 \text{ yrs}$  ✓ Same collateral, one 2.50-year average life, carved into a short 1.60-year A note (money funds) and a longer 3.40-year B note (insurers). A's WAL is also far *more stable* against prepayment shocks because B absorbs the timing tail.

## B5. Loss allocation and credit enhancement

**Structure:** \$500m pool. Senior A = \$400m, Mezzanine B = \$60m, Equity C = \$40m. Losses hit C, then B, then A.

**Credit enhancement (subordination) at close:** - A: cushion below  $= B + C = \$100m \rightarrow 100/500 = \mathbf{20\%}$ . - B: cushion below  $= C = \$40m \rightarrow 40/500 = \mathbf{8\%}$ . - C: **0%** — first-loss.

| Cumulative pool loss | Equity C (\$40m) | Mezz B (\$60m) | Senior A (\$400m) | Impaired                |
|----------------------|------------------|----------------|-------------------|-------------------------|
| \$30m                | -\$30m → \$10m   | intact         | intact            | C only                  |
| \$40m                | wiped            | intact         | intact            | C exactly exhausted     |
| \$70m                | wiped            | -\$30m → \$30m | intact            | C, B partially          |
| \$100m               | wiped            | wiped          | intact            | A whole (full 20% used) |
| \$110m               | wiped            | wiped          | -\$10m → \$390m   | A takes first loss      |

**Reading it:** Senior A stays money-good until cumulative losses exceed **20%** (\$100m) — exactly its subordination. At \$110m (22%) of losses, A loses \$110m – \$100m = \$10m, recovering \$390m of \$400m, i.e. **97.5 cents**. Tranching turned a mixed pool into a \$400m security that survives a 20% loss event untouched — the mechanical basis of its AAA. ✓

## Section C — Interview-Style Questions with Model Answers

**C1. "Explain to me why an MBS is negatively convex." Model answer:** "An MBS is a bond minus a call the borrower owns — the right to prepay. When rates fall, an ordinary bond's price rises, but here borrowers refinance, so principal comes back early and the price appreciation is capped; the prepay option I'm short is gaining value against me. When rates rise, prepayments slow and I'm stuck with a low coupon longer —

extension. So my upside is limited and my downside isn't, which bends the price-yield curve concave: negative convexity. Practically, that's why I use effective duration and OAS, not modified duration, and why MBS trade at a spread to Treasuries to pay me for that short option."

**C2. "Walk me through how a AAA tranche can sit on subprime loans. Where does the safety come from?"** *Model answer:* "The safety is structural subordination, not collateral quality. Say a \$500m pool has \$40m of equity and \$60m of mezzanine beneath a \$400m senior. The senior only takes a dollar of loss after cumulative pool losses exceed \$100m — 20% of the pool. Historically a 20% cumulative loss is a deep-tail event, so the senior earns AAA. The critical assumption is that defaults are weakly correlated, so that 20% cushion is enough. In 2008 that assumption broke: national house prices fell, defaults correlated, losses blew past 20%, and 'AAA' tranches took hits. So I'd always stress the correlation assumption, not just the headline enhancement number."

**C3. "A PAC bond and its support bond come from the same pool. Which would you buy and why?"** *Model answer:* "It depends on my mandate. The PAC has a promised principal schedule that holds as long as prepayment stays inside its collar, say 100–300 PSA, so it behaves almost like a bullet bond with a stable WAL — good for a liability-matching insurer who hates timing surprises, and it trades at a tight spread. The support tranche is the shock absorber: it eats excess principal when prepayments are fast and gets starved when slow, so it carries amplified contraction *and* extension risk. I'd buy the support only if I'm paid a wide enough spread and I have a view — say, that prepayments will stay moderate. The key point is that structuring didn't destroy the timing risk; it concentrated it in the support so the PAC could shed it."

**C4. "Why does CPR compound rather than just multiply SMM by 12?"** *Model answer:* "Because prepayment is a survival process, not additive. Each month a fraction SMM of the *remaining* balance prepays, so after 12 months the surviving fraction is  $(1 - \text{SMM})^{12}$ , and the annual prepaid fraction is  $1 - (1 - \text{SMM})^{12}$ . Multiplying by 12 would double-count — it ignores that the balance shrinks each month, so there's less left to prepay later. CPR is therefore always a bit *less* than  $12 \times \text{SMM}$ . It's the same compounding logic as an annual return built from monthly returns."

**C5. "What went wrong in 2008 from a securitization-structure standpoint?"** *Model answer:* "A chain of failures. Origination degraded to no-doc 'liar' loans under an originate-to-distribute model where originators kept no skin in the game. Rating agencies, paid by issuers and using low-correlation models, stamped AAA on senior tranches. ABS CDOs then re-securitized the mezzanine slices of subprime MBS — re-tranching already-thin risk and stacking correlation exposure everyone assumed away. The whole tower was funded short-term through repo and ABCP conduits, so it was a maturity mismatch on top of a credit problem. When house prices fell nationwide, defaults correlated, subordination was consumed, senior tranches took losses, mark-to-market collapsed, funding ran, and forced selling spread the fire. The post-crisis fixes — risk-retention rules, better correlation modelling, transparency, less rating reliance — target exactly those links."

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## Section D — Multiple Choice (with Reasoning)

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**D1.** In a cash-flow waterfall, losses are absorbed in which order? A) Senior → mezzanine → equity B) Equity → mezzanine → senior C) Pro-rata across all tranches D) Mezzanine → equity → senior **Answer: B.** Cash flows down the stack senior-first, but *losses* run in reverse — equity is first-loss, senior is last. This mirror image is what lets the senior be AAA while the equity is levered.

**D2.** A pool prepays at  $\text{SMM} = 0.5\%$  per month. The approximate CPR is: A) exactly 6.0% B) slightly below 6.0% C) slightly above 6.0% D) 0.5% **Answer: B.**  $\text{CPR} = 1 - (1 - 0.005)^{12} = 1 - 0.99^{12} \approx 5.84\%$ , which is

below  $12 \times 0.5\% = 6.0\%$  because the balance shrinks each month. CPR is always less than  $12 \times \text{SMM}$ .

**D3.** At 150 PSA, the CPR in month 20 is: A) 4% B) 6% C) 9% D) 3% **Answer: B.** Base 100 PSA CPR in month 20  $= 0.2\% \times 20 = 4\%$ ; at 150 PSA,  $1.5 \times 4\% = 6\%$ . (Note the base is still on its ramp at month 20 — it caps at 6% only at month 30 — so no cap binds here.)

**D4.** When interest rates fall sharply, an MBS holder most directly faces: A) Extension risk B) Contraction risk C) Credit risk D) Higher WAL **Answer: B.** Falling rates trigger refinancing, prepayments accelerate, WAL shortens (contraction), and cash returns when reinvestment rates are low. Extension risk (A, D) is the rates-rise case.

**D5.** An interest-only (IO) strip typically has: A) Large positive duration B) Zero duration C) Negative duration D) The same duration as the pool **Answer: C.** Fast prepayment destroys the balance the IO earns interest on, so the IO *loves* slow prepayment (high rates) and *hates* fast prepayment (low rates). Its price therefore *rises* when rates rise — negative duration. The PO is its mirror, with large positive duration.

**D6.** The senior tranche of a \$500m deal with \$40m equity and \$60m mezzanine below it has a credit enhancement of: A) 8% B) 12% C) 20% D) 100% **Answer: C.** Subordination below the senior  $= 40 + 60 = \$100\text{m}$ ;  $\$100 / 500 = 20\%$ . The senior is untouched until cumulative losses exceed 20% of the pool.

**D7.** In a PAC/support CMO, the support (companion) tranche exists primarily to: A) Provide external credit insurance B) Absorb prepayment timing variability so the PAC gets a stable schedule C) Increase the pool's total cash flow D) Guarantee the senior's principal **Answer: B.** The support soaks up excess principal when prepayments are fast and is starved when slow, stabilizing the PAC's WAL. It redistributes *timing* risk; it neither adds cash flow (C) nor provides credit protection (A, D).

**D8.** An ABS CDO of subprime mezzanine tranches was especially dangerous because: A) It held only Treasury collateral B) It re-tranched already-thin risk whose defaults were highly correlated C) It had no senior tranche D) It removed all prepayment risk **Answer: B.** Pooling mezzanine slices whose defaults share a common driver (national house prices) meant the layers defaulted together, so the CDO's "AAA" senior was wiped almost as fast as its equity — correlation was the hidden switch.

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*Self-verification note:* Every numerical result in Section B was cross-checked — the mortgage payment by confirming the interest/principal split reconciles to \$M\$; the  $\text{CPR} \rightleftharpoons \text{SMM}$  conversion by a full round-trip back to 4.00%; the dollar prepayment by confirming the SMM base excludes scheduled principal; the sequential WALs by rebuilding the pool WAL as their size-weighted average (2.50 yrs); and the loss table by confirming the senior's first loss occurs exactly at its 20% subordination. All figures reconcile to within rounding.



# Chapter 13 — Money Market Instruments and Yields

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## 1. The Problem / The Need

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Every serious participant in the financial system faces the same recurring headache: **the timing of cash inflows almost never matches the timing of cash outflows**. A corporation collects receivables on the 10th but must run payroll on the 1st. A bank takes in deposits that can be withdrawn tomorrow but has lent that money out for thirty years. A government collects most of its tax revenue in a few concentrated months but spends continuously. A money-market mutual fund promises investors they can redeem at any moment, yet it must earn a return on the cash it holds in the meantime.

Each of these players has two symmetric problems:

- **The cash-rich problem.** "I have money I don't need for a few days or weeks. Leaving it idle in a non-interest account is a pure loss. But I cannot lock it up for years, and I cannot afford to lose a cent of principal — I *will* need it back on a known date." This is the **surplus-unit** looking for a safe, short, liquid parking spot.
- **The cash-poor problem.** "I have a temporary gap. I am good for the money — I just need to bridge a few days or weeks until my own inflows arrive. Borrowing long-term is expensive and clumsy for a need this short." This is the **deficit-unit** looking for cheap, short-term funding.

The **money market** is the institutional answer to both. It is the wholesale market for **high-quality debt with an original maturity of one year or less** (in practice most activity is overnight to 90 days). It exists so that surplus units can lend idle cash safely and deficit units can borrow cheaply, both at maturities measured in days and weeks rather than years. The instruments are simple, standardized, and issued by borrowers of very high credit quality, precisely because the whole point is *safety and liquidity, not yield*.

The complication — and the reason this chapter needs real math — is that these instruments are quoted using several **incompatible yield conventions** that are historical accidents. A T-bill is quoted on a "discount" basis over a 360-day year; a CD is quoted on an "add-on" basis, also over 360 days; the "true" investor return that lets you compare a T-bill to a coupon bond uses 365 days. If you cannot convert fluently between **discount yield, money-market yield, bond-equivalent yield, and effective annual yield**, you will systematically misjudge which instrument is actually cheaper or richer — and in an interview, you will be exposed instantly.

## 2. The Core Idea

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Two ideas run through this entire chapter.

**Idea 1 — Money-market instruments come in exactly two structural flavors.**

- **Pure discount (single-payment) instruments.** You pay less than face today and receive the full face value at maturity. There is no coupon. The interest *is* the difference between what you pay and what you get back. T-bills, commercial paper, and banker's acceptances work this way. The instrument is bought at a **discount** to par.
- **Interest-bearing / add-on instruments.** You deposit (or lend) face value today and receive face value **plus** interest at maturity. The interest is *added on* to the principal. Negotiable certificates of deposit are the classic example.

Everything else — the quotes, the yields, the risks — is a variation on these two skeletons.

**Idea 2 — A "yield" is meaningless until you know the convention.** The same physical trade (pay \$98,000 today, get \$100,000 in 90 days) can be truthfully described as an 8.00% yield, an 8.16% yield, an 8.28% yield, or an 8.54% yield, depending on which convention you use. They differ on two dimensions: **which base** you divide the interest by (face value vs. price paid), and **which day-count** you annualize with (360 vs. 365 vs. compounding). Learn the four conventions and the conversions between them and this chapter is essentially solved.

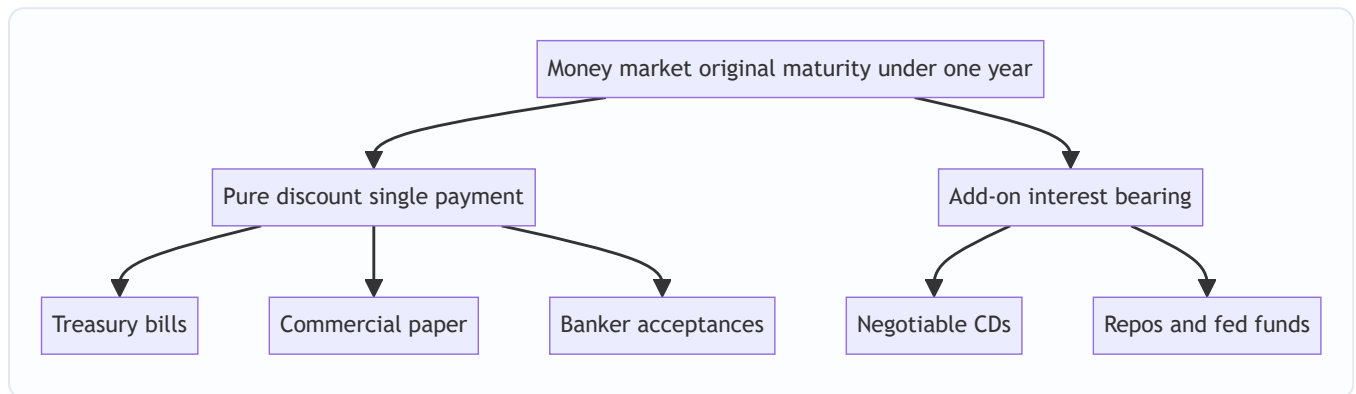


Figure 13.1 — The money-market universe splits cleanly into discount instruments and add-on instruments.

### 3. Why / How It Works

**Why so short?** Short maturity is the single most powerful risk-control device available. Interest-rate risk (duration) is tiny when the instrument matures in weeks; a 90-day bill barely moves in price when rates jump. Credit risk is contained because a borrower's ability to survive the next 30–90 days is far more predictable than its solvency in ten years. Liquidity is high because standardized, short paper is easy to value and easy to sell. This is why the money market is where cautious institutional cash lives.

**Why the odd quoting conventions?** History. The **bank discount** convention for T-bills dates to an era of hand computation: quoting interest as a percentage *of the round face value* over a *360-day* year (twelve 30-day months) made mental arithmetic and coupon tables easy. It stuck. It is admittedly a *worse* description of the investor's true return — it understates it — but it is the market standard and every U.S. T-bill, CP, and BA is quoted on it. The **add-on** convention for CDs mirrors how a bank actually credits interest to a deposit. And the **365-day** conventions exist so investors can compare these instruments to coupon-bearing notes and bonds, which use actual/365-type reasoning. The conventions are not "right" or "wrong"; they are dialects, and fluency means translating between them without thinking.

**Why credit quality is so high.** Because the buyers are cash managers whose mandate is capital preservation, the market only clears for very strong issuers: sovereigns (T-bills), highly rated corporations and financial firms (CP), and banks (CDs, BAs). Weak credits simply cannot sell short paper cheaply, which is itself a signal — a company that suddenly *cannot* roll its commercial paper is often in serious trouble (this is exactly what happened to several firms in 2008).



## 4. Full Content — Instruments and the Yield Mathematics

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### 4.1 The instruments

**Treasury bills (T-bills).** Short-term obligations of the national government (in the U.S., the Treasury), issued at a discount, in maturities of 4, 8, 13, 17, 26, and 52 weeks. Considered the closest thing to a risk-free instrument in domestic currency terms. Quoted on a **bank discount** basis. Deep, liquid, the benchmark against which all other money-market paper is priced.

**Commercial paper (CP).** Unsecured, short-term promissory notes issued by large, creditworthy corporations and financial institutions to fund working capital and bridge financing. Typical maturity 1–270 days (270 keeps it exempt from full securities registration in the U.S.). Issued at a discount, like a T-bill, but carries **credit and liquidity risk**, so it yields more than a T-bill. "Rollover risk" is central: issuers continuously re-issue maturing paper, and a market that refuses to buy new paper creates an instant funding crisis. **Asset-backed commercial paper (ABCP)** is CP collateralized by a pool of receivables.

**Certificates of deposit (CDs).** Time deposits at a bank with a fixed maturity and rate. A **negotiable CD** (typically \$1 million or more) can be sold in the secondary market before maturity, giving the holder liquidity while the bank keeps the funds for the full term. Structured as an **add-on** instrument: you deposit face, you receive face plus interest. Carries the bank's credit risk (mitigated by deposit insurance up to a limit, but large negotiable CDs exceed the insured cap).

**Banker's acceptances (BAs).** A time draft drawn on and *accepted* by a bank, historically used to finance international trade. When the bank stamps "accepted," it guarantees payment at maturity, converting the buyer's credit into the bank's credit. Sold at a discount, like a T-bill. Largely displaced by other instruments today but a classic exam topic because of the "two-name paper" credit-enhancement idea.

**Repurchase agreements (repos) and federal funds** are add-on, collateralized (repo) or uncollateralized (fed funds) overnight-ish lending between financial institutions. They are covered in their own right elsewhere but belong to the same family: overnight liquidity management.

### 4.2 The four yield conventions

Define, for a single-payment instrument:

- **F** = face (par) value received at maturity
- **P** = price paid today
- **D** = **F** – **P** = the dollar discount (the interest earned)
- **t** = days to maturity

**(a) Bank discount yield (BDY, also "discount yield" or "discount rate").** Interest as a fraction of **face**, annualized over **360** days:

$$\text{BDY} = \frac{D}{F} \times \frac{360}{t}$$

Note the two "flaws" from the investor's viewpoint: it divides by **F** (the amount you *don't* pay) rather than **P** (what you actually invest), and it uses a 360-day year. Both make BDY **understate** the true return. It is a quoting convention, not an investment return. To go from a quoted BDY back to price:  $P = F \times \left(1 - \text{BDY} \times \frac{t}{360}\right)$ .

**(b) Money-market yield (MMY, also "CD-equivalent yield").** Interest as a fraction of **price**, still annualized over **360** days:

$$\text{MMY} = \frac{D}{P} \times \frac{360}{t} = \text{HPY} \times \frac{360}{t}$$

where the **holding-period yield**  $\text{HPY} = D/P$  is the actual return earned over the  $t$  days. MMY fixes the "divide by face" flaw but keeps the 360-day convention, so it puts a discount instrument on the same footing as an add-on CD. Directly from BDY:

$$\text{MMY} = \frac{360 \times \text{BDY}}{360 - t \times \text{BDY}}$$

**(c) Bond-equivalent yield (BEY).** Interest as a fraction of **price**, annualized over **365** days:

$$\text{BEY} = \frac{D}{P} \times \frac{365}{t} = \text{HPY} \times \frac{365}{t}$$

This is the yield that lets you compare a discount instrument to a coupon bond on an apples-to-apples calendar basis (for maturities  $\leq 182$  days; longer bills need a semiannual-compounding adjustment, ignored here). Directly from BDY:  $\text{BEY} = \frac{365 \times \text{BDY}}{360 - t \times \text{BDY}}$ .

**(d) Effective annual yield (EAY, also EAR).** The HPY **compounded** over a 365-day year:

$$\text{EAY} = \left(1 + \text{HPY}\right)^{365/t} - 1 = \left(1 + \frac{D}{P}\right)^{365/t} - 1$$

This is the economically "truest" measure: it accounts for the ability to reinvest and compound, and it uses the actual calendar. It is always the **highest** of the four for a given trade because it adds compounding on top of the 365-day base.

**The universal ordering** for any discount instrument is:

$$\text{BDY} < \text{MMY} < \text{BEY} < \text{EAY}$$

BDY is lowest (smallest base, short year). MMY beats it by fixing the base. BEY beats MMY by stretching to 365 days. EAY beats BEY by compounding.

### 4.3 Add-on yields (for CDs)

For an add-on instrument, interest is quoted as a rate applied to face over a 360-day year:

$$\text{Interest} = F \times r_{\text{add-on}} \times \frac{t}{360}, \quad \text{Maturity value} = F + \text{Interest}$$

If you buy the CD at issue for its face value, your HPY is simply  $\text{Interest}/F$ , and you convert to BEY and EAY with the same 365-day formulas as above. The key contrast: a **discount** instrument's price is *below* face; an **add-on** instrument's initial investment *equals* face and you get more than face back.

## 5. Worked Examples

### Example 1 — A T-bill through all four conventions (the reconciliation)

**Setup.** A 90-day T-bill with face value **F = \$100,000** trades at price **P = \$98,000**. So **D = \$2,000** and **t = 90**.

First, the actual holding-period yield:  $\text{HPY} = \frac{2,000}{98,000} = 2.0408\%$

Now the four conventions:

| Convention             | Formula                                | Computation               | Result         |
|------------------------|----------------------------------------|---------------------------|----------------|
| Bank discount (BDY)    | $\frac{D}{F} \cdot \frac{360}{t}$      | $0.02 \times 4$           | <b>8.0000%</b> |
| Money-market (MMY)     | $\text{HPY} \cdot \frac{360}{t}$       | $0.020408 \times 4$       | <b>8.1633%</b> |
| Bond-equivalent (BEY)  | $\text{HPY} \cdot \frac{365}{t}$       | $0.020408 \times 4.05556$ | <b>8.2766%</b> |
| Effective annual (EAY) | $(1 + \text{HPY})^{\frac{365}{t}} - 1$ | $1.020408^{4.05556} - 1$  | <b>8.5383%</b> |

**Cross-checks (proving the direct BDY conversions agree):** -  $\text{MMY} = \frac{360 \times 0.08}{360 - 90 \times 0.08} = \frac{28.8}{352.8} = 8.1633\%$  ✓ -  $\text{BEY} = \frac{365 \times 0.08}{360 - 90 \times 0.08} = \frac{29.2}{352.8} = 8.2766\%$  ✓

Both routes land on the identical numbers, and the ordering **8.00 < 8.16 < 8.28 < 8.54** holds exactly as predicted. Notice the practical punchline: the *quoted* 8.00% understates the investor's true compounded return by **more than half a percentage point**. A cash manager comparing this bill to a CD quoted at, say, 8.20% add-on would be badly misled by the raw quotes and must convert both to a common basis first.

## Example 2 — From a commercial-paper quote to price and true yield

**Setup.** A dealer quotes 60-day commercial paper with face **F = \$1,000,000** at a discount yield of **BDY = 5.25%**.

**Step 1 — Price.**  $P = F \left(1 - \text{BDY} \cdot \frac{t}{360}\right) = 1,000,000 \left(1 - 0.0525 \times \frac{60}{360}\right) = 1,000,000 \times (1 - 0.00875) = \$991,250$

So the discount is **D = \$8,750** and  $\text{HPY} = 8,750 / 991,250 = 0.8827\%$ .

**Step 2 — The comparable yields.**

| Convention   | Computation                      | Result  |
|--------------|----------------------------------|---------|
| BDY (quoted) | given                            | 5.2500% |
| MMY          | $0.008827 \times \frac{360}{60}$ | 5.2963% |
| BEY          | $0.008827 \times \frac{365}{60}$ | 5.3699% |
| EAY          | $1.008827^{6.08333} - 1$         | 5.4918% |

**Reconciliation.** Because this CP yields **5.37% BEY** versus a comparable T-bill's BEY, the roughly one-percentage-point-plus pickup over a same-maturity bill is the compensation for the issuer's **credit and liquidity risk** — the money market pricing credit exactly as theory predicts. And again the four numbers respect the strict ordering, confirming the arithmetic is internally consistent.

## Example 3 — An add-on CD, contrasted with a discount instrument

**Setup.** A negotiable CD with face **F = \$1,000,000**, term **t = 180 days**, quoted add-on rate **r = 4.80%** (360-day basis). You buy it at issue for face value.

**Step 1 — Interest and maturity value.**  $\text{Interest} = 1,000,000 \times 0.048 \times \frac{180}{360} = \$24,000$ ,  $\text{Maturity value} = \$1,024,000$

You invest \$1,000,000 (equal to face — the add-on signature) and receive \$1,024,000.

**Step 2 — True yields.** Here HPY is on the face you invested:  $\text{HPY} = 24,000 / 1,000,000 = 2.400\%$ .  
 $\text{BEY} = 0.024 \times \frac{365}{180} = 4.8667\%$ ,  $\text{EAY} = 1.024^{365/180} - 1 = 1.024^{2.02778} - 1 = 4.9267\%$

**Reconciliation and the teaching point.** The CD's *quoted* 4.80% is already close to its true return because the add-on convention divides interest by the amount actually invested (face = price here). Contrast with a discount instrument, where the quoted BDY sat well *below* the true return. This is the whole reason MMY is also called the **CD-equivalent yield**: converting a discount instrument to MMY restates it as if it were an add-on CD, so the two can be compared directly. If this CD's issuer offered instead a 180-day discount instrument, you would convert its BDY to MMY before deciding which is richer.

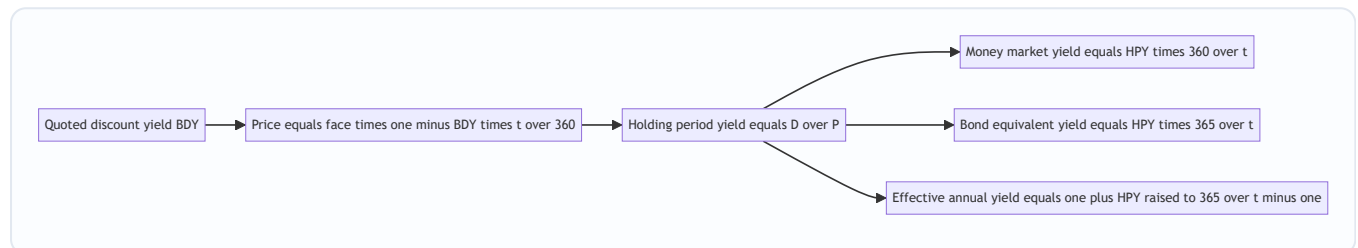


Figure 13.2 — The conversion ladder starts from a quote, recovers price, then produces the three comparable yields.

## 6. Money Market Funds and Liquidity Management

### 6.1 Money market funds (MMFs)

A **money market fund** is a mutual fund that holds a diversified pool of short, high-quality money-market instruments — T-bills, CP, CDs, repos, agency discount notes — and offers investors daily liquidity. It is the retail and corporate on-ramp to the wholesale money market: an individual cannot easily buy a \$1 million negotiable CD, but she can buy shares of a fund that holds hundreds of them.

The historically defining feature was the **stable \$1.00 net asset value (NAV)**. Because the underlying holdings are so short, funds were permitted to value them at **amortized cost** (accreting the discount toward par in a straight line) rather than marking to market, which kept the NAV pinned at exactly \$1.00 per share. Investors treated MMF shares almost like cash. This works only as long as the true market value stays within a whisker of \$1.00.

**Breaking the buck.** If losses push the true value below roughly \$0.995, the fund "breaks the buck" and the NAV drops below \$1.00 — a small percentage loss but a psychological earthquake, because investors were promised cash-like safety. This happened to the **Reserve Primary Fund in September 2008**, which held Lehman Brothers commercial paper that became nearly worthless when Lehman failed. The fund broke the buck, triggering a **run** on prime MMFs and a freeze in the CP market — a vivid demonstration that "safe" short paper is only as safe as its issuers and that liquidity can vanish in a panic.

**Post-crisis regulation** (in the U.S., the SEC's Rule 2a-7 framework, tightened after 2008 and again in 2014/2023) responded by:

- Tightening **credit quality** (only top-tier short-term paper), **maturity** (limits on weighted-average maturity, typically  $\leq 60$  days), and **diversification** limits.
- Requiring minimum **daily and weekly liquid assets** so the fund can meet redemptions without fire-selling.

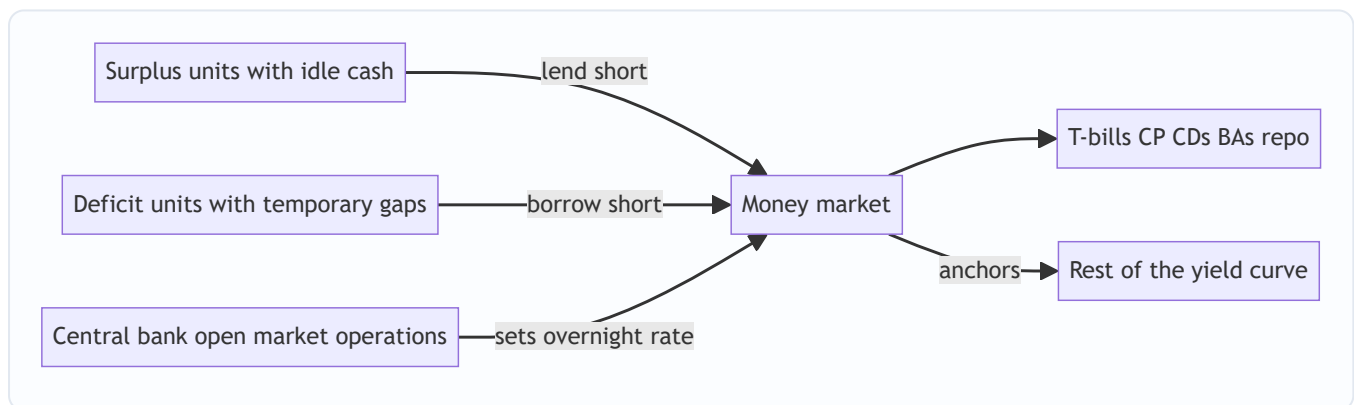
- Splitting funds into categories: **government MMFs** (T-bills, agencies, government repo — allowed to keep a stable \$1.00 NAV), **prime/institutional MMFs** (include CP and CDs — required to use a **floating NAV** so the price reflects true market value), and **tax-exempt MMFs** (municipal short paper).
- Permitting **liquidity fees** as a tool to slow runs.

Investors judge MMFs by the **7-day yield** (the annualized return over the past week) and its compounded cousin the **7-day effective yield** — a direct application of the annualization and compounding ideas from Section 4.

## 6.2 The money market and system-wide liquidity management

Zoom out and the money market is the **plumbing that keeps the financial system liquid**. Every major institution uses it to manage the gap between when cash arrives and when it is needed.

- **Banks** manage their reserve positions overnight: a bank short of reserves borrows in the **fed funds** or **repo** market; a bank with excess reserves lends. This is daily, and it is enormous.
- **Corporations** invest surplus operating cash in T-bills, CP, and MMFs for a few days or weeks, and issue their **own CP** to cover short gaps — cheaper and nimbler than drawing a bank line.
- **Governments** smooth the mismatch between lumpy tax receipts and continuous spending by issuing T-bills.
- **The central bank** conducts **monetary policy** here. Open-market operations, repos, and the policy rate all act on the money market first; the overnight rate is the anchor from which all other interest rates are built. When the central bank wants to ease, it injects reserves and pushes the short rate down; the effect ripples out along the curve.



*Figure 13.3 — The money market matches surplus and deficit units and is the point where central-bank policy enters the interest-rate system.*

The deep lesson: the money market is where **liquidity is priced**. In calm times the spread of CP over T-bills (a "credit spread" like the **TED spread**, CP or LIBOR over bills) is small and stable. When it blows out, it is one of the earliest and most reliable warnings of stress in the financial system — because it is precisely the market where the willingness to lend short-term is tested every single day.

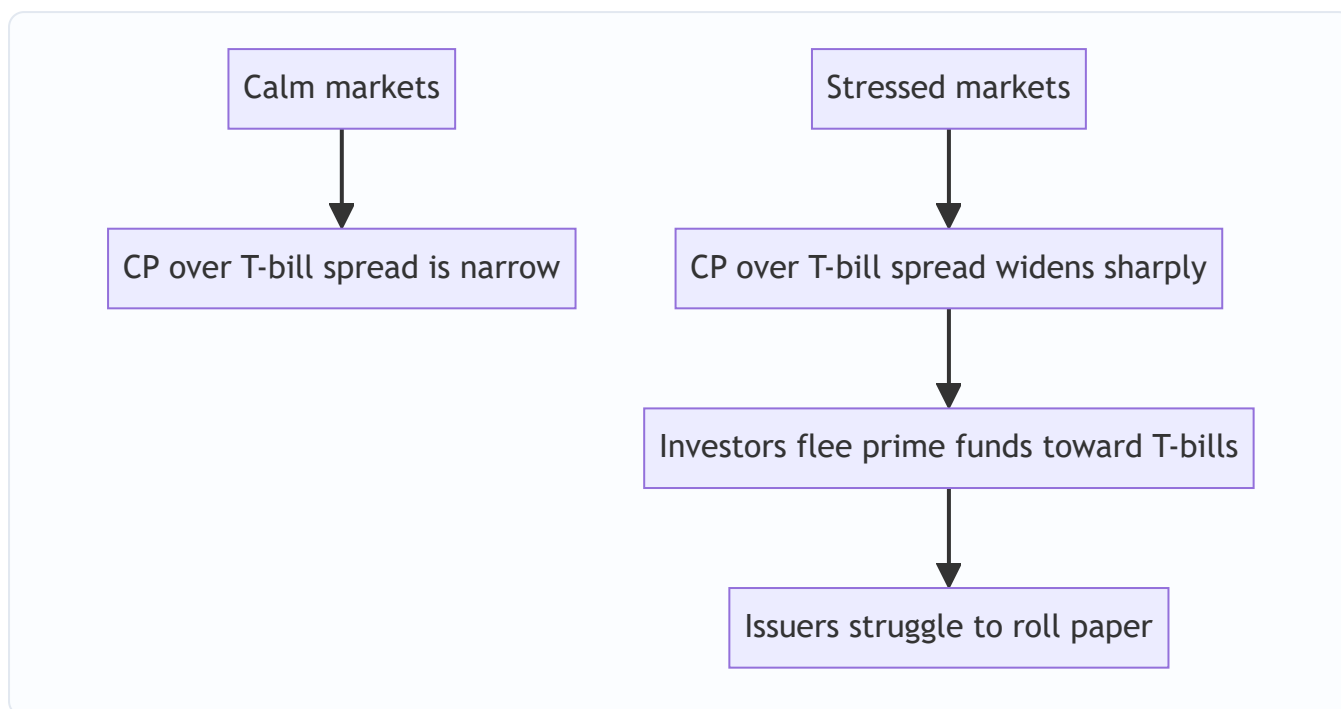


Figure 13.4 — The money-market credit spread is an early-warning gauge; a sudden widening signals a flight to quality and rollover stress.

## 7. Connections

- **To duration and interest-rate risk (Ch. on duration).** Money-market instruments have near-zero duration by construction, which is why they are the "safe" bucket. The entire appeal is the *absence* of the price sensitivity that dominates long bonds.
- **To the yield curve (Ch. on term structure).** The overnight rate set in the money market is the short end — the anchor — of the yield curve. Expectations of future short rates, formed here, drive the shape of the whole curve.
- **To credit spreads (Ch. on credit).** CP-over-bills and BA spreads are short-maturity credit spreads; the same risk-compensation logic that prices corporate bonds prices money-market paper, just over days instead of years.
- **To coupon-bond pricing.** BEY exists precisely to bridge money-market instruments and coupon bonds onto one comparable calendar; it is the same actual/365 reasoning used in accrued-interest and bond-yield calculations.
- **To liquidity and funding risk (Ch. on repo/financing).** Rollover risk in CP is the same mechanism as margin/haircut spirals in repo — short-term funding that must be continuously renewed and can evaporate.

## 8. Key Terms

- **Discount instrument** — bought below face, no coupon; interest = face – price (T-bill, CP, BA).
- **Add-on instrument** — invest face, receive face + interest (negotiable CD, repo, fed funds).
- **Bank discount yield (BDY)** —  $\text{interest} \div \text{face} \times 360/t$ ; the quoting convention, understates true return.
- **Money-market yield (MMY / CD-equivalent yield)** —  $\text{interest} \div \text{price} \times 360/t$ .
- **Bond-equivalent yield (BEY)** —  $\text{interest} \div \text{price} \times 365/t$ ; comparable to coupon bonds.

- **Effective annual yield (EAY / EAR)** — HPY compounded over 365 days; the truest return.
- **Holding-period yield (HPY)** — the un-annualized return, D/P, over the  $t$  days actually held.
- **Rollover / refinancing risk** — the danger that maturing short paper cannot be re-issued.
- **Negotiable CD** — a large time deposit that can be sold in the secondary market.
- **Banker's acceptance** — a time draft guaranteed ("accepted") by a bank; "two-name paper."
- **Money market fund (MMF)** — mutual fund of short, high-quality paper offering daily liquidity.
- **Breaking the buck** — an MMF's NAV falling below \$1.00.
- **Floating NAV** — a fund whose share price marks to market instead of holding at \$1.00.
- **7-day yield** — the standard annualized return quote for an MMF.
- **TED / CP spread** — short-term credit spread; a stress gauge.

## 9. Common Confusions

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**"Discount yield is the return I earn."** No. BDY is the *lowest* and least meaningful of the four measures. It divides by face (not what you invested) and uses a 360-day year. Your true annualized return is closer to the BEY, and your true compounded return is the EAY. In Example 1 the gap was 8.00% quoted vs. 8.54% true — over half a point.

**"360 vs. 365 is a rounding detail."** It shifts the yield by a factor of  $365/360 \approx 1.0139$ , i.e., ~1.4% *of the yield*. On an 8% instrument that is ~11 bps — enough to flip which of two instruments is cheaper. Never compare a 360-basis quote directly to a 365-basis quote.

**Confusing MMY and BEY.** Both divide by **price** (good). They differ *only* in the day-count: MMY uses 360, BEY uses 365. MMY compares a discount instrument to a CD (both 360); BEY compares it to a coupon bond (365).

**Thinking a CD is bought at a discount.** An add-on CD is bought at (or near) **face**; you get **more than face** back. Only discount instruments (T-bill, CP, BA) are bought below par. Mixing up the structure corrupts the yield base.

**"Money market funds are guaranteed / are cash."** They are *designed* to be cash-like but are not guaranteed. The Reserve Primary Fund broke the buck in 2008. Government MMFs are safest; prime funds carry credit and (now) floating-NAV risk.

**"Higher yield on CP than on a T-bill means CP is a better deal."** The extra yield is *compensation for credit and liquidity risk*, not free money. Convert both to BEY, then judge whether the spread adequately pays for the risk.

**Annualizing without compounding, then calling it the effective rate.** BEY annualizes linearly ( $\times 365/t$ ); EAY compounds ( $^{\wedge}365/t$ ).  $EAY > BEY$  always. Only EAY is the true effective annual rate.

## 10. Recap

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The money market is the wholesale market for high-quality debt maturing in a year or less. It exists to solve the universal mismatch between cash inflows and outflows: surplus units lend idle cash safely and short, deficit units borrow cheaply and short. Instruments come in two structures — **discount** (T-bills, commercial paper, banker's acceptances: buy below face, get face) and **add-on** (negotiable CDs: invest face, get face plus interest). Credit quality is uniformly high because the buyers' mandate is capital preservation.



The heart of the chapter is yield translation. A single trade can be quoted four ways: **bank discount yield** (interest ÷ face × 360/t — the quoting convention, and the lowest), **money-market yield** (interest ÷ price × 360/t — the CD-equivalent), **bond-equivalent yield** (interest ÷ price × 365/t — comparable to coupon bonds), and **effective annual yield** (HPY compounded over 365 — the truest, and always highest). The strict ordering BDY < MMY < BEY < EAY always holds, and every conversion in the worked examples reconciled exactly. Money market funds package this world for ordinary investors with daily liquidity and a (usually) stable \$1.00 NAV — a promise that can break, as it did in 2008. And system-wide, the money market is where liquidity is priced every day and where central-bank policy enters the rate structure, making its credit spreads an early warning of financial stress.

## 11. Quick-Reference — Interview Points

- **Two structures.** Discount (buy < face, no coupon): T-bill, CP, BA. Add-on (invest face, get face + interest): CD.
- **Price from a discount quote:**  $P = F(1 - \text{BDY} \times t/360)$ .
- **Four yields, memorize the table:**

| Yield | Base  | Day-count | Compounded? |
|-------|-------|-----------|-------------|
| BDY   | Face  | 360       | No          |
| MMY   | Price | 360       | No          |
| BEY   | Price | 365       | No          |
| EAY   | Price | 365       | Yes         |

- **Ordering, always:** BDY < MMY < BEY < EAY.
- **Fast conversions from BDY:**  $\text{MMY} = \frac{360 \times \text{BDY}}{360 - t \times \text{BDY}}$ ;  $\text{BEY} = \frac{365 \times \text{BDY}}{360 - t \times \text{BDY}}$ .
- **BDY understates the true return** — divides by face and uses 360 days. Never quote it as "the return."
- **BEY** is the money-market instrument restated to compare with **coupon bonds** (365-day calendar).
- **MMY = CD-equivalent yield** — restates a discount instrument as an add-on CD.
- **CP > T-bill yield** = credit + liquidity risk premium, priced daily. Rollover risk is the key CP danger.
- **Banker's acceptance** = "two-name paper" — bank's acceptance stamp adds its credit to the drawer's.
- **MMFs:** stable \$1.00 NAV via amortized cost; government funds keep stable NAV, prime/institutional funds now float NAV; "breaking the buck" = Reserve Primary Fund, Sept 2008, Lehman CP. Judge by the **7-day yield**.
- **Systemic role:** overnight money-market rate anchors the yield curve; central-bank open-market operations act here first; the **CP-over-bills (TED) spread** is a real-time stress gauge.
- **Numeric anchor to remember:** 90-day bill at \$98,000 face \$100,000 → BDY 8.00%, MMY 8.16%, BEY 8.28%, EAY 8.54%.



# Q&A — Money Market Instruments and Yields

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A practice bank for Chapter 13. Work each numerical item on paper before reading the solution. All yields follow the chapter conventions: BDY and MMY use a 360-day year, BEY and EAY use 365, and EAY compounds.

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## Section A — Concept Checks

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**A1. What defines the money market, and why is credit quality uniformly high?** The money market is the wholesale market for high-quality debt with an original maturity of one year or less (most activity is overnight to 90 days). Credit quality is uniformly high because the buyers are cash managers whose mandate is capital preservation, not yield. Only very strong issuers — sovereigns, top-rated corporates and financials, and banks — can sell short paper cheaply. A weak credit simply cannot place short paper at a good price, which is itself a warning signal (e.g., firms that lost access to the CP market in 2008).

**A2. Name the two structural flavors of money-market instruments and give examples of each.** (1) Pure discount / single-payment: you pay less than face today and receive full face at maturity, with no coupon; the interest is face minus price. Examples: T-bills, commercial paper, banker's acceptances. (2) Add-on / interest-bearing: you invest face today and receive face plus interest at maturity. Examples: negotiable CDs, repos, fed funds.

**A3. Why is short maturity such a powerful risk-control device?** It attacks all three risks at once. Interest-rate (duration) risk is tiny because a 90-day instrument barely moves in price when rates jump. Credit risk is contained because a borrower's ability to survive 30–90 days is far more predictable than its ten-year solvency. Liquidity is high because standardized short paper is easy to value and sell. This is why cautious institutional cash lives here.

**A4. On which two dimensions do the four yield conventions differ?** (1) The base you divide the interest by — face value (BDY) versus price paid (MMY, BEY, EAY). (2) The day-count / compounding used to annualize — 360 simple (BDY, MMY), 365 simple (BEY), or 365 compounded (EAY).

**A5. Why does the bank discount yield understate an investor's true return?** Two reasons. It divides the dollar discount by face value  $F$  — the amount you do NOT pay — rather than by price  $P$ , the amount you actually invest. And it annualizes over a 360-day year instead of the actual 365. Both push the number down, so BDY is a quoting convention, not an investment return.

**A6. State the universal ordering of the four yields and justify each inequality.**  $BDY < MMY < BEY < EAY$ . MMY beats BDY by fixing the base (price instead of face). BEY beats MMY by stretching the year from 360 to 365 days. EAY beats BEY by adding compounding on top of the same 365-day base.

**A7. Why is money-market yield also called "CD-equivalent yield"?** Converting a discount instrument to MMY restates its return on a price base over a 360-day year — exactly the basis on which an add-on CD is quoted. So MMY expresses a discount instrument's return "as if" it were an add-on CD, letting the two be compared directly on identical footing.

**A8. What is rollover (refinancing) risk, and why is it central to commercial paper?** CP issuers continuously re-issue maturing paper to fund ongoing needs. Rollover risk is the danger that when old paper matures, the market refuses to buy new paper. Because the issuer relies on constant re-issuance, a buyers' strike creates an instant funding crisis even for a fundamentally solvent firm.

**A9. Why is a banker's acceptance called "two-name paper"?** It starts as a time draft drawn on a bank by a trade counterparty. When the bank stamps "accepted," it guarantees payment at maturity, adding its own credit to the drawer's. The instrument then carries two obligors' credit — the original drawer and the accepting bank — hence "two-name paper."

**A10. What does "breaking the buck" mean and what caused the classic case?** A money-market fund normally holds a stable \$1.00 NAV. "Breaking the buck" is the NAV falling below \$1.00 (roughly below \$0.995). The classic case was the Reserve Primary Fund in September 2008, which held Lehman Brothers commercial paper that became nearly worthless when Lehman failed. The fund broke the buck, triggering a run on prime MMFs and a freeze in the CP market.

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## Section B — Numerical / Applied

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**B1. A 90-day T-bill with face \$100,000 trades at \$98,000. Compute HPY, BDY, MMY, BEY, and EAY.**  $D = 100,000 - 98,000 = 2,000$ ;  $t = 90$ . -  $HPY = 2,000 / 98,000 = 2.0408\%$ . -  $BDY = (D/F)(360/t) = (2,000/100,000)(360/90) = 0.02 \times 4 = 8.0000\%$ . -  $MMY = HPY \times 360/t = 0.020408 \times 4 = 8.1633\%$ . -  $BEY = HPY \times 365/t = 0.020408 \times 4.05556 = 8.2766\%$ . -  $EAY = (1.020408)^{(365/90)} - 1 = 1.020408^{4.05556} - 1 = 8.5383\%$ . Ordering holds:  $8.00 < 8.16 < 8.28 < 8.54$ . The quoted 8.00% understates the true compounded return by over half a point.

**B2. Verify B1's MMY and BEY using the direct-from-BDY formulas.** -  $MMY = 360 \cdot BDY / (360 - t \cdot BDY) = (360 \times 0.08)/(360 - 90 \times 0.08) = 28.8 / 352.8 = 8.1633\% \checkmark$  -  $BEY = 365 \cdot BDY / (360 - t \cdot BDY) = (365 \times 0.08)/352.8 = 29.2 / 352.8 = 8.2766\% \checkmark$  Both routes reconcile exactly to the HPY-based answers.

**B3. A dealer quotes 60-day CP, face \$1,000,000, at BDY = 5.25%. Find the price and the dollar discount.**  $P = F(1 - BDY \times t/360) = 1,000,000 \times (1 - 0.0525 \times 60/360) = 1,000,000 \times (1 - 0.00875) = \$991,250$ . Discount  $D = 1,000,000 - 991,250 = \$8,750$ .

**B4. Using B3, compute HPY, MMY, BEY, and EAY for the commercial paper.**  $HPY = 8,750 / 991,250 = 0.8827\%$ . -  $MMY = 0.008827 \times 360/60 = 0.008827 \times 6 = 5.2963\%$ . -  $BEY = 0.008827 \times 365/60 = 0.008827 \times 6.08333 = 5.3699\%$ . -  $EAY = 1.008827^{(365/60)} - 1 = 1.008827^{6.08333} - 1 = 5.4918\%$ . The BEY of ~5.37% versus a comparable T-bill's BEY is the credit-plus-liquidity risk premium the money market prices daily.

**B5. A negotiable CD has face \$1,000,000, term 180 days, add-on rate 4.80% (360 basis), bought at issue for face. Find interest, maturity value, BEY, and EAY.** Interest =  $F \times r \times t/360 = 1,000,000 \times 0.048 \times 180/360 = \$24,000$ . Maturity value = **\$1,024,000**.  $HPY = 24,000 / 1,000,000 = 2.400\%$  (base is face, since price = face). -  $BEY = 0.024 \times 365/180 = 0.024 \times 2.02778 = 4.8667\%$ . -  $EAY = 1.024^{(365/180)} - 1 = 1.024^{2.02778} - 1 = 4.9267\%$ . Note the CD's quoted 4.80% sits close to its true return, because the add-on convention already divides interest by the amount actually invested.

**B6. A 91-day T-bill is quoted at BDY = 4.00%, face \$100,000. Compute price, then MMY and BEY via the direct formulas.**  $P = 100,000 \times (1 - 0.04 \times 91/360) = 100,000 \times (1 - 0.0101111) = \$98,988.89$ . -  $MMY = (360 \times 0.04)/(360 - 91 \times 0.04) = 14.4 / (360 - 3.64) = 14.4 / 356.36 = 4.0409\%$ . -  $BEY = (365 \times 0.04)/356.36 = 14.6 / 356.36 = 4.0970\%$ .

**B7. Comparison decision: Instrument X is a 90-day discount bill quoted BDY = 5.00%. Instrument Y is a 90-day add-on CD quoted 5.05%. Which yields more on a comparable basis?** Put both on MMY (360, price base). For X:  $MMY = 360 \times 0.05 / (360 - 90 \times 0.05) = 18 / 355.5 = 5.0633\%$ . For Y, an add-on CD bought at face already has its quote on a price/360 basis, so its CD-equivalent (MMY) yield is simply 5.05%.

Compare:  $X = 5.0633\%$  vs  $Y = 5.0500\%$ . **The discount bill X is slightly richer** — a raw-quote glance (5.00 vs 5.05) would have wrongly favored the CD. This is exactly why you must convert before choosing.

**B8. Show that ignoring the 360-vs-365 difference on an 8% instrument mis-states the yield by roughly 11 bps.** The two conventions differ by the factor  $365/360 = 1.013889$ . On an 8% yield the gap is  $8\% \times (365/360 - 1) = 8\% \times 0.013889 = 0.11111\% \approx 11 \text{ bps}$ . Compare B1: MMY 8.1633% (360) vs BEY 8.2766% (365) — a difference of 11.3 bps, enough to flip which of two close instruments is cheaper.

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## Section C — Interview-Style

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**C1. "Walk me through why a T-bill quoted at 8% doesn't actually return 8% to the investor."** The 8% is a bank discount yield, a legacy convention. It divides the dollar discount by the bill's face value — but the investor never pays face; she pays the discounted price, a smaller number, so dividing by the larger face understates the return. It also annualizes over a 360-day year rather than the real 365. Fix the base (divide by price) and you get the money-market yield,  $\sim 8.16\%$ ; stretch to 365 days and you get the bond-equivalent yield,  $\sim 8.28\%$ ; add compounding and you get the effective annual yield,  $\sim 8.54\%$ . So the true economic return is about half a point above the quote. Never quote BDY as "the return."

**C2. "A client says commercial paper is a better deal than T-bills because it yields more. How do you respond?"** First, make sure we compare on a common basis — convert both quotes to bond-equivalent yield so the calendar and base match. Once that's done, the CP will still yield more, but that spread isn't free money: it's compensation for credit risk (CP is unsecured corporate debt, a T-bill is sovereign) and liquidity risk (CP is thinner and can seize up). The right question isn't "does it yield more?" — it's "does the spread adequately pay for the extra credit and rollover risk?" In calm markets that spread is small and stable; when it blows out, it's one of the earliest warnings of system stress.

**C3. "Explain the difference between a discount instrument and an add-on instrument, and why it matters for the yield math."** A discount instrument (T-bill, CP, BA) is bought below face with no coupon — the interest is face minus price, and your investment base is the discounted price. An add-on instrument (negotiable CD) is bought at face and pays face plus interest at maturity — your investment base is face itself. It matters because the yield base is the price you actually paid: for a discount instrument that's below face, which is why converting a quoted discount yield to a price-based yield raises it noticeably; for an add-on CD bought at face, the quote is already close to the true return. Mixing up the structure corrupts the denominator and the whole calculation.

**C4. "What happened to money market funds in 2008, and how did regulators respond?"** The Reserve Primary Fund held Lehman commercial paper. When Lehman failed in September 2008, that paper collapsed, pushing the fund's true value below \$1.00 — it "broke the buck." Investors, who had treated MMF shares as cash-equivalent, ran on prime funds, and the CP market froze because funds are major CP buyers. Regulators (via the SEC's Rule 2a-7 framework, tightened after 2008 and again in 2014 and 2023) responded by tightening credit-quality, maturity, and diversification limits, requiring minimum daily and weekly liquid assets, splitting funds so government funds keep a stable \$1.00 NAV while prime/institutional funds must float their NAV to reflect true market value, and permitting liquidity fees to slow runs.

**C5. "Why do central bankers care so much about the money market?"** Because it's where monetary policy enters the interest-rate system. Open-market operations, repos, and the policy rate all act on the money market first — the overnight rate is the anchor from which the rest of the yield curve is built. When the central bank injects reserves, the short rate falls and the effect ripples out along the curve. It's also where

liquidity is priced every single day, so its credit spreads (like the TED or CP-over-bills spread) are among the most reliable real-time gauges of financial stress.

**C6. "You see the CP-over-T-bill spread suddenly widen. What does it tell you?"** That spread is a short-maturity credit spread — the extra yield investors demand to lend to corporates rather than the sovereign for a few weeks. A sudden widening means investors are being paid much more to take short corporate credit, i.e., they're fleeing toward the safety of bills (a flight to quality). It typically signals that lenders are pulling back from short-term funding, which raises rollover risk for issuers who must continuously re-issue. Historically it's an early, reliable warning of a funding squeeze, precisely because it's the market where willingness to lend short is tested daily.

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## Section D — MCQs (with reasoning)

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**D1. Which yield always sits highest for a given discount trade?** (a) BDY (b) MMY (c) BEY (d) EAY **Answer: (d) EAY.** It uses the largest base (price), the longer 365-day year, AND adds compounding — every lever that raises the number. The fixed ordering is  $BDY < MMY < BEY < EAY$ .

**D2. The money-market yield differs from the bond-equivalent yield only in:** (a) the base (face vs price) (b) the day-count (360 vs 365) (c) whether it compounds (d) the instrument it applies to **Answer: (b).** Both divide by price; MMY uses 360 days (CD-comparable), BEY uses 365 (coupon-bond-comparable). Neither compounds.

**D3. A negotiable CD is best described as:** (a) bought at a discount to face (b) bought at face, redeemed above face (c) a coupon bond (d) a zero-coupon perpetual **Answer: (b).** It's an add-on instrument: invest face, receive face plus interest. Only discount instruments (T-bill, CP, BA) are bought below par. Assuming a CD is discounted is a classic trap.

**D4. The price of a discount instrument from its quoted BDY is:** (a)  $F(1 + BDY \times t/360)$  (b)  $F(1 - BDY \times t/360)$  (c)  $F / (1 + BDY \times t/365)$  (d)  $F \times BDY \times t/360$  **Answer: (b).** BDY quotes interest as a fraction of face over 360 days, so the discount is  $F \times BDY \times t/360$  and price is face minus that discount:  $F(1 - BDY \times t/360)$ .

**D5. Which spread is a widely used real-time gauge of money-market stress?** (a) the bid-ask spread on 30-year bonds (b) the TED / CP-over-bills spread (c) the dividend yield spread (d) the on-the-run/off-the-run Treasury spread **Answer: (b).** The TED spread (and CP-over-bills) is a short-term credit spread; it stays narrow in calm markets and widens sharply under stress, signaling a flight to quality and rollover strain.

**D6. After 2008 reforms, which category of MMF is REQUIRED to use a floating NAV?** (a) government MMFs (b) prime/institutional MMFs (c) tax-exempt retail MMFs (d) all MMFs **Answer: (b).** Prime/institutional funds hold CP and CDs (credit-sensitive paper) and must float their NAV to reflect true market value. Government MMFs (bills, agencies, government repo) may keep the stable \$1.00 NAV.

**D7. Converting a discount instrument to MMY makes it directly comparable to:** (a) a coupon Treasury bond (b) an add-on CD (c) a common stock (d) an effective annual rate **Answer: (b).** MMY is the "CD-equivalent yield" — price base, 360-day year, the same basis on which add-on CDs are quoted. BEY (365) is the one comparable to coupon bonds.

**D8. Which is a genuine reason BDY understates the true return?** (a) it compounds too aggressively (b) it divides interest by price (c) it divides interest by face and uses a 360-day year (d) it uses a 365-day year **Answer: (c).** BDY divides by face (larger than the price actually paid) and annualizes over 360 days — both flaws push it below the investor's true return.

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## Self-Verification of Key Calculations

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- B1 EAY:  $\ln(1.020408) = 0.0202029$ ;  $\times 4.05556 = 0.081934$ ;  $e^{0.081934} - 1 = 0.08538 \rightarrow 8.538\% \checkmark$
- B4 EAY:  $\ln(1.008827) = 0.0087882$ ;  $\times 6.08333 = 0.053461$ ;  $e^{0.053461} - 1 = 0.054916 \rightarrow 5.492\% \checkmark$
- B6:  $91 \times 0.04 = 3.64$ ;  $360 - 3.64 = 356.36$ ;  $14.4/356.36 = 0.040409 \checkmark$ ;  $14.6/356.36 = 0.040969 \checkmark$
- B7:  $90 \times 0.05 = 4.5$ ;  $360 - 4.5 = 355.5$ ;  $18/355.5 = 0.050633 \checkmark$  ( $> 0.0505$ , so X wins)
- All four-yield tables respect  $BDY < MMY < BEY < EAY$ .  $\checkmark$

# Chapter 14 — Repo and Funding Markets

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## 1. The Problem / The Need

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A bond, once bought, is not the end of a story — it is a position that must be *financed*. Imagine you run a bond desk. You buy ₹100 crore of a 10-year government security because you believe yields will fall. Where did the ₹100 crore come from? You did not have it lying idle in a vault; almost no trading desk does. You borrowed it. And you borrowed it, most cheaply of all, by pledging the very bond you just bought as security for the loan.

That single move — "*lend me cash against this bond, and I'll buy it back tomorrow*" — is the repurchase agreement, or **repo**. It is simultaneously the most boring and the most important instrument in fixed income. Boring because a one-day secured loan sounds like nothing. Important because virtually the entire edifice of modern fixed-income markets — dealer inventories, leverage, short-selling, central-bank policy transmission, money-market funds — sits on top of the repo market. When the repo market seizes, everything above it wobbles. In 2008 and again in September 2019 (the US "repo spike") and March 2020, the world was reminded that repo *is* the plumbing.

The problems repo solves are concrete:

- **A cash-rich lender** (a money-market fund, a corporate treasury, a pension fund) wants to earn a return on idle cash overnight, but with almost no credit risk. An unsecured deposit at a bank exposes them to that bank failing. They want *collateral*.
- **A securities-rich borrower** (a bank, a bond dealer, a hedge fund) owns bonds but needs cash — to fund inventory, to meet a margin call, to settle trades. Selling the bonds outright is disruptive and may trigger tax or accounting consequences. They want *temporary* cash without giving up the economic exposure.
- **A central bank** wants a precise, low-risk lever to inject or drain overnight liquidity from the banking system and to steer short-term interest rates toward its policy target.

Repo answers all three at once. It is a **collateralised loan dressed as a pair of trades**. Understanding it turns the abstract phrase "the money market" into machinery you can see turning.

## 2. The Core Idea

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A repurchase agreement is the sale of a security today combined with a binding agreement to buy it back at a fixed price on a future date. Economically it is a **secured loan**: the seller of the security is the *borrower of cash*, and the buyer of the security is the *lender of cash*. The security is the collateral. The difference between the sale price and the repurchase price is the *interest* on the loan, expressed as the **repo rate**.

The same transaction has two names depending on which side you stand:

- From the **cash borrower's** viewpoint (the one who owns the bond and needs cash), it is a **repo** — "I repo out my bonds."
- From the **cash lender's** viewpoint (the one who has cash and wants collateral), it is a **reverse repo** — "I reverse in the bonds."

One trade, two labels. This mirror is the single most common source of confusion, and we will nail it down in Section 8.

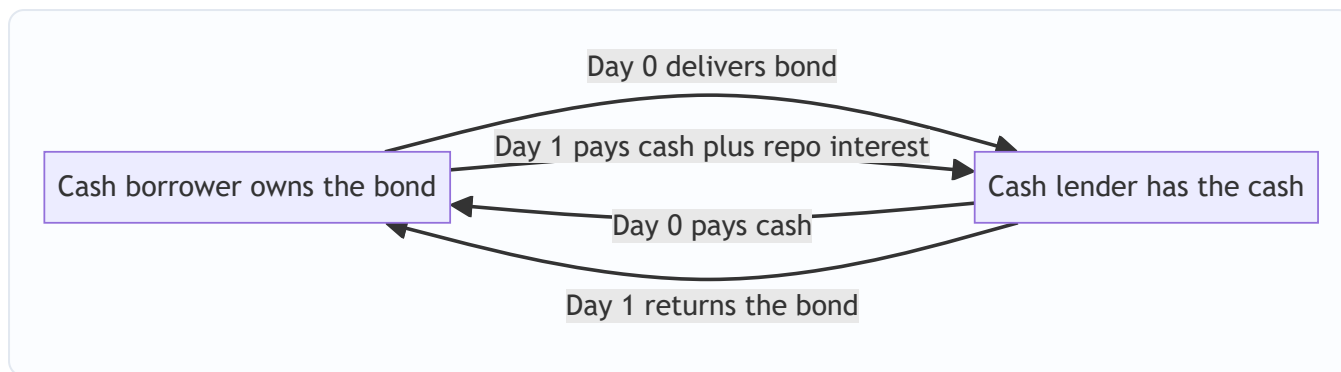


Figure 14.1 — The two legs of a repo: cash and collateral change hands on Day 0 and swap back on Day 1, with interest added on the return leg.

The elegance is that the cash lender holds the bond throughout the loan. If the borrower fails to repay, the lender simply keeps (and can sell) the bond. Credit risk is transformed into the far smaller risk that the *collateral loses value faster than you can liquidate it*. That residual risk is managed by the **haircut** (Section 4).

### 3. Why / How It Works

#### Why a secured loan is cheaper than an unsecured one

Interest compensates a lender for two things: the time value of money and the risk of not being repaid. Strip out the repayment risk with good collateral and the lender demands less. That is why the **repo rate sits below comparable unsecured rates** (e.g., below the uncollateralised interbank rate). The gap between an unsecured benchmark and the secured repo rate is a real-time market price of bank credit risk — it blew out spectacularly in 2008.

#### Why the "sale plus repurchase" structure rather than a plain pledged loan

Two reasons, one legal and one practical.

1. **Legal robustness on default.** In a plain pledged loan, if the borrower goes bankrupt, the lender may be caught in insolvency proceedings before it can seize and sell the collateral (an "automatic stay"). Because repo is legally a *sale* with an agreement to repurchase, in most major jurisdictions it enjoys **safe-harbour** treatment: the lender can immediately liquidate the collateral without waiting for the bankruptcy court. This legal certainty is *why* lenders accept such thin margins.
2. **Title transfer.** Ownership of the collateral actually passes to the cash lender for the life of the repo. The lender can, subject to the agreement, *re-use* (rehypothecate) that collateral — repo it out again to someone else. This re-use is what lets a single bond support a long chain of financing, multiplying the market's capacity to fund positions. It is powerful and, as 2008 showed, a channel through which stress propagates.

#### How the repo rate is determined

The repo rate is set by supply and demand for cash against collateral, anchored by the central bank's policy rate. Two forces pull on it:

- **General-collateral (GC) demand:** when the collateral is "any government bond," the rate reflects the general cost of overnight secured cash. This tracks the policy rate closely.



- **Specialness:** when *one particular* bond is in heavy demand — because many traders have shorted it and must borrow it to deliver — lenders of *that* specific bond can charge a premium. They charge the premium by accepting a *lower* repo rate on the cash (they are effectively being paid to lend the scarce bond). A bond "on special" therefore trades at a repo rate well below GC, sometimes near zero. This is the market's mechanism for pricing scarcity of a specific security. (We meet specialness again in Section 6.)

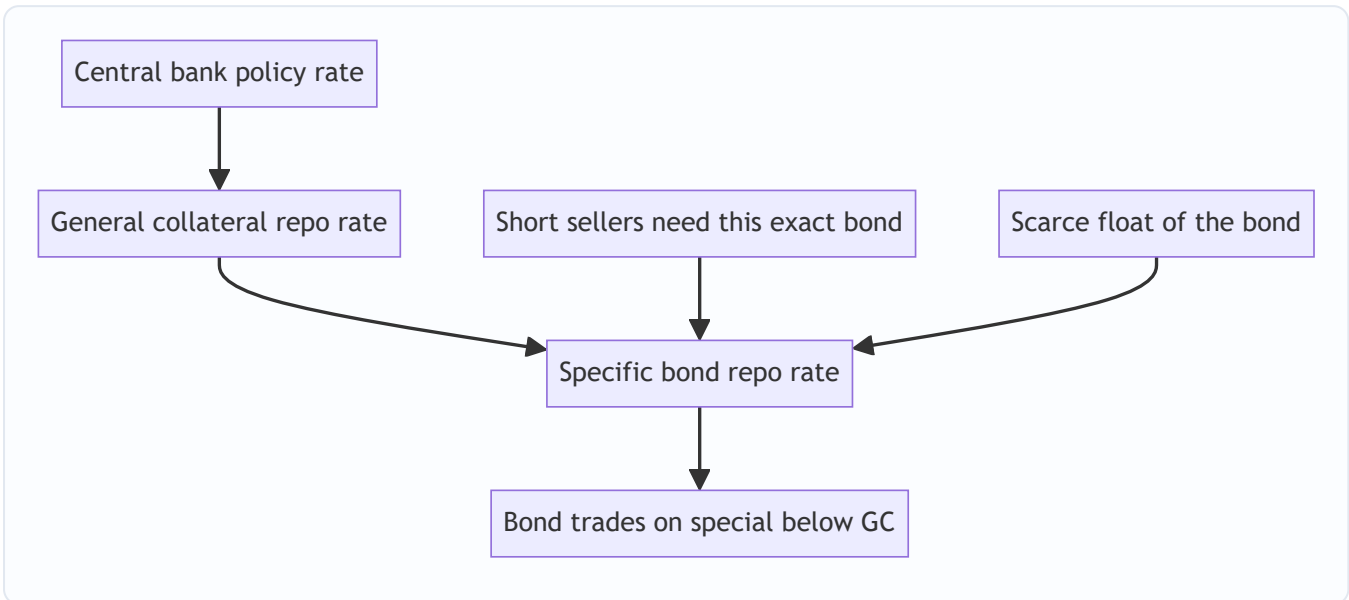


Figure 14.2 — The repo rate cascades from the policy rate to the general-collateral rate to the specific-bond rate, where scarcity can push a bond on special.

## 4. Full Content — Mechanics, Haircuts, Collateral, and the Math

### 4.1 The two prices and the repo interest

Let:

- $P_0$  = the cash the borrower receives on the near leg (the "purchase price" or start proceeds)
- $P_1$  = the cash the borrower repays on the far leg (the "repurchase price")
- $r$  = the repo rate (annualised, quoted as a money-market rate)
- $n$  = the term of the repo in days
- $B$  = the day-count basis (360 in USD/EUR money markets; **365 in India and GBP**)

The repurchase price is:

$$P_1 = P_0 \times \left(1 + r \times \frac{n}{B}\right)$$

The **repo interest** — the lender's earning, the borrower's cost — is:

$$\text{Interest} = P_1 - P_0 = P_0 \times r \times \frac{n}{B}$$

Note that repo interest uses **simple interest** on an actual/360 (or actual/365) money-market convention, *not* the bond-equivalent semi-annual compounding of Chapter 3.

### 4.2 The haircut (or initial margin)

The cash lender does not lend the full market value of the collateral. It lends *less*, keeping a cushion so that if the borrower defaults and the bond's price has fallen, the lender is still made whole after selling. That



cushion is the **haircut**.

If the collateral has market value  $M$  (including accrued interest — its *dirty* price) and the haircut is  $h$  (a fraction), then the cash advanced is:

$$P_0 = \frac{M}{1 + h} \quad \text{\textit{(margin-ratio convention)}}$$

or, under the simpler and very common **haircut-as-percentage-of-value** convention:

$$P_0 = M \times (1 - h)$$

The two conventions give slightly different numbers; documentation specifies which is used. Throughout this chapter, unless stated, we use  $P_0 = M \times (1 - h)$ .

Equivalently, market participants quote the **margin ratio** (also "initial margin"):

$$\text{Margin ratio} = \frac{\text{Collateral value}}{\text{Cash lent}} = \frac{M}{P_0}$$

A 2% haircut corresponds to a margin ratio of about 102%. The haircut protects the lender; a *larger* haircut means *less* cash for a given bond, i.e., worse funding for the borrower.

### What drives the size of the haircut?

| Factor                                                    | Effect on haircut                  | Intuition                                            |
|-----------------------------------------------------------|------------------------------------|------------------------------------------------------|
| Collateral price volatility                               | Higher volatility → larger haircut | Bigger price moves before you can liquidate          |
| Collateral liquidity                                      | Illiquid → larger haircut          | Harder to sell fast without moving the price         |
| Term of the repo                                          | Longer term → larger haircut       | More time for value to erode                         |
| Counterparty credit quality                               | Weaker borrower → larger haircut   | More likely to actually default                      |
| Correlation of collateral and borrower ("wrong-way risk") | Correlated → much larger haircut   | The collateral tanks exactly when the borrower fails |

Typical haircuts: near **0.5–2%** for government bonds, **3–8%** for investment-grade corporates, and much larger — sometimes 15%+ — for equities or lower-rated structured collateral.

## 4.3 Margining and marking to market

For repos longer than a day, the collateral is **marked to market** daily. If the bond's price falls, the lender is now under-collateralised and issues a **margin call**: the borrower must post additional collateral or cash (variation margin) to restore the agreed margin ratio. If the price rises, the borrower can call excess collateral back. This daily re-margining, governed by the **Global Master Repurchase Agreement (GMRA)** internationally (or the domestic master agreement), is what keeps the lender's exposure near zero throughout the life of the trade.

## 4.4 Collateral: general vs specific

- **General Collateral (GC) repo**: the lender will accept any bond from an agreed *basket* (typically all on-the-run and off-the-run government securities). The lender cares about the *cash rate*, not which bond it gets. GC is a pure money-market instrument for financing.

- **Specific / Special repo:** the lender wants *one particular* security (usually because it needs to deliver that bond into a short sale or a settlement fail). Here the trade is about *sourcing the bond*, and the cash rate becomes secondary — it drops below GC by the "specialness spread."

## 4.5 The full cost-of-carry picture

When you finance a bond in repo, your net carry over the holding period is:

$$Net\ carry = \underbrace{\text{Coupon accrued on the bond}}_{\text{you earn}} - \underbrace{\text{Repo interest paid}}_{\text{you pay}}$$

If the bond's coupon accrual exceeds the repo cost, you have **positive carry** — you are paid to hold the position. This is normal in an upward-sloping yield curve where long-bond coupons exceed short repo rates. Positive carry is a major reason leveraged bond positions are attractive; it is also why carry trades unwind violently when repo rates spike above coupon income.

## 4.6 Types of repo by structure

| Structure                           | Who holds the collateral                              | Key feature                                                                 |
|-------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Bilateral (deliverable) repo</b> | Cash lender receives the bond directly                | Simplest; used for specials and directional funding                         |
| <b>Hold-in-custody (HIC) repo</b>   | Borrower keeps the bond in a segregated account       | Risky for lender (collateral not delivered); rare post-crisis               |
| <b>Tri-party repo</b>               | A third-party agent bank holds and manages collateral | Agent handles valuation, margining, substitution; dominant for GC financing |
| <b>Term repo</b>                    | As above                                              | Fixed maturity beyond overnight (1 week to several months)                  |
| <b>Open repo</b>                    | As above                                              | No fixed end date; rolls daily until either side terminates; rate can reset |

**Tri-party repo** deserves emphasis: an agent bank (in the US, the clearing banks; in India, the CCIL platform) sits in the middle, holding collateral, valuing it, applying haircuts, and swapping collateral in and out as needed. This industrialises GC financing and removes the operational burden of delivery from the two principals.

## 5. Worked Examples

### Example 1 — A plain overnight government repo (the base case)

A dealer owns ₹100,00,00,000 (₹100 crore) face value of a government bond. Its *dirty* market price is 101.50 (per 100 face), so the market value is:

$$M = 100,00,00,000 \times \frac{101.50}{100} = ₹101,50,00,000$$

The lender applies a **1% haircut** and the overnight repo rate is **6.50%** (India, actual/365,  $n = 1$  day).

**Cash advanced (near leg):**

$$P_0 = M \times (1 - h) = 101,50,00,000 \times (1 - 0.01) = 101,50,00,000 \times 0.99 = ₹100,48,50,000$$

### Repo interest for 1 day:

$$\begin{aligned}\text{Interest} &= P_0 \times r \times \frac{n}{B} = 100,48,50,000 \times 0.065 \times \frac{1}{365} \\ &= 100,48,50,000 \times 0.065 \times 0.00273973 = ₹17,894.52\end{aligned}$$

### Repurchase price (far leg):

$$P_1 = P_0 + \text{Interest} = 100,48,50,000 + 17,894.52 = ₹100,48,67,894.52$$

**Reconciliation / self-check.** Let's verify the interest independently via the multiplicative form:

$$\begin{aligned}P_1 &= P_0 \times \left(1 + 0.065 \times \frac{1}{365}\right) = 100,48,50,000 \times 1.000178082 = \\ &= 100,48,67,894.5 \quad \checkmark\end{aligned}$$

The two methods agree to the rupee. Note the lender earns ₹17,894.52 for one night, fully secured, with a ₹1.015 crore collateral cushion (the 1% haircut) protecting it. The *lender's* return on cash outlaid is exactly 6.50% annualised:  $\frac{17,894.52}{100,48,50,000} \times 365 = 0.0650$ . ✓

### Example 2 — A term repo with a margin call

Same dealer, but now a **14-day term repo** at **6.75%**, collateral dirty value ₹101,50,00,000, haircut **2%**, margin ratio maintained at 102% (i.e., collateral must always be  $\geq 1.02 \times$  cash lent).

#### Near leg cash:

$$P_0 = 101,50,00,000 \times (1 - 0.02) = 101,50,00,000 \times 0.98 = ₹99,47,00,000$$

#### Repo interest over 14 days:

$$\begin{aligned}\text{Interest} &= 99,47,00,000 \times 0.0675 \times \frac{14}{365} = 99,47,00,000 \times 0.0675 \\ &\times 0.0383562\end{aligned}$$

$$= 99,47,00,000 \times 0.00258904 = ₹2,57,533.6$$

#### Repurchase price:

$$P_1 = 99,47,00,000 + 2,57,533.6 = ₹99,72,57,533.6$$

**Now a margin call on Day 5.** The bond's dirty price falls from 101.50 to 100.20. New collateral value:

$$M_{\text{new}} = 100,00,00,000 \times \frac{100.20}{100} = ₹100,20,00,000$$

Required collateral to keep the 102% margin ratio against the cash lent ( $P_0 = ₹99,47,00,000$ ):

$$\text{Required} = 1.02 \times 99,47,00,000 = ₹101,45,94,000$$

#### Shortfall (variation margin owed by the borrower):

$$101,45,94,000 - 100,20,00,000 = ₹1,25,94,000$$

So the dealer must post **₹1.2594 crore** of additional collateral (or cash) to the lender to cure the margin deficit.

**Reconciliation.** Sanity-check the margin logic: before the price drop the collateral was ₹101.50 cr against ₹99.47 cr cash, a ratio of  $101.50/99.47 = 1.0204$ , comfortably above 102%. After the drop, ₹100.20 cr / ₹99.47 cr = 1.00734, which is *below* 102% — hence a call is warranted, and topping up by ₹1.2594 cr restores  $(100.20 + 1.2594)/99.47 = 101.4594/99.47 = 1.0200$  exactly. ✓ The haircut plus daily margining together keep the lender's exposure near zero even as the collateral loses 1.28% of its value.

### Example 3 — Financing a levered position and computing net carry

A dealer buys ₹50 crore face of a bond carrying a **7.20% annual coupon** (paid semi-annually, but we accrue linearly for a 30-day hold), dirty price 100.00, and finances it overnight-rolled in GC repo at an average **6.40% for 30 days**, haircut **1%**.

**Cash the dealer must fund itself (its own capital / equity in the trade):** only the haircut portion. Market value = ₹50,00,00,000. Repo advances  $P_0 = 50,00,00,000 \times 0.99 = ₹49,50,00,000$ ; the dealer funds the remaining **₹50,00,000** from its own capital.

**Coupon earned over 30 days (accrued, actual/365):**

$$\text{Coupon} = 50,00,00,000 \times 0.072 \times \frac{30}{365} = 50,00,00,000 \times 0.072 \times 0.0821918 = ₹2,95,890.4$$

**Repo cost over 30 days on the borrowed ₹49.5 cr:**

$$\text{Repo cost} = 49,50,00,000 \times 0.064 \times \frac{30}{365} = 49,50,00,000 \times 0.064 \times 0.0821918 = ₹2,60,432.9$$

**Net carry over 30 days:**

$$\text{Net carry} = 2,95,890.4 - 2,60,432.9 = ₹35,457.5 \text{ ; (positive)}$$

**Return on the dealer's own capital (₹50 lakh) from carry alone, annualised:**

$$\frac{35,457.5}{50,00,000} \times \frac{365}{30} = 0.00709 \times 12.1667 = 0.0863 = 8.63\%$$

**Reconciliation and the leverage lesson.** The bond itself yields ~7.20% but the dealer earns **8.63%** on *its own capital* purely from carry — because it financed 99% of the position at 6.40% and pocketed the 0.80% coupon-minus-repo spread on a base 100× its equity. That is leverage: a 0.80% gross carry spread magnified into 8.63% on equity. Now flip the repo rate. If overnight repo spiked to **7.60%** (above the 7.20% coupon), the repo cost becomes  $49,50,00,000 \times 0.076 \times 0.0821918 = ₹3,09,264$ , net carry turns to  $2,95,890 - 3,09,264 = -₹13,374$  — **negative carry**, bleeding the position daily. This is exactly the dynamic that forces leveraged holders to dump bonds when funding markets tighten, and it is why the repo rate is not a sideshow but a driver of bond prices themselves. ✓

## 6. Connections

**To duration and directional views (Chapters 3–5).** Repo is *how* a view on rates gets expressed at scale.

Buying a bond you expect to rally, financed in repo, is the levered long; the short side (below) needs repo to source the bond.

**To short-selling.** You cannot sell a bond you do not own unless you can *deliver* it. You borrow the bond via a **reverse repo** (you lend cash, receive the specific bond), then sell it in the market. If everyone shorts the same bond, that bond goes **on special** — its repo rate collapses below GC, and the cost of maintaining the short (the specialness spread) rises. Specialness is thus the funding market's price signal for crowded short positions. It also drives the **cheapest-to-deliver** dynamics in bond futures (Chapter 15): the CTD bond often trades special precisely because shorts in the future need to deliver it.

**To the yield curve and money markets (Chapters 2, 13).** The overnight repo rate is the short-anchor of the money-market curve. Term repo rates (1-week, 1-month, 3-month) trace out a secured money-market curve that, alongside the OIS curve, defines the risk-free short end. Since the post-LIBOR transition, **SOFR**

(Secured Overnight Financing Rate) in the US is literally a *volume-weighted median of overnight Treasury repo rates* — the repo market is the risk-free benchmark now.

**To central-bank policy (Section 7).** Repo is the instrument through which the policy rate is imposed on the system.

**To credit risk (Chapter 10).** Repo converts counterparty credit risk into collateral (market + liquidity) risk. Wrong-way risk — collateral correlated with the counterparty — is the dangerous residue and a lesson from 2008's mortgage repo.

**To bond futures and swaps (Chapters 15–16).** The repo rate is an input to the futures fair-value (cost-of-carry) and defines the financing leg implicit in basis trades.

## 7. Key Terms

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- **Repo (repurchase agreement):** sale of a security with an agreement to repurchase it at a set price and date; economically a secured loan where the security seller *borrow cash*.
- **Reverse repo:** the same trade from the cash lender's side — you *lend cash and receive collateral*. One transaction, two names.
- **Repo rate:** the interest rate on the cash leg, i.e., the annualised return implied by  $\$(P_1 - P_0)/P_0$ .
- **Haircut / initial margin:** the percentage by which collateral value exceeds cash lent; the lender's protective cushion.
- **Margin ratio:** collateral value ÷ cash lent (e.g., 102%).
- **General Collateral (GC):** repo against any bond from an agreed basket; a pure financing rate.
- **Special / specialness:** state where a specific bond is scarce, so its repo rate falls below GC; the gap is the specialness spread.
- **Tri-party repo:** repo where an agent bank holds and administers collateral for both sides.
- **Rehypothecation / re-use:** the cash lender's right to re-pledge received collateral, enabling collateral chains.
- **GMRA:** Global Master Repurchase Agreement — the standard legal contract governing cross-border repo.
- **LAF (Liquidity Adjustment Facility):** the RBI's framework for daily repo/reverse-repo operations to manage system liquidity.
- **Repo rate (policy, India):** the RBI's benchmark rate at which it lends to banks against government securities — the anchor of the entire rate structure.
- **SDF (Standing Deposit Facility):** the RBI's uncollateralised overnight absorption facility, now the floor of the corridor.
- **MSF (Marginal Standing Facility):** the RBI's overnight lending facility above the repo rate — the ceiling of the corridor.
- **SOFR:** Secured Overnight Financing Rate — the US risk-free benchmark, built from overnight Treasury repo transactions.

## 8. Common Confusions

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**"Repo vs reverse repo — which am I doing?"** Anchor on **cash**. If you *give* cash and *take* the bond, you are doing a **reverse repo** (you are the cash lender). If you *take* cash and *give* the bond, you are doing a **repo**

(you are the cash borrower). A single trade is a repo to one party and a reverse repo to the other. The label depends entirely on the perspective.

**RBI terminology is the *opposite* of the market's perspective.** When the RBI conducts a **repo** operation under the LAF, *it lends cash to banks* against collateral — from the RBI's standpoint it is receiving securities and giving cash, which by the strict definition is a reverse repo, but the RBI names it "repo" because it is the banks (the counterparties) who are repo-ing out their securities. Conversely, the RBI's "reverse repo" absorbs liquidity — banks park cash with the RBI. **Rule of thumb for India:** *RBI repo = RBI injects cash (rate you hear about in policy). RBI reverse repo / SDF = RBI absorbs cash.* Do not try to force it into the market-side definition; learn the central-bank convention separately.

**Haircut direction.** A *bigger* haircut is *worse* for the borrower (less cash per bond) and *safer* for the lender. Students often invert this. Higher haircut = more protection = tighter funding.

**Repo rate is not the bond's yield.** The repo rate is a money-market financing rate (simple interest, actual/360 or 365). The bond's yield to maturity is a separate, compounded, longer-horizon measure. The interplay of the two produces carry, but they are different numbers on different conventions.

**"Special" means a *low* repo rate, not a *high* one.** Beginners assume a hotly-demanded bond commands a *high* rate. The opposite: to obtain the scarce bond you must *accept less interest on your cash*, so the repo rate goes *down*, sometimes to zero or negative. Demand for the *bond* depresses the *cash* rate.

**Repo interest accrues on the cash, not the collateral face.**  $\$P_1 - P_0\$$  is computed on  $\$P_0\$$  (the cash advanced), not on the bond's face value or market value. The collateral only determines *how much* cash ( $\$P_0\$$ ) via the haircut.

**Ownership actually transfers.** Unlike a pledge, in repo the collateral's legal title passes to the lender for the term. This is why the lender can re-use it and why default liquidation is fast — but it is also why collateral chains can amplify systemic stress.

## 9. Recap

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A repurchase agreement is a **collateralised loan structured as a sale-and-repurchase pair**. The party who owns the bond and needs cash *sells now, buys back later* — that is a **repo**; the party supplying the cash and receiving collateral is doing a **reverse repo**. The interest is the difference between the repurchase price  $\$P_1\$$  and the sale price  $\$P_0\$$ , set by the **repo rate** on a simple money-market basis:  $\$P_1 = P_0(1 + r \cdot n/B)\$$ .

The lender protects itself two ways: a **haircut** (lending less than the collateral is worth,  $\$P_0 = M(1-h)\$$ ) and **daily marking-to-market with margin calls**. These together drive the lender's credit exposure near zero, which is why repo rates sit *below* unsecured rates. Haircut size scales with collateral volatility, illiquidity, term, and counterparty weakness.

Collateral comes in two flavours: **General Collateral** (any bond in a basket — pure financing, rate tracks the policy rate) and **special** (one scarce bond — its repo rate falls below GC, pricing the scarcity created by short-sellers). Structurally, repo runs **bilaterally** or through **tri-party agents**, and can be **overnight, term, or open**.

Economically, repo lets dealers **finance inventory with high leverage**, capturing the **carry** between coupon income and repo cost — positive when the curve slopes up, negative (and dangerous) when funding spikes. Repo also enables **short-selling** (borrow the bond via reverse repo, sell it) and underpins **benchmark rates**

like SOFR. Central banks — the **RBI via its LAF** with the repo rate, SDF, and MSF forming a corridor — use repo as the primary lever to inject or drain overnight liquidity and steer the entire rate structure. When repo works, no one notices; when it jams, the whole market feels it.

## 10. Quick-Reference / Interview Points

**One-line definition to have ready:** "A repo is a secured overnight (or term) loan structured as a sale and repurchase of a bond; the security seller borrows cash, the buyer lends cash, and the repo rate is the interest."

### Core formulas:

$$P_1 = P_0 \left( 1 + r \frac{n}{B} \right) \quad \text{Interest} = P_0 \cdot r \frac{n}{B} \quad P_0 = M(1-h) \\ \text{Margin ratio} = \frac{M}{P_0}$$

Day-count  $B$ : **365 in India/GBP, 360 in USD/EUR** money markets.

**The perspective mirror (say this cleanly):** Give cash + take bond = reverse repo (cash lender). Take cash + give bond = repo (cash borrower). Same trade, opposite labels.

### RBI / LAF corridor (India, know cold):

| Facility                                | Direction                       | Role in corridor       |
|-----------------------------------------|---------------------------------|------------------------|
| <b>MSF</b> (Marginal Standing Facility) | RBI lends overnight, above repo | Ceiling                |
| <b>Repo rate</b> (LAF)                  | RBI lends against G-secs        | Policy rate / centre   |
| <b>SDF</b> (Standing Deposit Facility)  | RBI absorbs, uncollateralised   | Floor (since Apr 2022) |

The **weighted average call rate (WACR)** is the operating target the RBI keeps near the repo rate; the corridor is typically  $\pm 25$  bps (MSF = repo + 25 bps, SDF = repo – 25 bps).

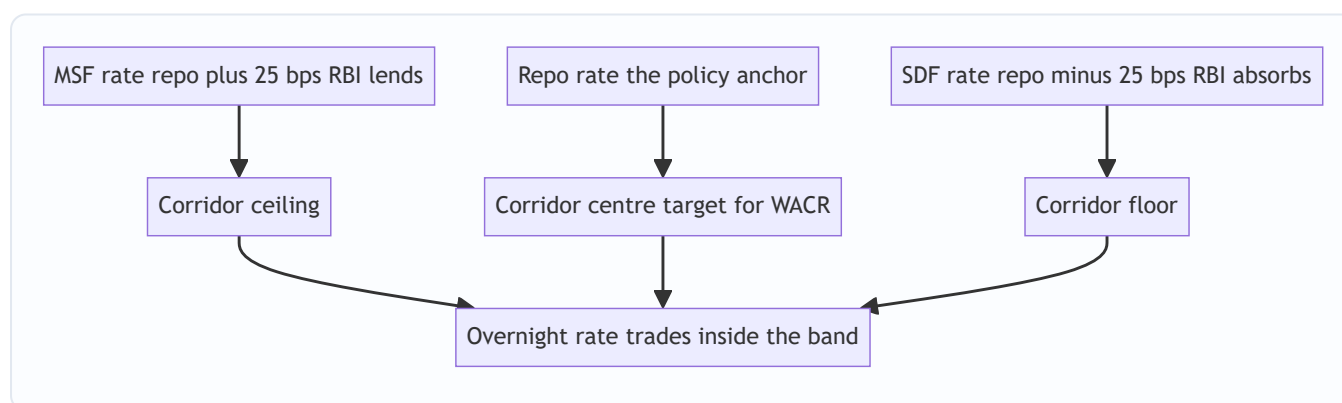


Figure 14.3 — The RBI's LAF corridor: MSF caps the overnight rate, SDF floors it, and the repo rate anchors the middle where the RBI steers the call rate.

### Punchy talking points interviewers love:

- Repo is the plumbing: dealer inventories, leverage, shorting, and money funds all sit on it. The 2019 US repo spike and March 2020 dash-for-cash showed what happens when it clogs.
- **SOFR is literally built from Treasury repo transactions** — the risk-free curve is now a repo curve. Know that LIBOR (unsecured, survey-based) was replaced by SOFR (secured, transaction-based) after the LIBOR scandal.

- **Specialness** prices crowded shorts: a bond on special has a repo rate *below* GC, and that spread is the cost of the short. Ties directly to **cheapest-to-deliver** in futures.
- **Carry = coupon accrual – repo cost.** Positive in an upward-sloping curve; leveraged carry trades unwind when repo spikes above coupon income.
- **Haircuts are procyclical:** they widen in stress exactly when funding is scarcest, forcing deleveraging — a documented amplifier of the 2008 crisis (the "haircut spiral").
- The genius of repo is **legal:** safe-harbour / title-transfer treatment lets the lender liquidate collateral instantly on default, which is *why* it can charge so little.
- **Wrong-way risk** — collateral whose value is correlated with the borrower's default — is the one place the "risk-free" story breaks; that was subprime mortgage repo in 2008.

**Quick numerical instinct:** overnight GC repo on ₹100 cr at 6.5% earns roughly  $₹100 \text{ cr} \times 0.065 / 365 \approx ₹17,800$  per night. Term and haircut scale from there.



# Q&A — Repo and Funding Markets

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Practice bank for Chapter 14. Work each question before reading the answer. Numerical answers use India money-market conventions (simple interest, actual/365) unless stated otherwise.

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## Section A — Concept Checks

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### A1. In one sentence, what is a repurchase agreement, and who is borrowing what?

A repo is the sale of a security today combined with a binding agreement to buy it back at a fixed price on a future date; economically it is a secured loan in which the *seller of the security borrows cash* and the *buyer of the security lends cash*, with the security serving as collateral.

### A2. Why does the same trade have two names — "repo" and "reverse repo"?

Because a single transaction looks different from each side. The party who owns the bond and needs cash is doing a **repo** ("I repo out my bonds"). The party who has cash and wants collateral is doing a **reverse repo** ("I reverse in the bonds"). One trade, two labels; the name depends purely on perspective. Anchor on cash: give cash + take bond = reverse repo; take cash + give bond = repo.

### A3. Why does a repo rate sit *below* the comparable unsecured interbank rate?

Interest compensates a lender for the time value of money and for repayment risk. Good collateral strips out most of the repayment risk, so the lender demands less. The gap between the unsecured benchmark and the secured repo rate is effectively a real-time market price of bank credit risk — it widened dramatically in 2008.

### A4. What is a haircut, and in which direction does a larger haircut help or hurt each party?

The haircut is the cushion by which collateral value exceeds cash lent — the lender advances less than the bond is worth. A *larger* haircut is *safer for the lender* (more protection if the collateral falls in value before liquidation) and *worse for the borrower* (less cash per bond, i.e., tighter funding). Higher haircut = more protection = tighter funding.

### A5. Name the five factors that push haircuts higher and give the intuition for one.

(1) Higher collateral price volatility, (2) lower collateral liquidity, (3) longer repo term, (4) weaker counterparty credit, and (5) wrong-way risk (collateral correlated with the borrower). Intuition for wrong-way risk: the collateral loses value at exactly the moment the borrower defaults, so the lender's protection evaporates when it is needed most — hence a much larger haircut.

### A6. What legally distinguishes a repo from an ordinary pledged loan, and why does that distinction matter on default?

A repo is structured as a true *sale* with an agreement to repurchase, so ownership (title) of the collateral actually transfers to the cash lender. In most major jurisdictions this earns **safe-harbour** treatment: on the borrower's bankruptcy the lender can immediately liquidate the collateral without waiting behind an automatic stay. This legal certainty is precisely why lenders accept such thin margins.

### A7. Distinguish General Collateral (GC) repo from special repo.

In **GC repo** the lender accepts any bond from an agreed basket and cares only about the *cash rate*; it is pure financing and the rate tracks the policy rate. In **special repo** the lender wants one *specific* scarce bond

(usually to deliver into a short sale or fail), so the trade is about sourcing the bond and the cash rate drops below GC by the "specialness spread."

**A8. Why does a bond "on special" trade at a lower repo rate, not a higher one?**

Demand is for the *bond*, not the cash. To obtain the scarce security, the cash lender must accept less interest on the cash it lends — they are effectively paying to borrow the scarce bond. So heavy demand for a specific bond depresses its repo *cash* rate, sometimes to near zero or negative.

**A9. Write the net-carry relationship for a repo-financed bond and state when carry is positive.**

Net carry = coupon accrued on the bond – repo interest paid. Carry is **positive** when coupon accrual exceeds repo cost — normal in an upward-sloping curve where long-bond coupons exceed short repo rates. It turns negative and dangerous when funding spikes above coupon income.

**A10. What is rehypothecation, and why is it both powerful and dangerous?**

Because title transfers to the cash lender, that lender can (subject to the agreement) re-use — rehypothecate — the collateral by repo-ing it out again to a third party. This lets a single bond support a long chain of financing, multiplying market funding capacity. It is dangerous because those same collateral chains transmit stress rapidly through the system, as 2008 demonstrated.

**A11. In tri-party repo, what does the agent bank do?**

A third-party agent bank sits between the two principals and holds, values, and manages the collateral — applying haircuts, marking to market, issuing margin calls, and swapping (substituting) collateral in and out. It industrialises GC financing and removes the operational burden of delivery from both principals (CCIL performs this role in India).

**A12. Why is SOFR relevant to a chapter on repo?**

SOFR (Secured Overnight Financing Rate), the US risk-free benchmark that replaced LIBOR, is literally a volume-weighted median of overnight Treasury repo transaction rates. The risk-free short-end benchmark is now a repo curve — LIBOR (unsecured, survey-based) was replaced by SOFR (secured, transaction-based) after the LIBOR scandal.

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## Section B — Numerical / Applied (full solutions)

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**B1. Plain overnight government repo.** A dealer owns ₹100 crore face of a G-sec; dirty price 101.50. Lender applies a 1% haircut; overnight repo rate 6.50% (actual/365,  $n = 1$ ). Find the cash advanced, the one-day interest, and the repurchase price.

*Solution.* Market value  $M = 100,00,00,000 \times 101.50/100 = ₹101,50,00,000$ . Cash advanced  $P_0 = M(1 - h) = 101,50,00,000 \times 0.99 = ₹100,48,50,000$ . Interest  $= P_0 \times r \times n/B = 100,48,50,000 \times 0.065 \times (1/365) = ₹17,894.52$ . Repurchase price  $P_1 = P_0 + \text{Interest} = ₹100,48,67,894.52$ .

*Self-check:* the lender's return on cash  $= 17,894.52 / 100,48,50,000 \times 365 = 0.0650 = 6.50\%$  annualised. ✓

**B2. Term repo.** Same collateral (dirty value ₹101,50,00,000), now a 14-day term repo at 6.75%, haircut 2%. Find near-leg cash, 14-day interest, and repurchase price.

*Solution.*  $P_0 = 101,50,00,000 \times 0.98 = ₹99,47,00,000$ . Interest  $= 99,47,00,000 \times 0.0675 \times (14/365) = 99,47,00,000 \times 0.00258904 = ₹2,57,533.60$ .  $P_1 = 99,47,00,000 + 2,57,533.60 = ₹99,72,57,533.60$ .

**B3. Margin call.** Continue B2 with a maintained margin ratio of 102% (collateral  $\geq 1.02 \times$  cash lent). On Day 5 the bond's dirty price falls from 101.50 to 100.20. How much variation margin must the borrower post?

*Solution.* New collateral value =  $100,00,00,000 \times 100.20/100 = ₹100,20,00,000$ . Required collateral =  $1.02 \times 99,47,00,000 = ₹101,45,94,000$ . Shortfall =  $101,45,94,000 - 100,20,00,000 = ₹1,25,94,000$  (₹1.2594 crore).

*Self-check:* after topping up,  $(100.20 + 1.2594) \text{ cr} / 99.47 \text{ cr} = 101.4594 / 99.47 = 1.0200$  exactly — the 102% ratio is restored. ✓

**B4. Net carry and return on equity.** A dealer buys ₹50 crore face of a 7.20% annual-coupon bond, dirty price 100.00, financed overnight-rolled in GC repo at an average 6.40% for 30 days, haircut 1%. Compute the dealer's own capital in the trade, coupon accrued, repo cost, net carry, and annualised return on equity.

*Solution.*  $M = ₹50,00,00,000$ .  $P_0 = 50,00,00,000 \times 0.99 = ₹49,50,00,000$ , so own capital =  $M - P_0 = ₹50,00,000$ . Coupon (actual/365, 30 days) =  $50,00,00,000 \times 0.072 \times 30/365 = ₹2,95,890.41$ . Repo cost =  $49,50,00,000 \times 0.064 \times 30/365 = ₹2,60,432.88$ . Net carry =  $2,95,890.41 - 2,60,432.88 = ₹35,457.53$  (positive). Return on equity, annualised =  $(35,457.53 / 50,00,000) \times (365/30) = 0.00709 \times 12.1667 = 8.63\%$ .

*Lesson:* a bond yielding ~7.20% delivers 8.63% on equity because 99% of the position was financed at 6.40% — a 0.80% gross carry spread magnified by ~100× leverage.

**B5. Negative carry stress.** In B4, suppose overnight repo spikes to 7.60% (above the 7.20% coupon). Recompute net carry.

*Solution.* Repo cost =  $49,50,00,000 \times 0.076 \times 30/365 = ₹3,09,264.66$ . Net carry =  $2,95,890.41 - 3,09,264.66 = -₹13,374.25$  (negative). The position now bleeds cash daily — exactly the dynamic that forces leveraged holders to dump bonds when funding tightens, transmitting repo stress into bond prices.

**B6. USD day-count contrast.** A US dealer does a \$200 million overnight Treasury repo at 5.30% (actual/360, haircut 0.5%). Find the cash advanced and one-day interest. Assume dirty value \$200m.

*Solution.*  $P_0 = 200,000,000 \times (1 - 0.005) = \$199,000,000$ . Interest =  $199,000,000 \times 0.0530 \times (1/360) = 199,000,000 \times 0.00014722 = \$29,297.22$ . Note the /360 basis: the same rate on the same day count earns slightly more interest per day than /365 because the denominator is smaller.

**B7. Back out the haircut.** A lender advances ₹98,00,00,000 against collateral with dirty value ₹100,00,00,000. What is the haircut (percentage-of-value convention) and the implied margin ratio?

*Solution.* Haircut  $h$ :  $P_0 = M(1 - h) \rightarrow 98 = 100(1 - h) \rightarrow h = 1 - 0.98 = 2\%$ . Margin ratio =  $M / P_0 = 100 / 98 = 1.0204 = 102.04\%$ .

**B8. Specialness spread as a cost.** GC repo is 6.50%. A short-seller must borrow a specific bond that is trading special at a 4.00% repo rate for 7 days on ₹20 crore of cash. What is the specialness spread in bps, and what does it cost the short over the week versus funding at GC?

*Solution.* Specialness spread =  $6.50\% - 4.00\% = 2.50\% = 250 \text{ bps}$ . The short lends ₹20 cr of cash to source the bond and earns only 4.00% instead of the 6.50% GC rate, forgoing 2.50%. Cost over 7 days =  $20,00,00,000 \times 0.025 \times 7/365 = ₹95,890.41$ . This forgone interest is the running cost of maintaining the short position.

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## Section C — Interview-Style (model answers)

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**C1. "Walk me through what actually happens, leg by leg, in an overnight repo."**

On Day 0 the cash borrower delivers the bond to the cash lender and simultaneously receives cash equal to the collateral's dirty market value less the haircut. The lender now holds legal title to the bond. On Day 1 the borrower pays back the cash plus repo interest (the repurchase price  $P_1 = P_0(1 + r \cdot n/B)$ ) and the lender returns the same bond. Throughout the loan the lender holds the collateral, so if the borrower fails to repay, the lender simply keeps and sells the bond. Credit risk has been transformed into the much smaller risk that the collateral loses value faster than it can be liquidated — and even that residual is managed by the haircut and daily margining.

## **C2. "The RBI's use of 'repo' seems backwards to me. Explain."**

The market defines repo from the cash borrower's side — the party giving up the bond and taking cash. The RBI, by contrast, names its operation from the *system's* function. Under the LAF, an RBI **repo** operation *injects* cash: the RBI lends cash to banks against G-secs. Strictly, the RBI is receiving securities and giving cash, which by the market definition is a reverse repo — but it is called "repo" because the banks (the counterparties) are the ones repo-ing out their securities. The RBI's "reverse repo" (and now the SDF) *absorbs* cash — banks park funds with the RBI. The clean rule for India: RBI repo = RBI injects cash (the policy rate you hear about); RBI reverse repo / SDF = RBI absorbs cash. Learn the central-bank convention separately rather than forcing it into the market-side definition.

## **C3. "Why do people say repo is the plumbing of the financial system?"**

Almost every fixed-income position is financed, and repo is the cheapest financing because it is secured. Dealer inventories, leverage, short-selling, money funds, and central-bank policy transmission all sit on top of the repo market. When it functions, no one notices. When it seizes — 2008, the September 2019 US repo spike, the March 2020 dash-for-cash — funding evaporates, leveraged holders are forced to sell, and stress propagates upward into every market above it.

## **C4. "What are haircut spirals and why are they dangerous?"**

Haircuts are procyclical: they widen in stress precisely when funding is scarcest. As collateral values fall and volatility rises, lenders demand bigger haircuts, so borrowers get less cash per bond and must sell assets to cover the shortfall. Those sales push prices down further, raising volatility and haircuts again — a self-reinforcing "haircut spiral." This feedback loop was a documented amplifier of the 2008 crisis: deleveraging fed on itself through the collateral market.

## **C5. "How does repo enable short-selling, and what is specialness telling you?"**

You cannot sell a bond you do not own unless you can deliver it. So the short-seller does a reverse repo — lends cash and receives the specific bond — then sells that bond in the market. If many traders short the same bond, demand to borrow it surges and the bond goes "on special": its repo rate falls below GC. The specialness spread (GC minus the special rate) is the running cost of holding the short and the market's price signal for crowded short positions. It also drives cheapest-to-deliver dynamics in bond futures, where the CTD bond often trades special because shorts in the future need to deliver it.

## **C6. "Give me the intuition for why leverage in a repo carry trade cuts both ways."**

The dealer funds nearly the whole position with borrowed cash and puts up only the haircut as equity, so a small gross carry spread — coupon accrual minus repo cost — is magnified into a large return on that thin equity (a 0.80% spread on a 100×-levered position became 8.63% on capital in my example). The same leverage works in reverse: if repo rises above the coupon, carry turns negative and the position bleeds daily on the full financed amount while equity is tiny. That asymmetry is why leveraged carry trades unwind violently when funding tightens.

### C7. "Where does the 'risk-free' story of repo actually break down?"

Wrong-way risk. Repo converts counterparty credit risk into collateral (market plus liquidity) risk, and that transformation is clean only when the collateral's value is independent of the borrower's solvency. When the collateral is correlated with the borrower — its value falls exactly when the borrower defaults — the lender's protection collapses at the worst moment. The canonical case is subprime mortgage repo in 2008: dealers financing mortgage collateral that was deteriorating for the same reasons the dealers were failing. That is why wrong-way risk commands a much larger haircut and is the one place the "risk-free" framing genuinely breaks.

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## Section D — MCQs (with reasoning)

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**D1. In a repo, the party that delivers the bond on the near leg is:** A) lending cash B) borrowing cash C) earning the specialness spread D) the tri-party agent

**Answer: B.** The bond-deliverer is the cash *borrower* (doing a repo); they give the bond and take cash. The bond-receiver lends the cash (reverse repo). The agent (D) never a principal.

**D2. A bond trading "on special" has a repo rate that is:** A) above GC B) equal to GC C) below GC D) equal to the coupon

**Answer: C.** Heavy demand for the specific bond forces the cash lender to accept *less* interest on the cash to obtain the scarce security, so the special rate sits below GC. Assuming "special = high rate" is the classic trap.

**D3. All else equal, increasing the haircut on a repo:** A) gives the borrower more cash B) reduces the lender's protection C) reduces the cash advanced to the borrower D) raises the repo interest rate

**Answer: C.** With  $P_0 = M(1 - h)$ , a larger  $h$  means less cash for a given collateral value — worse funding for the borrower and more protection for the lender. It does not by itself change the quoted repo rate.

**D4. Repo interest is calculated on:** A) the bond's face value B) the collateral's market value C) the cash advanced ( $P_0$ ) D) the repurchase price ( $P_1$ )

**Answer: C.** Interest =  $P_1 - P_0 = P_0 \cdot r \cdot n/B$ , computed on the cash advanced. The collateral value only determines *how much* cash via the haircut; it is not the interest base.

**D5. Which statement about an RBI LAF "repo" operation is correct?** A) It absorbs liquidity from banks B) It injects cash into the banking system C) It is the corridor floor D) It is uncollateralised

**Answer: B.** An RBI repo injects cash (RBI lends to banks against G-secs); it is the policy rate at the corridor centre and is collateralised. The SDF (uncollateralised, absorbs cash) is the floor.

**D6. SOFR is best described as:** A) an unsecured survey-based rate like LIBOR B) the RBI's overnight absorption rate C) a volume-weighted median of overnight Treasury repo rates D) a semi-annually compounded bond yield

**Answer: C.** SOFR is secured and transaction-based, built from overnight Treasury repo, and replaced the unsecured survey-based LIBOR after the LIBOR scandal.

**D7. Positive carry on a repo-financed bond position occurs when:** A) the repo rate exceeds the coupon accrual B) the coupon accrual exceeds the repo cost C) the haircut is zero D) the bond is on special

**Answer: B.** Net carry = coupon accrued – repo interest; it is positive when coupon accrual exceeds repo cost, typical in an upward-sloping curve.

**D8. The legal feature that lets a repo lender liquidate collateral immediately on the borrower's bankruptcy is:** A) rehypothecation B) the margin ratio C) safe-harbour / title-transfer treatment D) the GMRA close-out netting alone

**Answer: C.** Because repo is a true sale with title transfer, it enjoys safe-harbour treatment exempting it from the automatic stay, so the lender can liquidate at once. This legal certainty is why repo rates can be so thin. (The GMRA documents the terms, but the enabling feature is the sale/title-transfer structure.)

**D9. In the RBI's LAF corridor, the ceiling and floor are, respectively:** A) Repo rate and SDF B) MSF and SDF C) MSF and Repo rate D) SDF and MSF

**Answer: B.** MSF (repo + 25 bps, RBI lends) is the ceiling; SDF (repo – 25 bps, RBI absorbs, uncollateralised) has been the floor since April 2022; the repo rate anchors the centre where the RBI steers the WACR.

**D10. A dealer finances ₹100 cr of collateral (dirty) at a 2% haircut. The cash advanced is:** A) ₹102 cr B) ₹100 cr C) ₹98 cr D) ₹99 cr

**Answer: C.**  $P_0 = 100 \times (1 - 0.02) = ₹98$  cr. The 2% haircut is the lender's cushion; the margin ratio is  $100/98 \approx 102\%$ .

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*Self-verification note: all numerical answers (B1–B8) were recomputed by both the additive ( $\text{Interest} = P_0 \cdot r \cdot n/B$ ) and multiplicative ( $P_1 = P_0(1 + r \cdot n/B)$ ) forms where applicable, and margin-ratio checks reconcile to the stated tolerances. Day-count bases: 365 for India/GBP, 360 for USD/EUR.*

# Chapter 15 — Bond Portfolio Strategies

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## 1. The Problem / The Need

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Every chapter so far has been about a single bond, or a single risk sitting inside a single bond — its price, its yield, its duration, its spread. But nobody manages one bond in isolation. A pension fund holds ten thousand line items against a wall of future pension payments. An insurer holds a bond book calibrated to the payout profile of the policies it has written. A mutual fund holds a portfolio benchmarked to an index it must not drift too far from. A bank treasury holds a securities book that has to earn carry without blowing up when the Fed moves. In each case the question is no longer "*what is this bond worth?*" but "*how should I assemble and continually reshape a collection of bonds so that it does the job the money is there to do?*"

That is the portfolio problem, and it has a structure that a single-bond view completely hides. Three things collide:

- **A mandate.** What is the money *for*? Fund a stream of liabilities? Beat an index by 50 bp? Preserve capital? Generate income? The mandate fixes the objective function and the constraints. A liability-driven pension book and a total-return bond fund can hold the *same bonds* and still be run in opposite ways.
- **A view (or the deliberate absence of one).** Does the manager believe she can forecast rates, the shape of the curve, or credit spreads better than the market has already priced? If yes, she can try to *add* return by betting on that view. If no — or if the mandate forbids betting — she must build a portfolio that survives whatever happens.
- **Cost and governance.** Turnover costs money in bid-ask and taxes. Betting costs money when wrong and invites tracking error the trustees will question. Every strategy trades expected excess return against cost, risk, and the political reality of explaining a loss.

The strategic menu of fixed income is the set of disciplined answers to how those three collide. It divides, at the top, into **passive** strategies — which accept market pricing and aim to *match* something (an index, a liability, a horizon) — and **active** strategies — which reject the idea that the current curve and spreads are the last word and try to *beat* the market by positioning around a forecast. This chapter builds both halves from first principles, does the arithmetic that separates a laddered portfolio from an immunised one from a curve trade, and then closes the loop: how you actually *choose* a strategy by reading the mandate and the view together. This is the chapter that turns everything before it into a job.

## 2. The Core Idea

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The whole landscape hangs off one axis: **how much do you trust the market's current pricing, and how much freedom does your mandate give you to bet against it?**

At one end sits the **passive** philosophy. It says: the bond market is deep, liquid and fiercely competitive, so today's yield curve and spreads are the best available forecast of the future. Do not try to out-guess it. Instead, decide what outcome you need — replicate an index, defease a set of liabilities, lock a horizon return — and build the cheapest, most robust portfolio that *delivers that outcome regardless of what rates do*. Passive management is not "doing nothing"; laddering, immunisation and cash-flow matching are precise engineering. It is *doing nothing predictive*.

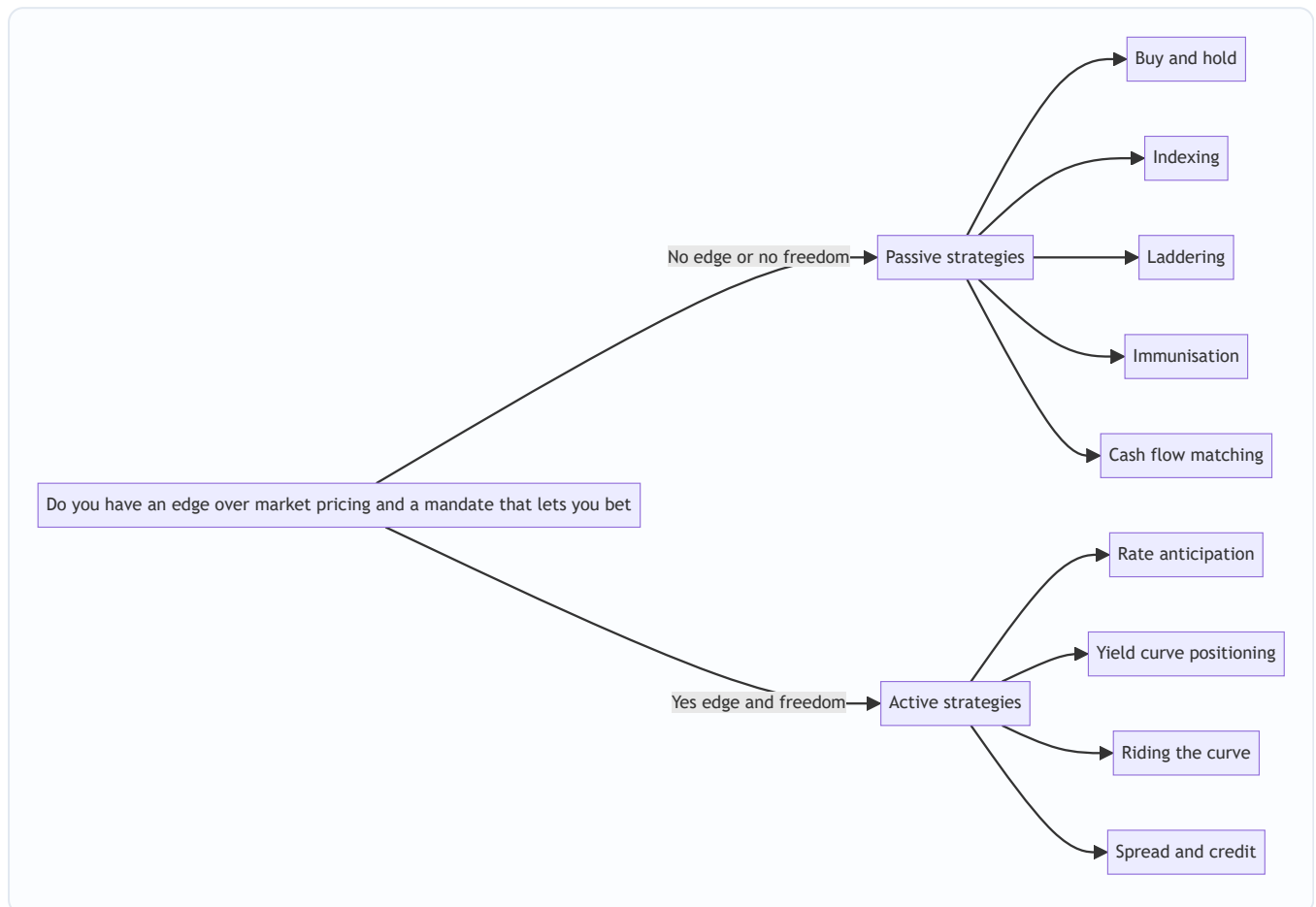
At the other end sits the **active** philosophy. It says: pricing is *usually* right but not *always* right, and a manager with a genuine edge — a rate forecast, a curve-shape view, a credit insight — can systematically



capture the gap between where a bond is priced and where it *should* be. Active management deliberately takes positions that *win if the view is right and lose if it is wrong*. Every active trade is a bet, sized against a benchmark, whose whole justification is a forecast that differs from the market's.

Between the poles lies a spectrum, not a binary — **enhanced indexing** that tilts a benchmark by a few bp of duration or spread; **contingent immunisation** that manages actively until a safety floor is threatened, then locks in. But the organising question never changes.

Figure 1 — the strategy spectrum organised by view and freedom.



The second core idea, which recurs everywhere below, is that fixed-income risk is *multi-dimensional*. A stock has essentially one market beta. A bond portfolio has, at minimum, exposure to the **level** of rates (duration), the **shape** of the curve (key-rate durations, curve slope and curvature), and **credit spreads** (spread duration). Passive strategies aim to *neutralise or match* these exposures; active strategies aim to *tilt* one of them on purpose while controlling the others. Knowing which dimension a strategy touches is the fastest way to understand it.

### 3. Why / How It Works

#### Why passive can be a rational choice, not a surrender

The efficient-market case for passive fixed income is actually stronger than for equities in one respect and weaker in another. Stronger: the largest, most liquid sectors (Treasuries, on-the-run issues, agency MBS) are traded by armies of well-capitalised professionals, so predictable mispricings are arbitrated away fast; after costs the median active government-bond manager struggles to beat the index. Weaker: bond indices are



genuinely hard to *replicate* — a broad aggregate index can contain thousands of illiquid issues that never trade, so pure "full replication" is impossible and even passive managers must *sample*. So passive bond management is rarely literal replication; it is **stratified sampling** — matching the index's exposures (duration, sector, quality, key-rate profile) with a tractable subset of bonds. The "why it works" is that *matching the risk factors matches the return* even if you don't hold every CUSIP.

### Why immunisation works — the duration offset

Immunisation is the deepest passive idea, and it rests on the price/reinvestment offset from the interest-rate-risk chapter. Recall the two-sided nature of rate risk: a rate rise *depresses* a bond's sale price but *raises* the rate at which coupons reinvest; a rate fall does the opposite. These forces cross at exactly one horizon — the **Macaulay duration**. If you set the duration of your asset portfolio equal to the horizon of your liability (or the duration of a liability stream), then to a first order a one-time parallel shift in rates leaves the *accumulated value at the horizon* unchanged. The loss on reinvested coupons is exactly offset by the gain on the bonds' terminal value, or vice versa. You have "immunised" the horizon outcome against the level of rates.

The mechanism is that immunisation *deliberately positions you at the crossover horizon* so that neither price risk nor reinvestment risk dominates. It converts an uncertain path into a locked terminal value — but only under assumptions (parallel shifts, one shift, periodic rebalancing) that we will stress-test in the confusions section.

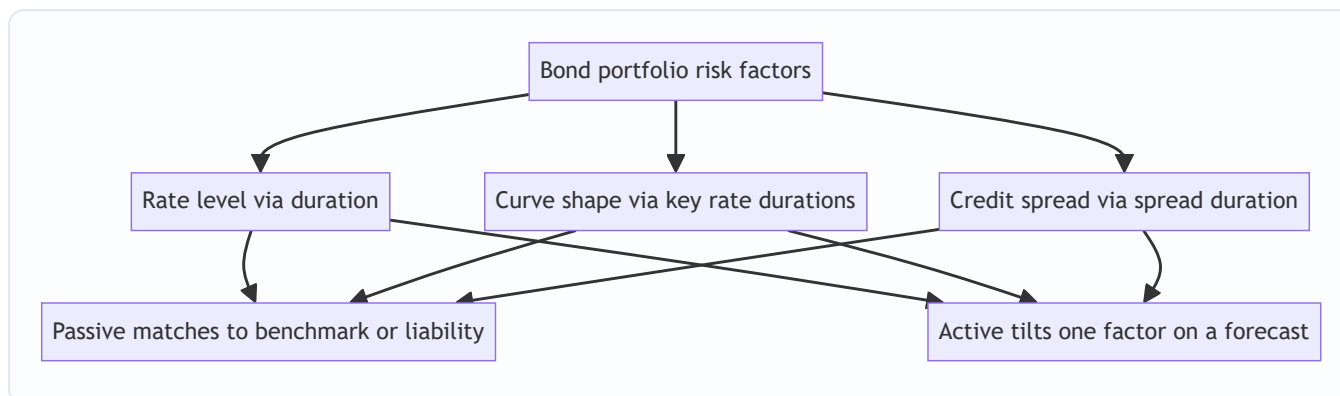
### Why cash-flow matching needs no such assumptions

Cash-flow matching sidesteps duration entirely. Instead of *offsetting* interest-rate risk, it *eliminates* it by construction: you buy bonds whose coupons and principal *land on the same dates and in the same amounts* as the liabilities. If the cash is already sitting there when the liability comes due, it does not matter what rates did in between — you are not selling anything and not reinvesting anything material. This is why cash-flow matching (also called **dedication**) is the most conservative passive strategy and needs no rebalancing. Its cost is that a perfectly matching bond set is expensive and constraining; you pay for certainty in yield give-up.

### Why active can add value — where the edge lives

Active management only "works" where the market's current pricing embeds a *forecast* that a manager can beat. The yield curve today implies a path of future short rates (via forward rates — see the spot/forward chapter). If a manager believes the *realised* path will differ, she can position for it: extend duration if she expects rates to fall by more than forwards imply, flatten or steepen if she expects the curve to reshape differently than forwards imply, overweight credit if she expects spreads to tighten more than the carry already compensates for. The edge is always the same shape: **a probability-weighted forecast that diverges from the one baked into current prices**. No divergence, no trade.

*Figure 2 — how the two philosophies attack the same three risk dimensions.*



## 4. Full Content — Strategies, Formulas and Mechanics

### 4.1 The passive family

**Buy-and-hold.** Buy bonds, clip coupons, hold to maturity, reinvest coupons, repeat. Turnover near zero; costs minimal; no view. Realised return over the holding period is the reinvested internal cash flow, sensitive only to *reinvestment* rates, not to interim price. Suited to an investor who genuinely holds to maturity and does not mark to market economically (e.g. a small held-to-maturity book). The weakness: no rebalancing means the portfolio's risk profile drifts, and there is no protection if you *are* forced to sell.

**Indexing.** Replicate the return of a bond benchmark (e.g. a broad aggregate). Because full replication is usually infeasible, managers use **stratified sampling / cell-matching**: partition the index into cells by sector × quality × maturity/duration bucket, then hold bonds so the portfolio's weight and *contribution to duration* in each cell matches the index. The objective is minimal **tracking error** — the standard deviation of the return difference between portfolio and index. Enhanced indexing permits small, controlled tilts (a few bp of duration, minor sector over/underweights) to add modest excess return while keeping tracking error tight.

The dollar contribution to duration of a position is the key matching quantity:

$$D_{i,\$} = D_i \times MV_i$$

and the portfolio duration is the market-value-weighted average:

$$D_P = \sum_i w_i D_i, \quad w_i = \frac{MV_i}{\sum_j MV_j}$$

Matching  $\sum_i w_i D_i$  and the key-rate-duration profile cell by cell is what makes a sample track the index.

**Laddering.** Spread maturities evenly across a horizon — e.g. equal amounts maturing in years 1, 2, ..., 10. Each year a bond matures and is reinvested at the long end of the ladder. A ladder is a *self-averaging* structure: it holds a roughly constant average maturity, continually reinvesting a slice at prevailing rates, so it dollar-cost-averages across the rate cycle. It blends price stability (short rungs) with yield (long rungs), needs almost no forecasting, and provides regular liquidity. Compared with a **barbell** (only short + long) or a **bullet** (all clustered at one maturity), a ladder is the diversified middle — lower curve-shape risk, no bet.

**Immunisation (single liability).** To fund one future payment of known amount at horizon  $H$ :

1. Set portfolio **Macaulay duration** =  $H$ .
2. Set portfolio **present value** = **present value of the liability**.
3. **Rebalance** periodically, because duration drifts as time passes and as yields change (duration does not fall one-for-one with calendar time), to keep  $D_P = H$  remaining horizon.

Under a one-time parallel shift, the terminal accumulated value is protected: reinvestment and price effects offset at  $\$H$ . Convexity refines this — for a given duration, a *more convex* (more dispersed) portfolio does *better* under large or non-parallel shifts, so immunising managers prefer, among duration-matched options, the one with dispersion close to the liability's (a **duration + convexity match**), not maximal convexity, to avoid non-parallel-shift risk.

**Multiple-liability immunisation** generalises this. To immunise a *stream* of liabilities the classical (Redington) conditions are:

1.  $PV_{\text{assets}} = PV_{\text{liabilities}}$
2.  $D_{\text{assets}} = D_{\text{liabilities}}$  (dollar durations equal)
3. Convexity (dispersion) of assets  $\geq$  convexity of liabilities, with asset cash flows *bracketing* the liability cash flows in time.

Condition 3 is what makes the surplus a local minimum that is *protected* against small parallel shifts.

**Cash-flow matching (dedication).** Choose a set of (usually non-callable) bonds whose combined coupons + principal reproduce the liability schedule date-by-date, working *backward* from the last liability: buy a bond maturing at the final date for that amount, net its coupons against earlier liabilities, then fill the next-earliest gap, and so on. No duration matching, no rebalancing, no interest-rate assumption. Most conservative, typically lowest yield (most constrained), highest certainty. **Horizon matching** is a hybrid: cash-flow match the near-term liabilities (where reinvestment/liquidity risk bites) and duration-immunise the long tail (where matching is expensive and shifts are more parallel).

Figure 3 — the passive strategies ranked by conservatism and reinvestment-risk exposure.



## 4.2 The active family

**Interest-rate (level) anticipation.** The purest active bet: forecast the *level* of rates and adjust **duration** accordingly. Expect rates to fall → *extend* duration (longer, lower-coupon, or use futures to add) to maximise price gain. Expect rates to rise → *shorten* duration (shift to short maturities, floaters, cash). The P&L of a duration tilt for a parallel yield change  $\Delta y$  is captured by:

$$\frac{\Delta P}{P} \approx -D_{\text{mod}} \Delta y + \frac{1}{2} C_{\text{mod}} (\Delta y)^2$$

The active manager sets portfolio duration *above* the benchmark's when bullish on price (bearish on rates falling), *below* when bearish. This is the highest-conviction, highest-variance active move — a wrong duration call is the fastest way to lose to a benchmark.

**Yield-curve positioning.** The curve moves in three roughly independent ways: **level (shift)**, **slope (steepen/flatten)**, and **curvature (butterfly)**. A manager with a *shape* view — not a level view — positions structure while holding total duration roughly constant:

- **Bullet** (concentrate near one maturity, say 10y): outperforms when the curve *flattens with a hump* / becomes less curved — benefits from the belly.

- **Barbell** (short + long, no belly): outperforms when the curve *steepens* or the belly cheapens; more convex than a duration-matched bullet, so it wins on large parallel moves too but gives up yield (the convexity is *paid for*).
- **Steeper / flattener** trades: to bet the 2s10s spread widens (steepens), go long the 2y and short the 10y in *duration-neutral* size so the trade is (approximately) immune to a parallel shift and exposed only to slope.

Key-rate (partial) durations  $D_{\{KR,k\}}$  decompose total duration by maturity point so the manager can target a *specific* section of the curve:

$$\frac{\Delta P}{P} \approx -\sum_k D_{\{KR,k\}} \Delta y_k$$

**Riding the curve (rolldown).** In a *normal, upward-sloping* curve that is expected to *stay put*, you can earn more than yield-to-maturity by buying a bond longer than your horizon and *selling it before maturity*. As time passes the bond "rolls down" the curve to a lower yield — and because yield falls, its price *rises* above pure pull-to-par. Total return = coupon/carry + **rolldown** (the price gain from the yield falling as maturity shortens). The strategy *only* works if the curve is upward-sloping and does not shift up; a rate rise can wipe out the rolldown. Return over horizon  $R_h$ :

$$R_h \approx \frac{\text{Coupon income} + (P_{\{\text{sold at lower yield}\}} - P_{\{\text{bought}\}})}{P_{\{\text{bought}\}}}$$

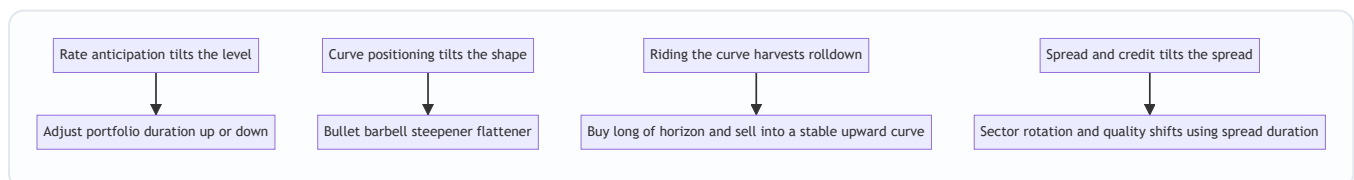
**Spread and credit strategies.** Move *down* (or up) the quality/liquidity spectrum to capture spread, and trade *changes* in spread:

- **Sector rotation / spread pickup:** overweight sectors (corporates, MBS, EM) whose spreads you expect to tighten or whose carry more than compensates for risk; underweight the expensive ones.
- **Credit up/down-grading anticipation:** buy a name before an expected upgrade (spread tightens, price rises), sell before a downgrade.
- **Spread duration** measures sensitivity to a change in the credit spread  $s$  (holding the risk-free curve fixed):

$$\frac{\Delta P}{P} \approx -D_{\{\text{spread}\}} \Delta s$$

A credit tilt is a bet on  $\Delta s$ ; the carry you earn while waiting is the spread itself, so even if spreads are unchanged you out-yield Treasuries — the classic "carry" argument, valid only until a default or a spread blow-out.

Figure 4 — active strategies mapped to the risk dimension each one tilts.



### 4.3 The blended middle

**Contingent immunisation.** Manage actively as long as the portfolio value stays above a **floor** — the amount that, immunised today at current rates, would still fund the liability at the horizon. The gap between current value and that floor is the **cushion** (safety margin). Trade actively while a cushion exists; the instant active losses erode the cushion to zero, *stop* and immunise, locking the minimum acceptable return. It is a stop-loss wrapped around active management, giving upside optionality with a hard floor.

## 5. Worked Examples

### Example 1 — Single-liability immunisation, and proving the offset

**Setup.** A fund owes **1,000,000 in 5 years**. The flat yield curve is **8%** annual. It wants to immunise with a portfolio whose Macaulay duration = 5.

- PV of liability =  $\$1,000,000 / 1.08^5 = 1,000,000 / 1.469328 = \mathbf{\$680,583}$ . Invest 680,583 today.
- Choose an 8% instrument with Macaulay duration 5. (A 5-year zero has duration 5 exactly; a coupon bond needs ~6-year maturity to reach duration 5. Use the **5-year zero** for a clean, exact demonstration — a zero held to its horizon has *no* reinvestment risk and its duration equals maturity, so it is the textbook perfect immuniser.)

**Buy the 5-year zero:** face value needed =  $\$680,583 \times 1.08^5 = 680,583 \times 1.469328 = \mathbf{\$1,000,000}$  at maturity. Trivially, at year 5 it pays exactly 1,000,000 — the liability is met.

**Now prove immunisation against a shift.** Suppose *immediately after purchase* the yield jumps to **10%** and stays there.

- The zero still matures at 1,000,000 in year 5 — its terminal value is unaffected by the intervening yield (no coupons to reinvest, held to maturity). Accumulated value at year 5 = **1,000,000**. Liability met. 

This is the limiting case: a zero-coupon bond matched to the horizon has duration = horizon *and* zero reinvestment risk, so it immunises exactly for *any* shift, not just small ones.

**Contrast with a coupon bond of the same duration.** Take a portfolio with duration 5 built from an 8% **6-year annual coupon bond** (its Macaulay duration at 8% is  $\approx 4.99 \approx 5$ ). Invest 680,583 (price  $\approx$  par-ish; assume we buy the right face so PV = 680,583). If yields jump to 10% right after purchase:

- *Coupons* (years 1–5) now reinvest at 10% instead of 8% → the reinvestment pot at year 5 is **larger** than planned.
- *The bond is sold at year 5* with 1 year remaining; its sale price is discounted at 10% instead of 8% → the sale price is **lower** than planned.

Because duration = horizon = 5, these two effects **offset** to first order, and the accumulated value at year 5 lands *very close* to the required 1,000,000 (within a rounding/convexity sliver, and slightly *above* thanks to positive convexity). If yields had instead *fallen* to 6%, the reinvestment pot shrinks but the sale price rises — again offsetting. That offset *is* immunisation. The zero is just the special case where each side is individually zero.

**Reconciliation check.** Required accumulation factor over 5 years at the original 8% =  $1.08^5 = 1.4693$ ;  $\$680,583 \times 1.4693 = 1,000,000$ .  Both constructions fund the liability; the zero does so exactly for any shift, the duration-matched coupon bond does so approximately for small shifts and needs periodic rebalancing as its duration drifts away from the shrinking horizon.

### Example 2 — Riding the curve vs. buy-and-hold

**Setup.** Upward-sloping curve, assumed **static** over a 1-year horizon. Zero-coupon yields:

| Maturity | Spot yield | Price per 100 face |
|----------|------------|--------------------|
| 1 year   | 3.0%       | 97.087             |
| 2 years  | 4.0%       | 92.456             |
| 3 years  | 5.0%       | 86.384             |

Prices:  $\$100/1.03 = 97.087\$$ ;  $\$100/1.04^2 = 92.456\$$ ;  $\$100/1.05^3 = 86.384\$$ .

**Strategy A — buy-and-hold a 1-year zero.** Buy at 97.087, redeem at 100 in one year. Return  $\$ = 100/97.087 - 1 = \mathbf{3.00\%}$  (by construction, the 1-year yield).

**Strategy B — ride the curve with a 3-year zero.** Buy the 3-year zero at **86.384**. One year later, *if the curve is unchanged*, it is now a **2-year** zero and is priced off the **2-year** point at 4.0%: price  $\$ = 100/1.04^2 = \mathbf{92.456\$}$ . Return  $\$ = 92.456/86.384 - 1 = \mathbf{7.03\%}$ .

**Reconcile the extra return.** The 3-year bond's yield *fell* from 5% (3-yr point) to 4% (2-yr point) simply because it aged and rolled down a static curve. Decompose:

- *Yield/carry portion* — one year of accretion at the original 5% purchase yield:  $\$86.384 \times 1.05 = 90.703\$$ , i.e. **+4.99%**, the yield-to-maturity you'd earn if the bond's own yield stayed 5%.
- *Rolldown portion* — the extra price gain because the yield dropped to 4%:  $\$92.456 - 90.703 = 1.753\$$ , i.e. **+2.03%** on the 86.384 cost.
- Total:  $4.99\% + 2.03\% = \mathbf{7.03\%}$  ✓ (matches the direct calculation).

**The catch (self-check the risk).** Suppose instead the whole curve shifts **up 1%** over the year, so the bond becomes a 2-year zero yielding **5.0%**: price  $\$ = 100/1.05^2 = 90.703\$$ . Return  $\$ = 90.703/86.384 - 1 = \mathbf{5.00\%}$  — still beats the 3% roll-hold, but the rolldown edge shrank. A larger shift up (curve to ~5.9% at the 2-yr point → price 89.2 → return  $\approx 3.3\%$ ) erases nearly the whole advantage. Riding the curve is an active bet that the curve stays put; the extra 4 percentage points in the static case is *compensation for the shift risk you are bearing*, not free money.

### Example 3 — Matching duration to an index (enhanced indexing tilt)

**Setup.** A benchmark aggregate has **modified duration 6.0**. A manager runs a 100 mn portfolio and is *mildly bearish* on rates — she wants a small level tilt but must keep tracking error tight (mandate: duration within  $\pm 0.5$  of benchmark). She holds two building blocks:

| Bond         | Modified duration | Market value (mn) |
|--------------|-------------------|-------------------|
| Short bond S | 2.0               | ?                 |
| Long bond L  | 9.0               | ?                 |

**Target portfolio duration = 5.6** (0.4 below benchmark — a defensive tilt, within the  $\pm 0.5$  band).

Let  $w$  = weight in the long bond L,  $(1-w)$  in short bond S. Market-value-weighted duration:

$$D_P = w(9.0) + (1-w)(2.0) = 5.6 \quad \$2.0 + 7.0w = 5.6 \quad \rightarrow w = \frac{3.6}{7.0} = 0.5143$$

So **51.43 mn in L, 48.57 mn in S**. Check:  $0.5143(9) + 0.4857(2) = 4.629 + 0.971 = \mathbf{5.60}$  ✓

**Interpretation and P&L check.** If rates rise **+50 bp** (parallel), the tilt's benefit vs. the benchmark:

- Portfolio:  $\Delta P/P \approx -5.6 \times 0.005 = -2.80\% \rightarrow \text{loss} \approx \mathbf{2.80 \text{ mn}}$ .

- Benchmark-duration portfolio (6.0):  $\$ \approx -6.0 \times 0.005 = -3.00\% \$ \rightarrow \text{loss} \approx 3.00 \text{ mn.}$
- The defensive tilt **saved  $\approx 0.20 \text{ mn}$**  (20 bp of relative return) — exactly  $\$ (6.0-5.6) \times 0.005 = 0.20\% \$$ .  


If instead rates *fell* 50 bp, the same tilt would *cost* 0.20% — the symmetric price of being wrong. The tilt is small precisely because the mandate caps tracking error; this is enhanced indexing, not a swing-for-the-fences duration bet. The arithmetic reconciles the relative return exactly to the duration gap times the yield move, which is the whole point of thinking in dollar duration.

## 6. Connections

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- **Duration & convexity (Ch. 07–08):** the master tools. Duration is the immunisation horizon and the unit of every active level/curve tilt; convexity is what makes a barbell beat a bullet on large moves and what makes immunisation robust (Redington's third condition).
- **Interest-rate risk (Ch. 09):** the price/reinvestment offset is the engine of immunisation and the risk that riding-the-curve accepts. This chapter is the *portfolio-level application* of that single-bond risk.
- **Spot & forward rates and term-structure theories (Ch. 05–06):** forward rates are the market's embedded forecast; every active trade is a bet that realised rates diverge from forwards. Riding the curve is profitable exactly when forwards *over-predict* the rate rise (the pure-expectations breakeven).
- **Yield measures (Ch. 04):** YTM assumes reinvestment at YTM — the assumption immunisation and cash-flow matching are designed to defuse. Roll-down return is precisely the gap between realised return and YTM on a static curve.
- **Credit risk & analysis (Ch. 10–11):** spread duration and credit rotation are the credit-dimension analogue of duration tilts; the carry-vs-default trade-off is the same expected-value logic.
- **Securitisation (Ch. 12):** MBS bring *negative convexity* and prepayment risk, which wreck naive immunisation — a caution that all the offsets here assume option-free bonds.
- **Liability-driven investing (LDI):** immunisation, cash-flow and horizon matching are the toolkit of pension and insurance ALM; the mandate side of this chapter is LDI.

## 7. Key Terms

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- **Passive management:** accept market pricing; match an index, liability or horizon rather than forecast.
- **Active management:** take positions based on a forecast that diverges from market-implied pricing to earn excess return.
- **Indexing / stratified sampling:** replicate a benchmark's return by matching its risk-factor cells (sector  $\times$  quality  $\times$  duration) with a subset of bonds; minimise tracking error.
- **Tracking error:** standard deviation of the portfolio-minus-benchmark return; the risk budget of an indexer/enhanced indexer.
- **Laddering:** evenly spaced maturities; self-averaging, low-forecast structure blending liquidity and yield.
- **Barbell / Bullet:** concentration at the two ends vs. at one point of the curve; the classic curve-shape structures (barbell = more convex, pays up).
- **Immunisation:** set asset duration = liability horizon/duration (and PVs equal, convexity  $\geq$ ) so a parallel shift leaves the horizon value unchanged.



- **Redington immunisation:** the three-condition multi-liability version (equal PV, equal duration, asset convexity  $\geq$  liability convexity).
- **Cash-flow matching / dedication:** buy bonds whose cash flows land exactly on the liabilities; eliminates rather than offsets rate risk; no rebalancing.
- **Horizon matching:** cash-flow match the near term, immunise the far term.
- **Rolldown / riding the curve:** buy longer than horizon and sell into a static upward-sloping curve to capture the price gain from falling yield-as-it-ages.
- **Spread duration:** price sensitivity to a change in credit spread, holding the risk-free curve fixed.
- **Dollar (money) duration:**  $\$D_{\text{mod}} \times MV\$$ ; the currency-unit sensitivity used to size duration-neutral and hedged trades.
- **Contingent immunisation:** manage actively above a floor; immunise the moment the cushion is exhausted.
- **Cushion / floor:** the safety margin between current value and the immunisable floor that funds the liability.

## 8. Common Confusions

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**"Passive means doing nothing."** Wrong. Buy-and-hold does little, but immunisation and cash-flow matching are precise engineering that require solving for exact face amounts and (for immunisation) *ongoing rebalancing*. Passive means *non-predictive*, not inactive.

**"Immunisation is set-and-forget."** No — cash-flow matching is set-and-forget; *immunisation is not*. Duration does **not** decline one-for-one with the passage of time, and it changes when yields move, so an immunised portfolio drifts off its target duration and must be **rebalanced** periodically to keep  $\$D_{\text{assets}} = \$$  remaining horizon. Skip rebalancing and the protection decays.

**"Immunisation protects against any rate move."** Only against *small parallel* shifts (first order). A *non-parallel* (twist) shift can break a duration-matched portfolio because two portfolios with the same duration can have very different key-rate profiles. This is why practitioners add convexity/dispersion matching and key-rate-duration matching, and why cash-flow matching (which needs no shift assumption) is the belt-and-braces alternative.

**"A barbell is a free lunch because it has more convexity."** No. For equal duration, the more convex barbell outperforms on *large* moves and on steepeners, but the market *charges* for that convexity via a lower yield. On a *stable* curve the lower-convexity bullet earns more carry. Convexity is bought, not free.

**"Riding the curve is arbitrage / guaranteed extra return."** No — it is an *active bet that the curve stays put*. If rates rise the rolldown gain shrinks or reverses. Its extra return over hold-to-maturity is compensation for the curve-shift risk you accept, and it needs an upward-sloping curve to exist at all.

**"Higher yield = better strategy."** Cash-flow matching usually has the *lowest* yield yet the *highest* certainty. Reaching for yield (down in credit, out in duration) buys expected return with risk. The mandate, not the yield, decides which trade-off is correct.

**"Duration matching = maturity matching."** A 10-year liability is not immunised by a 10-year coupon bond — that bond's *duration* is well below 10. You match **duration to horizon**, and for a coupon bond that means a longer maturity than the horizon.



"Active vs passive is a binary choice." It is a spectrum: enhanced indexing (tiny tilts), core-satellite (passive core + active satellites), and contingent immunisation all live in between.

## 9. Recap

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Bond portfolio management organises around one axis — **trust in market pricing × freedom in the mandate** — which splits strategy into passive (match) and active (beat).

- **Passive** accepts today's curve and spreads as the best forecast and engineers a robust outcome: **buy-and-hold** (minimal effort, reinvestment-driven), **indexing** via stratified sampling to minimise tracking error, **laddering** for self-averaging liquidity-plus-yield, **immunisation** (duration = horizon; offset price and reinvestment risk; rebalance; add convexity for robustness), and **cash-flow matching** (eliminate rate risk by dating cash to liabilities; no assumptions, lowest yield, highest certainty).
- **Active** bets that realised rates/curve/spreads will diverge from what prices imply: **rate anticipation** (tilt total duration), **curve positioning** (bullet/barbell/steepener-flattener on shape, duration-held-constant), **riding the curve** (harvest rolldown on a static upward curve), and **spread/credit** (rotate sectors and quality using spread duration for carry and spread-change gains).
- The **blended middle** (enhanced indexing, contingent immunisation) tilts modestly or bets under a hard floor.

The worked examples proved the mechanics: a horizon-matched zero immunises exactly; a duration-matched coupon bond immunises approximately via the price/reinvestment offset; rolldown decomposes cleanly into carry + yield-drop and evaporates if the curve shifts; and a duration tilt's relative P&L equals the duration gap times the yield move. **Matching the risk factors matches the return; tilting a factor is a forecast with a price for being wrong.**

## 10. Quick-Reference / Interview Points

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- **The one-liner:** passive = *match* (index/liability/horizon) accepting market pricing; active = *beat* by betting realised rates/curve/spreads diverge from what forwards and spreads imply.
- **Immunisation recipe:**  $PV(\text{assets}) = PV(\text{liab})$ ; Macaulay duration(assets) = horizon; convexity(assets)  $\geq$  convexity(liab); **rebalance** as duration drifts. Protects only small parallel shifts.
- **Duration  $\neq$  maturity.** Match duration to the liability horizon; a coupon bond needs a *longer* maturity than the horizon to hit a given duration.
- **Cash-flow matching vs immunisation:** matching *eliminates* rate risk (dated cash, no assumptions, no rebalancing, lower yield); immunisation *offsets* it (assumes parallel shifts, needs rebalancing, higher yield). Horizon matching hybridises them.
- **Barbell vs bullet:** equal duration — barbell is more convex, wins on large/steepener moves, *pays for it* in lower yield; bullet earns more carry on a stable curve.
- **Riding the curve** works only on an *upward-sloping, static* curve; return  $\approx$  carry + rolldown; the excess over YTM is pay for curve-shift risk. Be able to decompose it (Example 2).
- **Duration tilt P&L:** relative return vs benchmark  $\approx$  (benchmark duration – portfolio duration)  $\times \Delta y$ . Extend duration if bullish on price (expect rates down); shorten if bearish.
- **Curve trades are duration-neutral** by design — a 2s10s steepener sizes the two legs to equal dollar duration so only *slope*, not level, drives P&L.

- **Spread duration** measures credit-spread sensitivity; a credit overweight earns carry (the spread) while betting on tightening ( $\Delta s < 0$ ).
- **Contingent immunisation:** active while a cushion exists above the immunisable floor; lock in the instant the cushion hits zero.
- **Matching strategy to mandate:** liability-driven (pension/insurer) → immunise / cash-flow match / LDI; total-return fund vs benchmark → index or enhanced-index with controlled tilts; a house with a genuine rate/curve/credit view *and* an active mandate → the corresponding active tilt, sized by tracking-error budget. **The mandate sets the objective and constraints; the view justifies any deviation from passive. No view or no freedom → stay passive.**
- **Cardinal rule:** every active basis point of expected excess return is a forecast that diverges from market pricing — if you can't name the divergence, you don't have a trade, you have tracking error.

# Q&A — Bond Portfolio Strategies

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A companion practice bank for Chapter 15. Work each question before reading the answer. Numerical answers show full working so you can audit every step.

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## Section A — Concept Checks

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### A1. What single axis organises the entire menu of bond portfolio strategies, and how does it split the menu?

The organising axis is *how much you trust the market's current pricing* × *how much freedom the mandate gives you to bet against it*. It splits strategy into **passive** (accept today's curve and spreads as the best available forecast and *match* an index, liability or horizon) and **active** (reject the idea that current pricing is the last word and *beat* the market by positioning around a forecast that diverges from what prices imply). Between the poles is a spectrum — enhanced indexing, core-satellite, contingent immunisation — not a binary.

### A2. Why is passive fixed-income management rarely literal replication?

A broad aggregate index can contain thousands of illiquid issues that never trade, so buying every CUSIP ("full replication") is impossible. Passive managers instead use **stratified sampling / cell-matching**: partition the index by sector × quality × maturity/duration bucket and hold a tractable subset whose weight and contribution-to-duration match the index cell by cell. It works because *matching the risk factors matches the return* even without holding every bond.

### A3. Explain the mechanism by which immunisation protects a horizon value.

Rate risk is two-sided: a rate rise depresses a bond's sale price but raises the coupon reinvestment rate; a fall does the opposite. These forces cross at exactly one horizon — the **Macauley duration**. Set asset duration equal to the liability horizon (and PV assets = PV liability), and to first order a one-time parallel shift leaves the horizon accumulated value unchanged: the reinvestment loss/gain offsets the price gain/loss. You have deliberately positioned at the crossover point.

### A4. Why does cash-flow matching need no interest-rate assumption while immunisation does?

Cash-flow matching *eliminates* rate risk by construction — bonds whose coupons and principal land on the liabilities' dates and amounts mean the cash is already there when each liability falls due, so you neither sell nor reinvest and the intervening rate path is irrelevant. Immunisation only *offsets* risk via the duration crossover, which holds exactly only for small parallel shifts — hence it assumes parallel movement and needs rebalancing.

### A5. State the three Redington conditions for multiple-liability immunisation.

1.  $PV(\text{assets}) = PV(\text{liabilities})$ .
2. Duration of assets = duration of liabilities (dollar durations equal).
3. Convexity/dispersion of assets  $\geq$  convexity of liabilities, with asset cash flows *bracketing* the liability cash flows in time. Condition 3 makes the surplus a local minimum that is protected against small parallel shifts.

### A6. Where does the "edge" that justifies any active trade come from?

Always the same shape: a probability-weighted forecast of realised rates/curve/spreads that *diverges from the forecast baked into current prices* (forward rates for the curve, the spread itself for credit). Extend duration if you expect rates to fall by more than forwards imply; steepen/flatten if you expect the curve to reshape differently than forwards imply; overweight credit if you expect spreads to tighten more than the carry already pays for. No divergence, no trade — only tracking error.

**A7. What are the three roughly independent ways the yield curve moves, and which strategy family targets each?**

**Level (parallel shift)** — targeted by rate-anticipation duration tilts. **Slope (steepen/flatten)** — targeted by steepener/flattener and barbell/bullet trades held duration-neutral. **Curvature (butterfly/hump)** — targeted by bullet vs barbell positioning across the belly. Key-rate (partial) durations decompose total duration by maturity point so a manager can target a specific section.

**A8. Distinguish a ladder, a bullet, and a barbell.**

A **ladder** spreads maturities evenly across a horizon; it is self-averaging (constant average maturity, continually reinvesting a slice), needs no forecast, and has low curve-shape risk. A **bullet** clusters all maturities at one point — outperforms when the curve flattens/becomes less curved (benefits from the belly), earns more carry on a stable curve. A **barbell** holds only short + long ends (no belly) — more convex than a duration-matched bullet, so it wins on large parallel moves and steepeners, but the market charges for that convexity via a lower yield.

**A9. What is contingent immunisation?**

Active management wrapped in a stop-loss. You manage actively as long as portfolio value stays above a **floor** (the amount that, immunised today, would still fund the liability at horizon). The gap between current value and the floor is the **cushion**. Trade actively while a cushion exists; the instant active losses erode the cushion to zero, stop and immunise, locking the minimum acceptable return. It buys upside optionality with a hard floor.

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## Section B — Numerical / Applied

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**B1. Single-liability immunisation with a zero.** A fund owes 1,000,000 in 5 years. The flat curve is 8%. How much is invested today, what face of a 5-year zero is bought, and prove the liability is still met if yields jump to 10% immediately after purchase.

*Solution.* PV of liability =  $1,000,000 / 1.08^5 = 1,000,000 / 1.469328 = \mathbf{680,583}$ , invested today. A 5-year zero has Macaulay duration = 5 exactly, matching the horizon. Face needed =  $680,583 \times 1.469328 = \mathbf{1,000,000}$  at maturity. If yields jump to 10% right after purchase, the zero still matures at 1,000,000 in year 5 — it has no coupons to reinvest and is held to maturity, so its terminal value is unaffected by the intervening yield. Accumulated value at year 5 =  $\mathbf{1,000,000 = liability}$ . The zero immunises *exactly* for any shift because each side of the offset (price risk, reinvestment risk) is individually zero.

**B2. The offset for a coupon bond.** Explain qualitatively, for a duration-5 portfolio built from an 8% six-year coupon bond funding the same 1,000,000 liability, what happens to the two effects if yields jump to 10%, and why the horizon value still lands near 1,000,000.

*Solution.* Coupons in years 1–5 now reinvest at 10% not 8% → the reinvestment pot at year 5 is **larger** than planned. The bond is sold at year 5 with one year left, discounted at 10% not 8% → the sale price is **lower** than planned. Because duration = horizon = 5, these effects **offset** to first order, landing the accumulated

value very close to 1,000,000 (slightly above, thanks to positive convexity). Had yields fallen to 6%, the pot shrinks but the sale price rises — again offsetting. Unlike the zero, this is only approximate for small shifts and must be rebalanced as its duration drifts.

**B3. Riding the curve.** Static upward curve: 1y spot 3.0% (price 97.087), 2y 4.0% (92.456), 3y 5.0% (86.384). Compare buy-and-hold a 1-year zero versus buying the 3-year zero and selling in one year.

*Solution.* Buy-and-hold 1y zero:  $100 / 97.087 - 1 = 3.00\%$ . Ride the curve: buy 3y zero at 86.384; one year later, if the curve is unchanged, it is a 2-year zero priced at the 2y point =  $100 / 1.04^2 = 92.456$ . Return =  $92.456 / 86.384 - 1 = 7.03\%$ . The bond's own yield fell from 5% to 4% purely by aging down a static curve.

**B4. Decompose the rolldown return in B3.**

*Solution.* Carry/yield portion — one year of accretion at the original 5% purchase yield:  $86.384 \times 1.05 = 90.703$ , i.e. **+4.99%**. Rolldown portion — extra price gain because the yield dropped to 4%:  $92.456 - 90.703 = 1.753$ , i.e.  $1.753 / 86.384 = +2.03\%$ . Total  $4.99\% + 2.03\% = 7.03\%$ , matching B3.

**B5. The catch in B3.** If instead the whole curve shifts up 1% over the year (so the bond becomes a 2y zero yielding 5.0%), what is the ride-the-curve return, and what does the shrunk edge represent?

*Solution.* New price =  $100 / 1.05^2 = 90.703$ . Return =  $90.703 / 86.384 - 1 = 5.00\%$ . Still beats the 3% buy-and-hold, but the rolldown edge shrank from ~4 pts to ~2 pts. A larger rise (belly to ~5.9% → price ~89.2 → return ~3.3%) erases nearly the whole advantage. The extra return in the static case was **compensation for the curve-shift risk you bear**, not free money.

**B6. Duration tilt / enhanced indexing.** Benchmark modified duration 6.0. Portfolio of 100 mn; mandate keeps duration within  $\pm 0.5$  of benchmark. Manager is mildly bearish and targets duration 5.6 using short bond S (dur 2.0) and long bond L (dur 9.0). Find the weights.

*Solution.* Let  $w$  = weight in L.  $D_P = 9.0w + 2.0(1-w) = 5.6 \rightarrow 2.0 + 7.0w = 5.6 \rightarrow w = 3.6 / 7.0 = 0.5143$ . So **51.43 mn in L, 48.57 mn in S**. Check:  $0.5143(9) + 0.4857(2) = 4.629 + 0.971 = 5.60$ .

**B7. P&L of the B6 tilt.** Rates rise 50 bp parallel. What does the defensive tilt save versus a benchmark-duration portfolio?

*Solution.* Portfolio:  $\Delta P/P \approx -5.6 \times 0.005 = -2.80\% \rightarrow$  loss  $\approx 2.80$  mn. Benchmark-duration (6.0):  $\approx -3.00\% \rightarrow$  loss  $\approx 3.00$  mn. The tilt **saved  $\approx 0.20$  mn = 20 bp**, exactly  $(6.0 - 5.6) \times 0.005 = 0.20\%$ . Had rates *fallen* 50 bp, the tilt would have cost 0.20% — the symmetric price of being wrong. Relative return vs benchmark =  $(\text{benchmark duration} - \text{portfolio duration}) \times \Delta y$ .

**B8. Duration-neutral steepener sizing.** You want a 2s10s steepener: long the 2y (mod dur 1.9), short the 10y (mod dur 8.2), and you commit 50 mn face to the 2y leg. What 10y face makes the trade duration-neutral, and what remains as the exposure?

*Solution.* Duration-neutral means equal dollar duration on each leg. Taking price  $\approx$  par for both, dollar duration  $\propto$  face  $\times$  duration. Set  $50 \times 1.9 = F_{10} \times 8.2 \rightarrow F_{10} = 95 / 8.2 = 11.59$  mn face short in the 10y. The trade is (approximately) immune to a parallel level shift and is exposed only to a change in the **2s10s slope** — it profits if the curve steepens (2y yield falls relative to 10y). (In practice you match dollar durations using actual prices, not the par approximation.)

**B9. Spread duration / carry.** A corporate bond has spread duration 4.5 and trades at a 150 bp spread over Treasuries. (a) Price impact if the spread tightens 30 bp. (b) One-line statement of the carry argument.

*Solution.* (a)  $\Delta P/P \approx -D_{\text{spread}} \times \Delta s = -4.5 \times (-0.0030) = +1.35\%$ . (b) Even if the spread is unchanged, you out-yield Treasuries by the 150 bp spread (the carry) — valid only until a default or spread blow-out; the credit overweight is simultaneously earning carry and betting on tightening ( $\Delta s < 0$ ).

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## Section C — Interview-Style

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### C1. "A trustee says passive bond management is just being lazy — you're doing nothing. How do you respond?"

*Model answer.* Passive means *non-predictive*, not inactive. Immunisation and cash-flow matching are precise engineering: you solve for exact face amounts so PVs and durations match a liability, and immunisation requires *ongoing rebalancing* as duration drifts with time and yields. Indexing requires stratified sampling to match a benchmark's key-rate and sector profile within a tight tracking-error budget. The choice to be passive is a considered judgement that we have no reliable edge over deeply liquid, competitively priced markets — after costs, the median active government-bond manager struggles to beat the index — or that the mandate doesn't reward betting. That is discipline, not laziness.

### C2. "Walk me through why you'd choose cash-flow matching over immunisation for a pension book — and when you wouldn't."

*Model answer.* Cash-flow matching *eliminates* rate risk by dating the assets' cash flows to the liabilities, so it needs no interest-rate assumption, no rebalancing, and no parallel-shift hope — the highest certainty available. Immunisation only *offsets* risk via the duration crossover; it protects only small parallel shifts, breaks on twists, and must be rebalanced. So for a mandate that prizes certainty above all, cash-flow match. The cost is yield: a perfectly matching, usually non-callable bond set is expensive and constraining, so it typically has the lowest yield. When the yield give-up is unacceptable or the far-dated liabilities are hard to match precisely, I'd use **horizon matching** — cash-flow match the near term (where reinvestment and liquidity risk bite) and immunise the long tail (where shifts are more parallel and matching is expensive).

### C3. "Is a barbell a free lunch because it has more convexity than a bullet?"

*Model answer.* No. For equal duration, the more convex barbell does outperform on *large* moves and on steepeners — convexity is a positive-gamma benefit. But the market *charges* for that convexity via a lower yield, so on a *stable* curve the lower-convexity bullet earns more carry and wins. Convexity is bought, not free. The right structure depends on your view: barbell if you expect large moves or a steepening; bullet if you expect a calm curve or a flattening that benefits the belly.

### C4. "You put on a ride-the-curve trade and made 7% while the benchmark made 3%. Was that skill?"

*Model answer.* Partly — but I'd be honest that most of the excess was *risk compensation*, not alpha. Riding the curve is an active bet that the curve stays put. The 7% decomposed into ~5% carry at the purchase yield plus ~2% rolldown from the bond aging to a lower yield on a static curve. Had the curve shifted up meaningfully, the rolldown would have shrunk or reversed — a large enough rise erases the entire edge. So the trade needs an upward-sloping curve to exist at all and a stable-to-falling curve to pay off. The excess over hold-to-maturity is pay for the curve-shift risk I accepted; calling it pure skill would misrepresent the risk taken.

### C5. "How do you actually decide passive versus active for a new mandate?"

*Model answer.* Read the mandate and the view together. The mandate: liability-driven money (pension, insurer) → immunise, cash-flow match, or run LDI; a total-return fund benchmarked to an index → index or enhanced-index with controlled tilts; an explicit active remit → active tilts sized to a tracking-error budget.

The view: active deviation is only justified by a nameable forecast that diverges from market-implied pricing. So the mandate sets the objective and constraints; the view justifies any deviation from passive. No view or no freedom → stay passive. Cardinal rule: every active basis point of expected excess return must correspond to a divergence you can name — if you can't name it, you don't have a trade, you have tracking error.

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## Section D — Multiple Choice (with reasoning)

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**D1.** To immunise a single 10-year liability with coupon bonds, you should hold bonds whose:

A. maturity = 10 years B. Macaulay duration = 10 years C. yield = the liability discount rate D. convexity is maximised

**Answer: B.** You match *duration* to the horizon, not maturity — a 10-year coupon bond has duration well below 10, so it under-hedges. A is the classic "duration = maturity" trap. C is irrelevant to the offset mechanism. D is wrong: among duration-matched options you want convexity/dispersion *close to the liability's*, not maximal, to avoid non-parallel-shift risk.

**D2.** Cash-flow matching differs from immunisation primarily because it:

A. requires periodic rebalancing B. assumes only parallel yield shifts C. eliminates rate risk rather than offsetting it, needing no rate assumption D. always earns a higher yield

**Answer: C.** Cash-flow matching dates assets to liabilities so risk is eliminated by construction — no rate assumption, no rebalancing. A and B describe *immunisation*, not cash-flow matching. D is backwards: cash-flow matching usually has the *lowest* yield, the price of its certainty.

**D3.** An immunised (duration-matched) coupon portfolio is most vulnerable to:

A. a small parallel upward shift B. a small parallel downward shift C. a non-parallel twist of the curve D. the passage of time with no rate change

**Answer: C.** Duration matching protects only *small parallel* shifts to first order. Two portfolios with equal duration can have very different key-rate profiles, so a twist can break the hedge — hence key-rate-duration and convexity matching. A and B are exactly what immunisation is built to handle. D causes duration *drift* (handled by rebalancing) but is not the primary vulnerability.

**D4.** Riding the curve generates excess return over hold-to-maturity only when the curve is:

A. inverted and steepening B. flat and stable C. upward-sloping and stable (or falling) D. upward-sloping and rising sharply

**Answer: C.** Rolldown requires an upward-sloping curve (so the bond's yield falls as it ages down the curve) that stays put or falls (so the shift doesn't offset the rolldown). A inverted curve gives negative rolldown; a flat curve gives none; a sharply rising curve erases the edge.

**D5.** A manager expecting rates to fall by *more* than the forward curve implies should:

A. shorten duration below the benchmark B. extend duration above the benchmark C. hold duration equal and buy a bullet D. move down in credit quality

**Answer: B.** Bullish on price = extend duration to maximise the price gain when rates fall. A is the opposite (a rising-rate view). C is a *curve-shape* view, not a level view. D is a *spread* bet, unrelated to the rate-level forecast.

**D6.** In a duration-neutral 2s10s steepener, the two legs are sized to have equal:



A. face value B. market value C. dollar (money) duration D. yield to maturity

**Answer: C.** Sizing the legs to equal dollar duration makes the trade approximately immune to a parallel *level* shift, leaving only *slope* exposure. Equal face or market value would leave a residual duration mismatch and hence unwanted level exposure.

**D7.** "Enhanced indexing" is best described as:

A. full replication of every index constituent B. a benchmark match with small, controlled tilts within a tight tracking-error budget C. maximum-conviction duration betting D. cash-flow matching a liability stream

**Answer: B.** Enhanced indexing permits a few bp of duration or minor sector tilts to add modest excess return while keeping tracking error tight. A is (usually infeasible) pure indexing. C is aggressive active management. D is a liability-driven passive strategy, not benchmark-relative.

**D8.** Spread duration of 4.5 means a 25 bp *widening* in credit spread causes an approximate price change of:

A. +1.125% B. -1.125% C. -0.5625% D. +0.5625%

**Answer: B.**  $\Delta P/P \approx -D_{\text{spread}} \times \Delta s = -4.5 \times (+0.0025) = -1.125\%$ . Widening ( $\Delta s > 0$ ) hurts price. A has the wrong sign; C and D use the wrong magnitude.

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*Self-check note:* All numerical answers reconcile with the chapter's worked examples — B1–B2 (immunisation offset), B3–B5 (rolldown  $4.99\% + 2.03\% = 7.03\%$ ; shift-up return  $5.00\%$ ), B6–B7 (weight  $0.5143$ , 20 bp relative saving), and the first-order duration/spread-duration price formulas in B7–B9 and D8.



# Chapter 16 — Inflation-Linked and Floating-Rate Bonds

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## 1. The Problem / Need

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A plain "vanilla" bond promises a fixed stream of cash. A 10-year bond with a 5% coupon on ₹1,000 face pays ₹50 every year and returns ₹1,000 at maturity — no matter what happens in the world. That fixed-ness is exactly the problem. It exposes the investor to two very different risks that a fixed cash flow cannot defend against.

**Risk one: inflation erodes real purchasing power.** The ₹1,000 you get back in ten years buys far less than ₹1,000 does today. If inflation averages 6% a year, ₹1,000 in ten years is worth about ₹558 in today's money ( $1,000 / 1.06^{10}$ ). Your 5% coupon looked generous, but if inflation runs at 6% your *real* return is negative — you are getting poorer in purchasing-power terms while nominally "earning interest." A pensioner living off coupon income feels this directly: the grocery bill rises 6% a year, the coupon does not move at all.

**Risk two: rising interest rates crush the market price of a fixed coupon.** If you own a 5% bond and market yields jump to 8%, nobody will pay par for your stingy 5% stream. The price falls until the bond's yield matches the market. A fixed coupon is a *duration* liability — the longer the fixed stream, the more its price bleeds when rates rise.

Vanilla bonds bundle these two exposures and force the investor to eat both. The natural question is: *can we design a bond whose cash flows move with the thing that would otherwise hurt us?* If inflation is the enemy, build a bond whose payments grow with inflation. If rising rates are the enemy, build a bond whose coupon resets upward as rates rise. Those two design answers are the subject of this chapter:

- **Inflation-linked bonds (ILBs)** — TIPS in the US, Inflation Indexed Bonds (IIBs) in India — neutralise *inflation risk* by indexing principal (and therefore coupon) to a price index.
- **Floating-rate notes (FRNs)** — neutralise *interest-rate (price) risk* by resetting the coupon to a short-term reference rate every few months.

They protect against different dangers, and understanding *which* danger each one defuses is the whole point.

## 2. The Core Idea

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Both instruments share one principle: **make a cash flow float instead of fixing it, so the investor is not left holding a stale promise.** But they float *different* legs of the bond against *different* benchmarks.

- An **inflation-linked bond** fixes the *real* coupon rate but floats the *principal*. The principal is scaled up by a published inflation index. Because the coupon is a fixed percentage of an ever-growing principal, the coupon rupees also grow with inflation. The investor locks in a **real yield** — a guaranteed return *above inflation* — and lets the nominal cash flows take care of themselves.
- A **floating-rate note** keeps the principal fixed at par but floats the *coupon rate*. Every reset period the coupon is set to a short-term reference rate (SOFR, T-bill rate, MIBOR) plus a fixed spread. Because the coupon re-prices to the market every few months, the bond's price barely moves when rates change — it behaves almost like cash.

The mnemonic:

**Inflation-linked bonds protect the value of your money. Floating-rate notes protect the value of your bond.**

An ILB says "whatever inflation does, your purchasing power is preserved." An FRN says "whatever the market rate does, your bond stays near par and your coupon keeps up."

### 3. Why / How It Works

#### 3.1 Why indexing the principal defends purchasing power

Purchasing power is *nominal rupees divided by the price level*. If the price level (CPI) rises 6%, you need 6% more nominal rupees to buy the same basket. An ILB delivers exactly that by multiplying the principal by an **index ratio**:

$$\text{Index Ratio} = \frac{\text{CPI on the settlement/payment date}}{\text{CPI on the issue (base) date}}$$

The nominal principal is  $\text{Face} \times \text{Index Ratio}$ . When CPI is up 6%, the index ratio is 1.06, and the principal is up 6% — precisely cancelling the loss of purchasing power. Because the coupon is  $\text{real coupon rate} \times \text{inflation-adjusted principal}$ , the coupon rupees inherit the same 6% uplift. The *real* value of every cash flow is held constant by construction. The investor's guaranteed return is stated in real terms: a "2% real" TIPS returns 2% *plus whatever inflation turns out to be*.

#### 3.2 Why resetting the coupon defends price

A bond's price falls when rates rise because its coupon is now too low relative to the market. An FRN removes that gap. At each reset, the coupon is re-struck at the *current* market reference rate plus a spread. Right after a reset the bond's coupon equals the market's required return, so the bond is worth par. Its price can only drift away from par between resets, and only by the small amount that a single short period's mispricing is worth. This is why an FRN's **duration is tiny** — roughly the time to the next reset (e.g. ~0.25 years for a quarterly resetter), not the years-to-maturity. The investor holds something that looks like a money-market instrument wearing a long-maturity costume.

#### 3.3 The Fisher relationship — the bridge between the two worlds

Everything about real-vs-nominal rests on the **Fisher equation**, which links nominal yield, real yield, and expected inflation:

$$(1 + y_{\text{nominal}}) = (1 + y_{\text{real}}) \times (1 + \pi_{\text{expected}})$$

Rearranged, the **exact real yield** is:

$$y_{\text{real}} = \frac{1 + y_{\text{nominal}}}{1 + \pi} - 1$$

And the common approximation (good for small numbers) is:

$$y_{\text{nominal}} \approx y_{\text{real}} + \pi$$

This equation is the reason a nominal bond and an inflation-linked bond of the same maturity can be *compared*: the difference in their yields reveals what the market expects inflation to be — the **breakeven inflation rate**.

The diagram below shows how a rise in the price index propagates through an inflation-linked bond into every cash flow.

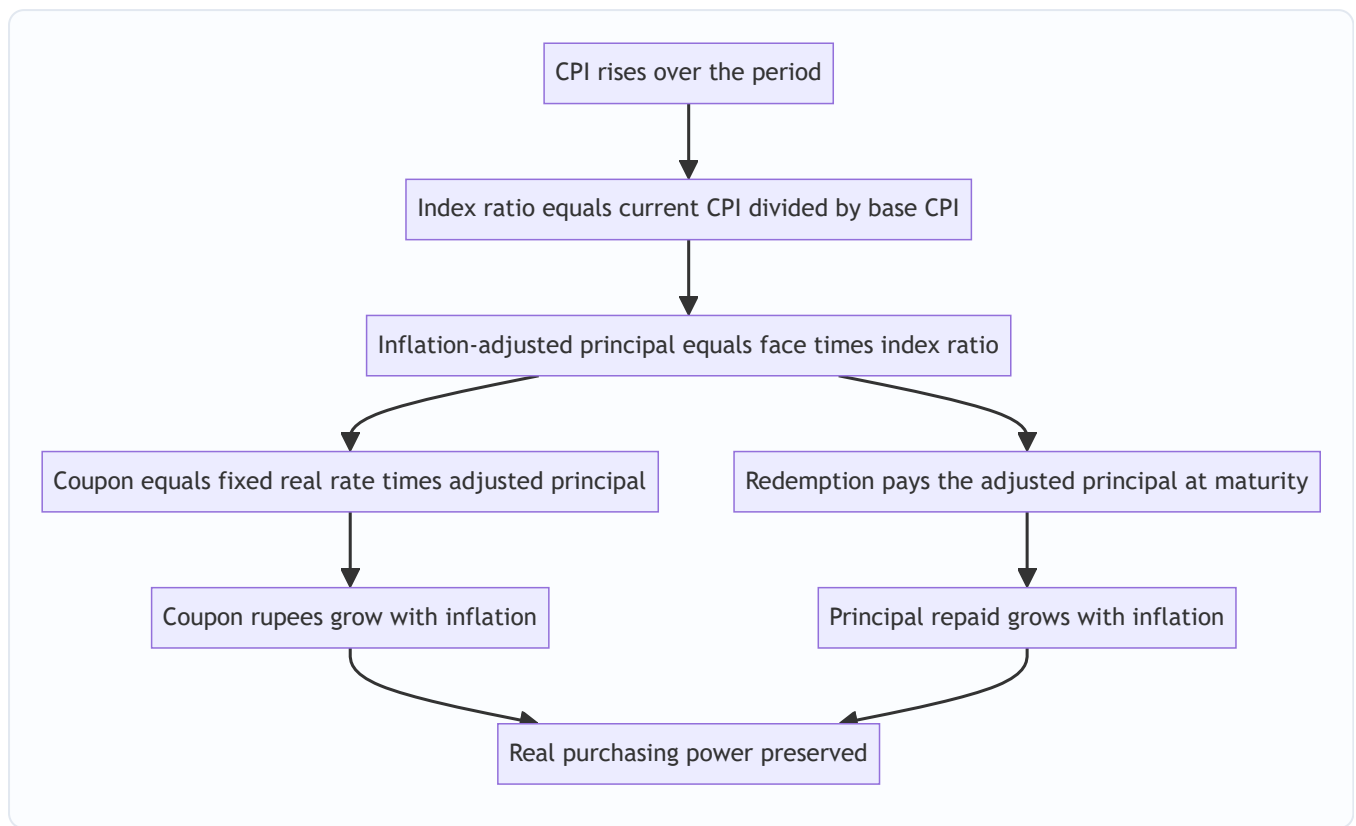


Figure 16.1 — How inflation flows through the principal into both coupon and redemption of an inflation-linked bond.

## 4. Full Content — Mechanics, Formulas, and Math

### 4.1 Inflation-linked bond mechanics (TIPS model)

The dominant global design is the **Canadian/US "capital-indexed" model**, used by US TIPS and Indian CPI-linked IIBs. Its features:

**(a) Inflation-adjusted principal.** On any date,

$$\text{Adjusted Principal} = \text{Face} \times \frac{\text{CPI}_{\text{ref, date}}}{\text{CPI}_{\text{base}}}$$

US TIPS use a *reference CPI* built by interpolating the CPI-U with a three-month lag (the CPI for a settlement date in a month is interpolated between the CPI values published for the months roughly two and three months earlier). Indian IIBs use the WPI or CPI with a similar lag. The lag exists because official price indices are published with a delay, so the bond references a slightly stale but *known* index value.

**(b) Coupon on the adjusted principal.** The coupon rate is *fixed and real*. Each payment is:

$$\text{Coupon}_t = \frac{c_{\text{real}}}{f} \times \text{Adjusted Principal}_t$$

where  $c_{\text{real}}$  is the annual real coupon rate and  $f$  is the payment frequency (2 for semiannual TIPS). Because the adjusted principal grows with inflation, the coupon rupees grow too.

**(c) Deflation floor (principal protection).** At *maturity*, TIPS repay the **greater of** the inflation-adjusted principal and the original face:

$$\text{Redemption} = \max(\text{Adjusted Principal}_{\text{maturity}}, \text{Face})$$

So cumulative deflation over the bond's life cannot pull the redemption below par. **Important nuance:** the floor applies only to the *final redemption*, not to coupons. Coupons are still computed on the (possibly sub-par) adjusted principal during a deflationary stretch, so coupon rupees *can* fall below the nominal-coupon-on-face level in deflation. Indian retail IINSS-C bonds and the sovereign IIBs likewise carried capital protection — redemption at face value if the adjusted value fell below it.

**(d) Phantom income / tax.** In the US the annual *principal accretion* (the inflation uplift) is taxable in the year it accrues, even though the investor receives no cash for it until maturity — hence "phantom income." This is why TIPS are often held in tax-deferred accounts.

## 4.2 Indian Inflation Indexed Bonds — the specifics

India has run two distinct products, and interviews reward knowing the difference:

| Feature              | Sovereign IIBs (Jun 2013)                                                   | IINSS-C, retail (Dec 2013)                    |
|----------------------|-----------------------------------------------------------------------------|-----------------------------------------------|
| Index used           | Wholesale Price Index (WPI) initially                                       | Consumer Price Index (CPI, combined)          |
| Design               | Capital-indexed (principal indexed; fixed real coupon on indexed principal) | Coupon = fixed 1.5% p.a. + CPI inflation rate |
| Inflation protection | Both principal and coupon protected                                         | Coupon protects; principal fixed at face      |
| Capital protection   | Redemption at face if indexed value falls below face                        | Principal returned at face                    |
| Target buyer         | Institutions / market                                                       | Retail savers                                 |

The sovereign IIBs are the "textbook TIPS" of India — indexed principal, fixed real coupon. The retail IINSS-C used a *simpler additive* structure: it did **not** index the principal; instead it paid **1.5% + realised CPI inflation** as the coupon. That additive design protects income against inflation but does not grow the principal — a meaningfully different mechanism worth calling out. (The switch from WPI to CPI for the sovereign series reflected CPI becoming the RBI's official inflation target measure.)

## 4.3 Real yield, nominal yield, breakeven inflation

The **breakeven inflation rate (BEI)** is the inflation rate at which a nominal bond and an inflation-linked bond of the same maturity deliver the *same* nominal return. From Fisher:

$$\text{BEI (exact)} = \frac{1 + y_{\text{nominal}}}{1 + y_{\text{real}}} - 1 \quad \text{BEI (approx)} = y_{\text{nominal}} - y_{\text{real}}$$

Interpretation:

- If **realised inflation** > **BEI**, the ILB wins (its cash flows grew faster than the nominal bond's fixed stream).
- If **realised inflation** < **BEI**, the nominal bond wins.
- The BEI is the market's *expected* inflation **plus an inflation risk premium minus a liquidity premium** — it is not a pure forecast, but it is the best market-implied read on inflation.

## 4.4 Floating-rate note mechanics

An FRN's coupon each period is:

$$\text{Coupon rate}_t = \text{Reference rate}_t + \text{Quoted Margin (spread)}$$

- **Reference rate:** SOFR (US), MIBOR / T-bill rate (India), historically LIBOR/EURIBOR. The rate is *observed at the start of the period* (or compounded over the period, for SOFR-in-arrears) and the coupon is *paid at the end* — this is "set in advance, pay in arrears."
- **Quoted margin:** a fixed spread reflecting the issuer's credit at issue. A weak issuer pays SOFR + 1.50%; a strong one pays SOFR + 0.10%.
- **Reset frequency:** monthly, quarterly, or semiannual. Between resets the coupon is fixed; at each reset it re-strikes.

**Discount margin (DM).** The FRN analogue of yield-to-maturity. It is the constant spread over the *projected* reference rate that discounts the FRN's cash flows back to its current price. If the issuer's credit is unchanged,  $\text{DM} \approx \text{quoted margin}$  and the price is near par. If the issuer's credit *deteriorates*, the market demands  $\text{DM} > \text{quoted margin}$ , and the price falls below par (a fixed quoted margin is now too thin). So an FRN removes *interest-rate* risk but **retains credit-spread risk** — that is the residual exposure.

**Why price stays near par.** Right after a reset the coupon equals reference + margin, which is what the market requires, so the discounted value is par. The price only wanders from par by the value of the spread mismatch over the remaining life, and resets keep pulling it back. Effective duration  $\approx$  time to next reset.

*The timeline below shows how an FRN coupon is fixed at the start of each period and paid at its end, re-striking every reset.*



Figure 16.2 — A floating-rate note sets each coupon in advance at reset and pays it in arrears, re-striking to the market every period.

## 5. Worked Examples

### Example 1 — TIPS cash flows and real-return reconciliation

**Setup.** A US TIPS: face \$1,000, real coupon 2.00% annual paid semiannually (so 1.00% per period). Base CPI at issue = 250.00. Suppose the reference CPI evolves as below over the first year.

| Period (6-mo) | Reference CPI | Index ratio (CPI / 250) | Adjusted principal | Coupon = 1.00% × adj. principal |
|---------------|---------------|-------------------------|--------------------|---------------------------------|
| Issue         | 250.00        | 1.0000                  | \$1,000.00         | —                               |
| 1             | 252.50        | 1.0100                  | \$1,010.00         | \$10.10                         |
| 2             | 256.00        | 1.0240                  | \$1,024.00         | \$10.24                         |

**Cash received in year 1:** \$10.10 + \$10.24 = **\$20.34** in coupons, and the principal has *accrued* from \$1,000 to \$1,024 (a \$24.00 uplift not yet paid in cash but taxable in the US as phantom income).

**Reconciliation — did the investor actually earn ~2% real?** Total one-year economic gain = coupons + principal accretion = 20.34 + 24.00 = **\$44.34** on the \$1,000 starting value, i.e. a nominal return of 4.434%.

One-year inflation was  $\text{CPI } 250 \rightarrow 256 = 2.40\%$ . The real return is:

$$\text{Real} = \frac{1.04434}{1.02400} - 1 = 1.01986 - 1 = 1.99\% \approx 2.00\%$$

The real return lands on the stated 2% real coupon, exactly as designed — inflation passed straight through to the nominal cash flows and left the real return untouched. ✓

**Deflation-floor check.** Suppose instead CPI had *fallen* to 245 by maturity (index ratio 0.98). The adjusted principal would be \$980, but the redemption floor pays  $\max(980, 1000) = \$1,000$ . The investor is protected against a below-par principal at maturity (though the coupons during the deflation would still have been computed on the \$980 base).

## Example 2 — Real vs nominal yield and the breakeven decision

**Setup.** The 10-year nominal Treasury yields **4.50%**. The 10-year TIPS (real) yields **2.00%**.

**Breakeven inflation:**

$$\text{BEI (exact)} = \frac{1.0450}{1.0200} - 1 = 1.02451 - 1 = 2.451\% \\ \text{BEI (approx)} = 4.50\% - 2.00\% = 2.50\%$$

**Verify with Fisher (the reverse check):** if real = 2.00% and inflation = 2.451%, the implied nominal is  $(1.0200)(1.02451) - 1 = 1.0450 - 1 = 4.50\%$ . ✓ It ties back exactly to the nominal Treasury yield — the two bonds break even at 2.451% inflation.

**Investor decision.** - If you expect inflation **above 2.451%** (say 3.5%), buy the **TIPS** — you'll earn 2% real + 3.5%  $\approx$  5.5% nominal, beating the 4.5% nominal bond. - If you expect inflation **below 2.451%** (say 1.5%), buy the **nominal bond** — the TIPS would give only  $\approx 2\% + 1.5\% = 3.5\%$  nominal, worse than 4.5%. - If you have *no view*, the TIPS still removes inflation *uncertainty* and may be worth a small give-up for that insurance.

**Quantifying the difference at 3.5% realised inflation** (approximate nominal returns): TIPS  $\approx 2.00\% + 3.50\% = 5.50\%$ ; nominal bond = 4.50% fixed. The TIPS outperforms by  $\approx 100$  bps per year — the payoff for having correctly bet inflation would exceed breakeven.

## Example 3 — Floating-rate note cash flows in a rising-rate world

**Setup.** A 2-year FRN, face \$1,000, coupon = **3-month SOFR + 0.40%**, reset quarterly, paid quarterly. Suppose SOFR at the four resets of year 1 comes in as below. (Quarterly coupon = annual rate  $\div$  4.)

| Quarter | SOFR at reset | Coupon rate = SOFR + 0.40% | Quarterly coupon = rate/4 $\times$ \$1,000 |
|---------|---------------|----------------------------|--------------------------------------------|
| Q1      | 3.00%         | 3.40%                      | \$8.50                                     |
| Q2      | 3.75%         | 4.15%                      | \$10.375                                   |
| Q3      | 4.50%         | 4.90%                      | \$12.25                                    |
| Q4      | 5.00%         | 5.40%                      | \$13.50                                    |

**Year-1 coupons total:**  $8.50 + 10.375 + 12.25 + 13.50 = \$44.625$ .

**Contrast with a fixed 3.40% bond.** A comparable fixed-rate bond struck at issue (3.40%) would have paid  $3.40\% \times 1,000 = \$34.00$  for the whole year — and its *price* would have **fallen** sharply as SOFR climbed from 3% to 5%. The FRN instead:

1. **Raised its coupon** from \$8.50 to \$13.50 per quarter as rates rose — the income kept pace.

2. **Held its price near \$1,000** throughout, because each reset re-struck the coupon to the market. Its duration was only  $\sim 0.25$  years (to the next reset), so the 200 bps rate move barely dented the price.

The fixed-rate holder was hurt twice (low income *and* capital loss); the FRN holder was protected on both counts. This is precisely the interest-rate protection an FRN is built for.

**Discount-margin nuance.** If, mid-life, the issuer's credit worsened so the market demanded  $\text{SOFR} + 0.90\%$  ( $\text{DM} = 0.90\%$  vs quoted margin  $0.40\%$ ), the FRN's price would drop below par by roughly  $(0.90\% - 0.40\%) \times \text{remaining duration}$ . The FRN neutralised *rate* risk but not *credit* risk — the residual exposure.

## 6. Connections

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- **Duration and interest-rate risk (Ch. on duration).** ILBs have *real-rate* duration — they are still long instruments and lose value if *real* yields rise, even though they are inflation-protected. FRNs have near-zero rate duration but full *spread* duration. The chapter's whole point is *which* duration you keep and which you shed.
- **Yield curve & the Fisher decomposition.** The nominal curve = real curve (from ILBs) + breakeven inflation curve. Central banks and traders read the TIPS market to extract *market-implied inflation expectations* — a direct application of Fisher across maturities.
- **Monetary policy.** FRNs are the natural instrument when rates are expected to *rise* (they track the policy rate). ILBs are the hedge against a central bank *losing control of inflation*. The RBI launched IIBs in 2013 precisely when Indian CPI inflation was in double digits.
- **Credit risk.** An FRN strips out rate risk but isolates and *concentrates* credit-spread risk into the discount margin — connecting to credit analysis and spread duration.
- **Liabilities / ALM.** Pension funds and insurers with inflation-linked liabilities (indexed pensions) buy ILBs to *match* liabilities. Banks with floating-rate assets fund with FRNs to match. The instruments exist to hedge specific liability shapes.
- **TIPS ↔ inflation swaps.** The breakeven rate is arbitrage-linked to the fixed leg of a zero-coupon inflation swap — the derivatives market and the cash ILB market police each other.

## 7. Key Terms

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- **Inflation-linked bond (ILB):** a bond whose principal (and hence coupon) is scaled by a price index to preserve real purchasing power.
- **TIPS:** Treasury Inflation-Protected Securities — US capital-indexed government ILBs, semiannual, CPI-U with a  $\sim 3$ -month lag, deflation floor at maturity.
- **IIB / IINSS-C:** India's sovereign Inflation Indexed Bonds (capital-indexed, WPI then CPI) and the retail IINSS-C (coupon =  $1.5\% + \text{CPI}$ , principal fixed).
- **Index ratio:** current reference CPI  $\div$  base CPI; the multiplier applied to face to get adjusted principal.
- **Adjusted (indexed) principal:** face  $\times$  index ratio; the inflation-uplifted principal on which coupons are computed.
- **Real yield:** the yield in purchasing-power terms; the ILB's stated/quoted yield.
- **Nominal yield:** the yield in money terms; the vanilla bond's quoted yield.
- **Breakeven inflation (BEI):** nominal yield  $-$  real yield (approx); the inflation rate that equalises returns on nominal and inflation-linked bonds.



- **Fisher equation:**  $(1 + \text{nominal}) = (1 + \text{real})(1 + \text{inflation})$ .
- **Phantom income:** the annual, unpaid principal accretion of an ILB that is nonetheless taxable in the accrual year (US).
- **Deflation floor:** ILB feature guaranteeing redemption at no less than original face.
- **Floating-rate note (FRN):** a bond whose coupon resets each period to a reference rate plus a fixed spread.
- **Reference rate:** the benchmark the FRN tracks (SOFR, MIBOR, T-bill rate).
- **Quoted margin / spread:** the fixed add-on to the reference rate, set at issue for the issuer's credit.
- **Discount margin (DM):** the FRN's yield measure — the spread over projected reference rates that prices the note;  $\text{DM} > \text{quoted margin} \Rightarrow \text{price below par}$ .
- **Set-in-advance, pay-in-arrears:** the coupon rate is fixed at the start of the period and paid at its end.

## 8. Common Confusions

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**"Inflation-linked bonds have no interest-rate risk."** False. ILBs are inflation-*protected*, not rate-immune. If *real* yields rise, a long ILB's price falls just like any long bond. They shed inflation risk, not duration. (The instrument that sheds duration is the FRN.)

**"Floating-rate notes protect against inflation."** Only indirectly and imperfectly. FRNs track *nominal* short rates. If inflation rises and the central bank hikes, the reference rate — and the FRN coupon — rises with it, giving *partial* inflation protection. But if inflation rises while the central bank keeps rates low (financial repression, negative real rates), the FRN does **not** protect real purchasing power. Only an ILB guarantees a real return.

**"Breakeven inflation is the market's inflation forecast."** Close but not exact.  $\text{BEI} = \text{expected inflation} + \text{inflation risk premium} - \text{ILB liquidity premium}$ . It is a market-implied number, not a clean forecast, and the premia can move it a few tenths either way.

**"The deflation floor protects every cash flow."** No — the `max(adjusted, face)` floor applies only to the **maturity redemption**. Coupons during a deflationary period are still computed on the (reduced) adjusted principal and can fall below the nominal-coupon-on-face level.

**"A TIPS coupon rate rises with inflation."** The *rate* is fixed (that's the real coupon). What grows is the *principal it's applied to*, so the coupon *rupees/dollars* grow while the *rate* stays constant. Contrast the Indian retail IINSS-C, where the *rate itself* ( $1.5\% + \text{CPI}$ ) moves.

**"An FRN always trades at par."** Only *at reset* and only if credit is unchanged. Between resets, and whenever the issuer's discount margin diverges from the quoted margin (credit deterioration), the price drifts from par.

**"FRN duration equals its maturity."** No — a 5-year quarterly-resetting FRN has effective duration  $\approx 0.25$  years (to next reset), not 5. Its *spread* duration, however, is close to full maturity.

**"Real yield can't be negative."** It can, and frequently is. In 2020–21 many TIPS had negative real yields, meaning investors *paid* for guaranteed inflation protection, accepting a return below inflation to lock in certainty.



## 9. Recap

- Vanilla fixed bonds expose investors to two enemies: **inflation** (erodes real value) and **rising rates** (crush price). Two instruments float different legs to fight each.
- **Inflation-linked bonds** fix the *real* coupon and float the *principal* via an index ratio (current CPI ÷ base CPI). Coupon = fixed real rate × adjusted principal; redemption carries a **deflation floor** at maturity. The investor locks a guaranteed **real yield**. TIPS (US, CPI, semiannual) and India's sovereign IIBs (WPI→CPI) are capital-indexed; India's retail IINSS-C used coupon = 1.5% + CPI with a fixed principal.
- The **Fisher equation** —  $(1 + \text{nominal}) = (1 + \text{real})(1 + \text{inflation})$  — bridges the two worlds and yields **breakeven inflation** = nominal yield – real yield. Buy the ILB if you expect inflation above breakeven; buy the nominal bond if below.
- **Floating-rate notes** fix the principal at par and float the *coupon* = reference rate + quoted margin, resetting every period. This keeps the price near par and duration tiny ( $\approx$  time to next reset). The residual risk is **credit spread**, measured by the **discount margin**.
- **Protection map**: ILBs defend *purchasing power* against inflation but keep real-rate duration; FRNs defend *price* against rising rates but keep credit risk. Neither is a free lunch — each swaps one exposure for another.

The decision tree below maps an investor's dominant fear to the instrument that neutralises it.

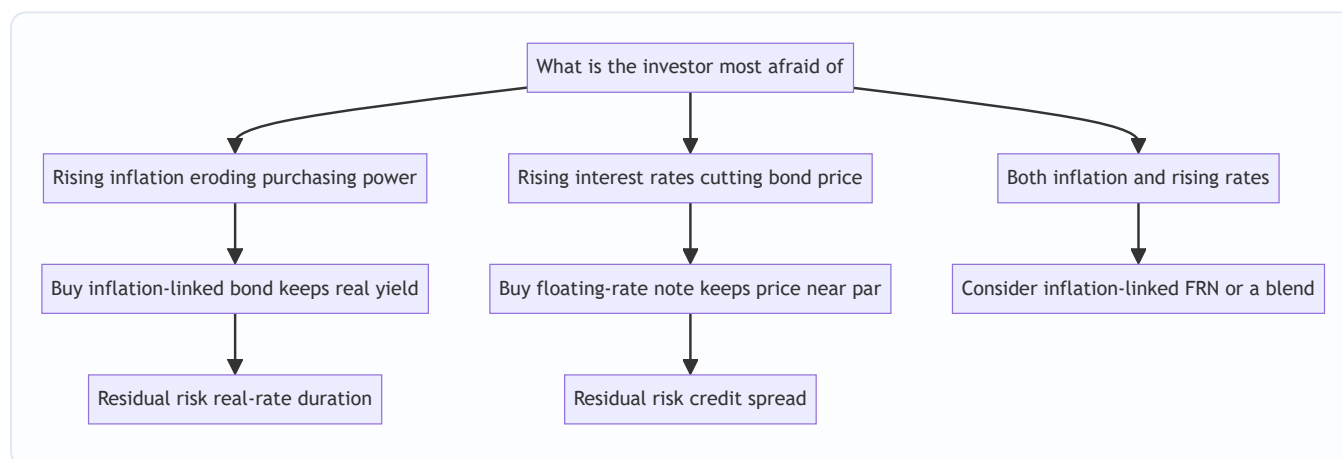


Figure 16.3 — Matching the investor's dominant fear to the protective instrument, and naming the exposure each one leaves behind.

## 10. Quick-Reference / Interview Points

### One-liners to have ready:

- "ILBs protect the value of your *money*; FRNs protect the value of your *bond*."
- "A TIPS fixes the real *rate* and floats the *principal*; an FRN fixes the *principal* at par and floats the *coupon rate*."
- "Breakeven inflation = nominal yield – real yield. Above it, the linker wins; below it, the nominal wins."

### Formulas to reproduce on demand:

| Concept             | Formula                                                                             |
|---------------------|-------------------------------------------------------------------------------------|
| Index ratio         | $\text{CPI}(\text{date}) \div \text{CPI}(\text{base})$                              |
| Adjusted principal  | $\text{Face} \times \text{index ratio}$                                             |
| ILB coupon          | $(\text{real rate} \div f) \times \text{adjusted principal}$                        |
| Redemption (TIPS)   | $\max(\text{adjusted principal}, \text{face})$                                      |
| Fisher (exact)      | $(1 + \text{nominal}) = (1 + \text{real})(1 + \text{inflation})$                    |
| Real yield          | $(1 + \text{nominal}) / (1 + \text{inflation}) - 1$                                 |
| Breakeven inflation | $(1 + \text{nominal}) / (1 + \text{real}) - 1 \approx \text{nominal} - \text{real}$ |
| FRN coupon rate     | reference rate + quoted margin                                                      |
| FRN price signal    | $\text{DM} > \text{quoted margin} \Rightarrow \text{price below par}$               |

### Likely interview questions and crisp answers:

- "Does a TIPS have interest-rate risk?" — Yes, **real**-rate duration. It's inflation-protected, not rate-immune; a long TIPS falls if real yields rise.
- "Why is an FRN's duration so low?" — The coupon resets to market every period, so its price is pulled back to par at each reset; effective duration  $\approx$  time to next reset ( $\sim 0.25y$  for quarterly). But spread duration  $\approx$  full maturity.
- "If inflation rises, is an FRN a good hedge?" — Only if the central bank hikes in response, since the FRN tracks nominal short rates. Under financial repression (rates held below inflation) the FRN fails to protect real value; only an ILB guarantees a real return.
- "What does breakeven inflation actually contain?" — Expected inflation + inflation risk premium – liquidity premium. It's market-implied, not a pure forecast.
- "Can real yields be negative?" — Yes; investors accept a sub-inflation return to lock in certainty (common 2020–21).
- "Difference between India's sovereign IIB and the retail IINSS-C?" — The IIB is capital-indexed (principal indexed to WPI/CPI, fixed real coupon on it); the IINSS-C paid coupon = 1.5% + CPI with the *principal* fixed at face — an additive, not a capital-indexed, design.
- "What risk does an FRN NOT remove?" — Credit-spread risk, captured by the discount margin; a widening DM above the quoted margin pushes the price below par.

**Portfolio takeaways:** Use ILBs to hedge inflation-linked liabilities (indexed pensions) and to lock real returns; use FRNs to reduce duration and ride rising-rate cycles while staying near par. Combine them — or use an inflation-linked FRN — when both inflation and rate risk are live concerns.

# Q&A — Inflation-Linked and Floating-Rate Bonds

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A practice bank covering inflation-linked bonds (ILBs/TIPS/IIBs) and floating-rate notes (FRNs). Work each question before reading the answer.

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## Section A — Concept Checks

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### A1. What two distinct risks does a plain vanilla fixed-coupon bond force the investor to bear, and which instrument neutralises each?

A vanilla bond bundles two exposures. First, **inflation risk**: a fixed coupon and fixed redemption lose real purchasing power as prices rise, so the investor grows poorer in real terms even while "earning interest." Second, **interest-rate (price) risk**: if market yields rise, the fixed stream is now too stingy and the bond's price falls until its yield matches the market. Inflation-linked bonds neutralise the first (they index principal to a price index, locking a real yield); floating-rate notes neutralise the second (they reset the coupon to a reference rate, keeping price near par). Mnemonic: **ILBs protect the value of your money; FRNs protect the value of your bond.**

### A2. In an inflation-linked bond, which leg is fixed and which floats? Contrast with an FRN.

An ILB fixes the **real coupon rate** and floats the **principal** (scaled by the index ratio). Because the coupon is a fixed percentage of an ever-growing principal, the coupon rupees also grow with inflation. An FRN does the opposite: it fixes the **principal at par** and floats the **coupon rate** (reference rate + spread) at each reset. So ILBs float the principal; FRNs float the coupon.

### A3. Define the index ratio and adjusted principal.

The index ratio is the current reference CPI divided by the base (issue-date) CPI:  $\text{CPI}(\text{date}) / \text{CPI}(\text{base})$ . The adjusted (indexed) principal is  $\text{Face} \times \text{index ratio}$ . If CPI has risen 6% since issue, the index ratio is 1.06 and the adjusted principal is 6% above face — exactly offsetting the 6% loss of purchasing power.

### A4. Why does indexing the principal automatically protect the coupon too?

Because the coupon is computed as  $(\text{real rate} / \text{frequency}) \times \text{adjusted principal}$ . The real *rate* never changes, but the principal it is applied to grows with inflation, so the coupon cash amount inherits the same uplift. The real value of every cash flow is held constant by construction.

### A5. State the Fisher equation and the approximation for the nominal yield.

Exact:  $(1 + y_{\text{nominal}}) = (1 + y_{\text{real}}) \times (1 + \pi_{\text{expected}})$ . Rearranged,  $y_{\text{real}} = (1 + y_{\text{nominal}}) / (1 + \pi) - 1$ . The small-number approximation is  $y_{\text{nominal}} \approx y_{\text{real}} + \pi$ . This is the bridge that lets a nominal bond and an ILB of the same maturity be compared.

### A6. What is breakeven inflation (BEI), and what does it actually contain?

BEI is the inflation rate at which a nominal bond and an inflation-linked bond of the same maturity deliver the same nominal return:  $\text{BEI} \approx y_{\text{nominal}} - y_{\text{real}}$ . It is *not* a clean forecast — it equals **expected inflation + inflation risk premium – ILB liquidity premium**. It is the best market-implied read on inflation, but the premia can shift it a few tenths either way.

### A7. Explain the deflation floor and its one key limitation.

At maturity, a TIPS repays  $\max(\text{adjusted principal}, \text{face})$ , so cumulative deflation cannot pull the redemption below original par. The limitation: the floor applies **only to the final redemption**, not to coupons. During a deflationary stretch, coupons are still computed on the (sub-par) adjusted principal, so coupon cash can fall below the nominal-coupon-on-face level.

#### A8. Why is an FRN's effective duration so small?

At each reset the coupon re-strikes to the current market rate + spread, so right after a reset the bond is worth par. The price can only wander from par between resets, by the small value of one period's mispricing. Effective (rate) duration  $\approx$  time to next reset ( $\sim 0.25$  years for a quarterly resetter), not years-to-maturity. Its **spread duration**, however, is close to full maturity.

#### A9. What is "phantom income"?

In the US, the annual principal accretion (the inflation uplift on a TIPS) is taxable in the year it accrues even though no cash is received until maturity. This unpaid-but-taxed accretion is "phantom income," which is why TIPS are often held in tax-deferred accounts.

#### A10. "Set in advance, pay in arrears" — what does it mean for an FRN?

The coupon rate is *fixed* at the start of each period using the reference rate observed at that reset, and the coupon is *paid* at the end of the period. So the investor always knows the rate for the coming period at its start.

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## Section B — Numerical / Applied (full solutions)

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**B1. TIPS cash flow.** Face \$1,000, real coupon 2.00% annual (1.00% semiannual), base CPI 250.00. Reference CPI after 6 months = 252.50, after 12 months = 256.00. Compute both coupons and the year-end adjusted principal.

*Solution.* Index ratios:  $252.50/250 = 1.0100$ ;  $256.00/250 = 1.0240$ . - Adjusted principal at period 1 =  $1,000 \times 1.0100 = \mathbf{\$1,010.00}$ ; coupon =  $1.00\% \times 1,010.00 = \mathbf{\$10.10}$ . - Adjusted principal at period 2 =  $1,000 \times 1.0240 = \mathbf{\$1,024.00}$ ; coupon =  $1.00\% \times 1,024.00 = \mathbf{\$10.24}$ . - Total year-1 coupons =  $10.10 + 10.24 = \mathbf{\$20.34}$ ; year-end adjusted principal =  $\mathbf{\$1,024.00}$  (a \$24 accretion, taxable in the US though unpaid).

**B2. Real-return reconciliation.** Using B1, verify the investor earned  $\approx 2\%$  real.

*Solution.* Total one-year economic gain = coupons + accretion =  $20.34 + 24.00 = \$44.34$  on a \$1,000 base  $\rightarrow$  nominal return 4.434%. One-year inflation =  $250 \rightarrow 256 = 2.40\%$ .  $y_{\text{real}} = 1.04434 / 1.02400 - 1 = 1.01986 - 1 = 1.99\% \approx 2.00\%$ .  $\checkmark$  The real return lands on the stated 2% real coupon; inflation passed straight through the nominal cash flows.

**B3. Deflation floor.** Same bond, but CPI falls to 245 by maturity. What is redeemed?

*Solution.* Index ratio =  $245/250 = 0.98$ , adjusted principal = \$980. Redemption =  $\max(980, 1,000) = \mathbf{\$1,000}$ . The investor is protected against sub-par principal at maturity — though coupons during the deflation would have been computed on the \$980 base.

**B4. Breakeven inflation.** 10-year nominal Treasury yields 4.50%; 10-year TIPS (real) yields 2.00%. Compute BEI exactly and approximately, then verify.

*Solution.* - Exact:  $\text{BEI} = 1.0450 / 1.0200 - 1 = 1.02451 - 1 = 2.451\%$ . - Approx:  $4.50\% - 2.00\% = 2.50\%$ . - Verify with Fisher:  $(1.0200)(1.02451) - 1 = 1.0450 - 1 = 4.50\%$  = the nominal Treasury yield.  $\checkmark$

The two bonds break even at 2.451% inflation.

**B5. Breakeven decision.** Using B4, you expect inflation of 3.5%. Which bond, and by roughly how much per year?

*Solution.* 3.5% > breakeven 2.451%, so buy the **TIPS**. Approximate nominal returns: TIPS  $\approx 2.00\% + 3.50\% = 5.50\%$ ; nominal bond = 4.50% fixed. The TIPS outperforms by  $\approx$  **100 bps per year**. (If instead you expected 1.5% inflation, below breakeven, the nominal bond's 4.50% beats the TIPS's  $\approx 3.5\%$ .)

**B6. FRN coupons in a rising-rate world.** 2-year FRN, face \$1,000, coupon = 3-month SOFR + 0.40%, reset and paid quarterly. SOFR at the four resets: 3.00%, 3.75%, 4.50%, 5.00%. Compute each quarterly coupon and the year-1 total.

*Solution.* Coupon rate = SOFR + 0.40%; quarterly coupon = rate/4  $\times$  1,000. - Q1: 3.40%  $\rightarrow 0.85\% \times 1,000 =$  **\$8.50** - Q2: 4.15%  $\rightarrow$  **\$10.375** - Q3: 4.90%  $\rightarrow$  **\$12.25** - Q4: 5.40%  $\rightarrow$  **\$13.50** - Year-1 total = 8.50 + 10.375 + 12.25 + 13.50 = **\$44.625**.

**B7. FRN vs fixed contrast.** Using B6, a fixed 3.40% bond struck at issue would have paid what for the year, and what happens to each bond's price as SOFR climbs 3% $\rightarrow$ 5%?

*Solution.* Fixed bond pays 3.40%  $\times$  1,000 = **\$34.00** for the year — far less than the FRN's \$44.625 — and its **price falls sharply** as rates rise (it has full duration). The FRN raised its coupon from \$8.50 to \$13.50 and held its price near \$1,000, because each reset re-struck the coupon to market; its duration was only  $\sim 0.25$  years, so the 200 bps move barely dented the price. The fixed holder is hurt twice (low income *and* capital loss); the FRN holder is protected on both counts.

**B8. Discount margin.** The FRN in B6 has quoted margin 0.40%. Mid-life the issuer's credit worsens and the market demands SOFR + 0.90%. What happens to the price, and what residual risk does this illustrate?

*Solution.* The discount margin (0.90%) now exceeds the quoted margin (0.40%), so the fixed 0.40% spread is too thin. Price falls below par by roughly  $(0.90\% - 0.40\%) \times \text{remaining spread duration}$ . This shows the FRN neutralised *interest-rate* risk but **retained credit-spread risk** — the residual exposure, measured by the discount margin.

**B9. Fisher reverse-check.** A linker offers 1.20% real. You forecast 4% inflation. What nominal return do you expect, exactly and approximately?

*Solution.* Exact:  $(1.0120)(1.0400) - 1 = 1.05248 - 1 = 5.25\%$ . Approx:  $1.20\% + 4.00\% = 5.20\%$ . The exact figure is slightly higher because of the cross-term (real  $\times$  inflation).

**B10. Negative real yield.** A TIPS is quoted at a real yield of  $-0.80\%$ . With expected inflation of 2.5%, what nominal return is implied, and what is the investor effectively doing?

*Solution.*  $(1 - 0.0080)(1.0250) - 1 = 0.9920 \times 1.0250 - 1 = 1.0168 - 1 = 1.68\%$  nominal. The investor accepts a return *below* inflation ( $-0.80\%$  real) to lock in guaranteed inflation protection — effectively paying an insurance premium for certainty, as was common in 2020–21.

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## Section C — Interview-Style (model answers)

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### C1. "Does a TIPS have interest-rate risk?"

Yes — it has **real-rate duration**. A TIPS is inflation-*protected*, not rate-immune. It is still a long-dated instrument, and if *real* yields rise, its price falls just like any long bond. What it sheds is inflation risk, not

duration. The instrument that sheds duration is the FRN, not the ILB.

### C2. "Why is an FRN's duration so low, and what duration does it keep?"

Because the coupon resets to the market reference rate + spread every period, the price is pulled back to par at each reset, so it can only drift within one reset window. Effective rate duration  $\approx$  time to next reset ( $\sim 0.25$  years for a quarterly note). However, its **spread duration is close to full maturity** — if the issuer's credit spread widens, the price still moves substantially. So an FRN keeps credit-spread risk while shedding rate risk.

### C3. "If inflation rises, is an FRN a good hedge?"

Only *indirectly and conditionally*. FRNs track *nominal* short rates. If inflation rises and the central bank hikes in response, the reference rate — and the coupon — rises with it, giving partial protection. But under financial repression (rates deliberately held below inflation, negative real rates), the FRN does **not** protect real purchasing power. Only an inflation-linked bond guarantees a real return regardless of policy.

### C4. "What does breakeven inflation actually contain — is it the market's forecast?"

It's market-*implied*, not a clean forecast. BEI = expected inflation + **inflation risk premium** – **ILB liquidity premium**. The risk premium reflects compensation for inflation uncertainty; the liquidity premium reflects that linkers are typically less liquid than nominals. These can move BEI a few tenths of a percent away from pure expectations.

### C5. "Can real yields be negative? What would justify buying at a negative real yield?"

Yes, and it happens — many TIPS traded at negative real yields in 2020–21. Buying at a negative real yield means accepting a guaranteed sub-inflation return in exchange for *certainty* of inflation protection. An investor with inflation-linked liabilities, or one who fears an inflation surprise, may rationally pay this premium rather than risk a nominal bond being eroded.

### C6. "Difference between India's sovereign IIB and the retail IINSS-C?"

The **sovereign IIB (2013)** is capital-indexed like TIPS: the principal is indexed (WPI initially, later CPI) and a fixed real coupon is paid on that indexed principal — so both principal and coupon are inflation-protected. The **retail IINSS-C (Dec 2013)** used a simpler *additive* design: coupon = 1.5% p.a. + realised CPI inflation, with the **principal fixed at face**. So the IINSS-C protects income against inflation but does not grow the principal — an additive structure, not a capital-indexed one. The sovereign series' switch from WPI to CPI reflected CPI becoming the RBI's official inflation target.

### C7. "What risk does an FRN NOT remove?"

Credit-spread risk. The FRN strips out interest-rate risk but isolates and concentrates credit exposure into the **discount margin**. If the issuer's credit deteriorates, the market demands a discount margin above the fixed quoted margin, and the price falls below par. The quoted margin is locked at issue; only the price can adjust.

### C8. "How would you use these two instruments to hedge liabilities?"

Match the instrument to the liability shape. Pension funds and insurers with **inflation-linked liabilities** (indexed pensions) buy ILBs to match real cash flows and lock a real return. Institutions with **floating-rate assets or a desire to minimise duration** in a rising-rate cycle use FRNs to stay near par and let income track the policy rate. If both inflation and rate risk are live, blend them or use an inflation-linked FRN.

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## Section D — MCQs (with reasoning)

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**D1.** In a capital-indexed TIPS, the coupon *rate* is: (a) reset each period to CPI (b) fixed in real terms (c) fixed in nominal terms (d) CPI + a fixed spread

**Answer: (b).** The real rate is fixed; what grows is the principal it's applied to, so coupon *cash* grows while the *rate* stays constant. (c) describes a vanilla bond; (a)/(d) describe the additive Indian IINSS-C, not a capital-indexed TIPS.

**D2.** A 5-year FRN resetting quarterly has an effective (rate) duration closest to: (a) 5 years (b) 2.5 years (c) 0.25 years (d) 0 exactly

**Answer: (c).** Duration  $\approx$  time to next reset. It is not literally zero because there is still a fraction of a period of rate exposure; it is nowhere near the 5-year maturity.

**D3.** Nominal yield is 5.0%, real yield is 2.0%. Approximate breakeven inflation is: (a) 2.5% (b) 3.0% (c) 7.0% (d) 10.0%

**Answer: (b).** BEI  $\approx$  nominal – real = 5.0% – 2.0% = 3.0%. (Exact:  $1.05/1.02 - 1 = 2.94\%$ .)

**D4.** The deflation floor on a TIPS guarantees: (a) every coupon is at least the nominal-on-face amount (b) redemption is at least original face (c) the index ratio never falls below 1 (d) the real yield stays positive

**Answer: (b).** Redemption = max(adjusted principal, face). It protects only the maturity redemption; coupons can still fall below the nominal-on-face level during deflation, so (a) is wrong.

**D5.** An FRN's price falls below par mainly when: (a) the reference rate rises (b) the reference rate falls (c) the issuer's discount margin exceeds the quoted margin (d) inflation rises

**Answer: (c).** Rate moves are absorbed by resets. Price drifts below par when credit deteriorates so the required discount margin exceeds the fixed quoted margin. (a) is largely neutralised by the reset mechanism.

**D6.** "Phantom income" on a TIPS refers to: (a) the coupon paid in kind (b) the inflation accretion taxed before it is received in cash (c) the deflation adjustment (d) the liquidity premium in BEI

**Answer: (b).** US tax rules tax the annual principal accretion in the accrual year, though the cash arrives only at maturity.

**D7.** You expect inflation of 2%. The market's breakeven inflation is 3%. You should prefer: (a) the TIPS (b) the nominal bond (c) indifferent (d) cannot tell

**Answer: (b).** Your expected inflation (2%) is below breakeven (3%), so the nominal bond is expected to outperform; the linker only wins if realised inflation exceeds breakeven.

**D8.** Which statement is TRUE? (a) ILBs have no interest-rate risk (b) FRNs fully hedge inflation (c) ILBs shed inflation risk but keep real-rate duration (d) FRNs have high rate duration and low spread duration

**Answer: (c).** ILBs remain long instruments exposed to real-rate moves. (a) and (b) are classic confusions; (d) is exactly backwards — FRNs have low rate duration and high spread duration.

**D9.** India's retail IINSS-C bond paid a coupon of: (a) a fixed real rate on indexed principal (b) 1.5% + realised CPI inflation, principal fixed (c) WPI-linked principal with a fixed coupon (d) SOFR + spread

**Answer: (b).** The IINSS-C was additive (1.5% + CPI) with principal fixed at face — distinct from the capital-indexed sovereign IIB in (a)/(c).

**D10.** In "set in advance, pay in arrears," the coupon rate is: (a) fixed at period end, paid at start (b) fixed at period start, paid at period end (c) fixed and paid both at end (d) unknown until maturity

**Answer: (b).** The rate is observed and fixed at the reset (start of period); the cash is paid at the end.

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*Self-verification note: B1–B10 solved and cross-checked against the chapter's worked examples and formula table; Fisher reverse-checks (B2, B4, B9) tie back exactly to the stated nominal yields. Formula usage — index ratio, adjusted principal,  $\max(\text{adjusted}, \text{face})$ , Fisher exact/approx, BEI, FRN coupon = reference + margin, and  $DM > \text{quoted margin} \Rightarrow \text{sub-par price}$  — matches Section 10's quick-reference table.*



# Chapter 17 — High-Yield and Distressed Debt

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## 1. The Problem / Need

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Not every company that borrows is a fortress balance sheet. For every Apple or Johnson & Johnson that can issue ten-year debt at a whisker over Treasuries, there are hundreds of leveraged buyout targets, turnaround stories, cyclical manufacturers, and young growth firms whose credit is genuinely uncertain. These companies still need capital — to refinance, to fund acquisitions, to bridge a rough patch. The bond market has to decide two things about them: *will they lend at all*, and *at what price*.

The investment-grade framework of Chapters 15 and 16 breaks down here. In investment grade, default is a tail event; you are mostly pricing liquidity, mild spread drift, and rating migration. You can, to a first approximation, treat the promised yield as the expected yield. In high yield you cannot. Default is a *base-rate* event — something like 3-5% of issuers default in an average year, and 10%+ in a bad one. The promised yield is a fiction that only materializes if the company survives; the *expected* return is a probability-weighted blend of "you get paid" and "you fight over the wreckage in bankruptcy court."

This creates a distinct discipline. You need a way to think about companies where the central question is not "how much extra yield for liquidity and volatility?" but "what is the probability this borrower fails to pay, and if it does, how much do I recover?" And at the far end of the spectrum — distressed debt — the question inverts entirely: the bonds already trade at 40 or 50 cents on the dollar, the market has *already* priced in serious trouble, and the investor is no longer a passive lender but an active participant in a restructuring, sometimes buying debt precisely *because* they intend to convert it into ownership of the reorganized company.

This chapter builds the toolkit for that world: what junk bonds are, why their spreads are so wide, how to analyze them, how default and recovery interact to produce credit returns, how distressed investing works, and how the whole thing pulses with the credit cycle.

## 2. Core Idea

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**High-yield bonds are the debt of below-investment-grade issuers, rated BB+/Ba1 or lower, where compensation for bearing default risk — not just liquidity or duration — is the dominant driver of return.** The higher promised yield is not a free lunch; it is a payment for accepting real losses that will occur in a predictable *fraction* of holdings, even if not in any single one.

Three ideas anchor everything:

1. **The spread is compensation for expected loss plus a risk premium.** Break the credit spread into (a) expected loss = probability of default × loss given default, and (b) everything left over — a premium for the *uncertainty* of that loss, illiquidity, and market risk aversion. In high yield, both pieces are large and both move a lot.
2. **Return is a portfolio phenomenon, driven by default and recovery.** You do not "earn the yield." You earn the yield on the survivors, minus the losses on the defaulters, net of recovery. Credit investing is fundamentally actuarial: you are underwriting a book of risks where the average outcome is what matters.
3. **It all breathes with the credit cycle.** Spreads, default rates, and recoveries are not constants — they swing together through repair, expansion, distribution/complacency, and downturn phases. Buying high

yield when spreads are 300 bps is a completely different trade from buying it at 900 bps, even though the "asset class" is nominally the same.

Distressed debt sits at the tail of this distribution: securities of issuers at or near default, trading at deep discounts, where the analysis shifts from cash-flow-and-coupon to *balance-sheet-and-recovery* — what the enterprise is worth in a restructuring and where your claim sits in the priority waterfall.

### 3. Why / How It Works

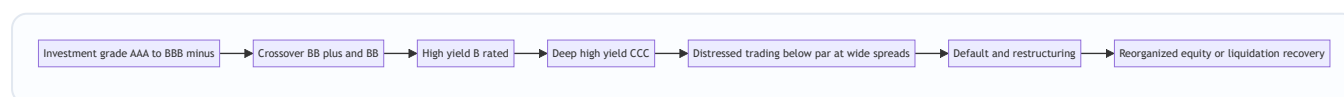
**Why does a below-investment-grade issuer exist as a category at all?** Ratings agencies draw a bright line at BBB–/Baa3. Above it, an issuer is "investment grade"; at BB+/Ba1 and below, "speculative grade," "high yield," or, less politely, "junk." The line matters enormously because of *institutional plumbing*: many insurers, pension funds, and bond index mandates are contractually or by regulation restricted to investment-grade paper. When an issuer falls below the line, a whole class of natural buyers is forced to sell (see "fallen angels," Chapter 15). That forced-selling dynamic, plus the genuine credit deterioration behind it, is why the yield jump across the BBB/BB boundary is so much larger than the jump between any two adjacent investment-grade notches.

**Why are spreads so much wider?** Because the *distribution of outcomes* is fundamentally different. Take a single-B issuer. Historically, its annual default probability might be ~3-4%, and if it defaults, senior unsecured holders might recover ~40 cents. Over a five-year horizon the cumulative default probability can exceed 20%. A lender who ignores this and prices the bond like a Treasury would systematically lose money. The spread must at minimum cover the expected credit loss, and rational investors demand more than that — a *credit risk premium* — because the losses are lumpy, correlated (many issuers default together in recessions, exactly when the investor's other assets are also hurting), and hard to diversify away entirely.

**How does the math connect spread to default?** A useful approximation, the "credit triangle," links three quantities:  $\text{spread} \approx \text{hazard rate} \times \text{loss given default}$ , i.e.,  $\text{spread} \approx (\text{annual default probability}) \times (1 - \text{recovery rate})$ . This is the break-even relationship: at this spread you are exactly compensated for expected loss and earn zero risk premium. Real spreads trade *wider* than this break-even because investors want compensation for uncertainty and illiquidity. Comparing the market spread to the break-even spread tells you how much risk premium you are being paid — the heart of relative-value credit analysis.

**How does distressed investing work mechanically?** When a company can no longer service its debt, it restructures — either out of court (an exchange offer, an amend-and-extend) or in court (Chapter 11 in the US). In a restructuring, the *absolute priority rule* governs who gets paid: secured creditors first, then senior unsecured, then subordinated, then equity last (often wiped out). Distressed investors buy claims at a discount, form or join creditor committees, and negotiate the reorganization plan. A common play is the *loan-to-own* or *fulcrum security* strategy: identify the layer of the capital structure that will convert into the new equity of the reorganized company (the "fulcrum"), buy it cheaply, and emerge owning the recapitalized business. The return comes not from coupons but from the gap between the distressed purchase price and the recovery value delivered by the restructuring.

Figure 17.1 — Where high-yield and distressed sit on the credit-quality spectrum.



## 4. Full Content — Formulas and Framework

### 4.1 The rating map

| Category             | S&P / Fitch     | Moody's        | Character                 |
|----------------------|-----------------|----------------|---------------------------|
| Investment grade     | BBB– and above  | Baa3 and above | Default a tail event      |
| Crossover / upper HY | BB+, BB, BB–    | Ba1, Ba2, Ba3  | Speculative but resilient |
| Mid high yield       | B+, B, B–       | B1, B2, B3     | Meaningful default risk   |
| Deep high yield      | CCC+, CCC, CCC– | Caa1-Caa3      | High default probability  |
| Near/at default      | CC, C, D        | Ca, C          | Distressed / defaulted    |

The single most important boundary is BBB–/BB+ (Baa3/Ba1) — the investment-grade / high-yield line.

### 4.2 The spread decomposition

The credit spread  $s$  over the risk-free rate can be decomposed:

$$s \approx \text{Expected Loss} + \text{Credit Risk Premium} + \text{Liquidity Premium}$$

where **Expected Loss (annualized)** =  $\text{PD} \times \text{LGD}$ , with: - **PD** = probability of default (annual hazard rate  $\lambda$ ) - **LGD** = loss given default =  $(1 - \text{Recovery Rate}) = (1 - \text{RR})$

So the pure **break-even (fair) spread** that just covers expected credit loss is:

$$s_{\text{breakeven}} \approx \lambda \times (1 - \text{RR})$$

This is the **credit triangle**. Any two of {spread, default rate, recovery} pin down the third under the zero-risk-premium assumption. The amount by which the *market* spread exceeds  $s_{\text{breakeven}}$  is the risk premium plus liquidity compensation you are being paid to bear.

### 4.3 Cumulative default and survival

If the annual (marginal, roughly constant) default hazard is  $\lambda$ , the probability of *surviving*  $T$  years is:

$$\begin{aligned} \text{Survival}(T) &= (1 - \lambda)^T && \text{(discrete)} \\ &\approx e^{(-\lambda T)} && \text{(continuous hazard)} \end{aligned}$$

and cumulative default probability:

$$\text{CDP}(T) = 1 - (1 - \lambda)^T$$

Real hazard rates are not constant — they typically *rise* with time from issuance for high-quality names (nowhere to go but down) and can *fall* for distressed survivors (the ones who make it past the crunch stabilize).

## 4.4 Expected return of a credit position

Over a one-year horizon, ignoring rating migration and price drift, the expected return of a bond yielding spread  $s$  above the risk-free rate  $r_f$ :

$$\begin{aligned} E[\text{return}] &\approx r_f + s - (\text{PD} \times \text{LGD}) \\ &= r_f + s - \lambda(1 - \text{RR}) \end{aligned}$$

The  $-\lambda(1-\text{RR})$  term is the *expected credit loss* haircut. If the spread only equals the break-even spread, expected return collapses to the risk-free rate — you took all that risk for nothing. The gap between spread and break-even is your *expected excess return* (the harvested risk premium).

## 4.5 Recovery and the priority waterfall

Recovery depends heavily on seniority and collateral. Rough long-run US averages (they vary by cycle and industry):

| Claim type                           | Typical recovery (cents on \$) |
|--------------------------------------|--------------------------------|
| Senior secured (1st lien) bank loans | 65-80                          |
| Senior secured bonds                 | 50-65                          |
| Senior unsecured bonds               | 35-45                          |
| Subordinated bonds                   | 20-30                          |
| Preferred / equity                   | 0-10                           |

Recoveries are **countercyclical to default rates**: when many firms default at once (recession), asset values are depressed and recoveries fall — a double hit. This positive correlation between PD and LGD makes bad years worse than a naive independent-risk model predicts.

## 4.6 High-yield covenants

Because default risk is real, high-yield indentures carry *incurrence covenants* that restrict the borrower from taking value-destroying actions. Unlike investment-grade "maintenance-lite" documents, HY covenants typically include:

- **Restricted payments** — limits on dividends, buybacks, and distributions to equity.
- **Debt incurrence test** — new debt only allowed if a leverage/coverage ratio is met (e.g., fixed-charge coverage  $\geq 2.0\times$ ).
- **Liens covenant** — limits on granting security to other creditors (protects unsecured holders from being *primed*/subordinated).
- **Change-of-control put** — holders can force repayment at 101 if the company is acquired.
- **Asset-sale covenant** — proceeds must repay debt or be reinvested.

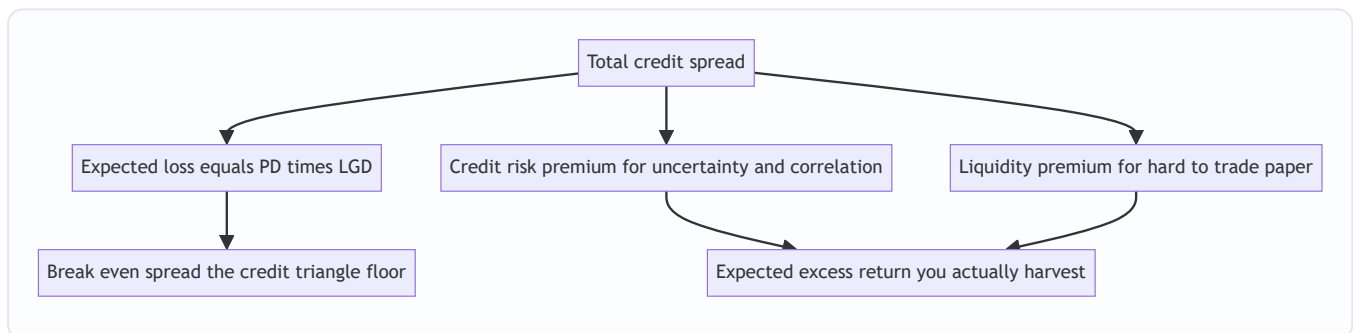
"Covenant-lite" (cov-lite) loans strip out *maintenance* tests (which are tested every quarter regardless of action) and rely only on incurrence tests. Cov-lite prevalence rises in frothy markets and is a classic late-cycle warning sign — it shifts negotiating leverage from lenders to sponsors/equity and tends to *lower* eventual recoveries.

## 4.7 Key credit ratios for HY analysis

| Ratio                 | Formula                                    | What it tells you                        |
|-----------------------|--------------------------------------------|------------------------------------------|
| Gross leverage        | Total Debt / EBITDA                        | Debt burden vs cash generation           |
| Net leverage          | (Total Debt – Cash) / EBITDA               | Leverage net of liquid buffer            |
| Interest coverage     | EBITDA / Interest expense                  | Cushion to service coupons               |
| Fixed-charge coverage | (EBITDA – Capex) / (Int + scheduled amort) | Ability to meet all fixed claims         |
| FCF conversion        | FCF / EBITDA                               | How much cash actually reaches creditors |
| Liquidity runway      | (Cash + Revolver availability) / burn      | Months of survival without refinancing   |

A high-yield analyst lives in these ratios plus the **maturity wall** (when does the debt come due, and can it be refinanced?) and the **liquidity picture** (can the company survive to that maturity?).

Figure 17.2 — The credit-spread decomposition.



## 5. Worked Examples

### Example 1 — The credit triangle and break-even spread

A single-B bond trades at a spread of 500 bps over Treasuries. Historical data for this rating: annual default probability  $\lambda = 4\%$ , recovery rate  $RR = 40\%$  (so  $LGD = 60\%$ ).

**Break-even spread:**

$$s_{\text{breakeven}} = \lambda \times (1 - RR) = 0.04 \times 0.60 = 0.024 = 240 \text{ bps}$$

**Interpretation:** The bond needs ~240 bps just to cover expected credit loss. It trades at 500 bps. So the *risk premium* + *liquidity compensation* being paid is:

$$500 - 240 = 260 \text{ bps}$$

**Expected one-year excess return over Treasuries:**

$$E[\text{excess}] \approx s - \lambda(1 - RR) = 500 - 240 = 260 \text{ bps}$$

**Reconciliation / sanity check:** Suppose you hold 100 identical such bonds, each \$100 face, coupon = risk-free + 5%. Say risk-free = 4%, coupon = 9%. In one year: - 96 bonds survive: each pays 9% → income =  $96 \times \$9 = \$864$ . - 4 bonds default (4%): recover 40% of \$100 = \$40 each, and assume they paid no coupon → \$160. Loss vs face on defaulters =  $4 \times \$60 = \$240$ . - Total value  $\approx (96 \times \$109 \text{ principal+coupon}) + (4 \times \$40) = \$10,464 + \$160 = \$10,624$  on \$10,000 invested.

Return  $\approx +6.24\%$ . Risk-free was 4%. Excess  $\approx \mathbf{2.24\%}$ , close to our 2.60% estimate (the small gap: defaulters also forfeit their coupon, a second-order effect the linear formula omits). The numbers reconcile — the approximation is slightly optimistic because it ignores lost coupon on defaulters.

## Example 2 — Cumulative default over a five-year holding

You buy a B-rated bond intending to hold five years. Annual hazard  $\lambda = 4\%$  (assumed constant).

### Survival probability over 5 years:

$$\text{Survival}(5) = (1 - 0.04)^5 = 0.96^5 = 0.8154$$

### Cumulative default probability:

$$\text{CDP}(5) = 1 - 0.8154 = 0.1846 \approx 18.5\%$$

So there is nearly a **one-in-five** chance this single bond defaults sometime in five years. That is why single-name concentration in high yield is dangerous and why the asset class is played as a diversified book.

### Reconciliation with continuous approximation:

$$\begin{aligned} \text{Survival}(5) &\approx e^{(-\lambda T)} = e^{(-0.04 \times 5)} = e^{(-0.20)} = 0.8187 \\ \text{CDP}(5) &\approx 1 - 0.8187 = 0.1813 \approx 18.1\% \end{aligned}$$

Discrete (18.5%) and continuous (18.1%) agree to within a few tenths of a percent — consistent, as expected for small  $\lambda$ .

## Example 3 — Distressed debt: buying the fulcrum security

Company XYZ is over-levered and heading into Chapter 11. Its capital structure (face values):

| Layer                     | Face (\$mm) | Cumulative claim (\$mm) |
|---------------------------|-------------|-------------------------|
| Senior secured (1st lien) | 300         | 300                     |
| Senior unsecured bonds    | 400         | 700                     |
| Subordinated notes        | 200         | 900                     |
| Equity                    | —           | —                       |

Independent valuation pegs the **reorganized enterprise value at \$550mm**.

**Apply the absolute priority waterfall:** - Senior secured (\$300mm) is fully covered → recovers 100% (\$300mm). Remaining value:  $\$550 - \$300 = \$250\text{mm}$ . - Senior unsecured (\$400mm claim) receives the

remaining \$250mm → recovery =  $250/400 = 62.5$  cents on the dollar. Remaining value: \$0. - Subordinated (\$200mm) receives **nothing** → 0 cents. Equity → wiped out.

**The fulcrum security is the senior unsecured bond** — it is the layer where value "runs out," so it converts partly to cash/new debt and partly to the *new equity* of the reorganized company.

**The trade:** Suppose senior unsecured bonds trade at **45 cents** (market fears EV is lower). If your \$550mm valuation is right, expected recovery is 62.5 cents.

Return if correct =  $(62.5 - 45) / 45 = 38.9\%$  on the position

If you buy \$400mm face at 45 = \$180mm invested, and recover \$250mm of value → gain \$70mm, i.e., +38.9%. **Reconciliation:**  $70/180 = 0.389$  ✓.

**Downside check:** If reorganized EV is only \$400mm (a bear case), senior secured takes \$300mm, leaving \$100mm for the unsecured → recovery =  $100/400 = 25$  cents. Buying at 45 → loss of  $(25-45)/45 = -44\%$ . This asymmetry — big upside if your EV estimate is right, big loss if it is low — is exactly why distressed investing is *valuation-driven credit work*, not coupon clipping. The entire edge is in estimating enterprise value and reading the priority waterfall correctly.

## 6. Connections

- **To credit spreads (Chapter 15/16):** High yield is the continuation of the same spread-analysis machinery, but where the *expected-loss* component dominates the spread rather than being a rounding error. Fallen angels (IG downgraded to HY) and rising stars (HY upgraded to IG) are the traffic across the boundary.
- **To duration and rates (Chapters 3-6):** High yield is *less* rate-sensitive than investment grade of the same maturity, because the spread is so large that credit dynamics swamp small rate moves, and because HY prices are partly driven by equity-like enterprise value. In risk-off episodes, HY spreads *widen* even as Treasuries rally — the two effects partly offset, giving HY lower effective duration.
- **To equity:** A high-yield bond behaves like a *short put on the firm's assets* (Merton model). Deep in distress, the bond trades on enterprise value and behaves almost like equity. Distressed debt and event-driven equity desks often analyze the same situations.
- **To CDS (Chapter 18):** Single-name and index CDS (CDX HY, iTraxx Crossover) let you take or hedge high-yield credit risk synthetically; the CDS-bond basis is a core relative-value trade.
- **To structured credit (Chapter 19):** CLOs are the largest buyers of leveraged loans; their bid drives loan-market technicals and the pace of new-issue.
- **To the business cycle (macro):** Default rates are the ultimate lagging-into-coincident indicator of corporate stress; HY spreads are a leading indicator watched by central banks as a financial-conditions gauge.

## 7. Key Terms

- **High-yield / junk / speculative-grade bond** — debt rated below BBB–/Baa3.
- **Fallen angel** — issuer downgraded from investment grade into high yield.
- **Rising star** — high-yield issuer upgraded into investment grade.

- **Default probability (PD) / hazard rate ( $\lambda$ )** — likelihood of default per period.
- **Loss given default (LGD)** — fraction of exposure lost in default =  $1 - \text{recovery rate}$ .
- **Recovery rate (RR)** — cents on the dollar recovered on a defaulted claim.
- **Credit triangle** — the break-even relation spread  $\approx \lambda \times (1 - \text{RR})$ .
- **Credit risk premium** — spread compensation above expected loss for bearing uncertain, correlated risk.
- **Absolute priority rule (APR)** — order in which claims are paid in bankruptcy: secured  $\rightarrow$  senior unsecured  $\rightarrow$  subordinated  $\rightarrow$  equity.
- **Fulcrum security** — the layer of the capital structure that converts to new equity in a restructuring; the "value breaks" here.
- **Loan-to-own** — distressed strategy of buying debt intending to convert it into ownership.
- **Incurrence vs maintenance covenant** — incurrence tested only on action; maintenance tested every period.
- **Covenant-lite (cov-lite)** — loan lacking maintenance covenants.
- **Priming** — new debt granted seniority above existing creditors.
- **Distressed exchange** — out-of-court restructuring swapping old debt for new (usually reduced) claims; counts as a default by rating agencies.
- **Maturity wall** — concentration of debt coming due in a given period.
- **Credit cycle** — the recurring expansion/contraction of credit availability, spreads, and defaults.

## 8. Common Confusions

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**"High yield means high return."** No — high *promised* yield. Expected return is the yield *minus* expected credit loss. A bond yielding 9% with 4% default risk and 40% recovery has an expected credit loss of  $\sim 2.4\%$ , so its expected return is closer to 6.6%. Promised yield is the ceiling you only reach if nothing defaults.

**"If I diversify, I eliminate default risk."** Diversification removes *idiosyncratic* (single-name) risk, so a portfolio's loss converges to the average default rate. But it does *not* remove *systematic* risk: defaults cluster in recessions, and recoveries fall at the same time, so a diversified HY book still suffers large correlated losses in a downturn. That residual systematic risk is precisely what the credit risk premium pays you for.

**"A wider spread is always a better deal."** A wider spread reflects higher perceived risk. It is only a better deal if it *exceeds* the compensation warranted by the true default/recovery outlook. At the depth of a recession, spreads can be enormous *and* still not wide enough if defaults are about to spike — or so wide that they overpay for risk. The relative-value question is always spread *versus* break-even.

**"Distressed = defaulted."** Distressed usually means *trading* at distressed levels (often defined as spread  $> 1000$  bps or price  $< \sim 70$ ), which can happen *before* any actual default. Defaulted debt is a subset; much distressed investing is in bonds of companies that have not yet — and may never — formally default.

**"Recovery is a fixed 40%."** The 40% figure is a long-run *average for senior unsecured*. Recovery varies enormously by seniority (secured 60-80%, subordinated 20-30%), by industry (asset-heavy vs asset-light), and by cycle (falls sharply when defaults spike). Using a single number blind is a classic modeling error.

**"HY has huge duration risk like long IG bonds."** HY typically has *lower* effective duration than same-maturity IG because credit dominates and because spreads move opposite to rates in stress. Its dominant risk is *spread/credit*, not rates.



**"Covenants protect me completely."** Incurrence covenants only bite when the company *takes an action*. A slowly deteriorating business can burn through cash for quarters without tripping any incurrence test — and cov-lite structures remove the maintenance tests that would have caught it. Covenant quality is a spectrum, and late-cycle documents are weakest exactly when you most need protection.

## 9. Recap

High-yield bonds are the debt of below-investment-grade issuers, where default risk is a base-rate reality rather than a tail event. The wide spreads compensate for two things: the *expected* credit loss (default probability  $\times$  loss given default) and a *risk premium* for the uncertainty and correlation of those losses. The credit triangle,  $\text{spread} \approx \lambda \times (1 - \text{RR})$ , gives the break-even floor; the market spread's excess over that floor is the risk premium you harvest. Because default is a portfolio phenomenon — you earn the yield on survivors minus losses on defaulters — high yield is fundamentally actuarial and demands diversification, though diversification cannot erase the systematic, recession-clustered component of default and the countercyclical collapse of recoveries.

High-yield analysis centers on leverage, coverage, free-cash-flow conversion, the maturity wall, liquidity runway, and covenant quality (incurrence vs maintenance, and the late-cycle drift toward cov-lite). Distressed debt is the tail: securities trading at deep discounts where analysis shifts from coupons to enterprise value and the absolute-priority waterfall. The signature move is identifying and buying the *fulcrum security* — the layer where value breaks and which converts to the reorganized equity — a loan-to-own strategy whose return is the gap between the distressed price and the restructuring recovery. All of this pulses with the credit cycle: spreads, defaults, and recoveries swing together through repair, expansion, complacency, and downturn, so the *timing* within the cycle matters as much as the *asset class*.

Figure 17.3 — The credit cycle and its four phases.

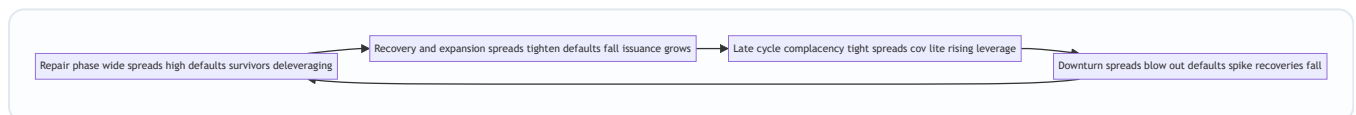
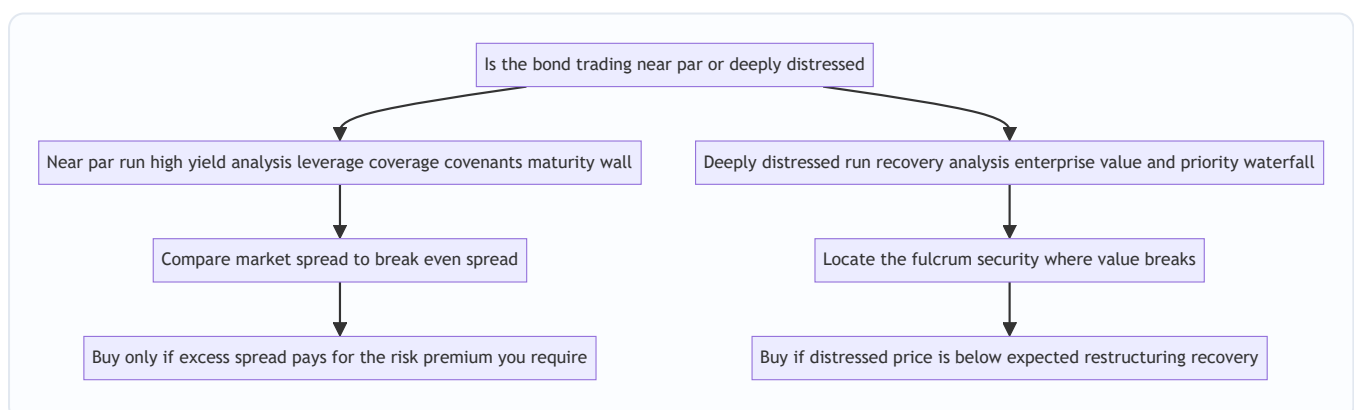


Figure 17.4 — Decision flow for evaluating a high-yield or distressed opportunity.



## 10. Quick-Reference / Interview Points

**One-liners to have ready:** - "High yield pays you the promised yield minus expected credit loss — the spread is compensation for PD times LGD plus a risk premium, not free money." - "The credit triangle:  $\text{spread} \approx \text{default probability} \times (1 - \text{recovery})$ . Market spread above that is the premium you harvest." - "Credit is actuarial — you

*earn the coupon on survivors and eat the losses on defaulters; diversification kills idiosyncratic risk but not the recession-clustered systematic risk, which is what the premium pays for."*

**Key formulas:** - Break-even spread:  $s \approx \lambda(1 - RR)$  - Cumulative default:  $CDP(T) = 1 - (1 - \lambda)^T$ , survival  $\approx e^{(-\lambda T)}$  - Expected excess return:  $E[\text{excess}] \approx s - \lambda(1 - RR)$

**Numbers to know (rough US long-run averages):** - IG/HY boundary: BBB-/Baa3 vs BB+/Ba1. - Long-run HY default rate: ~3-4%/yr average; ~1-2% in benign years; 10%+ at recession peaks. - Senior unsecured recovery: ~40 cents; secured ~60-80; subordinated ~20-30. - "Distressed" convention: spread > 1000 bps or price < ~70.

**Analysis checklist for a HY name:** 1. Leverage (Debt/EBITDA) and trend. 2. Interest / fixed-charge coverage. 3. Free-cash-flow conversion and liquidity runway. 4. Maturity wall and refinancing risk. 5. Covenant package (incurrence vs maintenance, cov-lite?). 6. Position in capital structure / seniority. 7. Spread vs break-even — am I paid for the risk?

**Distressed checklist:** 1. Estimate reorganized enterprise value. 2. Map the priority waterfall (secured → senior → sub → equity). 3. Identify the fulcrum security. 4. Compare distressed price to expected recovery; stress the EV. 5. Assess process risk (in-court vs out-of-court, creditor dynamics, timeline).

**Common interview traps:** - Confusing promised yield with expected return. - Forgetting that PD and LGD are positively correlated (recoveries fall when defaults spike). - Treating recovery as a constant 40% regardless of seniority. - Claiming HY has high rate duration — it is credit-dominated with *lower* effective duration than same-maturity IG. - Thinking distressed means already defaulted — it means trading at distressed levels, often pre-default. - Missing that cov-lite proliferation is a late-cycle red flag that lowers future recoveries.

**The mental model to leave with:** A high-yield bond is a *short put on the firm's assets*. You collect a premium (the spread) in exchange for absorbing losses if the firm's value falls below its debt. In good times you clip the premium; in bad times the put is exercised against you across many names at once. Distressed investing is what happens once that put is deep in the money — the game becomes valuing the wreckage and standing in the right place in the priority line to claim it.

# Q&A — High-Yield and Distressed Debt

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## Section A — Concept Check

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### A1. Where is the line between investment grade and high yield, and why does it matter so much?

The line sits at BBB–/Baa3 (last IG notch) versus BB+/Ba1 (first HY notch). It matters far more than any other adjacent-notch boundary because of *institutional plumbing*: many insurers, pension funds, and index mandates are restricted by regulation or contract to investment-grade paper. When an issuer is downgraded across the line (a "fallen angel"), a whole class of natural buyers is forced to sell. That forced selling, layered on top of genuine credit deterioration, makes the yield jump across BBB–/BB+ much larger than the jump between any two IG notches.

### A2. Why can't you treat the promised yield as the expected yield in high yield?

Because default is a base-rate event, not a tail event. Roughly 3–5% of HY issuers default in an average year and 10%+ in a bad one. The promised yield only materializes if the company survives; the *expected* return is a probability-weighted blend of "you get paid the coupon" and "you fight over the wreckage in bankruptcy." So expected return = promised yield minus expected credit loss, not the promised yield itself.

### A3. State the credit-triangle relationship and explain what it represents.

$s_{\text{breakeven}} \approx \lambda \times (1 - RR)$ , where  $\lambda$  is the annual default probability (hazard rate) and  $RR$  is the recovery rate. It is the *break-even* (fair) spread that exactly covers expected credit loss and delivers zero risk premium. Any two of {spread, default rate, recovery} pin down the third. The amount the *market* spread exceeds this break-even is the credit risk premium plus liquidity compensation you are actually being paid to harvest.

### A4. Decompose the credit spread into its components.

$s \approx \text{Expected Loss} + \text{Credit Risk Premium} + \text{Liquidity Premium}$ , where Expected Loss =  $PD \times LGD = \lambda \times (1 - RR)$ . Expected loss is the break-even floor; the risk premium compensates for the uncertainty and correlation (recession-clustering) of losses; the liquidity premium compensates for hard-to-trade paper. In HY, all three pieces are large and all move a lot.

### A5. Why are default losses only partly diversifiable?

Diversification removes *idiosyncratic* (single-name) risk, so a large book's loss converges to the average default rate. But it does not remove *systematic* risk: defaults cluster in recessions, and recoveries fall at exactly the same time (PD and LGD are positively correlated). A well-diversified HY book still suffers large correlated losses in a downturn — and that residual systematic risk is precisely what the credit risk premium pays for.

### A6. What is the fulcrum security, and why does the distressed investor care about it?

The fulcrum security is the layer of the capital structure where enterprise value "runs out" — the claim that is only partly covered and therefore converts partly to cash/new debt and partly to the *new equity* of the reorganized company. It is the target of loan-to-own investing: buy it cheaply and emerge owning the recapitalized business. Locating it correctly is the whole game, because the layer just above it recovers 100% and the layer just below it may get nothing.

### A7. Why is high yield typically less rate-sensitive than same-maturity investment grade?

Two reasons. First, the spread is so large that credit dynamics swamp small rate moves. Second, in risk-off episodes HY spreads *widen* while Treasuries *rally*, so the two effects partly offset — pushing HY's effective duration below that of same-maturity IG. HY's dominant risk is spread/credit, not rates; deep in distress it trades on enterprise value and behaves almost like equity.

#### **A8. Contrast incurrence and maintenance covenants, and explain why cov-lite is a late-cycle warning.**

An *incurrence* covenant is tested only when the company takes an action (issuing debt, paying a dividend). A *maintenance* covenant is tested every period regardless of action (e.g., leverage must stay below a threshold each quarter). Cov-lite loans strip out maintenance tests. This matters because a slowly deteriorating business can burn cash for quarters without tripping any incurrence test — the maintenance test would have caught it. Cov-lite prevalence rises in frothy markets, shifts leverage from lenders to sponsors, and tends to lower eventual recoveries; it is a classic late-cycle red flag.

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## **Section B — Numerical / Applied**

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**B1. Break-even spread and risk premium.** A single-B bond trades at 500 bps over Treasuries. For its rating,  $\lambda = 4\%$  and  $RR = 40\%$ .

Solution.  $LGD = 1 - 0.40 = 0.60$ . Break-even spread  $= \lambda \times LGD = 0.04 \times 0.60 = 0.024 = \mathbf{240 \text{ bps}}$ . Risk premium + liquidity comp = market – break-even =  $500 - 240 = \mathbf{260 \text{ bps}}$ . Expected one-year excess return over Treasuries  $\approx s - \lambda(1 - RR) = 500 - 240 = \mathbf{260 \text{ bps}}$ . You are being paid 260 bps above the expected-loss floor to bear the uncertainty and correlation of the risk.

**B2. Portfolio reconciliation of the excess return.** Take 100 identical B bonds, \$100 face, coupon = risk-free (4%) + 5% = 9%,  $\lambda = 4\%$ ,  $RR = 40\%$ . Verify the excess-return estimate.

Solution. - 96 survive: pay principal + coupon =  $96 \times \$109 = \$10,464$ . - 4 default: recover 40% of face =  $4 \times \$40 = \$160$  (assume they forfeit the coupon). - Total  $\approx \$10,624$  on \$10,000 invested  $\rightarrow$  return  $\approx +6.24\%$ . - Risk-free was 4%, so excess  $\approx \mathbf{+2.24\%}$ .

The linear formula gave 2.60%. The gap arises because defaulters also forfeit their coupon — a second-order effect the linear approximation omits, which makes the approximation slightly optimistic. The figures reconcile.

**B3. Cumulative default over five years.**  $\lambda = 4\%$  constant. Find the 5-year survival and cumulative default probabilities, discrete and continuous.

Solution. Discrete:  $\text{Survival}(5) = (1 - 0.04)^5 = 0.96^5 = 0.8154 \rightarrow \text{CDP}(5) = 1 - 0.8154 = \mathbf{18.5\%}$ . Continuous:  $\text{Survival}(5) \approx e^{(-\lambda T)} = e^{(-0.20)} = 0.8187 \rightarrow \text{CDP}(5) = \mathbf{18.1\%}$ . The two agree within a few tenths of a percent (as expected for small  $\lambda$ ). Nearly a one-in-five chance that a *single* B bond defaults within five years — which is why single-name concentration in HY is dangerous and the asset class is played as a diversified book.

**B4. The fulcrum security and waterfall.** Company XYZ, capital structure by face value: Senior secured \$300mm, Senior unsecured \$400mm, Subordinated \$200mm, Equity nil. Reorganized enterprise value = \$550mm. Find each layer's recovery and identify the fulcrum.

Solution (apply absolute priority): - Senior secured \$300mm  $\rightarrow$  fully covered, recovers 100% (\$300mm). Remaining:  $550 - 300 = \$250\text{mm}$ . - Senior unsecured \$400mm claim  $\rightarrow$  gets remaining \$250mm  $\rightarrow$  recovery =  $250/400 = \mathbf{62.5 \text{ cents}}$ . Remaining: \$0. - Subordinated \$200mm  $\rightarrow \mathbf{0 \text{ cents}}$ . Equity  $\rightarrow$  wiped out.

The **senior unsecured bond is the fulcrum** — value breaks here, so it converts partly to cash/new debt and partly to new equity.

**B5. The distressed trade and downside stress.** Senior unsecured from B4 trades at 45 cents. (a) Return if your \$550mm EV is right. (b) Return if EV is only \$400mm.

Solution. (a) Buy \$400mm face at 45 = \$180mm invested; recover \$250mm of value. Gain = \$70mm →  $70/180 = +38.9\%$ . Equivalently  $(62.5 - 45)/45 = +38.9\%$ . (b) At EV = \$400mm: secured takes \$300mm, leaving \$100mm for unsecured → recovery =  $100/400 = 25$  cents. Buying at 45 →  $(25 - 45)/45 = -44.4\%$ .

The asymmetry — large upside if the EV estimate is right, large loss if it is low — is why distressed investing is valuation-driven credit work, not coupon clipping. The edge is entirely in estimating enterprise value and reading the waterfall correctly.

**B6. Solving the triangle for an implied default rate.** A senior unsecured HY bond trades at 700 bps. Assume the entire spread is expected loss (zero risk premium) and RR = 35%. What annual default rate is implied? Then, if the true  $\lambda$  is only 5%, what excess return are you being paid?

Solution. Implied  $\lambda = s / (1 - RR) = 0.07 / 0.65 = 0.1077 = \sim 10.8\%$ . The market is pricing (under the no-premium assumption) roughly an 11% annual default rate. If true  $\lambda = 5\%$ : break-even =  $0.05 \times 0.65 = 0.0325 = 325$  bps. Expected excess  $\approx 700 - 325 = 375$  bps. You are being paid 375 bps above the expected-loss floor — attractive *if* your 5% default view is correct.

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## Section C — Interview Style

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**C1. "Walk me through why a wider spread is not automatically a better deal."**

Model answer. A wider spread reflects a higher perceived risk — more expected loss, more uncertainty, or less liquidity. Whether it is a good deal depends entirely on how it compares to the break-even spread implied by the true default and recovery outlook. Use the credit triangle: fair spread  $\approx \lambda \times (1 - RR)$ . If the market spread exceeds that break-even by more than the risk premium I require, it is cheap; if not, it is expensive despite looking fat. At the depth of a recession, spreads can be enormous and *still* not wide enough if defaults are about to spike, or so wide that they overpay. The question is never the spread level in isolation — it is spread *versus* break-even, and where we sit in the credit cycle.

**C2. "A bond yields 9%. Is that my expected return?"**

Model answer. No — 9% is the *promised* yield, the ceiling you reach only if nothing defaults. Expected return is promised yield minus expected credit loss = yield –  $\lambda \times (1 - RR)$ . If  $\lambda$  is 4% and recovery is 40%, expected loss is about 2.4%, so expected return is closer to 6.6%. Credit is actuarial: you earn the coupon on the survivors and eat the losses on the defaulters. Quoting the yield as the expected return is the single most common error in high yield.

**C3. "How would you analyze a high-yield name you'd never seen before?"**

Model answer. I run a structured checklist. (1) Leverage — Debt/EBITDA, gross and net, and its trend. (2) Coverage — interest coverage (EBITDA/interest) and fixed-charge coverage including capex and scheduled amortization. (3) Cash — FCF conversion (FCF/EBITDA) and liquidity runway (cash plus revolver availability over the burn rate). (4) The maturity wall — when does the debt come due and can it be refinanced? (5) The covenant package — incurrence versus maintenance, and is it cov-lite? (6) My position in the capital structure and seniority — what would I recover if it defaults? (7) Finally, spread versus break-even — am I

actually paid for the risk? The first six size the probability and severity of default; the last one tells me whether the price compensates me.

**C4. "Explain the mental model that a high-yield bond is a short put on the firm's assets."**

Model answer. As a lender I collect a premium — the spread — in exchange for absorbing losses if the firm's asset value falls below its debt. That payoff is exactly a short put on enterprise value: capped upside (I get my coupon and principal back), and a long tail of losses if the firm's value craters. In good times I clip the premium; in bad times the put is exercised against me across many names at once, because defaults cluster. This is the Merton view. Deep in distress the bond trades on enterprise value and behaves almost like equity — the put is deep in the money. Distressed investing is what happens then: the game becomes valuing the wreckage and standing in the right place in the priority line to claim it.

**C5. "Why do PD and LGD move together, and why does that make bad years worse?"**

Model answer. Defaults cluster in recessions, and recoveries are countercyclical: when many firms default at once, asset values are depressed, buyers are scarce, and forced sales happen into a weak market — so recoveries fall exactly when default rates spike. That positive correlation means a naive model treating PD and LGD as independent understates tail losses. In a bad year you get hit twice: more names default *and* each recovers less. This is a core reason the credit risk premium exists and why HY losses are lumpy rather than smooth.

**C6. "What is the difference between distressed and defaulted, and why does it matter for strategy?"**

Model answer. Defaulted means a payment was missed or a distressed exchange occurred. Distressed means the debt is *trading* at distressed levels — conventionally a spread above ~1000 bps or a price below ~70 — which frequently happens *before* any formal default and to companies that may never default. Much distressed investing is in bonds of stressed-but-not-defaulted issuers, betting either on a turnaround/refinancing or on a favorable restructuring outcome. The strategy differs: pre-default you may be underwriting a recovery or a coercive exchange; post-default you are negotiating a reorganization plan and locating the fulcrum. Conflating the two leads you to miss the pre-default opportunity set entirely.

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## Section D — MCQs with Reasoning

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**D1. The break-even (fair) credit spread is best approximated by:** A.  $\lambda + RR$  B.  $\lambda \times (1 - RR)$  C.  $\lambda \times RR$  D.  $(1 - \lambda) \times RR$

Answer: **B**. The credit triangle: break-even spread  $\approx$  hazard rate  $\times$  loss given default  $= \lambda \times (1 - RR)$ . A adds unrelated quantities; C multiplies by recovery instead of loss; D is not the expected-loss form.

**D2. Recoveries are described as countercyclical to default rates. This means:** A. Recoveries rise when defaults rise B. Recoveries and defaults are independent C. Recoveries fall when defaults rise D. Recoveries are fixed at 40%

Answer: **C**. When many firms default at once (recession), depressed asset values push recoveries down — PD and LGD are positively correlated, so bad years hit twice. A is backwards; B ignores the documented correlation; D is a modeling error (40% is only the long-run senior-unsecured average).

**D3. The fulcrum security is:** A. The most senior secured claim B. The layer where enterprise value runs out and which converts to new equity C. Always the subordinated notes D. The equity tranche

Answer: **B**. The fulcrum is where value "breaks" — partly covered, so it converts partly to new equity of the reorganized firm. The senior secured layer (A) is usually fully covered and recovers 100%; the fulcrum's identity depends on EV versus the waterfall, so it is not fixed at any layer (C, D wrong).

**D4. Relative to same-maturity investment-grade bonds, high-yield bonds typically have:** A. Higher effective duration B. Lower effective duration C. Identical duration D. Negative duration always

Answer: **B**. Credit dynamics dominate and, in stress, HY spreads widen while Treasuries rally, partly offsetting rate moves — lowering effective duration below same-maturity IG. HY's main risk is spread/credit, not rates.

**D5. Under the absolute priority rule, the correct payment order is:** A. Equity → subordinated → senior unsecured → secured B. Secured → senior unsecured → subordinated → equity C. Senior unsecured → secured → equity → subordinated D. Pro-rata across all claims equally

Answer: **B**. Secured creditors are paid first, then senior unsecured, then subordinated, with equity last (usually wiped out). The others violate the priority waterfall.

**D6. Cov-lite loan structures are a late-cycle warning sign primarily because they:** A. Add maintenance tests that trip too often B. Remove maintenance tests, delaying lender intervention and lowering recoveries C. Raise coupons for lenders D. Guarantee change-of-control puts

Answer: **B**. Cov-lite strips *maintenance* covenants (tested every period), so a deteriorating business can burn cash without tripping a test; lenders intervene later and recoveries tend to be lower. A misdescribes the mechanism; C and D are unrelated to the cov-lite definition.

**D7. A bond yields 10%, with  $\lambda = 5\%$  and  $RR = 30\%$ . Its approximate expected return is:** A. 10.0% B. 8.5% C. 6.5% D. 3.5%

Answer: **C**. Expected loss =  $\lambda \times (1 - RR) = 0.05 \times 0.70 = 3.5\%$ . Expected return  $\approx$  yield – expected loss =  $10\% - 3.5\% = 6.5\%$ . A is the promised yield (the trap); B and D use wrong loss figures.

**D8. "Distressed" debt is conventionally defined by which market signal?** A. Spread below 300 bps B. Price above par C. Spread above ~1000 bps or price below ~70 D. Rating of BBB–

Answer: **C**. The market convention for distressed is a spread wider than roughly 1000 bps or a price below about 70 — a level that can occur before any formal default. A and B describe healthy credit; D is the last investment-grade notch.



# Chapter 18 — International and Emerging-Market Bonds

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## 1. The Problem / The Need

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Every bond you have studied so far lived inside a single, tidy world: one issuer, one currency, one legal system, one yield curve. A US Treasury pays dollars under US law; an Indian government bond (G-Sec) pays rupees under Indian law. As long as you invest in your own country, in your own currency, the only questions are duration, credit and liquidity.

But the moment you step across a border, three new problems appear at once, and they are the reason this chapter exists.

**Problem 1 — The home market is too small or too concentrated.** An Indian pension fund that only buys G-Secs is betting its entire fixed-income book on one government, one central bank, one currency and one business cycle. A German insurer holding only Bunds earns near-zero yield. To earn more, or simply to spread risk, capital *must* travel. Roughly 60% of the world's investment-grade bonds are issued outside the United States; ignoring them means ignoring most of the asset class.

**Problem 2 — Issuers need capital that their home market cannot supply.** A Brazilian utility may need billions of dollars that the domestic real market simply does not have the depth to lend at a sensible tenor. A sovereign like Nigeria or Sri Lanka may find that global investors will lend in dollars but not in the local currency. So issuers reach *outside* their borders — and in doing so, they choose which currency and which legal jurisdiction to borrow under. Those choices create the whole taxonomy of eurobonds, foreign bonds and hard-currency sovereigns.

**Problem 3 — Returns now depend on the exchange rate, not just the coupon.** A 12% Turkish lira government bond sounds spectacular next to a 4% Treasury. But if the lira falls 25% against the dollar over the year, a dollar-based investor *loses* money despite the fat coupon. Cross-border fixed income welds together two risks that were previously separate: **interest-rate/credit risk** on the bond, and **currency risk** on the cash flows. Understanding — and pricing, and hedging — that second risk is the central skill of international bond investing.

The pay-off for solving these problems is large. Global bonds give diversification (different central banks ease and tighten at different times), higher yields (emerging markets pay a growth-and-risk premium), and access to the fastest-growing economies on earth. But the graveyard of fixed income is full of investors who bought a juicy foreign coupon and forgot they were also short a currency and long a sovereign that could default under a law they never read.

## 2. The Core Idea

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An international bond is defined by answering **three orthogonal questions**, and almost every confusing piece of jargon in this field is just a label for one combination of the answers:

1. **Who is the issuer, and where do they live?** (a US company, a Brazilian sovereign, a Japanese bank)
2. **What currency are the cash flows in?** (USD, EUR, the issuer's local currency)
3. **Under whose law and in whose market is it issued and regulated?** (domestic, foreign-domestic, or the offshore "euromarket")



From these three axes fall the core categories:

- A **domestic bond** — issuer, currency and market all the same country.
- A **foreign bond** — a foreign issuer sells into a *domestic* market, in *that market's* currency, under *that market's* rules (a Japanese firm selling USD bonds registered in the US = a **Yankee bond**).
- A **eurobond** — a bond issued in a currency *different* from the country where it is sold, in the loosely-regulated offshore market, typically under English law (a Brazilian firm selling USD bonds in London to global investors = a **eurodollar bond**). "Euro" here means *offshore*, not the euro currency — a eurobond can be in yen, dollars or anything.
- **Sovereign bonds** — issued by national governments; these split into **local-currency** debt (the sovereign borrows in its own money) and **hard-currency** debt (the sovereign borrows in USD or EUR).

The second core idea is the **currency decomposition of return**. For an unhedged foreign bond, your total return in your home currency is *approximately* the local return plus the currency move:

$$R_{\text{home}} \approx R_{\text{local}} + R_{\text{FX}}$$

Exactly, the two compound:

$$1 + R_{\text{home}} = (1 + R_{\text{local}}) \times (1 + R_{\text{FX}})$$

The third core idea is that **hard-currency and local-currency emerging-market debt are two completely different asset classes** that happen to share an issuer. Hard-currency EM debt is a *credit* bet (will the sovereign default on its dollars?). Local-currency EM debt is a *rates-plus-currency* bet (where will local yields and the exchange rate go?). They behave differently, are held by different investors, and default in different ways.

### 3. Why / How It Works

**Why does the euromarket exist at all?** History and regulation. In the 1960s, US regulations (Regulation Q interest-rate caps, the Interest Equalization Tax) made it unattractive to raise or hold dollars *inside* the US. Dollars held in European banks — "eurodollars" — could be lent out free of those rules. London became the hub of an offshore, lightly-regulated, bearer-bond market where issuers could raise money faster and cheaper, with less disclosure, and investors (often anonymous) avoided withholding tax. The regulatory arbitrage faded, but the market's efficiency, English-law standardisation and global investor base made it permanent. Today the eurobond market (formally the *international bond* market) is where most large cross-border issuance happens.

**Why do sovereigns borrow in a foreign currency at all**, given that a government controls its own printing press and could always print local money to repay local debt? Because for many emerging sovereigns, *global investors will not lend meaningful size in the local currency*. Currency risk, thin markets and a history of inflation scare foreign lenders away from, say, 10-year naira bonds. Issuing in dollars removes the currency risk *for the lender* — and transfers it onto the *borrower*. This is the famous "**original sin**" of emerging-market finance: the inability to borrow abroad in one's own currency forces sovereigns into hard-currency debt, which is exactly the debt they *cannot* print their way out of. That is why hard-currency sovereigns can — and do — default, while local-currency sovereigns rarely default outright (they inflate or devalue instead).

**How does currency risk actually bite?** Consider the mechanics. You are a dollar investor. You convert USD to local currency at today's spot rate, buy the bond, collect local coupons, and at the end convert local currency back to USD at the future spot rate. If the local currency has *depreciated* (you now get fewer dollars

per unit), your realised dollar return is dragged down — regardless of how the bond performed locally. The bond can rally, the coupon can be huge, and you can still lose in dollars. This is why the naive "carry" of a high local yield is only half the story: high-yielding currencies tend to be high-yielding *precisely because* the market expects them to depreciate.

**How is currency risk removed?** With an FX forward or an FX swap. **Covered interest rate parity (CIP)** says the forward exchange rate is pinned by the interest-rate differential — the forward already "prices in" the yield gap. So when you hedge, you lock in the forward rate and, in the process, you **give up the interest-rate differential**. The high foreign yield is (in theory) exactly offset by an unfavourable forward rate. Hedging converts a foreign bond back into something close to a home-currency instrument, isolating the *spread* and *duration* while neutralising the currency. This is the single most important practical insight of the chapter: **you cannot both hedge the currency and keep the yield pick-up — CIP takes the pick-up away**. What survives hedging is the *credit spread* and any *CIP basis*, not the raw yield difference.

The diagram below fixes the taxonomy before we go into the math.

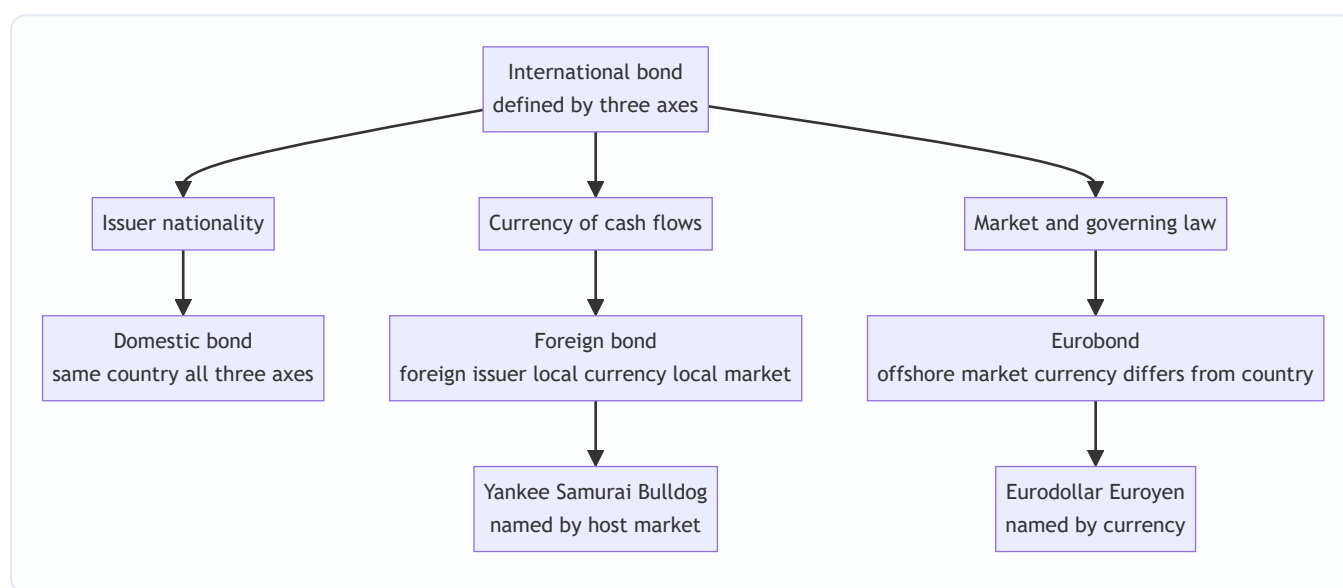


Figure 18.1 — The three defining axes of any cross-border bond and the names each combination earns.

## 4. Full Content — Instruments, Formulas and Frameworks

### 4.1 The naming zoo (foreign bonds)

Foreign bonds carry nicknames based on the *host* market they are sold into:

| Nickname           | Host market / currency             | Example issuer                                          |
|--------------------|------------------------------------|---------------------------------------------------------|
| Yankee bond        | USA / USD                          | European firm selling USD bonds registered with the SEC |
| Samurai bond       | Japan / JPY                        | Foreign issuer selling JPY bonds in Tokyo               |
| Bulldog bond       | UK / GBP                           | Foreign issuer selling GBP bonds in London              |
| Kangaroo / Matilda | Australia / AUD                    | Foreign issuer selling AUD bonds                        |
| Maple bond         | Canada / CAD                       | Foreign issuer selling CAD bonds                        |
| Panda / Dim sum    | China (onshore CNY) / offshore CNH | Foreign issuer selling renminbi bonds                   |

Contrast with **eurobonds**, named by *currency* not host market: **eurodollar**, **euroyen**, **eurosterling**. A eurobond is issued simultaneously across many countries, escapes any single national regulator, is usually in bearer form, and pays coupons gross (no withholding).

## 4.2 The two flavours of emerging-market debt

| Feature                                | Hard-currency EM debt             | Local-currency EM debt                    |
|----------------------------------------|-----------------------------------|-------------------------------------------|
| Currency                               | USD or EUR                        | Issuer's own currency                     |
| Primary risk for foreign investor      | Sovereign <b>credit</b> / default | <b>Currency</b> + local <b>rates</b>      |
| Governing law                          | Usually New York or English       | Local law                                 |
| Can the sovereign "print" its way out? | <b>No</b> — the killer feature    | Yes (inflate/devalue instead)             |
| Typical benchmark index                | JPMorgan <b>EMBI</b> Global       | JPMorgan <b>GBI-EM</b>                    |
| Default mode                           | Outright default / restructuring  | Rare default; inflation & FX loss instead |
| Recovery in distress                   | Legal restructuring, haircuts     | Erosion via devaluation                   |

## 4.3 Currency return decomposition

Let the spot exchange rate be quoted as **home currency per one unit of foreign currency** (\$\$ = HC/FC). If it rises, the foreign currency has *appreciated* (good for a home investor holding foreign assets).

$$R_{\text{FX}} = \frac{S_1 - S_0}{S_0} = \frac{S_1}{S_0} - 1$$

$$\boxed{1 + R_{\text{home}}} = (1 + R_{\text{local}})(1 + R_{\text{FX}})$$

$$R_{\text{home}} = R_{\text{local}} + R_{\text{FX}} + R_{\text{local}} \times R_{\text{FX}}$$

The cross-term  $R_{\text{local}} \times R_{\text{FX}}$  is small for modest moves but matters when either number is large (as in EM).

## 4.4 Covered Interest Rate Parity and the forward rate

The no-arbitrage forward exchange rate (again HC/FC) for tenor  $T$ :

$$\boxed{F_0 = S_0 \times \frac{1 + i_{\text{HC}}}{1 + i_{\text{FC}}}}$$

where  $i_{\text{HC}}$  and  $i_{\text{FC}}$  are the home and foreign interest rates for the period. If the foreign rate is *higher* ( $i_{\text{FC}} > i_{\text{HC}}$ ), then  $F_0 < S_0$ : the high-yield currency trades at a **forward discount**. Hedging a foreign bond means selling foreign currency forward at  $F_0$  — locking in that discount — which is exactly why the yield pick-up is cancelled.

#### Hedged return (approx.):

$$R_{\text{hedged}} \approx R_{\text{local}} - (i_{\text{FC}} - i_{\text{HC}}) = R_{\text{local}} - \text{(forward points as a rate)}$$

So a hedged foreign bond earns roughly the *local return minus the interest-rate differential*, i.e. you keep the bond's spread over the local risk-free rate, not its absolute yield.

### 4.5 Uncovered Interest Rate Parity (the theory that often fails)

UIP claims the market *expects* the high-yield currency to depreciate by exactly the rate differential, so expected hedged and unhedged returns are equal:

$$E[R_{\text{FX}}] \approx -(i_{\text{FC}} - i_{\text{HC}})$$

Empirically UIP fails at short horizons — high-yield currencies tend to *not* depreciate as much as UIP predicts, which is the source of the **FX carry trade** profit (and its periodic crashes). This gap between CIP (an arbitrage identity that holds) and UIP (an expectations theory that often does not) is a favourite interview probe.

### 4.6 Sovereign credit spread and default-recovery math

The **sovereign spread** is the extra yield of a hard-currency sovereign bond over the equivalent-maturity US Treasury (for USD debt) or Bund (for EUR debt):

$$\text{Spread} = y_{\text{sovereign}} - y_{\text{risk-free}}$$

A rough single-period breakeven linking spread, default probability  $p$  and recovery rate  $R$ :

$$\text{Spread} \approx p \times (1 - R) = p \times \text{LGD}$$

where LGD = loss given default. This lets you back out the market-implied default probability from a spread, given an assumed recovery.

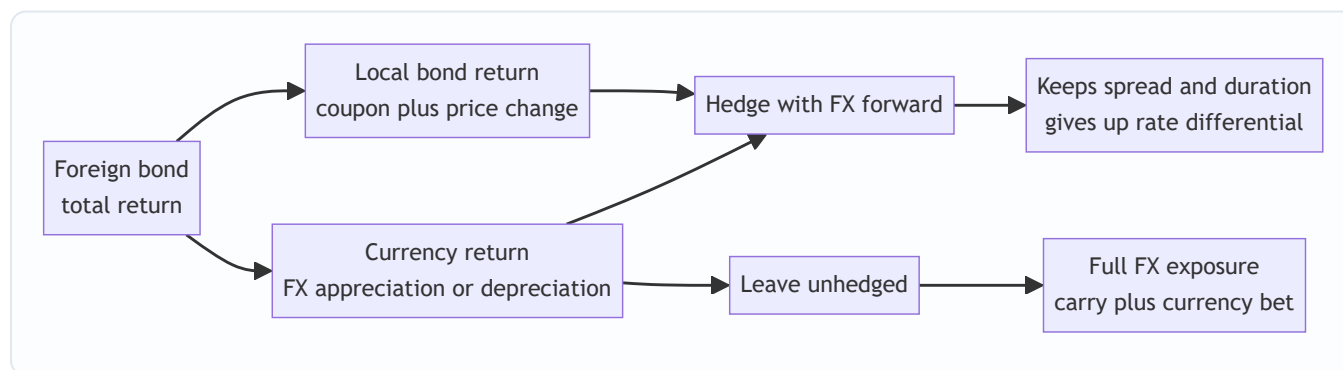


Figure 18.2 — How total return splits into local and currency components and what hedging keeps versus surrenders.

## 5. Worked Examples

### Example 1 — Unhedged vs hedged foreign bond return (the CIP reconciliation)

A US dollar investor buys a 1-year Brazilian local-currency (BRL) government note.

- Coupon / local yield held to the year:  $R_{\text{local}} = 11.00\%$  (assume bought and held, price flat, so local return = 11%).
- US 1-year rate:  $i_{\text{HC}} = 4.00\%$ .
- Brazil 1-year rate:  $i_{\text{FC}} = 11.00\%$ .
- Spot today:  $S_0 = 0.2000$  USD per BRL (i.e. 5.00 BRL per USD).

#### (a) The forward rate implied by CIP:

$$F_0 = 0.2000 \times \frac{1.04}{1.11} = 0.2000 \times 0.936937 = 0.187387 \text{ USD/BRL}$$

The BRL trades at a forward *discount* ( $0.18739 < 0.2000$ ), a drop of  $\frac{0.187387}{0.20} - 1 = -6.31\%$  — very close to the  $-(11\% - 4\%) = -7\%$  rate differential (the difference is the compounding/exact-vs-approx gap).

**(b) Hedged return.** Invest \$1,000. Convert to BRL:  $\$1,000 / 0.2000 = 5,000$  BRL. It grows at 11% to  $5,000 \times 1.11 = 5,550$  BRL at year-end. We sold BRL forward at  $F_0 = 0.187387$ . Convert back:

$$5,550 \times 0.187387 = 1,040.00 \text{ USD}$$

Return  $= 4.00\%$ . **This exactly equals the US rate of 4%** — confirming the core lesson: hedging away the currency leaves you with (essentially) the home risk-free rate, because CIP cancelled the 7% yield pick-up. (The BRL note here has no credit spread over the Brazil risk-free rate; if it had, say, +2% spread, the hedged return would be roughly 6%.)

**(c) Unhedged return, scenario where BRL falls 10%.** Suppose the spot at year-end is  $S_1 = 0.1800$  (BRL depreciated 10%). We still have 5,550 BRL, now converted at spot:

$$5,550 \times 0.1800 = 999.00 \text{ USD} \rightarrow R_{\text{home}} = -0.10\%$$

Check with the decomposition:  $R_{\text{FX}} = \frac{0.18}{0.20} - 1 = -10\%$ .  $\$1 + R_{\text{home}} = (1.11)(0.90) = 0.999 \rightarrow R_{\text{home}} = -0.10\% \checkmark$

The 11% coupon was more than wiped out by a 10% currency loss (plus the negative cross-term  $0.11 \times -0.10 = -1.1\%$ ). **Reconciliation:** local 11% + FX -10% + cross-term -1.1% = -0.1%. All three methods agree.

**(d) Unhedged, scenario where BRL rises 5%.**  $S_1 = 0.2100$ :  $\$1 + R_{\text{home}} = (1.11)(1.05) = 1.1655 \rightarrow R_{\text{home}} = 16.55\%$  Now the investor earns coupon *plus* currency gain — the upside of staying unhedged. The unhedged investor is long the bond *and* long the BRL.

### Example 2 — Backing out an implied default probability from a sovereign spread

A 5-year USD-denominated (hard-currency) sovereign eurobond issued by "Country X" yields 9.5%. The 5-year US Treasury yields 4.0%. Assume a recovery rate of 40% (LGD = 60%).

**Spread:**  $9.5\% - 4.0\% = 5.5\% = 550$  bps.

**Annual implied default probability (single-period approximation):**  $\text{Spread} \approx p \times \text{LGD} \rightarrow p \approx \frac{0.055}{0.60} = 9.17\% \text{ per year}$

**Cumulative 5-year survival** (assuming constant hazard, independence):  $\text{Survival} = (1 - 0.0917)^5 = (0.9083)^5 = 0.6182 \rightarrow \text{5-yr cumulative default} \approx 38.2\%$

Interpretation: the market is pricing roughly a 9% annual, ~38% cumulative chance that Country X defaults on its dollars over five years — consistent with a low-BB / high-B rating. Note this is a *risk-neutral* probability (embeds a risk premium), so it overstates the true real-world default odds; that gap is the sovereign risk premium the investor is paid to bear.

**Sensitivity check:** if we'd assumed a harsher 25% recovery (LGD = 75%),  $p \approx 0.055/0.75 = 7.33\%$  — lower default probability for the same spread, because each default now costs more. Recovery assumptions and default probabilities trade off against each other for a given observed spread; you cannot pin one without assuming the other.

### Example 3 — Hard-currency vs local-currency default behaviour

Country Y has both a 10% local-currency 1-year bond and an 8% USD 1-year bond outstanding. A crisis hits.

- **Local-currency bond:** The government does *not* default. Instead the central bank prints money; inflation runs to 30% and the currency devalues 25% against the dollar. A *domestic* holder is repaid in full in nominal local terms (10% coupon returned) but loses ~20% of *real* purchasing power. A *dollar* holder converts back at the devalued rate:  $\$1 + R = (1.10)(0.75) = 0.825 \rightarrow -17.5\%$  — a large loss, but *not* a legal default; no restructuring, no missed payment.
- **USD bond:** The government *cannot print dollars*. To pay 8% in USD it needs reserves it may not have. If reserves run dry, it misses the coupon → **legal default** → restructuring, litigation under New York law, and a haircut. Suppose bondholders recover 55 cents on the dollar: the dollar holder's loss is ~45% of principal.

**Reconciliation / lesson:** Same sovereign, same crisis, two utterly different loss mechanisms. Local-currency loss came through *inflation and FX* (no default event); hard-currency loss came through *outright default and restructuring*. This is precisely why credit-rating agencies assign a *higher* rating (lower default risk) to a sovereign's **local-currency** debt than to its **foreign-currency** debt — the government can always print its own money but never someone else's.

## 6. Connections

- **To duration and yield (Chapters on bond math):** everything about price sensitivity to rates still holds *within* the local market. A foreign bond just adds an FX layer on top of the duration/convexity you already know. Hedged foreign bonds are, in effect, a duration/spread instrument in your home currency.
- **To credit analysis (corporate credit chapters):** sovereign credit uses the same spread → default-probability → recovery logic, but the *causes* differ — willingness to pay (a political choice) matters as much as ability to pay, and there is no bankruptcy court for a country.
- **To derivatives (FX forwards, CDS, currency swaps):** hedging and CIP are pure applications of forward pricing; sovereign CDS lets you isolate default risk from currency risk directly.
- **To macro / monetary policy:** local-currency EM returns are ultimately a bet on a foreign central bank's inflation-fighting credibility and on the country's balance of payments.
- **To portfolio construction (Chapter on diversification):** global bonds lower portfolio volatility because rate cycles are imperfectly correlated across countries — a US recession-driven rally in Treasuries can coincide with an EM sell-off, or the reverse.

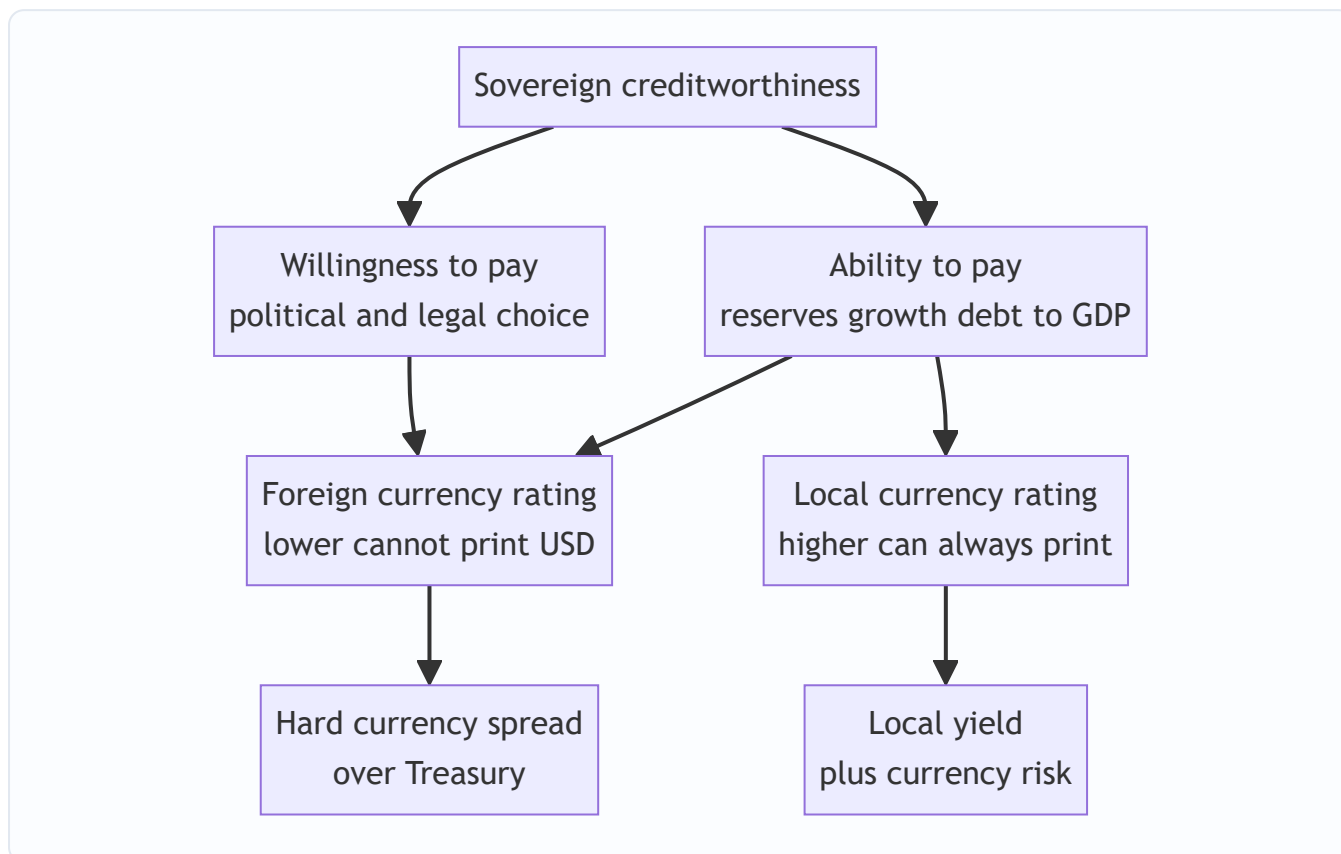


Figure 18.3 — Why one sovereign carries two different ratings and two different risk channels.

## 7. Key Terms

- **Eurobond:** a bond issued in a currency different from the country where it is sold, in the offshore international market, outside any single national regulator. "Euro" = offshore, not the euro currency.
- **Foreign bond:** a bond sold by a foreign issuer into a *domestic* market, in that market's currency and under its rules (Yankee, Samurai, Bulldog, Panda).
- **Sovereign bond:** debt issued by a national government.
- **Hard-currency debt:** EM/sovereign debt denominated in USD or EUR; a pure credit bet; the sovereign cannot print the currency.
- **Local-currency debt:** debt in the issuer's own currency; a rates-plus-FX bet; rarely defaulted, often inflated/devalued away.
- **Original sin:** an emerging economy's inability to borrow abroad in its own currency, forcing it into hard-currency debt.
- **Covered interest rate parity (CIP):** arbitrage identity fixing the forward FX rate to the interest-rate differential; makes hedged carry disappear.
- **Uncovered interest rate parity (UIP):** the (often-violated) expectation that high-yield currencies depreciate by the rate differential.
- **FX carry trade:** borrowing a low-yield currency to invest in a high-yield one, profiting when UIP fails.
- **Sovereign spread:** yield of a hard-currency sovereign bond over the matched-maturity risk-free (Treasury/Bund).
- **EMBI / GBI-EM:** JPMorgan's benchmark indices for hard-currency and local-currency EM debt respectively.



- **Brady bonds:** 1990s restructured, collateralised EM sovereign bonds; historical ancestor of today's EM hard-currency market.
- **Pari passu / collective action clause (CAC):** legal clauses in sovereign bonds governing equal treatment and how a supermajority of holders can bind all to a restructuring.
- **Forward discount/premium:** a currency trading below/above spot in the forward market; high-yield currencies trade at a forward discount.

## 8. Common Confusions

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1. **"Eurobond means a bond in euros."** No. A eurobond is any bond issued *offshore* in a currency foreign to the host market. A USD bond sold in London is a eurodollar bond; a yen bond sold offshore is a euroyen bond. The euro currency is irrelevant to the term.
2. **"High foreign yield = high return."** Only if the currency cooperates. CIP means that if you *hedge*, the yield pick-up vanishes and you keep only the spread. If you *don't* hedge, the high yield is compensation for expected depreciation. The 12% coupon and the currency loss are two sides of the same coin.
3. **"You can hedge the currency and still pocket the yield differential."** You cannot — that would be an arbitrage, and CIP forbids it (up to a small "cross-currency basis"). Hedging locks in the forward rate, which already bakes in the differential.
4. **"Sovereigns can't default because they print money."** True *only* for local-currency debt. Hard-currency (USD/EUR) sovereign debt cannot be printed and *is* regularly defaulted on — Argentina, Russia (1998), Sri Lanka (2022), Ghana, Zambia. This is why foreign-currency ratings sit below local-currency ratings.
5. **"Local-currency EM debt is safer because there's no default."** It rarely *defaults*, but it can inflict just-as-large losses through devaluation and inflation. A 25% currency crash hurts a dollar investor as much as a moderate default haircut — it just isn't called "default."
6. **"EM hard-currency and EM local-currency debt are basically the same trade."** They are different asset classes with different risk drivers (credit vs FX+rates), different benchmarks (EMBI vs GBI-EM), different investors, and low correlation between them. A portfolio can be long one and short the other.
7. **"Currency risk is diversified away in a global bond portfolio."** Currency exposure is *not* zero-mean over relevant horizons and adds volatility. Many global bond mandates deliberately hedge currency back to the base currency to isolate the diversification benefit of *rates* without importing FX noise.

## 9. Recap

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- Any cross-border bond is defined by three axes: **issuer nationality, currency, and market/law**. Foreign bonds (Yankee/Samurai) use the host market's currency and rules; eurobonds are offshore and named by currency.
- A foreign bond's home-currency return **compounds** the local return with the currency return:  $\$1 + R_{\text{home}} = (1 + R_{\text{local}})(1 + R_{\text{FX}})\$$ . The currency term can dominate.
- **CIP** pins the forward rate to the rate differential, so **hedging removes the currency risk but also removes the yield pick-up** — you keep spread and duration, not the raw yield. UIP (that FX moves offset the differential in expectation) often fails, which is why the carry trade exists.
- Emerging-market debt comes in two very different forms: **hard-currency** (a credit/default bet the sovereign can't print away — "original sin") and **local-currency** (a rates-plus-FX bet that rarely defaults).



but can devalue).

- Sovereign risk blends **ability** and **willingness** to pay; agencies rate **foreign-currency debt below local-currency debt** because a government can always print its own money.
- Global bonds earn their place through **diversification and yield**, but only if the investor consciously manages the currency and sovereign layers they inherit.

## 10. Quick-Reference / Interview Points

### Core formulas

| Concept                          | Formula                                                                        |
|----------------------------------|--------------------------------------------------------------------------------|
| Home-currency total return       | $1 + R_{\text{home}} = (1 + R_{\text{local}})(1 + R_{\text{FX}})$              |
| FX return ( $S$ = HC per FC)     | $R_{\text{FX}} = S_1/S_0 - 1$                                                  |
| CIP forward rate                 | $F_0 = S_0(1 + i_{\text{HC}})/(1 + i_{\text{FC}})$                             |
| Hedged return (approx)           | $R_{\text{hedged}} \approx R_{\text{local}} - (i_{\text{FC}} - i_{\text{HC}})$ |
| Spread $\leftrightarrow$ default | $\text{Spread} \approx p(1 - R) = p \times \text{LGD}$                         |
| Implied default prob             | $p \approx \text{Spread} / \text{LGD}$                                         |

### One-liners for the interview

- "Why doesn't hedging a 12% Brazilian bond leave me with 12%?" Because CIP makes the BRL trade at a forward discount equal to the rate gap; selling BRL forward gives back the pick-up, leaving me roughly my home rate plus any spread.
- "What's the difference between hard- and local-currency EM debt?" Hard-currency is a *credit* bet — the sovereign can't print USD, so it can default (original sin). Local-currency is a *rates + FX* bet — it rarely defaults but devalues/inflates. Different indices: EMBI vs GBI-EM.
- "Why is a country's local-currency rating higher than its foreign-currency rating?" It can always print its own money to service local debt; it cannot print dollars.
- "What is 'original sin'?" An EM economy's inability to borrow abroad in its own currency, forcing it into hard-currency debt it can't inflate away — the root cause of EM default crises.
- "CIP vs UIP?" CIP is an arbitrage identity that holds (forward = spot  $\times$  rate ratio). UIP is an expectations theory (FX moves offset the differential) that empirically fails at short horizons — hence the carry trade.
- "Is a eurobond in euros?" No — it's an offshore bond in any currency foreign to the host market; "euro" means offshore.
- "How do you get a market-implied default probability?" Take the sovereign spread, divide by LGD:  $p \approx \text{spread} / (1 - \text{recovery})$ . Remember it's risk-neutral, so it overstates true odds.
- "When would you leave EM currency unhedged?" When you have a positive view on the currency, want the full carry, and can tolerate the volatility — or when hedging costs (the rate differential) exceed the expected depreciation, i.e. you believe UIP fails in your favour.

**Numbers worth remembering:** ~60% of global IG bonds are issued outside the US; EM hard-currency sovereigns default outright while local-currency ones devalue; hedged carry  $\approx 0$  excess by CIP; recovery on sovereign defaults historically clusters around 30–55 cents.

# Q&A — International and Emerging-Market Bonds

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A practice bank for the concept chapter *International and Emerging-Market Bonds*. Work each question before reading the answer. Formulas used throughout:

- Home return:  $\$1 + R_{\text{home}} = (1 + R_{\text{local}})(1 + R_{\text{FX}})\$$ , with  $R_{\text{FX}} = S_1/S_0 - 1$  and  $\$ \$$  = home currency per unit of foreign currency.
  - CIP forward:  $\$F_0 = S_0 \cdot (1 + i_{\text{HC}}) / (1 + i_{\text{FC}})\$$ .
  - Hedged return (approx.):  $R_{\text{hedged}} \approx R_{\text{local}} - (i_{\text{FC}} - i_{\text{HC}})\$$ .
  - Spread  $\leftrightarrow$  default:  $\text{Spread} \approx p \times (1 - R) = p \times \text{LGD}\$, so  $p \approx \text{Spread} / \text{LGD}\$$ .$
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## Section A — Concept Checks

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**A1. What three orthogonal questions define any international bond?** (1) *Who is the issuer and where do they live?* (2) *What currency are the cash flows in?* (3) *Under whose law and in which market is it issued and regulated?* Every piece of jargon — Yankee, eurodollar, hard-currency sovereign — is just a label for one combination of answers to these three axes.

**A2. Distinguish a foreign bond from a eurobond.** A *foreign bond* is issued by a foreign issuer *into a domestic market*, in *that market's currency*, under *that market's rules* (e.g. a Japanese firm selling SEC-registered USD bonds in the US = a Yankee bond). A *eurobond* is issued in the offshore international market in a *currency different from the country where it is sold*, escapes any single national regulator, is usually in bearer form, and pays coupons gross. Foreign bonds are named by *host market* (Yankee, Samurai, Bulldog); eurobonds by *currency* (eurodollar, euroyen).

**A3. Does "eurobond" mean a bond denominated in euros?** No. "Euro" here means *offshore*, not the euro currency. A USD bond sold offshore in London is a eurodollar bond; a yen bond sold offshore is a euroyen bond. The euro currency is irrelevant to the term.

**A4. Name the host market and currency for: Yankee, Samurai, Bulldog, Kangaroo, Maple, Panda.**

Yankee = USA/USD; Samurai = Japan/JPY; Bulldog = UK/GBP; Kangaroo (Matilda) = Australia/AUD; Maple = Canada/CAD; Panda = China onshore/CNY (its offshore cousin is the Dim sum bond in CNH).

**A5. Why does the euromarket exist historically?** Regulatory arbitrage. In the 1960s US rules — Regulation Q interest-rate caps and the Interest Equalization Tax — made raising/holding dollars *inside* the US unattractive. Dollars held in European banks ("eurodollars") could be lent free of those rules. London became the hub of a lightly regulated, bearer-form, withholding-free offshore market. The original arbitrage faded, but English-law standardisation, efficiency, and a global investor base made the market permanent.

**A6. Write the home-currency return decomposition and identify the cross-term.**  $\$1 + R_{\text{home}} = (1 + R_{\text{local}})(1 + R_{\text{FX}})\$$ , which expands to  $R_{\text{home}} = R_{\text{local}} + R_{\text{FX}} + R_{\text{local}} \times R_{\text{FX}}\$$ . The cross-term  $R_{\text{local}} \times R_{\text{FX}}\$$  is negligible for small moves but material in EM where both numbers can be large.

**A7. State CIP and explain why hedging destroys the yield pick-up.** Covered interest rate parity:  $\$F_0 = S_0(1 + i_{\text{HC}}) / (1 + i_{\text{FC}})\$$ . If the foreign rate is higher,  $\$F_0 < S_0\$$  — the high-yield currency trades at a *forward discount*. Hedging means selling the foreign currency forward at  $\$F_0\$$ , locking in that discount. The

unfavourable forward rate exactly offsets the higher coupon, so what survives is the bond's *spread and duration*, not its raw yield. You cannot hedge the currency **and** keep the pick-up — that would be arbitrage.

**A8. Contrast CIP and UIP.** CIP is an *arbitrage identity* that holds (forward = spot × rate ratio) enforced by no-arbitrage. UIP is an *expectations theory*: the market *expects* the high-yield currency to depreciate by exactly the rate differential,  $E[R_{FX}] \approx -(i_{FC} - i_{HC})$ . UIP empirically fails at short horizons — high-yield currencies do not depreciate as much as predicted — which is the source of FX carry-trade profits (and their periodic crashes).

**A9. Distinguish hard-currency from local-currency EM debt across five dimensions.** Currency: USD/EUR vs issuer's own. Primary risk: sovereign credit/default vs currency+local rates. Governing law: New York/English vs local. Can the sovereign print its way out? No (the killer feature) vs Yes (inflate/devalue). Benchmark: JPMorgan EMBI vs GBI-EM. Default mode: outright default/restructuring vs erosion by devaluation.

**A10. What is "original sin"?** An emerging economy's inability to borrow abroad in its own currency. Global investors will lend in dollars but not in size in local currency, forcing the sovereign into hard-currency debt — exactly the debt it *cannot* print its way out of. This is the root cause of EM default crises.

**A11. Why is a sovereign's local-currency rating higher than its foreign-currency rating?** A government can always print its own money to service local-currency debt, so default risk there is low. It can never print dollars or euros, so foreign-currency debt genuinely can — and does — default. Ability to pay is unconditional in local currency but constrained by reserves in hard currency.

**A12. Write the spread-to-default relationship and state whether the resulting probability is real-world.**  $\text{Spread} \approx p \times (1 - R) = p \times \text{LGD}$ , so  $p \approx \frac{\text{Spread}}{\text{LGD}}$ . The result is a *risk-neutral* probability: it embeds a risk premium and therefore *overstates* the true real-world default probability. The gap is the sovereign risk premium the investor is paid to bear.

**A13. Distinguish ability to pay from willingness to pay.** Ability to pay is economic — reserves, growth, debt/GDP, access to hard currency. Willingness to pay is political and legal — a sovereign may *choose* to default even when it could pay (there is no bankruptcy court to force a country). Sovereign credit blends both; corporate credit is mostly about ability.

**A14. What are EMBI and GBI-EM?** JPMorgan benchmark indices: the EMBI (Global) tracks *hard-currency* EM sovereign debt; the GBI-EM tracks *local-currency* EM government debt. Different indices signal that these are two distinct asset classes with different risk drivers.

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## Section B — Numerical / Applied

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**B1. CIP forward rate.** Spot  $S_0 = 0.2000$  \$ USD/BRL, US 1-yr rate 4%, Brazil 1-yr rate 11%. Find the 1-year forward and the forward discount on the BRL.

*Solution.*  $F_0 = 0.2000 \times \frac{1.04}{1.11} = 0.2000 \times 0.936937 = 0.187387$  \$ USD/BRL. Forward discount  $\% = \frac{0.187387}{0.2000} - 1 = -6.31\%$ , close to the  $-(11\% - 4\%) = -7\%$  rate differential (the difference is the exact-vs-approximate/compounding gap). The high-yield currency trades at a forward discount.

**B2. Hedged return.** Same data, and the BRL note returns 11% locally (bought at par, held one year, price flat). Invest \$1,000 fully hedged. What USD return results?

*Solution.* Convert:  $\$1,000/0.2000 = \$5,000$  BRL. Grow at 11%:  $\$5,000 \times 1.11 = \$5,550$  BRL. Convert back at the forward  $\$0.187387$ :  $\$5,550 \times 0.187387 = \$1,040.00$  USD  $\rightarrow$  **+4.00%**. This equals the US risk-free rate exactly: CIP cancelled the 7% pick-up. If the note carried a +2% spread over the Brazilian risk-free rate, the hedged return would be roughly 6%.

**B3. Unhedged, currency falls.** Same 5,550 BRL at year-end; spot moves to  $\$S_1 = 0.1800$  (BRL – 10%). USD return?

*Solution.*  $\$5,550 \times 0.1800 = \$999.00$  USD  $\rightarrow$  **–0.10%**. Check:  $\$R_{\{FX\}} = 0.18/0.20 - 1 = -10\%$ ;  $\$(1.11)(0.90) = 0.999 \rightarrow -0.10\%$ . Decomposition: local +11%, FX – 10%, cross-term  $\$0.11 \times -0.10 = -1.1\%$ ; sum = –0.1%. The fat coupon was more than wiped out by the currency loss.

**B4. Unhedged, currency rises.** Same position; spot moves to  $\$S_1 = 0.2100$  (BRL + 5%). USD return?

*Solution.*  $\$(1.11)(1.05) = 1.1655 \rightarrow$  **+16.55%**. Now the unhedged investor collects coupon *plus* currency gain — the upside of being long the bond and long the BRL. Contrast B2/B3/B4: hedging locked +4%; unhedged ranged from –0.1% to +16.55% depending purely on the exchange rate.

**B5. Implied default probability.** A 5-yr USD sovereign eurobond yields 9.5%; the 5-yr Treasury yields 4.0%; assume recovery 40% (LGD 60%). Find the annual and cumulative 5-year implied default probability.

*Solution.* Spread  $\$ = 9.5\% - 4.0\% = 5.5\% = 550$  bps. Annual  $\$p \approx 0.055/0.60 = 9.17\%$ . Cumulative survival (constant hazard)  $\$ = (1 - 0.0917)^5 = (0.9083)^5 = 0.6182$ , so 5-yr cumulative default  $\$ \approx 1 - 0.6182 = 38.2\%$ . This is a risk-neutral probability, so it overstates true odds; a ~9% annual figure is consistent with a low-BB/high-B rating.

**B6. Recovery sensitivity.** Same 550 bps spread but assume harsher 25% recovery (LGD 75%). What annual  $\$p$  now?

*Solution.*  $\$p \approx 0.055/0.75 = 7.33\%$  — lower than in B5 for the same spread, because each default now costs more (higher LGD). Lesson: recovery and default probability trade off for a given observed spread; you cannot pin one without assuming the other.

**B7. Dollar-investor loss on a devaluing local-currency bond.** Country Y's 1-yr local bond pays a 10% coupon; over the year the currency devalues 25% vs the dollar and the government does *not* default. What is the dollar holder's return, and is it a default?

*Solution.*  $\$R_{\{FX\}} = -25\%$ ;  $\$(1.10)(0.75) = 0.825 \rightarrow$  **–17.5%**. A large loss but *not* a legal default — no missed payment, no restructuring. The loss arrived through inflation and FX, not a credit event.

**B8. Cross-term materiality.** Compare the approximation  $\$R_{\{home\}} \approx R_{\{local\}} + R_{\{FX\}}$  with the exact figure when  $\$R_{\{local\}} = 20\%$  and  $\$R_{\{FX\}} = -20\%$ .

*Solution.* Approximation:  $\$20\% - 20\% = 0\%$ . Exact:  $\$(1.20)(0.80) = 0.96 \rightarrow$  **–4.0%**. The cross-term  $\$0.20 \times -0.20 = -4\%$  is the entire difference. At EM-scale moves the simple sum is badly misleading; always compound.

**B9. Back out the recovery.** A sovereign USD bond trades at a 700 bps spread and the market believes the annual default probability is 14%. What recovery rate is implied?

*Solution.*  $\$\text{Spread} = p \times \text{LGD} \rightarrow \text{LGD} = 0.07/0.14 = 0.50$ , so recovery  $\$R = 1 - \text{LGD} = 50\%$ . Confirms the three quantities — spread,  $\$p$ , recovery — are mutually determined; fix any two and the third follows.

**B10. Hedged return with a spread.** A foreign 1-yr bond yields  $R_{\text{local}}=8\%$ ; foreign risk-free 6%, home risk-free 4%. Estimate the hedged return using the approximation.

*Solution.*  $R_{\text{hedged}} \approx R_{\text{local}} - (i_{\text{FC}} - i_{\text{HC}}) = 8\% - (6\% - 4\%) = 6\%$ . Equivalently, home rate 4% plus the bond's 2% spread over the foreign risk-free ( $8\% - 6\%$ ). Hedging keeps the *spread over the local risk-free*, not the absolute yield.

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## Section C — Interview-Style

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**C1. "Why doesn't hedging a 12% Brazilian bond leave me with 12%?"** Because CIP forces the BRL to trade at a forward discount roughly equal to the rate gap. To hedge, I sell BRL forward at that discounted rate, which hands back the yield pick-up. I'm left with approximately my home risk-free rate plus whatever *spread* the bond earns over the Brazilian risk-free rate — not the headline 12%. Keeping both the hedge and the pick-up would be a pure arbitrage that CIP rules out (bar a small cross-currency basis).

**C2. "What's the difference between hard- and local-currency EM debt, and why does it matter for a portfolio?"** Hard-currency debt (USD/EUR) is a *credit* bet: the sovereign can't print the currency, so it can genuinely default — that's original sin, benchmark EMBI. Local-currency debt is a *rates-plus-FX* bet: the sovereign rarely defaults but inflates/devalues instead, benchmark GBI-EM. They have different risk drivers, different investor bases, and low correlation — so a portfolio can be long one and short the other, and you must size and hedge them separately.

**C3. "Sovereigns print their own money — so how can they default?"** They can always print *local* currency, so local-currency default is rare. But they cannot print dollars or euros. Hard-currency debt must be serviced from reserves or export earnings; when those run dry, the sovereign misses a coupon and defaults legally — Argentina, Russia 1998, Sri Lanka 2022, Ghana, Zambia. That's precisely why foreign-currency ratings sit below local-currency ratings for the same issuer.

**C4. "Local-currency EM debt never defaults — so it's safe, right?"** It rarely *defaults*, but that's not the same as safe. Losses arrive through devaluation and inflation instead. A 25% currency crash costs a dollar investor about as much as a moderate default haircut — it just isn't labelled "default." The risk didn't disappear; it changed channels from credit to currency.

**C5. "How do you extract a market-implied default probability, and what's the catch?"** Take the sovereign spread over the matched-maturity Treasury and divide by LGD:  $\frac{\text{spread}}{(1 - \text{recovery})}$ . The catch is threefold: it's a *risk-neutral* probability that overstates real-world odds because it embeds a risk premium; it's only as good as your recovery assumption (which trades off directly against  $\text{spread}$ ); and the single-period formula ignores term structure of hazard. Useful for relative value, not a literal forecast.

**C6. "When would you deliberately leave EM currency exposure unhedged?"** When I have a positive view on the currency and want the full carry, and can tolerate the volatility — or, more precisely, when I believe hedging costs (the rate differential CIP makes me give up) exceed the currency's *expected* depreciation. That's a bet that UIP fails in my favour: the high-yield currency won't fall as much as the forward implies. If I only want the diversification benefit of foreign *rates* without FX noise, I hedge instead.

**C7. "Explain 'original sin' and its consequence."** It's an EM economy's inability to borrow abroad in its own currency because foreign lenders won't take the currency risk in size. It forces the sovereign to issue hard-currency debt — the one kind it cannot inflate or print away. The consequence is that EM crises become

*default* crises rather than mere inflation episodes, because the government's ultimate escape valve (the printing press) doesn't work on dollar liabilities.

**C8. "Why does a global bond allocation help a domestic fixed-income portfolio, and what's the hidden cost?"** Rate cycles are imperfectly correlated across countries — different central banks ease and tighten at different times — so foreign bonds lower portfolio volatility, and EM adds a growth-and-risk yield premium. The hidden cost is that unhedged foreign bonds import currency risk, which is not zero-mean over relevant horizons and adds volatility. Many global mandates therefore hedge FX back to the base currency to capture the *rates* diversification without the FX noise, and treat any unhedged currency exposure as a separate, conscious position.

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## Section D — Multiple Choice (with reasoning)

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**D1.** A eurodollar bond is best described as: A) A bond denominated in euros sold in the US B) A USD bond issued offshore, outside any single national regulator C) A US Treasury sold to European investors D) A bond paying coupons in both euros and dollars

**Answer: B.** "Euro" means offshore, not the euro currency. A eurodollar bond is a USD-denominated bond issued in the international market outside US regulation, typically bearer form, coupons paid gross. A is the classic trap.

**D2.** A Brazilian company sells SEC-registered USD bonds in the US. This is a: A) Eurodollar bond B) Samurai bond C) Yankee bond D) Panda bond

**Answer: C.** A foreign issuer selling into the US domestic market in USD under US rules is a Yankee bond. If it were sold *offshore* in USD it would be a eurodollar bond; foreign bonds are named by host market, eurobonds by currency.

**D3.** A US investor holds a foreign bond that returned +6% locally while the foreign currency fell 8%. The home-currency return is closest to: A) -2.0% B) -2.5% C) +6.0% D) -14%

**Answer: B.**  $\$(1.06)(0.92)=0.9752\$ \rightarrow -2.48\% \approx -2.5\%$ . The simple sum (-2%) omits the cross-term  $\$0.06 \times -0.08 = -0.48\%$ ; at this scale it's the difference between A and the correct B.

**D4.** Under CIP, if the foreign interest rate exceeds the home rate, the foreign currency's forward rate (home per foreign) is: A) Above spot (forward premium) B) Equal to spot C) Below spot (forward discount) D) Undefined

**Answer: C.**  $\$F_0 = S_0(1+i_{\{HC\}})/(1+i_{\{FC\}})$ ; with  $i_{\{FC\}} > i_{\{HC\}}$  the ratio is  $<1$ , so  $\$F_0 < S_0$  — a forward discount. Selling the high-yield currency forward at this discount is exactly what erases the hedged carry.

**D5.** A fully hedged foreign bond position primarily retains exposure to: A) The foreign currency B) The interest-rate differential C) The bond's spread and duration D) Nothing — it earns zero

**Answer: C.** Hedging with an FX forward neutralises the currency and, via CIP, surrenders the rate differential, but the bond's credit *spread* over the local risk-free and its *duration* survive. It behaves like a home-currency spread/duration instrument (plus any small cross-currency basis).

**D6.** A 5-yr USD sovereign bond has a 600 bps spread over Treasuries; assumed recovery is 25%. The implied annual default probability is closest to: A) 6.0% B) 8.0% C) 24% D) 1.5%

**Answer: B.**  $\text{LGD} \$ = 1 - 0.25 = 0.75\$$ ;  $\$p \approx 0.06 / 0.75 = 8.0\%$ . Choice A wrongly divides by 1 (ignores LGD); C confuses spread/LGD roles.



**D7.** Which statement about UIP is correct? A) It is a no-arbitrage identity that always holds B) It says forward = spot × rate ratio C) It predicts high-yield currencies depreciate by the rate differential, and empirically fails short-term D) It is enforced by FX forwards

**Answer: C.** UIP is an *expectations* theory, not an identity (that's CIP, described in B). It predicts  $E[R_{\text{FX}}] \approx (i_{\text{FC}} - i_{\text{HC}})$  but fails at short horizons — the failure is what makes the carry trade profitable.

**D8.** Rating agencies typically assign a sovereign's local-currency debt a *higher* rating than its foreign-currency debt because: A) Local debt has shorter maturities B) The government can always print its own currency but not foreign currency C) Local debt is collateralised D) Foreign debt has lower recovery by law

**Answer: B.** The government's unconditional ability to print local money makes local-currency default rare; it cannot print dollars/euros, so foreign-currency debt carries genuine default risk and a lower rating. This is the ratings expression of original sin.

**D9.** A sovereign faces a crisis. Its local-currency bondholders are most likely to suffer through \_\_, while its USD bondholders suffer through \_\_\_\_\_. A) Default; inflation B) Inflation/devaluation; outright default and restructuring C) Both through outright default D) Both through devaluation

**Answer: B.** Same sovereign, same crisis, two different loss channels: local-currency holders lose via inflation and FX devaluation (usually no default event); USD holders face legal default, litigation under New York law, and a haircut.

**D10.** Which pairing of asset class and benchmark index is correct? A) Local-currency EM debt → EMBI Global B) Hard-currency EM debt → GBI-EM C) Hard-currency EM debt → EMBI Global; local-currency EM debt → GBI-EM D) Both use the same index

**Answer: C.** JPMorgan's EMBI Global tracks hard-currency (USD/EUR) EM sovereigns; the GBI-EM tracks local-currency EM government debt. Distinct indices for distinct asset classes with distinct risk drivers.

**D11.** The FX carry trade earns money primarily when: A) CIP is violated B) UIP holds exactly C) UIP fails and high-yield currencies don't depreciate as much as implied D) Both currencies are hedged

**Answer: C.** Borrowing a low-yield currency to invest in a high-yield one profits precisely when UIP fails — the high-yield currency doesn't fall by the full rate differential. CIP (A) is an arbitrage identity that generally holds; if UIP held exactly (B), expected carry profit would be zero.

**D12.** An investor wants the diversification benefit of foreign interest-rate cycles without importing exchange-rate volatility. The appropriate action is to: A) Buy unhedged local-currency EM debt B) Hedge the currency back to the base currency, isolating rates and spread C) Buy only domestic bonds D) Short the foreign currency and the bond

**Answer: B.** Currency-hedging back to base currency strips out FX noise while retaining the foreign bond's rates/spread exposure — the diversification objective. Unhedged EM (A) adds exactly the FX volatility they want to avoid.

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## Self-Verification Notes

B1–B4 reproduce Example 1 (forward 0.187387; hedged +4.00%; unhedged –0.10% and +16.55%). B5–B6 match Example 2 ( $\$p \approx 9.17\%$ , ~38.2% cumulative; 7.33% at 25% recovery). B7 matches Example 3 (–17.5%, non-default). B8–B10 and MCQ arithmetic (D3: 0.9752; D6: 0.08) independently re-checked.